"God made the integers, all else is the work of man."

Leopold Kronecker

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NUMBER THEORY
Structures, Examples, and Problems

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Foreword

One of the oldest and liveliest branches of mathematics, Number Theory, is noted for its theoretical depth and applications to other fields, including representation theory, physics, and cryptography. The forefront of Number Theory is replete with sophisticated and famous open problems; at its foundation, however, are basic, elementary ideas that can stimulate and challenge beginning students. This textbook takes a problem-solving approach to Number Theory, situating each theoretical concept within the framework of some examples or some problems for readers to solve. Starting with the essentials, the text covers divisibility, powers of integers, floor function and fractional part, digits of numbers, basic methods of proof (extremal arguments, pigeonhole principle, induction, infinite descent, inclusion-exclusion), arithmetic function, important divisibility theorems and Diophantine equations. Emphasis is also placed on the presentation of some special problems involving quadratic residues, Fermat, Mersenne, and perfect numbers, as well as famous sequences of integers such as Fibonacci, Lucas, and other important ones defined by recursive relations. By thoroughly discussing interesting examples and applications and by introducing and illustrating every key idea, by relevant problems of various levels of difficulty, the book motivates, engages and challenges the

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reader. The exposition proceeds incrementally, intuitively and rigorously uncovers deeper properties.

A special feature of the book is an outstanding selection of genuine Olympiad and other important mathematical contest problems solved using the methods already presented. The book brings about the unique and vast experience of the authors. It captures the spirit of an important mathematical literature and distills the essence of a rich problem-solving culture.

"Number Theory: Structures, Examples and Problems" will appeal to senior high school and undergraduate students, their instructors, as well as to all who would like to expand their mathematical horizons. It is a source of fascinating problems for readers at all levels and widely opens the gate to further explorations in mathematics.

Acknowledgments

Many problems are either inspired by or adapted from various mathematical contests in different countries. We express our deepest appreciation to the original proposers of the problems. Special thanks are given to Gabriel Dospinescu (Ecole Normale Superieure Paris, France) for the careful proof reading of the manuscript and for many helpful suggestions.

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10 Acknowledgments

Notation

$\mathbb{Z}$	the set of integers
$\mathbb{Z}_n$	the set of integers modulo n
$\mathbb{N}$	the set of positive integers
$\mathbb{N}_0$	the set of nonnegative integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{Q}^+$	the set of positive rational numbers
$\mathbb{Q}^0$	the set of nonnegative rational numbers
$\mathbb{Q}^n$	the set of n-tuples of rational numbers
$\mathbb{R}$	the set of real numbers
$\mathbb{R}^+$	the set of positive real numbers
$\mathbb{R}^0$	the set of nonnegative real numbers
$\mathbb{R}^n$	the set of n-tuples of real numbers
$\mathbb{C}$	the set of complex numbers
$\|A\|$	the number of elements in the set A
$A \subset B$	A is a proper subset of B
$A \subseteq B$	A is a subset of B
$A \setminus B$	A without B (set difference)
$A \cap B$	the intersection of sets A and B
$A \cup B$	the union of sets A and B
$a \in A$	the element a belongs to the set A

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$n\|m$	n divides m
$gcd(m,n)$	the greatest common divisor of m,n
$lcm(m,n)$	the least common multiple of m,n
$\pi(n)$	the number of primes $\leq n$
$\tau(n)$	number of divisors of n
$\sigma(n)$	sum of positive divisors of n
$a \equiv b \pmod{m}$	a and b are congruent modulo m
φ	Euler's totient function
$ord_m(a)$	order of a modulo m
μ	Möbius function
$\overline{a_k a_{k-1} \dots a_0}_{(b)}$	base b representation
$S(n)$	the sum of digits of n
$(f_1, f_2, \dots, f_m)$	factorial base expansion
$\lfloor x \rfloor$	floor of x
$\lceil x \rceil$	celling of x
$\{x\}$	fractional part of x
e_p	Legendre's function
$p^k \| n$	p^k fully divides n
f_n	Fermat's number
M_n	Mersenne's number
$\left(\frac{a}{p}\right)$	Legendre's symbol
F_n	Fibonacci's number
L_n	Lucas' number
P_n	Pell's number
$\binom{n}{k}$	binomial coefficient

Part I

STRUCTURES, EXAMPLES, AND PROBLEMS

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1

Divisibility

1.1 Divisibility

For integers a and b, $a \neq 0$, we say that a *divides* b if $b = ac$ for some integer c. We denote this by $a|b$. We also say that b is divisible by a or that b is a multiple of a.

Because $0 = a \cdot 0$, it follows that $a|0$ for all integers a, $a \neq 0$.

Straight from the definition we can derive the following properties:

1. If $a|b$, $b \neq 0$, then $|a| \leq |b|$;
2. If $a|b$ and $a|c$, then $a|\alpha b + \beta c$ for any integers α and β;
3. If $a|b$ and $a|b \pm c$, then $a|c$;
4. $a|a$ (reflexivity);
5. If $a|b$ and $b|c$, then $a|c$ (transitivity);
6. If $a|b$ and $b|a$, then $|a| = |b|$.

The following result is called the Division Algorithm and it plays an important role:

Theorem. *For any positive integers a and b there exists a unique pair (q, r) of nonnegative integers such that*

$$b = aq + r, \quad r < a.$$

Proof. If $a > b$, then $q = 0$ and $r = b < a$.

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If $a = b$, then $q = 1$ and $r = 0 < a$.

If $a < b$, then there exist positive integers n such that $na > b$. Let q be the least positive integer for which $(q+1)a > b$. Then $qa \leq b$. Let $r = b - aq$. It follows that $b = aq + r$ and $0 \leq r < a$.

For the uniqueness, assume that $b = aq' + r'$, where q' and r' are also nonnegative integers satisfying $0 \leq r' < a$. Then $aq + r = aq' + r'$, implying $a(q - q') = r' - r$, and so $a|r' - r$. Hence $|r' - r| \geq a$ or $|r' - r| = 0$. Because $0 \leq r,\ r' < a$ yields $|r' - r| < a$, we are left with $|r' - r| = 0$, implying $r' = r$ and, consequently, $q' = q$. □

In the theorem above, when a is divided by b, q is called the *quotient* and r the *remainder*.

Remark. The Division Algorithm can be extended for integers as follows: For any integers a and b, $a \neq 0$, there exists a unique pair (q, r) of integers such that

$$b = aq + r, \quad 0 \leq r < |a|.$$

Example. Prove that for all positive integers n, the fraction

$$\frac{21n + 4}{14n + 3}$$

is irreducible.

(1st IMO)

Indeed, from the equality

$$2(21n + 4) - 3(14n + 3) = -1$$

it follows that $21n + 4$ and $14n + 3$ have no common divisor except for 1, hence the conclusion.

Problem 1.1.1. *Prove that for all integers n:*

a) $n^5 - 5n^3 + 4n$ is divisible by 120;

b) $n^2 + 3n + 5$ is not divisible by 121.

Solution. a) $n^5 - 5n^3 + 4n = n(n^2 - 1)(n^2 - 4)$

$$= n(n - 1)(n + 1)(n - 2)(n + 2),$$

the product of five consecutive integers: $n - 2$, $n - 1$, n, $n + 1$, $n + 2$.

If $n \in \{-2, -1, 0, 1, 2\}$ we get $n^5 - 5n^3 + 4n = 0$ and the property holds.

If $n \geq 3$ we can write

$$n^5 - 5n^3 + 4n = 5!\binom{n+2}{5} = 120\binom{n+2}{5},$$

and the conclusion follows.

If $n \leq -3$, write $n = -m$, where $m \geq 3$, and obtain

$$n^5 - 5n^3 + 4n = -120\binom{m+2}{5},$$

and we are done.

b) Observe that

$$n^2 + 3n + 5 = (n+7)(n-4) + 33,$$

so that $11|n^2+3n+5$ if and only if $11|(n+7)(n-4)$. Thus, if $11 \nmid (n+7)(n-4)$ then 11 (and hence 121) does not divide n^2+3n+5. So, assume 11 divides $(n+7)(n-4)$. Then $11|n+7$ or $11|n-4$; but then 11 must divide both of $n+7$ and $n-4$, since $(n+7)-(n-4) = 11$. Thus, $121|(n+7)(n-4)$. However, $121 \nmid 33$. So $121 \nmid n^2+3n+5 = (n+7)(n-4)+33$. Hence, in all cases, $121 \nmid n^2+3n+5$.

Problem 1.1.2. *Let $p > 2$ be an odd number and let n be a positive integer. Prove that p divides $1^{p^n} + 2^{p^n} + \cdots + (p-1)^{p^n}$.*

Solution. Define $k = p^n$ and note that k is odd. Then

$$d^k + (p-d)^k = p[d^{k-1} - d^{k-2}(p-d) + \cdots + (p-d)^{k-1}]$$

Summing up the equalities from $d = 1$ to $d = \dfrac{p-1}{2}$ implies that p divides $1^k + 2^k + \cdots + (p-1)^k$, as claimed.

Problem 1.1.3. *Prove that*

$$3^{4^5} + 4^{5^6}$$

is a product of two integers, each of which is larger than 10^{2002}.

Solution. The given number is of the form $m^4 + \dfrac{1}{4}n^4$, where $m = 3^{4^4}$ and

$$n = 4^{\frac{5^6+1}{4}} = 2^{\frac{5^6+1}{2}}.$$

The conclusion follows from the identity

$$m^4 + \frac{n^4}{4} = m^4 + m^2n^2 + \frac{1}{4}n^4 - m^2n^2 = \left(m^2 + \frac{1}{2}n^2\right)^2 - m^2n^2 =$$

$$= \left(m^2 + mn + \frac{1}{2}n^2\right)\left(m^2 - mn + \frac{1}{2}n^2\right),$$

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where n is even so $\frac{n^2}{2}$ is an integer, and from the inequalities:

$$m^2 - mn + \frac{1}{2}n^2 = \left(m - \frac{n}{2}\right)^2 + \frac{n^2}{4} > \frac{n^2}{4}$$

$$= 2^{5^6-1} > 2^{10008} > (2^4)^{2002} > 10^{2002}.$$

Problem 1.1.4. *Find all positive integers n such that for all odd integers a, if $a^2 \leq n$ then $a|n$.*

Solution. Let a be the greatest odd integer such that $a^2 < n$, hence $n \leq (a+2)^2$. If $a \geq 7$, then $a-4, a-2, a$ are odd integers which divide n. Note that any two of these numbers are relatively prime, so $(a-4)(a-2)a$ divides n. It follows that $(a-4)(a-2)a \leq (a+2)^2$ so $a^3 - 6a^2 + 8a \leq a^2 + 4a + 4$. Then $a^3 - 7a^2 + 4a - 4 \leq 0$ or $a^2(a-7) + 4(a-1) \leq 0$. This is false, because $a \geq 7$, hence $a = 1, 3$ or 5.

If $a = 1$, then $1^2 \leq n \leq 3^2$, so $n \in \{1, 2, \ldots, 8\}$.

If $a = 3$, then $3^2 \leq n \leq 5^2$ and $1 \cdot 3|n$, so $n \in \{9, 12, 15, 18, 21, 24\}$.

If $a = 5$, then $5^2 \leq n \leq 7^2$ and $1 \cdot 3 \cdot 5|n$ so $n \in \{30, 45\}$. Therefore $n \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 15, 18, 21, 24, 30, 45\}$.

Problem 1.1.5. *Find the elements of the set*

$$S = \left\{ x \in \mathbb{Z} |\ \frac{x^3 - 3x + 2}{2x+1} \in \mathbb{Z} \right\}.$$

Solution. Since $\frac{x^3 - 3x + 2}{2x+1} \in \mathbb{Z}$, then

$$\frac{8x^3 - 24x + 16}{2x+1} = 4x^2 - 2x - 11 + \frac{27}{2x+1} \in \mathbb{Z}.$$

It follows that $2x + 1$ divides 27, so

$$2x + 1 \in \{\pm 1, \pm 3, \pm 9, \pm 27\} \text{ and } x \in \{-14, -5, -2, -1, 0, 1, 4, 13\},$$

since $2x + 1$ is odd, $\frac{x^3 - 3x + 2}{2x+1} \in \mathbb{Z} \Leftrightarrow \frac{8x^3 - 24x + 16}{2x+1} \in \mathbb{Z}$, so all these are solutions.

Problem 1.1.6. *Find all positive integers n for which the number obtained by erasing the last digit is a divisor of n.*

Solution. Let b be the last digit of the number n and let a be the number obtained from n by erasing the last digit b. Then $n = 10a + b$. Since a is a divisor of n, we infer that a divides b. Any number n that ends in 0 is

therefore a solution. If $b \neq 0$, then a is a digit and n is one of the numbers 11, 12,..., 19, 22, 24, 26, 28, 33, 36, 39, 44, 48, 55, 56, 77, 88 or 99.

Problem 1.1.7. *Find the greatest positive integer x such that 23^{6+x} divides 2000!.*

Solution. The number 23 is prime and divides every 23rd number. In all, there are $\left\lfloor \frac{2000}{23} \right\rfloor = 86$ numbers from 1 to 2000 that are divisible by 23. Among those 86 numbers, three of them, namely 23, $2 \cdot 23$ and $3 \cdot 23^2$ are divisible by 23^3. Hence $23^{89} | 2000!$ and $x = 89 - 6 = 83$.

Problem 1.1.8. *Find the positive integers n with exactly 12 divisors $1 = d_1 < d_2 < \cdots < d_{12} = n$ such that the divisor with index d_4 (that is, $d_{d_4} - 1$) is $(d_1 + d_2 + d_4)d_8$.*

(1989 Russian Mathematical Olympiad)

Solution. Of course, there is $1 \leq i \leq 12$ such that $d_i = d_1 + d_2 + d_4$. Since $d_i > d_4$, we have $i \geq 5$. Also, observe that $d_j d_{13-j} = m$ for all j and since $d_i d_8 = d_{d_4 - 1} \leq n$, we must have $i \leq 5$, thus $i = 5$ and $d_1 + d_2 + d_4 = d_5$. Also, $d_{d_4 - 1} = d_5 d_8 = n = d_{12}$, thus $d_4 = 13$ and $d_5 = 14 + d_2$. Of course, d_2 is the smallest prime divisor of n and since $d_4 = 13$, we can only have $d_2 \in \{2, 3, 5, 7, 11\}$. Also, since n has 12 divisors, it has at most 3 prime divisors. If $d_2 = 2$ then $d_5 = 16$ and then 4 and 8 are divisors of n, smaller than $d_4 = 13$, impossible. A similar argument shows that $d_2 = 3$ and $d_5 = 17$. Since n has 12 divisors and is a multiple of $3 \cdot 13 \cdot 17$, the only possibilities are $9 \cdot 13 \cdot 17$, $3 \cdot 1697$ or $3 \cdot 13 \cdot 289$. One can easily check that only $9 \cdot 13 \cdot 17 = 1989$ is a solution.

Problem 1.1.9. *Let n be a positive integer. Show that any number greater than $n^4/16$ can be written in at most one way as the product of two of its divisors having difference not exceeding n.*

(1998 St. Petersburg City Mathematical Olympiad)

First Solution. Suppose, on the contrary, that there exist $a > c \geq d > b$ with $a - b \leq n$ and $ab = cd > n^4/16$. Put $p = a + b$, $q = a - b$, $r = c + d$, $s = c - d$. Now

$$p^2 - q^2 = 4ab = 4cd = r^2 - s^2 > n^4/4.$$

Thus $p^2 - r^2 = q^2 - s^2 \leq q^2 \leq n^2$. But $r^2 > n^4/4$ (so $r > n^2/2$) and $p > r$, so

$$p^2 - r^2 > (n^2/2 + 1)^2 - (n^2/2)^2 \geq n^2 + 1,$$

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a contradiction.

Second solution. Again, suppose that $ab = cd > n^4/16$, with $a > c, d$ and $n \geq a - b$. If we let $p = gcd(a, c)$, we can find positive integers p, q, r, s such that $a = pq$, $b = rs$, $c = pr$, $d = qs$. Then $a > c \Rightarrow q > r$ and $a > d \Rightarrow p > s$, so that

$$\begin{aligned} n &\geq pq - rs \\ &\geq (s+1)(r+1) - rs \\ &= r + s + 1 \\ &\geq 2\sqrt{b} + 1. \end{aligned}$$

Thus $b \leq \left(\frac{n-1}{2}\right)^2 < n^2/4$, and $a \leq \left(\frac{n-1}{2}\right)^2 + n = \left(\frac{n+1}{2}\right)^2$. Therefore $ab \leq \left(\frac{n^2-1}{4}\right)^2 < n^4/16$, a contradiction.

Proposed problems

Problem 1.1.10. Show that for any natural number n, between n^2 and $(n+1)^2$ one can find three distinct natural numbers a, b, c such that $a^2 + b^2$ is divisible by c.

(1998 St. Petersburg City Mathematical Olympiad)

Problem 1.1.11. Find all odd positive integers n greater than 1 such that for any relatively prime divisors a and b of n, the number $a + b - 1$ is also a divisor of n.

(2001 Russian Mathematical Olympiad)

Problem 1.1.12. Find all positive integers n such that $3^{n-1} + 5^{n-1}$ divides $3^n + 5^n$.

(1996 St. Petersburg City Mathematical Olympiad)

Problem 1.1.13. Find all positive integers n such that the set

$$\{n, n+1, n+2, n+3, n+4, n+5\}$$

can be split into two disjoint subsets such that the products of elements in these subsets are the same.

(12th IMO)

Problem 1.1.14. The positive integers $d_1, d_2, \ldots, d_n$ divide 1995. Prove that there exist d_i and d_j among them, such that the numerator of the reduced fraction d_i/d_j is at least n.

(1995 Israeli Mathematical Olympiad)

Problem 1.1.15. Determine all pairs (a, b) of positive integers such that $ab^2 + b + 7$ divides $a^2b + a + b$.

(39th IMO)

Problem 1.1.16. Find all integers a, b, c with $1 < a < b < c$ such that $(a-1)(b-1)(c-1)$ is a divisor of $abc - 1$.

(33rd IMO)

Problem 1.1.17. Find all pairs of positive integers (x, y) for which

$$\frac{x^2 + y^2}{x - y}$$

is an integer which divides 1995.

(1995 Bulgarian Mathematical Olympiad)

Problem 1.1.18. Find all positive integers (x, n) such that $x^n + 2^n + 1$ is a divisor of $x^{n+1} + 2^{n+1} + 1$.

(1998 Romanian IMO Team Selection Test)

Problem 1.1.19. Find the smallest positive integer K such that every K-element subset of $\{1, 2, \ldots, 50\}$ contains two distinct elements a, b such that $a + b$ divides ab.

(1996 Chinese Mathematical Olympiad)

1.2 Prime numbers

The integer $p > 1$ is called a *prime* if there is no integer $d > 1$ such that $d|p$. Any integer $n > 1$ has at least a prime divisor. If n is a prime, then that prime divisor is n itself. If n is not a prime, then let $a > 1$ be its least divisor. Then $n = ab$, where $1 < a \leq b$. If a were not a prime, then $a = a_1a_2$ with $1 < a_1 \leq a_2 < a$ and $a_1|n$, contradicting the minimality of a.

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An integer $n > 1$ that is not a prime is called *composite*. If n is a composite integer, then it has a prime divisor p not exceeding $\sqrt{n}$. Indeed, as above, $n = ab$, where $1 < a \leq b$ and a is the least divisor of n. Then $n \geq a^2$, hence $a \leq \sqrt{n}$.

The following result is known for more than 2000 years:

Theorem 1.2.1. (Euclid[1]) *There are infinitely many primes.*

Proof. Assume by way of contradiction that there are only a finite number of primes: $p_1 < p_1 < \cdots < p_m$. Consider the number $P = p_1 p_2 \ldots p_n + 1$.

If P is a prime, then $P > p_m$, contradicting the maximality of p_m. Hence P is composite and, consequently, it has a prime divisor $p > 1$ which is one of the primes $p_1, p_2, \ldots, p_m$, say p_k. It follows that $p_k | p_1 \ldots p_k \ldots p_m + 1$. This, together with $p_k | p_1 \ldots p_k \ldots p_m$, implies $p_k | 1$, a contradiction. □

Remark. The largest known prime is $2^{32582657} - 1$. It was discovered in 2006 and it has 9808358 digits.

The fundamental result in arithmetics pertains to the factorization of integers:

Theorem 1.2.2. (The prime factorization theorem) *Any integer $n > 1$ has a unique representation as a product of primes.*

Proof. The existence of such a representation can be obtained as follows: Let p_1 be a prime divisor (factor) of n. If $p_1 = n$, then $n = p_1$ is the prime factorization of n. If $p_1 < n$, then $n = p_1 r_1$, where $r_1 > 1$. If r_1 is a prime, then $n = p_1 p_2$ where $p_2 = r_1$, is the desired factorization of n. If r_1 is composite, then $r_1 = p_2 r_2$, where p_2 is a prime, $r_2 > 1$ and so $n = p_1 p_2 r_2$. If r_2 is a prime, then $n = p_1 p_2 p_3$ where $r_2 = p_3$ and we are done. If r_2 is composite, then we continue this algorithm, obtaining a sequence of integers $r_1 > r_2 > \cdots \geq 1$. After a finite number of steps, we reach $r_{k-1} = 1$, that is $n = p_1 p_2 \ldots p_k$.

For the uniqueness, let us assume that there is at least a positive integer n such that

$$n = p_1 p_2 \ldots p_k = q_1 q_2 \ldots q_h$$

where $p_1, p_2, \ldots, p_k, q_1, q_2, \ldots, q_h$ are primes. It is clear that $k \geq 2$ and $h \geq 2$. Let n be the minimal such integer. We claim that $p_i \neq q_j$ for any $i = 1, 2, \ldots, k$, $j = 1, 2, \ldots, h$. If, for example, $p_k = q_h = p$, then

[1] *Euclid of Alexandria* (about 325BC - about 365BC) is the most prominent mathematician of antiquity best known for his treatise on mathematics "The Elements". The long lasting nature of "The Elements" must make Euclid the leading mathematics teacher of all time.

$n' = n/p = p_1 \dots p_{k-1} = q_1 \dots q_{h-1}$ and $1 < n' < n$, contradicting the minimality of n. Assume without loss of generality that p_1 is the least prime factor of n in the above representations. By applying the Division Algorithm it follows that

$$\begin{aligned} q_1 &= p_1 c_1 + r_1 \\ q_2 &= p_1 c_2 + r_2 \\ &\dots \\ q_h &= p_1 c_h + r_h, \end{aligned}$$

where $1 \leq r_i < p_1$, $i = 1, \dots, h$.

We have

$$n = q_1 q_2 \dots q_h = (p_1 c_1 + r_1)(p_1 c_2 + r_2) \dots (p_1 c_h + r_h).$$

Expanding the last product we obtain $n = Ap_1 + r_1 r_2 \dots r_h$. Setting $n' = r_1 r_2 \dots r_h$ we have $n = p_1 p_2 \dots p_k = Ap_1 + n'$. It follows that $p_1 | n'$ and $n' = p_1 s_1 s_2 \dots s_i$, where $s_1, s_2, \dots, s_i$ are primes.

On the other hand, using the factorization of $r_1, r_2, \dots, r_h$ into primes, all their factors are less than $r_i < p_1$. From $n' = r_1 r_2 \dots r_h$, it follows that n' has a factorization into primes of the form $n' = t_1 t_2 \dots t_j$, where $t_s < p_1$, $s = 1, 2, \dots, j$. This factorization is different from $n' = p_1 s_1 s_2 \dots s_i$. But $n' < n$, contradicting the minimality of n. □

From the above theorem it follows that any integer $n > 1$ can be written uniquely in the form

$$n = p_1^{\alpha_1} \dots p_k^{\alpha_k},$$

where $p_1, \dots, p_k$ are distinct primes and $\alpha_1, \dots, \alpha_k$ are positive integers. This representation is called the *canonical factorization* of n.

An immediate application of the prime factorization theorem is an alternative way of proving that there are infinitely many primes.

As in the previous proof, assume that there are only finitely many primes: $p_1 < p_2 < \dots < p_m$. Let

$$N = \prod_{i=1}^{m} \left(1 + \frac{1}{p_i} + \dots + \frac{1}{p_i^k} + \dots\right) = \prod_{i=1}^{m} \frac{1}{1 - \frac{1}{p_i}}$$

On the other hand, by expanding and by using the canonical factorization of positive integers, we obtain

$$N = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

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yielding $\prod_{i=1}^{m} \frac{p_i}{p_i - 1} = \infty$, a contradiction. We have used the well-known fact that the harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \dots$$

diverges and the expansion formula

$$\frac{1}{1-x} = 1 + x + x^2 + \dots \quad (\text{for } |x| < 1)$$

which can also be interpreted as the summation formula for the infinite geometric progression $1, x, x^2, \dots$

From the formula

$$\prod_{i=1}^{\infty} \frac{p_i}{p_i - 1} = \infty,$$

using the inequality $1 + t \leq e^t$, $t \in \mathbb{R}$, we can easily derive

$$\sum_{i=1}^{\infty} \frac{1}{p_i} = \infty.$$

Even though there are no definitive ways to find primes, the density of primes (that is, the average appearances of primes among integers) has been determined for about 100 years. This was a remarkable result in the mathematical field of *Analytic Number Theory* showing that

$$\lim_{n \to \infty} \frac{\pi(n)}{\frac{n}{\log n}} = 1,$$

where $\pi(n)$ denotes the number of primes $\leq n$. The relation above is known as the Prime Number Theorem. It was proved by Hadamard[2] and de la Vallée Poussin[3] in 1896. An elementary, but difficult proof, was given by Erdös[4] and Selberg[5].

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[2] *Jacques Salomon Hadamard* (1865-1963), French mathematician whose most important result is the Prime Number Theorem which he proved in 1896.

[3] *Charles Jean Gustave Nicolas de la Vallée Poussin* (1866-1962), Belgian mathematician who proved the Prime Number Theorem independently of Hadamard in 1896.

[4] *Paul Erdös* (1913-1996), one of the greatest mathematician of the 20th century. Erdös posed and solved problems in number theory and other areas and founded the field of discrete mathematics.

[5] Atle Selberg (1917-), Norwegian mathematician known for his work in analytic number theory, and in the theory of automorphic forms.

The most important open problems in Number Theory involve primes. The recent book of David Wells [*Prime Numbers: The Most Mysterious Figures in Maths*, John Wiley and Sons, 2005] contains just few of them. We mention here only three such open problems:

1) Consider the sequence $(A_n)_{n\geq 1}$, $A_n = \sqrt{p_{n+1}} - \sqrt{p_n}$, where p_n denotes the nth prime. **Andrica's Conjecture** states that the following inequality holds

$$A_n < 1,$$

for any positive integer n. Results connected to this conjecture are given in D. Andrica [*On a Conjecture in Prime Number Theory*, Proc. Algebra Symposium, "Babeş-Bolyai" University of Cluj, 2005, pp.1-8]. The search given by H.J. Smith has gown past $n = 26 \cdot 10^{10}$, so it is highly likely the conjecture is true.

2) If p is prime such that $p+2$ is also a prime, then p and $p+2$ are called **twin primes**. It is not known if there are infinitely many twin primes. The largest such pair is $100314512544015 \cdot 2^{171960} \pm 1$ and it was found in 2006.

3) The following property is conjectured by Michael Th. Rassias, an IMO Silver Medail in 2003 in Tokyo: For any prime p greater than two there are two distinct primes p_1, p_2 such that

$$p = \frac{p_1 + p_2 + 1}{p_1}.$$

This is equivalent to the following statement: For any prime p greater than two there are two primes $p_1 < p_2$ such that $(p-1)p_1, p_2$ are consecutive integers [Octogon Mathematical Magazine, Vol.13, No.1.B, 2005, page 885].

For a prime p we say that p^k *fully divides* n and write $p^k \| n$ if k is the greatest positive integers such that $p^k | n$.

Problem 1.2.1. *Prove that for any integer $n > 1$ the number n^5+n^4+1 is not a prime.*

Solution. We have

$$\begin{aligned} n^5 + n^4 + 1 &= n^5 + n^4 + n^3 - n^3 - n^2 - n + n^2 + n + 1 \\ &= n^3(n^2 + n + 1) - n(n^2 + n + 1) + (n^2 + n + 1) \\ &= (n^2 + n + 1)(n^3 - n + 1), \end{aligned}$$

the product of two integers greater than 1. Hence n^5+n^4+1 is not a prime.

Problem 1.2.2. *Find all primes a, b, c such that*

$$ab + bc + ac > abc.$$

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Solution. Assume that $a \leq b \leq c$. If $a \geq 3$ then $ab+bc+ac \leq 3bc \leq abc$, a contradiction. Since a is prime, it is left that $a = 2$.

The inequality becomes $2b + 2c + bc > 2bc$, hence $\dfrac{1}{c} + \dfrac{1}{b} > \dfrac{1}{2}$.

If $b \geq 5$, then $c \geq 5$ and

$$\frac{1}{2} < \frac{1}{b} + \frac{1}{c} < \frac{1}{5} + \frac{1}{5} = \frac{2}{5},$$

false.

Therefore $b \leq 5$, that is

1° $b = 2$ and c is any prime;

2° $b = 3$ and c is 3 or 5.

Problem 1.2.3. *Find all the positive integers a, b for which $a^4 + 4b^4$ is a prime.*

Solution. Observe that

$$\begin{aligned} a^4 + 4b^4 &= a^4 + 4b^4 + 4a^2b^2 - 4a^2b^2 \\ &= (a^2 + 2b^2)^2 - 4a^2b^2 \\ &= (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab) \\ &= [(a+b)^2 + b^2][(a-b)^2 + b^2]. \end{aligned}$$

As $(a + b)^2 + b^2 > 1$, then $a^4 + 4b^4$ can be a prime number only if $(a - b)^2 + b^2 = 1$. This implies $a = b = 1$, which is the only solution of the problem.

Problem 1.2.4. *Let p, q be two distinct primes. Prove that there are positive integers a, b so that the arithmetic mean of all the divisors of the number $n = p^a \cdot q^b$ is also an integer.*

(2002 Romanian Mathematical Olympiad)

Solution. The sum of all divisors of n is given by the formula

$$(1 + p + p^2 + \cdots + p^a)(1 + q + q^2 + \cdots + q^b),$$

as it can be easily seen by expanding the brackets. The number n has $(a + 1)(b + 1)$ positive divisors and their arithmetic mean is

$$M = \frac{(1 + p + p^2 + \cdots + p^a)(1 + q + q^2 + \cdots + q^b)}{(a+1)(b+1)}.$$

If p and q are both odd numbers, we can take $a = p$ and $b = q$, and it is easy to see that m is an integer.

If $p = 2$ and q odd, choose again $b = q$ and consider $a+1 = 1+q+q^2+\cdots+q^{q-1}$. Then $m = 1+2+2^2+\cdots+2^a$, and it is an integer.

For p odd and $q = 2$, set $a = p$ and $b = p+p^2+p^3+\cdots+p^{p-1}$. The solution is complete.

Problem 1.2.5. *Let p, q, r be primes and let n be a positive integer such that*

$$p^n + q^n = r^2.$$

Prove that $n = 1$.

(2004 Romanian Mathematical Olympiad)

Solution. Clearly one of the primes p, q or r is equal to 2. If $r = 2$ then $p^n + q^n = 4$, false, so assume that $p > q = 2$.

Consider the case when $n > 1$ is odd; we have

$$(p+2)(p^{n-1} - 2p^{n-2} + 2^2p^{n-3} - \cdots + 2^{n-1}) = r^2.$$

Notice that

$$p^{n-1} - 2p^{n-2} + 2^2p^{n-3} - \cdots + 2^{n-1} = 2^{n-1} + (p-2)(p^{n-2} + 2^2p^{n-4} + \dots) > 1$$

and $p+2 > 1$ hence both factors are equal to r. This rewrites as $p^n + 2^n = (p+2)^2 = p^2 + 4p + 4$, which is false for $n \geq 3$.

Consider the case when $n > 1$ is even and let $n = 2m$. It follows that $p^m = a^2 - b^2$, $2^m = 2ab$ and $r = a^2 + b^2$, for some integers a, b with $(a, b) = 1$. Therefore, a and b are powers of 2, so $b = 1$ and $a = 2^{m-1}$. This implies $p^m = 4^{m-1} - 1 < 4^m$, so p must be equal to 3. The equality $3^m = 4^{m-1} - 1$ fails for $m = 1$ and also for $m \geq 2$, as $4^{m-1} > 3^m + 1$, by induction.

Consequently $n = 1$. Take for example $p = 23$, $q = 2$ and $r = 5$.

Problem 1.2.6. *Let a, b, c be non zero integers, $a \neq c$, such that*

$$\frac{a}{c} = \frac{a^2 + b^2}{c^2 + b^2}.$$

Prove that $a^2 + b^2 + c^2$ cannot be a prime.

(1999 Romanian Mathematical Olympiad)

Solution. The equality $\dfrac{a}{c} = \dfrac{a^2 + b^2}{c^2 + b^2}$ is equivalent to $(a-c)(b^2 - ac) = 0$. Since $a \neq c$, it follows that $b^2 = ac$ and therefore:

$$a^2 + b^2 + c^2 = a^2 + ac + c^2 = a^2 + 2ac + c^2 - b^2$$

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$$= (a+c)^2 - b^2 = (a+c-b)(a+c+b).$$

Now, clearly, $a^2+b^2+c^2 > 3$, so, if $a^2+b^2+c^2$ is a prime number, then only four cases are possible:

(1) $a+c-b = 1$ and $a+c-b = a^2+b^2+c^2$;

(2) $a+c+b = 1$ and $a+c+b = a^2+b^2+c^2$;

(3) $a+c-b = -1$ and $a+c+b = -(a^2+b^2+c^2)$, and finally:

(4) $a+c+b = -1$ and $a+c-b = -(a^2+b^2+c^2)$.

In the first two cases we are lead to: $a^2+b^2+c^2-2(a+c)+1 = 0$, or $(a-1)^2+(c-1)^2+b^2 = 1$, hence $a = c = 1$.

In other cases we obtain: $(a+1)^2+(c+1)^2+b^2 = 1$, hence $a = c = -1$. But $a = c$ is a contradiction.

Problem 1.2.7. *Show that each natural number can be written as the difference of two natural numbers having the same number of prime factors.*

(1999 Russian Mathematical Olympiad)

Solution. If n is even, then we can write it as $(2n)-(n)$. If n is odd, let d be the smallest odd prime that does not divide n. Then write $n = (dn)-((d-1)n)$. The number dn contains exactly one more prime factor than n. As for $(d-1)n$, it is divisible by 2 because $d-1$ is even. Its odd factors are less than d so they all divide n. Therefore $(d-1)n$ also contains exactly one more prime factor than n, and dn and $(d-1)n$ have the same number of prime factors.

Problem 1.2.8. *Let p be a prime number. Find all $k \in \mathbb{Z}$ such that $\sqrt{k^2-pk}$ is a positive integer.*

(1997 Spanish Mathematical Olympiad)

Solution. The values are $k = (p+1)^2/4$ for p odd (and none for $p = 2$).

We first consider $p = 2$, in which case we need $k^2-2k = (k-1)^2-1$ to be a positive square, which is impossible, as the only consecutive squares are 0 and 1.

Now assume p is odd. We first rule out the case where k is divisible by p: if $k = np$, then $k^2-pk = p^2n(n-1)$, and n and $n-1$ are consecutive numbers, so they cannot both be squares.

We thus assume k and p are coprime, in which case k and $k-p$ are coprime. Thus k^2-pk is a square if and only if k and $k-p$ are squares, say $k = m^2$ and $k-p = n^2$. Then $p = m^2-n^2 = (m+n)(m-n)$, which implies $m+n = p$, $m-n = 1$ and $k = (p+1)^2/4$.

Problem 1.2.9. *Let $p > 5$ be a prime number and*

$$X = \{p - n^2 | n \in \mathbb{N}, n^2 < p\}.$$

Prove that X contains two distinct elements x, y such that $x \neq 1$ and x divides y.

(1996 Balkan Mathematical Olympiad)

Solution. Take m such that $m^2 < p < (m+1)^2$ and write $p = k + m^2$, with $1 \leq k \leq 2m$. Since $p - (m-k)^2 = k(2m - k + 1)$ we have $p - m^2 | p - (m-k)^2$. Of course, $k \neq m$ since p is a prime number. Also, $m - k < m$ and $-m + k \neq m$ since p is not composite. The only case which remains is $k = 1$. Since m is even, $p - (m-1)^2$ divides $p - 1$ we are done again.

Proposed problems

Problem 1.2.10. For each integer n such that $n = p_1 p_2 p_3 p_4$, where p_1, p_2, p_3, p_4 are distinct primes, let

$$d_1 = 1 < d_2 < d_3 < \cdots < d_{16} = n$$

be the sixteen positive integers which divide n. Prove that if $n < 1995$, then $d_9 - d_8 \neq 22$.

(1995 Irish Mathematical Olympiad)

Problem 1.2.11. Prove that there are infinitely many positive integers a such that the sequence $(z_n)_{n \geq 1}$, $z_n = n^4 + a$, does not contain any prime number.

(11th IMO)

Problem 1.2.12. Let p, q, r be distinct prime numbers and let A be the set

$$A = \{p^a q^b r^c : \ 0 \leq a, b, c \leq 5\}.$$

Find the smallest integer n such that any n-element subset of A contains two distinct elements x, y such that x divides y.

(1997 Romanian Mathematical Olympiad)

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Problem 1.2.13. Prove Bonse's inequality:

$$p_1 p_2 \dots p_n > p_{n+1}^2$$

for $n \geq 4$, where $p_1 = 2$, $p_2 = 3, \dots$ is the increasing sequence of prime numbers.

Problem 1.2.14. Show that there exists a set A of positive integers with the following property: for any infinite set S of primes, there exist two positive integers $m \in A$ and $n \notin A$ each of which is a product of k distinct elements of S for some $k \geq 2$.

(35th IMO)

Problem 1.2.15. Let n be an integer number, $n \geq 2$. Show that if $k^2 + k + n$ is a prime number for any integer number k, $0 \leq k \leq \sqrt{\frac{n}{3}}$, then $k^2 + k + n$ is a prime number for any k, $0 \leq k \leq n - 2$.

(28th IMO)

Problem 1.2.16. A sequence $q_1, q_2, \dots$ of primes satisfies the following condition: for $n \geq 3$, q_n is the greatest prime divisor of $q_{n-1} + q_{n-2} + 2000$. Prove that the sequence is bounded.

(2000 Polish Mathematical Olympiad)

Problem 1.2.17. Let $a > b > c > d$ be positive integers and suppose

$$ac + bd = (b + d + a - c)(b + d - a + c).$$

Prove that $ab + cd$ is not prime.

(42nd IMO)

1.3 The greatest common divisor and the least common multiple

For a positive integer k we denote by D_k the set of all its positive divisors. It is clear that D_k is a finite set. For positive integers m, n the maximal element in the set $D_m \cap D_n$ is called the *greatest common divisor* of m and n and is denoted by $gcd(m, n)$.

In case when $D_m \cap D_n = \{1\}$, we have $gcd(m, n) = 1$ and we say that m and n are *relatively prime*.

The following properties can be directly derived from the definition above.

1) If $d = gcd(m, n)$, $m = dm'$, $n = dn'$, then $gcd(m', n') = 1$.

2) If $d = gcd(m, n)$, $m = d'm''$, $n = d'n''$, $gcd(m'', n'') = 1$, then $d' = d$.

3) If d' is a common divisor of m and n, then d' divides $gcd(m, n)$.

4) If $m = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ and $n = p_1^{\beta_1} \dots p_k^{\beta_k}$, $\alpha_i, \beta_i \geq 0$, $\alpha_i + \beta_i \geq 1$, $i = 1, \dots, k$, then

$$gcd(m, n) = p_1^{\min(\alpha_1, \beta_1)} \dots p_k^{\min(\alpha_k, \beta_k)}.$$

5) If $m = nq + r$, then $gcd(m, n) = gcd(n, r)$.

Let us prove the last property. Denote $d = gcd(m, n)$ and $d' = gcd(n, r)$. Because $d|m$ and $d|n$ it follows that $d|r$. Hence $d|d'$. Conversely, from $d'|n$ and $d'|r$ it follows that $d'|m$, so $d'|d$. Thus $d = d'$.

An useful algorithm for finding the greatest common divisor of two positive integers is the *Euclidean Algorithm*. It consists of repeated application of the Division Algorithm:

$$\begin{aligned}
m &= nq_1 + r_1, \quad 1 \leq r_1 < n \\
n &= r_1 q_2 + r_2, \quad 1 \leq r_2 < r_1 \\
&\dots \\
r_{k-2} &= r_{k-1} q_k + r_k, \quad 1 \leq r_k < r_{k-1} \\
r_{k-1} &= r_k q_{k+1} + r_{k+1}, \quad r_{k+1} = 0.
\end{aligned}$$

This chain of equalities is finite because $n > r_1 > r_2 > \dots > r_k$.

The last nonzero remainder, r_k, is the greatest common divisor of m and n. Indeed, by applying successively property 5) above we obtain

$$gcd(m, n) = gcd(n, r_1) = gcd(r_1, r_2) = \dots = gcd(r_{k-1}, r_k) = r_k.$$

Proposition 1.3.1. *For positive integers m and n, there exist integers a and b such that $am + bn = gcd(m, n)$.*

Proof. From the Euclidean Algorithm it follows that

$$r_1 = m - nq_1, \quad r_2 = -mq_2 + n(1 + q_1 q_2), \dots$$

In general, $r_i = m\alpha_i + n\beta_i$, $i = 1, \dots, k$. Because $r_{i+1} = r_{i-1} - r_i q_{i+1}$, it follows that

$$\begin{cases} \alpha_{i+1} = \alpha_{i-1} - q_{i+1}\alpha_i \\ \beta_{i+1} = \beta_{i-1} - q_{i+1}\beta_i, \end{cases}$$

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$i = 2, \ldots, k-1$. Finally, we obtain $gcd(m,n) = r_k = \alpha_k m + \beta_k n$. □

We can define the greatest common divisor of several positive integers $m_1, m_2, \ldots, m_s$ by considering

$$d_1 = gcd(m_1, m_2),\ d_2 = gcd(d_1, m_3), \ldots, d_{s-1} = gcd(d_{s-2}, m_s).$$

The integer $d = d_{s-1}$ is called the greatest common divisor of $m_1, \ldots, m_s$ and denoted by $gcd(m_1, \ldots, m_s)$. The following properties can be easily verified:

i) $gcd(gcd(m,n),p) = gcd(m, gcd(n,p))$; proving that $gcd(m,n,p)$ is well-defined.

ii) If $d|m_i$, $i = 1, \ldots, s$, then $d|gcd(m_1, \ldots, m_s)$.

iii) If $m_i = p_1^{\alpha_{1i}} \ldots p_k^{\alpha_{ki}}$, $i = 1, \ldots, s$, then

$$gcd(m_1, \ldots, m_s) = p_1^{\min(\alpha_{11}, \ldots, \alpha_{1k})} \ldots p_k^{\min(\alpha_{k1}, \ldots, \alpha_{kk})}.$$

For a positive integer k we denote by M_k the set of all multiples of k. Opposed to the set D_k defined earlier in this section, M_k is an infinite set.

For positive integers s and t the minimal element of the set $M_s \cap M_t$ is called the *least common multiple* of s and t and is denoted by $\text{lcm}(s,t)$.

The following properties are easily obtained from the definition above:

1') If $m = \text{lcm}(s,t)$, $m = ss' = tt'$, then $gcd(s',t') = 1$.

2') If m' is a common multiple of s and t and $m' = ss' = tt'$, $gcd(s',t') = 1$, then $m' = m$.

3') If m' is a common multiple of s and t, then $m|m'$.

4') If $s = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ and $t = p_1^{\beta_1} \ldots p_k^{\beta_k}$, $\alpha_i, b_i \geq 0$, $\alpha_i + \beta_i \geq 1$, $i = 1, \ldots, k$, then

$$\text{lcm}(s,t) = p_1^{\max(\alpha_1, \beta_1)} \ldots p_k^{\max(\alpha_k, \beta_k)}.$$

The following property establishes an important connection between *gcd* and lcm:

Proposition 1.3.2. *For any positive integers m, n the following relation holds:*

$$mn = gcd(m,n) \cdot \text{lcm}(m,n).$$

Proof. Let $m = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$, $n = p_1^{\beta_1} \ldots p_k^{\beta_k}$, $\alpha_i, \beta_i \geq 0$, $\alpha_i + \beta_i \geq 1$, $i = 1, \ldots, k$. From properties 4) and 4') we have

$$gcd(m,n) \cdot \text{lcm}(m,n) = p_1^{\min(\alpha_1, \beta_1) + \max(\alpha_1, \beta_1)} \ldots p_k^{\min(\alpha_k, \beta_k) + \max(\alpha_k, \beta_k)}$$

$$= p_1^{\alpha_1+\beta_1} \dots p_k^{\alpha_k+\beta_k} = mn. \qquad \square$$

It is also not difficult to see that if $m|s$ and $n|s$, then $\text{lcm}(m,n)|s$.

Problem 1.3.1. *Prove that for any odd integers n, a_1, $a_2, \dots, a_n$, the greatest common divisor of numbers $a_1, a_2, \dots, a_n$ is equal to the greatest common divisor of $\frac{a_1+a_2}{2}, \frac{a_2+a_3}{2}, \dots, \frac{a_n+a_1}{2}$.*

Solution. Let

$$a = gcd(a_1, a_2, \dots, a_n) \quad \text{and} \quad b = gcd\left(\frac{a_1+a_2}{2}, \frac{a_2+a_3}{2}, \dots, \frac{a_n+a_1}{2}\right)$$

Then $a_k = \alpha_k a$, for some integers α_k, $k = 1, 2, \dots, n$. It follows that

$$\frac{a_k + a_{k+1}}{2} = \frac{\alpha_k + \alpha_{k+1}}{2} a, \tag{1}$$

where $a_{n+1} = a_1$ and $\alpha_{n+1} = \alpha_1$. Since a_k are odd numbers, α_k are also odd, so $\frac{\alpha_k + \alpha_{k+1}}{2}$ are integers.

From relation (1) it follows that a divides $\frac{a_k + a_{k+1}}{2}$ for all so a divides b.

On the other hand, $\frac{a_k + a_{k+1}}{2} = \beta_k b$, for some integers β_k. Then

$$a_k + a_{k+1} \equiv 0 \pmod{2b}$$

for all $k \in \{1, 2, \dots, n\}$. Summing up from $k = 1$ to $k = n$ yields

$$2(a_1 + a_2 + \dots + a_n) \equiv 0 \pmod{2b},$$

hence

$$a_1 + a_2 + \dots + a_n \equiv 0 \pmod{b}. \tag{3}$$

Summing up for $k = 1, 3, \dots, n-2$ implies

$$a_1 + a_2 + \dots + a_{n-1} \equiv 0 \pmod{2b}$$

and furthermore

$$a_1 + a_2 + \dots + a_{n-1} \equiv 0 \pmod{b}. \tag{4}$$

Subtracting (4) from (3) implies $a_n \equiv 0 \pmod{b}$, then using relation (2) we obtain $a_k \equiv 0 \pmod{b}$ for all k. Hence $b|a$ and the proof is complete.

Problem 1.3.2. *Prove that for all nonnegative integers a, b, c, d such that a and b are relatively prime, the system*

$$ax - yz - c = 0$$
$$bx - yt + d = 0$$

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has at least a solution in nonnegative integers.

Solution. We start with a useful lemma.

Lemma. *If a and b are relatively prime positive integers, then there are positive integers u and v such that*

$$au - bv = 1.$$

Proof. Consider the numbers

$$1 \cdot 2,\ 2 \cdot a, \ldots, (b-1) \cdot a \tag{1}$$

When divided by b the remainders of these numbers are distinct. Indeed, otherwise we have $k_1 \neq k_2 \in \{1, 2, \ldots, b-1\}$ such that

$$k_1 a = p_1 b + r, \quad k_2 a = p_2 b + r$$

for some integers p_1, p_2. Hence

$$(k_1 - k_2)a = (p_1 - p_2)b \equiv 0 \pmod{b}.$$

Since a and b are relatively prime it follows that $|k_1 - k_2| \equiv 0 \pmod{b}$, which is false because $1 \leq |k_1 - k_2| < b$.

On the other hand, none of the numbers listed in (1) is divisible by b. Indeed, if so, then there is $k \in \{1, 2, \ldots, n-1\}$ such that

$$k \cdot a = p \cdot b \text{ for some integer } p.$$

Let d be the greatest common divisor of k and p. Hence $k = k_1 d$, $p = p_1 d$, for some integers p_1, k_1 with $gcd(p_1, k_1) = 1$. Then $k_1 a = p_1 b$ and since $gcd(a, b) = 1$, we have $k_1 = b$, $p_1 = a$. This is false, because $k_1 < b$.

It follows that one of the numbers from (1) has the remainder 1 when divided by b so there is $u \in \{1, 2, \ldots, b-1\}$ such that $au = bv + 1$ and the lemma is proved.

We prove now that the system

$$\begin{cases} ax - yz - c = 0 \\ bx - yt + d = 0 \end{cases}$$

with a, b, c, d nonnegative integers and $gcd(a, b) = 1$ has at least a solution in nonnegative integers.

Because $gcd(a, b) = 1$ using the lemma, there are positive integers u and v such that $au - bv = 1$. Hence

$$x = cu + dv, \quad y = ad + bc, \quad z = v, \quad t = u,$$

is a solution to the system.

Problem 1.3.3. *Find all the pairs of integers (m, n) so that the numbers $A = n^2+2mn+3m^2+2$, $B = 2n^2+3mn+m^2+2$, $C = 3n^2+mn+2m^2+1$ have a common divisor greater than 1.*

Solution. A common divisor of A, B and C is also a divisor for $D = 2A - B$, $E = 3A - C$, $F = 5E - 7D$, $G = 5D - E$, $H = 18A - 2F - 3E$, $I = nG - mF$ and $126 = 18nI - 5H + 11F = 2 \cdot 3^2 \cdot 7$. Since 2 and 3 do not divide A, B and C, then $d = 7$. It follows that (m, n) is equal to $(7a + 2, 7b + 3)$ or $(7c + 5, 7d + 4)$.

Problem 1.3.4. *Let n be an even positive integer and let a, b be positive coprime integers. Find a and b if $a + b$ divides $a^n + b^n$.*

(2003 Romanian Mathematical Olympiad)

Solution. As n is even, we have

$$a^n - b^n = (a^2 - b^2)(a^{n-2} + a^{n-4}b^2 + \cdots + b^{n-2}).$$

Since $a + b$ is a divisor of $a^2 - b^2$, it follows that $a + b$ is a divisor of $a^n - b^n$. In turn, $a+b$ divides $2a^n = (a^n + b^n) + (a^n - b^n)$, and $2b^n = (a^n + b^n) - (a^n - b^n)$. But a and b are coprime numbers, and so $gcd(2a^n, 2b^n) = 2$. Therefore $a + b$ is a divisor of 2, hence $a = b = 1$.

Problem 1.3.5. *M is the set of all values of the greatest common divisor d of the numbers $A = 2n + 3m + 13$, $B = 3n + 5m + 1$, $C = 6n + 8m - 1$, where m and n are positive integers. Prove that M is the set of all divisors of an integer k.*

Solution. If d is a common divisor of the numbers A, B and C, then d divides $E = 3A - C = m + 40$, $F = 2B - C = 2m + 3$ and $G = 2E - F = 77$. We prove that $k = 77$ satisfies the conditions.

Let d' be the greatest common divisor of the numbers E and F. Then $d' = 7u$ for $m = 7p + 2$. Moreover, $u = 1$ if $p \neq 11v + 5$ and $u = 11$ if $p = 11v + 5$. On the other hand, $d' = 11v$ for $m = 11q + 4$. Furthermore, $v = 1$ for $q \neq 7z + 3$ and $v = 7$ for $q = 7z + 3$.

The number d' is common divisor of the numbers A, B, C if and only if d' divides A.

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For $m = 7p + 2$, 7 divides $A = 2n + 21p + 19$ if and only if $n = 7p' + 1$.

For $m = 7(11v + 5)$, $A = 2(n + 59) + 3 \cdot 77v$ is divisible by 77 if and only if $n = 77t + 18$.

Problem 1.3.6. *Find the greatest common divisor of the numbers*

$$A_n = 2^{3n} + 3^{6n+2} + 5^{6n+2}$$

when $n = 0, 1, \ldots, 1999$.

(2001 Junior Balkan Mathematical Olympiad)

Solution. We have

$$A_0 = 1 + 9 + 25 = 35 = 5 \cdot 7.$$

Using congruence mod 5, it follows that

$$A_n \equiv 2^{3n} + 3^{6n+2} \equiv 2^{3n} + 9^{3n+1} \equiv 2^{3n} + (-1)^{3n+1} \pmod 5.$$

For $n = 1$, $A_1 \equiv 9 \neq 0 \pmod 5$, hence 5 is not a common divisor.

On the other hand,

$$\begin{aligned} A_n &= 8^n + 9 \cdot 9^{3n} + 25 \cdot 25^{3n} \\ &\equiv 1 + 2 \cdot 2^{3n} + 4 \cdot 4^{3n} \\ &\equiv 1 + 2 \cdot 8^n + 4 \cdot 64^n \\ &\equiv 1 + 2 \cdot 1^n + 4 \cdot 1^n \\ &\equiv 0 \pmod 7, \end{aligned}$$

therefore 7 divides A_n, for all integers $n \geq 0$.

Consequently, the greatest common divisor of the numbers A_0, $A_1, \ldots,$ A_{1999} is equal to 7.

Problem 1.3.7. *Let $m \geq 2$ be an integer. A positive integer n is called m-good if for every positive integer a, relatively prime to n, one has $n | a^m - 1$.*

Show that any m-good number is at most $4m(2^m - 1)$.

(2004 Romanian IMO Team Selection Test)

Solution. If m is odd then $n|(n-1)^m - 1$ implies $n|2$, hence $n \leq 2$.

Take now $m = 2^t q$, $t \geq 1$, q odd, If $n = 2^u(2v + 1)$ is m-good, then $(2v + 1)|(2v - 1)^m - 1$, hence $(2v + 1)|2^m - 1$. Also, if $a = 8v + 5$ then $(a, n) = 1$, so

$$2^u | (a^q)^{2^t} - 1 = (a^q - 1)(a^q + 1)(a^{2q} + 1) \ldots (a^{2^{t-1}q} + 1).$$

But $a^q \equiv 5 \pmod 8$ implies that the exponent of the factor 2 in the last product is $t+2$, therefore $u \leq t+2$, whence $n \leq 4 \cdot 2^t(2v+1) \leq 4m(2^m-1)$.

Remark. The estimation is optimal only for $m = 2$, $m = 4$.

Problem 1.3.8. *Find all triples of positive integers (a, b, c) such that $a^3 + b^3 + c^3$ is divisible by a^2b, b^2c, and c^2a.*

(2001 Bulgarian Mathematical Olympiad)

Solution. Answer: triples of the form (k, k, k) or $(k, 2k, 3k)$ or their permutations.

Let g be the positive greatest common divisor of a and b. Then g^3 divides a^2b, so g^3 divides $a^3 + b^3 + c^3$, and g divides c. Thus, the *gcd* of any two of a, b, c is the *gcd* of all three.

Let $(l, m, n) = (a/g, b/g, c/g)$. Then (l, m, n) is a triple satisfying the conditions of the problem, and l, m, n are pairwise relatively prime. Because l^2, m^2 and n^2 all divide $l^3 + m^3 + n^3$, we have

$$l^2m^2n^2|(l^3 + m^3 + n^3).$$

We will prove that (l, m, n) is either (1,1,1) or a permutation of (1,2,3).

Assume without loss of generality that $l \geq m \geq n$. We have

$$3l^3 \geq l^3 + m^3 + n^3 \geq l^2m^2n^2,$$

and, therefore, $l \geq m^2n^2/3$. Because $l^2|(m^3 + n^3)$, we also have

$$2m^3 \geq m^3 + n^3 \geq l^2 \geq m^4n^4/9.$$

If $n \geq 2$, then $m \leq 2 \cdot 9/2^4 < 2 \leq n$, which contradicts the assumption that $m \geq n$. Therefore, n must be 1. It is not difficult to see that (1,1,1) is the unique solution with $m = 1$.

If $m \geq 2$, then $l > m$ because l and m are relatively prime, so

$$2l^3 > l^3 + m^3 + 1 \geq l^2m^2,$$

and $l > m^2/2$, so

$$m^3 + 1 \geq l^2 > m^4/4,$$

and $m \leq 4$. It is not difficult to check that the only solution here is (3,2,1).

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Proposed problems

Problem 1.3.9. The sequence $a_1, a_2, \ldots$ of natural numbers satisfies

$$gcd(a_i, a_j) = gcd(i, j) \text{ for all } i \neq j.$$

Prove that $a_i = i$ for all i.

(1995 Russian Mathematical Olympiad)

Problem 1.3.10. The natural numbers a and b are such that

$$\frac{a+1}{b} + \frac{b+1}{a}$$

is an integer. Show that the greatest common divisor of a and b is not greater than $\sqrt{a+b}$.

(1996 Spanish Mathematical Olympiad)

Problem 1.3.11. The positive integers m, n, m, n are written on a blackboard. A generalized Euclidean algorithm is applied to this quadruple as follows: if the numbers x, y, u, v appear on the board and $x > y$, then $x-y$, y, $u+v$, v are written instead; otherwise x, $y-x$, u, $v+u$ are written instead. The algorithm stops when the numbers in the first pair become equal (they will equal the greatest common divisor of m and n). Prove that the arithmetic mean of the numbers in the second pair at that moment equals the least common multiple of m and n.

(1996 St. Petersburg City Mathematical Olympiad)

Problem 1.3.12. How many pairs (x, y) of positive integers with $x \leq y$ satisfy $gcd(x, y) = 5!$ and $\text{lcm}(x, y) = 50!$?

(1997 Canadian Mathematical Olympiad)

Problem 1.3.13. Several positive integers are written on a blackboard. One can erase any two distinct integers and write their greatest common divisor and least common multiple instead. Prove that eventually the numbers will stop changing.

(1996 St. Petersburg City Mathematical Olympiad)

Problem 1.3.14. (a) For which positive integers n do there exist positive integers x, y such that

$$\text{lcm}(x, y) = n!, \quad gcd(x, y) = 1998?$$

(b) For which n is the number of such pairs x, y with $x \leq y$ less than 1998?

(1998 Hungarian Mathematical Olympiad)

Problem 1.3.15. Determine all positive integers k for which there exists a function $f : \mathbb{N} \to \mathbb{Z}$ such that
(a) $f(1997) = 1998$;
(b) for all $a, b \in \mathbb{N}$, $f(ab) = f(a) + f(b) + kf(gcd(a, b))$.

(1997 Taiwanese Mathematical Olympiad)

Problem 1.3.16. Find all triples (x, y, n) of positive integers such that

$$gcd(x, n + 1) = 1 \text{ and } x^n + 1 = y^{n+1}.$$

(1998 Indian Mathematical Olympiad)

Problem 1.3.17. Find all triples (m, n, l) of positive integers such that

$$m + n = gcd(m, n)^2, \quad m + l = gcd(m, l)^2, \quad n + l = gcd(n, l)^2.$$

(1997 Russian Mathematical Olympiad)

1.4 Odd and even

The set $\mathbb{Z}$ of integers can be partitioned into two subsets, the set of odd integers and the set of even integers: $\{\pm 1, \pm 3, \pm 5, \dots\}$ and $\{0, \pm 2, \pm 4, \dots\}$, respectively. Although the concepts of odd and even integers appear straightforward, they come handly in various number theory problems. Here are some basic ideas:

1) an odd number is of the form $2k + 1$, for some integer k;
2) an even number is of the form $2m$, for some integer m;
3) the sum of two odd numbers is an even number;
4) the sum of two even numbers is an even number;
5) the sum of an odd and even number is an odd number;
6) the product of two odd numbers is an odd number;
7) a product of integers is even if and only if at least one of its factors is even.

Problem 1.4.1. *Let m and n be integers greater than 1. Prove that m^n is the sum of m odd consecutive integers.*

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Solution. The equality

$$m^n = (2k+1) + (2k+3) + \cdots + (2k+2m-1)$$

is equivalent to

$$m^n = 2km + (1 + 3 + \cdots + 2m - 1)$$

or $m^n = 2km + m^2$. It follows that $k = \dfrac{m(m^{n-2}-1)}{2}$ which is an integer, because m and $m^{n-2}-1$ have different parities.

Problem 1.4.2. *Let n be a positive integer. Find the sum of all even numbers between $n^2 - n + 1$ and $n^2 + n + 1$.*

Solution. We have $n^2-n+1 = n(n-1)+1$ and $n^2+n+1 = n(n+1)+1$, both odd numbers. It follows that the least even number to be considered is $n^2 - n + 2$ and the greatest is $n^2 + n$. The desired sum is

$$(n^2 - n + 2) + (n^2 - n + 4) + \cdots + (n^2 + n - 2) + (n^2 + n)$$

$$= (n^2 - n) + 2 + (n^2 - n) + 4 + \cdots + (n^2 - n) + 2n - 2 + (n^2 - n) + 2n$$

$$= n(n^2 - n) + 2(1 + 2 + \cdots + n) = n^3 - n^2 + n^2 + n = n^3 + n.$$

Problem 1.4.3. *Let n be a positive integer and let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n \in \{-1, 1\}$ such that $\varepsilon_1\varepsilon_2 + \varepsilon_2\varepsilon_3 + \cdots + \varepsilon_n\varepsilon_1 = 0$. Prove that n is divisible by 4.*

(Kvant)

Solution. The sum $\varepsilon_1\varepsilon_2+\varepsilon_2\varepsilon_3+\cdots+\varepsilon_n\varepsilon_1$ has n terms equal to 1 or -1, so n is even, say $n = 2k$. It is clear that k of the terms $\varepsilon_1\varepsilon_2, \varepsilon_2\varepsilon_3, \ldots, \varepsilon_n\varepsilon_1$ are 1 and k are -1. On the other hand, the product of the terms in the sum is

$$(\varepsilon_1\varepsilon_2)(\varepsilon_2\varepsilon_3)\ldots(\varepsilon_n\varepsilon_1) = \varepsilon_1^2\varepsilon_2^2\ldots\varepsilon_n^2 = 1,$$

hence $(+1)^k(-1)^k = 1$. That is k is even and the conclusion follows.

For any integer $n = 4m$ there exist $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ such that

$$\varepsilon_1\varepsilon_2 + \varepsilon_2\varepsilon_3 + \cdots + \varepsilon_n\varepsilon_1 = 0,$$

for example

$$\varepsilon_1 = \varepsilon_4 = \varepsilon_5 = \varepsilon_8 = \cdots = \varepsilon_{4m-3} = \varepsilon_{4m} = +1,$$

$$\varepsilon_2 = \varepsilon_3 = \varepsilon_6 = \varepsilon_7 = \cdots = \varepsilon_{4m-2} = \varepsilon_{4m-1} = -1.$$

Problem 1.4.4. *A table with m rows and n columns has all entries -1 or 1 such that for each row and each column the product of entries is -1. Prove that m and n have the same parity.*

Solution. We compute the product P of the $m \cdot n$ entries in two ways, by rows and by columns, respectively:

$$P = \underbrace{(-1)(-1)\dots(-1)}_{m\,times} = (-1)^m = (-1)^n = \underbrace{(-1)(-1)\dots(-1)}_{n\,times}.$$

The conclusion now follows.

We will show such a table for $m = 3$ and $n = 5$.

$$\begin{array}{rrrrr} -1 & 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 1 \end{array}$$

Remark. If m and n have the same parity, then the number of tables with the above property is $2^{(m-1)(n-1)}$.

Proposed problems

Problem 1.4.5. We are given three integers a, b, c such that a, b, c, $a+b-c$, $a+c-b$, $b+c-a$ and $a+b+c$ are seven distinct primes. Let d be the difference between the largest and smallest of these seven primes. Suppose that $800 \in \{a+b, b+c, c+a\}$. Determine the maximum possible value of d.

Problem 1.4.6. Determine the number of functions $f : \{1, 2, \dots, n\} \to \{1995, 1996\}$ which satisfy the condition that $f(1) + f(2) + \cdots + f(1996)$ is odd.

(1996 Greek Mathematical Olympiad)

Problem 1.4.7. Is it possible to place 1995 different natural numbers along a circle so that for any two these numbers, the ratio of the greatest to the least is a prime?

(1995 Russian Mathematical Olympiad)

Problem 1.4.8. Let a, b, c, d be odd integers such that $0 < a < b < c < d$ and $ad = bc$. Prove that if $a + d = 2^k$ and $b + c = 2^m$ for some integers k and m, then $a = 1$.

(25th IMO)

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1.5 Modular arithmetics

Let a, b, n be integers, with $n \neq 0$. We say that a and b are *congruent modulo* n if $n|a-b$. We denote this by $a \equiv b \pmod{n}$. The relation "$\equiv$" on the set $\mathbb{Z}$ of integers is called the *congruence relation*. If m does not divide $a-b$, then we say that integers a and b are not congruent modulo n and we write $a \not\equiv b \pmod{n}$. The following properties can be directly derived:

1) $a \equiv a \pmod{n}$ (reflexivity);

2) If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$ (transitivity);

3) If $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$;

4) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a+c \equiv b+d \pmod{n}$ and $a-c \equiv b-d \pmod{n}$;

5) If $a \equiv b \pmod{n}$, then for any integer k, $ka \equiv kb \pmod{n}$;

6) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $ac \equiv bd \pmod{n}$;

7) If $a_i \equiv b_i \pmod{n}$, $i = 1, \ldots, k$, then $a_1 \ldots a_k \equiv b_1 \ldots b_k \pmod{n}$. In particular, if $a \equiv b \pmod{n}$, then for any positive integer k, $a^k \equiv b^k \pmod{n}$.

8) We have $a \equiv b \pmod{m_i}$, $i = 1, \ldots, k$ if and only if $a \equiv b \pmod{\text{lcm}(m_1, \ldots m_k)}$.

In particular, if $m_1, \ldots, m_k$ are pairwise relatively prime, then $a \equiv b \pmod{m_i}$, $i = 1, \ldots, k$ if and only if $a \equiv b \pmod{m_1, \ldots, m_k}$.

Let us prove the last property. From $a \equiv b \pmod{m_i}$, $i = 1, \ldots, k$, it follows that $m_i|a-b$, $i = 1, \ldots, k$. Hence $a-b$ is a common multiple of $m_1, \ldots, m_k$, and so $\text{lcm}(m_1, \ldots, m_k)|a-b$. That is $a \equiv b \pmod{\text{lcm}(m_1, \ldots, m_k)}$. Conversely, from $a \equiv b \pmod{\text{lcm}(m_1, \ldots, _k)}$, and the fact that each m_i divides $\text{lcm}(m_1, \ldots, m_k)$ we obtain $a \equiv b \pmod{m_i}$, $i = 1, \ldots, k$.

Theorem 1.5.1. *Let a, b, n be integers, $n \neq 0$, such that $a = nq_1 + r_1$, $b = nq_2 + r_2$, $0 \leq r_1, r_2 < |n|$. Then $a \equiv b \pmod{n}$ if and only if $r_1 = r_2$.*

Proof. Because $a - b = n(q_1 - q_2) + (r_1 - r_2)$, it follows that $n|a-b$ if and only if $n|r_1 - r_2$. Taking into account that $|r_1 - r_2| < |n|$, we have $n|r_1 - r_2$ if and only if $r_1 = r_2$. □

Problem 1.5.1. *For all the positive integers $k \leq 1999$, let $S_1(k)$ be the sum of all the remainders of the numbers $1, 2, \ldots, k$ when divided by 4, and let $S_2(k)$ be the sum of all the remainders of the numbers $k+1, k+2, \ldots, 2000$ when divided by 3. Prove that there is an unique positive integer $m \leq 1999$ so that $S_1(m) = S_2(m)$.*

(1999 Romanian Mathematical Olympiad)

Solution. Let $A_k = \{1, 2, 3, \ldots, k\}$ and $B_k = \{k+1, k+2, \ldots, 2000\}$. From the division of integers we have

$$k = 4q_1 + r_1, \text{ with } r_1 \in \{0, 1, 2, 3\}. \tag{1}$$

If $s_1(k)$ is the sum of the remainders at the division by 4 of the last r_1 elements of A_k, then

$$S_1(k) = 6q_1 + s_1(k), \text{ with } 0 \leq s_1(k) \leq 6 \tag{2}$$

(if $r_1 = 0$, then set $s_1(k) = 0$).

Using again the division of integers there exist integers q_2, r_2 such that

$$2000 - k = 3q_2 + r_2, \text{ with } r_2 \in \{0, 1, 2\}. \tag{3}$$

If $s_2(k)$ is the sum of the remainders at the division by 3 of the last r_2 elements of B_k, then

$$S_2(k) = 3q_2 + s_2(k), \text{ with } 0 \leq s_2(k) \leq 3 \tag{4}$$

(again we set $s_2(k) = 0$, if $r_2 = 0$).

As $S_1(k) = S_2(k)$, $s_2(k) - s_1(k) = 3(2q_1 - q_2)$, so $3|2q_1 - q_2| = |s_2(k) - s_1(k)| \leq 6$, and $|2q_1 - q_2| \leq 2$. In other words, $|2q_1 - q_2| \in \{0, 1, 2\}$.

If $2q_1 = q_2$, then (1) and (3) imply $2000 - (r_1 + r_2) = 10q_1$, hence $10|(r_1 + r_2)$. Then $r_1 = r_2 = 0$ and $q_1 = 200$. From (1) it follows that $k = 800$, and from (2) and (4) we have $S_1(800) = S_2(800) = 1200$.

Furthermore $S_1(k) \leq S_1(k+1)$, and $S_2(k) \geq S_2(k+1)$ for all $k \in \{1, 2, \ldots, 1998\}$. Since $S_1(799) = S_1(800)$ and $S_2(799) = S_2(800) + 2 < S_1(800)$, we deduce that $S_1(k) < S_2(k)$ for all $k \in \{1, 2, \ldots, 799\}$. Since $S_1(801) = S_1(800) + 1 > S_2(800) \geq S_2(801)$, we derive that $S_1(k) > S_2(k)$ for all $k \in \{801, 802, \ldots, 1999\}$. Consequently, $S_1(m) = S_2(m)$ if and only if $m = 800$.

Problem 1.5.2. *Let n be a positive integer. Show that if a and b are integers greater than 1 such that $2^n - 1 = ab$, then $ab - (a - b) - 1$ can be written as $k \cdot 2^{2m}$ for some odd integer k and some positive integer m.*

(2001 Balkan Mathematical Olympiad)

Solution. Note that $ab - (a-b) - 1 = (a+1)(b-1)$. We shall show that the highest powers of two dividing $(a+1)$ and $(b-1)$ are the same. Let 2^s

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and 2^t be the highest powers of 2 dividing $(a+1)$ and $(b-1)$, respectively. Because $a+1, b+1 \leq ab+1 = 2^n$, we have $s, t \leq n$.

Note that 2^s divides $2^n = ab+1$ and $a+1$, so that

$$ab \equiv a \equiv -1 \pmod{2^s}.$$

Hence, $b \equiv 1 \pmod{2^s}$, or $2^s | b-1$, so that $s \leq t$.

Similarly, $ab \equiv -b \equiv -1 \pmod{2^t}$, so $a \equiv -1 \pmod{2^t}$, and $2^t | a+1$. Thus, $t \leq s$.

Therefore, $s = t$, the highest power of two dividing $(a+1)(b-1)$ is $2s$, and $ab-(a-b)-1 = k \cdot 2^{2s}$ for some odd k.

Problem 1.5.3. *Find all nonnegative integers m such that $(2^{2m+1})^2+1$ is divisible by at most two different primes.*

(2002 Baltic Mathematics Competition)

Solution. We claim $m = 0, 1, 2$ are the only such integers. It is easy to check that these values of m satisfy the requirement. Suppose some $m \geq 3$ works. Write

$$(2^{2m+1})^2 + 1 = (2^{2m+1}+1)^2 - 2 \cdot 2^{2m+1}$$

$$= (2^{2m+1} + 2^{m+1} + 1)(2^{2m+1} - 2^{m+1} + 1).$$

The two factors are both odd, and their difference is 2^{m+2}; hence, they are relatively prime. It follows that each is a prime power. We also know that $(2^{2m+1})^2 = 4^{2m+1} \equiv -1 \pmod{5}$, so one of the factors $2^{2m+1} \pm 2^{m+1} + 1$ must be a power of 5. Let $2^{2m+1} + 2^{m+1}s + 1 = 5^k$, where $s = \pm 1$ is the appropriate sign.

Taking the above equation modulo 8, and using the assumption $m \geq 3$, we obtain $5^k \equiv 1 \pmod{8}$, so that k is even. Writing $k = 2l$, we have

$$2^{m+1}(2^m + s) = (5^l - 1)(5^l + 1).$$

The factor $5^l + 1 \equiv 2 \pmod{4}$, so $5^l - 1 = 2^m a$ for some odd integer a. But if $a = 1$, then

$$2 = (5^l+1) - (5^l-1) = 2(2^m+s) - 2^m = 2^m + 2s \geq 2^3 - 2,$$

a contradiction, whereas if $a \geq 3$, then $5^l - 1 \geq 3 \cdot 2^m$ while $5^l + 1 \leq 2(2^m + s)$, another contradiction.

Problem 1.5.4. *Find an integer n with $100 \leq n \leq 1997$ such that n divides $2^n + 2$.*

(1997 Asian Pacific Mathematics Olympiad)

Solution. Note that 2 divides $2^n + 2$ for all n. Also, 11 divides $2^n + 2$ if and only if $n \equiv 6 \pmod{10}$, and 43 divides $2^n + 2$ if and only if $n \equiv 8 \pmod{14}$. Since $n = 946 = 2 \cdot 11 \cdot 43$ satisfies both congruences, n divides $2^n + 2$.

Remark. Actually, one can prove that there are infinitely many n such that $n|2^n + 2$. Also, any such n is even since by a theorem of W. Sierpinski we cannot have $n|2^{n-1} + 1$ unless $n = 1$.

Problem 1.5.5. *The number* $99\ldots99$ *(with 1997 nines) is written on a blackboard. Each minute, one number written on the blackboard is factored into two factors and erased, each factor is (independently) increased or diminished by 2, and the resulting two numbers are written. Is it possible that at some point all of the numbers on the blackboard equal 9?*

(1997 St. Petersburg City Mathematical Olympiad)

Solution. No, there is always a number congruent to 3 modulo 4: factoring such a number gives one factor congruent to 3 modulo 4, and changing that by 2 in either direction gives a number congruent to 3 modulo 4.

Problem 1.5.6. *Find the smallest positive integer which can be written both as (i) a sum of 2002 positive integers (not necessarily distinct), each of which has the same sum of digits and (ii) as a sum of 2003 positive integers (not necessarily distinct), each of which has the same sum of digits.*

(2002 Russian Mathematical Olympiad)

Solution. The answer is 10010. First observe that this is indeed a solution: $10010 = 2002 \cdot 5 = 1781 \cdot 4 + 222 \cdot 13$, so we may express 10010 as the sum of 2002 fives or of 1781 fours and 222 thirteens, where $1781 + 222 = 2003$. To prove minimality, observe that a number is congruent modulo 9 to the sum of its digits, so two positive integers with the same digit sum are in the same residue class modulo 9. Let k_1 be the digit sum of the 2002 numbers and k_2 the digit sum of the 2003 numbers. Then $4k_1 \equiv 2002k_1 \equiv 2003k_2 \equiv 5k_2 \pmod{9}$. If $k_1 \geq 5$, the sum of the 2002 numbers is at least 10010; if $k_2 \geq 5$, the sum of the 2003 numbers is greater than 10010. However, the solutions $k_1 \equiv 1, 2, 3, 4 \pmod{9}$ give $k_2 \equiv 8, 7, 6, 5$, respectively, so that at least one of k_1 or k_2 is greater than or equal to 5, and the minimal integer is 10010.

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Proposed problems

Problem 1.5.7. Find all integers $n > 1$ such that any prime divisor of $n^6 - 1$ is a divisor of $(n^3 - 1)(n^2 - 1)$.

(2002 Baltic Mathematics Competition)

Problem 1.5.8. Let $f(n)$ be the number of permutations $a_1, \ldots, a_n$ of the integers $1, \ldots, n$ such that

(i) $a_1 = 1$;

(ii) $|a_i - a_{i+1}| \leq 2,\ i = 1, \ldots, n-1$.

Determine whether $f(1996)$ is divisible by 3.

(1996 Canadian Mathematical Olympiad)

Problem 1.5.9. For natural numbers m, n, show that $2^n - 1$ is divisible by $(2^m - 1)^2$ if and only if n is divisible by $m(2^m - 1)$.

(1997 Russian Mathematical Olympiad)

Problem 1.5.10. Suppose that n is a positive integer and let

$$d_1 < d_2 < d_3 < d_4$$

be the four smallest positive integer divisors of n. Find all integers n such that

$$n = d_1^2 + d_2^2 + d_3^2 + d_4^2.$$

(1999 Iranian Mathematical Olympiad)

Problem 1.5.11. Let p be an odd prime. For each $i = 1, 2, \ldots, p-1$ denote by r_i the remainder when i^p is divided by p^2. Evaluate the sum

$$r_1 + r_2 + \cdots + r_{p-1}.$$

(Kvant)

Problem 1.5.12. Find the number of integers x with $|x| \leq 1997$ such that 1997 divides $x^2 + (x+1)^2$.

(1998 Indian Mathematical Olympiad)

1.6 Chinese remainder theorem

In many concrete situations we need to find a solution to some system of linear congruences of the form

$$a_1x \equiv b_1 \pmod{m_1}, \ldots, a_nx \equiv b_n \pmod{m_n}.$$

Under some additional hypothesis (for instance $gcd(a_k, m_k) = 1$, $k = 1, \ldots, n$) this system reduces to the form

$$x \equiv c_1 \pmod{m_1}, \ldots, x \equiv c_n \pmod{m_n}.$$

In solving this class of systems an important part is played by the following result:

Theorem 1.6.1. (Chinese Remainder Theorem) *Let $m_1, \ldots, m_n$ be positive integers different from 1 and pairwise relatively prime. Then for any nonzero integers $a_1, \ldots, a_r$ the system of linear congruences*

$$x \equiv a_1 \pmod{m_1}, \ldots, x \equiv a_r \pmod{m_r}$$

has solutions and any two such solutions are congruent modulo $m = m_1 \ldots m_r$.

Proof. It is clear that $gcd\left(\frac{m}{m_j}, m_j\right) = 1$, $j = 1, \ldots, r$. Applying Proposition 1.3.1 it follows that there is an integer b_j such that

$$\frac{m}{m_j}b_j \equiv 1 \pmod{m_j}, \quad j = 1, \ldots, r.$$

Then

$$\frac{m}{m_j}b_ja_j \equiv a_j \pmod{m_j}, \quad j = 1, \ldots, r.$$

Now consider the integer

$$x_0 = \sum_{j=1}^{r} \frac{m}{m_j}b_ja_j.$$

We have

$$x_0 \equiv \left(\sum_{j=1}^{r} \frac{m}{m_j}b_ja_j\right) \pmod{m_i} \equiv \frac{m}{m_i}b_ia_i \pmod{m_i}$$

$$\equiv a_i \pmod{m_i}, \quad i = 1, \ldots, r,$$

that is x_0 is a solution to the system of linear congruences.

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If x_1 is another solution, then $x_1 \equiv x_0 \pmod{m_i}$, $i = 1, \ldots, r$. Applying property 8) in Section 1.5, the conclusion follows. □

Example. Let us find the solutions to the system of linear congruences

$$x \equiv 2 \pmod 3, \quad x \equiv 1 \pmod 4, \quad x \equiv 3 \pmod 5.$$

We proceed as in the proof of the theorem. Because in this case $m = 3 \cdot 4 \cdot 5 = 60$, we have to find a solution to each of the congruences

$$\frac{60}{3}b_1 \equiv 1 \pmod 3, \quad \frac{60}{4}b_2 \equiv 1 \pmod 4, \quad \frac{60}{5}b_3 \equiv 1 \pmod 5.$$

This is equivalent to finding solutions to the congruences

$$2b_1 \equiv 1 \pmod 3, \quad 3b_2 \equiv 1 \pmod 4, \quad 2b_3 \equiv 1 \pmod 5.$$

We obtain $b_1 = 2$, $b_2 = 3$, $b_3 = 3$. Then

$$x_0 = 20 \cdot 2 \cdot 2 + 15 \cdot 3 \cdot 1 + 12 \cdot 3 \cdot 3 = 233.$$

Taking into account that all solutions are congruent modulo 60 it follows that it suffices to take $x_0 = 53$. All solutions are given by $x = 53 + 60k$, $k \in \mathbb{Z}$.

Problem 1.6.1. *We call a lattice point X in the plane visible from the origin O if the segment $\overline{OX}$ does not contain any other lattice points besides O and X. Show that for any positive integer n, there exists a square of n^2 lattice points (with sides parallel to the coordinate axes) such that none of the lattice points inside the square is visible from the origin.*

(2002 Taiwanese Mathematical Olympiad)

Solution. Suppose that the lower-left lattice point of such a square has coordinates (x_1, y_1). We shall show that it is possible to select (x_1, y_1) such that the square of lattice points with (x_1, y_1) at its corner and n points on a side contains only invisible points. This can be accomplished by ensuring that each point has both coordinates divisible by some prime number; this would imply that by dividing both coordinates by this prime we could find another lattice point that is between the origin and this point.

Select n^2 distinct prime numbers and call them $p_{i,j}$, $1 \leq 1, j \leq n$. Now find x_1 satisfying the following congruences:

$$x_1 \equiv 0 \pmod{p_{1,1}, p_{1,2} \ldots p_{1,n}},$$

$$x_1 + \equiv 0 \pmod{p_{2,1}p_{2,2}\dots p_{2,n}},$$

$$\dots$$

$$x_1 + n - 1 \equiv 0 \pmod{p_{n,1}p_{n,2}\dots p_{n,n}}.$$

Likewise select y_1 satisfying:

$$y_1 \equiv 0 \pmod{p_{1,1}p_{2,1}\dots p_{n,1}},$$

$$y_1 + 1 \equiv 0 \pmod{p_{1,2}p_{2,2}\dots p_{n,2}},$$

$$\dots$$

$$y_1 + n - 1 \equiv 0 \pmod{p_{1,n}p_{2,n}\dots, p_{n,n}}.$$

Both values must exist by the Chinese Remainder Theorem. Thus we have proved that it is possible to determine a position for (x_1, y_1) such that every point in the square of n^2 lattice points with (x_1, y_1) at it's lower left corner is associated with some prime by which both of its coordinates are divisible, thus all points in this square are not visible from the origin.

Problem 1.6.2. *Show that there exists an increasing sequence $\{a_n\}_{n=1}^{\infty}$ of natural numbers such that for any $k \geq 0$, the sequence $\{k+a_n\}$ contains only finitely many primes.*

(1997 Czech and Slovak Mathematical Olympiad)

Solution. Let p_k be the k-th prime number, $k \geq 1$. Set $a_1 = 2$. For $n \geq 1$, let a_{n+1} be the least integer greater than a_n that is congruent to $-k$ modulo p_{k+1} for all $k \leq n$. Such an integer exists by the Chinese Remainder Theorem. Thus, for all $k \geq 0$, $k + a_n \equiv 0 \pmod{p_{k+1}}$ for $n \geq k+1$. Then at most $k+1$ values in the sequence $\{k+a_n\}$ can be prime; from the $k+2$-th term onward, the values are nontrivial multiples of p_{k+1} and must be composite. This completes the proof.

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Problem 1.6.3. Let $P(x)$ be a polynomial with integer coefficients. Suppose that the integers $a_1, a_2, \dots, a_n$ have the following property: For any integer x there exists an $i \in \{1, 2, \dots, n\}$ such that $P(x)$ is divisible by a_i. Prove that there is an $i_0 \in \{1, 2, \dots, n\}$ such that a_{i_0} divides $P(x)$ for any integer x.

(St. Petersburg City Mathematical Olympiad)

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Problem 1.6.4. For any positive integer set $\{a_1, a_2, \ldots, a_n\}$ there exists a positive integer b such that the set $\{ba_1, ba_2, \ldots, ba_n\}$ consists of perfect powers.

1.7 Numerical systems

1.7.1 Representation of integers in an arbitrary base

The fundamental result in this subsection is given by the following theorem:

Theorem 1.7.1. *Let b be an integer greater than 1. For any integer $n \geq 1$ there is a unique system $(k, a_0, a_1, \ldots, a_k)$ of integers such that $0 \leq a_i \leq b-1$, $i = 0, 1, \ldots, k$, $a_k \neq 0$, and*

$$n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0. \tag{1}$$

Proof. For the existence, we repeatedly apply the Division Algorithm:

$$n = q_1 b + r_1, \quad 0 \leq r_1 \leq b-1$$

$$q_1 = q_2 b + r_2, \quad 0 \leq r_2 \leq b-1$$

$$\cdots$$

$$q_{k-1} = q_k b + r_k, \quad 0 \leq r_k \leq b-1$$

where q_k is the last nonzero quotient.

Let

$$q_0 = n,\ a_0 = n - q_1 b,\ a_1 = q_1 - q_2 b, \ldots, a_{k-1} = q_{k-1} - q_k b,\ a_k = q_k.$$

Then

$$\sum_{i=0}^{k} a_i b^i = \sum_{i=0}^{k-1} (q_i - q_{i+1} b) b^i + q_k b^k = q_0 + \sum_{i=1}^{k} q_i b^i - \sum_{i=1}^{k} q_i b^i = q_0 = n.$$

For the uniqueness, assume that $n = c_0 + c_1 b + \cdots + c_h b^h$ is another such representation.

If $h \neq k$, for example $h > k$, then $n \geq b^k \geq b^{k+1}$, But

$$n = a_0 + a_1 b + \cdots + a_k b^k \leq (b-1)(1 + b + \cdots + b^k) = b^{k+1} - 1 < b^{k+1},$$

a contradiction.

If $h = k$, then

$$a_0 + a_1 b + \cdots + a_k b^k = c_0 + c_1 b + \cdots + c_k b^k$$

and so $b|a_0 - c_0$. On the other hand, $|a_0 - c_0| < b$, hence $a_0 = c_0$, Therefore

$$a_1 + a_2 b + \cdots + a_k b^{k-1} = c_1 + c_2 b + \cdots + c_k b^{k-1}.$$

Repeating the procedure above, it follows that $a_1 = c_1$, $a_2 = c_2$,..., $a_k = c_k$. □

Relation (1) is called *the base b representation* of n and is denoted by

$$n = \overline{a_k a_{k-1} \ldots a_0}_{(b)}$$

The usual *decimal representation* corresponds to $b = 10$.

Examples. 1) $4567 = 4 \cdot 10^3 + 5 \cdot 10^2 + 6 \cdot 10 + 7 = \overline{4567}_{(10)}$.

2) Let us write $\overline{1010011}_{(2)}$ in base 10. We have

$$\overline{1010011}_{(2)} = 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2 + 1 = 64 + 16 + 2 + 1 = 83.$$

3) Let us write 1211 in base 3. As above, dividing by 3 successively, the remainders give the digits of the base 3 representation, beginning with the last. The first digit is the last nonzero quotient. We can arrange the computations as follows:

1211	3				
1209	403	3			
2	402	134	3		
	1	132	44	3	
		2	42	14	3
			2	12	4	3
				2	3	**1**
					1	

Hence $1211 = \overline{1122212}_{(3)}$.

1.7.2 Divisibility criteria in the decimal system

We will prove some divisibility criteria for integers in decimal representation. In this subsection, we will denote $n = \overline{a_h a_{h-1} \ldots a_0}$ with the understanding that we operate in base 10.

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Criterion 1. *a) The integer* $n = \overline{a_h a_{h-1} \dots a_0}$ *is divisible by* 3 *if and only if the sum* $s(n)$ *of its digits is divisible by* 3.

b) The integer $n = \overline{a_h a_{h-1} \dots a_0}$ *is divisible by* 9 *if and only if* $s(n)$ *is divisible by* 9.

Proof. We have $10^k \equiv 1 \pmod 9$ since $10 \equiv 1 \pmod 9$, hence

$$n = \sum_{k=0}^{h} a_k 10^k s(n) \pmod 9.$$

Both conclusions follow. □

Criterion 2. *The integer* $n = \overline{a_h a_{h-1} \dots a_0}$ *is divisible by* 11 *if and only if* $a_0 - a_1 + \dots + (-1)^h a_h$ *is divisible by* 11.

Proof. We have $10^k = (11-1)^k \equiv (-1)^k \pmod{11}$, hence

$$n = \sum_{k=0}^{h} a_k 10^k \equiv \sum_{k=0}^{h} (-1)^k a_k \pmod{11},$$

and the conclusion follows. □

Criterion 3. *The integer* $n = \overline{a_h a_{h-1} \dots a_0}$ *is divisible by* $7, 11,$ *or* 13 *if and only if* $\overline{a_h a_{h-1} \dots a_3} - \overline{a_2 a_1 a_0}$ *has this property.*

Proof. We have

$$n = \overline{a_2 a_1 a_0} + (1001 - 1)\overline{a_h a_{h-1} \dots a_3}$$

$$= 7 \cdot 11 \cdot 13 \overline{a_h a_{h-1} \dots a_3} - (\overline{a_h a_{h-1} \dots a_3} - \overline{a_2 a_1 a_0})$$

hence the desired conclusion. □

Criterion 4. *The integer* $n = \overline{a_h a_{h-1} \dots a_0}$ *is divisible by* 27 *or* 37 *if and only if* $\overline{a_h a_{h-1} \dots a_3} + \overline{a_2 a_1 a_0}$ *has this property.*

Proof. We have

$$n = \overline{a_2 a_1 a_0} + (999 + 1)\overline{a_h a_{h-1} \dots a_3}$$

$$= 27 \cdot 37 \overline{a_h a_{h-1} \dots a_3} + (\overline{a_h a_{h-1} \dots a_3} + \overline{a_2 a_1 a_0})$$

and the conclusion follows. □

Examples. 1) The integer 123456789 is divisible by 9 because the sum of its digits $1 + 2 + \dots + 9 = 45$ has this property (Criterion 1b)).

2) The integer $\underbrace{20 \dots 04}_{2004}$ is not a perfect square because the sum of its digits is 6, a multiple of 3 but not of 9, hence the integer itself has these properties (Criteria 1a) and 1b)).

3) All integers of the form $\overline{abcdef}$ where $a+c+e=8$ and $b+d+f=19$ are divisible by 99, because $a+b+c+d+f=8+19$, a multiple of 9, and $f-e+d-c+b-a=19-8$, a multiple of 11 and the conclusion follows from Criteria 1b) and 2.

4) For any nonzero digit a, the integer $\overline{a1234567}$ is not divisible by 37. Indeed, applying Criterion 4 we have $\overline{a1234}+567=\overline{a1801}$ and $\overline{a1}+801=\overline{8a2}=800+10a+2=37\cdot 21+10a+25$. The integer $10a+25=5(2a+5)$ is not divisible by 37 because $7\leq 2a+5\leq 23$.

Problem 1.7.1. *Find all integers written as $\overline{abcd}$ in decimal representation and $\overline{dcba}$ in base 7.*

Solution. We have

$$\overline{abcd}_{(10)}=\overline{dcba}_{(7)} \Leftrightarrow 999a+93b=39c+342d \Leftrightarrow 333a+31b=13c+114d,$$

hence $b\equiv c \pmod 3$. As $b,c\in\{0,1,2,3,4,5,6\}$, the possibilities are:

i) $b=c$;

ii) $b=c+3$;

iii) $b+3=c$.

In the first case we must have $a=2a'$, $d=3d'$, $37a'+b=19d'$, $d'=2$. Hence $a'=1$, $a=2$, $d=6$, $b=1$, $c=1$, and the number $\overline{abcd}$ is 2116.

In the other cases a has to be odd. Considering $a=1$, 3 or 5 we obtain no solutions.

Problem 1.7.2. *Prove that every integer $k>1$ has a multiple less than k^4 whose decimal expansion has at most four distinct digits.*

(1996 German Mathematical Olympiad)

Solution. Let n be the integer such that $2^{n-1}\leq k<2^n$. For $n\leq 6$ the result is immediate, so assume $n>6$.

Let S be the set of nonnegative integers less than 10^n whose decimal digits are all 0 or 1. Since $|S|=2^n>k$, we can find two elements $a<b$ of S which are congruent modulo k, and $b-a$ only has the digits 8, 9, 0, 1 in its decimal representation. On the other hand,

$$b-a\leq 1+10+\cdots+10^{n-1}<10^n<16^{n-1}\leq k^4,$$

hence $b-a$ is the desired multiple.

Problem 1.7.3. *A positive integer is written on a board. We repeatedly erase its unit digit and add 5 times that digit to what remains. Starting with 7^{1998}, can we ever end up at 1998^7?*

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(1998 Russian Mathematical Olympiad)

Solution. The answer is no. Let a_n be the n-th number written on the board; let u_n be the unit digit and $a_n = 10t_n + u_n$. We have

$$a_{n+1} = t_n + 5u_n \equiv 50t_n + 5u_n = 5(10t_n + u_n) = 5a_n \pmod 7.$$

Since $a_1 = 7^{1998} \equiv 0 \not\equiv 1998^7 \pmod 7$, we can never obtain 1998^7 from 7^{1998}.

Problem 1.7.4. *Find all the three digit numbers $\overline{abc}$ such that the 6003-digit number $\overline{abcabc\ldots abc}$ is divisible by 91 ($\overline{abc}$ occurs 2001 times).*

Solution. The number is equal to

$$\overline{abc}(1 + 10^3 + 10^6 + \cdots + 10^{6000}).$$

Since 91 is a divisor of $1001 = 1 + 10^3$ and the sum $S = 1 + 10^3 + 10^6 + \cdots + 10^{6000}$ has 2001 terms, it follows that 91 and $(1+10^3)+10^6(1+10^3)+\cdots+10^{1999}(1+10^3)+10^{6000}$ are relatively prime. Thus $\overline{abc}$ is divisible by 91. The numbers are

$$182,\ 273,\ 364,\ 455,\ 546,\ 637,\ 728,\ 819,\ 910.$$

Problem 1.7.5. *Let n be an integer greater than 10 such that each of its digits belongs to the set $S = \{1, 3, 7, 9\}$. Prove that n has some prime divisor greater than or equal to 11.*

(1999 Iberoamerican Mathematical Olympiad)

Solution. Note that any product of any two numbers from $\{1, 3, 7, 9\}$ taken modulo 20 is still in $\{1, 3, 7, 9\}$. Therefore any finite product of such numbers is still in this set. Specifically, any number of the form 3^j7^k is congruent to 1, 3, 7, or 9 (mod 20).

Now if all the digits of $n \geq 10$ are in S, then its tens digit is odd and we cannot have $n \equiv 1, 3, 7,$ or $9 \pmod{20}$. Thus, n cannot be of the form 3^j7^k. Nor can n be divisible by 2 or 5 (otherwise, its last digit would not be 1, 3, 7, or 9). Hence n must be divisible by some prime greater than or equal to 11, as desired.

Problem 1.7.6. *Find all natural numbers with the property that, when the first digit is moved to the end, the resulting number is $3\frac{1}{2}$ times the original one.*

(1997 South African Mathematical Olympiad)

Solution. Such numbers are those of the form

$$153846153846153846\ldots153846.$$

Obviously, since the number has the same number of digits when multiplied by 3.5, it must begin with either 1 or 2.

Case 1. The number is of the form 10^N+A, $A<10^N$. So $7/2*(10^N+A) = 10A+1 \rightarrow A = (7*10^N-2)/13$. The powers of 10 repeat with a period of 6 mod 13 (10,9,12,3,4,1) so A will be an integer iff $n \equiv 5 \pmod 6$. This gives the family of solutions above.

Case 2. The number is of the form $2*10^N+A$, $A<10^N$. Then, as before, $A = (14*10^N-4)/13$. But as $A<10^N$, this implies $10^N<4$, which is impossible.

Problem 1.7.7. *Any positive integer m can be written uniquely in base 3 form as a string of 0's, 1's and 2's (not beginning with a zero). For example,*

$$98 = 81+9+2\times 3+2\times 1 = (10122)_3.$$

Let $c(m)$ denote the sum of the cubes of the digits of the base 3 form of m; thus, for instance

$$c(98) = 1^3+0^3+1^3+2^3+2^3 = 18.$$

Let n be any fixed positive integer. Define the sequence $\{u_r\}$ as

$$u_1 = n, \text{ and } u_r = c(u_{r-1}) \text{ for } r \geq 2.$$

Show that there is a positive integer r such that $u_r = 1, 2,$ or 17.

(1999 United Kingdom Mathematical Olympiad)

Solution. If m has $d \geq 5$ digits then we have $m \geq 3^{d-1} = (80+1)^{(d-1)/4} \geq 80\cdot\dfrac{d-1}{4}+1 > 8d$ by Bernoulli's inequality. Thus $m > c(m)$.

If $m>32$ has 4 digits in base 3, then $c(m) \leq 2^3+3^3+2^3+2^3 = 32 < m$. On the other hand, if $27 \leq m \leq 32$, then m starts with the digits 10 in base 3 and $c(m) < 1^3+0^3+2^3+2^3 = 17 < m$.

Therefore $0 < c(m) < m$ for all $m \geq 27$. Hence, eventually, we have $u_s < 27$. Because u_s has at most three digits, u_{s+1} can only equal 8, 16, 24, 1, 9, 17, 2, 10, or 3. If it equals 1, 2, or 17 we are already done; if it equals 3 or 9 then $u_{s+2} = 1$. Otherwise a simple check shows that u_r will

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eventually equal 2:

$$\left.\begin{array}{l} 8 = (22)_3 \\ 24 = (220)_3 \end{array}\right\} \to 16 = (121)_3 \to 10 = (101_3 \to 2.$$

Problem 1.7.8. *Do there exist n-digit numbers M and N such that all of the digits of M are even, all of the digits of N are odd, each digit from 0 to 9 occurs exactly once among M and N, and N divides M?*

(1998 Russian Mathematical Olympiad)

Solution. The answer is no. We proceed by indirect proof. Suppose that such M and N exist and let $a = M/N$. Then $M \equiv 0 + 2 + 4 + 6 + 8 \equiv 2 \pmod 9$ and $N \equiv 1 + 3 + 5 + 7 + 9 \equiv 7 \pmod 9$; they are both relatively prime to 9. Now $a \equiv M/N \equiv 8 \pmod 9$ and so $a \geq 8$. But $N \geq 13579$ so $M = aN \geq 8(13579) > 99999$, a contradiction.

Problem 1.7.9. *Let $k \geq 1$ be an integer. Show that there are exactly 3^{k-1} positive integers n with the following properties:*

(a) The decimal representation of n consists of exactly k digits.

(b) All digits of k are odd.

(c) The number n is divisible by 5.

(d) The number $m = n/5$ has k (decimal) digits.

(1996 Austrian-Polish Mathematics Competition)

Solution. The multiplication in each place must produce an even number of carries, since these will be added to 5 in the next place and an odd digit must result. Hence all of the digits of m must be 1, 5 or 9, and the first digit must be 1, since m and n have the same number of decimal digits. Hence there are 3^{k-1} choices for m and hence for n.

Problem 1.7.10. *Can the number obtained by writing the numbers from 1 to n in order $(n > 1)$ be the same when read left-to-right and right-to-left?*

(1996 Russian Mathematical Olympiad)

Solution. This is not possible. Suppose $N = 123\ldots321$ is an m-digit symmetric number, formed by writing the numbers from 1 to n in succession. Clearly $m > 18$. Also let A and B be the numbers formed from the first and last k digits, respectively, of N, where $k = \lfloor m/2 \rfloor$. If 10^p is the largest power of 10 dividing A, then $n < 2 \cdot 10^{p+1}$, that is, n has at most

$p+2$ digits. Moreover, A and B must contain the fragments

$$\underbrace{99\ldots9}_{p}\underbrace{100\ldots0}_{p}1 \quad \text{and} \quad 1\underbrace{00\ldots0}_{p}\underbrace{99\ldots9}_{p},$$

respectively, which is impossible.

Problem 1.7.11. *Three boxes with at least one marble in each are given. In a step we choose two of the boxes, doubling the number of marbles in one of the boxes by taking the required number of marbles from the other box. Is it always possible to empty one of the boxes after a finite number of steps?*

(1999 Slovenian Mathematical Olympiad)

Solution. Without loss of generality suppose that the number of marbles in the boxes are a, b, and c with $a \leq b \leq c$. Write $b = qa+r$ where $0 \leq r < a$ and $q \geq 1$. Then express q in binary:

$$q = m_0 + 2m_1 + \cdots + 2^k m_k,$$

where each $m_i \in \{0, 1\}$ and $m_k = 1$. Now for each $i = 0, 1, \ldots, k$, add $2^i a$ marbles to the first box: if $m_i = 1$ take these marbles from the second box; otherwise take them from this third box. In this way we take at most $(2^k - 1)a < qa \leq b \leq c$ marbles from the third box and exactly qa marbles from the second box altogether.

In the second box there are now $r < a$ marbles left. Thus the box with the least number of marbles now contains less than a marbles. Then by repeating the described procedure, we will eventually empty one of the boxes.

Proposed problems

Problem 1.7.12. The natural number A has the following property: the sum of the integers from 1 to A, inclusive, has decimal expansion equal to that of A followed by three digits. Find A.

(1999 Russian Mathematical Olympiad)

Problem 1.7.13. A positive integer is said to be *balanced* if the number of its decimal digits equals the number of its distinct prime factors. For instance, 15 is balanced, while 49 is not. Prove that there are only finitely many balanced numbers.

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(1999 Italian Mathematical Olympiad)

Problem 1.7.14. Let $p \geq 5$ be a prime and choose $k \in \{0, \ldots, p-1\}$. Find the maximum length of an arithmetic progression, none of whose elements contain the digit k when written in base p.

(1997 Romanian Mathematical Olympiad)

Problem 1.7.15. How many 10-digit numbers divisible by 66667 are there whose decimal representation contains only the digits 3, 4, 5, and 6?

(1999 St. Petersburg City Mathematical Olympiad)

Problem 1.7.16. Call positive integers *similar* if they are written using the same set of digits. For example, for the set 1, 1, 2, the similar numbers are 112, 121 and 211. Prove that there exist 3 similar 1995-digit numbers containing no zeros, such that the sum of two them equals the third.

(1995 Russian Mathematical Olympiad)

Problem 1.7.17. Let k and n be positive integers such that

$$(n+2)^{n+2}, (n+4)^{n+4}, (n+6)^{n+6}, \ldots, (n+2k)^{n+2k}$$

end in the same digit in decimal representation. At most how large is k?

(1995 Hungarian Mathematical Olympiad)

Problem 1.7.18. Let

$$\prod_{n=1}^{1996}(1+nx^{3^n}) = 1 + a_1x^{k_1} + a_2x^{k_2} + \cdots + a_mx^{k_m},$$

where $a_1, a_2, \ldots, a_m$ are nonzero and $k_1 < k_2 < \cdots < k_m$, Find a_{1996}.

(1996 Turkish Mathematical Olympiad)

Problem 1.7.19. For any positive integer k, let $f(k)$ be the number of element in the set $\{k+1, k+2, \ldots, 2k\}$ whose base 2 representation has precisely three 1s.

a) Prove that, for each positive integer m, there exists at least one positive integer k, such that $f(k) = m$.

b) Determine all positive integers m for which there exists exactly one k with $f(k) = m$.

(35th IMO)

Problem 1.7.20. For each positive integer n, let $S(n)$ be the sum of digits in the decimal representation of n. Any positive integer obtained by removing several (at least one) digits from the right-hand end of the decimal representation of n is called a stump of n. Let $T(n)$ be the sum of all stumps of n. Prove that $n = S(n) + 9T(n)$.

(2001 Asian Pacific Mathematical Olympiad)

Problem 1.7.21. Let p be a prime number and m be a positive integer. Show that there exists a positive integer n such that there exist m consecutive zeroes in the decimal representation of p^n.

(2001 Japanese Mathematical Olympiad)

Problem 1.7.22. Knowing 2^{29} is an 9-digit number whose digits are distinct, without computing the actual number determine which of the ten digits is missing. Justify your answer.

Problem 1.7.23. It is well known that the divisibility tests for division by 3 and 9 do not depend on the order of the decimal digits. Prove that 3 and 9 are the only positive integers with this property. More exactly, if an integer $d > 1$ has the property that $d|n$ implies $d|n_1$, where n_1 is obtained from n through an arbitrary permutation of its digits, then $d = 3$ or $d = 9$.

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2

Powers of Integers

An integer n is a *perfect square* if $n = m^2$ for some integer m. Taking into account the prime factorization, if $m = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, then $n = p_1^{2\alpha_1} \dots p_k^{2\alpha_k}$. That is, n is a perfect square if and only if all exponents in its prime factorization are even.

An integer n is a *perfect power* if $n = m^s$ for some integers m and s, $s \geq 2$. Similarly, n is an s-th perfect power if and only if all exponents in its prime factorization are divisible by s.

We say that the integer n is *squarefree* if for any prime divisor p, p^2 does not divide n. Similarly, we can define the s-th power-free integers.

These preliminary considerations seem trivial but as you will see shortly they have significant rich applications in solving various problems.

2.1 Perfect squares

Problem 2.1.1. *Find all nonnegative integers n such that there are integers a and b with the property:*

$$n^2 = a + b \text{ and } n^3 = a^2 + b^2.$$

(2004 Romanian Mathematical Olympiad)

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Solution. From the inequality $2(a^2+b^2) \geq (a+b)^2$ we get $2n^3 \geq n^4$, that is $n \leq 2$. Thus:

- for $n=0$, we choose $a=b=0$,
- for $n=1$, we take $a=1$, $b=0$ and
- for $n=2$, we may take $a=b=2$.

Problem 2.1.2. *Find all integers n such that $n-50$ and $n+50$ are both perfect squares.*

Solution. Let $n-50=a^2$ and $n+50=b^2$. Then $b^2-a^2=100$, so $(b-a)(b+a)=2^2\cdot 5^2$. Because $b-a$ and $b+a$ are distinct and of the same parity, the only possibility is $b-a=2$ and $b+a=50$, yielding $b=26$ and $a=24$. Hence there is only one n with this property, namely $n=626$.

Problem 2.1.3. *Let $n \geq 3$ be a positive integer. Show that it is possible to eliminate at most two numbers among the elements of the set $\{1,2,\ldots,n\}$ such that the sum of remaining numbers is a perfect square.*

(2003 Romanian Mathematical Olympiad)

Solution. Let $m=\left\lfloor \sqrt{\dfrac{n(n+1)}{2}} \right\rfloor$. From $m^2 \leq \dfrac{n(n+1)}{2} < (m+1)^2$ we obtain

$$\frac{n(n+1)}{2} - m^2 < (m+1)^2 - m^2 = 2m+1.$$

Therefore, we have:

$$\frac{n(n+1)}{2} - m^2 \leq 2m \leq \sqrt{2n^2+2n} \leq 2n-1.$$

Since, any number k, $k \leq 2n-1$ can be obtained by adding at most two numbers from $\{1,2,\ldots,n\}$, we obtain the result.

Problem 2.1.4. *Let k be a positive integer and $a=3k^2+3k+1$.*

(i) Show that $2a$ and a^2 are sums of three perfect squares.

(ii) Show that if a is a divisor of a positive integer b and b is a sum of three perfect squares then any power b^n is a sum of three perfect squares.

(2003 Romanian Mathematical Olympiad)

Solution. (i) $2a=6k^2+6k+2=(2k+1)^2+(k+1)^2+k^2$ and $a^2=9k^2+19k^3+15k^2+6k+1=(k^2+k)^2+(2k^2+3k+1)^2+k^2(2k+1)=a_1^2+a_2^2+a_3^2$.

(ii) Let $b=ca$. Then $b=b_1^2+b_2^2+b_3^2$ and $b^2=c^2a^2=c^2(a_1^2+a_2^2+a_3^2)$. To end the proof, we proceed as follows: for $n=2p+1$ we have $b^{2p+1}=(b^p)^2(b_1^2+b_2^2+b_3^2)$ and for $n=2p+2$, $b^n=(b^p)^2b^2=(b^p)^2c^2(a_1^2+a_2^2+a_3^2)$.

Problem 2.1.5. *a) Let k be an integer number. Prove that the number:*

$$(2k+1)^3-(2k-1)^3$$

is the sum of three squares.

b) Let n be a positive number. Prove that the number $(2n+1)^3-2$ can be represented as the sum of $3n-1$ squares greater than 1.

(2000 Romanian Mathematical Olympiad)

Solution. a) It is easy to check that

$$(2k+1)^3-(2k-1)^3=(4k)^2+(2k+1)^2+(2k-1)^2.$$

b) Observe that

$$(2n+1)^3-1=(2n+1)^3-(2n-1)^3+(2n-1)^3-(2n-3)^3+\cdots+3^3-1^3.$$

Each of the n differences in the right hand side can be written as a sum of three squares greater than 1, except for the last one:

$$3^3-1^3=4^2+3^2+1^2.$$

It follows that

$$(2n+1)^3-2=3^2+4^2+\sum_{k=2}^{n}[(4k)^2+(2k+1)^2+(2k-1)^2]$$

as desired.

Problem 2.1.6. *Prove that for any positive integer n the number*

$$\frac{\left(17+12\sqrt{2}\right)^n-\left(17-12\sqrt{2}\right)^n}{4\sqrt{2}}$$

is an integer but not a perfect square.

Solution. Note that $17+12\sqrt{2}=\left(\sqrt{2}+1\right)^4$ and $17-12\sqrt{2}=\left(\sqrt{2}-1\right)^4$, so

$$\frac{\left(17+12\sqrt{2}\right)^n-\left(17-12\sqrt{2}\right)^n}{4\sqrt{2}}=\frac{\left(\sqrt{2}+1\right)^{4n}-\left(\sqrt{2}-1\right)^{4n}}{4\sqrt{2}}=$$

$$=\frac{\left(\sqrt{2}+1\right)^{2n}+\left(\sqrt{2}-1\right)^{2n}}{2}\cdot\frac{\left(\sqrt{2}+1\right)^{2n}-\left(\sqrt{2}-1\right)^{2n}}{2\sqrt{2}}$$

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Define

$$A = \frac{\left(\sqrt{2}+1\right)^{2n} + \left(\sqrt{2}-1\right)^{2n}}{2} \quad \text{and} \quad B = \frac{\left(\sqrt{2}+1\right)^{2n} - \left(\sqrt{2}-1\right)^{2n}}{2\sqrt{2}}$$

Using the binomial expansion formula we obtain positive integers x and y such that

$$\left(\sqrt{2}+1\right)^{2n} = x + y\sqrt{2}, \quad \left(\sqrt{2}-1\right)^{2n} = x - y\sqrt{2}$$

Then

$$x = \frac{\left(\sqrt{2}+1\right)^{2n} + \left(\sqrt{2}-1\right)^{2n}}{2} = A$$

and

$$y = \frac{\left(\sqrt{2}+1\right)^{2n} - \left(\sqrt{2}-1\right)^{2n}}{2\sqrt{2}} = B$$

and so AB is as integer, as claimed.

Observe that

$$A^2 - 2B^2 = (A + \sqrt{2}B)(A - \sqrt{2}B) = (\sqrt{2}+1)^{2n}(\sqrt{2}-1)^{2n} = 1$$

so A and B are relatively prime. It is sufficient to prove that at least one of them is not a perfect square.

We have

$$A = \frac{\left(\sqrt{2}+1\right)^{2n} + \left(\sqrt{2}-1\right)^{2n}}{2} = \left[\frac{\left(\sqrt{2}+1\right)^{n} + \left(\sqrt{2}-1\right)^{n}}{\sqrt{2}}\right]^2 - 1 \quad (1)$$

and

$$A = \frac{\left(\sqrt{2}+1\right)^{2n} + \left(\sqrt{2}-1\right)^{2n}}{2} = \left[\frac{\left(\sqrt{2}+1\right)^{n} - \left(\sqrt{2}-1\right)^{n}}{\sqrt{2}}\right]^2 + 1 \quad (2)$$

Since only one of the numbers

$$\frac{\left(\sqrt{2}+1\right)^{n} + \left(\sqrt{2}-1\right)^{n}}{\sqrt{2}}, \quad \frac{\left(\sqrt{2}+1\right)^{n} - \left(\sqrt{2}-1\right)^{n}}{\sqrt{2}}$$

is an integer – depending on the parity of n – from the relations (1) and (2) we derive that A is not a square. This completes the proof.

Problem 2.1.7. *The integers a and b have the property that for every nonnegative integer n, the number $2^n a + b$ is a perfect square. Show that $a = 0$.*

(2001 Polish Mathematical Olympiad)

Solution. If $a \neq 0$ and $b = 0$, then at least one of $2^1a + b$ and $2^2a + b$ is not a perfect square, a contradiction.

If $a \neq 0$ and $b \neq 0$, then each $(x_n, y_n) = (2\sqrt{2^n a + b}, \sqrt{2^{n+2}a + b})$ satisfies

$$(x_n + y_n)(x_n - y_n) = 3b.$$

Hence, $x + n + y_n | 3b$ for each n. But thus is impossible because $3b \neq 0$ but $|x_n + y_n| > |3b|$ for large enough n.

Therefore, $a = 0$.

Remark. We invite the courageous reader to prove that if $f \in \mathbb{Z}[X]$ is a polynomial and $f(2^n)$ is a perfect square for all n, then there is $g \in \mathbb{Z}[X]$ such that $f = g^2$.

Problem 2.1.8. *Prove that the number*

$$\underbrace{11\ldots11}_{1997}\underbrace{22\ldots22}_{1998}5$$

is a perfect square.

Solution.

$$\begin{aligned} N &= \underbrace{11\ldots11}_{1997} \cdot 10^{1999} + \underbrace{22\ldots22}_{1998} \cdot 10 + 5 \\ &= \frac{1}{9}(10^{1997} - 1) \cdot 10^{1999} + \frac{2}{9}(10^{1998} - 1) \cdot 10 + 5 \\ &= \frac{1}{9}(10^{3996} + 2 \cdot 5 \cdot 10^{1998} + 25) = \left[\frac{1}{3}(10^{1998} + 5)\right]^2 \\ &= \left(\frac{1\overbrace{00\ldots00}^{1997}5}{3}\right)^2 = \underbrace{33\ldots33}_{1997}5^2. \end{aligned}$$

Problem 2.1.9. *Find all the positive integers n, $n \geq 1$, such that $n^2 + 3^n$ is a perfect square.*

Solution. Let m be a positive integer such that

$$m^2 = n^2 + 3^n.$$

Since $(m - n)(m + n) = 3^n$, there is $k \geq 0$ such that $m - n = 3^k$ and $m + n = 3^{n-k}$. From $m - n < m + n$ follows $k < n - k$, and so $n - 2k \geq 1$.

If $n - 2k = 1$, then $2n = (m+n) - (m-n) = 3^{n-k} - 3^k = 3^k(3^{n-2k} - 1) = 3^k(3^1 - 1) = 2 \cdot 3^k$, so $n = 3^k = 2k + 1$. We have $3^m = (1 + 2)^m =$

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$1+2m+2^2\binom{m}{2}+\cdots > 2m+1$, therefore $k=0$ or $k=1$ and consequently $n=1$ or $n=3$.

If $n-2k>1$, then $n-2k\geq 2$ and $k\leq n-k-2$. It follows that $3^k\leq 3^{n-k-2}$, and consequently

$$\begin{aligned}2n &= 3^{n-k}-3^k \geq 3^{n-k}-3^{n-k-2}=3^{n-k-2}(3^2-1)=8\cdot 3^{n-k-2}\\ &\geq 8[1+2(n-k-2)]=16n-16k-24,\end{aligned}$$

which implies $8k+12\geq 7n$.

On the other hand, $n\geq 2k+2$, hence $7n\geq 14k+14$, contradiction.

In conclusion, the only possible values for n are 1 and 3.

Problem 2.1.10. *Find the number of five-digit perfect squares having the last two digits equal.*

Solution. Suppose $n=\overline{abcdd}$ is a perfect square. Then $n=100\overline{abc}+11d=\mathcal{M}4+3d$, and since all the squares have the form $\mathcal{M}4$ or $\mathcal{M}4+1$ and $d\in\{0,1,4,5,6,9\}$, as the last digit of a square, it follows that $d=0$ or $d=4$.

If $d=0$, then $n=100\overline{abc}$ is a square if $\overline{abc}$ is a square.

Hence $\overline{abc}\in\{10^2,11^2,\ldots,31^2\}$, so there are 22 numbers.

If $d=4$, then $100\overline{abc}+44=n=k^2$ implies $k=2p$ and $\overline{abc}=\dfrac{p^2-11}{25}$.

1) If $p=5x$, then $\overline{abc}$ is not an integer, false.

2) If $p=5x+1$, then $\overline{abc}=\dfrac{25x^2+10x-1}{25}=x^2+\dfrac{2(x-1)}{5}\ \Rightarrow$ $x\in\{11,16,21,26,31\}$, so there are 5 solutions.

3) If $p=5x+2$, then $\overline{abc}=x^2+\dfrac{20x-7}{25}\notin\mathbb{N}$, false.

4) If $p=5x+3$, then $\overline{abc}=x^2+\dfrac{30x-2}{25}\notin\mathbb{N}$, false.

5) If $p=5x+4$ then $\overline{abc}=x^2+\dfrac{8x+1}{5}$, hence $x=\mathcal{M}5+3\ \Rightarrow$ $x\in\{13,18,23,28\}$, so there are 4 solutions.

Finally, there are $22+5+4=31$ squares.

Problem 2.1.11. *The last four digits of a perfect square are equal. Prove they are all zero.*

(2002 Romanian Team Selection Test for JBMO)

Solution. Denote by k^2 the perfect square and by a the digit that appears in the last four position. It easily follows that a is one of the numbers 0, 1, 4, 5, 6, 9. Thus $k^2\equiv a\cdot 1111 \pmod{16}$.

1) If $a=0$, we are done.

2) Suppose that $a \in \{1,5,9\}$. Since $k^2 \equiv 0 \pmod 8$, $k^2 \equiv 1 \pmod 8$ or $k^2 \equiv 4 \pmod 8$ and $1111 \equiv 7 \pmod 8$, we obtain $1111 \equiv 7 \pmod 8$, $5 \cdot 1111 \equiv 3 \pmod 8$ and $9 \cdot 1111 \equiv 7 \pmod 8$. Thus the congruence $k^2 \equiv a \cdot 1111 \pmod{16}$ cannot hold.

3) Suppose $a \in \{4,6\}$. As $1111 \equiv 7 \pmod{16}$, $4 \cdot 1111 \equiv 12 \pmod{16}$ and $6 \cdot 1111 \equiv 10 \pmod{16}$, we conclude that in this case the congruence $k^2 \equiv a \cdot 1111 \pmod{16}$ cannot hold. Thus $a = 0$.

Problem 2.1.12. *Let $1 < n_1 < n_2 < \cdots < n_k < \ldots$ be a sequence of integers such that no two are consecutive. Prove that for all positive integers m between $n_1 + n_2 + \cdots + n_m$ and $n_2 + n_2 + \cdots + n_{m+1}$ there is a perfect square.*

Solution. It is easy to prove that between numbers $a > b \geq 0$ such that $\sqrt{a} - \sqrt{b} > 1$ there is a perfect square - take for example $([\sqrt{b}] + 1)^2$.

It suffices to prove that

$$\sqrt{n_1 + \cdots + n_{m+1}} - \sqrt{n_1 + \cdots + n_m} > 1, \quad m \geq 1.$$

This is equivalent to

$$n_1 + \cdots + n_m + n_{m+1} > (1 + \sqrt{n_1 + n_2 + \cdots + n_m})^2$$

and then

$$n_{m+1} > 1 + 2\sqrt{n_1 + n_2 + \cdots + n_m}, \quad m \geq 1.$$

We induct on m. For $m = 1$ we have to prove that $n_2 > 1 + 2\sqrt{n_1}$. Indeed, $n_2 > n_1 + 2 = 1 + (1 + n_1) > 1 + 2\sqrt{n_1}$. Assume that the claim holds for some $m \geq 1$. Then

$$n_{m+1} - 1 > \sqrt{n_1 + \cdots + n_m}$$

so $(n_{m+1} - 1)^2 > 4(n_1 + \cdots + n_m)$ hence

$$(n_{m+1} + 1)^2 > 4(n_1 + \cdots + n_{m+1}).$$

This implies

$$n_{m+1} > 2\sqrt{n_1 + \cdots + n_{m+1}},$$

and since $n_{m+2} - n_{m+1} \geq 2$, it follows that

$$n_{m+2} > 1 + 2\sqrt{n_1 + \cdots + n_{m+1}},$$

as desired.

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Problem 2.1.13. *Find all the integers x, y, z so that $4^x + 4^y + 4^z$ is a square.*

Solution. It is clear that there are no solutions with $x < 0$. Without loss of generality assume that $x \leq y \leq z$ and let $4^x + 4^y + 4^z = u^2$. Then $2^{2x}(1 + 4^{y-x} + 4^{z-x}) = u^2$. We have two situations.

Case 1. $1 + 4^{y-x} + 4^{z-x}$ is odd, i.e. $1 + 4^{y-x} + 4^{z-x} = (2a+1)^2$. It follows

$$4^{y-x-1} + 4^{z-x-1} = a(a+1)$$

and then

$$4^{y-x-1}(1 + 4^{z-y}) = a(a+1).$$

We consider two cases.

1) The number a is even. Then $a+1$ is odd, so $4^{y-x-1} = a$ and $1+4^{z-y} = a+1$. It follows that $4^{y-x-1} = 4^{z-y}$, hence $y - x - 1 = z - y$. Thus $z = 2y - x - 1$ and

$$4^x + 4^y + 4^z = 4^x + 4^y + 4^{2y-x-1} = (2^x + 2^{2y-x-1})^2.$$

2) The number a is odd. Then $a+1$ is even, so $a = 4^{z-y}+1$, $a+1 = 4^{y-x-1}$ and $4^{y-x-1} - 4^{z-y} = 2$. It follows that $2^{2y-2x-3} = 2^{2x-2y-1} + 1$, which is impossible since $2x - 2y - 1 \neq 0$.

Case 2. $1 + 4^{y-x} + 4^{z-x}$ is even, thus $y = x$ or $z = x$. Anyway, we must have $y = x$ and then $2+4^{z-x}$ is a square, impossible since it is $\equiv 2 \pmod 4$ or $\equiv 3 \pmod 4$.

Proposed problems

Problem 2.1.14. Let x, y, z be positive integers such that

$$\frac{1}{x} - \frac{1}{y} = \frac{1}{z}.$$

Let h be the greatest common divisor of x, y, z. Prove that $hxyz$ and $h(y-x)$ are perfect squares.

(1998 United Kingdom Mathematical Olympiad)

Problem 2.1.15. Let b an integer greater than 5. For each positive integer n, consider the number

$$x_n = \underbrace{11\ldots1}_{n-1}\underbrace{22\ldots2}_{n}5,$$

written in base b. Prove that the following condition holds if and only if $b = 10$: There exists a positive integer M such that for every integer n greater than M, the number x_n is a perfect square.

(44th IMO Shortlist)

Problem 2.1.16. Do there exist three natural numbers greater than 1, such that the square of each, minus one, is divisible by each of the others?

(1996 Russian Mathematical Olympiad)

Problem 2.1.17. (a) Find the first positive integer whose square ends in three 4's.

(b) Find all positive integers whose squares end in three 4's.

(c) Show that no perfect square ends with four 4's.

(1995 United Kingdom Mathematical Olympiad)

Problem 2.1.18. Let m, n be a natural numbers and $m + i = a_i b_i^2$ for $i = 1, 2, \ldots, n$, where a_i and b_i are natural numbers and a_i is squarefree. Find all values of n for which there exists m such that $a_1 + a_2 + \cdots + a_n = 12$.

(1997 Bulgarian Mathematical Olympiad)

Problem 2.1.19. For each positive integer n, denote by $s(n)$ the greatest integer such that for all positive integer $k \leq s(n)$, n^2 can be expressed as a sum of squares of k positive integers.

(a) Prove that $s(n) \leq n^2 - 14$ for all $n \geq 4$.

(b) Find a number n such that $s(n) = n^2 - 14$.

(c) Prove that there exist infinitely many positive integers n such that

$$s(n) = n^2 - 14.$$

(33rd IMO)

Problem 2.1.20. Let A be the set of positive integers representable in the form $a^2 + 2b^2$ for integers a, b with $b \neq 0$. Show that if $p^2 \in A$ for a prime p, then $p \in A$.

(1997 Romanian IMO Team Selection Test)

Problem 2.1.21. Is it possible to find 100 positive integers not exceeding 25000 such that all pairwise sums of them are different?

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(42nd IMO Shortlist)

Problem 2.1.22. Do there exist 10 distinct integers, the sum of any 9 of which is a perfect square?

(1999 Russian Mathematical Olympiad)

Problem 2.1.23. Let n be a positive integer such that n is a divisor of the sum

$$1 + \sum_{i=1}^{n-1} i^{n-1}$$

Prove that n is square-free.

(1995 Indian Mathematical Olympiad)

Problem 2.1.24. Let n, p be integers such that $n > 1$ and p is a prime. If $n|(p-1)$ and $p|(n^3-1)$, show that $4p-3$ is a perfect square.

(2002 Czech-Polish-Slovak Mathematical Competition)

Problem 2.1.25. Show that for any positive integer $n > 10000$, there exists a positive integer m that is a sum of two squares and such that $0 < m - n < 3\sqrt[4]{n}$.

(Russian Mathematical Olympiad)

Problem 2.1.26. Show that a positive integer m is a perfect square if and only if for each positive integer n, at least one of the differences

$$(m+1)^2 - m, (m+2)^2 - m, \ldots, (m+n)^2 - m$$

is divisible by n.

(2002 Czech and Slovak Mathematical Olympiad)

2.2 Perfect cubes

Problem 2.2.1. *Prove that if n is a perfect cube, then n^2+3n+3 cannot be a perfect cube.*

Solution. Suppose by way of contradiction that $n^2 + 3n + 3$ is a cube. Hence $n(n^2 + 3n + 3)$ is a cube. Note that

$$n(n^2 + 3n + 3) = n^3 + 3n^2 + 3n = (n+1)^3 - 1$$

and since $(n+1)^3 - 1$ is not a cube, we obtain a contradiction.

Problem 2.2.2. *Let m be a given positive integer. Find a positive integer n such that $m+n+1$ is a perfect square and $mn+1$ is a perfect cube.*

Solution. Choosing $n = m^2 + 3m + 3$, we have

$$m + n + 1 = m^2 + 4m + 4 = (m+2)^2$$

and

$$mn + 1 = m^3 + 3m^2 + 3m + 1 = (m+1)^3.$$

Problem 2.2.3. *Which are there more of among the natural numbers from 1 to 1000000, inclusive: numbers that can be represented as the sum of a perfect square and a (positive) perfect cube, or numbers that cannot be?*

(1996 Russian Mathematical Olympiad)

Solution. There are more numbers not of this form. Let $n = k^2 + m^3$, where $k, m, n \in \mathbb{N}$ and $n \leq 1000000$. Clearly $k \leq 1000$ and $m \leq 100$. Therefore there cannot be more numbers in the desired form than the 100000 pairs (k, m).

Problem 2.2.4. *Show that no integer of the form $\overline{xyxy}$ in base 10 can be the cube of an integer. Also find the smallest base $b > 1$ in which there is a perfect cube of the form $xyxy$.*

(1998 Irish Mathematical Olympiad)

Solution. If the 4-digit number $\overline{xyxy} = 101 \times \overline{xy}$ is a cube, then $101 | \overline{xy}$, which is a contradiction.

Convert $\overline{xyxy} = 101 \times \overline{xy}$ from base b to base 10. We find $\overline{xyxy} = (b^2+1) \times (bx+y)$ with $x, y < b$ and $b^2 + 1 > bx + y$. Thus for $\overline{xyxy}$ to be a cube, $b^2 + 1$ must be divisible by a perfect square. We can check easily that $b = 7$ is the smallest such number, with $b^2 + 1 = 50$. The smallest cube divisible by 50 is 1000 which is $\overline{2626}$ is base 7.

Proposed problems

Problem 2.2.5. Find all the positive perfect cubes that are not divisible by 10 so that the number obtained by erasing the last three digits is also a perfect cube.

Problem 2.2.6. Find all positive integers n less than 1999 such that n^2 is equal to the cube of the sum of n's digits.

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(1999 Iberoamerican Mathematical Olympiad)

Problem 2.2.7. Prove that for any non-negative integer n the number

$$A = 2^n + 3^n + 5^n + 6^n$$

is not a perfect cube.

Problem 2.2.8. Prove that any integer is a sum of five cubes.

Problem 2.2.9. Show that any rational number can be written as a sum of three cubes.

2.3 k^{th} powers of integers, $k \geq 4$

Problem 2.3.1. *Given* 81 *natural numbers whose prime divisors belong to the set* $\{2, 3, 5\}$*, prove there exist* 4 *numbers whose product is the fourth power of an integer.*

(1996 Greek Mathematical Olympiad)

Solution. It suffices to take 25 such numbers. To each number, associate the triple (x_2, x_3, x_5) recording the parity of the exponents of 2, 3, and 5 is its prime factorization. Two numbers have the same triple if and only if their product is a perfect square. As long as there are 9 numbers left, we can select two whose product is a square; in so doing, we obtain 9 such pairs. Repeating the process with the square roots of the products of the pairs, we obtain four numbers whose product is a fourth power.

Problem 2.3.2. *Find all collections of* 100 *positive integers such that the sum of the fourth powers of every four of the integers is divisible by the product of the four numbers.*

(1997 St. Petersburg City Mathematical Olympiad)

Solution. Such sets must be $n, n, \ldots, n$ or $3n, n, n, \ldots, n$ for some integer n. Without loss of generality, we assume the numbers do not have a common factor. If u, v, w, x, y are five of the numbers, then uvw divides $u^4 + v^4 + w^4 + x^4$ and $u^4 + v^4 + w^4 + y^4$, and so divides $x^4 - y^4$. Likewise, $v^4 \equiv w^4 \equiv x^4 \pmod{u}$, and from above, $3v^4 \equiv 0 \pmod{u}$. If u has a prime divisor not equal to 3, we conclude that every other integer is divisible by the same prime, contrary to assumption. Likewise, if u is divisible by 9, then every other integer is divisible by 3. Thus all of the numbers equal 1 or 3. Moreover, if one number is 3, the others are all congruent modulo 3, so are all 3 (contrary to assumption) or 1. This completes the proof.

Problem 2.3.3. *Let M be a set of 1985 distinct positive integers, none of which has a prime divisor greater than 26. Prove that M contains at least one subset of four distinct elements whose product is the fourth power of an integer.*

(26th IMO)

Solution. There are nine prime numbers less than 26: $p_1 = 2, p_2 = 3, \ldots, p_9 = 23$. Any element x of M has a representation $x = \prod_{i=1}^{9} p_i^{a_i}$, $a_i \geq 0$. If $x, y \in M$ and $y = \prod_{i=1}^{9} p_i^{b_i}$, the product $xy = \prod_{i=1}^{9} p_i^{a_i+b_i}$ is a perfect square if and only if $a_i + b_i \equiv 0 \pmod 2$. Equivalently, $a_i \equiv b_i \pmod 2$ for all $i = 1, 2, \ldots, 9$. Because there are $2^9 = 512$ elements in $(\mathbb{Z}/2\mathbb{Z})^9$ any subset of M having at least 513 elements contains two elements x, y such that xy is a perfect square. Starting from M and eliminating such pairs one finds $\frac{1}{2}(1985 - 513) = 736 > 513$ distinct two-element subsets of M having a square as the product of elements. Reasoning as above, we find among these squares at least one pair (in fact many pairs) whose product is a fourth power.

Problem 2.3.4. *Let A be a subset of $\{0, 1, \ldots, 1997\}$ containing more than 1000 elements. Prove that A contains either a power of 2, or two distinct integers whose sum is a power of 2.*

(1997 Irish Mathematical Olympiad)

Solution. Suppose A did not verify the conclusion. Then A would contain at most half of the integers from 51 to 1997, since they can be divided into pairs whose sum is 2048 (with 1024 left over); likewise, A contains at most half of the integers from 14 to 50, at most half of the integers from 3 to 13, and possibly 0, for a total of

$$973 + 18 + 5 + 1 = 997$$

integers.

Problem 2.3.5. *Show that in the arithmetic progression with first term 1 and ratio 729, there are infinitely many powers of 10.*

(1996 Russian Mathematical Olympiad)

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Solution. We will show that for all natural numbers n, $10^{81n}-1$ is divisible by 729. In fact,

$$10^{81n}-1=(10^{81})^n-1^n=(10^{81}-1)\cdot A,$$

and

$$\begin{aligned}10^{81}-1&=\underbrace{9\ldots9}_{81}\\&=\underbrace{9\ldots9}_{9}\ldots\underbrace{10\ldots01}_{8}\underbrace{10\ldots01}_{8}\ldots\underbrace{10\ldots01}_{8}\\&=9\underbrace{1\ldots1}_{9}\ldots\underbrace{10\ldots01}_{8}\underbrace{10\ldots01}_{8}\ldots\underbrace{10\ldots01}_{8}.\end{aligned}$$

The second and third factors are composed of 9 units, so the sum of their digits is divisible by 9, that is, each is a multiple of 9. Hence $10^{81}-1$ is divisible by $9^3=729$, as is $10^{81n}-1$ for any n.

Remark. An alternative solution uses Euler's Theorem (see Section 7.2). We have $10^{\varphi(729)}\equiv 1 \pmod{7)29}$, thus $10^{n\varphi(729)}$ is in this progression for any positive integer n.

Proposed problems

Problem 2.3.6. Let p be a prime number and a, n positive integers. Prove that if

$$2^p+3^p=a^n,$$

then $n=1$.

(1996 Irish Mathematical Olympiad)

Problem 2.3.7. Let x, y, p, n, k be natural numbers such that

$$x^n+y^n=p^k.$$

Prove that if $n>1$ is odd, and p is an odd prime, then n is a power of p.

(1996 Russian Mathematical Olympiad)

Problem 2.3.8. Prove that a product of three consecutive integers cannot be a power of an integer.

Problem 2.3.9. Show that there exists an infinite set A of positive integers such that for any finite nonempty subset $B\subset A$, $\sum_{x\in B} x$ is not a perfect power.

(Kvant)

Problem 2.3.10. Prove that there is no infinite arithmetic progression consisting only of powers ≥ 2.

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3

Floor Function and Fractional Part

3.1 General problems

For a real number x there is a unique integer n such that $n \leq x < n+1$. We say that n is the *greatest integer less than or equal to* x or the *floor* of x. We denote $n = \lfloor x \rfloor$. The difference $x - \lfloor x \rfloor$ is called the *fractional part* of x and is denoted by $\{x\}$.

The integer $\lfloor x \rfloor + 1$ is called the *ceiling* of x and is denoted by $\lceil x \rceil$.

Examples. 1) $\lfloor 2.1 \rfloor = 2$, $\{2.1\} = .1$, and $\lceil 2.1 \rceil = 3$.

2) $\lfloor -3.9 \rfloor = -4$, $\{-3.9\} = .1$, and $\lceil -3.9 \rceil = -3$.

The following properties are useful:

1) If a and b are integers, $b > 0$, and q is the quotient when a is divided by b, then $q = \left\lfloor \frac{a}{b} \right\rfloor$.

2) For any real number x and any integer n, $\lfloor x + n \rfloor = \lfloor x \rfloor + n$ and $\lceil x + n \rceil = \lceil x \rceil + n$.

3) For any positive real number x and any positive integer n the number of positive multiples of n not exceeding x is $\left\lfloor \frac{x}{n} \right\rfloor$.

4) For any real number x and any positive integer n, $\left\lfloor \frac{\lfloor x \rfloor}{n} \right\rfloor = \left\lfloor \frac{x}{n} \right\rfloor$.

We will prove the last two properties. For 3) consider all multiples

$$1 \cdot n,\ 2 \cdot n, \ldots, k \cdot n,$$

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where $k \cdot n \leq x < (k+1)n$. That is $k \leq \dfrac{x}{n} < k+1$ and the conclusion follows. For 4) denote $\lfloor x \rfloor = m$ and $\{x\} = \alpha$. From the Division Algorithm and property 1) above it follows that $m = n\left\lfloor \dfrac{m}{n} \right\rfloor + r$, where $0 \leq r \leq n-1$. We obtain $0 \leq r + \alpha \leq n - 1 + \alpha < n$, that is $\lfloor \dfrac{r+\alpha}{n} \rfloor = 0$ and

$$\left\lfloor \frac{x}{n} \right\rfloor = \left\lfloor \frac{m+\alpha}{n} \right\rfloor = \left\lfloor \left\lfloor \frac{m}{n} \right\rfloor + \frac{r+\alpha}{n} \right\rfloor = \left\lfloor \frac{m}{n} \right\rfloor + \left\lfloor \frac{r+\alpha}{n} \right\rfloor = \left\lfloor \frac{m}{n} \right\rfloor = \left\lfloor \frac{\lfloor x \rfloor}{n} \right\rfloor.$$

Problem 3.1.1. *Find all positive integers n such that $\lfloor \sqrt[n]{111} \rfloor$ divides* 111.

Solution. The positive divisors of 111 are 1, 3, 37, 111. So we have the following cases:

1) $\lfloor \sqrt[n]{111} \rfloor = 1$ or $1 \leq 111 < 2^n$, hence $n \geq 7$.

2) $\lfloor \sqrt[n]{111} \rfloor = 3$, or $3^n \leq 111 < 4^n$, so $n = 4$.

3) $\lfloor \sqrt[n]{111} \rfloor = 37$, or $37^n \leq 111 < 38^n$, impossible.

4) $\lfloor \sqrt[n]{111} \rfloor = 111$, or $111^n \leq 111 < 112^n$, and so $n = 1$.

Therefore $n = 1$, $n = 4$ or $n \geq 7$.

Problem 3.1.2. *Solve in $\mathbb{R}$ the equation:*

$$\lfloor x \lfloor x \rfloor \rfloor = 1.$$

Solution. By definition,

$$\lfloor x \lfloor x \rfloor \rfloor = 1$$

implies

$$1 \leq x \lfloor x \rfloor < 2.$$

We consider the following cases:

a) $x \in (-\infty, -1)$. Then $\lfloor x \rfloor \leq -2$ and $x \lfloor x \rfloor > 2$, a contradiction.

b) $x = -1 \Rightarrow \lfloor x \rfloor = -1$. Then $x \lfloor x \rfloor = (-1) \cdot (-1) = 1$ and $\lfloor x \lfloor x \rfloor \rfloor = 1$, so $x = -1$ is a solution.

c) $x \in (-1, 0)$. We have $\lfloor x \rfloor = -1$ and $x \lfloor x \rfloor = -x < 1$, false.

d) If $x \in [0, 1)$, then $\lfloor x \rfloor = 0$ and $x \lfloor x \rfloor = 0 < 1$, so we have no solution in this case.

e) For $x \in [1, 2)$ we obtain $\lfloor x \rfloor = 1$ and $x \lfloor x \rfloor = \lfloor x \rfloor = 1$, as needed.

f) Finally, for $x \geq 2$ we have $\lfloor x \rfloor \geq 2$ and $x \lfloor x \rfloor = 2x \geq 4 \cdot 2$, a contradiction with (1).

Consequently, $x \in \{-1\} \cup [1, 2)$.

Problem 3.1.3. *Prove that for any integer n one can find integers a and b such that*

$$n = \lfloor a\sqrt{2} \rfloor + \lfloor b\sqrt{3} \rfloor.$$

Solution. For any integer n, one can find an integer b so that

$$\sqrt{2} + b\sqrt{3} - 2 < n \leq \sqrt{2} + b\sqrt{3}.$$

We consider the cases:
1) If $n = \lfloor \sqrt{2} \rfloor + \lfloor b\sqrt{3} \rfloor$, we are done.
2) If $n = \lfloor \sqrt{2} \rfloor + \lfloor b\sqrt{3} \rfloor + 1$, then $n = \lfloor 2\sqrt{2} \rfloor + \lfloor b\sqrt{3} \rfloor$.
3) If $n = \lfloor \sqrt{2} \rfloor + \lfloor b\sqrt{3} \rfloor - 1$, then $n = \lfloor 0\sqrt{2} \rfloor + \lfloor b\sqrt{3} \rfloor$.

Problem 3.1.4. *Find all real numbers $x > 1$, such that $\sqrt[n]{\lfloor x^n \rfloor}$ is an integer for all positive integers n, $n \geq 2$.*

(2004 Romanian Regional Mathematical Contest)

Solution. Put $\sqrt[n]{\lfloor x^n \rfloor} = a_n$. Then $\lfloor x^n \rfloor = a_n^n$ and $a_n^n \leq x^n < a_n^n + 1$. Taking roots, one obtains $a_n \leq x < \sqrt[n]{a_n^n + 1}$. This shows that $\lfloor x \rfloor = a_n$.

We will show that all positive integers x, $x \geq 2$, satisfy the condition. Assume, by way of contradiction, that there is a solution x which is not a nonnegative integer. Put $x = a + \alpha$, $a \in \mathbb{Z}$, $a \geq 1$, $0 < \alpha < 1$.

It follows that $a^n < (a + \alpha)^n < a^n + 1$, and therefore,

$$1 < \left(1 + \frac{\alpha}{a}\right)^n < 1 + \frac{1}{a^n} \leq 2.$$

On the other hand, by Bernoulli inequality,

$$\left(1 + \frac{\alpha}{a}\right)^n \geq 1 + n\frac{\alpha}{a} > 2,$$

for sufficiently large n, a contradiction.

Problem 3.1.5. *Let $r \geq 1$ be a real number such that for all m, n such that m divides n, $\lfloor mr \rfloor$ divides $\lfloor nr \rfloor$. Prove that r is an integer.*

(1997 Iberoamerican Mathematical Olympiad)

Solution. Fix the positive integer m and observe that $\lfloor mr \rfloor$ divides $\lfloor k\{mr\} \rfloor$ for any positive integer k. If $\{mr\} \neq 0$ take a positive integer k such that

$$\frac{1}{\{mr\}} \leq k < \frac{2}{\{mr\}},$$

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possible since $\dfrac{2}{\{mr\}} > \dfrac{1}{\{mr\}} + 1$. Then $\lfloor mr \rfloor | 1$, thus it is 1, which means $m = 1$. This shows that if $m \geq 2$, then $\{mr\} = 0$, thus $mr \in \mathbb{Z}$ for any integer $m \geq 2$ and clearly $r \in \mathbb{Z}$.

Problem 3.1.6. *Find the number of different terms of the finite sequence* $\left\lfloor \dfrac{k^2}{1998} \right\rfloor$, *where* $k = 1, 2, \ldots, 1997$.

(1998 Balkan Mathematical Olympiad)

Solution. Note that

$$\left\lfloor \frac{998^2}{1998} \right\rfloor = 498 < 499 = \left\lfloor \frac{999^2}{1998} \right\rfloor,$$

so we can compute the total number of distinct terms by considering $k = 1, \ldots, 998$ and $k = 999, \ldots, 1997$ independently. Observe that for $k = 1, \ldots, 997$,

$$\frac{(k+1)^2}{1998} - \frac{k^2}{1998} = \frac{2k+1}{1998} < 1,$$

so for $k = 1, \ldots, 998$, each of the numbers

$$\left\lfloor \frac{1^2}{1998} \right\rfloor = 0, 1, \ldots, 498 = \left\lfloor \frac{998^2}{1998} \right\rfloor$$

appears at least once in the sequence $\lfloor k^2/1998 \rfloor$ for a total of 499 distinct terms. For $k = 999, \ldots, 1996$, we have

$$\frac{(k+1)^2}{1998} - \frac{k^2}{1998} = \frac{2k+1}{1998} > 1,$$

so the numbers $\lfloor k^2/1998 \rfloor$ ($k = 999, \ldots, 1997$) are all distinct, giving $1997 - 999 + 1 = 999$ more terms. Thus the total number of distinct terms is 1498.

Problem 3.1.7. *Determine the number of real solutions* a *of the equation*

$$\left\lfloor \frac{a}{2} \right\rfloor + \left\lfloor \frac{a}{3} \right\rfloor + \left\lfloor \frac{a}{5} \right\rfloor = a.$$

(1998 Canadian Mathematical Olympiad)

Solution. There are 30 solutions. Since $\lfloor a/2 \rfloor$, $\lfloor a/3 \rfloor$, and $\lfloor a/5 \rfloor$ are integers, so is a. Now write $a = 30p + q$ for integers p and q, $0 \leq q < 30$. Then

$$\left\lfloor \frac{a}{2} \right\rfloor + \left\lfloor \frac{a}{3} \right\rfloor + \left\lfloor \frac{a}{5} \right\rfloor = a$$

$$\Leftrightarrow\ 31p + \left\lfloor \frac{q}{2} \right\rfloor + \left\lfloor \frac{q}{3} \right\rfloor + \left\lfloor \frac{q}{5} \right\rfloor = 30p + q$$

$$\Leftrightarrow\ p = q - \left\lfloor \frac{q}{2} \right\rfloor - \left\lfloor \frac{q}{3} \right\rfloor - \left\lfloor \frac{q}{5} \right\rfloor.$$

Thus, for each value of q, there is exactly one value of p (and one value of a) satisfying the equation. Since q can equal any of thirty values, there are exactly 30 solutions, as claimed.

Problem 3.1.8. *Let λ be the positive root of the equation $t^2 - 1998t - 1 = 0$. Define the sequence $x_0, x_1, \ldots$ by setting*

$$x_0 = 1, \quad x_{n+1} = \lfloor \lambda x_n \rfloor, \quad n \geq 0.$$

Find the remainder when x_{1998} is divided by 1998.

(1998 Iberoamerican Mathematical Olympiad)

Solution. We have

$$1998 < \lambda = \frac{1998 + \sqrt{1998^2 + 4}}{2}$$

$$= 999 + \sqrt{999^2 + 1} < 1999,$$

$x_1 = 1998$, $x_2 = 1998^2$. Since $\lambda^2 - 1998\lambda - 1 = 0$,

$$\lambda = 1998 + \frac{1}{\lambda} \text{ and } x\lambda = 1998x + \frac{x}{\lambda}$$

for all real number x. Since $x_n = \lfloor x_{n-1}\lambda \rfloor$ and x_{n-1} is an integer and λ is irrational, we have

$$x_n < x_{n-1}\lambda < x_n + 1 \text{ or } \frac{x_n}{\lambda} < x_{n-1} < \frac{x_n + 1}{\lambda}.$$

Since $\lambda > 1998$, $\lfloor x_n/\lambda \rfloor = x_{n-1} - 1$. Therefore,

$$x_{n+1} = \lfloor x_n \lambda \rfloor = \left\lfloor 1998x_n + \frac{x_n}{\lambda} \right\rfloor = 1998x_n + x_{n-1} - 1,$$

i.e., $x_{n+1} \equiv x_{n-1} - 1 \pmod{1998}$. Therefore by induction $x_{1998} \equiv x_0 - 999 \equiv 1000 \pmod{1998}$.

Problem 3.1.9. *Let n be a positive integer. Prove that for any real number x,*

$$\lfloor nx \rfloor = \lfloor x \rfloor + \left\lfloor x + \frac{1}{n} \right\rfloor + \cdots + \left\lfloor x + \frac{n-1}{n} \right\rfloor$$

(Hermite[1]).

[1] *Charles Hermite* (1822-1901), French mathematician who did brilliant work in many branches of mathematics.

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Solution. Let $f(x)$ be the difference between the right-hand side and the left-hand side of (1). Then

$$f\left(x+\frac{1}{n}\right) = \left\lfloor x+\frac{1}{n}\right\rfloor + \cdots + \left\lfloor x+\frac{1}{n}+\frac{n-1}{n}\right\rfloor - \left\lfloor n\left(x+\frac{1}{n}\right)\right\rfloor$$
$$= \left\lfloor x+\frac{1}{n}\right\rfloor + \cdots + \left\lfloor x+\frac{n-1}{n}\right\rfloor + \lfloor x+1\rfloor - \lfloor nx+1\rfloor,$$

and since $\lfloor x+k\rfloor = \lfloor x\rfloor + k$ for each integer k, it follows that

$$f\left(x+\frac{1}{n}\right) = f(x)$$

for all real x. Hence f is periodic with period $1/n$. Thus it suffices to study $f(x)$ for $0 \le x < 1/n$. But $f(x) = 0$ for all these values, hence $f(x) = 0$ for all real x, and the proof is complete.

Proposed problems

Problem 3.1.10. Let n be a positive integer. Find with proof a closed formula for the sum:

$$\left\lfloor\frac{n+1}{2}\right\rfloor + \left\lfloor\frac{n+2}{2^2}\right\rfloor + \cdots + \left\lfloor\frac{n+2^k}{2^{k+1}}\right\rfloor + \ldots$$

(10th IMO)

Problem 3.1.11. Compute the sum

$$\sum_{0\le i<j\le n}\left\lfloor\frac{x+i}{j}\right\rfloor,$$

where x is a real number.

Problem 3.1.12. Evaluate the difference between the numbers

$$\sum_{k=0}^{2000}\left\lfloor\frac{3^k+2000}{3^{k+1}}\right\rfloor \quad \text{and} \quad \sum_{k=0}^{2000}\left\lfloor\frac{3^k-2000}{3^{k+1}}\right\rfloor.$$

Problem 3.1.13. a) Prove that there are infinitely many rational positive numbers x such that:

$$\{x^2\} + \{x\} = 0,99.$$

b) Prove that there are no rational numbers $x > 0$ such that:

$$\{x^2\} + \{x\} = 1.$$

(2004 Romanian Mathematical Olympiad)

Problem 3.1.14. Show that the fractional part of the number $\sqrt{4n^2+n}$ is not greater than 0.25.

(2003 Romanian Mathematical Olympiad)

Problem 3.1.15. Prove that for every natural number n,

$$\sum_{k=1}^{n^2}\{\sqrt{k}\} \leq \frac{n^2-1}{2}.$$

(1999 Russian Mathematical Olympiad)

Problem 3.1.16. The rational numbers $\alpha_1, \ldots, \alpha_n$ satisfy

$$\sum_{i=1}^{n}\{k\alpha_i\} < \frac{n}{2}$$

for any positive integer k.

(a) Prove that at least one of $\alpha_1, \ldots, \alpha_n$ is an integer.

(b) Do there exist $\alpha_1, \ldots, \alpha_n$ that satisfy

$$\sum_{i=1}^{n}\{k\alpha_i\} \leq \frac{n}{2},$$

such that no α_i is an integer?

(2002 Belarus Mathematical Olympiad)

3.2 Floor function and integer points

The following results are helpful in proving many relations involving the floor function.

Theorem 3.2.1. *Let a, c be nonnegative real numbers and let $f : [a, b] \to [c, d]$ be a bijective increasing function.*

Then

$$\sum_{a\leq k\leq b}\lfloor f(k)\rfloor + \sum_{c\leq k\leq d}\lfloor f^{-1}(k)\rfloor - n(G_f) = \lfloor b\rfloor\lfloor d\rfloor - \alpha(a)\alpha(c), \qquad (1)$$

where k is integer, $n(G_f)$ is the number of points with nonnegative integer coordinates on the graph of f and $\alpha : \mathbb{R} \to \mathbb{Z}$ is defined by

$$\alpha(x) = \begin{cases} \lfloor x\rfloor & \text{if} \quad x \in \mathbb{R}\setminus\mathbb{Z} \\ 0 & \text{if} \quad x = 0 \\ x-1 & \text{if} \quad x \in \mathbb{Z}\setminus\{0\} \end{cases}$$

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Proof. For a bounded region M of the plane we denote by $n(M)$ the number of points with nonnegative integral coordinates in M.

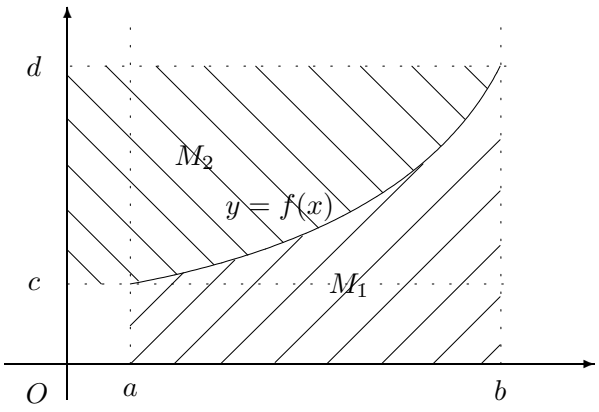
Function f is increasing and bijective, hence continuous. Consider the sets

$$M_1 = \{(x,y) \in \mathbb{R}^2 |\ a \le x \le b,\ 0 \le y \le f(x)\},$$

$$M_2 = \{(x,y) \in \mathbb{R}^2 |\ c \le y \le d,\ 0 \le x \le f^{-1}(y)\},$$

$$M_3 = \{(x,y) \in \mathbb{R}^2 |\ 0 \le x \le b,\ 0 \le y \le d\},$$

$$M_4 = \{(x,y) \in \mathbb{R}^2 |\ 0 \le x \le a,\ 0 \le y \le c\}.$$



Then

$$n(M_1) = \sum_{a \le k \le b} [f(k)], \quad n(M_2) = \sum_{c \le k \le d} [f^{-1}(k)],$$

$$n(M_3) = [b][d], \quad n(M_4) = \alpha(a)\alpha(c).$$

We have

$$n(m_1) + n(M_2) - n(M_1 \cap M_2) = n(M_1 \cup M_2),$$

hence

$$n(M_1) + n(M_2) - n(G_f) = n(M_3) - n(M_4),$$

and the conclusion follows. □

Theorem 3.2.2. *Let m, n, s be positive integers, $m \leq n$, Then*

$$\sum_{k=1}^{s} \left\lfloor \frac{km}{n} \right\rfloor + \sum_{1 \leq k \leq \frac{ms}{n}} \left\lfloor \frac{kn}{m} \right\rfloor = s \left\lfloor \frac{ms}{n} \right\rfloor + \left\lfloor \frac{gcd(m,n) \cdot s}{n} \right\rfloor. \tag{2}$$

Proof. We first prove the following lemma.

Lemma. *The array*

$$\frac{1 \cdot m}{n}, \frac{2 \cdot m}{n}, \ldots, \frac{s \cdot m}{n}$$

contains exactly $\left\lfloor \frac{gcd(m,n) \cdot s}{n} \right\rfloor$ *integers.*

Proof of the lemma. Let d be the greatest common divisor of m and n. Hence $m = m_1 d$ and $n = n_1 d$ for some integers m_1 and n_1.

The numbers in the array are

$$\frac{1 \cdot m_1}{n_1}, \frac{2 \cdot m_1}{n_1}, \ldots, \frac{p \cdot m_1}{n_1}$$

and, since m_1, n_1 are relatively prime, there are $\left\lfloor \frac{p}{n_1} \right\rfloor$ integers among them. Because $n_1 = \frac{n}{d} = \frac{n}{gcd(m,n)}$ it follows that there are $\left\lfloor \frac{gcd(m,n)p}{n} \right\rfloor$ integers in the array. □

In order to prove the desired result, let us consider the function $f : [1, s] \to \left[\frac{m}{n}, \frac{ms}{n}\right]$, $f(x) = \frac{m}{n}x$ in Theorem 3.2.1. Using the lemma above we have $n(G_f) = \left\lfloor \frac{gcd(m,n) \cdot s}{n} \right\rfloor$ and the conclusion follows. □

Remark. The special case $s = n$ leads to an important result:

$$\sum_{k=1}^{n} \left\lfloor \frac{km}{n} \right\rfloor + \sum_{k=1}^{m} \left\lfloor \frac{kn}{m} \right\rfloor = mn + gcd(m,n). \tag{3}$$

Theorem 3.2.3. *Let a, c be nonnegative real numbers and let $f : [a, b] \to [c, d]$ be a bijective decreasing function.*

Then

$$\sum_{a \leq k \leq b} \lfloor f(k) \rfloor - \sum_{c \leq k \leq d} [f^{-1}(k)] = \lfloor b \rfloor \alpha(c) - \lfloor d \rfloor \alpha(a),$$

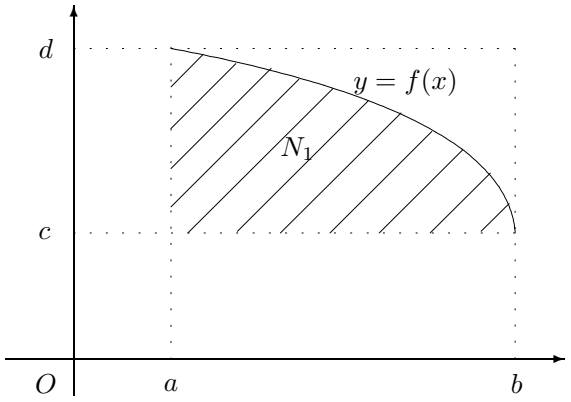
where k is integer and α is the function defined in Theorem 3.2.1.

Proof. Function f is decreasing and bijective, hence continuous. Consider the sets

$$N_1 = \{(x, y) \in \mathbb{R}^2 |\ a \leq x \leq b,\ c \leq y \leq f(x)\},$$

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$$N_2 = \{(x,y) \in \mathbb{R}^2 |\ c \le y \le d,\ a \le x \le f^{-1}(y)\},$$
$$N_3 = \{(x,y) \in \mathbb{R}^2 |\ a \le x \le b,\ 0 \le y \le c\},$$
$$N_4 = \{(x,y) \in \mathbb{R}^2 |\ 0 \le x \le a,\ c \le y \le d\}.$$



Then

$$\sum_{a \le k \le b} \lfloor f(k) \rfloor = n(N_1) + n(N_3),$$
$$\sum_{c \le k \le d} \lfloor f^{-1}(k) \rfloor = n(N_2) + n(N_4),$$

$n(N_1) = n(N_2)$, and

$$n(N_3) = (\lfloor b \rfloor - \alpha(a))\alpha(c), \quad n(N_4) = (\lfloor d \rfloor - \alpha(c))\alpha(a)$$

It follows that

$$\sum_{a \le k \le b} \lfloor f(k) \rfloor - \sum_{c \le k \le d} \lfloor f^{-1}(k) \rfloor = n(N_3) - n(N_4) =$$
$$= \lfloor b \rfloor \alpha(c) - \lfloor d \rfloor \alpha(a),$$

as desired. □

Remark. Combining the result in Theorem 3.2.3 and the relation (3) for the function $f : [1,n] \to \left[0, m - \frac{m}{n}\right]$, $f(x) = -\frac{m}{n}x + m$, $m \le n$, yields after some computations:

$$\sum_{k=1}^{n} \left\lfloor \frac{km}{n} \right\rfloor = \frac{1}{2}(mn + m - n + gcd(m,n)). \tag{4}$$

From the above relation we obtain

$$gcd(m,n) = 2\sum_{k=1}^{n-1}\left\lfloor \frac{km}{n}\right\rfloor + m - n - mn,$$

i.e. a 1998 Taiwanese Mathematical Olympiad problem.

From here we get

$$\sum_{k=1}^{n-1}\left\{\frac{km}{n}\right\} = \sum_{k=1}^{n-1}\frac{lm}{n} - \sum_{k=1}^{n-1}\left\lfloor\frac{km}{n}\right\rfloor$$

$$= \frac{m}{n}\cdot\frac{(n-1)n}{2} - \frac{1}{2}(mn - m - n + gcd(m,n)) = \frac{1}{2}(n - gcd(m,n)),$$

that is a 1995 Japanese Mathematical Olympiad problem.

Problem 3.2.1. *Express* $\sum_{k=1}^{n}\lfloor\sqrt{k}\rfloor$ *in terms of* n *and* $a = \lfloor\sqrt{n}\rfloor$.

(1997 Korean Mathematical Olympiad)

Solution. We apply Theorem 3.2.1 for the function $f : [1,n] \to [1,\sqrt{n}]$, $f(x) = \sqrt{x}$. Because $n(G_f) = \lfloor\sqrt{n}\rfloor$, we have

$$\sum_{k=1}^{n}\lfloor\sqrt{k}\rfloor + \sum_{k=1}^{\lfloor\sqrt{n}\rfloor}\lfloor k^2\rfloor - \lfloor\sqrt{n}\rfloor = n\lfloor\sqrt{n}\rfloor,$$

hence

$$\sum_{k=1}^{n}\lfloor\sqrt{k}\rfloor = (n+1)a - \frac{a(a+1)(2a+1)}{6}.$$

Problem 3.2.2. *Compute*

$$S_n = \sum_{k=1}^{\frac{n(n+1)}{2}}\left\lfloor\frac{-1+\sqrt{1+8k}}{2}\right\rfloor$$

Solution. Consider the function $f : [1,n] \to \left[1, \frac{n(n+1)}{2}\right]$,

$$f(x) = \frac{x(x+1)}{2}.$$

Function f is increasing and bijective. Note that $n(G_f) = n$ and $f^{-1}(x) = \frac{-1+\sqrt{1+8x}}{2}$. Applying formula in Theorem 3.2.1 we obtain

$$\sum_{k=1}^{n}\left\lfloor\frac{k(k+1)}{2}\right\rfloor + \sum_{k=1}^{\frac{n(n+1)}{2}}\left\lfloor\frac{-1+\sqrt{1+8k}}{2}\right\rfloor - n = \frac{n^2(n+1)}{2},$$

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hence

$$\sum_{k=1}^{\frac{n(n+1)}{2}}\left[\frac{-1+\sqrt{1+8k}}{2}\right]=\frac{n^2(n+1)}{2}+n-\frac{1}{2}\sum_{k=1}^{n}k(k+1)=$$

$$=\frac{n^2(n+1)}{2}+n-\frac{n(n+1)}{4}-\frac{n(n+1)(2n+1)}{12}=\frac{n(n^2+2)}{3}$$

Proposed problems

Problem 3.2.3. Prove that

$$\sum_{k=1}^{n}\left\lfloor\frac{n^2}{k^2}\right\rfloor=\sum_{k=1}^{n^2}\left\lfloor\frac{n}{\sqrt{k}}\right\rfloor$$

for all integers $n\geq 1$.

Problem 3.2.4. Let θ be a positive irrational number. Then, for any positive integer m,

$$\sum_{k=1}^{m}\lfloor k\theta\rfloor+\sum_{k=1}^{\lfloor m\theta\rfloor}\left\lfloor\frac{k}{\theta}\right\rfloor=m\lfloor m\theta\rfloor.$$

Problem 3.2.5. Let p and q be relatively prime positive integers and let m be a real number such that $1\leq m<p$.

1) If $s=\left\lfloor\frac{mq}{p}\right\rfloor$, then

$$\sum_{k=1}^{\lfloor m\rfloor}\left\lfloor\frac{kq}{p}\right\rfloor+\sum_{k=1}^{s}\left\lfloor\frac{kp}{q}\right\rfloor=\lfloor m\rfloor s.$$

2) (Landau[2]) If p and q are odd, then

$$\sum_{k=1}^{\frac{p-1}{2}}\left\lfloor\frac{kq}{p}\right\rfloor+\sum_{k=1}^{\frac{q-1}{2}}\left\lfloor\frac{kp}{q}\right\rfloor=\frac{(p-1)(q-1)}{4}.$$

3.3 An useful result

The following theorem is also helpful in proving some relations involving floor function.

[2] *Edmond Georg Hermann Landau* (1877-1838), German mathematician who gave the the first systematic presentation of analytic number theory and wrote important work on the theory of analytic functions of single variable.

Theorem 3.3.1. *Let p be an odd prime and let q be an integer that is not divisible by p. If $f : \mathbb{Z}_+^* \to \mathbb{R}$ is a function such that:*

i) $\dfrac{f(k)}{p}$ is not an integer, $k = 1, 2, \ldots, p-1$;

ii) $f(k) + f(p-k)$ is an integer divisible by p, $k = 1, 2, \ldots, p-1$, then

$$\sum_{k=1}^{p-1} \left\lfloor f(k)\frac{q}{p} \right\rfloor = \frac{q}{p}\sum_{k=1}^{p-1} f(k) - \frac{p-1}{2}. \tag{1}$$

Proof. From ii) it follows that

$$\frac{qf(k)}{p} + \frac{qf(p-k)}{p} \in \mathbb{Z} \tag{2}$$

and from i) we obtain that $\dfrac{qf(k)}{p} \notin \mathbb{Z}$ and $\dfrac{qf(p-k)}{p} \notin \mathbb{Z}$, $k = 1, \ldots, p-1$, hence

$$0 < \left\{\frac{qf(k)}{p}\right\} + \left\{\frac{qf(p-k)}{p}\right\} < 2.$$

But, from (2), $\left\{\dfrac{qf(k)}{p}\right\} + \left\{\dfrac{qf(p-k)}{p}\right\} \in \mathbb{Z}$, thus

$$\left\{\frac{qf(k)}{p}\right\} + \left\{\frac{qf(p-k)}{p}\right\} = 1, \quad k = 1, \ldots, p-1.$$

Summing up and dividing by 2 yields

$$\sum_{k=1}^{p-1} \left\{\frac{q}{p}f(k)\right\} = \frac{p-1}{2}.$$

It follows that

$$\sum_{k=1}^{p-1} \frac{q}{p}f(k) - \sum_{k=1}^{p-1} \left\lfloor \frac{q}{p}f(k) \right\rfloor = \frac{p-1}{2}$$

and the conclusion follows. □

Problem 3.3.1. *Let p and q be two relatively prime integers. The following identity holds:*

$$\sum_{k=1}^{p-1} \left\lfloor k\frac{q}{p} \right\rfloor = \frac{(p-1)(q-1)}{2} \quad \textit{(Gauss)}.$$

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Solution. The function $f(x) = x$ satisfies both i) and ii) in Theorem 3.3.1, hence

$$\sum_{k=1}^{p-1} \left\lfloor k\frac{q}{p} \right\rfloor = \frac{q}{p}\frac{(p-1)p}{2} - \frac{p-1}{2},$$

hence the desired relation follows.

Problem 3.3.2. *Let p be an odd prime. Prove that*

$$\sum_{k=1}^{p-1} \left\lfloor \frac{k^3}{p} \right\rfloor = \frac{(p-2)(p-1)(p+1)}{4}.$$

(2002 German Mathematical Olympiad)

Solution. The function $f(x) = x^3$ also satisfies conditions i) and ii), hence

$$\sum_{k=1}^{p-1} \left\lfloor k^3\frac{q}{p} \right\rfloor = \frac{q}{p}\cdot\frac{(p-1)^2p^2}{4} - \frac{p-1}{2} = \frac{(p-1)(p^2q-pq-2)}{4}.$$

For $q = 1$ the identity in our problem follows.

Proposed problems

Problem 3.3.3. Let p be an odd prime and let q be an integer that is not divisible by p. Shows that

$$\sum_{k=1}^{p-1} \left\lfloor (-1)^k k^2\frac{q}{p} \right\rfloor = \frac{(p-1)(q-1)}{2}.$$

Problem 3.3.4. Let p be an odd prime. Show that

$$\sum_{k=1}^{p-1} \frac{k^p-k}{p} \equiv \frac{p+1}{2} \pmod{p}.$$

4

Digits of Numbers

4.1 The last digits of a number

Let $\overline{a_n a_1 \dots a_0}$ be the decimal representation of the positive integer N. The last digit of N is $l(N) = a_0$ and for $k \geq 2$, the last k digits of N are $l_k(N) = \overline{a_{k-1} \dots a_0}$. These simple concepts appear in numerous situations.

It is useful to point out the last digit of k^n, where $k = 2, 3, \dots, 9$:

$$l(2^n) = \begin{cases} 6, n \equiv 0 \pmod 4 \\ 2, n \equiv 1 \pmod 4 \\ 4, n \equiv 2 \pmod 4 \\ 8, n \equiv 3 \pmod 4 \end{cases}, \quad l(3^n) = \begin{cases} 1, n \equiv 0 \pmod 4 \\ 3, n \equiv 1 \pmod 4 \\ 9, n \equiv 2 \pmod 4 \\ 7, n \equiv 3 \pmod 4 \end{cases}$$

$$l(4^n) = \begin{cases} 6, n \equiv 0 \pmod 2 \\ 4, n \equiv 1 \pmod 2 \end{cases}, \quad l(5^n) = 5, \quad l(6^n) = 6,$$

$$l(7^n) = \begin{cases} 1, n \equiv 0 \pmod 4 \\ 7, n \equiv 1 \pmod 4 \\ 9, n \equiv 2 \pmod 4 \\ 3, n \equiv 3 \pmod 4 \end{cases}, \quad l(8^n) = \begin{cases} 6, n \equiv 0 \pmod 4 \\ 8, n \equiv 1 \pmod 4 \\ 4, n \equiv 2 \pmod 4 \\ 2, n \equiv 3 \pmod 4 \end{cases}$$

$$l(9^n) = \begin{cases} 1, n \equiv 0 \pmod 2 \\ 9, n \equiv 1 \pmod 2 \end{cases}$$

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It is clear that if $l(N) = 0$, then $l_n(N^n) = \underbrace{\overline{0\ldots0}}_{n\,times}$ and if $l(N) = 1$, then $l(N^n) = 1$ for all $n \geq 2$.

Problem 4.1.1. *What is the final digit of* $(\ldots(((7^7)^7)^7)\ldots^7)$.

There are 1001 7s in the formula.

Solution. The final digit of a (decimal) number is its remainder modulo 10. Now $7^2 = 49 \equiv -1 \pmod{10}$. So $7^7 = (7^2)^3 \cdot 7 \equiv -7 \pmod{10}$, and

$$(7^7)^7 \equiv (-7)^7 \equiv -(7^7) \equiv -(-7) \equiv 7 \pmod{10}.$$

Proceeding in this way, we see that $((7^7)^7)^7 \equiv 7 \pmod{10}$, and in general

$$(\ldots(((7^7)^7)^7)\ldots^7) \equiv \pm 7 \pmod{10},$$

where the sign is $+$ if alltogether there is an odd number of 7s in the formula, and $-$ if there is an even number of 7s. Now, 1001 is odd. So the final digit of the given formula is 7.

Problem 4.1.2. *Prove that every positive integer has at least as many (positive) divisors whose last decimal digit is 1 or 9 as divisors whose last digit is 3 or 7.*

(1997 St. Petersburg City Mathematical Olympiad)

Solution. Let $d_1(m), d_3(m), d_7(m), d_9(m)$ be the number of divisors of m ending in 1, 3, 7, 9, respectively. We prove the claim by induction on m; it holds obviously for m a prime power, and if m is composite, write $m = pq$ with p, q coprime, and note that

$$d_1(m) - d_3(m) - d_7(m) + d_9(m)$$

$$= (d_1(p) - d_3(p) - d_7(p) + d_9(p))(d_1(q) - d_3(q) - d_7(q) + d_9(q)).$$

For instance,

$$d_3(m) = d_1(p)d_3(q) + d_3(p)d_1(q) + d_7(p)d_9(q) + d_9(p)d_7(q).$$

Problem 4.1.3. *Find the least positive integer n with the following properties:*

a) the last digit of its decimal representation is 6;

b) by deleting the last digit 6 and replacing it in front of the remaining digits one obtains a number four times greater than the given number.

(4th IMO)

Solution. Let $n = 10^k a_k + 10^{k-1} a_{k-1} + \cdots + 10a_1 + 6$ be the required number. Writing n under the form $n = 10N + 6$, where $10^{k-1} < N < 10^k$, the condition b) becomes:

$$4(10N + 6) = 6 \cdot 10^k + N.$$

Thus, we obtain

$$39N = 6 \cdot 10^k - 24,$$

and equivalently

$$13N = 2(10^k - 4).$$

Thus, we obtain that $10^k \equiv 4 \pmod{13}$.

It is more convenient to write:

$$(-3)^k \equiv 4 \pmod{13}.$$

From the conditions of the problem it is required the least k with this property. We have:

$$(-3)^2 = 9 \pmod{13}, \quad (-3)^3 \equiv -27 \pmod{13} \equiv -1 \pmod{13}$$

$$(-3)^5 \equiv (-3)^2(-3)^3 \equiv -9 \equiv 4 \pmod{13}.$$

Then, $k = 5$ is the least positive solution of the equation. Thus,

$$13N = 2 \cdot 99996 \ \Rightarrow \ N = 15384 \ \Rightarrow \ n = 153846.$$

This number verifies b).

Proposed problems

Problem 4.1.4. In how may zeroes can the number $1^n + 2^n + 3^n + 4^n$ end for $n \in \mathbb{N}$?

(1998 St. Petersburg City Mathematical Olympiad)

Problem 4.1.5. Find the last 5 digits of the number 5^{1981}.

Problem 4.1.6. Consider all pairs (a, b) of natural numbers such that the product $a^a b^b$, written in base 10, ends with exactly 98 zeroes. Find the pair (a, b) for which the product ab is smallest.

(1998 Austrian-Polish Mathematics Competition)

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4.2 The sum of the digits of a number

For a positive integer $N = \overline{a_n a_{n-1} \dots a_0}$ in decimal representation we denote by $S(N)$ the sum of its digits $a_0 + \cdots + a_{n-1} + a_n$. Problems involving the function S defined above appear frequently in various contexts. We present a few basic properties.

1) $S(N) = N - 9 \sum_{k \geq 1} \left\lfloor \frac{N}{10^k} \right\rfloor$;

2) $9 | S(N) - N$;

3) (subadditivity): $S(N_1 + N_2) \leq S(N_1) + S(N_2)$;

4) $S(N_1 N_2) \leq \min(N_1 S(N_2), N_2(S(N_1))$;

5) (submultiplicity): $S(N_1 N_2) \leq S(N_1) S(N_2)$.

Let us prove the last three properties. Using 1) and the inequality $\lfloor x + y \rfloor \geq \lfloor x \rfloor + \lfloor y \rfloor$ we have

$$\begin{aligned} S(N_1 + N_2) &= N_1 + N_2 - 9 \sum_{k \geq 1} \left\lfloor \frac{N_1 + N_2}{10^k} \right\rfloor \\ &\leq N_1 + N_2 - 9 \sum_{k \geq 1} \left(\left\lfloor \frac{N_1}{10^k} \right\rfloor + \left\lfloor \frac{N_2}{10^k} \right\rfloor \right) \\ &= S(N_1) + S(N_2). \end{aligned}$$

Because of the symmetry, in order to prove 4) it suffices to prove that $S(N_1 N_2) \leq N_1 S(N_2)$.

The last inequality follows by applying the subadditivity property repeatedly. Indeed,

$$S(2N_2) = S(N_2 + N_2) \leq S(N_2) + S(N_2) = 2S(N_2)$$

and after N_1 steps we obtain

$$S(N_1 N_2) = S(\underbrace{N_2 + N_2 + \cdots + N_2}_{N_1 \, times})$$

$$\leq \underbrace{S(N_2) + S(N_2) + \cdots + S(N_2)}_{N_1 \, times} = N_1 S(N_2).$$

For 5) observe that

$$S(N_1 N_2) = S\left(N_1 \sum_{i=0}^{h} b_i 10^i \right) = S\left(\sum_{i=0}^{h} N_1 b_i 10^i \right) \leq \sum_{i=0}^{h} S(N_1 b_i 10^i)$$

$$= \sum_{i=0}^{h} S(N_2 b_i) S(10^i) = \sum_{i=0}^{h} S(N_1 b_i) \leq \sum_{i=0}^{h} b_i S(N_1)$$

$$= S(N_1)\sum_{i=0}^{h} b_i = S(N_1)S(N_2).$$

Examples. 1) *In the decimal expansion of N, the digits occur in increasing order. What is $S(9N)$?*

(1999 Russian Mathematical Olympiad)

Solution. Write $N = \overline{a_k a_{k-1} \dots a_0}$. By performing the subtraction

$$\begin{array}{cccccccc} & a_k & a_{k-1} & \dots & a_1 & a_0 & 0 \\ - & & a_k & \dots & a_2 & a_1 & a_0 \\ \hline \end{array}$$

we find that the digits of $9N = 10N - N$ are

$$a_k, a_{k-1}, -a_k, \dots, a_1 - a_2, a_0 - a_1 - 1, 10 - a_0$$

These digits sum to $10 - 1 = 9$.

2) *Find a positive integer N such that $S(N) = 1996S(3N)$.*

(1996 Irish Mathematical Olympiad)

Solution. Consider $N = 1\ \underbrace{33\dots3}_{5986\,times}\ 5$. Then $3N = 4\ \underbrace{00\dots0}_{5986\,times}\ 5$ and

$$S(N) = 3 \cdot 5986 + 1 + 5 = 17964 = 1996 \cdot 9 = 1996S(N).$$

Problem 4.2.1. *Determine all possible values of the sum of the digits of a perfect square.*

(1995 Iberoamerican Olympiad)

Solution. The sum of the digits of a number is congruent to the number modulo 9, and so for a perfect square this must be congruent to 0, 1, 4 or 7. We show that all such numbers occur. The cases $n = 1$ and $n = 4$ are trivial, so assume $n > 4$.

If $n = 9m$, then n is the sum of the digits of $(10^m-1)^2 = 10^m(10^m-2)+1$, which looks like $9\dots980\dots01$. If $n = 9m + 1$, consider $(10^m - 2)^2 = 10^m(10^m - 4) + 4$, which looks like $9\dots960\dots04$. If $n = 9m + 4$, consider $(10^m - 3)^2 = 10^m(10^m - 6) + 9$, which looks like $9\dots94\dots09$. Finally, if $n = 9m - 2$, consider $(10^m - 5)^2 = 10^m(10^m - 10) + 25$, which looks like $9\dots900\dots025$.

Problem 4.2.2. *Find the number of positive 6 digit integers such that the sum of their digits is 9, and four of its digits are $1, 0, 0, 4$.*

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(2004 Romanian Mathematical Olympiad)

Solution. The pair of missing digits must be 1, 2 or 0, 3.

In the first case the first digit can be 1, 2 or 4. When 1 is the first digit, the remaining digits, (1, 2, 0, 0, 4), can be arranged in 60 ways. When 4 or 2 is the first digit, the remaining ones can be arranged in 30 ways.

In the same way, when completing with the pair (0,3), the first digit can be 1, 3 or 4. In each case, the remaining ones (three zeros and two distinct non-zero digits) can be arranged in 20 ways.

In conclusion, we have $60 + 2 \cdot 30 + 3 \cdot 20 = 180$ numbers which satisfy the given property.

Problem 4.2.3. *Find the sum of the digits of the numbers from 1 to 1,000,000.*

Solution. Write the numbers from 0 to 999,999 in a rectangular array as follows:

$$\begin{array}{cccccc}
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 2 \\
\dots & \dots & \dots & \dots & \dots & \dots \\
0 & 0 & 0 & 0 & 0 & 9 \\
0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 1 \\
\dots & \dots & \dots & \dots & \dots & \dots \\
0 & 0 & 0 & 0 & 1 & 9 \\
0 & 0 & 0 & 0 & 2 & 0 \\
\dots & \dots & \dots & \dots & \dots & \dots \\
9 & 9 & 9 & 9 & 9 & 9
\end{array}$$

There are 1,000,000 six-digits numbers, hence 6,000,000 digits are used. In each column every digit is equally represented, as in the units column each digit appears from 10 to 10, in the tens column each digit appears successively in blocks of 10 and so on. Thus each digit appears 600,000 times, so the required sum is

$$600,000 \cdot 45 + 1 = 27,000,001$$

(do not forget to count 1 from 1,000,000).

Problem 4.2.4. *Find all the positive integers n which are equal to the sum of its digits added to the product of its digits.*

Solution. Let $\overline{a_1a_2\ldots a_n}$, $a_1 \neq 0$ and $a_2, \ldots, a_n \in \{0, 1, \ldots, 9\}$, be a number such that

$$\overline{a_1a_2\ldots a_n} = a_1 + a_2 + \cdots + a_n + a_1a_2\ldots a_n.$$

The relation is equivalent to

$$a_1(10^{n-1} - 1) + a_2(10^{n-2} - 1) + \cdots + 9a_{n-1} = a_1a_2\ldots a_n$$

and

$$a_2(10^{n-2} - 1) + \cdots + 9a_{n-1} = a_1(a_2a_3\ldots a_n - \underbrace{99\ldots9}_{n-1\ digits}).$$

The left-hand side of the equality is nonnegative, whole the right-hand side is nonpositive, hence both are equal to zero. The left-hand side is zero if $n = 0$ or

$$a_2 = a_3 = \cdots = a_{n-1} = 0.$$

For $a_2 = a_3 = \cdots = a_{n-1} = 0$ the left-hand side do not equal zero, hence $n = 2$. Then $a_1(a_2 - 9) = 0$, so $a_2 = 0$ and $a_1 \in \{1, 2, \ldots, 9\}$. The number are 19, 29, 39, 49, 59, 69, 79, 89, 99.

Problem 4.2.5. *What is the smallest multiple of 99 whose digits sum to 99 and which begins and ends with 97?*

(1997 Rio Platense Mathematical Olympiad)

Solution. We refer to the digits of the number besides the two 97s as interior digits; the sum of these digits is $99-2(9+7) = 67$. Since each digit is at most 9, there are at least 8 such digits.

Note that the sum of digits being 99 forces the number to be divisible by 9; thus it suffices to ensure that the number be divisible by 11, which is to say, the alternating sum of digits must be divisible by 11.

Suppose the number has exactly 8 interior digits. If a is the sum of the odd interior places and b the sum of the even places, we have $a + b = 67$ and $a - b \equiv -3 \pmod{11}$. Since $a - b$ must also be odd, we have $a - b \geq 7$ or $a - b \leq -15$, and so either $a \geq 37$ or $b \geq 41$, contradicting the fact that a and b are each the sum of four digits.

Now suppose the number has 9 interior digits. In this case, $a - b \equiv 0 \pmod{11}$, so $a - b \geq 11$ or $a - b \leq -11$. In the latter case, $b \geq 39$, again a contradiction, but in the former case, we have $a \geq 39$, which is possible because a is now the sum of five digits. To minimize the original number,

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we take the odd digits to be 3, 9, 9, 9, 9 and the even digits to be 1, 9, 9, 9, making the minimal number 9731999999997.

Problem 4.2.6. *Find all the positive integers n such that there are non-negative integers a and b with*

$$S(a) = S(b) = S(a+b) = n.$$

(1999 Romanian Selection Test for JBMO)

Solution. We prove that the required numbers are all multiples of 9.

a) Let n be an integer such that there are positive integers a and b so that

$$S(a) = S(b) - S(a+b).$$

We prove that $9|n$.

We have the property

$$9|k - S(k). \tag{1}$$

Using the relation (1) we obtain

$$9|a - S(a) \tag{2}$$

$$9|b - S(b) \tag{3}$$

and

$$9|(a+b) - S(a+b). \tag{4}$$

From (2) and (3) follows that

$$9|a + b - (S(a) + S(b)) \tag{5}$$

hence

$$9|S(a) + S(b) - S(a+b) = n + n - n = n, \tag{6}$$

as desired.

b) Conversely, we prove that if $n = 9p$ is a multiple of 9, then integers $a, b > 0$ with $S(a) = S(b) = S(a+b)$ can be found. Indeed, set $a = \underbrace{\overline{531531\ldots531}}_{3p\ digits}$ and $b = \underbrace{\overline{171171\ldots171}}_{3p\ digits}$. Then $a + b = \underbrace{\overline{702702\ldots702}}_{3p\ digits}$ and

$$S(a) = S(b) = S(a+b) = 9p = n,$$

as claimed.

Proposed problems

Problem 4.2.7. Show that there exist infinitely many natural numbers n such that $S(3^n) \geq S(3^{n+1})$.

(1997 Russian Mathematical Olympiad)

Problem 4.2.8. Do there exist three natural numbers a, b, c such that $S(a+b) < 5$, $S(b+c) < 5$, $S(c+a) < 5$, but $S(a+b+c) > 50$?

(1998 Russian Mathematical Olympiad)

Problem 4.2.9. Prove that there exist distinct positive integers $\{n_i\}_{1 \leq i \leq 50}$ such that

$$n_1 + S(n_1) = n_2 + S(n_2) = \cdots = n_{50} + S(n_{50}).$$

(1999 Polish Mathematical Olympiad)

Problem 4.2.10. The sum of the decimal digits of the natural number n is 100, and that of $44n$ is 800. What is the sum of the digits of $3n$?

(1999 Russian Mathematical Olympiad)

Problem 4.2.11. Consider all numbers of the form $3n^2 + n + 1$, where n is a positive integer.

(a) How small can the sum of the digits (in base 10) of such a number be?

(b) Can such a number have the sum of its digits (in base 10) equal to 1999?

(1999 United Kingdom Mathematical Olympiad)

Problem 4.2.12. Consider the set A of all positive integers n with the following properties: the decimal expansion contains no 0, and the sum of the (decimal) digits of n divides n.

(a) Prove that there exist infinitely many elements in A with the following property: the digits that appear in the decimal expansion of A appear the same number of times.

(b) Show that for each positive integer k, there exists an element in A with exactly k digits.

(2001 Austrian-Polish Mathematics Competition)

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4.3 Other problems involving digits

Problem 4.3.1. *Prove that there are at least* 666 *positive composite numbers with* 2006 *digits, having a digit equal to* 7 *and all the rest equal to* 1.

Solution. The given numbers are

$$n_k = 111\ldots17\underbrace{11\ldots1}_{k\ digits} = \underbrace{111\ldots1}_{2006\ digits} + 6\underbrace{000\ldots0}_{k\ digits}$$

$$= \frac{1}{9}(10^{2006}-1) + 6\cdot 10^k, \quad k = \overline{0,2005}.$$

It is obvious that none of these numbers is a multiple of 2, 3, 5 or 11, as 11 divides $\underbrace{111\ldots1}_{2006\ digits}$, but not $6\cdot 10^k$.

So we are lead to the idea of counting multiples of 7 and 13. We have $9n_k = 100\cdot 1000^{668} - 1 + 54\cdot 10^k \equiv 2\cdot(-1)^{668} - 1 + (-2)\cdot 10^k \equiv 1 - 2\cdot 10^k \pmod 7$, hence $7|n_k$ if $10^k \equiv 3^k \equiv 4 \pmod 7$. This happens for $k = 4, 10, 16, \ldots, 2002$ so there are 334 multiples of 7. Furthermore, $9n_k = 7\cdot(-1)^{668} - 1 + 2\cdot 10^k = 6 + 2\cdot 10^k \pmod{13}$, hence $13|n_k$ if $10^k \equiv 10 \pmod{13}$. This happens for $k = 1, 7, 13, 19, \ldots, 2005$, so there are 335 multiples of 13. In all we have found 669 non-prime numbers.

Problem 4.3.2. *Let* $a_1, a_2, \ldots, a_{10^6}$ *be nonzero integers between 1 and 9, inclusive. Prove that at most 100 of the numbers* $\overline{a_1a_2\ldots a_k}$ $(1 \le k \le 10^6)$ *are perfect squares.*

(2001 Russian Mathematical Olympiad)

Solution. For each positive integer x, let $d(x)$ be the number of decimal digits in x.

Lemma. *Suppose that* $y > x$ *are perfect squares such that* $y = 10^{2b}x + c$ *for some positive integers* b, c *with* $c < 10^{2b}$. *Then*

$$d(y) - 1 \ge 2(d(x) - 1).$$

Proof. Because $y > 10^{2b}x$, we have $\sqrt{y} > 10^b\sqrt{x}$. Because $\sqrt{y}$ and $10^b\sqrt{x}$ are both integers, $\sqrt{y} \ge 10^b\sqrt{x} + 1$, so that $10^{2b}x + c = y \ge 10^{2b}x + 2\cdot 10^b\sqrt{x} + 1$. Thus, $c \ge 2\cdot 10^b\sqrt{x} + 1$.

Also, $10^{2b} > c$ by assumption, implying that

$$10^{2b} < c \ge 2\cdot 10^b\sqrt{x} + 1.$$

Hence, $10^b > 2\sqrt{x}$. It follows that

$$y > 10^{2b}x > 4x^2.$$

Therefore,

$$d(y) \geq 2d(x) - 1,$$

as desired.

We claim that there are at most 20 perfect squares $\overline{a_1a_2\ldots a_k}$ with an even (resp. odd) number of digits. Let $s_1 < s_2 < \cdots < s_n$ be these perfect squares. Clearly $d(s_n) \leq 10^6$. We now prove that if $n > 1$, then $d(s_n) \geq 1 + 2^{n-1}$.

Because $s_1, s_2, \ldots, s_n$ all have an even (resp. odd) number of digits, for each $i = 1, 2, \ldots, n-1$, we can write $s_{i+1} = 10^{2b}s_i + c$ for some integers $b > 0$ and $0 \leq c < 10^{2b}$. Because no a_i equals 0, we further know that $0 < c$. Hence, by our lemma,

$$d(s_{i+1}) - 1 \geq 2(d(s_i) - 1)$$

for each $i = 1, 2, \ldots, n-1$. Because $d(s_2) - 1 \geq 2$, we thus have $d(s_n) - 1 \geq 2^{n-1}$, as desired.

Thus, if $n > 1$,

$$1 + 2^{n-1} \leq d(s_n) \leq 10^6,$$

and

$$n \leq \left\lfloor \frac{\log(10^6 - 1)}{\log 2} \right\rfloor + 1 = 20.$$

Hence, there are at most 20 perfect squares $\overline{a_1a_2\ldots a_k}$ with an even (resp. odd) number of digits.

Therefore, there are at most $40 < 100$ perfect squares $\overline{a_1a_2\ldots a_k}$.

Proposed problems

Problem 4.3.3. A *wobbly* number is a positive integer whose digits in base 10 are alternately non-zero and zero, the units digit being non-zero. Determine all positive integers which do not divide any wobbly number.

(35th IMO Shortlist)

Problem 4.3.4. A positive integer is called *monotonic* if its digits in base 10, read from left right, are in nondecreasing order. Prove that for each $n \in \mathbb{N}$, there exists an n-digit monotonic number which is a perfect square.

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(2000 Belarussian Mathematical Olympiad)

5

Basic Principles in Number Theory

5.1 Two simple principles

5.1.1 Extremal arguments

In many problems it is useful to consider the least or the greatest element with a certain property. Very often such a choice leads to the construction of other elements or to a contradiction.

Problem 5.1.1. *Show that there exist infinitely many positive integers n such that the largest prime divisor of n^4+1 is greater than $2n$.*

(2001 St. Petersburg City Mathematical Olympiad)

Solution. First we prove the following result.

Lemma. *There are infinitely many numbers that are prime divisors of m^4+1 for some m.*

Proof. Suppose that there are only finite number of such primes. Let $p_1, p_2, \ldots, p_k$ be all of them. Let p be any prime divisor of $(p_1 p_2 \ldots p_k)^4+1$. This number cannot equal to any p_i. It makes a contradiction with our assumption, and proves the lemma. □

Let $\mathcal{P}$ be the set of all numbers being prime divisors of m^4+1 for some m. Pick any p from $\mathcal{P}$ and m from $\mathbb{Z}$, such that p divides m^4+1. Let r be the residue of m modulo p. We have $r < p$, $p|r^4+1$ and $p|(p-r)^4+1$. Let

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n be the minimum of r and $p-r$. It follows that $n < p/2$ and $p > 2n$ and of course $p|n^4+1$. Thus we have found for each $p \in \mathcal{P}$ a good number n_p. Since $n_p \geq \sqrt[4]{p} - 1$, and $\mathcal{P}$ is infinite, the set $\{n_p : p \in \mathcal{P}\}$ is also infinite.

Problem 5.1.2. *Let $a_1, a_2, \ldots$ be strictly increasing sequence of positive integers such that $gcd(a_m, a_n) = a_{gcd(m,n)}$ for all positive integers m and n. There exists a least positive integer k for which there exist positive integers $r < k$ and $s > k$ such that $a_k^2 = a_r a_s$. Prove that r divides k and that k divides s.*

(2001 Indian Mathematical Olympiad)

Solution. We begin by proving a lemma.

Lemma. *If positive integers a, b, c satisfy $b^2 = ac$, then*

$$gcd(a,b)^2 = gcd(a,c) \cdot a.$$

Proof. Consider any prime p. Let e be the highest exponent such that p^e divides b, and let e_1 and e_2 be the corresponding highest exponents for a and c, respectively. Because $b^2 = ac$, we have $2e = e_1 + e_2$. If $e_1 \geq e$, then the highest powers of p that divide $gcd(a,b)$, $gcd(a,c)$, and a are e, e_2 and e_1, respectively. Otherwise, these highest powers are all e_1. Therefore, in both cases, the exponent of p on the left side of the desired equation is the same as the exponent of p on the right side. The desired result follows. □

Applying the lemma to the given equation $a_k^2 = a_r a_s$, we have

$$gcd(a_r, a_k)^2 = gcd(a_r, a_s) a_r.$$

It now follows from the given equation that

$$a_{gcd(r,k)}^2 = a_{gcd(r,s)} a_r.$$

Assume, for sake of contradiction, that $gcd(r,k) < r$, so that $a_{gcd(r,k)} < a_r$. Then from the above equation, it follows that $a_{gcd(r,k)} > a_{gcd(r,s)}$, so that $gcd(r,k) > gcd(r,s)$. But then we have $(k_0, r_0, s_0) = (gcd(r,k), gcd(r,s), r)$ satisfies $a_{k_0}^2 = a_{r_0} a_{s_0}$ with $r_0 < k_0 < s_0$ and $k_0 < r < k$, contradicting the minimality of k.

Thus, we must have $gcd(r,k) = r$, implying that $r|k$. Then

$$gcd(a_r, a_k) = a_{gcd(r,k)} = a_r,$$

so $a_r|a_k$. Thus $a_s = a_k\dfrac{a_k}{a_r}$ is an integer multiple of a_k, and

$$a_{gcd(k,s)} = gcd(a_k, a_s) = a_k.$$

Because $a_1, a_2, \ldots$ is increasing, it follows that $gcd(k,s) = k$. Therefore, $k|s$, completing the proof.

Problem 5.1.3. *Determine all pairs (n,p) of positive integers such that p is a prime, $n \leq 2p$ and $(p-1)^n + 1$ is divisible by n^{p-1}.*

(40th IMO)

Solution. All pairs $(1,p)$, where p is a prime number, satisfy the conditions. When $p = 2$, it follows $n = 2$ and thus the pair $(2,2)$ is also a solution of the problem. Thus, we may suppose $p \geq 3$ and let n be such that $n \leq 2p$ and n^{p-1} divides $(p-1)^n+1$. Since $(p-1)^n+1$ is odd number, it follows that $n < 2p$. We shall prove that $n = p$.

Let q be a minimal prime divisor of n. Since $q|n$ and $n^{p-1}|(p-1)^n + 1$, it follows $(p-1)^n \equiv -1 \pmod{q}$. Since n and $q-1$ are relatively prime numbers we may express $an + b(q-1) = 1$.

We have

$$p-1 \equiv (p-1)^{an+b(q-1)} \equiv (p-1)^{na}(p-1)^{(q-1)b} \equiv (-1)^a 1^b \equiv -1 \pmod{q},$$

because a must be odd. This shows that $q|p$, and therefore $q = p$. Since $n < 2p$, by the consideration of q, we have $n = p$.

Let consider in these conditions the original divisibility:

$$p^{p-1}|(p-1)^p + 1 = p^p - \binom{p}{1}p^{p-1} + \binom{p}{2}p^{p-2} - \cdots + \binom{p}{p-1}p - 1 + 1$$

$$= p^2\left[p^{p-2} - \binom{p}{1}p^{p-3} + \binom{p}{2}p^{p-4} - \cdots + 1\right].$$

Therefore $p - 1 = 2$, $p = 3$ and then obtain the pair (3,3).

The conclusion is: the required solutions are $(1,p)$, $(2,2)$ and $(3,3)$, where p is an arbitrary prime.

Remark. With a little bit more work, we can even erase the condition $n \leq 2p$.

5.1.2 Pigeonhole principle

Let S be a nonempty set and let $S_1, S_2, \ldots, S_n$ be a partition of S (that is $S_1 \cup S_2 \cup \cdots \cup S_n = S$ and $S_i \cap S_j = \emptyset$ for $i \neq j$). If $a_1, a_2, \ldots, a_{n+1}$ are

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distinct elements in S, then there is a $k \in \{1, 2, \ldots, n+1\}$ such that at least two of these elements belong to S_k.

This simple observation is called the *Pigeonhole Principle* (or the *Dirichlet's Principle*).

Examples. 1) Let $m_1, m_2, \ldots, m_{n+1}$ be distinct integers. Then $m_i \equiv m_j \pmod{n}$ for some $i, j \in \{1, 2, \ldots, n+1\}$, $i \neq j$.

Indeed, let $S_t = \{x \in \mathbb{Z} |\ x \equiv t \pmod{n}\}$, $t = 1, 2, \ldots, n$. There is a $k \in \{1, 2, \ldots, n+1\}$ such that S_k contains at least two of the given integers, say m_i and m_j. Then $m_i \equiv m_j \pmod{n}$.

2) (Erdös) Given $n+1$ distinct positive integers $m_1, m_2, \ldots, m_{n+1}$ not exceeding $2n$, prove that there are two of them m_i and m_j such that $m_i | m_j$.

Indeed for each $s \in \{1, 2, \ldots, n+1\}$ write $m_s = 2^{e_s} q_s$, where e_s is a nonnegative integer and q_s is an odd positive integer. Because $q_1, q_2, \ldots, q_{n+1} \in \{1, 2, \ldots, 2n\}$ and the set $\{1, 2, \ldots, 2n\}$ has exactly n odd elements, it follows that $q_i = q_j$ for some i and j. Without loss of generality, assume that $e_i < e_j$. Then $m_i | m_j$, as desired.

Problem 5.1.4. *Prove that among any integers $a_1, a_2, \ldots, a_n$, there are some whose sum is a multiple of n.*

Solution. Let $s_1 = a_1$, $s_2 = a_1 + a_2, \ldots$, $s_n = a_1 + a_2 + \cdots + a_n$. If at least one of the integers $s_1, s_2, \ldots, s_n$ is divisible by n, then we are done. If not, there are $n-1$ possible remainders when $s_1, s_2, \ldots, s_n$ are divided by n. It follows that $s_i \equiv s_j \pmod{n}$ for some i and j, $i < j$. Then $s_j - s_i = a_{i+1} + \cdots + a_j$ is a multiple of n (see also Example 1) above).

Problem 5.1.5. *In a 10×10 table are written natural numbers not exceeding 10. Any two numbers that appear in adjacent or diagonally adjacent spaces of the table are relatively prime. Prove that some number appears in the table at least 17 times.*

(2001 St. Petersburg City Mathematical Olympiad)

Solution. In any 2×2 square, only one of the numbers can be divisible by 2 and only one can be divisible by 3, so if we tile the table with these 2×2 squares, at most 50 of the numbers in the table are divisible by 2 or 3. The remaining 50 numbers must be divided among the integers not divisible by 2 or 3, and thus only ones available are 1, 5, and 7. By the Pigeonhole Principle, one of these numbers appears at least 17 times.

Problem 5.1.6. *Prove that from any set of 117 pairwise distinct three-digit numbers, it is possible to select 4 pairwise disjoint subsets such that the sums of the numbers in each subset are equal.*

(2001 Russian Mathematical Olympiad)

Solution. We examine subsets of exactly two numbers. Clearly, if two distinct subsets have the same sum, they must be disjoint. The number of two-element subsets is $\binom{117}{2} = 6786$. Furthermore, the lowest attainable sum is $100 + 101 = 201$, while the highest sum is $998 + 999 = 1997$, for a maximum of 1797 different sums. By the Pigeonhole Principle and the fact that $1797 \cdot 3 + 1 = 5392 < 6786$, we see that there are 4 two-element subsets with the required property.

Proposed problems

Problem 5.1.7. Let $n_1 < n_2 < \cdots < n_{2000} < 10^{100}$ be positive integers. Prove that one can find two nonempty disjoint subsets A and B of $\{n_1, n_2, \ldots, n_{2000}\}$ such that

$$|A| = |B| \quad , \sum_{x \in A} x = \sum_{x \in B} x, \quad \text{and} \quad \sum_{x \in A} x^2 = \sum_{x \in B} x^2.$$

(2001 Polish Mathematical Olympiad)

Problem 5.1.8. Find the greatest positive integer n for which there exist n nonnegative integers $x_1, x_2, \ldots, x_n$, not all zero, such that for any sequence $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ of elements $\{-1, 0, 1\}$, not all zero, n^3 does not divide $\varepsilon_1 x_1 + \varepsilon_2 x_2 + \cdots + \varepsilon_n x_n$.

(1996 Romanian Mathematical Olympiad)

Problem 5.1.9. Given a positive integer n, prove that there exists $\varepsilon > 0$ such that for any n positive real numbers $a_1, a_2, \ldots, a_n$, there exists $t > 0$ such that

$$\varepsilon < \{ta_1\}, \{ta_2\}, \ldots, \{ta_n\} < \frac{1}{2}.$$

(1998 St. Petersburg City Mathematical Olympiad)

Problem 5.1.10. We have 2^n prime numbers written on the blackboard in a line. We know that there are less than n different prime numbers on

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the blackboard. Prove that there is a compact subsequence of numbers in that line whose product is a perfect square.

Problem 5.1.11. Let $x_1 = x_2 = x_3 = 1$ and $x_{n+3} = x_n + x_{n+1}x_{n+2}$ for all positive integers n. Prove that for any positive integer m there is an integer $k > 0$ such that m divides x_k.

5.2 Mathematical induction

Mathematical induction is a powerful and elegant method for proving statements depending on nonnegative integers.

Let $(P(n))_{n\geq 0}$ be a sequence of propositions. The method of mathematical induction assists us in proving that $P(n)$ is true for all $n \geq n_0$, where n_0 is a given nonnegative integer.

Mathematical Induction (weak form): *Suppose that:*

- $P(n_0)$ *is true;*
- *For all* $k \geq n_0$, $P(k)$ *is true implies* $P(k+1)$ *is true.*

Then $P(n)$ *is true for all* $n \geq n_0$.

Mathematical Induction (with step s): *Let* s *be a fixed positive integer. Suppose that:*

- $P(n_0), P(n_0+1), \ldots, P(n_0+s-1)$ *are true;*
- *For all* $k \geq n_0$, $P(k)$ *is true implies* $P(k+s)$ *is true.*

Then $P(n)$ *is true for all* $n \geq n_0$.

Mathematical Induction (strong form): *Suppose that*

- $P(n_0)$ *is true;*
- *For all* $k \geq n_0$, $P(m)$ *is true for all* m *with* $n_0 \leq m \leq k$ *implies* $P(k+1)$ *is true.*

Then $P(n)$ *is true for all* $n \geq n_0$.

This method of proof is widely used in various areas of Mathematics, including Number Theory.

Problem 5.2.1. *Prove that, for any integer* $n \geq 2$, *there exist positive integers* $a_1, a_2, \ldots, a_n$ *such that* $a_j - a_i$ *divides* $a_i + a_j$ *for* $1 \leq i < j \leq n$.

(Kvant)

Solution. We will prove the statement by induction on the number of terms n. For $n = 2$, we can choose $a_1 = 1$ and $a_2 = 2$.

We assume that we can find integers $a_1, a_2, \ldots, a_n$ such that $a_j - a_i$ divides $a_i + a_j$ for $1 \leq i < j \leq n$, where n is a positive integer greater than 1. Let m be the least common multiple of numbers $a_1, a_2, \ldots, a_n, a_j - a_i$, for all $1 \leq i < j \leq n$. Then

$$(a_1', a_2', a_3', \ldots, a_{n+1}) = (m, m + a_1, m + a_2, \ldots, m + a_n)$$

is a $n + 1$ term sequence satisfying the conditions of the problem. Indeed, $a_i' - a_1' = a_{i-1}$ divides m and $a_i' + a_1' = 2m + a_{i-1}$ by the definition of m and $a_j' - a_i' = a_{j-1} - a_{i-1}$ $(2 \leq i < j \leq n + 1)$ divides m. Also, $a_j' + a_i' = 2m + (a_{j-1} + a_{i-1})$ by the definition of m and by the inductive hypothesis. Therefore our induction is complete.

Problem 5.2.2. *Prove that, for each $n \geq 2$, the number $n!$ can be represented as the sum of n distinct divisors of itself.*

(Erdös)

Solution. Strengthening the statement, by imposing the condition that one of the n divisors should be 1, puts us in a winning position. The question here is how we came to think of this. Well, there is just about one way to go in using the induction hypothesis $n! = d_1 + d_2 + \cdots + d_n$ (where $d_1, d_2, \ldots, d_n$ are the n divisors arranged in increasing order); namely, multiplying the above relation by $n + 1$. This yields

$$(n+1)! = (n+1)d_1 + (n+1)d_2 + \cdots + (n+1)d_n$$

$$= d_1 + nd_1 + (n+1)d_2 + \cdots + (n+1)d_n.$$

We split $(n+1)d_1$ into $d_1 + nd_1$, thus getting $n+1$ summands, as needed. Of them, only the second one might not be a divisor of $(n+1)!$. We would like to ensure that it is such a divisor, too. Hence the idea of insisting that $d_1 = 1$.

Problem 5.2.3. *Prove that there are infinitely many numbers not containing the digit 0, that are divisible by the sum of their digits.*

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Solution. Let us prove by induction that $\underbrace{11\ldots1}_{3^n}$ is a good choice. The base case is clearly verified and for the inductive step we have

$$\begin{aligned}\underbrace{11\ldots1}_{3^{n+1}} &= \frac{10^3-1}{9} = \frac{(10^{3^n})^3-1}{9}\\ &= \frac{10^{3^n}-1}{9}(10^{2\cdot 3^n}+10^{3^n}+1)\\ &= \underbrace{11\ldots1}_{3^n}\cdot N,\end{aligned}$$

where N is a multiple of 3 and the conclusion follows.

Problem 5.2.4. *Let n be a positive integer. Let O_n be the number of $2n$-tuples $(x_1,\ldots,x_n,y_1,\ldots,y_n)$ with values in 0 or 1 for which the sum $x_1y_1+\cdots+x_ny_n$ is odd, and let E_n be the number of $2n$-tuples for which the sum is even. Prove that*

$$\frac{O_n}{E_n} = \frac{2^n-1}{2^n+1}.$$

(1997 Iberoamerican Mathematical Olympiad)

Solution. We prove by induction that $O_n = 2^{2n-1}-2^{n-1}$ and $E_n = 2^{2n-1}+2^{n-1}$, which will give the desired ratio.

The base case is $n=1$. This case works because $O_1 = 1 = 2^1-2^0$, and $E_1 = e = 2^1+2^0$.

For the inductive step, we assume this is true for $n=k$; then $x_1y_1+\cdots+x_ky_k$ is even for $2^{2k-1}+2^{k-1}2k$-tuples and odd for $2^{2k-1}-2^{2k-1}2k$-tuples. Now, $x_1y_1+\cdots+x_{k+1}y_{k+1}$ is odd if and only if either $x_1y_1+\cdots+x_ky_k$ is odd and is even or $x_1y_1+\cdots+x_ky_k$ is even and $x_{k+1}y_{k+1}$ is odd. $x_{k+1}y_{k+1}$ can be odd one way and even three ways, so

$$O_{k+1} = 3(2^{2k-1}-2^{k-1})+2^{2k-1}+2^{k-1} = 2^{2(k+1)-1}-2(k+1)-1$$

and $E_{k+1} = 2^{2(k+1)}-O_{k+1}$, which completes the induction.

Problem 5.2.5. *Prove that for all integers $n\geq 3$, there exist odd positive integers x,y, such that $7x^2+y^2=2^n$.*

(1996 Bulgarian Mathematical Olympiad)

Solution. We will prove that there exist odd positive integers x_n,y_n such that $7x_n^2+y_n^2=2^n$, $n\geq 3$.

For $n = 3$, we have $x_3 = y_3 = 1$. Now suppose that for a given integer $n \geq 3$ we have odd integers x_n, y_n satisfying $7x_n^2 + y_n^2 = 2^n$. We shall exhibit a pair (x_{n+1}, y_{n+1}) of odd positive integers such that $7x_{n+1}^2 + y_{n+1}^2 = 2^{n+1}$. In fact,

$$7\left(\frac{x_n \pm y_n}{2}\right)^2 + \left(\frac{7x_n \mp y_n}{2}\right)^2 = 2(7x_n^2 + y_n^2) = 2^{n+1}$$

Precisely one of the numbers $\frac{x_n + y_n}{2}$ and $\frac{|x_n - y_n|}{2}$ is odd (as their sum is the larger of x_n and y_n, which is odd). If, for example, $\frac{x_n + y_n}{2}$ is odd, then

$$\frac{7x_n - y_n}{2} = 3x_n + \frac{x_n - y_n}{2}$$

is also odd (as a sum of an odd and an even number), hence in this case we may choose

$$x_{n+1} = \frac{x_n + y_n}{2} \quad \text{and} \quad y_{n+1} = \frac{7x_n - y_n}{2}$$

If $\frac{x_n - y_n}{2}$ is odd, then

$$\frac{7x_n + y_n}{2} = 3x_n + \frac{x_n + y_n}{2}$$

so we can choose

$$x_{n+1} = \frac{|x_n - y_n|}{2} \quad \text{and} \quad y_{n+1} = \frac{7x_n + y_n}{2}.$$

Remark. Actually, the problem goes back to Euler.

Problem 5.2.6. *Let $f(x) = x^3 + 17$. Prove that for each natural number n, $n \geq 2$, there is a natural number x for which $f(x)$ is divisible by 3^n but not by 3^{n+1}.*

(1999 Japanese Mathematical Olympiad)

Solution. We prove the result by induction on n. If $n = 2$, then $x = 1$ suffices. Now suppose that the claim is true for $n \geq 2$, that is, there is a natural number y such that $y^3 + 17$ is divisible by 3^n but not 3^{n+1}. We prove that the claim is true for $n + 1$.

Suppose we have integers a, m such that a is not divisible by 3 and $m \geq 2$. Then $a^2 \equiv 1 \pmod 3$ and thus $3^m a^2 \equiv 3^m \pmod{3^{m+1}}$. Also, because $m \geq 2$ we have $3m - 3 \geq 2m - 1 \geq m + 1$. Hence

$$(a + 3^{m-1})^3 \equiv a^3 + 3^m a^2 + 3^{2m-1}a + 3^{3m-3} \equiv a^3 + 3^m \pmod{3^{m+1}}.$$

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Because $y^3 + 17$ is divisible by 3^n, it is congruent to either 0, 3^n, or $2 \cdot 3^n$ modulo 3^{n+1}. Because 3 does not divide 17, 3 cannot divide y either. Hence applying our result from the previous paragraph twice, once with $(a, m) = (y, n)$ and once with $(a, m) = (y + 3^{n-1}, n)$, we find that 3^{n+1} must divide either $(y + 3^{n-1})^3 + 17$ or $(y + 2 \cdot 3^{n-1})^3 + 17$.

Hence there exists a natural number x' not divisible by 3 such that $3^{n+1} | x'^3 + 17$. If 3^{n+2} does not divide $x'^3 + 17$, we are done. Otherwise, we claim the number $x = x' + 3^n$ suffices. Because $x = x' + 3^{n-1} + 3^{n-1} + 3^{n-1}$, the result from previous the paragraphs tells us that $x^3 \equiv x'^3 + 3^n + 3^n + 3^n \equiv x'^3 \pmod{3^{n+1}}$. Thus $3^{n+1} | x^3 + 17$ as well. On the other hand, because $x = x' + 3^n$, we have $x^3 \equiv x'^3 + 3^{n+1} \not\equiv x'^3 \pmod{3^{n+2}}$. It follows that 3^{n+2} does not divide $x^3 + 17$, as desired. This completes the inductive step.

Proposed problems

Problem 5.2.7. Let p be an odd prime. The sequence $(a_n)_{n \geq 0}$ is defined as follows: $a_0 = 0$, $a_1 = 1, \ldots, a_{p-2} = p - 2$ and, for all $n \geq p - 1$, a_n is the least positive integer that does not form an arithmetic sequence of length p with any of the preceding terms. Prove that, for all n, a_n is the number obtained by writing n in base $p - 1$ and reading the result in base p.

(1995 USA Mathematical Olympiad)

Problem 5.2.8. Suppose that x, y and z are natural numbers such that $xy = z^2 + 1$. Prove that there exist integers a, b, c and d such that $x = a^2 + b^2$, $y = c^2 + d^2$, and $z = ac + bd$.

(Euler's problem)

Problem 5.2.9. Find all pairs of sets A, B, which satisfy the conditions:
(i) $A \cup B = \mathbb{Z}$;
(ii) if $x \in A$, then $x - 1 \in B$;
(iii) if $x \in B$ and $y \in B$, then $x + y \in A$.

(2002 Romanian IMO Team Selection Test)

Problem 5.2.10. Find all positive integers n such that

$$n = \prod_{k=0}^{m} (a_k + 1),$$

where $\overline{a_m a_{m-1} \dots a_0}$ is the decimal representation of n.

(2001 Japanese Mathematical Olympiad)

Problem 5.2.11. The sequence $(u_n)_{n\geq 0}$ is defined as follows: $u_0 = 2$, $u_1 = \dfrac{5}{2}$ and

$$u_{n+1} = u_n(u_{n-1}^2 - 2) - u_1 \text{ for } n = 1, 2, \dots$$

Prove that $[u_n] = 2^{\frac{2^n - (-1)^n}{3}}$, for all $n > 0$ ($[x]$ denotes the integer part of x).

(18th IMO)

5.3 Infinite descent

Fermat[1] was the first mathematicians to use a method of proof called the *infinite descent*.

Let P be a property concerning the nonnegative integers and let $(P(n))_{n\geq 1}$ be the sequence of propositions,

$$P(n): \text{"}n \text{ satisfies property } P\text{"}$$

The following method is useful in proving that proposition $P(n)$ is false for all large enough n.

Let k be a nonnegative integer. Suppose that:

- *$P(k)$ is not true;*
- *if $P(m)$ is true for a positive integer $m > k$, then there is some smaller j, $m > j \geq k$ for which $P(j)$ is true.*

Then $P(n)$ is false for all $n \geq k$.

This is just the contrapositive of strong induction, applied to the negation of proposition $P(n)$. In the language of the ladder metaphor, if you know you cannot reach any rung without first reaching a lower rung, and you also know you cannot reach the bottom rung, then you cannot reach any rungs.

[1] *Pierre de Fermat* (1601-1665), French lawyer and government official most remembered for his work in number theory, in particular for Fermat's Last Theorem. He is also important in the foundations of the calculus.

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The above is often called the *finite descent method*.

The *Fermat's method of infinite descent* (FMID) can be formulated as follows:

Let k be a nonnegative integer. Suppose that:

- *if $P(m)$ is true for an integer $m > k$, then there must be some smaller integer j, $m > j > k$ for which $P(j)$ is true.*

Then $P(n)$ is false for all $n > k$.

That is, if there where an n for which $P(n)$ was true, one could construct a sequence $n > n_1 > n_2 > \dots$ all of which would be greater than k, but for the nonnegative integers, no such descending is possible.

Two special cases of FMID are particularly useful in solving Number Theory problems.

FMID Variant 1. *There is no sequence of nonnegative integers $n_1 > n_2 > \dots$*

In some situations it is convenient to replace FMID Variant 1 by the following equivalent form: If n_0 is the smallest positive integer n for which $P(n)$ is true, then $P(n)$ is false for all $n < n_0$. In fact, this is equivalent to an extremal argument.

FMID Variant 2. *If the sequence of nonnegative integers $(n_i)_{i\geq 1}$ satisfies the inequalities $n_1 \geq n_2 \geq \dots$, then there exists i_0 such that $n_{i_0} = n_{i_0+1} = \dots$*

Problem 5.3.1. *Find all triples (x, y, z) of nonnegative integers such that*

$$x^3 + 2y^3 = 4z^3.$$

Solution. Note that (0,0,0) is such a triple. We will prove that there is no other. Assume that (x_1, y_1, z_1) is a nontrivial solution to the given equation. Because $\sqrt[3]{2}$, $\sqrt[3]{4}$ are both irrational, it is not difficult to see that $x_1 > 0$, $y_1 > 0$, $z_1 > 0$.

From $x_1^3 + 2y_1^3 = 4z_1^3$ it follows that $2|x_1$, so $x_1 = 2x_2$, $x_2 \in \mathbb{Z}_+$. Then $4x_2^3 + y_1^3 = 2z_1^3$, hence $y_1 = 2y_2$, $y_2 \in \mathbb{Z}_+$. Similarly, $z_1 = 2z_2$, $z_2 \in \mathbb{Z}_+$. We obtain the "new" solution (x_2, y_2, z_2) with $x_1 > x_2$, $y_1 > y_2$, $z_1 > z_2$. Continuing this procedure, we construct a sequence of positive integral triples $(x_n, y_n, z_n)_{n\geq 1}$ such that $x_1 > x_2 > x_3 > \dots$ But this contradicts FMID Variant 1.

Proposed problems

Problem 5.3.2. Find all primes p for which there exist positive integers x, y and n such that $p^n = x^3 + y^3$.

(2000 Hungarian Mathematical Olympiad)

5.4 Inclusion-exclusion

The main result in this section is contained in the following theorem.

Theorem 5.4.1. *Let $S_1, S_2, \ldots, S_n$ be the finite sets. Then*

$$\left|\bigcup_{i=1}^{n} S_i\right| = \sum_{i=1}^{n} |S_i| - \sum_{1\le i<j\le n} |S_i \cap S_j| + \sum_{1\le i<j<k\le n} |S_i \cap S_j \cap S_k|$$

$$- \cdots + (-1)^{n-1} \left|\bigcap_{i=1}^{n} S_i\right|,$$

where $|S|$ denotes the number of elements in S.

Proof. We proceed by induction. For $n = 2$, we have to prove that $|S_1 \cup S_2| = |S_1| + |S_2| - |S_1 \cap S_2|$. This is clear because the number of elements in $S_1 \cup S_2$ is the number of elements in S_1 and S_2 less the ones in $S_1 \cap S_2$, since the latter elements were counted twice.

The inductive step uses the formula above for $S_1 \to \bigcup_{i=1}^{k} S_k$ and $S_2 \to S_{k+1}$. □

The formula in the Theorem is called the *Inclusion-Exclusion Principle.*

Example. *How many positive integers not exceeding 1000 are divisible by 2, or 3, or 5?*

Solution. Consider the sets

$$S_1 = \{2m | 1 \le m \le 500\},\ S_2 = \{3n | 1 \le n \le 333\},\ S_3 = \{5p | 1 \le p \le 200\}.$$

Then

$$S_1 \cap S_2 = \{6q | 1 \le q \le 166\}, \quad S_1 \cap S_3 = \{10r | 1 \le r \le 100\},$$

$$S_2 \cap S_3 = \{15s | 1 \le s \le 66\} \quad \text{and} \quad S_1 \cap S_2 \cap S_3 = \{30u | 1 \le u \le 33\}.$$

Applying the Inclusion-Exclusion Principle we obtain

$$|S_1 \cup S_2 \cup S_3| = |S_1| + |S_2| + |S_3| - |S_1 \cap S_2| - |S_1 \cap S_3| = |S_2 \cap S_3| - |S_1 \cap S_2 \cap S_3|$$

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$$= 500 + 333 + 200 - 166 - 100 - 66 + 33 = 734.$$

The dual version of Theorem 5.4.1 is the following:

Theorem 5.4.2. *Let $S_1, S_2, \ldots, S_n$ be subsets of the finite set S and let $\overline{S}_i = S - S_i$ be the complementary set of S_i, $i = 1, 2, \ldots, n$. Then*

$$\left|\bigcap_{i=1}^{n} \overline{S}_i\right| = |S| - \sum_{i=1}^{n} |S_i| + \sum_{1\le i<j\le n} |S_i \cap S_j| - \sum_{1\le i<j<k\le n} |S_i \cap S_j \cap S_k|$$

$$+ \cdots + (-1)^n \left|\bigcap_{i=1}^{n} S_i\right|.$$

Proof. Let

$$A = \bigcap_{i=1}^{n} \overline{S}_i \quad \text{and} \quad B = \bigcup_{i=1}^{n} S_i.$$

It is clear that $A \cup B = S$ and $A \cap B = \emptyset$. Hence $|S| = |A| + |B|$ and the conclusion follows from Theorem 5.4.1. □

Example. *How many positive integers exceeding 120 are divisible by neither 2, nor 3, nor 5?*

Solution. Consider the sets

$$S_1 = \{2m | 1 \le m \le 60\}, \quad S_2 = \{3n | 1 \le n \le 40\}, \quad S_3 = \{5p | 1 \le p \le 24\}.$$

We have

$$S_1 \cap S_2 = \{6q | 1 \le q \le 20\}, \quad S_1 \cap S_3 = \{10r | 1 \le r \le 12\},$$

$$S_2 \cap S_3 = \{15s | 1 \le s \le 8\} \quad \text{and} \quad S_1 \cap S_2 \cap S_3 = \{30u | 1 \le u \le 4\}.$$

Applying the formula in Theorem 5.3.2, we get

$$|\overline{S}_1 \cap \overline{S}_2 \cap \overline{S}_3| = 120 - (|S_1| + |S_2| + |S_3|) + |S_1 \cap S_2| + |S_1 \cap S_3| + |S_2 \cap S_3|$$

$$-|S_1 \cap S_2 \cap S_3| = 120 - (60 + 40 + 24) + 20 + 12 + 8 - 4 = 32.$$

Problem 5.4.1. *Let $S = \{1, 2, 3, \ldots, 280\}$. Find the smallest integer n such that each n-element subset of S contains five numbers which are pairwise relatively prime.*

(32nd IMO)

Solution. The solutions is given in two steps.

First step. Let consider the sets

$$M_2 = \{2, 4, 6, \ldots, 280\}, \quad M_3 = \{3, 6, 9, \ldots, 279\},$$

$$M_5 = \{5, 10, 15, \ldots, 280\}, \quad M_7 = \{7, 14, \ldots, 280\}$$

and let $M = M_2 \cup M_3 \cup M_5 \cup M_7$. The following cardinalities are obvious:

$$|M_2| = 140, \quad |M_3| = 93, \quad |M_5| = 56 \quad \text{and} \quad |M_7| = 40.$$

It is easy to prove that:

$$|M_2 \cap M_3| = \left[\frac{280}{6}\right] = 46, \quad |M_2 \cap M_5| = \left[\frac{280}{10}\right] = 28,$$

$$|M_2 \cap M_7| = \left[\frac{280}{14}\right] = 20, \quad |M_3 \cap M_5| = \left[\frac{280}{15}\right] = 18,$$

$$|M_3 \cap M_7| = \left[\frac{280}{21}\right] = 13, \quad |M_5 \cap M_7| = \left[\frac{280}{35}\right] = 8,$$

$$|M_2 \cap M_3 \cap M_5| = \left[\frac{280}{30}\right] = 9, \quad |M_2 \cap M_3 \cap M_7| = \left[\frac{280}{42}\right] = 6,$$

$$|M_2 \cap M_5 \cap M_7| = \left[\frac{280}{70}\right] = 4, \quad |M_4 \cap M_5 \cap M_7| = \left[\frac{28}{105}\right] = 2$$

and

$$|M_2 \cap M_3 \cap M_5 \cap M_7| = \left[\frac{280}{210}\right] = 1.$$

By the Principle of Inclusion-Exclusion we obtain:

$$|M| = |M_2 \cup M_3 \cup M_5 \cup M_7|$$

$$= 140 + 93 + 56 + 40 - (46 + 28 + 20 + 18 + 13 + 8) + (9 + 6 + 4 + 2) - 1 = 216.$$

By the Pigeonhole Principle, any five-element subset of M contains at least two elements from the same subset M_i, $i \in \{2, 3, 5, 7\}$. These elements are not relatively prime numbers. Thus, we prove that $n > 216$.

Second step. We will prove that $n = 217$.

The set $S \setminus M$ contains $280 - 216 = 64$ elements. It contains prime numbers and composite numbers. Taking account that $[\sqrt{280}] = 16$, we may state that the composite numbers in $S \setminus M$ are precisely the elements of the set

$$C = \{11^2; 11 \cdot 13; 11 \cdot 17; 11 \cdot 19; 11 \cdot 23; 13^2; 13 \cdot 17; 13 \cdot 19\}.$$

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Observe that $|C| = 8$. Thus, the set $S \setminus M$ contains 1, 8 composite numbers are 55 prime numbers. Also, taking in account the prime numbers 2, 3, 5, 7 we infer that the set S contains 59 prime numbers in all.

Let $p_1 = 2$, $p_2 = 3$, $p_3 = 5, \ldots, p_{59}$ be all these prime numbers and let denote $P = \{1, p_2, p_2, \ldots, p_{59}\}$. Thus, $|P| = 60$.

Let T be a subset containing 217 elements of S. If $|T \cap P| \geq 5$ it follows that T contains 5 elements which are relatively prime numbers. So, let suppose $|T \cap P| \leq 4$. In this case, $|T \cap (S \setminus P)| \geq 217 - 4 = 213$. Since S contains 220 composite numbers, it follows that at most 7 composite numbers are not in T.

Consider the following five-element subsets of $S \setminus P$:

$$A_1 = \{2^2; 3^2; 5^2; 7^2; 13^2\}$$

$$A_2 = \{2 \cdot 23; 3 \cdot 19; 5 \cdot 17; 7 \cdot 13; 11 \cdot 11\}$$

$$A_3 = \{2 \cdot 29; 3 \cdot 23; 5 \cdot 19; 7 \cdot 17; 11 \cdot 13\}$$

$$A_4 = \{2 \cdot 31; 3 \cdot 29; 5 \cdot 23; 7 \cdot 19; 11 \cdot 17\}$$

$$A_5 = \{2 \cdot 37; 3 \cdot 31; 5 \cdot 29; 7 \cdot 23; 11 \cdot 19\}$$

$$A_6 = \{2 \cdot 41; 3 \cdot 37; 5 \cdot 31; 7 \cdot 29; 11 \cdot 23\}$$

$$A_7 = \{2 \cdot 43; 3 \cdot 41; 5 \cdot 37; 7 \cdot 23; 13 \cdot 17\}$$

$$A_8 = \{2 \cdot 47; 3 \cdot 43; 5 \cdot 41; 7 \cdot 37; 12 \cdot 19\}.$$

By the Pigeonhole Principle, there exists a set A_i, $1 \leq i \leq 8$, such that $A_i \subset T$; if not, the set $S \setminus T$ would contain 8 composite numbers. Each A_i contains five relatively prime numbers and we are done.

Proposed problems

Problem 5.4.2. The numbers from 1 to 1000000 can be colored black or white. A permissible move consists of selecting a number from 1 to 1000000 and changing the color of that number and each number not relatively prime to it. Initially all of the numbers are black. Is it possible to make a sequence of moves after which all of the numbers are colored white?

(1999 Russian Mathematical Olympiad)

6

Arithmetic Functions

6.1 Multiplicative functions

Arithmetic functions are defined on the positive integers and are complex valued. The arithmetic function $f \neq 0$ is called *multiplicative* if for any relatively prime positive integers m and n,

$$f(mn) = f(m)f(n)$$

The arithmetic function $f \neq 0$ is called *completely multiplicative* if the relation above holds for any positive integers m and n.

Remarks. 1) If $f : \mathbb{Z}_+^* \to \mathbb{C}$ is multiplicative, then $f(1) = 1$. Indeed, if a is a positive integer for which $f(a) \neq 0$, then $f(a) = f(a \cdot 1) = f(a)f(1)$ and simplify by $f(a)$ yields $f(1) = 1$.

2) If f is multiplicative and $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ is the prime factorization of the positive integer n, then $f(n) = f(p_1^{\alpha_1}) \dots f(p_k^{\alpha_k})$, that is in order to compute $f(n)$ it suffices to compute $f(p_i^{\alpha_i})$, $i = 1, \dots, k$.

3) If f is completely multiplicative and $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ is the prime factorization of n, then $f(n) = f(p_1)^{\alpha_1} \dots f(p_k)^{\alpha_k}$, that is in order to compute $f(n)$ it suffices to compute $f(p_i)$, $i = 1, \dots, k$.

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An important arithmetic function is the *Möbius*[1] *function* defined by

$$\mu(n) = \begin{cases} 1 & \text{if} \quad n = 1 \\ 0 & \text{if} \quad p^2|n \text{ for some prime } p > 1 \\ (-1)^k & \text{if} \quad n = p_1 \dots p_k, \text{ where } p_1, \dots, p_k \text{ are distinct primes} \end{cases}$$

For example, $\mu(2) = -1$, $\mu(b) = 1$, $\mu(12) = \mu(2^2 \cdot 3) = 0$.

Theorem 6.1.1. *The Möbius function μ is multiplicative.*

Proof. Let m, n be positive integers such that $gcd(m, n) = 1$. If $p^2|m$ for some $p > 1$, then $p^2|mn$ and so $\mu(m) = \mu(mn) = 0$ and we are done. Consider now $m = p_1 \dots p_k$, $n = q_1 \dots q_h$, where $p_1, \dots, p_k, q_1, \dots, q_h$ are distinct primes. Then $\mu(m) = (-1)^k$, $\mu(n) = (-1)^h$, and $mn = p_1 \dots p_k q_1 \dots q_h$. It follows that $\mu(mn) = (-1)^{k+h} = (-1)^k(-1)^h = \mu(m)\mu(n)$. □

For an arithmetic function f we define its *summation function* F by

$$F(n) = \sum_{d|n} f(d).$$

The connection between f and F is given by the following result.

Theorem 6.1.2. *If f is multiplicative, then so is its summation function F.*

Proof. Let m, n be positive integers such that $gcd(m, n) = 1$ and let d be a divisor of mn. Then d can be uniquely represented as $d = kh$, where $k|m$ and $h|n$. Because $gcd(m, n) = 1$, we have $gcd(k, h) = 1$, so $f(kh) = f(k)f(h)$. Hence

$$F(mn) = \sum_{d|mn} f(d) = \sum_{\substack{k|m \\ h|n}} f(k)f(h)$$

$$= \left(\sum_{k|m} f(k) \right) \left(\sum_{h|n} f(h) \right) = F(m)F(n). \qquad \square$$

Remark. If f is a multiplicative function and $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, then

$$F(n) = \prod_{i=1}^{k} (1 + f(p_i) + \dots + f(p_i^{\alpha_i})) \tag{1}$$

[1] *August Ferdinand Möbius* (1790-1868), German mathematician best known for his work in topology, especially for his conception of the Möbius strip, a two dimensional surface with only one side.

Indeed, after multiplication in the right hand side we get a sum having terms of the form $f(p_1^{\beta_1})\dots f(p_k^{\beta_k}) = f(p_1^{\beta_1}\dots p_k^{\beta_k})$, where $0 \le \beta_1 \le \alpha_1, \dots, 0 \le \beta_k \le \alpha_k$. This sum is obviously $F(n)$.

From (1) we can derive the following formula

$$\sum_{d|n} \mu(d)f(d) = (1 - f(p_1))\dots(1 - f(p_k)). \tag{2}$$

The function $g(n) = \mu(n)f(n)$ is multiplicative, hence applying (1) we get for its summation function G

$$G(n) = \prod_{i=1}^{k}(1 + \mu(p_i)f(p_i)) = \prod_{i=1}^{k}(1 - f(p_i)).$$

Theorem 6.1.3. (Möbius inversion formula) *Let f be an arithmetic function and let F be its summation function. Then*

$$f(n) = \sum_{d|n} \mu(d)F\left(\frac{n}{d}\right). \tag{3}$$

Proof. We have

$$\sum_{d|n} \mu(d)F\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d)\left(\sum_{c|\frac{n}{d}} f(c)\right) = \sum_{d|n}\left(\sum_{c|\frac{n}{d}} \mu(d)f(c)\right)$$

$$= \sum_{c|n}\left(\sum_{d|\frac{n}{c}} \mu(d)f(c)\right) = \sum_{c|n} f(c)\left(\sum_{d|\frac{n}{c}} \mu(d)\right) = f(n),$$

since for $\dfrac{n}{c} > 1$ we have $\displaystyle\sum_{d|\frac{n}{c}} \mu(d) = 0$.

We have used the fact that sets

$$\left\{(d,c)|\ d|n \text{ and } c|\frac{n}{d}\right\} \quad \text{and} \quad \left\{(d,c)|\ c|n \text{ and } d|\frac{n}{c}\right\}$$

are equal. □

Theorem 6.1.4. *Let f be an arithmetic function and let F be its summation function. If F is multiplicative, then so is f.*

Proof. Let m, n be positive integers such that $gcd(m, n) = 1$ and let d be a divisor of mn. Then $d = kh$ where $k|m$, $h|n$ and $gcd(k, h) = 1$. Applying Möbius inversion formula it follows

$$f(mn) = \sum_{d|mn} \mu(d)F\left(\frac{mn}{d}\right) = \sum_{\substack{k|m\\ h|n}} \mu(kh)F\left(\frac{mn}{kh}\right)$$

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$$= \sum_{\substack{k|m \\ h|n}} \mu(k)\mu(h) F\left(\frac{m}{k}\right) F\left(\frac{n}{h}\right) = \left(\sum_{k|m} \mu(k) F\left(\frac{m}{k}\right)\right)\left(\sum_{h|n} \mu(h) F\left(\frac{n}{h}\right)\right)$$

$$= f(m)f(n). \qquad \square$$

Let f and g be two arithmetic functions. Define their *convolution product* or *Dirichlet*[2] *product* $f * g$ by

$$(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right)$$

Problem 6.1.1. *1) Prove that the convolution product is commutative and associative.*

2) Prove that for any arithmetic function f,

$$f * \varepsilon = \varepsilon * f = f,$$

where $\varepsilon(n) = 1$ if $n = 1$ and 0 otherwise.

Solution. Let f and g be two arithmetic functions. Then

$$(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right) = \sum_{d_1|n} f\left(\frac{n}{d_1}\right) g(d_1) = (g * f)(n),$$

since if d runs through all divisors of, then so does $d_1 = \dfrac{n}{d}$. Therefore $f * g = g * f$.

Let f, g, h be arithmetic functions. To prove the associativity law, let $u = g * h$ and consider $f * u = f * (g * h)$. We have

$$(f * u)(n) = \sum_{a|n} f(a) u\left(\frac{n}{a}\right) = \sum_{ad=n} f(a) \sum_{bc=d} g(b)h(c)$$

$$= \sum_{abc=n} f(a)g(b)h(c)$$

Similarly, if we set $v = f * g$ and consider $v * h$, we have

$$(v * h)(n) = \sum_{dc=n} v(d)h(c) = \sum_{dc=n} \sum_{ab=d} f(a)g(b)h(c)$$

[2] *Johann Peter Gustav Lejeune Dirichlet* (1805-1859), German mathematician who proved in 1837 that there are infinitely many primes in any arithmetic progression of integers for which the common difference is relatively prime to the terms. Dirichlet has essential contributions in number theory, probability theory, functional analysis and Fourier series.

$$= \sum_{abc=n} f(a)g(b)h(c),$$

hence $f * (g * h) = (f * g) * h$.

2) We have

$$(\varepsilon * f)(n) = \sum_{d|n} \varepsilon(d) f\left(\frac{n}{d}\right) = f(n),$$

and we get $\varepsilon * f = f * \varepsilon = f$.

Problem 6.1.2. *Let f be an arithmetical function. If $f(1) \neq 0$, then there is a unique arithmetical function g such that*

$$f * g = \varepsilon.$$

Solution. We show by induction on n that $(f*g)(n) = \varepsilon(n)$ has a unique solution $g(1), \ldots, g(n)$.

For $n = 1$, we have $f(1)g(1) = 1$, hence $g(1) = \dfrac{1}{f(1)}$.

Suppose $n > 1$ and assume $g(1), \ldots, g(n-1)$ have been determined such that $(f * g)(k) = \varepsilon(k)$ holds for $k = 1, 2, \ldots, n-1$. Then

$$f(1)g(n) + \sum_{\substack{d|n \\ d>1}} f(d) g\left(\frac{n}{d}\right) = 0,$$

and we get

$$g(n) = -\frac{1}{f(1)} \sum_{\substack{d|n \\ d>1}} f(d) g\left(\frac{n}{d}\right)$$

i.e. the function g is unique.

Remark. The unique function g satisfying $f * g = \varepsilon$ where $f(1) \neq 0$ is called the *convolution inverse* of f.

Problem 6.1.3. *If f and g are multiplicative, so is their convolution product.*

Solution. Let $h = f * g$. We have

$$h(mn) = \sum_{c|mn} f(c) g\left(\frac{mn}{c}\right)$$

Set $c = ab$, where $a|m$ and $b|n$. Since $gcd(m, n) = 1$, we have

$$h(mn) = \sum_{a|m} \sum_{b|n} f(ab) g\left(\frac{m}{a}\frac{n}{b}\right)$$

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$$= \left(\sum_{a|m} f(a)g\left(\frac{m}{a}\right)\right)\left(\sum_{b|n} f(b)g\left(\frac{n}{b}\right)\right) = h(m)h(n).$$

Problem 6.1.4. *1) If both g and $f * g$ are multiplicative, then f is also multiplicative.*

2) If g is multiplicative, then so is its convolution inverse.

Solution. 1) We shall prove by contradiction. Suppose f is not multiplicative. Let $h = f * g$. Since f is not multiplicative, there exist m and n, $gcd(m, n) = 1$ such that $f(mn) \neq f(m)f(n)$. We choose mn as small as possible. If $mn = 1$, then we get $f(1) \neq f(1)f(1)$ so $f(1) \neq 1$. Since $h(1) = f(1)g(1) = f(1) \neq 1$, h is not multiplicative, a contradiction. If $mn > 1$, we have $f(ab) = f(a)f(b)$ for all $ab < mn$ with $gcd(a, b) = 1$. Now

$$h(mn) = f(mn)g(1) + \sum_{\substack{a|m \\ b|n}} f(ab)g\left(\frac{mn}{ab}\right)$$

$$= f(mn) + \sum_{\substack{a|m \\ b|n \\ ab<mn}} f(a)f(b)g\left(\frac{m}{a}\right)g\left(\frac{n}{b}\right) = f(mn) - f(m)f(n) + h(m)h(n).$$

Since $f(mn) \neq f(m)f(n)$, $h(mn) \neq h(m)h(n)$. Therefore, h is not multiplicative, a contradiction.

2) Denote by g^{-1} the convolution inverse of g. Then $\varepsilon = g * g^{-1} = g^{-1} * g$ and g are both multiplicative. From the previous result it follows that g^{-1} is multiplicative.

Problem 6.1.5. *Prove that the arithmetic function f is completely multiplicative if and only if $f * f = f\tau$, where $\tau(n)$ is the number of divisors of n.*

(American Mathematical Monthly)

Solution. If f is completely multiplicative, we have

$$(f * f)(n) = \sum_{d|n} f(d)f\left(\frac{n}{d}\right) = \sum_{d|n} f\left(d\frac{n}{d}\right) = \sum_{d|n} f(n)$$

$$= f(n)\sum_{d|n} 1 = f(n)\tau(n) = (f\tau)(n),$$

and the relation follows.

Conversely, take $n = 1$, and it follows $f(1) = 0$ or $f(1) = 1$. Now suppose that $n \geq 2$ and let $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ the prime factorization of

n. Put $\alpha(n) = \alpha_1 + \cdots + \alpha_k$. It suffice to show that for any positive integer $n \geq 2$, the following relation holds

$$f(n) = f(1)f(p_1)^{\alpha_1} \dots f(p_k)^{\alpha_k}.$$

We proceed by induction on α. If $\alpha(n) = 1$, then n is a prime, say $n = p$, and the property follows from the fact that

$$2f(p) = \tau(p)f(p) = f(1)f(p) + f(p)f(1) = 2f(1)f(p)$$

Suppose then that the property holds for all n with $\alpha(n) \leq k$. Take any n with $\alpha(n) = k + 1$. Then

$$\tau(n)f(n) = 2f(1)f(n) + \sum f(a)f(b),$$

where the sum runs over all a, b with $ab = n$ and $1 < a, b < n$. It follows that $\alpha(a) \leq k$, $\alpha(b) \leq k$ and from the inductive assumption we get

$$\tau(n)f(n) = 2f(1)f(n) + (\tau(n) - 2)f^2(1)f(p_1)^{\alpha_1} \dots f(p_k)^{\alpha_k}$$

Since n is not a prime, certainly $\tau(n) > 2$ and so, for both $f(1) = 0$ and $f(1) = 1$, the desired result follows.

Proposed problems

Problem 6.1.6. Let f be a function from the positive integers to the integers satisfying $f(m + n) \equiv f(n) \pmod{m}$ for all $m, n \geq 1$ (e.g., a polynomial with integer coefficients). Let $g(n)$ be the number of values (including repetitions) of $f(1), f(2), \dots, f(n)$ divisible by n, and let $h(n)$ be the number of these values relatively prime to n. Show that g and h are multiplicative functions related by

$$h(n) = n \sum_{d|n} \mu(d) \frac{g(d)}{d} = n \prod_{j=1}^{k} \left(1 - \frac{g(p_j)}{p_j}\right),$$

where $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ is the prime factorization of n.

(American Mathematical Monthly)

Problem 6.1.7. Define $\lambda(1) = 1$, and if $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, define

$$\lambda(n) = (-1)^{\alpha_1 + \cdots + \alpha_k}$$

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1) Show that λ is completely multiplicative.

2) Prove that

$$\sum_{d|n} \lambda(d) = \begin{cases} 1 & \text{if } n \text{ is a square} \\ 0 & \text{otherwise} \end{cases}$$

3) Find the convolutive inverse of λ.

Problem 6.1.8. Let an integer $n > 1$ be factored into primes: $n = p_1^{\alpha_1} \dots p_m^{\alpha_m}$ (p_i distinct) and let its own positive integral exponents be factored similarly. The process is to be repeated until it terminates with a unique "constellation" of prime numbers. For example, the constellation for 192 is $192 = 2^{2^2 \cdot 3} \cdot 3$ and for 10000 is $10000 = 2^{2^2} \cdot 5^2$. Call an arithmetic function g generally multiplicative if $g(ab) = g(a)g(b)$ whenever the constellations for a and b have no prime in common.

1) Prove that every multiplicative function is generally multiplicative. Is the converse true?

2) Let h be an additive function (i.e. $h(ab) = h(a) + h(b)$ whenever $gcd(a, b) = 1$). Call a function k generally additive if $k(ab) = k(a) + k(b)$ whenever the constellations for a and b have no prime in common. Prove that every additive function is generally additive. Is the converse true?

(American Mathematical Monthly)

6.2 Number of divisors

For a positive integer n denote by $\tau(n)$ the number of its divisors. It is clear that

$$\tau(n) = \sum_{d|n} 1,$$

that is τ is the summation function of the multiplicative function $f(m) = 1$, $m \in \mathbb{Z}_+^*$. Applying Theorem 6.1.2 it follows that τ is multiplicative.

Theorem 6.2.1. *If $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ is the prime factorization of n, then*

$$\tau(n) = (\alpha_1 + 1) \dots (\alpha_k + 1). \tag{4}$$

Proof. Using the fact that τ is multiplicative, we have

$$\tau(n) = \tau(p_1^{\alpha_1}) \dots \tau(p_k^{\alpha_k}) = (\alpha_1 + 1) \dots (\alpha_k + 1),$$

because $p_i^{\alpha_i}$ has exactly $\alpha_i + 1$ divisors, $i = 1, \dots, k$. □

Problem 6.2.1. *For any $n \geq 2$*

$$\tau(n) = \sum_{k=1}^{n} \left(\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor \right).$$

Solution. Note that

$$\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor = \begin{cases} 1 & \text{if } k|n \\ 0 & \text{otherwise} \end{cases}$$

Hence

$$\sum_{k=1}^{n} \left(\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor \right) = \sum_{k|n} 1 = \tau(n).$$

Remark. It is clear that n is a prime if and only if $\tau(n) = 2$. Hence

$$\sum_{k=1}^{n} \left(\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor \right) = 2$$

if and only if n is a prime.

Problem 6.2.2. *Find all positive integers d that have exactly 16 positive integral divisors $d_1, d_2, \ldots, d_{16}$ such that*

$$1 = d_1 < d_2 < \cdots < d_{16} = d,$$

$d_6 = 18$ and $d_9 - d_8 = 17$.

(1998 Irish Mathematical Olympiad)

Solution. Let $d = p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_m^{\alpha_m}$ with $p_1, \ldots, p_m$ distinct primes. Then n has $(a_1 + 1)(a_2 + 1) \ldots (a_n + 1)$ divisors. Since $18 = 2 \cdot 3^2$, it has 6 divisors: 1, 2, 3, 6, 9, 18. Since d has 16 divisors, we know that $d = 2 \cdot 3^3 p$ or $d = 2 \cdot 3^7$. If $b = 2 \cdot 3^7$, $d_8 = 54$, $d_9 = 81$ and $d_9 - d_8 \neq 17$. Thus $d = 2 \cdot 3^3 p$ for some prime $p > 18$. If $p < 27$, then $d_7 = p$, $d_8 = 27$, $d_9 = 2p = 27 + 17 + 44 \Rightarrow p = 22$, a contradiction. Thus $p > 27$. If $p < 54$, $d_7 = 27$, $d_8 = p$, $d_9 = 54 = d_8 + 17 \Rightarrow p = 37$. If $p > 54$, then $d_7 = 27$, $d_8 = 54$, $d_9 = d_8 + 17 = 71$. We obtain two solutions for the problem: $2 \cdot 3^3 \cdot 37 = 1998$ and $2 \cdot 3^3 \cdot 71 = 3834$.

Problem 6.2.3. *For how many a) even and b) odd numbers n, does n divide $3^{12} - 1$, yet n does not divide $3^k - 1$ for $k = 1, 2, \ldots, 11$.*

(1995 Austrian Mathematical Olympiad)

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Solution. We note

$$\begin{aligned} 3^{12}-1 &= (3^6-1)(3^6+1) \\ &= (3^2-1)(3^4+3^2+1)(3^2+1)(3^4-3^2+1) \\ &= (2^3)(7\cdot 13)(2\cdot 5)(73). \end{aligned}$$

Recall that the number of divisors of $p_1^{e_1}\dots p_k^{e_k}$ is $(e_1+1)\dots(e_k+1)$. Therefore $3^{12}-1$ has $2\cdot 2\cdot 2\cdot 2 = 16$ odd divisors and $4\cdot 16 = 64$ even divisors.

If $3^{12}\equiv 1 \pmod m$ for some integer m, then the smallest integer d such that $3^d \equiv 1 \pmod m$ divides 12. (Otherwise we could write $12 = pq + r$ with $0 < r < d$ and find $3^r \equiv 1 \pmod m$). Hence to ensure $n \nmid 3^k - 1$ for $k = 1, \dots, 11$, we need only check $k = 1, 2, 3, 4, 6$. But

$$\begin{aligned} 3^1 - 1 &= 2 \\ 3^2 - 1 &= 2^3 \\ 3^3 - 1 &= 2\cdot 13 \\ 3^4 - 1 &= 2^4\cdot 5 \\ 3^6 - 1 &= 2^3\cdot 7\cdot 13. \end{aligned}$$

The odd divisors we throw out are 1, 5, 7, 13, 91, while the even divisors are 2^i for $1 \le i \le 4$, $2^i\cdot 5$ for $1 \le i \le 4$, and each of $2^j\cdot 7$, $2^j\cdot 13$, and $2^j\cdot 7\cdot 13$ for $1 \le i \le 3$. As we are discarding 17 even divisors and 5 odd ones, we remain with 47 even divisors and 11 odd ones.

Problem 6.2.4. *Let $\tau(n)$ denote the number of divisors of the natural number n. Prove that the sequence $\tau(n^2+1)$ does not become monotonic from any given point onwards.*

(1998 St. Petersburg City Mathematical Olympiad)

Solution. We first note that for n even, $\tau(n^2+1) \le n$. Indeed, exactly half of the divisors of n^2+1 are less than n, and all are odd, so there are at most $2(n/2)$ in all.

Now if $\tau(n^2+1)$ becomes strictly monotonic for $n \ge N$, then

$$\tau((n+1)^2+1) \ge \tau(n^2+1) + 2$$

for $n \ge N$ (since $\tau(k)$ is even for k not a perfect square). Thus

$$\tau(n^2+1) \ge \tau(N^2+1) + 2(n-N)$$

which exceeds n for large, contradiction.

Proposed problems

Problem 6.2.5. Does there exist a positive integer such that the product of its proper divisors ends with exactly 2001 zeroes?

(2001 Russian Mathematical Olympiad)

Problem 6.2.6. Prove that the number of divisors of the form $4k+1$ of each positive integer is not less than the number of its divisors of the form $4k+3$.

Problem 6.2.7. Let $d_1, d_2, \ldots, d_l$ be all positive divisors of a positive integer. For each $i = 1, 2, \ldots, l$ denote by a_i the number of divisors of d_i. Then

$$a_1^3 + a_2^3 + \cdots + a_l^3 = (a_1 + a_2 + \cdots + a_l)^2.$$

6.3 Sum of divisors

For a positive integer n denote by $\sigma(n)$ the sum of its divisors. It is clear that

$$\sigma(n) = \sum_{d|n} d,$$

that is σ is the summation function of the multiplicative function $d(m) = m$, $m \in \mathbb{Z}_+^*$. Applying Theorem 6.1.2 it follows that σ is multiplicative.

Theorem 6.3.1. *If $n = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ is the prime factorization of n, then*

$$\sigma(n) = \frac{p_1^{\alpha_1+1} - 1}{p_1 - 1} \cdots \frac{p_k^{\alpha_k+1} - 1}{p_k - 1}.$$

Proof. Because σ is multiplicative, it suffices to compute $\sigma(p_i^{\alpha_i})$, $i = 1, \ldots, k$. The divisors of $p_i^{\alpha_i}$ are $1, p_i, \ldots, p_i^{\alpha_i}$, hence

$$\sigma(p_i^{\alpha_i}) = 1 + p_i + \cdots + p_i^{\alpha_i} = \frac{p_i^{\alpha_1+1} - 1}{p_i - 1}$$

and the conclusion follows. □

Problem 6.3.1. *For any $n \geq 2$,*

$$\sigma(n) = \sum_{k=1}^{n} k \left(\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor \right).$$

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Solution. We have

$$\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor = \begin{cases} 1 & \text{if } k|n \\ 0 & \text{otherwise} \end{cases}$$

hence

$$\sum_{k=1}^{n} k \left(\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor \right) = \sum_{k|n} k = \sigma(n)$$

Remark. It is clear that n is a prime if and only if $\sigma(n) = n+1$. Hence

$$\sum_{k=1}^{n} k \left(\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor \right) = n+1$$

if and only if n is a prime.

Problem 6.3.2. *If n is a composite positive integer, then*

$$\sigma(n) \geq n + \sqrt{n} + 1.$$

Solution. The integer n has a divisor d such that $d \neq 1$ and $d \leq \sqrt{n}$. Because $\frac{n}{d}$ is also a divisor of n, it follows that $\frac{n}{d} \geq \sqrt{n}$, therefore

$$\sigma(n) = \sum_{k|n} k \geq 1 + n + \frac{n}{d} \geq n + \sqrt{n} + 1.$$

Problem 6.3.3. *For any $n \geq 7$,*

$$\sigma(n) < n \ln n.$$

Solution. Let $d_1, d_2, \ldots, d_k$ all divisors of n. They can be also written as

$$\frac{n}{d_1}, \frac{n}{d_2}, \ldots, \frac{n}{d_k},$$

hence

$$\sigma(n) = n \left(\frac{1}{d_1} + \frac{1}{d_2} + \cdots + \frac{1}{d_k} \right) \leq n \left(1 + \frac{1}{2} + \cdots + \frac{1}{k} \right),$$

where $k = \tau(n)$. Inducting on k we prove that for any $k \geq 2$,

$$1 + \frac{1}{2} + \cdots + \frac{1}{k} < .81 + \ln k.$$

Using the inequality $\tau(n) \leq 2\sqrt{n}$ (Problem 2, Section 6.2) it follows that

$$1 + \frac{1}{2} + \cdots + \frac{1}{k} < .81 + \ln(2\sqrt{n}) < 1.51 + \frac{1}{2}\ln n.$$

For $n \geq 21$ we have $\ln n > 1.51 + \frac{1}{2}\ln n$ and checking directly the desired inequality for $n = 7, \ldots, 20$, the conclusion follows.

Problem 6.3.4. *For any $n \geq 2$,*

$$\frac{\sigma(n)}{\tau(n)} \geq \sqrt{n}.$$

Solution. Let $d_1, d_2, \ldots, d_{\tau(n)}$ ne the divisors of n. They can be rewritten as

$$\frac{n}{d_1}, \frac{n}{d_2}, \ldots, \frac{n}{d_{\tau(n)}}$$

Hence

$$\sigma(n)^2 = n(d_1 + d_2 + \cdots + d_{\tau(n)}) \left(\frac{1}{d_1} + \frac{1}{d_2} + \cdots + \frac{1}{d_{\tau(n)}} \right) \geq n\tau(n)^2$$

and the conclusion follows.

Proposed problems

Problem 6.3.5. For any $n \geq 2$,

$$\sigma(n) < n\sqrt{2\tau(n)}.$$

(1999 Belarusian Mathematical Olympiad)

Problem 6.3.6. Find all the four-digit numbers so that when decomposed in prime factors have the sum of the prime factors equal to the sum of the exponents.

Problem 6.3.7. Let m, n, k be positive integers with $n > 1$. Show that $\sigma(n)^k \neq n^m$.

(2001 St. Petersburg City Mathematical Olympiad)

6.4 Euler's totient function

For any positive integer n we denote by $\varphi(n)$ the number of all integers m such that $m \leq n$ and $gcd(m, n) = 1$. The arithmetic function φ is called

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the *Euler*[3] *'s totient function*. It is clear that $\varphi(1) = 1$ and for any prime p, $\varphi(p) = p - 1$. Moreover, if n is a positive integer such that $\varphi(n) = n - 1$ then n is a prime.

Theorem 6.4.1. (Gauss) *For any positive integer n,*

$$\sum_{d|n} \varphi(d) = n.$$

Proof. Let $d_1, d_2, \ldots, d_k$ be the divisors of n and let $S_i = \{m|\ m \leq n \text{ and } gcd(m, n) = d_i\}$, $i = 1, \ldots, k$. If $m \in S_i$, then $m = d_i m'$, where $gcd\left(m', \frac{n}{d_i}\right) = 1$. Because $m' \leq \frac{n}{d_i}$, from the definition of φ it follows that $|S_i| = \varphi\left(\frac{n}{d_i}\right)$. The sets $S_1, \ldots, S_k$ give a partition of $\{1, 2, \ldots, n\}$, hence

$$\sum_{i=1}^{k} \varphi\left(\frac{n}{d_i}\right) = \sum_{i=1}^{k} |S_i| = n.$$

But $\left\{\frac{n}{d_1}, \ldots, \frac{n}{d_k}\right\} = \{d_1, \ldots, d_k\}$, so $\sum_{d|n} \varphi(d) = n$. □

Theorem 6.4.2. *The function φ is multiplicative.*

Proof. From Theorem 6.4.1 we obtain that the summation function of φ is $F(n) = n$, which is multiplicative.

The conclusion now follows from Theorem 6.1.4. □

Theorem 6.4.3. *If $n = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ is the prime factorization of $n > 1$, then*

$$\varphi(n) = n\left(1 - \frac{1}{p_1}\right) \ldots \left(1 - \frac{1}{p_k}\right).$$

Proof. We first notice that for any prime p and for any positive integer α,

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^\alpha\left(1 - \frac{1}{p}\right).$$

Indeed, the number of all positive integers not exceeding n that are divisible by p is $p^{\alpha-1}$, hence $\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$.

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[3] *Leonhard Euler* (1707-1783), Swiss mathematician which worked at the Petersburg Academy and Berlin Academy of Science. Euler systematized mathematics by introducing the symbols e and i, and $f(x)$ for a function of x. He also made major contributions in optics, mechanics, electricity, and magnetism. Euler did important work in number theory, proving that the divergence of the harmonic series implies an infinite number of primes, factoring the fifth Fermat number, and introducing the totient function φ.

Using Theorem 6.4.3 we have

$$\varphi(n) = \varphi(p_1^{\alpha_1} \dots p_k^{\alpha_k}) = \varphi(p_1^{\alpha_1}) \dots \varphi(p_k^{\alpha_k})$$

$$= p_1^{\alpha_1}\left(1 - \frac{1}{p_1}\right) \dots p_k^{\alpha_k}\left(1 - \frac{1}{p_k}\right) = p_1^{\alpha_1} \dots p_k^{\alpha_k}\left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

$$= n\left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_k}\right).$$

Alternative proof. We employ the inclusion-exclusion principle. Let

$$T_i = \{d | d \leq n \text{ and } p_i | d\}, \quad i = 1, \dots, k.$$

It follows that

$$T_1 \cup \dots \cup T_k = \{m | m \leq n \text{ and } gcd(m, n) > 1\}.$$

Hence

$$\varphi(n) = n - |T_1 \cup \dots \cup T_k| = n - \sum_{i=1}^{k} |T_i| + \sum_{1 \leq i < j \leq k} |T_i \cap T_j|$$

$$- \dots + (-1)^k |T_1 \cap \dots \cap T_k|.$$

We have

$$|T_i| = \frac{n}{p_i}, \ |T_i \cap T_j| = \frac{n}{p_i p_j}, \dots, |T_1 \cap \dots \cap T_k| = \frac{n}{p_1 \dots p_k}$$

Finally,

$$\varphi(n) = n\left(1 - \sum_{i=1}^{n} \frac{1}{p_i} + \sum_{1 \leq i < j \leq k} \frac{1}{p_i p_j} - \dots + (-1)^k \frac{1}{p_1 \dots p_k}\right)$$

$$= n\left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_k}\right). \qquad \square$$

Problem 6.4.1. *Prove that there are infinitely many even positive integers k such that the equation $\varphi(n) = k$ has no solution.*

(Schinzel[4])

[4] *Andrzej Schinzel*, Polish mathematician with important work on exponential congruences, Euler's φ-function, Diophantine equations, applications of transcendental number theory to arithmetic problems.

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Solution. Take $k = 2 \cdot 7^m$, $m \geq 1$. If $n = p_1^{\alpha_1} \dots p_h^{\alpha_h}$, then

$$\varphi(n) = p_1^{\alpha_1}\left(1 - \frac{1}{p_1}\right) \dots p_h^{\alpha_h}\left(1 - \frac{1}{p_h}\right)$$
$$= p_1^{\alpha_1 - 1} \dots p_h^{\alpha_h - 1}(p_1 - 1) \dots (p_h - 1).$$

If at least two of the primes $p_1, \dots, p_h$ are odd, then $4|\varphi(n)$ and $\varphi(n) \neq k$. If $n = 2^\alpha p^\beta$, with $p \geq 3$, then

$$\varphi(n) = 2^\alpha p^\beta \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{p}\right) = 2^{\alpha-1} p^{\beta-1}(p-1).$$

In this case the equality $\varphi(n) = k$ is also impossible.

Problem 6.4.2. *Prove that there are infinitely many positive integers n such that*

$$\varphi(n) = \frac{n}{3}.$$

Solution. Let $n = 2 \cdot 3^m$, where m is a positive integer. Then

$$\varphi(n) = \varphi(2 \cdot 3^m) = \varphi(2)\varphi(3^m) = 3^m - 3^{m-1} = 2 \cdot 3^{m-1} = \frac{n}{3}$$

for infinitely many values of n, as desired.

Problem 6.4.3. *If n is a composite positive integer, then*

$$\varphi(n) \leq n - \sqrt{n}.$$

Solution. because n is composite, it has a prime factor $p_j \leq \sqrt{n}$. We have

$$\varphi(n) = n\left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_k}\right) \leq n\left(1 - \frac{1}{p_j}\right) \leq n\left(1 - \frac{1}{\sqrt{n}}\right) = n - \sqrt{n}.$$

Problem 6.4.4. *For any positive integer n, $n \neq 2$, $n \neq 6$,*

$$\varphi(n) \geq \sqrt{n}.$$

Solution. Let $m \geq 2$. If $n = 2^m$, then

$$\varphi(n) = 2^m - 2^{m-1} = 2^{m-1} \geq \sqrt{2^m} = \sqrt{n}.$$

If $n = p^m$, where p is an odd prime then

$$\varphi(n) = p^m - p^{m-1} = p^{m-1}(p-1) \geq \sqrt{p^m} = \sqrt{n}.$$

If $n = p^m$, where p is a prime greater than or equal to 5, then $\varphi(n) \geq \sqrt{2n}$.

If n is odd or $4|n$, then

$$\varphi(n) = \varphi(p_1^{\alpha_1}) \dots \varphi(p_k^{\alpha_k}) \geq \sqrt{p_1^{\alpha_1}} \dots \sqrt{p_k^{\alpha_k}} = \sqrt{n}.$$

If $n = 2t$, with t odd, $t \neq 1$, $t \neq 3$, then all prime factors of t are greater than or equal to 5, hence $\varphi(n) = \varphi(t) \geq \sqrt{2t}$. It remains to settle the case $n = 2 \cdot 3^i$, $i \geq 2$. For $i = 2$, $\varphi(12) = 6 > \sqrt{18}$ and for $i \geq 3$, $\varphi(n) = 2 \cdot 3^{i-1}$ and the inequality reduces to $\sqrt{2} \cdot 3^{\frac{i}{2}-1} > 1$, which is clear.

Proposed problems

Problem 6.4.5. For a positive integer n, let $\psi(n)$ be the number of prime factors of n. Show that if $\varphi(n)$ divides $n-1$ and $\psi(n) \leq 3$, then n is prime.

(1998 Korean Mathematical Olympiad)

Problem 6.4.6. Show that the equation $\varphi(n) = \tau(n)$ has only the solutions $n = 1, 3, 8, 10, 18, 24, 30$.

Problem 6.4.7. Let $n > 6$ be an integer and $a_1, a_2, \dots, a_k$ be all positive integers less than n and relatively prime to n. If

$$a_2 - a_1 = a_3 - a_2 = \dots = a_k - a_{k-1} > 0,$$

prove that n must be either a prime number or a power of 2.

(32nd IMO)

6.5 Exponent of a prime and Legendre's formula

Let p be a prime and let us denote by $v_p(a)$ the exponent of p in the decomposition of a. Of course, if p doesn't divide a, then $v_p(a) = 0$.

It is easy to prove the following properties of v_p:

1) $\min\{v_p(a), v_p(b)\} \leq v_p(a+b) \leq \max\{v_p(a), v_p(b)\}$;

2) $v_p(ab) = v_p(a) + v_p(b)$;

3) $v_p(gcd(a_1, a_2, \dots, a_n)) = \min\{v_p(a_1), v_p(a_2), \dots, v_p(a_n)\}$;

4) $v_p(lcm(a_1, a_2, \dots, a_n)) = \max\{v_p(a_1), v_p(a_2), \dots, v_p(a_n)\}$.

If we have to prove that $a|b$, then it is enough to prove that the exponent of any prime number in decomposition of a is at least the exponent of

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that prime in the decomposition of b. Now, let us repeat the above idea in terms of function v_p. We have $a|b$ if and only if for any prime p we have $v_p(a) \leq v_p(b)$. Also, we have $a = b$ if and only if for any prime p, $v_p(a) = v_p(b)$.

For any positive integer n, let $e_p(n)$ be the exponent of prime p in the prime factorization of $n!$.

The arithmetic function e_p is called the *Legendre*[5] *'s function* associated with the prime p and it is connected to function v_p by the relation $e_p(n) = v_p(n!)$.

The following result gives a formula for the computation of $e_p(n)$.

Theorem 6.5.1. (Legendre's formula) *For any prime p and any positive integer n,*

$$e_p(n) = \sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - S_p(n)}{p-1},$$

where $S_p(n)$ is the sum of digits of n when written in base p.

Proof. For $n < p$ it is clear that $e_p(n) = 0$. If $n \geq p$, then in order to determine $e_p(n)$ we need to consider only the multiples of p in the product $1 \cdot 2 \ldots n$, that is $(1 \cdot p)(2 \cdot p) \ldots (kp) = p^k k!$, where $k = \left\lfloor \frac{n}{p} \right\rfloor$. Hence

$$e_p(n) = \left\lfloor \frac{n}{p} \right\rfloor + e_p\left(\left\lfloor \frac{n}{p} \right\rfloor \right).$$

Replacing n by $\left\lfloor \frac{n}{p} \right\rfloor$ and taking into account that

$$\left\lfloor \frac{\left\lfloor \frac{n}{p} \right\rfloor}{p} \right\rfloor = \left\lfloor \frac{n}{p^2} \right\rfloor$$

we obtain

$$e_p\left(\left\lfloor \frac{n}{p} \right\rfloor \right) = \left\lfloor \frac{n}{p^2} \right\rfloor + e_p\left(\left\lfloor \frac{n}{p^2} \right\rfloor \right)$$

Continuing this procedure we get

$$e_p\left(\left\lfloor \frac{n}{p^2} \right\rfloor \right) = \left\lfloor \frac{n}{p^3} \right\rfloor + e_p\left(\left\lfloor \frac{n}{p^3} \right\rfloor \right)$$

[5] *Adrien-Marie Legendre* (1752-1833), French mathematician who was a disciple of Euler and Lagrange. In number theory, he studied the function e_p, and he proved the unsolvability of Fermat's last theorem for $n = 5$.

$$\dots$$

$$e_p\left(\left\lfloor \frac{n}{p^{m-1}} \right\rfloor\right) = \left\lfloor \frac{n}{p^m} \right\rfloor + e_p\left(\left\lfloor \frac{n}{p^m} \right\rfloor\right)$$

where m is the least positive integer such that $n < p^{m+1}$, that is $m = \left\lfloor \frac{\ln n}{\ln p} \right\rfloor$. Summing up the relations above yields

$$e_p(n) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{p^m} \right\rfloor.$$

The other relation is not difficult. Indeed, let us write

$$n = a_0 + a_1 p + \cdots + a_k p^k,$$

where $a_0, a_1, \dots, a_k \in \{0, 1, \dots, p-1\}$ and $a_k \neq 0$. Then

$$\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \cdots = a_1 + a_2 p + \cdots + a_k p^{k-1} + a_2 + a_3 p + \cdots + a_k p^{k-2} + \cdots + a_k,$$

and now using the formula

$$1 + p + \cdots + p^i = \frac{p^{i+1} - 1}{p - 1},$$

we find exactly the second part in expression of $e_p(n)$. □

Examples. 1) Let us find the exponent of 7 in 400!. Applying Legendre's formula, we have

$$e_7(400) = \left\lfloor \frac{400}{7} \right\rfloor + \left\lfloor \frac{400}{7^2} \right\rfloor + \left\lfloor \frac{400}{7^3} \right\rfloor = 57 + 8 + 1 = 66.$$

2) Let us determine the exponent of 3 in ((3!)!)!. We have ((3!)!)! = (6!)! = 720!. Applying Legendre's formula yields

$$e_3(720) = \left\lfloor \frac{720}{3} \right\rfloor + \left\lfloor \frac{720}{3^2} \right\rfloor + \left\lfloor \frac{720}{3^3} \right\rfloor + \left\lfloor \frac{720}{3^4} \right\rfloor + \left\lfloor \frac{720}{3^5} \right\rfloor$$

$$= 240 + 80 + 26 + 8 + 2 = 356.$$

Problem 6.5.1. *Let p be a prime. Find the exponent of p in the prime factorization of $(p^m)!$.*

Solution. Using Legendre's formula, we have

$$e_p(p^m) = \sum_{i \geq 1} \left\lfloor \frac{p^m}{p^i} \right\rfloor = p^{m-1} + p^{m-2} + \cdots + p + 1 = \frac{p^m - 1}{p - 1}$$

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Problem 6.5.2. *Find all positive integers n such that $n!$ ends in exactly 1000 zeros.*

Solution. There are clearly more 2's than 5's in the prime factorization of $n!$, hence it suffices to solve the equation

$$\left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{5^2} \right\rfloor + \cdots = 1000.$$

But

$$\left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{5^2} \right\rfloor + \cdots < \frac{n}{5} + \frac{n}{5^2} + \cdots = \frac{n}{5}\left(1 + \frac{1}{5} + \dots\right)$$

$$= \frac{n}{5} \cdot \frac{1}{1 - \frac{1}{5}} = \frac{n}{4},$$

hence $n > 4000$.

On the other hand, using the inequality $\lfloor a \rfloor > a - 1$, we have

$$1000 > \left(\frac{n}{5} - 1\right) + \left(\frac{n}{5^2} - 1\right) + \left(\frac{n}{5^3} - 1\right) + \left(\frac{n}{5^4} - 1\right) + \left(\frac{n}{5^5} - 1\right)$$

$$= \frac{n}{5}\left(1 + \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \frac{1}{5^4}\right) - 5 = \frac{n}{5} \cdot \frac{1 - \left(\frac{1}{5}\right)^5}{1 - \frac{1}{5}} - 5,$$

so

$$n < \frac{1005 \cdot 4 \cdot 3125}{3124} < 4022.$$

We narrowed n down to $\{4001, 4002, \dots, 4021\}$. Using Legendre's formula we find that 4005 is the first positive integer with the desired property and that 4009 is the last. Hence $n = 4005$, 4006, 4007, 4008, 4009.

Problem 6.5.3. *Prove that for any positive integer n, 2^n does not divide $n!$.*

Solution. The exponent of 2 in the prime factorization of $n!$ is

$$k = e_2(n) = \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{2^2} \right\rfloor + \dots$$

We have

$$k < \frac{n}{2} + \frac{n}{2^2} + \cdots = \frac{n}{2}\left(1 + \frac{1}{2} + \dots\right) = \frac{n}{2} \cdot \frac{1}{1 - \frac{1}{2}} = n$$

and we are done.

Remark. Similarly, for any prime p, p^n does not divide $((p-1)n)!$.

Problem 6.5.4. *Find all positive integers n such that 2^{n-1} divides $n!$.*

Solution. If $n = 2^s$, $s = 0, 1, 2, \ldots$, then

$$e_2(n) = 2^{s-1} + \cdots + 2 + 1 = 2^s - 1,$$

hence 2^{n-1} divides $n!$.

Assume that n is odd, $n = 2n_1 + 1$. Then from $2^{n-1} = 2^{2n_1} | (2n_1 + 1)! = (2n_1)!(2n_1+1)$ it follows $2^{2n_1} | (2n_1)!$ which is not possible by Problem 6.5.3. We get $n = 2m_1$. If m_1 is odd, $m_1 = 2n_2 + 1$, we have

$$2^{n-1} = 2^{4n_2+1} | (4n_2 + 2)! = (4n_2)!(4n_2 + 1) \cdot 2 \cdot (2n_2 + 1)$$

and we obtain $2^{4n_2} | (4n_2)!$, a contradiction. Continuing this procedure we get $n = 2^s$.

Problem 6.5.5. *Let p be an odd prime. Prove that the exponent of p in the prime factorization of $1 \cdot 3 \cdot 5 \ldots (2m + 1)$ is*

$$\sum_{k \geq 1} \left(\left\lfloor \frac{2m+1}{p^k} \right\rfloor - \left\lfloor \frac{m}{p^k} \right\rfloor \right).$$

Solution. We have

$$1 \cdot 3 \cdot 5 \ldots (2m + 1) = \frac{(2m + 1)!}{m! \cdot 2^m}.$$

Because p is odd, the desired exponent is

$$e_p(2m + 1) - e_p(m) = \sum_{k \geq 1} \left\lfloor \frac{2m+1}{p^k} \right\rfloor - \sum_{k \geq 1} \left\lfloor \frac{m}{p^k} \right\rfloor$$

and the conclusion follows.

Problem 6.5.6. *If p is a prime and $p^\alpha | \binom{n}{m}$, then $p^\alpha \leq n$.*

Solution. Because

$$\binom{n}{m} = \frac{n!}{m!(n-m)!},$$

the exponent of p in the prime factorization of $\binom{n}{m}$ is

$$\beta = e_p(n) - e_p(m) - e_p(n - m) = \sum_{k \geq 1} \left(\left\lfloor \frac{n}{p^k} \right\rfloor - \left\lfloor \frac{m}{p^k} \right\rfloor - \left\lfloor \frac{n-m}{p^k} \right\rfloor \right).$$

This sum has at most s nonzero terms, where $p^s \leq n < p^{s+1}$. Using the inequality $\lfloor x + y \rfloor - \lfloor x \rfloor - \lfloor y \rfloor \leq 1$ for $x = \dfrac{m}{p^k}$ and $y = \dfrac{n-m}{p^k}$, it follows that $\beta \leq s$. Because $p^\alpha | \binom{n}{m}$, we obtain $\alpha \leq \beta \leq s$, hence $p^\alpha \leq p^s \leq n$.

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Proposed problems

Problem 6.5.7. a) If p is a prime, prove that for any positive integer n,

$$-\left\lfloor \frac{\ln n}{\ln p} \right\rfloor + n \sum_{k=1}^{\left\lfloor \frac{\ln n}{\ln p} \right\rfloor} \frac{1}{p^k} < e_p(n) < \frac{n}{p-1}.$$

b) Prove that

$$\lim_{n\to\infty} \frac{e_p(n)}{n} = \frac{1}{p-1}.$$

Problem 6.5.8. Show that for all nonnegative integers m, n the number

$$\frac{(2m)!(2n)!}{m!n!(m+n)!}$$

is also an integer.

(14th IMO)

Problem 6.5.9. Prove that $\dfrac{(3a+3b)!(2a)!(3b)!(2b)!}{(2a+3b)!(a+2b)!(a+b)!a!(b!)^2}$ is an integer for any positive integers a, b.

(American Mathematical Monthly)

Problem 6.5.10. Prove that there exists a constant c such that for any positive integers a, b, n that verify $a! \cdot b! | n!$ we have $a + b < n + c \ln n$.

(Paul Erdös)

Problem 6.5.11. Prove that the equation

$$\frac{1}{10^n} = \frac{1}{n_1!} + \frac{1}{n_2!} + \cdots + \frac{1}{n_k!}$$

does not have integer solutions such that $1 \leq n_1 < n_2 < \cdots < n_k$.

(Tuymaada Olimpiad)

7

More on Divisibility

7.1 Fermat's Little Theorem

Theorem 7.1.1. (Fermat's Little Theorem) *Let a be a positive integer and let p be a prime. Then*

$$a^p \equiv a \pmod{p}.$$

Proof. We induct on a. For $a = 1$ every thing is clear. Assume that $p|a^p - a$. Then

$$(a+1)^p - (a+1) = (a^p - a) + \sum_{k=1}^{p-1} \binom{p}{k} a^k.$$

Using the fact that $p|\binom{p}{k}$ for $1 \leq k \leq p-1$ and the inductive hypothesis, it follows that $p|(a+1)^p - (a+1)$, that is $(a+1)^p \equiv (a+1) \pmod{p}$.

Alternative proof. Suppose that $gcd(a,p) = 1$ and let us show that $a^{p-1} \equiv 1 \pmod{p}$. Consider the integers $a, 2a, \ldots, (p-1)a$, whose remainders when divided by p are distinct (otherwise, if $ia \equiv ja \pmod{p}$, then $p|(i-j)a$, that is $p|i-j$, which holds only if $i = j$). Hence

$$a \cdot (2a) \ldots (p-1)a \equiv 1 \cdot 2 \ldots (p-1) \pmod{p},$$

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i.e.

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Because p and $(p-1)!$ are relatively prime, the conclusion follows. □

Remark. The converse is not true. For example, $3{\cdot}11{\cdot}17$ divides $a^{3\cdot 11\cdot 17}-a$, since 3, 11, 17 each divides $a^{3\cdot 11\cdot 17}-a$ (for instance, if 11 did not divide a, then from Fermat's Little Theorem, we have $11|a^{10}-1$, hence $11|a^{10\cdot 56}-1$, i.e. $11|a^{561}-a$ and $561 = 3\cdot 11\cdot 17$).

We saw that the composite integers n satisfying $a^n \equiv a \pmod{n}$ for any integer a are called Carmichael's integers. There are also even such integers, for example $n = 2{\cdot}73{\cdot}1103$. For other comments see Remark after Problem 1.6.5.

Problem 7.1.1. *1) Let a be a positive integer. Prove that any prime factor > 2 of a^2+1 is of the form $4m+1$.*

2) Prove that there are infinitely many primes of the form $4m+1$.

Solution. 1) Assume that $p|a^2+1$ and $p = 4m+3$ for some integer m. Then $a^2 \equiv -1 \pmod{p}$ and $a^{p-1} = (a^2)^{2m+1} \equiv (-1)^{2m+1} \equiv -1 \pmod{p}$, contradicting Fermat's Little Theorem.

2) The integer $(n!)^2+1$ is of the form $4m+1$, hence all its prime factors are of this form. It follows that for any prime p of the form $4m+1$, $(p!)^2+1$ is a prime or has a prime factor $p_1 > p$ and we are done.

Problem 7.1.2. *For any prime p, $p^{p+1}+(p+1)^p$ is not a perfect square.*

Solution. For $p = 2$ the property holds. Assume by way of contradiction that $p \geq 3$ and $p^{p+1}+(p+1)^p = t^2$ for some positive integer t. It follows that $(t+p^{\frac{p+1}{2}})(t-p^{\frac{p+1}{2}}) = (p+1)^p$, hence $t \pm p^{\frac{p+1}{2}} = 2^{p-1}u^p$ and $t \mp p^{\frac{p+1}{2}} = 2v^p$, for some positive integers u, v such that $2uv = p+1$ and $gcd(u,v) = 1$. We obtain $p^{\frac{p+1}{2}} = |2^{p-2}u^p - v^p|$. Using Fermat's Little Theorem we have $u^p \equiv u \pmod{p}$, $v^p \equiv v \pmod{p}$ and $2^{p-1} \equiv 1 \pmod{p}$, so $u \equiv 2v \pmod{p}$. From $2uv = p+1$ we get $u = 2v$ and finally $v = 1$ and $p = 3$. This leads to $t^2 = 145$, a contradiction.

Problem 7.1.3. *Let $n \geq 2$, $a > 0$ be integers and p a prime such that $a^p \equiv 1 \pmod{p^n}$. Show that if $p > 2$, then $a \equiv 1 \pmod{p^{n-1}}$, and if $p = 2$, then $a \equiv \pm 1 \pmod{2^{n-1}}$.*

(1995 UNESCO Mathematical Contest)

Solution. We have $a^p \equiv 1 \pmod{p}^n$ with $n \geq 2$, so $a^p \equiv 1 \pmod{p}$. But, from Fermat's Little Theorem, $a^p \equiv a \pmod{p}$, hence $a \equiv 1 \pmod{p}$. For $a = 1$, the result is obvious; otherwise, put $a = 1 + kp^d$, where $d \geq 1$

and $p \nmid k$. Then for $p > 2$, $a^p = 1 + kp^{d+1} + Mp^{2d+1}$ for M an integer. Therefore $d + 1 \geq n$ and so $s \equiv 1 \pmod{p^{n-1}}$. In case $p = 2$, we have $2^n | a^2 - 1 = (a-1)(a+1)$. Since these differ by 2, both cannot be multiples of 4. Hence either $a+1$ or $a-1$ is divisible by 2^{n-1}, i.e. $a \equiv \pm 1 \pmod{2^{n-1}}$, as desired.

Problem 7.1.4. *Find the smallest integer n such that among any n integers, there exist 18 integers whose sum is divisible by 18.*

(1997 Ukrainean Mathematical Olympiad)

Solution. The minimum is $n = 35$; the 34-element set of 17 zeroes and 17 ones shows that $n \geq 35$, so it remains to show that among 35 integers, there are 18 whose sum is divisible by 18. In fact, one can show that for any n, among $2n - 1$ integers there are n whose sum is divisible by n.

We show this claim by induction on n; it's clear for $n = 1$. If n is composite, say $n = pq$, we can assemble sets of p integers whose sum is divisible by p as long as at least $2p - 1$ numbers remain; this gives $2q - 1$ sets, and again by the induction hypothesis, some q of these have sum divisible by q.

Now suppose $n = p$ is prime. The number x is divisible by p if and only if $x^{p-1} \not\equiv 1 \pmod{p}$. Thus if the claim is false, then the sum of $(a_1 + \cdots + a_p)^{p-1}$ over all subsets $\{a_1, \ldots, a_p\}$ of the given numbers is congruent to $\binom{2p-1}{p-1} \equiv 1 \pmod{p}$. On the other hand, the sum of $a_1^{e_1} \ldots a_p^{e_p}$ for $e_1 + \cdots + e_p \leq p - 1$ is always divisible by p: if $k \leq p - 1$ of the e_i are nonzero, then each product is repeated $\binom{2p-1-k}{p-k}$ times, and the latter is a multiple of p. This contradiction shows that the claim holds in this case. (Note: to solve the original problem, of course it suffices to prove the cases $p = 2, 3$ directly).

Remark. The fact that for any n, among $2n - 1$ integers there are n whose sum is divisible by n is a famous theorem of Erdös and Ginzburg.

Problem 7.1.5. *Several integers are given (some of them may be equal) whose sum is equal to 1492. Decide whether the sum of their seventh powers can equal*

(a) 1996;

(b) 1998.

(1997 Czech-Slovak Match)

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Solution. (a) Consider a set of 1492 1's, 4 2's, and 8 -1's. Their sum is 1492, and the sum of their seventh powers is $1482(1) + 4(128) + 8(-1) = 1996$.

(b) By Fermat's Little Theorem, $x^7 \equiv x \pmod 7$. Thus, the sum of the numbers' seventh powers must be congruent to the sum of the numbers, modulo 7. But $1998 \not\equiv 1492 \pmod 7$, so the numbers' seventh powers cannot add up to 1998.

Problem 7.1.6. *Find the number of integers $n > 1$ for which the number $a^{25} - a$ is divisible by n for each integer a.*

(1995 Bulgarian Mathematical Olympiad)

Solution. Let n have the required property. Then p^2 (p a prime) does not divide n since p^2 does not divide $p^{25} - p$. Hence n is the multiple of different prime numbers. On the other hand $2^{25} - 2 = 2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241$. But n is not divisible by 17 and 241 since $3^{25} \equiv -3 \pmod{17}$ and $3^{25} \equiv 32 \pmod{241}$. The Fermat Theorem implies that $a^{25} \equiv a \pmod p$ when $p = 2, 3, 5, 7, 13$. Thus n should be equal to the divisors of $2 \cdot 3 \cdot 5 \cdot 7 \cdot 13$, which are different from 1 and there are $2^5 - 1 = 31$ of them.

Problem 7.1.7. *a) Find all positive integers n such that 7 divides $2^n - 1$.*

b) Prove that for any positive integer n the number $2^n + 1$ cannot be divisible by 7.

(6th IMO)

Solution. Fermat's Little Theorem gives:

$$2^6 \equiv 1 \pmod 7.$$

It follows from the divisibility $7|(2^3 - 1)(2^3 + 1)$ that $2^3 \equiv 1 \pmod 7$. Hence all numbers n which are divisible by 3 answer to the question.

Let $n = 3k + r$ where $r = 1$ or $r = 2$. Then

$$2^n \equiv 2^{3k+r} \equiv (2^3)^k \cdot 2^r = 2 \text{ or } 4 \pmod 7.$$

Hence, we cannot obtain $2^n \equiv -1 \pmod 7$.

Problem 7.1.8. *Prove that the following are equivalent.*

(a) For any positive integer a, n divides $a^n - a$.

(b) For any prime divisor p of n, p^2 does not divides n and $p - 1$ divides $n - 1$.

(1995 Turkish Mathematical Olympiad)

Solution. First assume (a). If $p^2|n$ for some prime p, we must have $p^2|(p+1)^{p^2}-(p+1)$. However,

$$(p+1)^{p^2}-(p+1)=p^2-p+\sum_{k=2}^{p^2}\binom{p^2}{k}p^k.$$

All terms but the first are divisible by p^2, contradicting the assumption. Therefore $p^2 \nmid n$. Moreover, if a is a primitive root modulo p, then $a^{n-1}\equiv 1 \pmod{p}$ implies $p-1|n-1$.

On the other hand, if n is square-free and $p-1|n-1$ for all primes $p|n$, then for any a, either $p|a$ or $a^{p-1}\equiv 1 \pmod{p}$; in either case $a^n \equiv a \pmod{p}$ for all p dividing n. Hence the conditions are equivalent.

Problem 7.1.9. *Prove that the sequence* $\{2^n-3|n=2,3,\dots\}$ *contains infinitely many pairs of relatively prime numbers.*

(13th IMO)

Solution. We use the induction. The numbers 2^2-3, 2^3-3, 2^4-3 are pairwise relatively prime numbers. We shall prove that if $n_1, n_2, \dots, n_k$ are positive integers such that the members of the sequence

$$2^{n_1}-3, 2^{n_2}-3, \dots, 2^{n_k}-3 \tag{1}$$

are relatively prime to each other, then there exists n_{k+1} such that $2^{n_{k+1}}-3$ is relatively prime to each number of the sequence (1).

Let $\{p_1, p_2, \dots, p_r\}$ be the set of all prime divisors of numbers from the sequence (1). Then $p_1, p_2, \dots, p_r$ are odd prime numbers and by Fermat's Little Theorem

$$2^{p_i-1}\equiv 1 \pmod{p_i}.$$

It follows that

$$2^{(p_1-1)(p_2-1)\dots(p_r-1)}\equiv 1 \pmod{p_i},\ \forall\ i=1,\dots,r.$$

Let $n_{k+1}=\prod_{i=1}^{r}(p_i-1)$. We shall prove that $2^{n_i}-3$ and $2^{n_{k+1}}-3$, are relatively prime, for all $i=1,\dots,r$. Let p be a common prime divisor of $2^{n_i}-3$ and $2^{n_{k+1}}-3$. Then $2^{n_{k+1}}-3\equiv 1-3 \pmod{p} \equiv 0 \pmod{p}$; this is a contradiction.

Problem 7.1.10. *Let* $p>2$ *be a prime number such that* $3|(p-2)$. *Let*

$$S=\{y^2-x^3-1|\ x \text{ and } y \text{ are integers},\ 0\le x,y\le p-1\}.$$

Prove that at most p *elements of* S *are divisible by* p.

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(1999 Balkan Mathematical Olympiad)

Solution. We need the following

Lemma. *Given a prime p and a positive integer $k > 1$, if k and $p-1$ are relatively prime then $x^k \equiv y^k \pmod{p} \Rightarrow x \equiv y \pmod{p}$ for all x, y.*

Proof. If $y \equiv 0 \pmod{p}$ the claim is obvious. Otherwise, note that $x^k \equiv y^k \Rightarrow (xy^{-1})^k \equiv 1 \pmod{p}$, so it suffices to prove that $a^k \equiv 1 \pmod{p} \Rightarrow a \equiv 1 \pmod{p}$.

Because $gcd(p-1, k) = 1$, there exist integers b and c such that $b(p-1) + ck = 1$. Thus, $a^k \equiv 1 \pmod{p} \Rightarrow a^c \equiv 1 \pmod{p} \Rightarrow a^{1-b(p-1)} \equiv 1 \pmod{p}$. If $a = 0$ this is impossible. Otherwise, by Fermat's Little Theorem, $(a^{-b})^{p-1} \equiv 1 \pmod{p}$ so that $a \equiv 1 \pmod{p}$, as desired.

Alternatively, again note that clearly $a \not\equiv 0 \pmod{p}$. Then let d be the order of a, the smallest positive integer such that $a^d \equiv 1 \pmod{p}$; we have $d|k$. Take the set $\{1, a, a^2, \ldots, a^{d-1}\}$. If it does not contain all of $1, 2, \ldots, p-1$ then pick some other element b and consider the set $\{b, ba, ba^2, \ldots, ba^{d-1}\}$. These two sets are disjoint, because otherwise $ba^i \equiv a^j \Rightarrow b \equiv a^{j-1} \pmod{p}$, a contradiction. Continuing similarly, we can partition $\{1, 2, \ldots, p-1\}$ into d-element subsets, and hence $d|p-1$. However, $d|k$ and $gcd(k, p-1) = 1$, implying that $d = 1$. Therefore $a \equiv a^d \equiv 1 \pmod{p}$, as desired. □

Because $3|p-2$, $gcd(3, p-1) = 1$. Then from the claim, it follows that the set of elements $\{1^3, 2^3, \ldots, p^3\}$ equals $\{1, 2, \ldots, p\}$ modulo p. Hence, for each y with $0 \le y \le p-1$, there is exactly one x between 0 and $p-1$ such that $x^3 \equiv y^2 - 1 \pmod{p}$: that is, such that $p|y^2 - x^3 - 1$. Therefore S contains at most p elements divisible by p, as desired.

Proposed problems

Problem 7.1.11. Let $3^n - 2^n$ be a power of a prime for some positive integer n. Prove that n is a prime.

Problem 7.1.12. Let $f(x_1, \ldots, x_n)$ be a polynomial with integer coefficients of total degree less than n. Show that the number of ordered n-tuples $(x_1, \ldots, x_n)$ with $0 \le x_i \le 12$ such that $f(x_1, \ldots, x_n) \equiv 0 \pmod{13}$ is divisible by 13.

(1998 Turkish Mathematical Olympiad)

Problem 7.1.13. Find all pairs (m, n) of positive integers, with $m, n \ge 2$, such that $a^n - 1$ is divisible by m for each $a \in \{1, 2, \ldots, n\}$.

(2001 Romanian IMO Team Selection Test)

Problem 7.1.14. Let p be a prime and b_0 an integer, $0 < b_0 < p$. Prove that there exists a unique sequence of base p digits $b_0, b_1, b_2, \ldots, b_n, \ldots$ with the following property: If the base p representation of a number x ends in the group of digits $b_n b_{n-1} \ldots b_1 b_0$ then so does the representation of x^p.

Problem 7.1.15. Determine all integers $n > 1$ such that $\dfrac{2^n + 1}{n^2}$ is an integer.

(31st IMO)

Problem 7.1.16. Let p be a prime number. Prove that there exists a prime number q such that for every integer n, the number $n^p - p$ is not divisible by q.

(44th IMO)

Problem 7.1.17. Prove that for any $n > 1$ we cannot have $n | 2^{n-1} + 1$.

(Sierpinski)

Problem 7.1.18. Prove that for any natural number n, $n!$ is a divisor of

$$\prod_{k=0}^{n-1} (2^n - 2^k).$$

7.2 Euler's Theorem

Theorem 7.2.1. (Euler's Theorem) *Let a and n be relatively prime positive integers. Then $a^{\varphi(n)} \equiv 1 \pmod{n}$.*

Proof. Consider the set $S = \{a_1, a_2, \ldots, a_{\varphi(n)}\}$ consisting of all positive integer less than n that are relatively prime to n. Because $gcd(a, n) = 1$, it follows that $aa_1, aa_2, \ldots, aa_{\varphi(n)}$ is a permutation of $a_1, a_2, \ldots, a_{\varphi(n)}$. Then

$$(aa_1)(aa_2)\ldots(aa_{\varphi(n)}) \equiv a_1 a_2 \ldots a_{\varphi(n)} \pmod{n}.$$

Using that $gcd(a_k, n) = 1$, $k = 1, 2, \ldots, \varphi(n)$, the conclusion now follows. □

Remark. The Euler's Theorem also follows from Fermat's Little Theorem. Indeed, let $n = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ be the prime factorization of n. We have $a^{p_i - 1} \equiv 1 \pmod{p_i}$, hence $a^{p_i(p_1 - 1)} \equiv 1 \pmod{p_i^2}$, $a^{p_i^2(p_i - 1)} \equiv 1$

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$(\text{mod } p_i^3), \ldots, a^{p_i^{\alpha_i-1}(p_i-1)} \equiv 1 \ (\text{mod } p_i^{\alpha_i})$. That is $a^{\varphi(p_i^{\alpha_i})} \equiv 1 \ (\text{mod } p_i^{\alpha_i})$, $i = 1, \ldots, k$. Applying this property for each prime factor, the conclusion follows.

Problem 7.2.1. *Prove that for any positive integer s, there is a positive integer n whose sum of digits is s and $s|n$.*

(Sierpinski[1])

Solution. Let $n = 10^{s\varphi(s)} + 10^{(s-1)\varphi(s)} + \cdots + 10^{\varphi(s)}$. It is clear that the sum of digits of n is s and that

$$n = (10^{s\varphi(s)} - 1) + (10^{(s-1)\varphi(s)} - 1) + \cdots + (10^{\varphi(s)} - 1) + s$$

is divisible by s, by Euler's Theorem.

Problem 7.2.2. *Let $n > 3$ be an odd integer with prime factorization $n = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ (each p_i is prime). If*

$$m = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\ldots\left(1 - \frac{1}{p_k}\right),$$

prove that there is a prime p such that p divides $2^m - 1$, but does not divide m.

(1995 Iranian Mathematical Olympiad)

Solution. Because $m = \varphi(n)$ is Euler's phi-function and n is odd, we know by Euler Theorem that n divides $2^m - 1$. We consider two cases.

First let $n = p^r > 3$ for some odd prime p. Then $m = p^r - p^{r-1}$ is even and $m \geq 4$. Since p divides

$$2^m - 1 = (2^{m/2} - 1)(2^{m/2} + 1),$$

is must also divide one of the factors on the right. Any prime divisor of the other factor (note this factor exceeds 1) will also divide $2^m - 1$ but will not divide $n = p^r$.

If n has at least two distinct prime factors, then $m \equiv 0 \ (\text{mod } 4)$ and $p - 1$ divides $m/2$ for each prime factor of n. Hence, by Fermat's Theorem, p also divides $2^{m/2} - 1$. It follows that no prime factor of n divides $2^{m/2} + 1$. Hence any prime factor of $2^{m/2} + 1$ is a factor of $2^m - 1$ but not a factor of n.

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[1] *Waclaw Sierpinski* (1882-1969), Polish mathematician with important work in the area of set theory, point set topology and number theory.

Problem 7.2.3. *Let $a > 1$ be an integer. Show that the set*

$$\{a^2 + a - 1, a^3 + a^2 - 1, \dots\}$$

contains an infinite subset, any two members of which are relatively prime.

(1997 Romanian IMO Team Selection Test)

Solution. We show that any set of n elements of the set which are pairwise coprime can be extended to a set of $n + 1$ elements. For $n = 1$, note that any two consecutive terms in the sequence are relatively prime. For $n > 1$, let N be the product of the numbers in the set so far; then $a^{\varphi(N)+1} + a^{\varphi(N)} - 1 \equiv a \pmod{N}$, and so can be added (since every element of the sequence is coprime to a, N is as well).

Problem 7.2.4. *Let X be the set of integers of the form*

$$a_{2k}10^{2k} + a_{2k-2}10^{2k-2} + \cdots + a_2 10^2 + a_0,$$

where k is a nonnegative integer and $a_{2i} \in \{1, 2, \dots, 9\}$ for $i = 0, 1, \dots, k$. Show that every integer of the form $2^p 3^q$, for p and q nonnegative integers, divides some element of X.

(1997 Taiwanese Mathematical Olympiad)

Solution. In fact, every integer that is not divisible by 10 divides some element of X. We first note that there exists a multiple of 4^p in X with $2p - 1$ digits for all nonnegative integer p. This follows by induction on p: it's obvious for $p = 0, 1$, and if x is such a multiple for $p = k$, then we can choose a_{2k} so that $x + a_{2k}10^{2k} \equiv 0 \pmod{4^{k+1}}$ since $10^{2k} \equiv 0 \pmod{4^k}$.

Now we proceed to show that any integer n that isn't divisible by 10 divides some element of X. Let $n = 2^p k$, where k is odd. Then by the lemma above one can find a multiple of 2^p in X. Let m be the multiple, d be the number of digits of m, and $f = 10^{d+1} - 1$. By Euler's Theorem, $10^{\varphi(fk)} \equiv 1 \pmod{fk}$. Therefore $m(10^{(d+1)\varphi(fk)} - 1)/(10^{d+1} - 1)$ is divisible by $2^p k$ and lies in X (since it is the concatenation of m's).

Proposed problems

Problem 7.2.5. Prove that, for every positive integer n, there exists a polynomial with integer coefficients whose values at $1, 2, \dots, n$ are different powers of 2.

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(1999 Hungarian Mathematical Olympiad)

Problem 7.2.6. Let $a > 1$ be an odd positive integer. Find the least positive integer n such that 2^{2000} is a divisor of $a^n - 1$.

(2000 Romanian IMO Team Selection Test)

Problem 7.2.7. Let $n = p_1^{r_1} \dots p_k^{r_k}$ be the prime factorization of the positive integer n and let $r \geq 2$ be an integer. Prove that the following are equivalent:

(a) The equation $x^r \equiv a \pmod{n}$ has a solution for every a.

(b) $r_1 = r_2 = \cdots = r_k = 1$ and $(p_i - 1, r) = 1$ for every $i \in \{1, 2, \dots, k\}$.

(1995 UNESCO Mathematical Contest)

7.3 The order of an element

Given are the positive integer $n > 1$ and the integer a such that $gcd(a, n) = 1$, the smallest possible positive integer d for which $n|a^d - 1$ is called the *order of a modulo n*. Observe first of all that the definition is connect, since from Euler's theorem we have $n|a^{\varphi(n)} - 1$, so such numbers d indeed exist. In what follows we will denote by $o_n(a)$ the order of a modulo n. The following properties hold:

1) If $a^m \equiv 1 \pmod{n}$, then $o_n(a)|m$;

2) $o_n(a)|\varphi(n)$;

3) If $a^l \equiv a^m \pmod{n}$, then $l \equiv m \pmod{o_n(a)}$.

In order to prove property 1) let us consider $d = o_n(a)$. Indeed, because $n|a^m - 1$ and $n|a^d - 1$, we find that $n|a^{gcd(m,d)} - 1$. But from the definition of d it follows that $d \leq gcd(m, d)$, which cannot hold unless $d|m$.

The positive integer a is called a *primitive root modulo n* if we have $gcd(a, n) = 1$ and $o_n(a) = \varphi(n)$. One can show that there are primitive roots modulo n if and only if $n \in \{2, 4, p^\alpha, 2p^\alpha\}$, where $p \geq 3$ is any prime and α is any positive integer.

Problem 7.3.1. *Prove that $n|\varphi(a^n - 1)$ for all positive integers a, n.*

(Saint Petersburg Mathematical Olympiad)

Solution. What is $o_{a^n-1}(a)$? It may seem a silly question, since of course $o_{a^n-1}(a) = n$. Using the observation in the introduction, we obtain exactly $n|\varphi(a^n - 1)$.

Problem 7.3.2. *Prove that any prime factor of the nth Fermat number* $2^{2^n}+1$ *is congruent to 1 modulo* 2^{n+1}*. Show that there are infinitely many prime numbers of the form* $2^n k+1$ *for any fixed* n*.*

Solution. Let us consider a prime p such that $p|2^{2^n}+1$. Then $p|2^{2^{n+1}}-1$ and consequently $o_p(2)|2^{n+1}$. This ensures the existence of a positive integer $k \leq n+1$ such that $o_p(2)+2^k$. We will prove that in fact $k=n+1$. The proof is easy. Indeed, if this is not the case, then $o_p(2)|2^n$ and so $p|2^{o_p(2)}-1|2^{2^n}-1$. But this is impossible, since $p|2^{2^n}+1$. Therefore, we have found that $o_p(2)=2^{n+1}$ and we have to prove that $o_p(2)|p-1$ to finish the first part of the question. But this follows from the introduction.

The second part is a direct consequence of the first. Indeed, it is enough to prove that there exists an infinite set of Fermat's numbers $(2^{2^{n_k}}+1)_{n_k>a}$ any two relatively prime. Then we could take a prime factor of each such Fermat's number and apply the first part to obtain that each such prime is of the form $2^n k+1$. But not only it is easy to find such a sequence of Fermat's coprime numbers, but in fact any two different Fermat's numbers are relatively prime. Indeed, suppose that $d|gcd(2^{2^n}+1, 2^{2^{n+k}}+1)$. Then $d|2^{2^{n+1}}-1$ and so $d|2^{2^{n+k}}-1$. Combining this with $d|2^{2^{n+k}}+1$, we obtain a contradiction. Hence both parts of the problem are solved.

Problem 7.3.3. *For a prime* p*, let* $f_p(x)=x^{p-1}+x^{p-2}+\cdots+x+1$*.*

a) If $p|m$*, prove that there exists a prime factor of* $f_p(m)$ *that is relatively prime with* $m(m-1)$*.*

b) Prove that there are infinitely many numbers n *such that* $pn+1$ *is prime.*

(2003 Korean IMO Team Selection Test)

Solution.

a) is straightforward. In fact, we will prove that any prime factor of $f_p(m)$ is relatively prime with $m(m-1)$. Take such a prime divisor q. Because $q|1+m+\cdots+m^{p-1}$, it is clear that $gcd(q,m)=1$. Moreover, if $gcd(q,m-1)\neq 1$, then $q|m-1$ and because $q|1+m+\cdots+m^{p-1}$, it follows that $q|p$. But $p|m$ and we find that $q|m$, which is clearly impossible.

More difficult is b). But we are tempted to use a) and to explore the properties of $f_p(m)$, just like in the previous problem. So, let us take a prime $q|f_p(m)$ for a certain positive integer m divisible by p. Then we have of course $q|m^p-1$. But this implies that $o_q(m)|q$ and consequently $o_q(m)\in\{1,p\}$. If $o_q(m)=p$, then $q\equiv 1 \pmod p$. Otherwise, $q|m-$

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1 and because $q|f_p(m)$, we deduce that $q|p$, hence $q = p$. But we have seen while solving a) that this is not possible, so the only choice is $p|q-1$. Now, we need to find a sequence $(m_k)_{k\geq 1}$ of multiples of p such that $f_p(m_k)$ are pairwise relatively prime. This is not as easy as in the first example. Anyway, just by trial and error, it is not difficult to find such a sequence. There are many other approaches, but we like the following one: take $m_1 = p$ and $m_k = pf(m_1)f_p(m_2)\dots f_p(m_{k-1})$. Let us prove that $f_p(m_k)$ is relatively prime to $f_p(m_1), f_p(m_2), \dots, f_p(m_{k-1})$. Fortunately, this is easy, since $f_p(m_1)f_p(m_2)\dots f_p(m_{k-1})|f_p(m_k) - f_p(0)|f_p(m_k) - 1$. The solution ends here.

Problem 7.3.4. *Find the smallest number n with the property that*

$$2^{2005}|17^n - 1.$$

Solution. The problem actually asks for $o_{2^{2005}}(17)$. We know that $o_{2^{2005}}(17)|\varphi(2^{2005}) = 2^{2004}$, so $o_{2^{2005}}(17) = 2^k$, where $k \in \{1, 2, \dots, 2004\}$. The order of an element has done its job. Now, it is time to work with exponents. We have $2^{2005}|17^{2^k} - 1$. Using the factoring

$$17^{2^k} - 1 = (17-1)(17+1)(17^2+1)\dots(17^{2^{k-1}}+1),$$

we proceed by finding the exponent of 2 in each factor of this product. But this is not difficult, because for all $i \geq 0$ the number $17^{2^t} + 1$ is a multiple of 2, but not a multiple of 4. Thus, $v_2(17^{2^k} - 1) = 4 + k$ and the order is found by solving the equation $k + 4 = 2005$. Thus, $o_{2^{2005}}(17) = 2^{2001}$ is the answer to the problem.

Problem 7.3.5. *Find all prime numbers p, q such that $p^2 + 1|2003^q + 1$ and $q^2 + 1|2003^p + 1$.*

Solution. Let us suppose that $p \leq q$. We discuss first the trivial case $p = 2$. In this case, $5|2003^q + 1$ and it is easy to deduce that q is even, hence $q = 2$, which is a solution of the problem. Now, suppose that $p > 2$ and let r be a prime factor of $p^2 + 1$. Because $r|2003^{2q} - 1$, it follows that $o_r(2003)|2q$. Suppose that $(q, o_r(2003)) = 1$. Then $o_r(2003)|2$ and $r|2003^2 - 1 = 2^3 \cdot 3 \cdot 7 \cdot 11 \cdot 13 \cdot 167$. It seems that this is a dead end, since there are too many possible values for r. Another simple observation narrows the number of possible cases: because $r|p^2+1$, must be of the form $4k+1$ or equal to 2 and now we do not have many possibilities: $r \in \{2, 13\}$. The case $r = 13$ is also impossible, because $2003^q + 1 \equiv 2 \pmod{13}$ and

$r|2003^q+1$. So, we have found that for any prime factor r of p^2+1, we have either $r=2$ or $q|o_r(2003)$, which in turn implies $q|r-1$. Because p^2+1 is even, but not divisible by 4 and because any odd prime factor of it is congruent to 1 modulo q, we must have $p^2+1 \equiv 2 \pmod{q}$. This implies that $p^2+1 \equiv 2 \pmod{q}$, that is $q|(p-1)(p+1)$. Combining this with the assumption that $p \leq q$ yields $q|p+1$ and in fact $q=p+1$. It follows that $p=2$, contradicting the assumption $p>2$. Therefore the only pair is (2,2).

Proposed problems

Problem 7.3.6. Find all ordered triples of primes (p,q,r) such that

$$p|q^r+1, q|r^p+1, r|p^q+1.$$

(2003 USA IMO Team Selection Test)

Problem 7.3.7. Find all primes p,q such that $pq|2^p+2^q$.

Problem 7.3.8. Prove that for any positive integer n, 3^n-2^n is not divisible by n.

Problem 7.3.9. Find all positive integers m,n such that $n|1+m^{3^n}+m^{2\cdot 3^n}$.

(Bulgarian IMO Team Selection Test)

Problem 7.3.10. Let $a,n>2$ be positive integers such that $n|a^{n-1}-1$ and n does not divide any of the numbers a^x-1, where $x<n-1$ and $x|n-1$. Prove that n is a prime number.

Problem 7.3.11. Find all prime numbers p,q for which the congruence

$$\alpha^{3pq} \equiv \alpha \pmod{3pq}$$

holds for all integers α.

(1996 Romanian Mathematical Olympiad)

7.4 Wilson's Theorem

Theorem 7.4.1. (Wilson[2]'s Theorem) *For any prime p, $p|(p-1)!+1$.*

[2] *John Wilson* (1741-1793), English mathematician who published this results without proof. It was first proved by Lagrange in 1773 who showed that the converse is also true.

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Proof. The property holds for $p = 2$ and $p = 3$, so we may assume that $p \geq 5$. Let $S = \{2, 3, \ldots, p-2\}$. For any h in S, the integers $h, 2h, \ldots, (p-1)h$ yield distinct remainders when divided by p. Hence there is a unique $h' \in \{1, 2, \ldots, p-1\}$ such that $hh' \equiv 1 \pmod{p}$. Moreover, $h' \neq 1$ and $h' \neq p-1$, hence $h' \in S$. In addition, $h' \neq h$, otherwise $h^2 \equiv 1 \pmod{p}$, implying $p|h-1$ or $p|h+1$, which is not possible, since $h+1 < p$. It follows that we can group the elements of S in $\dfrac{p-3}{2}$ distinct pairs (h, h') such that $hh' \equiv 1 \pmod{p}$. Multiplying these congruences gives $(p-2)! \equiv 1 \pmod{p}$ and the conclusion follows. □

Remark. The converse is true, that is if $n|(n-1)! + 1$ for an integer $n \geq 2$, then n is a prime. Indeed, if n were equal to $n_1 n_2$ for some integers $n_1, n_2 \geq 2$, we would have $n_1|1 \cdot 2 \ldots n_1 \ldots (n-1)+1$, which is not possible.

Problem 7.4.1. *If p is an odd prime, then the remainder when $(p-1)!$ is divided by $p(p-1)$ is $p-1$.*

Solution. We need to show that $(p-1)! \equiv p-1 \pmod{p(p-1)}$.

From Wilson's Theorem we obtain $(p-1)!-(p-1) \equiv 0 \pmod{p}$. Because $(p-1)! - (p-1) \equiv 0 \pmod{p-1}$ and $gcd(p, p-1) = 1$ we get

$$(p-1)! - (p-1) \equiv 0 \pmod{p(p-1)}.$$

Problem 7.4.2. *Let p be an odd prime and $a_1, a_2, \ldots, a_p$ an arithmetic sequence whose common difference is not divisible by p. Prove that there is an $i \in \{1, 2, \ldots, p\}$ such that $a_i + a_1 a_2 \ldots a_p \equiv 0 \pmod{p^2}$.*

Solution. Note that $a_1, a_2, \ldots, a_p$ give distinct remainders when divided by p. Take i such that $a_i \equiv 0 \pmod{p}$. It follows that

$$\frac{a_1 a_2 \ldots a_p}{a_i} \equiv (p-1)! \pmod{p}.$$

From Wilson's Theorem, we have $(p-1)! \equiv -1 \pmod{p}$ and the conclusion follows.

Problem 7.4.3. *Let a and n be positive integers such that $n \geq 2$ and $gcd(a, n) = 1$. Prove that*

$$a^{n-1} + (n-1)! \equiv 0 \pmod{n}$$

if and only if n is a prime.

Solution. If n is a prime the conclusion follows from Fermat's Little Theorem and Wilson's Theorem.

For the converse, assume by way of contradiction that $n = n_1 n_2$, where $n_1 \geq n_2 \geq 2$.

Because $n|a^{n-1}+(n-1)!$, it follows that $n_1|a^{n-1}+(n-1)!$, that is $n_1|a^{n-1}$, contradicting the hypothesis $gcd(a,n)=1$.

Problem 7.4.4. *If p is an odd prime, then for any positive integer $n<p$,*

$$(n-1)!(p-n)!\equiv(-1)^n \pmod p.$$

Solution. From Wilson's Theorem, $(p-1)!\equiv -1 \pmod p$, hence

$$(n-1)!n(n+1)\dots(p-1)\equiv -1 \pmod p.$$

This is equivalent to

$$(n-1)!(p-(p-n))(p-(p-n-1))\dots(p-1)\equiv -1 \pmod p.$$

But $p-k\equiv -k \pmod p$, $k=1,2,\dots,p-n$, hence

$$(n-1)!(-1)^{p-n}(p-n)!\equiv -1 \pmod p,$$

and taking into account that p is odd, the conclusion follows.

Proposed problems

Problem 7.4.5. Let p be an odd prime. Prove that

$$1^2\cdot 3^2\dots(p-2)^2\equiv(-1)^{\frac{p+1}{2}} \pmod p$$

and

$$2^2\cdot 4^2\dots(p-1)^2\equiv(-1)^{\frac{p+1}{2}} \pmod p.$$

Problem 7.4.6. Show that there do not exist nonnegative integers k and m such that $k!+48=48(k+1)^m$.

(1996 Austrian-Polish Mathematics Competition)

Problem 7.4.7. For each positive integer n, find the greatest common divisor of $n!+1$ and $(n+1)!$.

(1996 Irish Mathematical Olympiad)

Problem 7.4.8. Let $p\geq 3$ be a prime and let σ be a permutation of $\{1,2,\dots,p-1\}$. Prove that there are $i\neq j$ such that $p|i\sigma(i)-j\sigma(j)$.

(1986 Romanian IMO Team Selection Test)

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8

Diophantine Equations

8.1 Linear Diophantine equations

An equation of the form

$$a_1x_1 + \cdots + a_nx_n = b \tag{1}$$

where $a_1, a_2, \ldots, a_n, b$ are fixed integers, is called *linear Diophantine*[1] *equation*. We assume that $n \geq 1$ and that coefficients $a_1, \ldots, a_n$ are all different from zero.

The main result concerning linear Diophantine equations is the following:

Theorem 8.1.1. *The equation (1) is solvable if and only if*

$$gcd(a_1, \ldots, a_n)|b.$$

In case of solvability, all integer solutions to (1) can be expressed in terms of $n-1$ integral parameters.

Proof. Let $d = gcd(a_1, \ldots, a_n)$.

If b is not divisible by d, then (1) is not solvable, since for any integers $x_1, \ldots, x_n$ the left-hand side of (1) is divisible by d and the right-hand side is not.

[1] *Diophantus of Alexandria* (about 200 - about 284), Greek mathematician sometimes known as "the father of algebra" who is the best known for his book "Arithmetica". This had an enormous influence on the development of number theory.

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If $d|b$, then we obtain the equivalent equation

$$a_1'x_1 + \cdots + a_n'x_n = b',$$

where $a_i' = a_i/d$ for $i = 1, \ldots, n$ and $b' = b/d$. Clearly, we have $gcd(a_1', \ldots, a_n') = 1$.

We use induction on the number n of the variables. In the case $n = 1$ the equation has the form $x_1 = b$ or $-x_1 = b$, and thus the unique solution does not depend on any parameter.

Actually, we need to prove that $gcd(x_1, x_2, \ldots, x_n)$ is a linear combination with integer coefficients of $x_1, x_2, \ldots, x_n$. For $n = 2$ this follows from Proposition 1.3.1. Since

$$gcd(x_1, \ldots, x_n) = gcd(gcd(x_1, \ldots, x_{n-1}), x_n),$$

we obtain that $gcd(x_1, \ldots, x_n)$ is a linear combination of x_n and $gcd(x_1, \ldots, x_{n-1})$, thus by induction hypothesis, a linear combination of $x_1, \ldots, x_{n-1}, x_n$. □

Corollary 8.1.2. *Let a_1, a_2 be relatively prime integers. If (x_1^0, x_2^0) is a solution to the equation*

$$a_1x_1 + a_2x_2 = b, \tag{2}$$

then all of its solutions are given by

$$\begin{cases} x_1 = x_1^0 + a_2t \\ x_2 = x_2^0 - a_1t \end{cases} \tag{3}$$

where $t \in \mathbb{Z}$.

Example. *Solve the equation*

$$3x + 4y + 5z = 6.$$

Solution. Working modulo 5 we have $3x + 4y \equiv 1 \pmod 5$, hence

$$3x + 4y = 1 + 5s, \quad s \in \mathbb{Z}.$$

A solution to this equation is $x = -1 + 3s$, $y = 1 - s$. Applying (3) we obtain $x = -1 + 3s + 4t$, $y = 1 - s - 3t$, $t \in \mathbb{Z}$, and substituting back into the original equation yields $z = 1 - s$. Hence all solutions are

$$(x, y, z) = (-1 + 3s + 4t, 1 - s - 3t, 1 - s), \quad s, t \in \mathbb{Z}.$$

Problem 8.1.1. *Solve in nonnegative integers the equation*

$$x + y + z + xyz = xy + yz + zx + 2.$$

Solution. We have

$$xyz - (xy + yz + zx) + x + y + z - 1 = 1,$$

and, consequently,

$$(x - 1)(y - 1)(z - 1) = 1.$$

Because x, y, z are integers, we obtain

$$x - 1 = y - 1 = z - 1 = 1,$$

so $x = y = z = 2$.

Problem 8.1.2. *Find all triples (x, y, z) of integers such that*

$$x^2(y - z) + y^2(z - x) + z^2(x - y) = 2.$$

Solution. The equation is equivalent to

$$(x - y)(x - z)(y - z) = 2.$$

Observe that $(x - y) + (y - z) = x - z$. On the other hand, 2 can be written as a product of three distinct integers in the following ways

i) $2 = (-1) \cdot (-1) \cdot 2$,

ii) $2 = 1 \cdot 1 \cdot 2$,

iii) $2 = (-1) \cdot 1 \cdot (-2)$.

Since in the first case any two factors do not add up to the third, we only have three possibilities:

a) $\begin{cases} x - y = 1 \\ x - z = 2 \\ y - z = 1 \end{cases}$ so $(x, y, z) = (k + 1, k, k - 1)$ for some integer k;

b) $\begin{cases} x - y = -2 \\ x - z = -1 \\ y - z = 1 \end{cases}$ so $(x, y, z) = (k - 1, k + 1, k)$ for some integer k;

c) $\begin{cases} x - y = 1 \\ x - z = -1 \\ y - z = -2. \end{cases}$ so $(x, y, z) = (k, k - 1, k + 1)$ for some integer k.

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Problem 8.1.3. *Let p and q be prime numbers. Find all positive integers x and y such that*

$$\frac{1}{x}+\frac{1}{y}=\frac{1}{pq}.$$

Solution. The equation is equivalent to

$$(x-pq)(y-pq)=p^2q^2.$$

We have the cases:
1) $x-pq=1$, $y-pq=p^2q^2$, so $x=1+pq$, $y=pq(1+pq)$.
2) $x-pq=p$, $y-pq=pq^2$, so $x=p(1+q)$, $y=pq(1+q)$.
3) $x-pq=q$, $y-pq=p^2q$, so $x=q(1+p)$, $y=pq(1+p)$.
4) $x-pq=p^2$, $y-pq=q^2$, so $x=p(p+q)$, $y=q(p+q)$.
5) $x-pq=pq$, $y-pq=pq$, so $x=2pq$, $y=2pq$.
The equation is symmetric, so we have also:
6) $x=pq(1+pq),\ y=1+pq$.
7) $x=pq(1+q),\ y=p(1+q)$.
8) $x=pq(1+p),\ y=q(1+p)$.
9) $x=q(1+q),\ y=p(p+q)$.

Proposed problems

Problem 8.1.4. Solve in integers the equation

$$(x^2+1)(y^2+1)+2(x-y)(1-xy)=4(1+xy).$$

Problem 8.1.5. Determine the side lengths of a right triangle if they are integers and the product of the legs' lengths equals three times the perimeter.

(1999 Romanian Mathematical Olympiad)

Problem 8.1.6. Let a, b and c be positive integers, each two of them being relatively prime. Show that $2abc-ab-bc-ca$ is the largest integer which cannot be expressed in the form $xbc+yca+zab$ where x, y and z are nonnegative integers.

(24th IMO)

8.2 Quadratic Diophantine equations

8.2.1 Pythagorean equation

One of the most celebrated Diophantine equation is the so-called *Pythagorean equation*

$$x^2 + y^2 = z^2 \tag{1}$$

Studied in detail by Pythagoras[2] in connection with the right-angled triangles whose sidelengths are all integers, this equation was known even to the ancient Babylonians.

Note first that if the triple of integers (x_0, y_0, z_0) satisfies the equation (1), then all triples of the form (kx_0, ky_0, kz_0), $k \in \mathbb{Z}$, also satisfy (1). That is why it is sufficient to find solutions (x, y, z) to (1) with $gcd(x, y, z) = 1$. This is equivalent to the fact that x, y, z are pairwise relatively prime.

A solution (x_0, y_0, z_0) to (1) where x_0, y_0, z_0 are pairwise relatively prime is called *primitive solution.*

Theorem 8.2.1. *Any primitive solution (x, y, z) in positive integers to the equation (1) is of the form*

$$x = m^2 - n^2, \quad y = 2mn, \quad z = m^2 + n^2 \tag{2}$$

where m and n are relatively prime positive integers such that $m > n$.

Proof. The integers x and y cannot be both odd, for otherwise

$$z^2 = x^2 + y^2 \equiv 2 \pmod 4,$$

a contradiction. Hence exactly one of the integers x and y is even.

The identity

$$(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$$

shows that the triple given by (2) is indeed a solution to the equation (1) and y is even.

Moreover, if $gcd(x, y, z) = d \geq 2$, then d divides

$$2m^2 = (m^2 + n^2) + (m^2 - n^2)$$

[2] *Pythagoras of Samos* (about 569BC - about 475BC), Greek philosopher who made important developments in mathematics, astronomy, and the theory of music. The theorem now known as Pythagoras's theorem was known to Babylonians 1000 years earlier but he may have been the first to prove it.

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and d divides

$$2n^2 = (m^2 + n^2) - (m^2 - n^2).$$

Since m and n are relatively prime it follows that $d = 2$. Hence $m^2 + n^2$ is even, in contradiction with m odd and n even. It follows that $d = 1$, so the solution (2) is primitive.

Conversely, let (x, y, z) be a primitive solution to (1) with $y = 2a$. Then x and z are odd and consequently the integers $z + x$ and $z - x$ are even. Let $z + x = 2b$ and $z - x = 2c$. We may assume that b and c are relatively prime, for otherwise z and x would have a nontrivial common divisor. On the other hand, $4a^2 = y^2 = z^2 - z^2 = (z + x)(z - x) = 4bc$, i.e. $a^2 = bc$. Since b and c are relatively prime, it follows that $b = m^2$ and $c = n^2$ for some positive integers m and n. We obtain

$$x = b - c = m^2 - n^2, \quad y = 2mn, \quad z = b + c = m^2 + n^2. \qquad \square$$

A triple (x, y, z) of the form (2) is called a *Pythagorean triple*.

In order to list systematically all the primitive solutions to the equation (1), we assign values 2,3,4,... for the number m successively and then for each of these values we take those integers n which are relatively prime to m, less than m and even whenever m is odd.

Here is the table of the first twenty primitive solutions listed according to the above-mentioned rule.

m	n	x	y	z	area
2	1	3	4	5	6
3	2	5	12	13	30
4	1	15	8	17	60
4	3	7	24	25	84
5	2	21	20	29	210
5	4	9	40	41	180
6	1	35	12	37	210
6	5	11	60	61	330
7	2	45	28	53	630
7	4	33	56	65	924

m	n	x	y	z	area
7	6	13	84	85	546
8	1	63	16	65	504
8	3	55	48	73	1320
8	5	39	80	89	1560
8	7	15	112	113	840
9	2	77	36	85	1386
9	4	65	72	97	2340
9	8	17	144	145	1224
10	1	99	20	101	990
10	3	91	60	109	2730

Corollary 8.2.2. *The general integral solution to (1) is given by*

$$x = k(m^2 - n^2), \quad y = 2kmn, \quad z = k(m^2 + n^2), \tag{3}$$

where $k, m, n \in \mathbb{Z}$.

Problem 8.2.1. *Solve the following equation in positive integers*

$$x^2 + y^2 = 1997(x - y).$$

(1998 Bulgarian Mathematical Olympiad)

Solution. The solutions are

$$(x, y) = (170, 145) \text{ or } (1827, 145).$$

We have

$$\begin{gathered} x^2 + y^2 = 1997(x - y) \\ 2(x^2 + y^2) = 2 \times 1997(x - y) \\ x^2 + y^2 + (x^2 + y^2 - 2 \times 1997(x - y)) = 0 \\ (x + y)^2 + ((x - y)^2 - 2 \times 1997(x - y)) = 0 \\ (x + y)^2 + (1997 - x + y)^2 = 1997^2. \end{gathered}$$

Since x and y are positive integers, $0 < x + y < 1997$ and $0 < 1997 - x + y < 1997$. Thus the problem reduces to solving $a^2 + b^2 = 1997^2$ in positive integers. Since 1997 is a prime, $gcd(a, b) = 1$. By Pythagorean substitution, there are positive integers $m > n$ such that $gcd(m, n) = 1$ and

$$1997 = m^2 + n^2, \quad a = 2mn, \quad b = m^2 - n^2.$$

Since $m^2, n^2 \equiv 0, 1, -1 \pmod 5$ and $1997 \equiv 2 \pmod 5$, $m, n \equiv \pm 1 \pmod 5$. Since $m^2, n^2 \equiv 0, 1 \pmod 3$ and $1997 \equiv 2 \pmod 3$, $m, n \equiv \pm 1 \pmod 3$. Therefore $m, n \equiv 1, 4, 11, 14 \pmod{15}$. Since $m > n$, $1997/2 \leq m^2 \leq 1997$. Thus we only need to consider $m = 34, 41, 44$. The only solution is $(m, n) = (34, 29)$. Thus

$$(a, b) = (1972, 315),$$

which leads to our final solutions.

Proposed problems

Problem 8.2.2. Find all Pythagorean triangles whose areas are numerically equal to their perimeters.

Problem 8.2.3. Prove that for every positive integer n there is a positive integer k such that k appears in exactly n nontrivial Pythagorean triples.

(American Mathematical Monthly)

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8.2.2 Pell's equation

A special quadratic equation is

$$u^2 - Dv^2 = 1 \tag{1}$$

where D is a positive integer that is not a perfect square. Equation (1) is called *Pell*[3] *'s equation* and it has numerous applications in various fields of mathematics. We will present an elementary approach to solving this equation, due to Lagrange.

Theorem 8.2.3. *If D is a positive integer that is not a perfect square, then the equation (1) has infinitely many solutions in positive integers and the general solution is given by $(u_n, v_n)_{n\geq 1}$,*

$$u_{n+1} = u_0u_n + Dv_0v_n, \quad v_{n+1} = v_0u_n + u_0v_n, \quad u_1 = u_0, \quad v_1 = v_0 \tag{2}$$

where (u_0, v_0) is its fundamental solution, i.e. the minimal solution different from $(1, 0)$.

Proof. First, we will prove that the equation (1) has a fundamental solution.

Let c_1 be an integer greater than 1. We will show that there exist integers $t_1, w_1 \geq 1$ such that

$$|t_1 - w_1\sqrt{D}| < \frac{1}{c_1}, \quad w_1 \leq c_1.$$

Indeed, considering $l_k = [k\sqrt{D} + 1]$, $k = 0, 1, \ldots, c_1$, yields $0 < l_k - k\sqrt{D} \leq 1$, $k = 0, 1, \ldots, c_1$, and since $\sqrt{D}$ is an irrational number, it follows that $l_{k'} \neq l_{k''}$ whenever $k' \neq k''$.

There exist $i, j, p \in \{0, 1, 2, \ldots, c_1\}$, $i \neq j$, $p \neq 0$, such that

$$\frac{p-1}{c_1} < l_i - i\sqrt{D} \leq \frac{p}{c_1} \quad \text{and} \quad \frac{p-1}{c_1} < l_j - j\sqrt{D} \leq \frac{p}{c_1}$$

because there are c_1 intervals of the form $\left(\frac{p-1}{c_1}, \frac{p}{c_1}\right)$, $p = 0, 1, \ldots, c_1$ and $c_1 + 1$ numbers of the form $l_k - k\sqrt{D}$, $k = 0, 1, \ldots, c_1$.

From the inequalities above it follows that $|(l_i - l_j) - (j - i)\sqrt{D}| < \frac{1}{c_1}$ and setting $|l_i - l_j| = t_1$ and $|j - i| = w_1$ yields $|t_1 - w_1\sqrt{D}| < \frac{1}{c_1}$ and $w_1 \leq c_1$.

[3] *John Pell* (1611-1685), English mathematician best known for "Pell's equation" which in fact he had litlle to do with.

Multiplying this inequality by $t_1 + w_1\sqrt{D} < 2w_1\sqrt{D} + 1$ gives

$$|t_1^2 - Dw_1^2| < 2\frac{w_1}{c_1}\sqrt{D} + \frac{1}{c_1} < 2\sqrt{D} + 1.$$

Choosing a positive integer $c_2 > c_1$ such that $|t_1 - w_1\sqrt{D}| > \dfrac{1}{c_2}$, we obtain positive integers t_2, w_2 with the properties

$$|t_2^2 - Dw_2^2| < 2\sqrt{D} + 1 \quad \text{and} \quad |t_1 - t_2| + |w_1 - w_2| \neq 0.$$

By continuing this procedure, we find a sequence of distinct pairs $(t_n, w_n)_{n\geq 1}$ satisfying the inequalities $|t_n^2 - Dw_n^2| < 2\sqrt{D} + 1$ for all positive integers n. It follows that the interval $(-2\sqrt{D} - 1, 2\sqrt{D} + 1)$ contains a nonzero integer k such that there exists a subsequence of $(t_n, w_n)_{n\geq 1}$ satisfying the equation $t^2 - Dw^2 = k$. This subsequence contains at least two pairs (t_s, w_s), (t_r, w_r) for which $t_s \equiv t_r (\mathrm{mod}|k|)$, $w_s \equiv w_r (\mathrm{mod}|k|)$, and $t_s w_r - t_r w_s \neq 0$, otherwise $t_s = t_r$ and $w_s = w_r$, in contradiction with $|t_s - t_r| + |w_s - w_r| \neq 0$.

Let $t_0 = t_s t_r - Dw_s w_r$ and let $w_0 = t_s w_r - t_r w_s$. Then

$$t_0^2 - Dw_0^2 = k^2. \tag{3}$$

On the other hand, $t_0 = t_s t_r - Dw_s w_r \equiv t_s^2 - Dw_0^2 \equiv 0 (\mathrm{mod}|k|)$, and it follows immediately that $w_0 \equiv 0 (\mathrm{mod}|k|)$. The pair (t, w) where $t_0 = t|k|$ and $w_0 = w|k|$ is a nontrivial solution to equation (1).

We show now that the pair (u_n, v_n) defined by (2) satisfies Pell's equation (1). We proceed by induction with respect to n. Clearly, (u_0, v_0) is a solution to the equation (1). If (u_n, v_n) is a solution to this equation, then

$$u_{n+1}^2 - Dv_{n+1}^2 = (u_0 u_n + Dv_0 v_n)^2 - D(v_0 u_n + u_0 v_n)^2 =$$

$$= (u_0^2 - Dv_0^2)(u_n^2 - Dv_n^2) = 1,$$

i.e. the pair (u_{n+1}, v_{n+1}) is also a solution to the equation (1).

It is not difficult to see that for all positive integer n,

$$u_{n-1} + v_{n-1}\sqrt{D} = (u_0 + v_0\sqrt{D})^n. \tag{4}$$

Let $z_n = u_{n-1} + v_{n-1}\sqrt{D} = (u_0 + v_0\sqrt{D})^n$ and note that $z_1 < z_2 < \cdots < z_n < \ldots$. We will prove now that all solutions to the equation (1) are of the form (4). Indeed, if the equation (1) had a solution (u, v) such that $z = u + v\sqrt{D}$ is not of the form (4), then $z_m < z < z_{m+1}$ for some

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integer m. Then $1 < (u+v\sqrt{D})(u_m - v_m\sqrt{D}) < u_0 + v_0\sqrt{D}$, and therefore $1 < (uu_m - Dvv_m) + (u_m v - uv_m)\sqrt{D} < u_0 + v_0\sqrt{D}$. On the other hand, $(uu_m - Dvv_m)^2 - D(u_m v - uv_m)^2 = (u^2 - Dv^2)(u_m^2 - Dv_m^2) = 1$, i.e. $(uu_m - Dvv_m, u_m v - uv_m)$ is a solution of (1) smaller than (u_0, v_0), in contradiction with the assumption that (u_0, v_0) was the minimal one. □

Remarks. 1) The relations (1) could be written in the following useful matrix form

$$\begin{pmatrix} u_{n+1} \\ v_{n+1} \end{pmatrix} = \begin{pmatrix} u_0 & Dv_0 \\ v_0 & u_0 \end{pmatrix} \begin{pmatrix} u_n \\ v_n \end{pmatrix}$$

from where

$$\begin{pmatrix} u_n \\ v_n \end{pmatrix} = \begin{pmatrix} u_0 & Dv_0 \\ v_0 & u_0 \end{pmatrix}^n \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}. \tag{5}$$

If

$$\begin{pmatrix} u_0 & Dv_0 \\ v_0 & u_0 \end{pmatrix}^n = \begin{pmatrix} a_n & b_n \\ c_n & d_n \end{pmatrix}$$

then it is well-known that each of a_n, b_n, c_n, d_n is a linear combination of λ_1^n, λ_2^n, where λ_1, λ_2 are the eigenvalues of the matrix $\begin{pmatrix} u_0 & Dv_0 \\ v_0 & u_0 \end{pmatrix}$. By using (5) after an easy computation it follows that

$$\begin{aligned} u_n &= \frac{1}{2}[(u_0 + v_0\sqrt{D})^n + (u_0 - v_0\sqrt{D})^n], \\ v_n &= \frac{1}{2\sqrt{D}}[(u_0 + v_0\sqrt{D})^n - (u_0 - v_0\sqrt{D})^n] \end{aligned} \tag{6}$$

2) The solutions to Pell's equation given in one of the forms (4) or (6) may be used in the approximation of the square roots of positive integers that are not perfect squares. Indeed, if (u_n, v_n) are the solutions of the equation (1), then

$$u_n - v_n\sqrt{D} = \frac{1}{u_n + v_n\sqrt{D}}$$

and so

$$\frac{u_n}{v_n} - \sqrt{D} = \frac{1}{v_n(u_n + v_n\sqrt{D})} < \frac{1}{\sqrt{D}v_n^2} < \frac{1}{v_n^2}.$$

It follows that

$$\lim_{n\to\infty} \frac{u_n}{v_n} = \sqrt{D}, \tag{7}$$

i.e. the fractions $\frac{u_n}{v_n}$ approximate $\sqrt{D}$ with an error less than $\frac{1}{v_n^2}$.

Problem 8.2.4. *Consider the sequences* $(u_n)_{n\geq 1}$, $(v_n)_{n\geq 1}$ *defined by* $u_1 = 3$, $v_1 = 2$ *and* $u_{n+1} = 3u_n + 4v_n$, $v_{n+1} = 2u_n + 3v_n$, $n \geq 1$. *Define* $x_n = u_n + v_n$, $y_n = u_n + 2v_n$, $n \geq 1$. *Prove that* $y_n = [x_n\sqrt{2}]$ *for all* $n \geq 1$.

Solution. We prove by induction that

$$u_n^2 - 2v_n^2 = 1, \quad n \geq 1. \tag{1}$$

For $n = 1$ the claim is true. Assuming that the equality is true for some n, we have

$$u_{n+1}^2 - 2v_{n+1}^2 = (3u_n + 4v_n)^2 - 2(2u_n + 3v_n)^2 = u_n^2 - 2v_n^2 = 1$$

hence (1) is true for all $n \geq 1$.

We prove now that

$$2x_n^2 - y_n^2 = 1, \quad n \geq 1 \tag{2}$$

Indeed,

$$2x_n^2 - y_n^2 = 2(u_n + v_n)^2 - (u_n + 2v_n)^2 = u_n^2 - 2v_n^2 = 1,$$

as claimed. It follows that

$$\left(x_n\sqrt{2} - y_n\right)\left(x_n\sqrt{2} + y_n\right) = 1, \quad n \geq 1.$$

Notice that $x_n\sqrt{2} + y_n > 1$ so

$$0 < x_n\sqrt{2} - y_n < 1, \quad n \geq 1.$$

Hence $y_n = \left[x_n\sqrt{2}\right]$, as claimed.

Problem 8.2.5. *Show that there exists infinitely many systems of positive integers* (x, y, z, t) *which have no common divisor greater than 1 and such that*

$$x^3 + y^2 + z^2 = t^4.$$

(2000 Romanian IMO Team Selection Test)

First Solution. Let consider the identity:

$$[1^3 + 2^3 + \cdots + (n-2)^3] + (n-1)^3 + n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

We may write it under the form:

$$(n-1)^3 + n^3 + \left(\frac{(n-1)(n-2)}{2}\right)^2 = \left(\frac{n(n+1)}{2}\right)^2.$$

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It is sufficient to find positive integers n for which $\frac{n(n+1)}{2}$ is a perfect square. Such a goal can be attained.

Let us remark that the equality

$$(2n+1)^2 - 2(2x)^2 = 1$$

can be realized by taking the solutions (u_k, v_k) of the Pell equation $u^2 - 2v^2 = 1$, where $u_0 = 3$, $v_0 = 2$ and u_k, v_k are obtained from the identity:

$$(u_0 + \sqrt{2}v_0)^k (u_0 - \sqrt{2}v_0)^k = (u_k + \sqrt{2}v_k)(u_k - \sqrt{2}v_k) = 1.$$

Alternative Solution. Let consider the following identity:

$$(a+1)^4 - (a-1)^4 = 8a^3 + 8a,$$

where a is a positive integer. Take $a = b^3$, where b is an even integer number. From the above identity one obtains:

$$(b^3+1)^4 = (2b^3)^3 + (2b)^3 + [(b^3-1)^2]^2.$$

Since b is even number, $b^3 + 1$ and $b^3 - 1$ are odd numbers. It follows that the numbers $x = 2b^3$, $y = 2b$, $z = (b^3-1)^2$ and $t = b^3 + 1$ have no common divisor greater than 1.

Proposed problems

Problem 8.2.6. Let p be a prime number congruent to 3 modulo 4. Consider the equation

$$(p+2)x^2 - (p+1)y^2 + px + (p+2)y = 1.$$

Prove that this equation has infinitely many solutions in positive integers, and show that if $(x, y) = (x_0, y_0)$ is a solution of the equation in positive integers, then $p|x_0$.

(2001 Bulgarian Mathematical Olympiad)

Problem 8.2.7. Determine all integers a for which the equation

$$x^2 + axy + y^2 = 1$$

has infinitely many distinct integer solutions (x, y).

(1995 Irish Mathematical Olympiad)

Problem 8.2.8. Prove that the equation

$$x^3 + y^3 + z^3 + t^3 = 1999$$

has infinitely many integral solutions.

(1999 Bulgarian Mathematical Olympiad)

8.2.3 Other quadratic equations

There are many other general quadratic equations that appear in concrete situations. Here is an example.

Consider the equation

$$axy + bx + cy + d = 0, \tag{1}$$

where a is a nonzero integer and b, c, d are integers such that $ad - bc \neq 0$.

Theorem 8.2.4. *If $gcd(a, b) = gcd(a, c) = 1$, then equation (1) is solvable if and only if there is a divisor m of $ad - bc$ such that $a|m - b$ or $a|m - c$.*

Proof. We can write (1) in the following equivalent form:

$$(ax + c)(ay + b) = bc - ad. \tag{2}$$

If such a divisor m exists and $a|m - c$, then we take $ax + c = m$ and $ay + b = m'$, where $mm' = bc - ad$. In order to have solutions it suffices to show that $a|m' - b$. Indeed, the relation $mm' = bc - ad$ implies $(ax + c)m' = bc - ad$, which is equivalent to $a(m'x + d) = -c(m' - b)$. Taking into account that $gcd(a, c) = 1$, we get $a|m' - b$.

The converse is clearly true. □

Remarks. 1) In case of solvability, equation (1) has only finitely many solutions. These solutions depend upon the divisors m of $ad - bc$.

2) If a does not divide $b - c$, then equation (1) is not solvable because from the proof of the theorem it follows that conditions $a|m - b$ and $a|m - c$ are equivalent and, in case of solvability, they hold simultaneously. In this case a must divide $(m - c) - (m - b) = b - c$.

Example. Solve the equation

$$3xy + 4x + 7y + 6 = 0.$$

Solution. We have $ad - bc = -10$, whose integer divisors are -10, -5, -2, -1, 1, 2, 5, 10. The conditions in Theorem 8.2.4 are satisfied only for

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$m = -5, -2, 1, 10$. We obtain the solutions $(x, y) = (-4, -2)$, $(-3, -3)$, $(-2, 2)$, $(1, -1)$, respectively.

In what follows you can find several nonstandard quadratic equations.

Problem 8.2.9. *For any given positive integer n, determine (as a function of n) the number of ordered pairs (x, y) of positive integers such that*

$$x^2 - y^2 = 10^2 \cdot 30^{2n}.$$

Prove further that the number of such pairs is never a perfect square.

(1999 Hungarian Mathematical Olympiad)

Solution. Because $10^2 \cdot 30^{2n}$ is even, x and y must have the same parity. Then (x, y) is a valid solution if and only if $(u, v) = \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$ is a pair of positive integers that satisfies $u > v$ and $uv = 5^2 \cdot 30^{2n}$. Now $5^2 \cdot 30^{2n} = 2^{2n} \cdot 3^{2n} \cdot 5^{2n+2}$ has exactly $(2n+1)^2(2n+3)$ factors. Thus without the condition $u > v$ there are exactly $(2n+1)^2(2n+3)$ such pairs (u, v). Exactly one pair has $u = v$, and by symmetry half of the remaining pairs have $u > v$. It follows that there are $\frac{1}{2}((2n+1)^2(2n+3) - 1) = (n+1)(4n^2+6n+1)$ valid pairs.

Now suppose that $(n+1)(4n^2+6n+1)$ were a square. Because $n+1$ and $4n^2+6n+1 = (4n+2)(n+1) - 1$ are coprime, $4n^2+6n+1$ must be a square as well. However, $(2n+1)^2 < 4n^2+6n+1 < (2n+2)^2$, a contradiction.

Problem 8.2.10. *Prove that the equation $a^2 + b^2 = c^2 + 3$ has infinitely many integer solutions $\{a, b, c\}$.*

(1996 Italian Mathematical Olympiad)

Solution. Let a be any odd number, let $b = (a^2-5)/2$ and $c = (a^2-1)/2$. Then

$$c^2 - b^2 = (c+b)(c-b) = a^2 - 3.$$

Remark. Actually one can prove that any integer n can be represented in infinitely many ways in the form $a^2 + b^2 - c^2$ with $a, b, c \in \mathbb{Z}$.

Proposed problems

Problem 8.2.11. Prove that the equation

$$x^2 + y^2 + z^2 + 3(x + y + z) + 5 = 0$$

has no solutions in rational numbers.

(1997 Bulgarian Mathematical Olympiad)

Problem 8.2.12. Find all integers x, y, z such that $5x^2 - 14y^2 = 11z^2$.

(2001 Hungarian Mathematical Olympiad)

Problem 8.2.13. Let n be a nonnegative integer. Find the nonnegative integers a, b, c, d such that

$$a^2 + b^2 + c^2 + d^2 = 7 \cdot 4^n.$$

(2001 Romanian JBMO Team Selection Test)

Problem 8.2.14. Prove that the equation

$$x^2 + y^2 + z^2 + t^2 = 2^{2004},$$

where $0 \leq x \leq y \leq x \leq t$, has exactly two solutions in the set of integers.

(2004 Romanian Mathematical Olympiad)

Problem 8.2.15. Let n be a positive integer. Prove that the equation

$$x + y + \frac{1}{x} + \frac{1}{y} = 3n$$

does not have solutions in positive rational numbers.

8.3 Nonstandard Diophantine equations

8.3.1 Cubic equations

Problem 8.3.1. *Find all pairs (x, y) of nonnegative integers such that $x^3 + 8x^2 - 6x + 8 = y^3$.*

(1995 German Mathematical Olympiad)

Solution. Note that for all real x,

$$0 < 5x^2 - 9x + 7 = (x^3 + 8x^2 - 6x + 8) - (x + 1)^3.$$

Therefore if (x, y) is a solution, we must have $y \geq x + 2$. In the same vein, we note that for $x \geq 1$,

$$0 > -x^2 - 33x + 15 = (x^3 + 8x^2 - 6x + 8) - (x^3 + 9x^2 + 27x + 27).$$

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Hence we either have $x = 0$, in which case $y = 2$ is a solution, or $x \geq 1$, in which case we must have $y = x + 2$. But this means

$$0 = (x^3 + 8x^2 - 6x + 8) - (x^3 + 6x^2 + 12x + 8) = 2x^2 - 18x.$$

Hence the only solutions are (0,2), (9,11).

Problem 8.3.2. *Find all pairs (x, y) of integers such that*

$$x^3 = y^3 + 2y^2 + 1.$$

(1999 Bulgarian Mathematical Olympiad)

Solution. When $y^2 + 3y > 0$, $(y + 1)^3 > x^3 > y^3$. Thus we must have $y^2 + 3y \leq 0$, and $y = -3, -2, -1$, or 0, yielding the solutions $(x, y) = (1, 0)$, $(1, -2)$, and $(-2, -3)$.

Problem 8.3.3. *Find all the triples (x, y, z) of positive integers such that*

$$xy + yz + zx - xyz = 2.$$

Solution. Let $x \leq y \leq z$. We consider the following cases:

1) For $x = 1$, we obtain $y + z = 2$, and then

$$(x, y, z) = (1, 1, 1).$$

2) If $x = 2$, then $2y + 2z - yz = 2$, which gives $(z - 2)(y - 2) = 2$. The solutions are $z = 4$, $y = 3$ or $z = 3$, $y = 4$. Due to the symmetry of the relations the solutions (x, y, z) are

$$(2, 3, 4),\ (2, 4, 3),\ (3, 2, 4),\ (4, 2, 3),\ (3, 4, 2),\ (4, 3, 2).$$

3) If $x \geq 3$, $y \geq 3$, $z \geq 3$ then $xyz \geq 3yz$, $xyz \geq 3xz$, $xyz \geq 3xy$. Thus $xy + xz + yz - xyz \leq 0$, so there are no solutions.

Problem 8.3.4. *Determine a positive constant c such that the equation*

$$xy^2 - y^2 - x + y = c$$

has exactly three solutions (x, y) in positive integers.

(1999 United Kingdom Mathematical Olympiad)

Solution. When $y = 1$ the left-hand side is 0. Thus we can rewrite our equation as

$$x = \frac{y(y-1)+c}{(y+1)(y-1)}.$$

The numerator is congruent to $-1(-2)+c$ modulo $(y+1)$, and it is also congruent to c modulo $(y-1)$. Hence we must have $c \equiv -2 \pmod{(y+1)}$ and $c \equiv 0 \pmod{(y-1)}$. Because $c = y-1$ satisfies these congruences, we must have $c \equiv y-1 \pmod{\text{lcm}(y-1, y+1)}$. When y is even, $\text{lcm}(y-1, y+1) = y^2-1$; when y is odd, $\text{lcm}(y-1, y+1) = \frac{1}{2}(y^2-1)$.

Then for $y = 2, 3, 11$ we have $c \equiv 1 \pmod 3$, $c \equiv 2 \pmod 4$, $c \equiv 10 \pmod{60}$. Hence, we try setting $c = 10$. For x to be an integer we must have $(y-1)|10 \Rightarrow y = 2, 3, 6$, or 11. These values give $x = 4, 2, \frac{2}{7}$, and 1, respectively. Thus there are exactly three solutions in positive integers, namely $(x, y) = (4, 2)$, (2,3), and (1,11).

Proposed problems

Problem 8.3.5. Find all triples (x, y, z) of natural numbers such that y is a prime number, y and 3 do not divide z, and $x^3 - y^3 = z^2$.

(1999 Bulgarian Mathematical Olympiad)

Problem 8.3.6. Find all the positive integers a, b, c such that

$$a^3 + b^3 + c^3 = 2001.$$

(2001 Junior Balkan Mathematical Olympiad)

Problem 8.3.7. Determine all ordered pairs (m, n) of positive integers such that

$$\frac{n^3+1}{mn-1}$$

is an integer.

(35th IMO)

8.3.2 High-order polynomial equations

Problem 8.3.8. *Prove that there are no integers x, y, z so that*

$$x^4 + y^4 + z^4 - 2x^2y^2 - 2y^2z^2 - 2z^2x^2 = 2000.$$

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Solution. Suppose by way of contradiction that such numbers exist. Assume without loss of generality that x, y, z are nonnegative integers.

At first we prove that the numbers are distinct. For this, consider that $y = z$. Then $x^4 - 4x^2y^2 = 2000$, hence x is even.

Setting $x = 2t$ yields $t^2(t^2 - y^2) = 125$. It follows that $t^2 = 25$ and $y^2 = 20$, a contradiction.

Let now $x > y > z$. Since $x^4 + y^4 + z^4$ is odd, at least one of the numbers x, y, z is even and the other two have the same parity. Observe that

$$x^4 + y^4 + z^4 - 2x^2y^2 - 2y^2z^2 - 2z^2x^2$$

$$= (x^2 - y^2)^2 - 2(x^2 - y^2)z^2 + z^4 - 4y^2z^2$$

$$= (x^2 - y^2 - z^2 - 2yz)(x^2 - y^2 - z^2 + 2yz)$$

$$= (x + y + z)(x - y - z)(x - y + z)(x + y - z),$$

each of the four factors being even. Since $2000 = 16 \cdot 125 = 2^4 \cdot 125$ we deduce that each factor is divisible by 2, but not by 4. Moreover, the factors are distinct

$$x + y + z > x + y - z > x - y + z > x - y - z.$$

The smallest even divisors of 2000 that are not divisible by 4 are 2, 10, 50, 250. But $2 \cdot 10 \cdot 50 \cdot 250 > 2000$, a contradiction.

Problem 8.3.9. *Find the smallest value for n for which there exist the positive integers $x_1, \ldots, x_n$ with*

$$x_1^4 + x_2^4 + \cdots + x_n^4 = 1998.$$

Solution. Observe that for any integer x we have $x^4 = 16k$ or $x^4 = 16k + 1$ for some k.

As $1998 = 16 \cdot 124 + 14$, it follows that $n \geq 14$.

If $n = 14$, all the numbers $x_1, x_2, \ldots, x_{14}$ must be odd, so let $x_k^4 = 16a_k + 1$. Then $a_k = \dfrac{x_k^4 - 1}{16}$, $k = \overline{1,14}$ hence $a_k \in \{0, 5, 39, 150, \ldots\}$ and $a_1 + a_2 + \cdots + a_{14} = 124$. It follows that $a_k \in \{0, 5, 39\}$ for all $k = \overline{1,14}$, and since $124 = 5 \cdot 24 + 4$, the number of the terms a_k equal to 39 is 1 or at least 6. A simple analysis show that the claim fails in both cases, hence

$n \geq 15$. Any of the equalities

$$\begin{aligned}1998 &= 5^4+5^4+3^4+3^4+3^4+3^4+3^4+3^4+3^4+3^4+3^4+2^4\\ &\quad +1^4+1^4+1^4\\ &= 5^4+5^4+4^4+3^4+3^4+3^4+3^4+3^4+3^4\\ &\quad +1^4+1^4+1^4+1^4+1^4+1^4\end{aligned}$$

proves that $n = 15$.

Problem 8.3.10. *Find all integer and positive solutions (x, y, z, t) of the equation*

$$(x+y)(y+z)(z+x) = txyz$$

such that $(x, y) = (y, z) = (z, x) = 1$.

(1995 Romanian Mathematical IMO Team Selection Test)

Solution. It is obvious that $(x, x+y) = (x, x+z) = 1$, then x divides $y+z$, y divides $z+x$ and z divides $x+y$. Let a, b, and c be integers such that

$$x+y = cz$$

$$y+z = ax$$

$$z+x = by.$$

We may assume that $x \geq y \geq z$. If $y = z$, then $y = z = 1$ and then $x \in \{1, 2\}$. If $x = y$, then $x = y = 1$ and $z = 1$. So, assume that $x > y > z$. Since $a = \dfrac{y+z}{x} < 2$, we have $a = 1$ and $x = y+z$. Thus, $y|y+2z$ and $y|2z$. Since $y > z$, $y = 2z$ and since $gcd(y, z) = 1$, one has $z = 1$, $y = 2$, $x = 3$.

Finally the solutions are $(1, 1, 1, 8)$, $(2, 1, 1, 9)$, $(3, 2, 1, 10)$ and those obtained by permutations of x, y, z.

Problem 8.3.11. *Determine all triples of positive integers a, b, c such that a^2+1, b^2+1 are prime and $(a^2+1)(b^2+1) = c^2+1$.*

(2002 Polish Mathematical Olympiad)

Solution. Of course, we may assume that $a \leq b$. Since $a^2(b^2+1) = (c-b)(c+b)$ and b^2+1 is a prime, we have $b^2+1|c-b$ or $b^2+1|c+b$. If $b^2+1|c-b$, then $a^2 \geq c+b \geq b^2+2b+1$, impossible since $a \leq b$. So there is k such that $c+b = k(b^2+1)$ and $a^2 = k(b^2+1)-2b$. Thus, $b^2 \geq k(b^2+1)-2b > kb^2-2b$, from where $k \leq 2$. If $k = 2$, then $b^2 \geq 2b^2-2b+2$, thus $(b-1)^2+1 \leq 0$,

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false. Thus $k = 1$ and $a = b-1$. But then b^2+1 and $(b-1)^2+1$ are primes and at least one of them is even, forcing $b-1 = 1$ and $b = 2$, $a = 1$, $c = 3$. By symmetry, we find $(a, b, c) = (1, 2, 3)$ or $(2, 1, 3)$.

Proposed problems

Problem 8.3.12. Prove that there are no positive integers x and y such that

$$x^5 + y^5 + 1 = (x + 2)^5 + (y - 3)^5.$$

Problem 8.3.13. Prove that the equation $y^2 = x^5 - 4$ has no integer solutions.

(1998 Balkan Mathematical Olympiad)

Problem 8.3.14. Let $m, n > 1$ be integer numbers. Solve in positive integers the equation

$$x^n + y^n = 2^m.$$

(2003 Romanian Mathematical Olympiad)

Problem 8.3.15. For a given positive integer m, find all pairs (n, x, y) of positive integers such that m, n are relatively prime and $(x^2+y^2)^m = (xy)^n$, where n, x, y can be represented in terms of m.

(1995 Korean Mathematical Olympiad)

8.3.3 Exponential Diophantine equations

Problem 8.3.16. *Find the integer solutions to the equation*

$$9^x - 3^x = y^4 + 2y^3 + y^2 + 2y.$$

Solution. We have successively

$$4((3^x)^2 - 3^x) + 1 = 4y^4 + 8y^3 + 4y^2 + 8y + 1,$$

then

$$(2t - 1)^2 = 4y^4 + 8y^3 + 4y^2 + 8y + 1,$$

where $3^x = t \geq 1$ since it is clear that there are no solutions with $x < 0$.

Observe that

$$(2y^2+2y)^2 < E \le (2y^2+2y+1)^2.$$

Since $E=(2t-1)^2$ is a square, then

$$E=(2y^2+2y+1)^2$$

if and only if

$$4y(y-1)=0,$$

so $y=0$ or $y=1$.

If $y=0$ then $t=1$ and $x=0$.

If $y=1$, then $t=3$ and $x=1$.

Hence the solutions (x,y) are (0,0) and (1,1).

Problem 8.3.17. *The positive integers x,y,z satisfy the equation $2x^x = y^y+z^z$. Prove that $x=y=z$.*

(1997 St. Petersburg City Mathematical Olympiad)

Solution. We note that $(x+1)^{x+1} \ge x^{x+1}+(x+1)x^x > 2x^x$. Thus we cannot have $y>x$ or $z>x$, else the right side of the equation will exceed the left. But then $2x^x \ge y^y+z^z$, with equality if and only if $x=y=z$.

Problem 8.3.18. *Find all solutions in nonnegative integers x,y,z of the equation*

$$2^x+3^y=z^2.$$

(1996 United Kingdom Mathematical Olympiad)

Solution. If $y=0$, then $2^x=z^2-1=(z+1)(z-1)$, so $z+1$ and $z-1$ are powers of 2. The only powers of 2 which differ by 2 are 4 and 2, so $(x,y,z)=(3,0,3)$.

If $y>0$, then taking the equation mod 3 it follows that x is even. Now we have $3^y=z^2-2^x=(z+2^{x/2})(z-2^{x/2})$. The factors are powers of 3, say $z+2^{x/2}=3^m$ and $z-2^{x/2}=3^n$, but then $3^m-3^n=2^{x/2+1}$. Since the right side is not divisible by 3, we must have $n=0$ and

$$3^m-1=2^{x/2+1}.$$

If $x=0$, we have $m=1$, yielding $(x,y,z)=(0,1,2)$. Otherwise, 3^m-1 is divisible by 4, so m is even and $2^{x/2+1}=(3^{m/2}+1)(3^{m/2}-1)$. The two factors on the right are powers of 2 differing by 2, so they are 2 and 4, giving $x=4$ and $(x,y,z)=(4,2,5)$.

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Proposed problems

Problem 8.3.19. Determine all triples (x, k, n) of positive integers such that

$$3^k - 1 = x^n.$$

(1999 Italian Mathematical Olympiad)

Problem 8.3.20. Find all pairs of nonnegative integers x and y which satisfy the equation

$$p^x - y^p = 1$$

where p is a given odd prime.

(1995 Czech-Slovak Match)

Problem 8.3.21. Let x, y, z be integers with $z > 1$. Show that

$$(x+1)^2 + (x+2)^2 + \cdots + (x+99)^2 \neq y^z.$$

(1998 Hungarian Mathematical Olympiad)

Problem 8.3.22. Determine all solutions (x, y, z) of positive integers such that

$$(x+1)^{y+1} + 1 = (x+2)^{z+1}.$$

(1999 Taiwanese Mathematical Olympiad)

9

Some special problems in number theory

9.1 Quadratic residues. Legendre's symbol

Let a and m be positive integers such that $m \neq 0$ and $gcd(a, m) = 1$. We say that a is a *quadratic residue* mod m if the congruence $x^2 \equiv a \pmod{m}$ has a solution. Otherwise we say that a is a nonquadratic residue.

Let p be an odd prime and let a be a positive integer not divisible by p. The *Legendre's symbol* of a with respect to p is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ quadratic residue pmod } p \\ -1 & \text{otherwise} \end{cases}$$

It is clear that the perfect squares are quadratic residues mod p. It is natural to ask how many integers among $1, 2, \ldots, p-1$ are quadratic residues. The answer is given in the following theorem.

Theorem 9.1.1. *Let p be an odd prime. There are $\frac{p-1}{2}$ quadratic residues in the set $\{1, 2, \ldots, p-1\}$.*

Proof. Consider the numbers k^2, $k = 1, 2, \ldots, \frac{p-1}{2}$. These are quadratic residues and moreover, they are distinct. Indeed, if $i^2 \equiv j^2 \pmod{p}$, then it follows that $p|(i-j)(i+j)$ and, since $i + j < p$, this implies $p|i - j$, hence $i = j$.

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Conversely, if $gcd(a,p)=1$ and the congruence $x^2 \equiv a \pmod p$ has a solution x, then $x = qp+i$, where $-\dfrac{p-1}{2} \le i \le \dfrac{p-1}{2}$ and so $i^2 \equiv q \pmod p$. □

The basic properties of Legendre's symbol are:

1) (Euler's criterion) If p is an odd prime and a an integer not divisible by p, then

$$a^{\frac{p-1}{2}} \equiv \left(\frac{a}{p}\right) \pmod p.$$

2) If $a \equiv b \pmod p$, then $\left(\dfrac{a}{p}\right) = \left(\dfrac{b}{p}\right)$.

3) (multiplicity) $\left(\dfrac{a_1 \dots a_n}{p}\right) = \left(\dfrac{a_1}{p}\right) \dots \left(\dfrac{a_n}{p}\right)$.

4) $\left(\dfrac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$.

For Euler's criterion, suppose that $\left(\dfrac{a}{p}\right) = 1$. Then $i^2 \equiv a \pmod p$ for some integer i. We have $gcd(i,p)=1$ and from Fermat's Little Theorem, $i^{p-1} \equiv 1 \pmod p$. Hence $a^{\frac{p-1}{2}} \equiv 1 \pmod p$ and we are done.

If $\left(\dfrac{a}{p}\right) = -1$, then each of the congruences

$$x^{\frac{p-1}{2}} - 1 \equiv 0 \pmod p \quad \text{and} \quad x^{\frac{p-1}{2}} + 1 \equiv 0 \pmod p$$

has $\dfrac{p-1}{2}$ distinct solutions in the set $\{1,2,\dots,p-1\}$. The $\dfrac{p-1}{2}$ quadratic residues correspond to the first congruence and the $\dfrac{p-1}{2}$ nonquadratic residues correspond to the second. Hence if a is nonquadratic residue, we have $a^{\frac{p-1}{2}} \equiv -1 \pmod p$ and we are done.

Remark. From Fermat's Little Theorem, $a^{p-1} \equiv 1 \pmod p$, hence $p | (a^{\frac{p-1}{2}} - 1)(a^{\frac{p-1}{2}} + 1)$. From Euler's criterion, $p | a^{\frac{p-1}{2}} - 1$ if and only if a is a quadratic residue mod p.

Property 2) is clear. For 3) we apply Euler's criterion:

$$\left(\frac{a_i}{p}\right) \equiv a_i^{\frac{p-1}{2}} \pmod p, \quad i = 1, \dots, n.$$

Therefore

$$\left(\frac{a_1}{p}\right) \dots \left(\frac{a_n}{p}\right) \equiv a_1^{\frac{p-1}{2}} \dots a_n^{\frac{p-1}{2}} = (a_1 \dots a_n)^{\frac{p-1}{2}}$$

$$\equiv \left(\frac{a_1 \dots a_n}{p}\right) \pmod p.$$

In order to prove 4) note that $(-1)^{\frac{p-1}{2}}, \left(\frac{-1}{p}\right) \in \{-1,1\}$. Hence $p|(-1)^{\frac{p-1}{2}} - \left(\frac{-1}{p}\right)$ reduces to Euler's criterion.

The following theorem gives necessary and sufficient conditions under which 2 is a quadratic residue.

Theorem 9.1.2. *For any odd prime p,*

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

Proof. We need the following lemma.

Lemma. (Gauss[1]) *If a is a positive integer that is not divisible by p, then from the Division Algorithm,*

$$ka = pq_k + r_k, \quad k = 1, \ldots, \frac{p-1}{2}.$$

Let $b_1, \ldots, b_m$ be the distinct remainders $r_1, \ldots, r_{\frac{p-1}{2}}$ that are less than $\frac{p}{2}$ and let $c_1, \ldots, c_n$ be the distinct remaining remainders. Then

$$\left(\frac{a}{p}\right) = (-1)^n.$$

Proof of Lemma. We have

$$\prod_{i=1}^{m} b_i \prod_{j=1}^{n} c_j = \prod_{k=1}^{\frac{p-1}{2}} r_k = \prod_{k=1}^{\frac{p-1}{2}} (ka - pq_k) \equiv \prod_{k=1}^{\frac{p-1}{2}} ka = a^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \pmod{p}.$$

Because $\frac{p}{2} < c_j \leq p-1$, $j = 1, \ldots, n$, we have $1 \leq p - c_j \leq \frac{p-1}{2}$. It is not possible to have $p - c_j = b_i$ for some i and j. Indeed, if $b_i + c_j = p$, then $p = as - pq_s + at - pq_t$, so $p|s+t$, which is impossible, since $1 \leq s, t \leq \frac{p-1}{2}$.

Therefore the integers $b_1, \ldots, b_m, p - c_1, \ldots, p - c_n$ are distinct and

$$\{b_1, \ldots, b_m, p - c_1, \ldots, p - c_n\} = \left\{1, 2, \ldots, \frac{p-1}{2}\right\}$$

[1] *Karl Friedrich Gauss* (1777-1855), German mathematician who is sometimes called the "prince of mathematics". Gauss proved in 1801 the fundamental theorem of arithmetic and he published one of the most brilliant achievements in mathematics, "Disquisitiones Arithmeticae". In this book he systematized the study of number theory and developed the algebra of congruences.

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We obtain

$$\prod_{i=1}^{m} b_i \prod_{j=1}^{n}(p-c_j) = \left(\frac{p-1}{2}\right)!$$

Finally,

$$(-1)^n \prod_{i=1}^{m} b_i \prod_{j=1}^{n} c_j \equiv \left(\frac{p-1}{2}\right)! \pmod{p}$$

hence $a^{\frac{p-1}{2}} \equiv (-1)^n \pmod{p}$. The conclusion now follows from Euler's criterion. □

In order to prove the theorem we use Gauss Lemma for $a = 2$. We have $\{r_1, r_2, \ldots, r_{\frac{p-1}{2}}\} = \{2, 4, \ldots, p-1\}$. The number of integers k such that $\frac{p}{2} < 2k < p$ is $n = \left\lfloor \frac{p}{2} \right\rfloor - \left\lfloor \frac{p}{4} \right\rfloor$.

If $p = 4u+1$, then $n = 2u - u = u$ and $\frac{p^2-1}{8} = 2u^2 + u$. We have $n \equiv \frac{p^2-1}{8} \pmod 2$ and we are done.

If $p = 4v+3$, then $n = 2v + 1 - v = v + 1$ and $\frac{p^2-1}{8} = 2v^2 + 3v + 1$ and again $n \equiv \frac{p^2-1}{8} \pmod 2$. □

The central result concerning Legendre's symbol is the so-called *Quadratic Reciprocity Law* of Gauss.

Theorem 9.1.3. *If p and q are distinct odd primes, then*

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

Proof. In Gauss Lemma we take $a = q$ and we get $\left(\frac{q}{p}\right) = (-1)^n$. Let $\sum_{i=1}^{m} b_i = b$ and $\sum_{j=1}^{n} c_j = c$. Then using the equality

$$\{b_1, \ldots, b_m, p-c_1, \ldots, p-c_n\} = \left\{1, 2, \ldots, \frac{p-1}{2}\right\}$$

it follows that

$$b + np - c = \sum_{k=1}^{\frac{p-1}{2}} k = \frac{p^2-1}{8}.$$

But from Gauss Lemma we have $q_k = \left\lfloor \frac{kq}{p} \right\rfloor$, $k = 1, 2, \ldots, p-1$, hence

$$q\frac{p^2-1}{8} = p\sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor + b + c.$$

Summing up the last two relations gives

$$2c + p\sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor + \frac{p^2-1}{8}(1-q) - np = 0.$$

Because $2c$ and $1-q$ are even, it follows that

$$n \equiv \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor \pmod 2$$

and applying Gauss Lemma again we obtain

$$\left(\frac{q}{p}\right) = (-1)^{\sum\limits_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor}$$

Similarly, we derive the relation

$$\left(\frac{p}{q}\right) = (-1)^{\sum\limits_{j=1}^{\frac{p-1}{2}} \left\lfloor \frac{jp}{q} \right\rfloor}$$

Multiplying the last two equalities and taking into account the Landau's identity in Problem 18 of Chapter 3, the conclusion follows. □

Problem 9.1.1. *Let $k = 2^{2^n} + 1$ for some positive integer n. Show that k is a prime if and only if k is a factor of $3^{(k-1)/2} + 1$.*

(1997 Taiwanese Mathematical Olympiad)

Solution. Suppose k is a factor of $3^{(k-1)/2} + 1$. This is equivalent to $3^{(k-1)/2} \equiv -1 \pmod k$. Hence $3^{k-1} \equiv 1 \pmod k$. Let d be the order of 3 mod k. Then $d \nmid (k-1)/2$ but $d|(k-1)$, hence $(k-1)|d$, so $d = k-1$ (since d must be smaller than k). Therefore k is prime.

Conversely, suppose k is prime. By Quadratic Reciprocity Law

$$\left(\frac{3}{k}\right) = \left(\frac{k}{3}\right) = \left(\frac{2}{3}\right) = -1.$$

By Euler's criterion, $3^{(k-1)/2} \equiv \left(\frac{3}{k}\right) \equiv -1 \pmod k$, as claimed.

Problem 9.1.2. *Prove that if n is a positive integer such that the equation $x^3 - 3xy^2 + y^3 = n$ has an integer solution (x, y) then it has at least three such solutions.*

(23rd IMO)

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Solution. The idea of the solution is to find a non-singular change of coordinates with integer coefficients

$$(x, y) \to (ax + by, cx + dy),$$

such that the polynomial $x^3 - 3xy^2 + y^3$ does not change after changing coordinates. Such a transformation can be found after remarking the identity:

$$x^3 - 3xy^2 + y^3 = (y - x)^3 - 3x^2y + 2x^3 = (y - x)^3 - 3(y - x)x^2 + (-x)^3.$$

Thus, such a transformation is $T(x, y) = (y - x, -x)$. It can be represented like a linear transformation

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x + y \\ -x \end{pmatrix}.$$

We have

$$T^2 = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$$

and

$$T^3 = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus, $T^2(x, y) = (-y, x - y)$. Moreover, it is easy to see that if $x^3 - 3xy^2 + y^3 = n$, $n \geq 0$, then the pairs (x, y), $(-y, x - y)$ are distinct.

For the second part, observe that $2819 = 7^2 \cdot 59$. Suppose that x, y are integer numbers such that $x^3 - 3xy^2 + y^3 = 2891$. Then x, y are pairwise prime, because from $d = (x, y)$ we obtain $d^3|2891$. The numbers x, y are not divisible by 7, then they are inversible modulo 7. Thus, from the equation we obtain

$$\left(\frac{y}{x}\right)^3 - 3\left(\frac{y}{x}\right)^2 + 1 \equiv 0 \pmod 7.$$

This proves that the congruence

$$a^3 - 3a^2 + 1 \equiv 0 \pmod 7$$

has a solution, $a \in \mathbb{Z}$. Since 7 is not a divisor of a, by Fermat's Little Theorem one has $a^6 \equiv 1 \pmod 7$. There are two possibilities: $a^3 \equiv 1 \pmod 7$ or $a^3 \equiv -1 \pmod 7$. When $a^3 \equiv 1 \pmod 7$ we obtain:

$$a^3 - 3a^2 + 1 \equiv 0 \pmod 7 \Rightarrow 3a^2 \equiv 2 \pmod 7 \Rightarrow a^2 \equiv 3 \pmod 7.$$

Using Legendre's symbol and the Quadratic Reciprocity Law:

$$\left(\frac{3}{7}\right) = (-1)^{\frac{3-1}{2}\cdot\frac{7-1}{2}}\left(\frac{7}{3}\right) = (-1)\left(\frac{1}{3}\right) = -1.$$

This proves that 3 is not a square modulo 7. When $a^3 \equiv -1 \pmod 7$ we obtain the contradiction from: $3a^2 \equiv 0 \pmod 7$. Thus, the equation $x^3 - 3xy^2 + y^3 = 2891$ has no solution in integers (x, y).

Problem 9.1.3. *Let m, n be positive integers such that*

$$A = \frac{(m+3)^n + 1}{3m}$$

is an integer. Prove that A is odd.

(1998 Bulgarian Mathematical Olympiad)

Solution. If m is odd, then $(m+3)^n + 1$ is odd and A is odd. Now we suppose that m is even. Since A is an integer,

$$0 \equiv (m+3)^n + 1 \equiv m^n + 1 \pmod 3,$$

so $n = 2k+1$ is odd and $m \equiv -1 \pmod 3$. We consider the following cases.

(a) $m = 8m'$ for some positive integer m'. Then

$$(m+3)^n + 1 \equiv 3^{2k+1} + 1 \equiv 4 \pmod 8$$

and $3m \equiv 0 \pmod 8$. So A is not an integer.

(b) $m = 2m'$ for some odd positive integer m', i.e., $m \equiv 2 \pmod 4$. Then

$$(m+3)^n + 1 \equiv (2+3) + 1 \equiv 2 \pmod 4$$

and $3m \equiv 2 \pmod 4$. So A is odd.

(c) $m = 4m'$ for some odd positive integer m'. Because $m \equiv -1 \pmod 3$, there exists an odd prime p such that $p \equiv -1 \pmod 3$ and $p|m$. Since A is an integer,

$$0 \equiv (m+3)^n + 1 \equiv 3^{2k+1} + 1 \pmod m$$

and $3^{2k+1} \equiv -1 \pmod p$. Let a be a primitive root modulo p; let b be a positive integer such that $3 \equiv a^b \pmod p$. Thus $a^{(2k+1)b} \equiv -1 \pmod p$. Note that $(p/3) = (-1/3) = -1$. We consider the following cases.

(i) $p \equiv 1 \pmod 4$. From the Quadratic Reciprocity Law, $(-1/p) = 1$, so

$$a^{2c} \equiv -1 \equiv a^{(2k+1)b} \pmod p$$

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for some positive integer c. Therefore b is even and $(3/p) = 1$. Again, from the Quadratic Reciprocity Law,

$$-1 = (3/p)(p/3) = (-1)^{(3-1)(p-1)/4} = 1,$$

a contradiction.

(ii) $p \equiv 3 \pmod 4$. From the Quadratic Reciprocity Law, $(-1/p) = -1$, so

$$a^{2c+1} \equiv -1 \equiv a^{(2k+1)b} \pmod p$$

for some positive integer c. Therefore b is odd and $(3/p) = -1$. Again, from the Quadratic Reciprocity Law,

$$1 = (3/p)(p/3) = (-1)^{(3-1)(p-1)/4} = -1,$$

a contradiction.

Thus for $m = 4m'$ and m' is odd, A is not an integer.

From the above, we see that if A is an integer, A is odd.

Problem 9.1.4. *Prove that $2^n + 1$ has no prime factors of the form $8k + 7$.*

(2004 Vietnamese IMO Team Selection Test)

Solution. Assume that we have a prime p such that $p|2^n+1$ and $p \equiv -1 \pmod 8$. If n is even, then $p \equiv 3 \pmod 4$ and $\left(\dfrac{-1}{p}\right) = 1$, a contradiction. If n is odd, then $\left(\dfrac{-2}{p}\right) = 1$ and we get $(-1)^{\frac{p^2-1}{8}}(-1)^{\frac{p-1}{2}} = 1$, again a contradiction.

Problem 9.1.5. *Prove that $2^{3^n} + 1$ has at least n prime divisors of the form $8k + 3$.*

Solution. Using the result of the previous problem, we deduce that 2^n+1 does not have prime divisors of the form $8k + 7$. We will prove that if n is odd, then it has no prime divisors of the form $8k + 5$ either. Indeed, let p be a prime divisor of $2^n + 1$. Then $2^n \equiv -1 \pmod p$ and so $-2 \equiv (2^{\frac{n+1}{2}})^2 \pmod p$. Using the same argument as the one in the previous problem, we deduce that $\dfrac{p^2-1}{8} + \dfrac{p-1}{2}$ is even, which cannot happen if p is of the form $8k + 5$.

Now, let us solve the proposed problem. We will assume $n > 2$ (otherwise the verification is trivial). The essential observation is the identity:

$$2^{3^n} + 1 = (2 + 1)(2^2 - 2 + 1)(2^{2\cdot 3} - 2^3 + 1) \dots (2^{2\cdot 3^{n-1}} - 2^{3^{n-1}} + 1)$$

Now, we will prove that for all $1 \leq i < j \leq n-1$,

$$gcd(2^{2\cdot 3^i} - 2^{3^i} + 1, 2^{2\cdot 3^j} - 2^{3^j} + 1) = 3.$$

Indeed, assume that p is a prime number dividing $gcd(2^{2\cdot 3^i} - 2^{3^i} + 1, 2^{2\cdot 3^j} - 2^{3^j} + 1)$ We will then have $p|2^{3^{i+1}} + 1$. Thus,

$$2^{3^j} \equiv (2^{3^{i+1}})^{3^{j-i-1}} \equiv (-1)^{3^{j-i-1}} \equiv -1 \pmod{p},$$

implying

$$0 \equiv 2^{2\cdot 3^j} - 2^{3^j} + 1 \equiv 1 - (-1) + 1 \equiv 3 \pmod{p}.$$

This cannot happen unless $p = 3$. But since

$$v_3(gcd(2^{2\cdot 3^i} - 2^{3^i} + 1, 2^{2\cdot 3^j} - 2^{3^j} + 1)) = 1$$

(as one can immediately check), it follows that

$$gcd(2^{2\cdot 3^i} - 2^{3^i} + 1, 2^{2\cdot 3^j} - 2^{3^j} + 1) = 3$$

and the claim is proved.

It remains to show that each of the numbers $2^{2\cdot 3^i} - 2^{3^i} + 1$, with $1 \leq i \leq n-1$ has at least a prime divisor of the form $8k+3$ different from 3. It would follow in this case that $2^{3^n} + 1$ has at least $n-1$ distinct prime divisors of the form $8k+3$ (from the previous remarks) and since it is also divisible by 3, the conclusion would follow. Fix $i \in \{1, 2, \ldots, n-1\}$ and observe that any prime factor of $2^{2\cdot 3^i} - 2^{3^i} + 1$ is also a prime factor of $2^{3^n} + 1$ and thus, from the first remark, it must be of the form $8k+1$ or $8k+3$. Because $v_3(2^{2\cdot 3^i} - 2^{3^i} + 1) = 1$, it follows that if all prime divisors of $2^{2\cdot 3^i} - 2^{3^i} + 1$ except for 3 are of the form $8k+1$, then $2^{2\cdot 3^i} - 2^{3^i} + 1 \equiv 8 \pmod{8}$, which is clearly impossible. Thus at least a prime divisor of $2^{2\cdot 3^i} - 2^{3^i} + 1$ is different from 3 and is of the form $8k+3$ and so the claim is proved. The conclusion follows.

Problem 9.1.6. *Find a number n between 100 and 1997 such that $n|2^n + 2$.*

(1997 Asian-Pacific Mathematical Olympiad)

Solution. The first step would be choosing $n = 2p$, for some prime number p. Unfortunately this cannot work by Fermat's little theorem. So let us try setting $n = 2pq$, with p, q different prime numbers. We need

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$pq|2^{2pq-1}+1$ and so we must have $\left(\frac{-2}{p}\right) = \left(\frac{-2}{q}\right) = 1$. Also, using Fermat's little theorem, $p|2^{2q-1}+1$ and $q|2^{2p-1}+1$. A small verification shows that $q = 3, 5, 7$ are not good choices, so let us try $q = 11$. In this case we find $p = 43$ and so it suffices to show that $pq|2^{2pq-1}+1$ for $q = 11$ and $p = 43$. This is immediate, since the hard work has already been completed: we have shown that it suffices to have $p|q^{2q-1}$, $q|2^{2p-1}+1$, and $\left(\frac{-2}{p}\right) = \left(\frac{-2}{q}\right) = 1$ in order to have $pq|2^{2pq-1}+1$. But as one can easily check, all these conditions are verified and the number $2 \cdot 11 \cdot 43$ is a valid answer.

Proposed problems

Problem 9.1.7. Let $f, g : Z^+ \to Z^+$ functions with the properties:

i) g is surjective;

ii) $2f^2(n) = n^2 + g^2(n)$ for all positive integers n.

If, moreover, $|f(n) - n| \leq 2004\sqrt{n}$ for all n, prove that f has infinitely many fixed points.

(2005 Moldavian IMO Team Selection Test)

Problem 9.1.8. Suppose that the positive integer a is not a perfect square. Then $\left(\frac{a}{p}\right) = -1$ for infinitely many primes p.

Problem 9.1.9. Suppose that $a_1, a_2, \ldots, a_{2004}$ are nonnegative integers such that $a_1^n + a_2^n + \cdots + a_{2004}^n$ is a perfect square for all positive integers n. What is the minimal number of such integers that must equal 0?

(2004 Mathlinks Contest)

Problem 9.1.10. Find all positive integers n such that $2^n - 1|3^n - 1$.

(American Mathematical Monthly)

Problem 9.1.11. Find the smallest prime factor of $12^{2^{15}} + 1$.

9.2 Special numbers

9.2.1 Fermat's numbers

Trying to find all primes of the form $2^m + 1$, Fermat noticed that m must be a power of 2. Indeed, if m equaled $k \cdot h$ with k an odd integer greater

than 1, then

$$2^m + 1 = (2^h)^k + 1 = (2^h + 1)(2^{h(k-1)} - 2^{h(k-2)} + \cdots - 2^h + 1),$$

and so $2^m + 1$ would not be a prime.

The integers $f_n = 2^{2^n} + 1$, $n \geq 0$, are called *Fermat's numbers*. We have

$$f_0 = 3, \quad f_1 = 5, \quad f_2 = 17, \quad f_3 = 65,573, \quad f_4 = 4,294,967,297.$$

After checking that these five numbers are primes, Fermat conjectured that f_n is a prime for all n. But Euler proved that $641|f_5$. His argument was the following:

$$f_5 = 2^{32} + 1 = 2^{28}(5^4 + 2^4) - (5 \cdot 2^7)^4 + 1 = 2^{28} \cdot 641 - (640^4 - 1)$$

$$= 641(2^{28} - 639(640^2 + 1)).$$

It is still an open problem if there are infinitely many Fermat primes. The answer to this question is important because Gauss proved that a regular polygon $Q_1Q_2 \ldots Q_n$ can be constructed by using only a ruler and a compass if and only if $n = 2^h p_1 \ldots p_k$, where $k \geq 0$ and $p_1, \ldots, p_k$ are distinct Fermat primes. Gauss was the first to construct such a polygon for $n = 17$.

Problem 9.2.1. *Prove that*

(i) $f_n = f_0 \ldots f_{n-1} + 2$, $n \geq 1$;

(ii) $gcd(f_k, f_h) = 1$ *if* $k \neq h$;

(iii) f_n *ends in* 7 *for all* $n \geq 2$.

Solution. (i) We have

$$f_k = 2^{2^k} + 1 = (2^{2^{k-1}})^2 + 1 = (f_{k-1} - 1)^2 + 1 = f_{k-1}^2 - 2f_{k-1} + 2,$$

hence

$$f_k - 2 = f_{k-1}(f_{k-1} - 2), \quad k \geq 1. \tag{1}$$

Multiplying relations (1) for $k = 1, \ldots, n$ yields

$$f_n - 2 = f_0 \ldots f_{n-1}(f_0 - 2)$$

and the conclusion follows.

For a different proof we can use directly the identity

$$\frac{x^{2^n} - 1}{x - 1} = \prod_{k=0}^{n-1} (x^{2^k} + 1)$$

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(ii) From (i) we have

$$gcd(f_n, f_0) = gcd(f_n, f_1) = \cdots = gcd(f_n, f_{n-1}) = 1$$

for all $n \geq 1$, hence $gcd(f_k, f_h) = 1$ for all $k \neq h$.

(iii) Because $f_1 = 5$ and $f_0 \dots f_{n-1}$ is odd, using (i) it follows that f_n ends in $5 + 2 = 7$ for all $n \geq 2$.

Problem 9.2.2. *Find all Fermat's numbers that can be written as a sum of two primes.*

Solution. All Fermat's numbers are odd. If $f_n = p + q$ for some primes p and q, $p \leq q$, then $p = 2$ and $q > 2$. We obtain

$$q = 2^{2^n} - 1 = (2^{2^{n-1}})^2 - 1 = (2^{2^{n-1}} - 1)(2^{2^{n-1}} + 1),$$

hence $2^{2^{n-1}} - 1$ must equal 1. That is $n = 1$ and $f_1 = 2 + 3$ is the unique Fermat's number with this property.

An alternative solution uses Problem 1 (iii): if $n \geq 2$, then f_n ends in 7, so q must end in 5. Hence $q = 5$ and $2 + 5 \neq f_n$ for $n \geq 2$. The only Fermat's number with the given property is f_1.

Problem 9.2.3. *Show that for any $n \geq 2$ the prime divisors p of f_n are of the form $p = s \cdot 2^{n+2} + 1$.*

Solution. Because $p | f_n$, it follows that $2^{2^n} \equiv -1 \pmod{p}$. Let i be the least positive integer such that $2^i \equiv 1 \pmod{p}$. By squaring the congruence $2^{2^n} \equiv -1 \pmod{p}$ we get $2^{2^{n+1}} \equiv 1 \pmod{p}$, hence $i | 2^{n+1}$, that is $i = 2^k$ for some nonnegative integer k. If $k \leq n$, then $2^{2^k} \equiv 1 \pmod{p}$, implying $2^{2^n} \equiv 1 \pmod{p}$, which contradicts the congruence $2^{2^n} \equiv -1 \pmod{p}$. Therefore $k = n + 1$. From Fermat's Little Theorem, $2^{p-1} \equiv 1 \pmod{p}$, hence $2^{n+1} | p - 1$, that is $p = h \cdot 2^{n+1} + 1$ for some positive integer h. We get $p = 8t + 1$ and $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = 1$, so 2 is a quadratic residue mod p. Using Euler's criterion, it follows that $2^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. Finally, $2^{n+1} | \frac{p-1}{2}$, hence $\frac{p-1}{2} = s \cdot 2^{n+1}$, that is $p = s \cdot 2^{n+2} + 1$.

Proposed problems

Problem 9.2.4. Find all positive integers n such that $2^n - 1$ is a multiple of 3 and $\frac{2^n - 1}{3}$ is a divisor of $4m^2 + 1$ for some integer m.

(1999 Korean Mathematical Olympiad)

Problem 9.2.5. Prove that the greatest prime factor of f_n, $n \geq 2$, is greater than $2^{n+2}(n+1)$.

(2005 Chinese IMO Team Selection Test)

9.2.2 Mersenne's numbers

The integers $M_n = 2^n - 1$, $n \geq 1$, are called *Mersenne*[2] *'s numbers*. It is clear that if n is composite, then so is M_n. Moreover, if $n = ab$, where a and b are integers greater than 1, then M_a and M_b both divide M_n. But there are primes n for which M_n is composite. For example $47|M_{23}$, $167|M_{83}$, $263|M_{13}$, and so on.

It is not known if there are infinitely many primes with this property. The largest known prime is

$$2^{32582657} - 1$$

and it a Mersenne's number. Presently, we know 42 Mersenne's numbers which are primes.

Theorem 9.2.1. *Let p be an odd prime and let q be a prime divisor of M_p. Then $q = 2kp + 1$ for some positive integer k.*

Proof. From the congruence $2^p \equiv 1 \pmod{q}$ and from the fact that p is a prime, it follows that p is the least positive integer satisfying this property. By using Fermat's Little Theorem, we have $2^{q-1} \equiv 1 \pmod{q}$, hence $p|q-1$. But $q-1$ is an even integer, so $q-1 = 2kp$ and the conclusion follows.

Problem 9.2.6. *Let p be a prime of the form $4k+3$. Then $2p+1$ is a prime if and only if $2p+1$ divides M_p.*

Solution. Suppose that $q = 2p + 1$ is a prime. Then

$$\left(\frac{2}{q}\right) = (-1)^{\frac{q^2-1}{8}} = (-1)^{\frac{p(p+1)}{2}} = (-1)^{2(k+1)(4k+3)} = 1,$$

hence 2 is a quadratic residue mod q.

Using Euler's criterion it follows that $2^{\frac{q-1}{2}} \equiv 1 \pmod{q}$, that is $2^p \equiv 1 \pmod{q}$ and the conclusion follows.

[2] *Marin Mersenne* (1588-1648), French monk who is best known for his role as a clearing house for correspondence eminent philosophers and scientists and for his work in number theory.

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If q is composite, then it has a prime divisor q_1 such that $q_1 \leq \sqrt{q}$. Using Fermat's Little Theorem, we have $2^{q_1-1} \equiv 1 \pmod{q_1}$. But $2^p \equiv 1 \pmod{q_1}$ with p prime implies that p is the least positive integer with the property. Hence $p|q_1 - 1$, thus $q_1 \geq p+1 > \sqrt{p}$, contradicting the choice of q_1. Therefore q must be a prime and the conclusion follows.

Proposed problems

Problem 9.2.7. Let P^* denote all the odd primes less than 10000, and suppose $p \in P^*$. For each subset $S = \{p_1, p_2, \ldots, p_k\}$ of P^*, with $k \geq 2$ and not including p, there exists a $q \in P^* \setminus S$ such that

$$(q+1)|(p_1+1)(p_2+1)\ldots(p_k+1).$$

Find all such possible values of p.

(1999 Taiwanese Mathematical Olympiad)

9.2.3 Perfect numbers

An integer $n \geq 2$ is called *perfect* if the sum of its divisors is equal to $2n$. That is $\sigma(n) = 2n$. For example, the numbers 6, 28, 496 are perfect. The even perfect numbers are closely related to Mersenne's numbers.

Theorem 9.2.2. (Euclid) *If M_k is a prime, then $n = 2^{k-1}M_k$ is a perfect number.*

Proof. Because $gcd(2^{k-1}, 2^k - 1) = 1$, and the fact that σ is a multiplicative function, it follows that

$$\sigma(n) = \sigma(2^{k-1})\sigma(2^k - 1) = (2^k - 1) \cdot 2^k = 2n. \qquad \square$$

There is also a partial converse, due to Euler.

Theorem 9.2.3. *If the even positive integer n is perfect, then $n = 2^{k-1}M_k$ for some positive integer k for which M_k is a prime.*

Proof. Let $n = 2^t u$, where $t \geq 1$ and u is odd. Because n is perfect, we have $\sigma(n) = 2n$, hence $\sigma(2^t u) = 2^{t+1}u$. Using again that σ is multiplicative, we get

$$\sigma(2^t u) = \sigma(2^t)\sigma(u) = (2^{t+1} - 1)\sigma(u).$$

This is equivalent to

$$(2^{t+1} - 1)\sigma(u) = 2^{t+1}u.$$

Because $gcd(2^{t+1}-1, 2^{t+1}) = 1$, it follows that $2^{t+1}|\sigma(u)$, hence $\sigma(u) = 2^{t+1}v$ for some positive integer v. We obtain $u = (2^{t+1}-1)v$.

The next step is to show that $v = 1$. If $v > 1$, then

$$\sigma(u) \geq 1 + v + 2^{t+1} - 1 + v(2^{t+1}-1) = (v+1)2^{t+1} > v \cdot 2^{t+1} = \sigma(u),$$

a contradiction. We get $v = 1$, hence $u = 2^{t+1} - 1 = M_{t+1}$ and $\sigma(u) = 2^{t+1}$. If M_{t+1} is not a prime, then $\sigma(u) > 2^{t+1}$, which is impossible. Finally, $n = 2^{k-1}M_k$, where $k = t+1$. □

Remark. Recall that M_k is a prime only if k is a prime. This fact reflects also in Theorem 9.2.2 and Theorem 9.2.3.

Problem 9.2.8. *Show that any even perfect number is triangular.*

Solution. Using Theorem 9.2.3, we have

$$n = 2^{k-1}M_k = \frac{2^k}{2}(2^k - 1) = \frac{m(m+1)}{2},$$

where $m = 2^k - 1$ and we are done.

Proposed problems

Problem 9.2.9. Prove that if n is an even perfect number, then $8n+1$ is a perfect square.

Problem 9.2.10. Show that if k is an odd positive integer, then $2^{k-1}M_k$ can be written as the sum of the cubes of the first $2^{\frac{k-1}{2}}$ odd positive integers. In particular, any perfect number has this property.

9.3 Sequences of integers

9.3.1 Fibonacci and Lucas sequences

Leonardo Fibonacci[3] introduced in 1228 the sequence $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$, $n \geq 2$. It is not difficult to prove by induction that the closed form for F_n is given by the Binet's formula

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right] \tag{1}$$

[3] *Leonardo Pisano Fibonacci* (1170-1250) played an important role in reviving ancient mathematics and made significant contributions of his "Liber abaci" introduced the Hindu-Arabic place-valued decimal system and the use of Arabic numerals into Europe.

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for all $n \geq 1$. As a consequence of the recursive definition or of formula above, it is a convention to define $F_0 = 0$.

In what follows we give some arithmetical properties of the Fibonacci numbers.

1) If $m|n$, then $F_m|F_n$. If $n \geq 5$ and F_n is a prime, then so is n.

2) For any $m, n \geq 1$, $gcd(F_m, F_n) = F_{gcd(m,n)}$.

3) If $gcd(m, n) = 1$, then $F_m F_n | F_{mn}$.

In order to prove 1) suppose that $n = mk$ for some integer $k > 1$ and denote $\alpha = \dfrac{1+\sqrt{5}}{2}$, $\beta = \dfrac{1-\sqrt{5}}{2}$. Using (1), we have

$$\frac{F_n}{F_m} = \frac{\alpha^n - \beta^n}{\alpha^m - \beta^m} = \frac{(\alpha^m)^k - (\beta^m)^k}{\alpha^m - \beta^m} = \alpha^{m(k-1)} + \alpha^{m(k-2)}\beta^m + \cdots + \beta^{m(k-1)}.$$

Because $\alpha + \beta = 1$ and $\alpha\beta = -1$ it follows by induction that $\alpha^i + \beta^i$ is an integer for all integers $i \geq 1$ and the conclusion follows.

It is now clear that if $n = kh$, $k \geq 3$, then F_k divides F_n hence F_n is not a prime.

For 2) let $d = gcd(m, n)$ and suppose that $n > m$. Applying Euclid's Algorithm, we get

$$n = mq_1 + r_1$$

$$m = r_1 q_2 + r_2$$

$$r_1 = r_2 q_3 + r_3$$

$$\cdots$$

$$r_{i-1} = r_i q_{i+1}$$

and so $d = r_i$. We have

$$gcd(F_m, F_n) = gcd(F_m, F_{mq_1+r_1}) = gcd(F_m, F_{mq_1-1}F_{r_1} + F_{mq_1}F_{r_1+1})$$

$$= gcd(F_m, F_{mq_1-1}F_{r_1}) = gcd(F_m, F_{r_1})$$

because it is not difficult to check that for any positive integers m, n,

$$F_{m+n} = F_{m-1}F_n + F_m F_{n+1} \tag{2}$$

property 1), and the fact that $gcd(F_{mk-1}, F_m) = 1$.

By applying repeatedly this procedure, we arrive at

$$gcd(F_m, F_n) = gcd(F_m, F_{r_1}) = gcd(F_{r_1}, F_{r_2})$$

$$= \cdots = gcd(F_{r_i-1}, F_{r_i}) = F_{r_i} = F_d.$$

Property 3) follows from 2) by observing that

$$gcd(F_m, F_n) = F_{gcd(m,n)} = F_1 = 1$$

and then by using 1).

Lucas' sequence is defined by $L_0 = 2$, $L_1 = 1$, and $L_{n+1} = L_n + L_{n-1}$, $n \geq 1$. The Lucas numbers are the companions to the Fibonacci numbers because they satisfy the same recurrence.

The analog of Binet's Fibonacci number formula for Lucas numbers is

$$L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n, \quad n \geq 0. \tag{3}$$

Problem 9.3.1. *Show that there is a positive number in the Fibonacci sequence which is divisible by 1000.*

(1999 Irish Mathematical Olympiad)

Solution. In fact, for any natural number n, there exist infinitely many positive Fibonacci numbers divisible by n.

Consider ordered pairs of consecutive Fibonacci numbers (F_0, F_1), $(F_1, F_2), \ldots$ taken modulo n. Because the Fibonacci sequence is infinite and there are only n^2 possible ordered pairs of integers modulo n, two such pairs (F_j, F_{j+1}) must be congruent: $F_i \equiv F_{i+m}$ and $F_{i+1} \equiv F_{i+m+1} \pmod{n}$ for some i and m.

If $i \geq 1$ then $F_{i-1} \equiv F_{i+1} - F_i \equiv F_{i+m+1} - F_{i+m} \equiv F_{i+m-1} \pmod{n}$. Likewise, $F_{i+2} \equiv F_{i+1} + F_i \equiv F_{i+m+1} + F_{i+m} \equiv F_{i+2+m} \pmod{n}$. Continuing similarly, we have $F_j \equiv F_{j+m} \pmod{n}$ for all $j \geq 0$. In particular, $0 = F_0 \equiv F_m \equiv F_{2m} \equiv \pmod{n}$, so the numbers $F_m, F_{2m}, \ldots$ are all positive Fibonacci numbers divisible by n. Applying this to $n = 1000$, we are done.

Problem 9.3.2. *Prove that*

(i) The statement "$F_{n+k} - F_n$ is divisible by 10 for all positive integers n" is true if $k = 60$ and false for any positive integer $k < 60$;

(ii) The statement "$F_{n+t} - F_n$ is divisible by 100 for all positive integers n" is true if $t = 300$ and false for any positive integer $t < 300$.

(1996 Irish Mathematical Olympiad)

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Solution. A direct computation shows that the Fibonacci sequence has period 3 modulo 2 and 20 modulo 5 (compute terms until the initial terms 0, 1 repeat, at which time the entire sequence repeats), yielding (a). As for (b), one computes that the period mod 4 is 6. The period mod 25 turns out to be 100, which is awfully many terms to compute by hand, but knowing that the period must be a multiple of 20 helps, and verifying the recurrence $F_{n+8} = tF_{n+4} + F_n$, where t is an integer congruent to 2 modulo 5, shows that the period divides 100; finally, an explicit computation shows that the period is not 20.

Problem 9.3.3. *Let $(a_n)_{n\geq 0}$ be the sequence defined by $a_0 = 0$, $a_1 = 1$ and*

$$\frac{a_{n+1} - 3a_n + a_{n-1}}{2} = (-1)^n$$

for all integers $n > 0$. Prove that a_n is a perfect square for all $n \geq 0$.

Solution. Note that $a_2 = 1$, $a_3 = 4$, $a_4 = 9$, $a_5 = 25$, so $a_0 = F_0^2$, $a_1 = F_1^2$, $a_2 = F_2^2$, $a_3 = F_3^2$, $a_4 = F_4^2$, $a_5 = F_5^2$, where $(F_n)_{n\geq 0}$ is the Fibonacci sequence.

We induct on n to prove that $a_n = F_n^2$ for all $n \geq 0$. Assume that $a_k = F_k^2$ for all $k \leq n$. Hence

$$a_n = F_n^2, \quad a_{n-1} = F_{n-1}^2, \quad a_{n-2} = F_{n-2}^2. \tag{1}$$

From the given relation we obtain

$$a_{n+1} - 3a_n + a_{n-1} = 2(-1)^n$$

and

$$a_n - 3a_{n-1} + a_{n-2} = 2(-1)^{n-1}, \quad n \geq 2.$$

Summing up these equalities yields

$$a_{n+1} - 2a_n - 2a_{n-1} + a_{n-2} = 0, \quad n \geq 2. \tag{2}$$

Using the relations (1) and (2) we obtain

$$a_{n+1} = 2F_n^2 + 2F_{n-1}^2 - F_{n-2}^2 = (F_n + F_{n-1})^2 + (F_n - F_{n-1})^2 - F_{n-2}^2 =$$

$$= F_{n+1}^2 + F_{n-2}^2 - F_{n-2}^2 = F_{n+1}^2,$$

as desired.

Problem 9.3.4. *Define the sequence $(a_n)_{n\geq 0}$ by $a_0 = 0$, $a_1 = 1$, $a_2 = 2$, $a_3 = 6$ and*

$$a_{n+4} = 2a_{n+3} + a_{n+2} - 2a_{n+1} - a_n, \quad n \geq 0.$$

Prove that n divides a_n for all $n > 0$.

Solution. From the hypothesis it follows that $a_4 = 12$, $a_5 = 25$, $a_6 = 48$. We have $\frac{a_1}{1}$, $\frac{a_2}{2} = 1$, $\frac{a_3}{3} = 2$, $\frac{a_4}{4} = 3$, $\frac{a_5}{5} = 5$, $\frac{a_6}{6} = 8$ so $\frac{a_n}{n} = F_n$ for all $n = 1, 2, 3, 4, 5, 6$, where $(F_n)_{n\geq 1}$ is the Fibonacci's sequence.

We prove by induction that $a_n = nF_n$ for all n. Indeed assuming that $a_k = kF_k$ for $k \leq n+3$, we have

$$a_{n+4} = 2(n+3)F_{n+3} + (n+2)F_{n+2} - 2(n+1)F_{n+1} - nF_n =$$

$$= 2(n+3)F_{n+3} + (n+2)F_{n+2} - 2(n+1)F_{n+1} - n(F_{n+2} - F_{n+1}) =$$

$$= 2(n+3)F_{n+3} + 2F_{n+2} - (n+2)F_{n+1} =$$

$$= 2(n+3)F_{n+3} + 2F_{n+2} - (n+2)(F_{n+3} - F_{n+2}) =$$

$$= (n+4)(F_{n+3} + F_{n+2}) = (n+4)F_{n+4},$$

as desired.

Proposed problems

Problem 9.3.5. Determine the maximum value of $m^2 + n^2$, where m and n are integers satisfying $1 \leq m, n \leq 1981$ and $(n^2 - mn - m^2)^2 = 1$.

(22nd IMO)

Problem 9.3.6. Prove that for any integer $n \geq 4$, $F_n + 1$ is not a prime.

Problem 9.3.7. Let k be an integer greater than 1, $a_0 = 4$, $a_1 = a_2 = (k^2 - 2)^2$ and

$$a_{n+1} = a_n a_{n-1} - 2(a_n + a_{n-1}) - a_{n-2} + 8 \text{ for } n \geq 2.$$

Prove that $2 + \sqrt{a_n}$ is a perfect square for all n.

9.3.2 Problems involving linear recursive relations

A sequence $x_0, x_1, x_2, \ldots$ of complex numbers is defined recursively by a *linear recurrence of order k* if

$$x_n = a_1 x_{n-1} + a_2 x_{n-2} + \cdots + a_k x_{n-k}, \quad n \geq k \tag{1}$$

where $a_1, a_2, \ldots, a_k$ are given complex numbers and $x_0 = \alpha_0$, $x_1 = \alpha_1, \ldots,$ $x_{k-1} = \alpha_{k-1}$ are also given.

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The main problem is to find a general formula for x_n in terms of $a_1, a_2, \ldots, a_k,\ \alpha_0, \alpha_1, \ldots, \alpha_{k-1}$ and n. In order to solve this problem we attach to (1) the algebraic equation

$$t^k - a_1 t^{k-1} - a_2 t^{k-2} - \cdots - a_k = 0, \tag{2}$$

which is called the characteristic equation of (1).

Theorem 9.3.1. *If the characteristic equation (2) has distinct roots $t_1, t_2, \ldots, t_k$, then*

$$x_n = c_1 t_1^n + c_2 t_2^n + \cdots + c_k t_k^n \tag{3}$$

where the constants $c_1, c_2, \ldots, c_k$ are determined by the initial conditions $x_0 = \alpha_0,\ x_1 = \alpha_1, \ldots,\ x_{k-1} = \alpha_{k-1}$.

Proof. Consider the sequence $y_0, y_1, y_2, \ldots$ given by

$$y_n = c_1 t_1^n + c_2 t_2^n + \cdots + c_n t_n^n.$$

It is not difficult to prove that the sequence $(y_n)_{n\geq 0}$ satisfies the linear recurrence (1), since $t_1, t_2, \ldots, t_k$ are the roots of the characteristic equation (2). Consider the following system of linear equations:

$$\begin{aligned} &c_1 + c_2 + \cdots + c_k = \alpha_0 \\ &c_1 t_1 + c_2 t_2 + \cdots + c_k t_k = \alpha_1 \\ &\ldots \\ &c_1 t_1^{k-1} + c_2 t_2^{k-1} + \cdots + c_k t_k^{k-1} = \alpha_{k-1} \end{aligned} \tag{4}$$

whose determinant is the so-called Vandermonde determinant

$$V(t_1, t_2, \ldots, t_k) = \prod_{1\leq i<j\leq k} (t_j - t_i).$$

This determinant is nonzero, because $t_1, t_2, \ldots, t_k$ are distinct.

Hence $c_1, c_2, \ldots, c_k$ are uniquely determined as solution to system (4). Moreover, $y_0 = \alpha_0 = x_0$, $y_1 = \alpha_1 = x_1, \ldots,\ y_{k-1} = \alpha_{k-1} = x_{k-1}$. Using strong induction, from (1) it follows that $y_n = x_n$ for all n. □

The case when the roots of the characteristic equation (2) are not distinct is addressed in the following theorem.

Theorem 9.3.2. *Suppose that the equation (2) has the distinct roots $t_1, \ldots, t_h$, with multiplicities $s_1, \ldots, s_h$, respectively. Then x_n is a linear combination of*

$$\begin{aligned} &t_1^n, nt_1^n, \ldots n^{s_1-1} t_1^n \\ &\ldots \\ &t_h^n, nt_h^n, \ldots, n^s h^{-1} t_h^n \end{aligned}$$

The proof of this result uses the so-called Hermite's interpolation polynomial or formal series.

The most frequent situation is when $k = 2$. Then the linear recurrence becomes

$$x_n = a_1 x_{n-1} + a_2 x_{n-2}, \quad n \geq 2$$

where a_1, a_2 are given complex numbers and $x_0 = \alpha_0$, $x_1 = \alpha_1$.

If the characteristic equation $t^2 - a_1 t - a_2 = 0$ has distinct roots t_1, t_2, then

$$x_n = c_1 t_1^n + c_2 t_2^n, \quad n \geq 0,$$

where c_1, c_2 are solutions to the system of linear equations

$$c_1 + c_2 = \alpha_0, \ c_1 t_1 + c_2 t_2 = \alpha_1,$$

that is

$$c_1 = \frac{\alpha_1 - \alpha_0 t_2}{t_1 - t_2}, \quad c_2 = \frac{\alpha_0 t_1 - \alpha_1}{t_1 - t_2}.$$

If the characteristic equation has the nonzero double root t_1, then

$$x_n = c_1 t_1^n + c_2 n t_1^n = (c_1 + c_2 n) t_1^n,$$

where c_1, c_2 are determined from the system of equations $x_0 = \alpha_0$, $x_1 = \alpha_1$, that is

$$c_1 = \alpha_0, \quad c_2 = \frac{\alpha_1 - \alpha_0 t_1}{t_1}.$$

Example. Let us find the general term of the sequence

$$P_0 = 0, \ P_1 = 1, \ldots, P_n = 2P_{n-1} + P_{n-2}, \quad n \geq 2.$$

The characteristic equation is $t^2 - 2t - 1 = 0$, whose roots are $t_1 = 1 + \sqrt{2}$ and $t_2 = 1 - \sqrt{2}$. We have $P_n = c_1 t_1^n + c_2 t_2^n$, $n \geq 0$, where $c_1 + c_2 = 0$ and $c_1(1 + \sqrt{2}) + c_2(1 - \sqrt{2}) = 1$, hence

$$P_n = \frac{1}{2\sqrt{2}}[(1 + \sqrt{2})^n - (1 - \sqrt{2})^n], \quad n \geq 0.$$

This sequence is called the *Pell's sequence* and it plays an important part in Diophantine equations.

In some situations we encounter nonhomogeneous recurrences of order k of the form

$$x_n = a_1 x_{n-1} + a_2 x_{n-2} + \cdots + a_k x_{n-k} + b, \quad n \geq k,$$

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where $a_1, a_2, \ldots, a_k, b$ are given complex numbers and $x_1 = \alpha_1$, $x_2 = \alpha_2, \ldots, x_{k-1} = \alpha_{k-1}$. The method of attack consists of performing a translation $x_n = y_n + \beta$, where β is the solution to the equation $(1 - a_1 - a_2 - \cdots - a_k)\beta = b$ when $a_1 + a_2 + \cdots + a_k \neq 1$. The sequence $(y_n)_{n\geq 0}$ satisfies the linear recurrence (1).

Example. Let us find x_n if $x_0 = \alpha$, $x_n = ax_{n-1} + b$, $n \geq 1$.

If $a = 1$, we have an arithmetical sequence whose first term is α and whose common difference is b. In this case $x_n = \alpha + nb$.

If $a \neq 1$, we perform the translation $x_n = y_n + \beta$, where $\beta = \dfrac{b}{1-a}$. In this case $(y_n)_{n\geq 0}$ satisfies the recurrence $y_0 = \alpha - \beta$, $y_n = ay_{n-1}$, $n \geq 1$, which is a geometric sequence whose first term is $\alpha - \beta$ and whose ratio is a. We obtain $y_n = (\alpha - \beta)a^n$, hence

$$x_n = \left(\alpha - \frac{b}{1-a}\right) a^n + \frac{b}{1-a}, \quad n \geq 0.$$

Problem 9.3.8. *Let a and b be positive integers and let the sequence $(x_n)_{n\geq 0}$ be defined by $x_0 = 1$ and $x_{n+1} = ax_n + b$ for all nonnegative integers n. Prove that for any choice of a and b, the sequence $(x_n)_{n\geq 0}$ contains infinitely many composite numbers.*

(1995 German Mathematical Olympiad)

Solution. Assume on the contrary that x_n is composite for only finitely many n. Take N large than all such n, so that x_m is prime for all $n > N$. Choose such a prime $x_m = p$ not dividing $a - 1$ (this excludes only finitely many candidates). Let t be such that $t(1 - a) \equiv b \pmod{p}$; then

$$x_{n+1} - t \equiv ax_n + b - b = a(x_n - t) \pmod{p}.$$

In particular,

$$x_{m+p-1} = t + (x_{m+p-1} - t) \equiv t + a^{p-1}(x_m - t) \equiv 0 \pmod{p}.$$

However, x_{m+p-1} is a prime greater than p, yielding a contradiction. Hence infinitely many of the x_n are composite.

Problem 9.3.9. *Find a_n if $a_0 = 1$ and $a_{n+1} = 2a_n + \sqrt{3a_n^2 - 2}$, $n \geq 0$.*

Solution. We have $(a_{n+1} - 2a_n)^2 = 3a_n^2 - 2$, so

$$a_{n+1}^2 - 4a_{n+1}a_n + a_n^2 + 2 = 0, \quad n \geq 0.$$

Then

$$a_n^2 - 4a_n a_{n-1} + a_{n-1}^2 + 2 = 0, \quad n \geq 1,$$

hence, by subtraction,

$$a_{n+1}^2 - a_{n-1}^2 - 4a_n(a_{n+1} - a_{n-1}) = 0$$

for all $n \geq 1$. Because it is clear that $(a_n)_{n\geq 0}$ is increasing we have $a_{n+1} - a_{n-1} \neq 0$, for all $n \geq 1$, so

$$a_{n+1} + a_{n-1} - 4a_n = 0, \quad n \geq 1,$$

that is $a_{n+1} = 4a_n - a_{n-1}$, $n \geq 1$. Moreover, $a_0 = 1$ and $a_1 = 3$. The characteristic equation is $t^2 - 4t + 1 = 0$, whose roots are $t_1 = 2 + \sqrt{3}$ and $t_2 = 2 - \sqrt{3}$. We obtain

$$a_n = \frac{1}{2\sqrt{3}}[(1+\sqrt{3})(2+\sqrt{3})^n - (1-\sqrt{3})(2-\sqrt{3})^n], \quad n \geq 0.$$

We can also write a_n as follows:

$$a_n = \frac{1}{\sqrt{3}}\left[\left(\frac{1+\sqrt{3}}{2}\right)^{2n+1} - \left(\frac{1-\sqrt{3}}{2}\right)^{2n+1}\right], \quad n \geq 0.$$

Note that from $a_0 = 1$, $a_1 = 3$, and $a_{n+1} = 4a_n - a_{n-1}$ it follows by strong induction that a_n is a positive integer for all n.

Problem 9.3.10. *Consider the sequence $\{a_n\}$ such that $a_0 = 4$, $a_1 = 22$ and $a_n - 6a_{n-1} + a_{n-2} = 0$ for $n \geq 2$. Prove that there exist sequences $\{x_n\}$ and $\{y_n\}$ of positive integers such that*

$$a_n = \frac{y_n^2 + 7}{x_n - y_n}$$

for any $n \geq 0$.

(2001 Bulgarian Mathematical Olympiad)

Solution. Consider the sequence $\{c_n\}$ of positive integers such that $c_0 = 2$, $c_1 = 1$ and $c_n = 2c_{n-1} + c_{n-2}$ for $n \geq 2$.

We prove by induction that $a_n = c_{2n+2}$ for $n \geq 0$. We check the base cases of $a_0 = 4 = c_2$ and $a_1 = 9 = c_4$. Then, for any $k \geq 2$, assuming the claim holds for $n = k-2$ and $n = k-1$,

$$\begin{aligned} c_{2k+2} &= 2c_{2k+1} + c_{2k} \\ &= 2(2c_{2k} + c_{2k-1}) + a_{k-1} \\ &= 4c_{2k} + (c_{2k} - c_{2k-2}) + a_{k-1} \\ &= 6a_{k-1} - a_{k-2} \\ &= a_k, \end{aligned}$$

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so the claim holds for $n = k$ as well, and the induction is complete.

For $n \geq 1$,

$$\begin{pmatrix} a_{n+1} & a_n \\ a_{n+2} & a_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a_n & a_{n-1} \\ a_{n+1} & a_n \end{pmatrix},$$

and

$$\begin{vmatrix} a_{n+1} & a_n \\ a_{n+2} & a_{n+1} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} \begin{vmatrix} a_n & a_{n-1} \\ a_{n+1} & a_n \end{vmatrix} = - \begin{vmatrix} a_n & a_{n-1} \\ a_{n+1} & a_n \end{vmatrix}.$$

Thus, for $n \geq 0$,

$$c_{n+1}^2 - c_n c_{n+2} = (-1)^n (c_1^2 - c_0 c_2) = (-1)^n (1^2 - 2 \cdot 4) = (-1)^n (-7).$$

In particular, for all $n \geq 0$,

$$c_{2n+1}^2 - c_{2n} a_n = c_{2n+1}^2 - c_{2n} c_{2n+2} = (-1)^{2n} (-7) = -7,$$

and

$$a_n = \frac{c_{2n+1}^2 + 7}{c_{2n}}.$$

We may therefore take $y_n = c_{2n+1}$ and $x_n = c_{2n} + y_n$.

Problem 9.3.11. *The sequence $a_1, a_2, \ldots$ is defined by the initial conditions $a_1 = 20$, $a_2 = 30$ and the recursion $a_{n+2} = 3a_{n+1} - a_n$ for $n \geq 1$. Find all positive integers n for which $1 + 5a_n a_{n+1}$ is a perfect square.*

(2002 Balkan Mathematical Olympiad)

Solution. The only solution is $n = 3$. We can check that $20 \cdot 30 \cdot 5 + 1 = 3001$ and $30 \cdot 70 \cdot 5 + 1 = 10501$ are not perfect squares, while $70 \cdot 180 \cdot 5 + 1 = 63001 = 251^2$ is a perfect square. Then we must only prove that $1 + 5a_n a_{n+1}$ is not a perfect square for $n \geq 4$. First, we will prove a lemma.

Lemma. *For any integer $n \geq 2$,*

$$a_n^2 + 500 = a_{n-1} a_{n+1}.$$

Proof. We will prove this by induction on n. In the base case, $30^2 + 500 = 1400 = 20 \cdot 70$. Now assume that $a_n^2 + 500 = a_{n-1} a_{n+1}$. Then

$$a_n a_{n+2} = (3a_{n+1} - a_n)(a_n) = 3a_{n+1} a_n - a_n^2$$

$$= 3a_{n+1} a_n - (a_{n-1} a_{n+1} - 500) = 500 + a_{n+1}(3a_n - a_{n-1}) = 500 + a_{n+1}^2,$$

proving the inductive step. Therefore the desired statement is true from induction. □

Now, for $n \geq 4$, $(a_n + a_{n+1})^2 = a_n^2 + a_{n+1}^2 + 2a_n a_{n+1}$. But

$$a_{n+1}^2 = 9a_n^2 + a_{n-1}^2 - 6a_{n-1}a_n,$$

so

$$\begin{aligned}(a_n + a_{n+1})^2 &= 2a_n a_{n+1} + 3a_n(3a_n - a_{n-1}) + a_{n-2}^2 + a_n^2 - 3a_n a_{n-1} \\ &= 5a_n a_{n+1} + a_{n-1}^2 - a_n a_n - 2 \\ &= 5a_n a_{n+1} + a_{n-1}^2 - (a_{n-1}^2 + 500) = 5a_n a_{n+1} - 500,\end{aligned}$$

by the lemma and the definition of a.

Therefore $(a_n + a_{n+1})^2 = 5a_n a_{n+1} - 500 < 5a_n a_{n+1} + 1$. Since a_n is increasing and $n \geq 4$,

$$a_n + a_{n+1} \geq 180 + 470 = 650,$$

so

$$\begin{aligned}(a_n + a_{n+1} + 1)^2 &= (a_n + a_{n+1})^2 + 2(a_n + a_{n+1}) + 1 \\ &> (a_n + a_{n+1})^2 + 501 = 5a_n a_{n+1} + 1.\end{aligned}$$

Because two adjacent integers have squares above and below $5a_n a_{n+1}+1$, that value is not a perfect square for $n \geq 4$.

Proposed problems

Problem 9.3.12. Let a, b be integers greater than 1. The sequence x_1, $x_2, \ldots$ is defined by the initial conditions $x_0 = 0$, $x_1 = 1$ and the recursion

$$x_{2n} = ax_{2n-1} - x_{2n-2}, \quad x_{2n+1} = bx_{2n} - x_{2n-1}$$

for $n \geq 1$. Prove that for any natural numbers m and n, the product $x_{n+m}x_{n+m-1}\ldots x_{n+1}$ is divisible by $x_m x_{m-1}$.

(2001 St. Petersburg City Mathematical Olympiad)

Problem 9.3.13. Let m be a positive integer. Define the sequence $\{a_n\}_{n\geq 0}$ by $a_0 = 0$, $a_1 = m$ and $a_{n+1} = m^2 a_n - a_{n-1}$ for $n \geq 1$. Prove that an ordered pair (a, b) of nonnegative integers, with $a \leq b$, is a solution of the equation

$$\frac{a^2 + b^2}{ab + 1} = m^2$$

if and only if $(a, b) = (a_n, a_{n+1})$ for some $n \geq 0$.

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(1998 Canadian Mathematical Olympiad)

Problem 9.3.14. Let b, c be positive integers, and define the sequence $a_1, a_2, \ldots$ by $a_1 = b$, $a_2 = c$, and

$$a_{n+2} = |3a_{n+1} - 2a_n|$$

for $n \geq 1$. Find all such (b, c) for which the sequence $a_1, a_2, \ldots$ has only a finite number of composite terms.

(2002 Bulgarian Mathematical Olympiad)

9.3.3 Nonstandard sequences of integers

Problem 9.3.15. *Let k be a positive integer. The sequence a_n is defined by $a_1 = 1$, and a_n is the n-th positive integer greater than a_{n-1} which is congruent to n modulo k. Find a_n is closed form.*

(1997 Austrian Mathematical Olympiad)

Solution. We have $a_n = \dfrac{n(2 + (n-1)k}{2}$. If $k = 2$, then $a_n = n^2$. First, observe that $a_1 \equiv 1 \pmod{k}$. Thus, for all n, $a_n \equiv n \pmod{k}$, and the first positive integer greater than a_{n-1} which is congruent to n modulo k must be $a_{n-1} + 1$. The n-th positive integer greater than a_{n-1} that is congruent to n modulo k is simply $(n-1)k$ more than the first positive integer greater than a_{n-1} which satisfies that condition. Therefore, $a_n = a_{n-1}+1+(n-1)k$. Solving this recursion gives

$$a_n = n + \frac{(n-1)n}{2}k.$$

Problem 9.3.16. *Let $a_1 = 19$, $a_2 = 98$. For $n \geq 1$, define a_{n+2} to be the remainder of $a_n + a_{n+1}$ when it is divided by 100. What is the remainder when*

$$a_1^2 + a_2^2 + \cdots + a_{1998}^2$$

is divided by 8?

(1998 United Kingdom Mathematical Olympiad)

Solution. The answer is 0. Consider $a_n \pmod 4$ which is not changed by taking the remainder divided by 100, there's the cycle 3, 2, 1, 3, 0, 3 which repeats 333 times. Then

$$a_1^2 + a_2^2 + \cdots + a_{1998}^2 \equiv 333(1 + 4 + 1 + 1 + 0 + 1) \equiv 0 \pmod{8},$$

as claimed.

Problem 9.3.17. *A sequence of integers $\{a_n\}_{n\geq 1}$ satisfies the following recursive relation*

$$a_{n+1} = a_n^3 + 1999 \text{ for } n = 1, 2, \ldots$$

Prove that there exists at most one n for which a_n is a perfect square.

(1999 Austrian-Polish Mathematics Competition)

Solution. Consider the possible values of (a_n, a_{n+1}) modulo 4:

a_n	0	1	2	3
a_{n+1}	3	0	3	2

No matter what a_1 is, the terms $a_3, a_4, \ldots$ are all 2 or 3 (mod 4). However, all perfect squares are 0 or 1 (mod 4), so at most two terms (a_1 and a_2) can be perfect squares. If a_1 and a_2 are both perfect squares, then writing $a_1 = a^2$, $a_2 = b^2$ we have $a^6 + 1999 = b^2$ or $1999 = b^2 - (a^3)^2 = (b+a^3)(b-a^3)$. Because 1999 is prime, $b - a^3 = 1$ and $b + a^3 = 1999$. Thus $a^3 = \dfrac{1999-1}{2} = 999$, which is impossible. Hence at most one term of the sequence is a perfect square.

Problem 9.3.18. *Determine if there exists an infinite sequence of positive integers such that*

(i) no term divides any other term;

(ii) every pair of terms has a common divisor greater than 1, but no integer greater than 1 divides all the terms.

(1999 Hungarian Mathematical Olympiad)

Solution. The desired sequence exists. Let $p_0, p_1, \ldots$ be the primes greater than 5 in order, and let $q_{3i} = 6$, $q_{3i+1} = 10$, $q_{3i+2} = 15$ for each nonnegative integer i. Then let $s_i = p_i q_i$ for all $i \geq 0$. The sequence $s_0, s_1, s_2, \ldots$ clearly satisfies (i) because s_i is not even divisible by p_j for $i \neq j$. For the first part of (ii), any two terms have their indices both in $\{0, 1\}$, both in $\{0, 2\}$, or both in $\{1, 2\}$ (mod 3), so they have a common divisor of 2, 3, or 5, respectively. For the second part, we just need to check that no prime divides all the s_i. Indeed, $2 \nmid s_2$, $3 \nmid s_1$, $5 \nmid s_0$, and no prime greater than 5 divides more than one s_i.

Problem 9.3.19. *Let $a_1, a_2, \ldots$ be a sequence satisfying $a_1 = 2$, $a_2 = 5$ and*

$$a_{n+2} = (2 - n^2)a_{n+1} + (2 + n^2)a_n$$

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for all $n \geq 1$. Do there exist indices p, q and r such that $a_p a_q = a_r$?

(1995 Czech-Slovak Match)

Solution. No such p, q, r exist. We show that for all n, $a_n \equiv 2 \pmod 3$. This holds for $n = 1$ and $n = 2$ by assumption and follows for all n by induction:

$$a_{n+2} = (2 - n^2)a_{n+1} + (2 + n^2)a_n$$
$$\equiv 2(2 - n^2) + 2(2 + n^2) = 8 \equiv 2 \pmod 3.$$

Hence for any p, q, r, $a_p a_q \equiv 1 \pmod 3$ while $a_r \equiv 2 \pmod 3$, so $a_p a_q \neq a_r$.

Problem 9.3.20. *Is there a sequence of natural numbers in which every natural number occurs just once and moreover, for any $k = 1, 2, 3, \ldots$ the sum of the first k terms is divisible by k?*

(1995 Russian Mathematical Olympiad)

Solution. We recursively construct such a sequence. Suppose $a_1, \ldots, a_m$ have been chosen, with $s = a_1 + \cdots + a_m$, and let n be the smallest number not yet appearing. By the Chinese Remainder Theorem, there exists t such that $t \equiv -s \pmod{m+1}$ and $t \equiv -s - n \pmod{m+2}$. We can increase t by a suitably large multiple of $(m+1)(m+2)$ to ensure it does not equal any of $a_1, \ldots, a_m$. Then $a_1, \ldots, a_m, t, n$ also has the desired property, and the construction assures that $1, \ldots, m$ all occur among the first $2m$ terms.

Proposed problems

Problem 9.3.21. Let $\{a_n\}$ be a sequence of integers such that for $n \geq 1$

$$(n-1)a_{n+1} = (n+1)a_n - 2(n-1).$$

If 2000 divides a_{1999}, find the smallest $n \geq 2$ such that 2000 divides a_n.

(1999 Bulgarian Mathematical Olympiad)

Problem 9.3.22. The sequence $(a_n)_{n \geq 0}$ is defined by $a_0 = 1$, $a_1 = 3$ and

$$a_{n+2} = \begin{cases} a_{n+1} + 9a_n & \text{if } n \text{ is even,} \\ 9a_{n+1} + 5a_n & \text{if } n \text{ is odd.} \end{cases}$$

Prove that

(a) $\sum_{k=1995}^{2000} a_k^2$ is divisible by 20,
(b) a_{2n+1} is not a perfect square for every $n = 0, 1, 2, \ldots$

(1995 Vietnamese Mathematical Olympiad)

Problem 9.3.23. Prove that for any natural number $a_1 > 1$, there exists an increasing sequence of natural numbers $a_1, a_2, \ldots$ such that $a_1^2 + a_2^2 + \cdots + a_k^2$ is divisible by $a_1 + a_2 + \cdots + a_k$ for all $k \geq 1$.

(1995 Russian Mathematical Olympiad)

Problem 9.3.24. The sequence $a_0, a_1, a_2, \ldots$ satisfies

$$a_{m+n} + a_{m-n} = \frac{1}{2}(a_{2m} + a_{2n})$$

for all nonnegative integers m and n with $m \geq n$. If $a_1 = 1$, determine a_n.

(1995 Russian Mathematical Olympiad)

Problem 9.3.25. The sequence of real numbers $a_1, a_2, a_3, \ldots$ satisfies the initial conditions $a_1 = 2$, $a_2 = 500$, $a_3 = 2000$ as well as the relation

$$\frac{a_{n+2} + a_{n+1}}{a_{n+1} + a_{n-1}} = \frac{a_{n+1}}{a_{n-1}}$$

for $n = 2, 3, 4, \ldots$ Prove that all the terms of this sequence are positive integers and that 2^{2000} divides the number a_{2000}.

(1999 Slovenian Mathematical Olympiad)

Problem 9.3.26. Let k be a fixed positive integer. We define the sequence $a_1, a_2, \ldots$ by $a_1 = k + 1$ and the recursion $a_{n+1} = a_n^2 - ka_n + k$ for $n \geq 1$. Prove that a_m and a_n are relatively prime for distinct positive integers m and n.

Problem 9.3.27. Suppose the sequence of nonnegative integers a_1, $a_2, \ldots, a_{1997}$ satisfies

$$a_i + a_j \leq a_{i+j} \leq a_i + a_j + 1$$

for all $i, j \geq 1$ with $i + j \leq 1997$. Show that there exists a real number x such that $a_n = \lfloor nx \rfloor$ for all $1 \leq n \leq 1997$.

(1997 USA Mathematical Olympiad)

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Problem 9.3.28. The sequence $\{a_n\}$ is given by the following relation:

$$a_{n+1} = \begin{cases} \dfrac{a_n - 1}{2}, & \text{if} \quad a_n \geq 1, \\ \dfrac{2a_n}{1 - a_n}, & \text{if} \quad a_n < 1. \end{cases}$$

Given that a_0 is a positive integer, $a_n \neq 2$ for each $n = 1, 2, \ldots, 2001$, and $a_{2002} = 2$. Find a_0.

(2002 St. Petersburg City Mathematical Olympiad)

Problem 9.3.29. Let $x_1 = x_2 = x_3 = 1$ and $x_{n+3} = x_n + x_{n+1}x_{n+2}$ for all positive integers n. Prove that for any positive integer m there is an integer $k > 0$ such that m divides x_k.

Problem 9.3.30. Find all infinite bounded sequences $a_1, a_2, \ldots$ of positive integers such that for all $n > 2$,

$$a_n = \frac{a_{n-1} + a_{n-2}}{gcd(a_{n-1}, a_{n-2})}.$$

(1999 Russian Mathematical Olympiad)

Problem 9.3.31. Let $a_1, a_2, \ldots$ be a sequence of positive integers satisfying the condition $0 < a_{n+1} - a_n \leq 2001$ for all integers $n \geq 1$. Prove that there exist an infinite number of ordered pairs (p, q) of distinct positive integers such that a_p is a divisor of a_q.

(2001 Vietnamese Mathematical Olympiad)

Problem 9.3.32. Define the sequence $\{x_n\}_{n \geq 0}$ by $x_0 = 0$ and

$$x_n = \begin{cases} x_{n-1} + \dfrac{3^{r+1} - 1}{2}, & \text{if} \quad n = 3^r(3k + 1), \\ \\ x_{n-1} - \dfrac{3^{r+1} + 1}{2}, & \text{if} \quad n = 3^r(3k + 2), \end{cases}$$

where k and r are nonnegative integers. Prove that every integer appears exactly once in this sequence.

(1999 Iranian Mathematical Olympiad)

Problem 9.3.33. Suppose that $a_1, a_2, \ldots$ is a sequence of natural numbers such that for all natural numbers m and n, $gcd(a_m, a_n) = a_{gcd(m,n)}$. Prove that there exists a sequence $b_1, b_2, \ldots$ of natural numbers such that $a_n = \prod_{d|n} b_d$ for all integers $n \geq 1$.

(2001 Iranian Mathematical Olympiad)

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10

Problems Involving Binomial Coefficients

10.1 Binomial coefficients

One of the main problems leading to considering binomial coefficients is the expansion of $(a+b)^n$, where a, b are complex numbers and n is a positive integer. It is well-known that

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{n-1}ab^{n-1} + \binom{n}{n}b^n,$$

where $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$, $k = 0, 1, \ldots, n$ with the convention $0! = 1$. The integers $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$ are called *binomial coefficients*. They can be obtained recursively by using *Pascal*[1] *'s triangle*:

[1] *Blaise Pascal* (1623-1662) was a very influencial French mathematician and philosophers who contributed to many areas of mathematics.

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$$
\begin{array}{ccccccccccccccccccccc}
 & & & & & & & & 1 & & & & & & & & & & \\
 & & & & & & & 1 & & & 1 & & & & & & & & \\
 & & & & & & 1 & & & 2 & & & 1 & & & & & & \\
 & & & & & 1 & & & 3 & & & 3 & & & 1 & & & & \\
 & & & & 1 & & & 4 & & & 6 & & & 4 & & & 1 & & \\
 & & & 1 & & & 5 & & & 10 & & & 10 & & & 5 & & & 1 \\
\cdots & & \cdots & & \cdots & & \cdots & & \cdots & & \cdots & & \cdots & & \cdots & & \cdots & & \cdots
\end{array}
$$

in which every entry different from 1 is the sum of the two entries above adjacent to it.

The fundamental properties of the binomial coefficients are the following:

1) (symmetry) $\binom{n}{k} = \binom{n}{n-k}$;

2) (Pascal's triangle property) $\binom{n}{k+1} = \binom{n-1}{k+1} + \binom{n-1}{k}$;

3) (monotonicity) $\binom{n}{0} < \binom{n}{1} < \cdots < \binom{n}{\lfloor \frac{n-1}{2} \rfloor + 1} = \binom{n}{\lfloor \frac{n}{2} \rfloor}$;

4) (sum of binomial coefficients) $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$;

5) (alternating sum) $\binom{n}{0} - \binom{n}{1} + \cdots + (-1)^n \binom{n}{n} = 0$;

6) (Vandermonde property) $\sum_{i=0}^{k} \binom{m}{i}\binom{n}{k-i} = \binom{m+n}{k}$;

7) If p is a prime, then $p \mid \binom{p}{k}$, $k = 1, \ldots, p-1$.

Problem 10.1.1. *Let n be an odd positive integer. Prove that the set*

$$\left\{ \binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{\frac{n-1}{2}} \right\}$$

contains an odd number of odd numbers.

Solution. For $n = 1$ the claim is clear, so let $n \geq 3$.

Define $S_n = \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{\frac{n-1}{2}}$. Then

$$2S_n = \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} = 2^n - 2$$

or $S_n = 2^{n-1} - 1$. Because S_n is odd it follows that the sum S_n contains an odd number of odd terms, as desired.

Problem 10.1.2. *Determine all the positive integers $n \geq 3$, such that 2^{2000} is divisible by*

$$1 + \binom{n}{1} + \binom{n}{2} + \binom{n}{3}.$$

(1998 Chinese Mathematical Olympiad)

Solution. The solutions are $n = 3, 7, 23$. Since 2 is a prime,

$$1 + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} = 2^k$$

for some positive integer $k \leq 2000$. We have

$$1 + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} = (n+1)(n^2 - n + 6)/6,$$

i.e., $(n+1)(n^2 - n + 6) = 3 \times 2^{k+1}$. Let $m = n + 1$, then $m \geq 4$ and $m(m^2 - 3m + 8) = 3 \times 2^{k+1}$. We consider the following two cases.

(a) $m = 2^s$. Since $m \geq 4$, $s \geq 2$. We have

$$2^{2s} - 3 \times 2^s + 8 = m^2 - 3m + 8 = 3 \times 2^t$$

for some positive integer t. If $s \geq 4$, then

$$8 \equiv 3 \times 2^t \pmod{16} \Rightarrow 2^t = 8 \Rightarrow m^2 - 3m + 8 = 24 \Rightarrow m(m-3) = 16,$$

which is impossible. Thus either $s = 3$, $m = 8$, $t = 4$, $n = 7$, or $s = 2$, $m = 4$, $t = 2$, $n = 3$.

(b) $m = 3 \times 2^u$. Since $m \geq 4$, $m > 4$ and $u \geq 1$. We have

$$9 \times 2^{2u} - 9 \times 2^u + 8 = m^2 - 3m + 8 = 2^v$$

for some positive integer v. It is easy to check that there is no solution for v when $u = 1, 2$. If $u \geq 4$, we have $8 \equiv 2^v \pmod{16} \Rightarrow v = 3$ and $m(m-3) = 0$, which is impossible. So $u = 3$, $m = 3 \times 2^3 = 24$, $v = 9$, $n = 23$.

Problem 10.1.3. *Let m and n be integers such that $1 \leq m \leq n$. Prove that m is a divisor of*

$$n \sum_{k=0}^{m-1} (-1)^k \binom{n}{k}.$$

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(2001 Hungarian Mathematical Olympiad)

Solution. We can write the given expression as follows:

$$\begin{aligned} n\sum_{k=0}^{m-1}(-1)^k\binom{n}{k} &= n\sum_{k=0}^{m-1}(-1)^k\left(\binom{n-1}{k}+\binom{n-1}{k-1}\right) \\ &= n\sum_{k=0}^{m-1}(-1)^k\binom{n-1}{k}+n\sum_{k=1}^{m-1}(-1)^k\binom{n-1}{k-1} \\ &= n\sum_{k=0}^{m-1}(-1)^k\binom{n-1}{k}-n\sum_{k=0}^{m-2}(-1)^k\binom{n-1}{k} \\ &= n(n-1)^{m-1}\binom{n-1}{m-1} \\ &= m(-1)^{m-1}\binom{n}{m}. \end{aligned}$$

The final expression is clearly divisible by m.

Problem 10.1.4. *Show that for any positive integer n, the number*

$$S_n=\binom{2n+1}{0}\cdot 2^{2n}+\binom{2n+1}{2}\cdot 2^{2n-2}\cdot 3+\cdots+\binom{2n+1}{2n}\cdot 3^n$$

is the sum of two consecutive perfect squares.

(1999 Romanian IMO Team Selection Test)

Solution. It is easy to see that:

$$S_n=\frac{1}{4}[(2+\sqrt{3})^{2n+1}+(2-\sqrt{3})^{2n+1}].$$

The required property says: there exists $k>0$ such that $S_n=(k-1)^2+k^2$, or, equivalently,

$$2k^2-2k+1-S_n=0.$$

The discriminant of this equation is $\Delta=4(2S_n-1)$, and, after usual computations, we obtain

$$\Delta=\left(\frac{(1+\sqrt{3})^{2n+1}+(1-\sqrt{3})^{2n+1}}{2^n}\right)^2.$$

After solving the equation, we find that

$$k=\frac{2^{n+1}+(1+\sqrt{3})^{2n+1}+(1-\sqrt{3})^{2n+1}}{2^{n+2}}.$$

Therefore, it is sufficient to prove that k is an integer number. Let us denote $E_m = (1+\sqrt{3})^m + (1-\sqrt{3})^m$, where m is a positive integer. Clearly, E_m is an integer. We shall prove that $2^{\left[\frac{m}{2}\right]}$ divides E_m, For $E_0 = 2$, $E_1 = 2$, $E_2 = 8$, the assertion is true. Moreover, the numbers E_m satisfy the relation:

$$E_m = 2E_{m-1} + 2E_{m-2}.$$

The property now follows by induction.

Problem 10.1.5. *Prove that for every pair m, k of natural numbers, m has a unique representation in the form*

$$m = \binom{a_k}{k} + \binom{a_{k-1}}{k-1} + \cdots + \binom{a_t}{t},$$

where

$$a_k > a_{k-1} > \cdots > a_t \geq t \geq 1.$$

(1996 Iranian Mathematical Olympiad)

Solution. We first show uniqueness. Suppose m is represented by two sequences $a_k, \ldots, a_t$ and $b_k, \ldots, b_t$. Find the first position in which they differ; without loss of generality, assume this position is k and that $a_k > b_k$. Then

$$m \leq \binom{b_k}{k} + \binom{b_k - 1}{k-1} + \cdots + \binom{b_k - k + 1}{1} < \binom{b_k + 1}{k} \leq m,$$

a contradiction.

To show existence, apply the greedy algorithm: find the largest a_k such that $\binom{a_k}{k} \leq m$, and apply the same algorithm with m and k replaced by $m - \binom{a_k}{k}$ and $k-1$. We need only make sure that the sequence obtained is indeed decreasing, but this follows because by assumption, $m < \binom{a_k + 1}{m}$, and so $m - \binom{a_k}{k} < \binom{a_k}{k-1}$.

Problem 10.1.6. *Show that for any positive integer $n \geq 3$, the least common multiple of the numbers $1, 2, \ldots, n$ is greater than 2^{n-1}.*

(1999 Czech-Slovak Match)

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Solution. For any $n \geq 3$ we have

$$2^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} < \sum_{k=0}^{n-1} \binom{n-1}{\lfloor \frac{n-1}{2} \rfloor} = n \binom{n-1}{\lfloor \frac{n-1}{2} \rfloor}.$$

Hence it suffices to show that $n\binom{n-1}{\lfloor \frac{n-1}{2} \rfloor}$ divides $\operatorname{lcm}(1, 2, \ldots, n)$. Using an argument involving prime factorizations, we will prove the more general assertion that for each $k < n$, $\operatorname{lcm}(n, n-1, \ldots, n-k)$ is divisible by $n\binom{n-1}{k}$.

Let k and n be fixed natural numbers with $k < n$, and let $p \leq n$ be an arbitrary prime. Let p^α be the highest power of p which divides $\operatorname{lcm}(n, n-1, \ldots, n-k)$, where $p^\alpha | n-l$ for some l. Then for each $i \leq \alpha$, we know that $p^i | n-l$. Thus exactly $\left\lfloor \frac{l}{p^i} \right\rfloor$ of $\{n-l+1, n-l+2, \ldots, n\}$ and exactly $\left\lfloor \frac{k-l}{p^i} \right\rfloor$ of $\{n-l-1, n-l-2, \ldots, n-k\}$ are multiples of p^i, so p^i divides $\left\lfloor \frac{l}{p^i} \right\rfloor + \left\lfloor \frac{k-l}{p^i} \right\rfloor \leq \left\lfloor \frac{k}{p^i} \right\rfloor$ of the remaining k numbers, that is, at most the number of multiples of p^i between 1 and k. It follows that p divides

$$n\binom{n-1}{k} = \frac{n(n-1)\ldots(n-l+1)(n-l-1)\ldots(n-k)}{k!}(n-l)$$

at most α times, so that indeed $n\binom{n-1}{k} | \operatorname{lcm}(n, n-1, \ldots, n-k)$.

Proposed problems

Problem 10.1.7. Show that the sequence

$$\binom{2002}{2002}, \binom{2003}{2002}, \binom{2004}{2002}, \ldots$$

considered modulo 2002, is periodic.

(2002 Baltic Mathematical Competition)

Problem 10.1.8. Prove that

$$\binom{2p}{p} \equiv 2 \pmod{p^2}$$

for any prime number p.

Problem 10.1.9. Let k, m, n be positive integers such that $m + k + 1$ is a prime number greater than $n + 1$. Let us denote $C_s = s(s + 1)$. Show that the product

$$(C_{m+1} - C_k)(C_{m+2} - C_k) \dots (C_{m+n} - C_k)$$

is divisible by $C_1 C_2 \dots C_n$.

(18th IMO)

Problem 10.1.10. Let n, k be arbitrary positive integers. Show that there exists positive integers $a_1 > a_2 > a_3 > a_4 > a_5 > k$ such that

$$n = \pm\binom{a_1}{3} \pm \binom{a_2}{3} \pm \binom{a_3}{3} \pm \binom{a_4}{3} \pm \binom{a_5}{3}.$$

(2000 Romanian IMO Team Selection Test)

Problem 10.1.11. Prove that if n and m are integers, and m is odd, then

$$\frac{1}{3^m n} \sum_{k=0}^{m} \binom{3m}{3k} (3n - 1)^k$$

is an integer.

(2004 Romanian IMO Team Selection Test)

Problem 10.1.12. Show that for any positive integer n the number

$$\sum_{k=0}^{n} \binom{2n+1}{2k+1} 2^{3k}$$

is not divisible by 5.

(16th IMO)

Problem 10.1.13. Prove that for a positive integer k there is an integer $n \geq 2$ such that $\binom{n}{1}, \dots, \binom{n}{n-1}$ are all divisible by k if and only if k is a prime.

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10.2 Lucas' and Kummer's Theorems

The following theorems by E. Lucas[2] (1878) and E. Kummer[3] (1852) are very useful in number theory. Let n be a positive integer, and let p be a prime. Let $\overline{n_m n_{m-1} \dots n_0}_p$ denote the base p representation of n; that is,

$$n = \overline{n_m n_{m-1} \dots n_0}_p = n_0 + n_1 p + \cdots + n_m p^m,$$

where $0 \le n_0, n_1, \dots, n_m \le p-1$ and $n_m \neq 0$.

Theorem 10.2.1. (Lucas) *Let p be a prime, and let n be a positive integer with $n = \overline{n_m n_{m-1} \dots n_0}_p$. Let i be a positive integer less than n. If $i = i_0 + i_1 p + \cdots + i_m p^m$, where $0 \le i_0, i_1, \dots, i_m \le p-1$, then*

$$\binom{n}{i} \equiv \prod_{j=0}^{m} \binom{n_j}{i_j} \pmod{p}. \tag{1}$$

Here $\binom{0}{0} = 1$ and $\binom{n_j}{i_j} = 0$ if $n_j < i_j$.

To prove this theorem, we need some additional techniques. Let p be a prime, and let $f(x)$ and $g(x)$ be two polynomials with integer coefficients. We say that $f(x)$ is congruent to $g(x)$ modulo p, and write $f(x) \equiv g(x) \pmod{p}$ if all of the coefficients of $f(x) - g(x)$ are divisible by p. (Note that the congruence of polynomials is different from the congruence of the values of polynomials. For example, $x(x+1) \not\equiv 0 \pmod 2$ even though $x(x+1)$ is divisible by 2 for all integers x.) The following properties can be easily verified:

(a) $f(x) \equiv f(x) \pmod{p}$;

(b) if $f(x) \equiv g(x) \pmod{p}$, then $g(x) \equiv f(x) \pmod{p}$;

(c) if $f(x) \equiv g(x) \pmod{p}$ and $g(x) \equiv h(x) \pmod{p}$, then

$$f(x) \equiv h(x) \pmod{p};$$

(d) if $f(x) \equiv g(x) \pmod{p}$ and $f_1(x) \equiv g_1(x) \pmod{p}$, then

$$f(x) \pm f_1(x) \equiv g(x) \pm g_1(x) \pmod{p}$$

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[2] *Ernst Eduard Kummer* (1810-1893), German mathematician who's main achievement was the extension of results about integers to other integral domains by introducing the concept of an ideal.

[3] *François Edouard Anatole Lucas* (1842-1891), French mathematician best known for his results in number theory. He studied the Fibonacci sequence and divised the test for Mersenne primes.

and

$$f(x)f_1(x) \equiv g(x)g_1(x) \pmod{p}.$$

Proof. By property 7), the binomial coefficients $\binom{p}{k}$, where $1 \leq k \leq p-1$, are divisible by p. Thus,

$$(1+x)^p \equiv 1 + x^p \pmod{p}$$

and

$$(1+x)^p = [(1+x)^p]^p \equiv [1+x^p]^p \equiv 1 + x^{p^2} \pmod{p},$$

and so on; so that for any positive integer r,

$$(1+x)^{p^r} \equiv 1 + x^{p^r} \pmod{p}$$

by induction.

We have

$$\begin{aligned}(1+x)^n &= (1+x)^{n_0+n_1p+\cdots+n_mp^m} \\ &= (1+x)^{n_0}[(1+x)^p]^{n_1}\ldots[(1+x)^{p^m}]^{n_m} \\ &\equiv (1+x)^{n_0}(1+x^p)^{n_1}\ldots(1+x^{p^m})^{n_m} \pmod{p}.\end{aligned}$$

The coefficient of x^i in the expansion of $(1+x)^n$ is $\binom{n}{i}$. On the other hand, because $i = i_0 + i_1p + \cdots + i_mp^m$, the coefficient of x^i is the coefficient of $x^{i_0}(x^p)^{i_1}\ldots(x^{p^m})^{i_m}$, which is equal to $\binom{n_0}{i_0}\binom{n_1}{i_1}\ldots\binom{n_m}{i_m}$. Hence

$$\binom{n}{i} \equiv \binom{n_0}{i_0}\binom{n_1}{i_1}\ldots\binom{n_m}{i_m} \pmod{p},$$

as desired. □

Theorem 10.2.2. (Kummer) *Let n and i be positive integers with $i \leq n$, and let p be a prime. Then p^t divides $\binom{n}{i}$ if and only if t is less than or equal to the number of carries in the addition $(n-i)+i$ in base p.*

Proof. We will use the formula

$$e_p(n) = \frac{n - S_p(n)}{p-1}, \tag{2}$$

where e_p is the Legendre's function and $S_p(n)$ is the sum of digits of n in base p (see Section 6.5). We actually prove that the largest nonnegative

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integer t such that p^t divides $\binom{n}{i}$ is exactly the number of carries in the addition $(n-i)+i$ in base p.

Let $n! = \overline{a_m a_{m-1} \dots a_0}_p$, $i! = \overline{b_k b_{k-1} \dots b_0}_p$, $(n-i)! = (\overline{c_l c_{l-1} \dots c_0})_p$. Because $1 \le i \le n$, it follows that $k, l \le m$. Without loss of generality, we assume that $k \le l$. Let a, b, c, and t' be integers such that $p^a \| n!$, $p^b \| i!$, $p^c \| (n-i)!$, and $p^{t'} \| \binom{n}{i}$. Then $t' = a - b - c$.

From formula (2) we have

$$a = \frac{n - (a_m + a_{m-1} + \cdots + a_0)}{p-1},$$

$$b = \frac{i - (b_k + b_{k-1} + \cdots + b_0)}{p-1},$$

$$c = \frac{(n-i) - (c_l + c_{l-1} + \cdots + c_0)}{p-1}.$$

Thus

$$t' = \frac{-(a_m + \cdots + a_0) + (b_k + \cdots + b_0) + (c_l + \cdots + c_0)}{p-1}. \tag{3}$$

On the other hand, if we add $n-i$ and i in base p, we have

$$\begin{array}{cccccccccccc}
 & & & & & & b_k & b_{k-1} & \dots & b_1 & b_0 \\
 & & & c_l & c_{l-1} & \dots & c_k & c_{k-1} & \dots & c_1 & c_0 \\
\hline
a_m & a_{m-1} & \dots & a_l & a_{l-1} & \dots & a_k & a_{k-1} & \dots & a_1 & a_0
\end{array}$$

Then we have either $b_0 + c_0 = a_0$ (with no carry) or $b_0 + c_0 = a_0 + p$ (with a carry of 1). More generally, we have

$$b_0 + c_0 = a_0 + \alpha_1 p,$$

$$b_1 + c_1 + \alpha_1 = a_1 + \alpha_2 p,$$

$$b_2 + c_2 + \alpha_2 = a_2 + \alpha_3 p,$$

$$\dots$$

$$b_m + c_m + \alpha_m = a_m,$$

where α_i denotes the carry at the $(i-1)^{th}$ digit from the right. (Note also that $b_j = 0$ for $j > k$ and that $c_j = 0$ for $j > l$.) Adding the above equations together yields

$$(b_0 + \cdots + b_k) + (c_0 + \cdots + c_l) = (a_0 + \cdots + a_m) + (p-1)(\alpha_1 + \cdots + \alpha_m).$$

Thus, equation (3) becomes

$$t' = \alpha_1 + \cdots + \alpha_m,$$

as desired. □

Problem 10.2.1. *Let n be a positive integer. Prove that the number of $k \in \{0, 1, \ldots, n\}$ for which $\binom{n}{k}$ is odd is a power of 2.*

Solution. Let the base 2 expansion of n be $2^0 n_0 + 2^1 n_1 + \cdots + 2^a n_a$, where $n_i \in \{0, 1\}$ for each i. Then for any $k = 2^0 k_0 + 2^1 k_1 + \cdots + 2^a k_a$, we have

$$\binom{n}{k} \equiv \binom{n_0}{k_0}\binom{n_1}{k_1}\ldots\binom{n_a}{k_a} \pmod 2$$

by Lucas' theorem. Thus $\binom{n}{k}$ is odd if and only if $k_i \leq n_i$ for each i. Let m be the number of n_i's equal to 1. Then the values of $k \in \{0, 1, \ldots, 2^{a+1} - 1\}$ for which $\binom{n}{k}$ is odd are obtained by setting $k_i = 0$ or 1 for each of the m values of i such that $n_i = 1$, and $k_i = 0$ for the other values of i. Thus there are 2^m values of k in $\{0, 1, \ldots, 2^{a+1} - 1\}$ for which $\binom{n}{k}$ is odd. Finally, note that for $k > n$, $\binom{n}{k} = 0$ is never odd, so the number of $k \in \{0, 1, \ldots, n\}$ for which $\binom{n}{k}$ is odd is 2^m, a power of 2.

Problem 10.2.2. *Determine all positive integers n, $n \geq 2$, such that $\binom{n-k}{k}$ is even for $k = 1, 2, \ldots, \left\lfloor \frac{n}{2} \right\rfloor$.*

(1999 Belarussian Mathematical Olympiad)

Solution. Suppose that $p = 2$, $a = 2^s - 1$, and $a_{s-1} = a_{s-2} = \cdots = a_0 = 1$. For any b with $0 \leq b \leq 2^s - 1$, each term $\binom{a_i}{b_i}$ in the above equation equals 1. Therefore, $\binom{a}{b} \equiv 1 \pmod 2$.

This implies that $n+1$ is a power of two. Otherwise, let $s = \lfloor \log_2 n \rfloor$ and let

$$k = n - (2^s - 1) = n - \frac{2^{s+1} - 2}{2} \leq n - \frac{n}{2} = \frac{n}{2}.$$

Then $\binom{n-k}{k} = \binom{2^s - 1}{k}$ is odd, a contradiction.

Conversely, suppose that $n = 2^s - 1$ for some positive integer s. For $k = 1, 2, \ldots, \left\lfloor \frac{n}{2} \right\rfloor$, there is at least one 0 in the binary representation of

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$a = n - k$ (not counting leading zeros, of course). Whenever there is a 0 in the binary representation of $n - k$, there is a 1 in the corresponding digit of $b = k$. Then the corresponding $\binom{a_i}{b_i}$ equals 0, and by Lucas' Theorem, $\binom{n-k}{k}$ is even.

Therefore, $n = 2^s - 1$ for integers $s \geq 2$.

Problem 10.2.3. *Prove that* $\binom{2^n}{k}$, $k = 1, 2, \ldots, 2^n - 1$, *are all even and that exactly one of them is not divisible by 4.*

Solution. All these numbers are even, since

$$\binom{2^n}{k} = \frac{2^n}{k}\binom{2^n - 1}{k - 1}$$

and $2^n/k$ is different from 1 for all $k = 1, 2, \ldots, 2^n - 1$.

From the same relation it follows that $\binom{2^n}{k}$ is a multiple of 4 for all k different from 2^{n-1}. For $k = 2^{n-1}$ we have

$$\binom{2^n}{2^{n-1}} = 2\binom{2^n - 1}{2^{n-1} - 1}.$$

But from Lucas' theorem it follows that $\binom{2^n - 1}{2^{n-1} - 1}$ is odd, since $2^n - 1$ contains only 1's in its binary representation and $\binom{1}{k} = 1$ if $k = 0$ or 1. This solves the problem.

Proposed problems

Problem 10.2.4. Let p be an odd prime. Find all positive integers n such that $\binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n-1}$ are all divisible by p.

Problem 10.2.5. Let p be a prime. Prove that p does not divide any of $\binom{n}{1}, \ldots, \binom{n}{n-1}$ if and only if $n = sp^k - 1$ for some positive integer k and some integer s with $1 \leq s \leq p - 1$.

Problem 10.2.6. Prove or disprove the following claim: For any integer $k \geq 2$, there exists an integer $n \geq 2$ such that the binomial coefficient $\binom{n}{i}$ is divisible by k for any $1 \leq i \leq n - 1$.

(1999 Hungarian-Israel Mathematical Competition)

11

Miscellaneous Problems

Problem 11.1. *Find all positive integers x, y, z which satisfy conditions: $x + y \geq 2z$ and $x^2 + y^2 - 2z^2 = 8$.*

(2003 Romanian Mathematical Olympiad)

Solution. There are two possible cases:

Case I. $x \geq y \geq z$.

We denote $x - z = a \geq 0$, $y - z = b \geq 0$, $a \geq b$. One obtains the equation $2z(a + b) + a^2 + b^2 = 8$. When $z \geq 3$, there are no solutions. For $z = 2$, we get $(a + 2)^2 + (b + 2)^2 = 16$, which again has no solution. When $z = 1$ we obtain solutions $(x, y, z) = (3, 1, 1)$ or $(x, y, z) = (1, 3, 1)$. When $z = 0$, $a^2 + b^2 = 8$ and we get the solution $(x, y, z) = (2, 2, 0)$.

Case II. $x \geq z \geq y$.

Note again that $x - z = a$, $y - z = b$ and obtain the solution $(x, y, z) = (n + 2, n - 2, n)$ or $(x, y, z) = (n - 2, n + 2, n)$.

Problem 11.2. *Let n be a positive integer. Find all integers that can be written as:*

$$\frac{1}{a_1} + \frac{2}{a_2} + \cdots + \frac{n}{a_n},$$

for some positive integers $a_1, a_2, \ldots, a_n$.

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Solution. First, observe that $k = \dfrac{1}{a_1} + \dfrac{1}{a_2} + \cdots + \dfrac{n}{a_n}$, then

$$k \leq 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

We prove that any integer $k \in \left\{1, 2, \ldots, \dfrac{n(n+1)}{2}\right\}$ can be written as requested.

For $k = 1$, put $a_1 = a_2 = \cdots = a_n = \dfrac{n(n+1)}{2}$.

For $k = n$, set $a_1 = 1$, $a_2 = 2, \ldots, a_n = n$.

For $1 < k < n$, let $a_{k-1} = 1$ and $a_i = \dfrac{n(n+1)}{2} - k + 1$ for $i \neq k-1$.

Thus

$$\frac{1}{a_1} + \frac{2}{a_2} + \cdots + \frac{n}{a_n} = \frac{k-1}{1} + \sum_{\substack{i=1 \\ i \neq k-1}} \frac{i}{a_i} = k - 1 + \frac{\dfrac{n(n+1)}{2} - k + 1}{\dfrac{n(n+1)}{2} - k + 1} = k.$$

For $n < k < \dfrac{n(n+1)}{2}$, write k as

$$k = n + p_1 + p_2 + \cdots + p_i,$$

with $1 \leq p_i \leq \cdots \leq p_2 \leq p_1 \leq n - 1$.

Setting $a_{p_1+1} = a_{p_2+1} = \cdots = a_{p_i+1} = 1$ and else $a_j = j$ we are done.

Problem 11.3. *Find all the positive integers $a < b < c < d$ with the property that each of them divides the sum of the other three.*

Solution. Since $d|(a+b+c)$ and $a+b+c < 3d$, it follows that $a+b+c = d$ or $a + b + c = 2d$.

Case i). If $a + b + c = d$, as $a|(b + c + d)$, we have $a|2d$ and similarly $b|2d$, $c|2d$.

Let $2d = ax = by = cz$, where $2 < z < y < x$. Thus $\dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z} = \dfrac{1}{2}$.

1° If $z = 3$, then $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{1}{6}$. The solutions are

$$(x, y) = \{(42, 7), (24, 8), (18, 9), (15, 10)\},$$

hence

$$(a, b, c, d) \in \{(k, 6k, 14k, 21k), (k, 3k, 8k, 12k), (k, 2k, 6k, 9k),$$

$$(2k, 3k, 10k, 15k), (k, 3k, 8k, 12k)\},$$

for $k > 0$.

2° If $z = 4$, then $\frac{1}{x} + \frac{1}{y} = \frac{1}{4}$, and

$$(x, y) = \{(20, 5), (12, 6)\}.$$

The solutions are

$$(a, b, c, d) = (k, 4k, 5k, 10k) \text{ and } (a, b, c, d) = (k, 2k, 3k, 6k),$$

for $k > 0$.

3° If $z = 5$, then $\frac{1}{x} + \frac{1}{y} = \frac{3}{10}$, and $(3x - 10)(3y - 10) = 100$.

As $3x - 10 \equiv 2 \pmod 3$, it follows that $3x - 10 = 20$ and $3y - 10 = 5$. Thus $y = 3$, false.

4° If $z \geq 6$ then $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} < \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$ so there are no solutions.

Case ii). If $a + b + c = 2d$, we obtain $a|3d$, $b|3d$, $c|3d$.

Then $3d = ax = by = cz$, with $x > y > z > 3$ and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{2}{3}$. Since $x \geq 4$, $y \geq 5$, $z \geq 6$ we have $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \leq \frac{1}{6} + \frac{1}{5} + \frac{1}{4} = \frac{37}{60} < \frac{2}{3}$, so there are no solutions in this case.

Problem 11.4. *Find the greatest number that can be written as a product of some positive integers with the sum* 1976.

(18th IMO)

Solution. Let $x_1, x_2, \ldots, x_n$ be the numbers having the sum $x_1 + x_2 + \cdots + x_n = 1976$ and the maximum value of the product $x_1 \cdot x_2 \cdots x_n = p$.

If one of the numbers, say x_1, is equal to 1, then $x_1 + x_2 = 1 + x_2 > x_2 = x_1 x_2$. Hence the product $(x_1 + x_2) \cdot x_3 \cdots x_n$ is greater than $x_1 \cdot x_2 \ldots x_n = p$, false. Therefore $x_k \geq 2$ for all k.

If one of the numbers is equal to 4 we can replace him with two numbers 2 without changing the sum or the product.

Suppose that $x_k \geq 5$ for some k. Then $x_k < 3(x_k - 3)$, so replacing the number x_k with the numbers 3 and $x_k - 3$, the sum remains constant while the product increases, contradiction.

Therefore all the numbers are equal to 2 or 3. If there are more than 3 numbers equal to 2, we can replace them by two numbers equal to 3, preserving the sum and increasing the product (as $2 \cdot 2 \cdot 2 < 3 \cdot 3$). Hence at most two terms equal to 2 are allowed. Since $1976 = 3 \cdot 658 + 2$ the maximum product is equal to $2 \cdot 3^{658}$.

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Problem 11.5. *Prove that there exist infinitely many positive integers that cannot be written in the form*

$$x_1^3 + x_2^5 + x_3^7 + x_4^9 + x_5^{11}$$

for some positive integers x_1, x_2, x_3, x_4, x_5.

(2002 Belarussian Mathematical Olympiad)

Solution. For each integer N, we consider the number of integers in $[1, N]$ that can be written in the above form. Because $x_1 \leq N^{\frac{1}{3}}$, there are at most $N^{\frac{1}{3}}$ ways to choose x_1. Similar argument applies to the other x_is. Therefore, there are at most $N^{\frac{1}{3}} N^{\frac{1}{5}} N^{\frac{1}{7}} N^{\frac{1}{9}} N^{\frac{1}{11}} = N^{\frac{3043}{3465}}$ combinations. So there are at least $N - N^{\frac{3043}{3465}}$ integers not covered. It is easy to see that this value can be arbitrarily large as N approaches infinity. Therefore, there exist infinitely many positive integers that cannot be written in the form $x_1^3 + x_2^5 + x_3^7 + x_4^9 + x_5^{11}$.

Proposed problems

Problem 11.6. Let a, b be positive integers. By integer division of a^2+b^2 to $a+b$ we obtain the quotient q and the remainder r. Find all pairs (a, b) such that $q^2 + r = 1977$.

(19th IMO)

Problem 11.7. Let m, n be positive integers. Show that $25^n - 7^m$ is divisible by 3 and find the least positive integer of the form $|25^n - 7^m - 3^m|$, when m, n run over the set of non-negative integers.

(2004 Romanian Mathematical Regional Contest)

Problem 11.8. Given an integer d, let

$$S = \{m^2 + dn^2 | m, n \in \mathbb{Z}\}.$$

Let $p, q \in S$ be such that p is a prime and $r = \dfrac{q}{p}$ is an integer. Prove that $r \in S$.

(1999 Hungary-Israel Mathematical Competition)

Problem 11.9. Prove that every positive rational number can be represented in the form

$$\frac{a^3 + b^3}{c^3 + d^3}$$

where a, b, c, d are positive integers.

(1999 IMO Shortlist)

Problem 11.10. Two positive integers are written on the board. The following operation is repeated: if $a < b$ are the numbers on the board, then a is erased and $ab/(b-a)$ is written in its place. At some point the numbers on the board are equal. Prove that again they are positive integers.

(1998 Russian Mathematical Olympiad)

Problem 11.11. Let $f(x) + a_0 + a_1x + \cdots + a_mx^m$, with $m \geq 2$ and $a_m \neq 0$, be a polynomial with integer coefficients. Let n be a positive integer, and suppose that:

i) $a_2, a_3, \ldots, a_m$ are divisible by all the prime factors of n;

ii) a_1 and n are relatively prime.

Prove that for any positive integer k, there exists a positive integer c such that $f(c)$ is divisible by n^k.

(2001 Romanian IMO Team Selection Test)

Problem 11.12. Let x, a, b be positive integers such that $x^{a+b} = a^b b$. Prove that $a = x$ and $b = x^x$.

(1998 Iranian Mathematical Olympiad)

Problem 11.13. Let m, n be integers with $1 \leq m < n$. In their decimal representations, the last three digits of 1978^m are equal, respectively, to the last three digits of 1978^n. Find m and n such that $m + n$ is minimal.

(20th IMO)

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Part II

SOLUTIONS TO PROPOSED PROBLEMS

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12

Divisibility

12.1 Divisibility

Problem 1.1.10. *Show that for any natural number n, between n^2 and $(n+1)^2$ one can find three distinct natural numbers a, b, c such that a^2+b^2 is divisible by c.*

(1998 St. Petersburg City Mathematical Olympiad)

Solution. (We must assume $n > 1$.) Take

$$a = n^2 + 2, \quad b = n^2 + n + 1, \quad c = n^2 + 1.$$

Then $a^2 + b^2 = (2n^2 + 2n + 5)c$.

Problem 1.1.11. *Find all odd positive integers n greater than 1 such that for any relatively prime divisors a and b of n, the number $a + b - 1$ is also a divisor of n.*

(2001 Russian Mathematical Olympiad)

Solution. We will call a number "good" if it satisfies the given conditions. It is not difficult to see that all prime powers are good. Suppose n is a good number that has at least two distinct prime factors. Let $n = p^r s$, where p is the smallest prime dividing n and s is not divisible by p. Because n is good, $p + s - 1$ must divide n. For any prime q dividing s,

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$s < p + s - 1 < s + q$, so q does not divide $p + s - 1$. Therefore, the only prime factor of $p + s - 1$ is p. Then $s = p^c - p + 1$ for some $c > 1$. Because p^c must also divide n, $p^c + s - 1 = 2p^c - p$ divides n. Because $2p^{c-1} - 1$ has no factors of p, it must divide s. But a simple computation shows that $\frac{p-1}{2} < \frac{p^c - p + 1}{2p^{c-1} - 1} < \frac{p+1}{2}$, therefore $2p^{c-1} - 1$ cannot divide s.

Problem 1.1.12. *Find all positive integers n such that $3^{n-1} + 5^{n-1}$ divides $3^n + 5^n$.*

(1996 St. Petersburg City Mathematical Olympiad)

Solution. This only occurs for $n = 1$. Let $s_n = 3^n + 5^n$ and note that

$$s_n = (3 + 5)s_{n-1} - 3 \cdot 5 \cdot s_{n-2}$$

so s_{n-1} must also divide $3 \cdot 5 \cdot s_{n-2}$. If $n > 1$, then s_{n-1} is coprime to 3 and 5, so s_{n-1} must divide s_{n-2}, which is impossible since $s_{n-1} > s_{n-2}$.

Remark. Alternatively, note that $1 < \frac{3^n + 5^n}{3^{n-1} + 5^{n-1}} < 5$ so we can only have $\frac{3^n + 5^n}{3^{n-1} + 5^{n-1}} \in \{2, 3, 4\}$ cases which are easily checked.

Problem 1.1.13. *Find all positive integers n such that the set*

$$\{n, n+1, n+2, n+3, n+4, n+5\}$$

can be split into two disjoint subsets such that the products of elements in these subsets are the same.

(12th IMO)

Solution. At least one of six consecutive numbers is divisible by 5. From the given condition it follows that two numbers must be divisible by 5. These two numbers are necessarily n and $n + 5$. Therefore n and $n + 5$ are in distinct subsets. Since $n(n + 1) > n + 5$, it follows that a required partition cannot be considered with subsets of different cardinality. Thus each subset must contain three numbers. The following possibilities have to be considered:

a) $\{n, n + 2, n + 4\} \cup \{n + 1, n + 3, n + 5\}$

b) $\{n, n + 3, n + 4\} \cup \{n + 1, n + 2, n + 5\}$.

In case a), $n < n + 1$, $n + 2 < n + 3$ and $n + 4 < n + 5$.

In case b), the condition of the problem gives:

$$n(n + 3)(n + 4) = (n + 1)(n + 3)(n + 5).$$

We obtain $n^2 + 5n + 10 = 0$ and this equation has no real solution.

Remark. One can prove that if p is a prime of the form $4k + 3$, then one cannot partition $p - 1$ consecutive integers in two classes with equal product. This problem is the particular case $p = 7$.

Problem 1.1.14. *The positive integers $d_1, d_2, \ldots, d_n$ divide 1995. Prove that there exist d_i and d_j among them, such that the numerator of the reduced fraction d_i/d_j is at least n.*

(1995 Israeli Mathematical Olympiad)

Solution. Note that $3 \cdot 5 \cdot 7 \cdot 19 = 1995$. If the chosen divisors include one divisible by 19 and another not divisible by 19, the quotient of the two has numerator divisible by 19, solving the problem since $n \leq 16$. If this is not the case, either all divisors are or divisible by 19 or none of them has this property, and in particular $n \leq 8$. Without loss of generality, assume the divisors are all not divisible by 19.

Under this assumption, we are done if the divisors include one divisible by 7 and another not divisible by 7, unless $n = 8$. In the latter case all of the divisors not divisible by 19 occur, including 1 and $3 \cdot 5 \cdot 7$, so this case also follows. We now assume that none of the chosen divisors is divisible by 4, so that in particular $n \leq 4$.

Again, we are done if the divisors include one divisible by 5 and another not divisible by 5. But this can only fail to occur if $n = 1$ or $n = 2$. The former case is trivial, while in the latter case we simply divide the larger divisor by the smaller one, and the resulting numerator has at least one prime divisor and so is at least 3. Hence the problem is solved in all cases.

Problem 1.1.15. *Determine all pairs (a, b) of positive integers such that $ab^2 + b + 7$ divides $a^2b + a + b$.*

(39th IMO)

Solution. From the divisibility $ab^2 + b + 7 | a^2b + a + b$ we obtain:

$$ab^2 + b + 7 | b(a^2b + a + b) - a(ab^2 + b + 7) \Rightarrow ab^2 + b + 7 | b^2 - 7a.$$

When $b^2 - 7a = 0$, it follows $b^2 = 7k$, $a = 7k^2$. Observe that all pairs $(7k^2, 7k)$, $k \geq 1$ are solutions for the problem.

Suppose $b^2 - 7a > 0$. Then $ab^2 + b + 7 \leq b^2 - 7a$ and we get a contradiction:

$$b^2 - 7a < b^2 < ab^2 + b + 7.$$

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Suppose $b^2 - 7a < 0$. Then $ab^2 + b + 7 \leq 7a - b^2$. This is possible only for $b^2 < 7$, i.e. either $b = 1$ or $b = 2$. If $b = 1$, we obtain $a = 11$ or $a = 49$.

If $b = 2$, we obtain $4a + 9 | a + 22 \ \Rightarrow \ 4a + 9 \leq a + 22 \ \Rightarrow \ 3a \leq 13$. This case cannot give a solution.

Hence, the solutions of the problem are: $(7k^2, 7k)$, $(11, 1)$ and $(49, 1)$.

Problem 1.1.16. *Find all integers a, b, c with $1 < a < b < c$ such that $(a-1)(b-1)(c-1)$ is a divisor of $abc - 1$.*

(33rd IMO)

Solution. It is convenient to note $a-1 = x$, $b-1 = y$ and $c-1 = z$. Then we have the conditions: $1 \leq x < y < z$ and $xyz | xy + yz + zx + x + y + z$.

The idea of the solution is to point out that we cannot have $xyz \leq xy + yz + zx + x + y + z$ for infinitely many triples (x, y, z) of positive integers. Let $f(x, y, z)$ be the quotient of the required divisibility.

From the algebraic form:

$$f(x, y, z) = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} + \frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx}$$

we can see that f is decreasing function in one of the variables x, y, z. By symmetry and because x, y, z are distinct numbers,

$$f(x, y, z) \leq f(1, 2, 3) = 2 + \frac{5}{6} < 3.$$

Thus, if the divisibility is fulfilled we can have either $f(x, y, z) = 1$ or $f(x, y, z) = 2$. So, we have to solve in positive integers the equations

$$xy + yz + zx + x + y + z = kxyz \tag{1}$$

where $k = 1$ or $k = 2$.

Observe that $f(3, 4, 5) = \dfrac{59}{60} < 1$. Thus $x \in \{1, 2\}$. Also $f(2, 3, 4) = \dfrac{35}{24} < 2$. Thus, for $x = 2$, we necessarily have $k = 1$. The conclusion is that only three equations have to be considered in (1).

Case 1. $x = 1$ and $k = 1$. We obtain the equation:

$$1 + 2(y + z) + yz = yz.$$

It has no solutions.

Case 2. $x = 1$ and $k = 2$. We obtain the equation:

$$1 + 2(y + z) = yz.$$

Write it under the form: $(y-2)(z-2)=5$ and obtain $y-2=1$, $z-2=5$. It has a unique solution: $y=3$, $z=7$.

Case 3. $x=2$ and $k=1$. We obtain the equation:

$$2+3(y+z)=yz.$$

By writing it under the form: $(y-3)(z-3)=11$ we find $y-3=1$, $z-3=11$. Thus, it has a unique solution: $y=4$, $z=15$.

From Case 2 and Case 3 we obtain respectively: $a=2$, $b=4$, $c=8$ and $a=3$, $b=5$, $c=16$. These are the solutions of the problem.

Problem 1.1.17. *Find all pairs of positive integers* (x,y) *for which*

$$\frac{x^2+y^2}{x-y}$$

is an integer which divides 1995.

(1995 Bulgarian Mathematical Olympiad)

Solution. It is enough to find all pairs (x,y) for which $x>y$ and $x^2+y^2=k(x-y)$, where k divides $1995=3\cdot 5\cdot 7\cdot 19$. We shall use the following well-known fact: if p is prime of the form $4q+3$ and if it divides x^2+y^2 then p divides x and y. (For $p=3$, 7, 19 the last statement can be proved directly). If k is divisible by 3, then x and y are divisible by 3 too. Simplifying by 9 we get an equality of the form $x_1^2+y_1^2=k_1(x_1-y_1)$, where k_1 divides $5\cdot 7\cdot 19$. Considering 7 and 19, analogously we get an equality of the form $a^2+b^2=5(a-b)$, where $a>b$. (It is not possible to get an equality of the form $a^2+b^2=a-b$). From here $(2a-5)^2+(2b+5)^2=50$, i.e. $a=3$, $b=1$, or $a=2$, $b=1$. The above consideration implies that the pairs we are looking for are of the form $(3c,c)$, $(2c,c)$, $(c,3c)$, $(c,2c)$, where $c=1,\ 3,\ 7,\ 19,\ 3\cdot 7,\ 3\cdot 19,\ 7\cdot 19,\ 3\cdot 7\cdot 19$.

Problem 1.1.18. *Find all positive integers* (x,n) *such that* x^n+2^n+1 *is a divisor of* $x^{n+1}+2^{n+1}+1$.

(1998 Romanian IMO Team Selection Test)

Solution. The solutions are $(x,n)=(4,1)$ and (11,1). If $n=1$, we need $x+3=x+2+1|x^2+4+1=x^2+5=(x+3)(x-3)+14$, so $x+3$ divides 14 and $x=4$ or 11. Suppose $n\geq 2$. For $x\in\{1,2,3\}$ we have

$$1+2^n+1<1+2^{n+1}+1<2(1+2^n+1),$$

$$2^n+2^n+1<2^{n+1}+2^{n+1}+1<2(2^n+2^n+1),$$

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$$2(3^n + 2^n + 1) < 3^{n+1} + 2^{n+1} + 1 < 3(3^n + 2^n + 1),$$

so x^n+2^n+1 does not divide $x^{n+1}+2^{n+1}+1$. For $x \geq 4$, $x^n = x^n/2+x^n/2 \geq 2^{2n}/2 + x^2/2$, so

$$(2^n + 1)x \leq ((2^n + 1)^2 + x^2)/2$$

$$= (2^{2n} + 2^{n+1} + 1 + x^2)/2 < 2^{n+1} + x^n + 2^n + 2.$$

Therefore

$$(x - 1)(x^n + 2^n + 1) = x^{n+1} + 2^n x + x - x^n - 2^n - 1$$

$$< x^{n+1} + 2^{n+1} + 1 < x(x^n + 2^n + 1);$$

again $x^n + 2^n + 1$ does not divide $x^{n+1} + 2^{n+1} + 1$. So the only solutions are (4,1) and (11,1).

Problem 1.1.19. *Find the smallest positive integer K such that every K-element subset of $\{1, 2, \ldots, 50\}$ contains two distinct elements a, b such that $a + b$ divides ab.*

(1996 Chinese Mathematical Olympiad)

Solution. The minimal value is $k = 39$. Suppose $a, b \in S$ are such that $a+b$ divides ab. Let $c = gcd(a, b)$, and put $a = ca_1$, $b = cb_1$, so that a_1 and b_1 are relatively prime. Then $c(a_1 + b_1)$ divides $c^2a_1b_1$, so $a_1 + b_1$ divides ca_1b_1. Since a_1 and b_1 have no common factor, neither do a_1 and $a_1 + b_1$, or b_1 and $a_1 + b_1$. In short, $a_1 + b_1$ divides c.

Since $S \subseteq \{1, \ldots, 50\}$, we have $a + b \leq 99$, so $c(a_1 + b_1) \leq 99$, which implies $a_1 + b_1 \leq 9$; on the other hand, of course $a_1 + b_1 \geq 3$. An exhaustive search produces 23 pairs a, b satisfying the condition:

$a_1 + b_1 = 3$	$(6, 3), (12, 6), (18, 9), (24, 12),$
	$(30, 15), (36, 18), (42, 21), (48, 24)$
$a_1 + b_1 = 4$	$(12, 4), (24, 8), (36, 12), (48, 16)$
$a_1 + b_1 = 5$	$(20, 5), (40, 10), (15, 10), (30, 20), (45, 30)$
$a_1 + b_1 = 6$	$(30, 6)$
$a_1 + b_1 = 7$	$(42, 7), (35, 14), (28, 21)$
$a_1 + b_1 = 8$	$(40, 24)$
$a_1 + b_1 = 9$	$(45, 36)$

12.2 Prime numbers

Problem 1.2.10. *For each integer n such that $n = p_1p_2p_3p_4$, where p_1, p_2, p_3, p_4 are distinct primes, let*

$$d_1 = 1 < d_2 < d_3 < \cdots < d_{16} = n$$

be the sixteen positive integers which divide n. Prove that if $n < 1995$, then $d_9 - d_8 \neq 22$.

(1995 Irish Mathematical Olympiad)

Solution. Note that $35 \cdot 57 = 1995 = 2 \cdot 3 \cdot 7 \cdot 19$. Suppose that $n < 1995$ and $d_9 - d_8 = 22$; then $d_8d_9 = n$, so $d_8 < 35$. Moreover, d_8 cannot be even since that would make n divisible by 4, whereas n has distinct prime factors. Hence d_8, d_9 and n are odd.

The divisors $d_1, \ldots, d_8$ each are the product of distinct odd primes, since they divide n. Since $3 \cdot 5 \cdot 7 > 35$, none of $d_1, \ldots, d_8$ is large enough to have three odd prime factors, so each is either prime or the product of two primes. Since n only has four prime factors, four of the d_i must be the product of two odd primes. But the smallest such numbers are

$$15, 21, 33, 35, \ldots$$

and so we must have $d_8 \geq 35$, contrary to assumption.

Problem 1.2.11. *Prove that there are infinitely many positive integers a such that the sequence $(z_n)_{n\geq 1}$, $z_n = n^4 + a$, does not contain any prime number.*

(11th IMO)

Solution. To consider all positive integers of the form $n^4 + a$, $n \geq 1$, means to consider all values of the polynomial $P(X) = X^4 + a$ in the positive integers. A decomposition of the polynomial $P(X)$ gives us decompositions of the numbers $n^4 + a$, unless the case of factors taking values 1.

The polynomial $P(X)$ can have a decomposition in integer polynomials only into quadratic factors:

$$P(X) = (X^2 + mX + n)(X^2 + m'X + n').$$

Such a decomposition is possible if and only if:

$$m + m' = 0,\ mm' + n + n' = 0,\ mn' + m'n = 0 \text{ and } nn' = a.$$

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We obtain: $m' = -m$, $n = n'$, $m^2 - 2n = 0$ and $n^2 = a$.

Therefore, there is a unique possibility:

$$X^4 + a = (X^2 + mX + n)(X^2 - mX + n).$$

This case may fulfill when $m = 2k$, $n = 2k^2$ and $a = 4k^4$, with $k > 1$.

Problem 1.2.12. *Let p, q, r be distinct prime numbers and let A be the set*

$$A = \{p^a q^b r^c : \ 0 \leq a, b, c \leq 5\}.$$

Find the smallest integer n such that any n-element subset of A contains two distinct elements x, y such that x divides y.

(1997 Romanian Mathematical Olympiad)

Solution. Define an order relation on A by setting $p^a q^b r^c \leq p^{a_1} q^{b_1} r^{c_1}$ iff $a \leq a_1$, $b \leq b_1$, $c \leq c_1$. We must find thus the longest antichain with respect to this relation, that is the maximal number n such that there is $B \subset A$ with $|B| = n$ and no two elements of B are comparable. The answer will then be $n + 1$.

From now on, identity $p^a q^b r^c$ with (a, b, c) and regard it as a laticial point in $\mathbb{R}^3$. One can easily check that the set

$$B = \{(a, b, c) \mid a, b, c \in \{0, 1, \ldots, 5\}, \ a + b + c = 8\}$$

has 27 elements and that it is an antichain. We will prove that any set with 28 elements contains two comparable elements. Of course, it suffices to find 27 chains which partition $\{(a, b, c) \mid 0 \leq a, b, c \leq 5\}$ and such that each chain has a unique representation from B. Take $A = \{(a, b) \mid 0 \leq a, b \leq 5\}$ and partition it into 6 chains (draw a picture!)

$$A_1 = \{(0,0), (0,1), \ldots, (0,5), (1,5), \ldots, (5,5)\},$$

$$A_2 = \{(1,0), (1,1), \ldots, (1,4), (2,4), \ldots, (5,4)\},$$

$$A_3 = \{(2,0), (2,1), \ldots, (2,3), (3,3), \ldots, (5,8)\},$$

$$A_4 = \{(3,0), (3,1), (3,2), (4,2), (5,2)\},$$

$$A_5 = \{(4,0), (4,1), (5,1)\},$$

$$A_6 = \{(5,0)\}.$$

Next define $A_{1j} = \{(a, b, j) \mid (a, b) \in A_1\}$ and similarly for A_2, A_3. We have found 18 chains till now.

For $(a,b) \in A_4 \cup A_5 \cup A_6$ we define the chain $A_{(a,b)} = \{(a,b,j) \mid 0 \leq j \leq 5\}$ and we have 9 chains, for a total of 27 chains.

Problem 1.2.13. *Prove Bonse's inequality:*

$$p_1 p_2 \dots p_n > p_{n+1}^2$$

for $n \geq 4$, where $p_1 = 2$, $p_2 = 3, \dots$ is the increasing sequence of prime numbers.

Solution. Let us define $A_k = p_1 p_2 \dots p_k$ and $a_k = kA_{n-1} - p_n$ for $1 \leq k \leq p_n - 1$. Observe that these numbers are relatively prime. Indeed, a prime common divisor of a_{k_1} and a_{k_2} would divide $(k_1 - k_2)A_{n-1}$ and since $gcd(a_{k_1}, p_n) = 1$, this divisor would be $p_1, \dots, p_{n-1}$, which is clearly impossible. Of course, this implies that $a_k \geq p_{n+k}$ (since a_k is relatively prime with $p_1, \dots, p_{n-1}$). Thus for $k = p_n - 1$ we have $A_n - A_{n-1} - p_n > p_{p_n+n-1}$ and so $p_1 p_2 \dots p_n > p_{p_n+n-1} > p_{3n-1}$ for $n \geq 5$. From here we find that for $n \geq 6$ we have $p_1 \dots p_n > \left(p_1 \dots p_{\left[\frac{n}{2}\right]}\right)^2 > p^2_{3\left[\frac{n}{2}\right]-1} > p_{n+1}^2$. For $n = 5$ one can easily check the inequality.

Problem 1.2.14. *Show that there exists a set A of positive integers with the following property: for any infinite set S of primes, there exist two positive integers $m \in A$ and $n \notin A$ each of which is a product of k distinct elements of S for some $k \geq 2$.*

(35th IMO)

Solution. There are several constructions for such A, involving different ideas about the decomposition of integer numbers.

First example. Let $p_1 < p_2 < \dots < p_n < \dots$ be the increasing sequence of all prime numbers. Define A being the set of numbers of the form $p_{i_1} p_{i_2} \dots p_{i_k}$ where $i_1 < i_2 < \dots < i_k$ and $i_k = p_{i_1}$. For example $3 \cdot 5 \cdot 7 \in A$; $3 \cdot 11 \cdot 13 \in A$ and $5 \cdot 7 \cdot 11 \notin A$; $3 \cdot 5 \cdot 7 \cdot 11 \notin A$.

We will see that A satisfies the required condition. Let S be an infinite set of prime numbers, say $q_1 < q_2 < \dots < q_n < \dots$ Take $m = q_1 q_2 \dots q_{q_1}$ and $n = q_1 q_2 \dots q_{q_1+1}$. Then $m \in A$ and $n \notin A$.

Second example. Define $A = \bigcup_{i=1}^{\infty} A_i$ where A_i is the set of numbers which are product of $i+1$ distinct primes which are different from p_i. For example $3 \cdot 5 \cdot 7 \in A_2$, $2 \cdot 3 \cdot 7 \cdot 11 \in A_3$ and $2 \cdot 3 \cdot 7 \notin A_2$, $3 \cdot 5 \cdot 7 \cdot 13 \notin A_3$.

Let S be an infinite set of prime numbers, say $q_1 < q_2 < \dots < q_n < \dots$ Suppose that $q_1 = p_{i_1}$. If $i_1 > 1$, note $i_1 = k$. Then $n = q_1 q_2 \dots q_{k+1} \notin$

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A, because it contains prime factor $q_1 = p_{i_1} = p_k$. The number $m = q_2 q_3 \dots q_{k+2}$ contains $k+1$ factors, all different from $p_k = q_1$. Thus $m \in A$. If $i_1 = 1$, take $k = i_2$ and the same construction will answer the question.

Third example. Let P be the set of all positive primes and let $P_1 \subset P_2 \subset \dots \subset P_n \subset \dots$ be an ascending chain of finite distinct subsets of P, such that $P = \bigcup_{i=1}^{\infty} P_i$. Define A to be the set of elements of the form

$$a = p_1 p_2 \dots p_k$$

$k = i_1 < i_2 < \dots < i_k$ and $p_1 \in P_{i_1} \setminus P_{i_1 - 1}$, $p_2 \in P_{i_2}, \dots, p_k \in P_{i_k}$.

Let S be an infinite set of prime numbers and let $S_i = S \cap P_i$. It is obvious that $S_1 \subset S_2 \subset \dots \subset S_n \subset \dots$ This chain is not stationary because $S = \bigcup_{i=1}^{\infty} S_i$. Then, it contains an infinite subchain with distinct sets:

$$S_{i_1} \subset S_{i_2} \subset \dots \subset S_{i_n} \subset \dots$$

Suppose that $S_{i_n} = S_{i_{n+1}} = \dots = S_{i_{n+1}-1} \subset S_{i_{n+1}}$. Set $i_1 = k > 1$ and choose $p_1 \in S_{i_1} \setminus S_{i_1 - 1}$, $p_2 \in S_{i_2} \setminus S_{i_2 - 1}, \dots$, $p_k \in S_{i_k} \setminus S_{i_k - 1}$ and $p_{k+1} \in S_{i_{k+1}} \setminus S_{i_k}$. Then $m = p_1 p_2 \dots p_k \in A$ and $n = p_2 p_3 \dots p_{k+1} \notin A$ because $p_2 \notin S_{i_1} = S_k$.

Problem 1.2.15. *Let n be an integer number, $n \geq 2$. Show that if $k^2 + k + n$ is a prime number for any integer number k, $0 \leq k \leq \sqrt{\frac{n}{3}}$, then $k^2 + k + n$ is a prime number for any k, $0 \leq k \leq n - 2$.*

(28th IMO)

Solution. It is not difficult to check that the property is verified for $n = 2, 3$. So, we may suppose $n \geq 5$ and assume the contrary: $k^2 + k + n$ is prime for $0 \leq k \leq \sqrt{\frac{n}{3}}$ and there exists l, $\sqrt{\frac{n}{3}} < l \leq n - 2$ such that $l^2 + l + n$ is not prime. Let p be the least number l such that $l^2 + l + n$ is composed number and let $p^2 + p + n = ab$ be a nontrivial decomposition, such that $1 < a \leq b$. Then, $l^2 + l + n$ is a prime for all l, $l < p$.

We prove first that $a \geq p + 1$.

Assuming the contrary, one obtains $0 \leq p - a < p$ and

$$(p-a)^2 + (p-a) + n = p^2 + p + n + a(a - 2p - 1) = a(b + a - 2p - 1).$$

Since $(p-a)^2+(p-a)+n$ is a prime number, it follows that $b+a-2p-1=1$ and then, $a+b=2(p+1)$. By AM-GM inequality,

$$ab \leq \frac{(a+b)^2}{4} = (p+1)^2.$$

Since $ab = p^2+p+n$, it follows that: $p^2+p+n \leq (p+1)^2$. From the last inequality, $p \geq n-1$ and this contradicts the choice of p. The conclusion is $a \geq p+1$.

Since $n < 3p^2$, $p^2+p+n < 4p^2+p < (2p+1)^2$. Taking in account that $p^2+p+n = ab$, we have $ab < (2p+1)^2$ and thus, $a < 2p+1$.

We may repeat the previous argument: $0 \leq a-p-1 < p$ and then $(a-p-1)^2+(a-p-1)+n$ is prime a number. A standard computation gives:

$$(a-p-1)^2+(a-p-1)+n = p^2+p+n+a(a-2p-1) = a(b+a-2p-1).$$

We obtain, again $b+a = 2p+2 = 2(p+1)$.

On the other hand:

$$a+b \geq 2\sqrt{ab} = 2\sqrt{p^2+p+n} > 2\sqrt{p^2+p+p+1} = 2(p+1).$$

Thus, we obtain a contradiction, so our initial assumption was incorrect.

Remark. The problem is related to the famous example of Euler of a polynomial generator of primes: x^2+x+41 produces primes for $0 \leq x \leq 39$. The problem shows that it suffices to check the primality only for the first 4 values of x.

Problem 1.2.16. *A sequence $q_1, q_2, \ldots$ of primes satisfies the following condition: for $n \geq 3$, q_n is the greatest prime divisor of $q_{n-1}+q_{n-2}+2000$. Prove that the sequence is bounded.*

(2000 Polish Mathematical Olympiad)

Solution. Let $b_n = \max\{q_n, q_{n+1}\}$ for $n \geq 1$. We first prove that $b_{n+1} \leq b_n + 2002$ for all such n. Certainly $q_{n+1} \leq b_n$, so it suffices to show that $q_{n+2} \leq b_n + 2002$. If either q_n or q_{n+1} equals 2, then we have $q_{n+2} \leq q_n + q_{n+1} + 2000 = b_n + 2002$. Otherwise, q_n and q_{n+1} are both odd, so $q_n + q_{n+1} + 2000$ is even. Because $q_{n+2} \neq 2$ divides this number, we have

$$q_{n+2} \leq \frac{1}{2}(q_n + q_{n+1} + 2000) = \frac{1}{2}(q_n + q_{n+1}) + 1000 \leq b_n + 1000.$$

This proves the claim.

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Choose k large enough so that $b_1 \leq k \cdot 2003! + 1$. We prove by induction that $b_n \leq k \cdot 2003! + 1$ for all n. If this statement holds for some n, then $b_{n+1} \leq b_n + 2002 \leq k \cdot 2003! + 2003$. If $b_{n+1} > k \cdot 2003! + 1$, then let $m = b_{n+1} - k \cdot 2003!$. We have $1 < m \leq 2003$, implying that $m|2003!$. Hence, m is a proper divisor of $k \cdot 2003! + m = b_{n+1}$, which is impossible because b_{n+1} is prime. Thus, $q_n \leq b_n \leq k \cdot 2003! + 1$ for all n.

Problem 1.2.17. *Let $a > b > c > d$ be positive integers and suppose*

$$ac + bd = (b + d + a - c)(b + d - a + c).$$

Prove that $ab + cd$ is not prime.

(42nd IMO)

Solution. The given equality is equivalent to $a^2 - ac + c^2 = b^2 + bd + d^2$. Hence

$$(ab + cd)(ad + bc) = ac(b^2 + bd + d^2) + bd(a^2 - ac + c^2),$$

or equivalently,

$$(ab + cd)(ad + bc) = (ac + bd)(a^2 - ac + c^2). \tag{1}$$

Now suppose that $ab + cd$ is prime. It follows from $a > b > c > d$ that

$$ab + cd > ac + bd > ad + bc; \tag{2}$$

hence $ac + bd$ is relatively prime with $ab + cd$. But then (1) implies that $ac + bd$ divides $ad + bc$, which is impossible by (2).

12.3 The greatest common divisor and the least common multiple

Problem 1.3.9. *The sequence $a_1, a_2, \ldots$ of natural numbers satisfies*

$$gcd(a_i, a_j) = gcd(i, j) \text{ for all } i \neq j.$$

Prove that $a_i = i$ for all i.

(1995 Russian Mathematical Olympiad)

Solution. For any integer m, we have $(a_m, a_{2m}) = (2m, m)$ and so $m|a_m$. This means that for any other integer n, m divides a_n if and only if it divides

$(a_m, a_n) = (m, n)$. Hence a_n has exactly the same divisors as n and so must equal n for all n.

Problem 1.3.10. *The natural numbers a and b are such that*

$$\frac{a+1}{b} + \frac{b+1}{a}$$

is an integer. Show that the greatest common divisor of a and b is not greater than $\sqrt{a+b}$.

(1996 Spanish Mathematical Olympiad)

Solution. Let $d = gcd(a, b)$ and put $a = md$ and $b = nd$. Then we have

$$\frac{md+1}{nd} + \frac{nd+1}{md} = \frac{m^2d + m + n^2d + n}{mnd}$$

is an integer, so that in particular, d divides $m^2d + m + n^2d + n$ and also $m + n$. However, this means $d \leq m + n$, and so $d \leq \sqrt{d(m+n)} = \sqrt{a+b}$.

Problem 1.3.11. *The positive integers m, n, m, n are written on a blackboard. A generalized Euclidean algorithm is applied to this quadruple as follows: if the numbers x, y, u, v appear on the board and $x > y$, then $x - y$, y, $u + v$, v are written instead; otherwise x, $y - x$, u, $v + u$ are written instead. The algorithm stops when the numbers in the first pair become equal (they will equal the greatest common divisor of m and n). Prove that the arithmetic mean of the numbers in the second pair at that moment equals the least common multiple of m and n.*

(1996 St. Petersburg City Mathematical Olympiad)

Solution. Note that $xv + yu$ does not change under the operation, so it remains equal to $2mn$ throughout. Thus when the first two numbers both equal $gcd(m, n)$, the sum of the latter two is $2mn/gcd(m, n) = 2\text{lcm}(m, n)$.

Problem 1.3.12. *How many pairs (x, y) of positive integers with $x \leq y$ satisfy $gcd(x, y) = 5!$ and $\text{lcm}(x, y) = 50!$?*

(1997 Canadian Mathematical Olympiad)

Solution. First, note that there are 15 primes from 1 to 50:

$$(2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47).$$

To make this easier, let us define $f(a, b)$ to be greatest power of b dividing a. (Note $g(50!, b) > g(5!, b)$ for all $b < 50$.) Therefore, for each prime p, we

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have either $f(x,p) = f(5!,p)$ and $f(y,p) = f(50!,p)$ or $f(y,p) = f(5!,p)$ and $f(x,p) = f(50!,p)$. Since we have 15 primes, this gives 2^{15} pairs, and clearly $x \neq y$ in any such pair (since the gcd and lcm are different), so there are 2^{14} pairs with $x \leq y$.

Problem 1.3.13. *Several positive integers are written on a blackboard. One can erase any two distinct integers and write their greatest common divisor and least common multiple instead. Prove that eventually the numbers will stop changing.*

(1996 St. Petersburg City Mathematical Olympiad)

Solution. If a, b are erased and $c < d$ are written instead, we have $c \leq \min(a,b)$ and $d \geq \max(a,b)$; moreover, $ab = cd$. From this we may conclude $a + b \leq c + d$ writing $ab + a^2 = cd + a^2 \leq ac + ad$ (the latter since $(d-a)(c-a) \leq 0$) and dividing both sides by a. Thus the sum of the numbers never decrease, and it is obviously bounded (e.g. by n times the product of the numbers, where n is the number of numbers on the board); hence it eventually stops changing, at which time the numbers never change.

Problem 1.3.14. *(a) For which positive integers n do there exist positive integers x, y such that*

$$\text{lcm}(x,y) = n!, \quad gcd(x,y) = 1998?$$

(b) For which n is the number of such pairs x, y with $x \leq y$ less than 1998?

(1998 Hungarian Mathematical Olympiad)

Solution. (a) Let $x = 1998a$, $y = 1998b$. So a, b are positive integers such that $a < b$, $gcd(a,b) = 1$. We have $\text{lcm}(x,y) = 1998ab = 2 \cdot 3^3 \cdot 37ab = n!$. Thus $n \geq 37$ and it is easy to see that this condition is also sufficient.

(b) The answers are $n = 37, 38, 39, 40$. We only need to consider positive integers $n \geq 37$. For $37 \leq n < 41$, let $k = ab = n!/1998$. Since $gcd(a,b) = 1$, any prime factor of k that occurs in a cannot occur in b, and vice-versa. There are 11 prime factors of k, namely 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31. For each of those prime factors, one must decide only whether it occurs in a or in b. These 11 decisions can be made in a total of $2^{11} = 2048$ ways. However, only half of these ways will satisfy the condition $a < b$. Thus there will be a total of 1024 such pairs of (x,y) for $n = 37$, 38, 39, 40. Since

41 is a prime, we can see by a similar argument that there will be at least 2048 such pairs of (x, y) for $n \geq 41$.

Problem 1.3.15. *Determine all positive integers k for which there exists a function $f : \mathbb{N} \to \mathbb{Z}$ such that*

(a) $f(1997) = 1998$;

(b) for all $a, b \in \mathbb{N}$, $f(ab) = f(a) + f(b) + kf(gcd(a, b))$.

(1997 Taiwanese Mathematical Olympiad)

Solution. Such f exists for $k = 0$ and $k = -1$. First take $a = b$ in (b) to get $f(a^2) = (k + 2)f(a)$. Applying this twice, we get

$$f(a^4) = (k + 2)f(a^2) = (k + 2)^2 f(a).$$

On the other hand,

$$\begin{aligned} f(a^4) &= f(a) + f(a^3) + kf(a) = (k + 1)f(a) + f(a^3) \\ &= (k + 1)f(a) + f(a) + f(a^2) + kf(a) \\ &= (2k + 2)f(a) + f(a^2) = (3k + 4)f(a). \end{aligned}$$

Setting $a = 1997$ so that $f(a) \neq 0$, we deduce $(k + 2)^2 = 3k + 4$, which has roots $k = 0, -1$. For $k = 0$, an example is given by

$$f(p_1^{e_1} \dots p_n^{e_n}) = e_1 g(p_1) + \dots + e_n g(p_n),$$

where m is a prime factor of 1997, $g(m) = 1998$ and $g(p) = 0$ for all primes $p \neq m$. For $k = 1$, an example is given by

$$f(p_1^{e_1} \dots p_n^{e_n}) = g(p_1) + \dots + g(p_n).$$

Problem 1.3.16. *Find all triples (x, y, n) of positive integers such that*

$$gcd(x, n + 1) = 1 \text{ and } x^n + 1 = y^{n+1}.$$

(1998 Indian Mathematical Olympiad)

Solution. All solutions are of the form $(a^2 - 1, a, 1)$ with a even. We have $x^n = y^{n+1} - 1 = (y - 1)m$ with $m = y^n + y^{n-1} + \dots + y + 1$. Thus $m|x^n$ and $gcd(m, n + 1) = 1$. Rewrite m as

$$m = (y - 1)(y^{n-1} + 2y^{n-2} + 3y^{n-3} + \dots + (n - 1)y + n) + (n + 1).$$

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Thus we have $gcd(m, y-1)|n+1$. But $gcd(m, n+1) = 1$, so $gcd(m, y-1) = 1$. Since $x^n = (y-1)m$, m must be a perfect n-th power. But

$$(y+1)^n = y^n + \binom{n}{1}y^{n-1} + \cdots + \binom{n}{n-1}y + 1 > m > y^n,$$

for $n > 1$. So m can be a perfect n-th power only if $n = 1$ and $x = y^2 - 1$. Since x and $n + 1 = 2$ are relatively prime, y must be even, yielding the presented solutions.

Problem 1.3.17. *Find all triples (m, n, l) of positive integers such that*

$$m + n = gcd(m, n)^2, \quad m + l = gcd(m, l)^2, \quad n + l = gcd(n, l)^2.$$

(1997 Russian Mathematical Olympiad)

Solution. The only solution is $l = m = n = 2$. Let $d = gcd(l, m, n)$, and put $l = dl_1$, $m = dm_1$, $n = dn_1$. Then $d(m_1 + n_1) = d^2 d_{mn}^2$, where $d_{mn} = gcd(m_1, n_1)$, so $m_1 + n_1 = dd_{mn}^2$. Defining d_{ln} and d_{lm} likewise, we get

$$2(l_1 + m_1 + n_1) = d(d_{lm}^2 + d_{ln}^2 + d_{mn}^2).$$

Since $\dfrac{d}{gcd(d, 2)}$ divides $l_1 + m_1 + n_1$ as well as $m_1 + n_1$, it divides l_1 and likewise m_1 and n_1. As these three numbers are relatively prime, we have $\dfrac{d}{gcd(d, 2)} = 1$ and so $d \leq 2$.

Note that d_{lm}, d_{ln}, d_{mn} are pairwise relatively prime; therefore we can write $l_1 = l_2 d_{lm} d_{ln}$, $m_1 = m_2 d_{lm} d_{mn}$, $n_1 = n_2 d_{ln} d_{mn}$. Then we have

$$d_{lm} d_{mn} m_2 + d_{ln} d_{mn} n_2 = dd_{mn}^2$$

and so $m_2 d_{lm} + n_2 d_{ln} = dd_{mn}$ and so forth. Assuming without loss of generality that d_{mn} is no larger than d_{lm}, d_{ln}, we get

$$2d_{mn} \geq dd_{mn} = d_{lm} m_2 + d_{ln} n_2 \geq d_{lm} + d_{ln} \geq 2d_{mn}.$$

Thus we have equality throughout: $d = 2$, $m_2 = n_2 = 1$ and $d_{lm} = d_{ln} = d_{mn}$. But these three numbers are pairwise relatively prime, so they are all 1. Then $m_1 = n_1 = 1$ and from $l_1 + m_1 = dd_{lm}^2$, $l_1 = 1$ as well. Therefore $l = m = n = 2$.

12.4 Odd and even

Problem 1.4.5. *We are given three integers a, b, c such that a, b, c, $a+b-c$, $a+c-b$, $b+c-a$ and $a+b+c$ are seven distinct primes. Let d be the difference between the largest and smallest of these seven primes. Suppose that $800 \in \{a+b, b+c, c+a\}$. Determine the maximum possible value of d.*

Solution. Answer: 1594.

First, observe that a, b, c must all be odd primes; this follows from the assumption that the seven quantities listed are distinct primes and the fact that there is only one even prime, 2. Therefore, the smallest of the seven primes is at least 3. Next, assume without loss of generality that $a+b = 800$. Because $a+b-c > 0$, we must have $c < 800$. We also know that c is prime; therefore, since $799 = 17 \cdot 47$, we have $c \leq 797$. It follows that the largest prime, $a+b+c$, is no more than 1597. Combining these two bounds, we can bound d by $d \leq 1597 - 3 = 1594$. It remains to observe that we can choose $a = 13$, $b = 787$, $c = 797$ to achieve this bound. The other four primes are then 3, 23, 1571 and 1597.

Problem 1.4.6. *Determine the number of functions $f : \{1, 2, \ldots, n\} \to \{1995, 1996\}$ which satisfy the condition that $f(1) + f(2) + \cdots + f(1996)$ is odd.*

(1996 Greek Mathematical Olympiad)

Solution. We can send $1, 2, \ldots, n-1$ anywhere, and the value of $f(n)$ will then be uniquely determined. Hence there are 2^{n-1} such functions.

Problem 1.4.7. *Is it possible to place 1995 different natural numbers along a circle so that for any two these numbers, the ratio of the greatest to the least is a prime?*

(1995 Russian Mathematical Olympiad)

Solution. No, this is impossible. Let $a_0, \ldots, a_{1995} = a_0$ be the integers. Then for $i = 1, \ldots, 1995$, a_{k-1}/a_k is either a prime or the reciprocal of a prime; suppose the former occurs m times and the latter $1995 - m$ times. The product of all of these ratios is $a_0/a_{1995} = 1$, but this means that the product of some m primes equals the product of some $1995-m$ primes. This can only occurs when the primes are the same (by unique factorization), and in particular there have to be the same number on both sides. But $m = 1995 - m$ is impossible since 1995 is odd, contradiction.

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Problem 1.4.8. *Let a, b, c, d be odd integers such that $0 < a < b < c < d$ and $ad = bc$. Prove that if $a + d = 2^k$ and $b + c = 2^m$ for some integers k and m, then $a = 1$.*

(25th IMO)

Solution. Since $ad = bc$, we have

$$a((a + d) - (b + c)) = (a - b)(a - c) > 0.$$

Thus $a + d > b + c$, $2^k > 2^m$ and $k > m$. Since $ad = a(2^k - a) = bc = b(2^m - b)$ we obtain

$$2^m b - 2^k a = b^2 - a^2 = (b - a)(b + a).$$

By the equality $2^m(b - 2^{k-m}a) = (b - a)(b + a)$, we infer that $2^m|(b - a)(b + a)$. But $b - a$ and $b + a$ differ by $2a$, an odd multiple of 2, so either $b - a$ or $b + a$ is not divisible by 4. Hence, either $2^{m-1}|b - a$ or $2^{m-1}|b + a$. But $0 < b - a < b < 2^{m-1}$, so it must be that $2^{m-1}|b + a$.

Since $0 < b+a < b+c = 2^m$, it follows that $b+a = 2^{m-1}$ and $b = 2^{m-1}-a$. Then $c = 2^{m-1}$ and $ad = bc = (2^{m-1} - a)(2^{m-1} + a)$.

From this equality we obtain $a(a + d) = 2^{2m-2}$, hence $a = 1$.

12.5 Modular arithmetics

Problem 1.5.7. *Find all integers $n > 1$ such that any prime divisor of $n^6 - 1$ is a divisor of $(n^3 - 1)(n^2 - 1)$.*

(2002 Baltic Mathematics Competition)

Solution. We show that $n = 2$ is the only such integer. It is clear that $n = 2$ satisfies the conditions. For $n > 2$, write

$$n^6 - 1 = (n^3 - 1)(n^3 + 1) = (n^3 - 1)(n + 1)(n^2 - n + 1);$$

hence, all prime factors of $n^2 - n + 1$ must divide $n^3 - 1$ or $n^2 - 1 = (n-1)(n+1)$. Note, however, that $(n^2-n+1, n^3-1) \leq (n^3+1, n^3-1) \leq 2$; on the other hand, $n^2 - n + 1 = n(n - 1) + 1$ is odd, so all prime factors of $n^2 - n + 1$ must divide $n + 1$. But $n^2 - n + 1 = (n + 1)(n - 2) + 3$, so we must have $n^2 - n + 1 = 3^k$ for some k. Because $n > 2$, we have $k \geq 2$. Now $3|(n^2 - n + 1)$ gives $n \equiv 2 \pmod 3$; but for each of the cases $n \equiv 2, 5, 8 \pmod 9$, we have $n^2 - n + 1 \equiv 3 \pmod 9$, a contradiction.

Problem 1.5.8. *Let $f(n)$ be the number of permutations $a_1, \ldots, a_n$ of the integers $1, \ldots, n$ such that*

(i) $a_1 = 1$;

(ii) $|a_i - a_{i+1}| \leq 2,\ i = 1, \ldots, n-1$.

Determine whether $f(1996)$ is divisible by 3.

(1996 Canadian Mathematical Olympiad)

Solution. Let $g(n)$ be the number of permutations of the desired form with $a_n = n$. Then either $a_{n-1} = n-1$ or $a_{n-1} = n-2$; in the latter case we must have $a_{n-2} = n-1$ and $a_{n-3} = n-3$. Hence $g(n) = g(n-1) + g(n-3)$ for $n \geq 4$. In particular, the values of $g(n)$ modulo 3 are $g(1) = 1, 1, 1, 2, 0, 1, 0, 0, \ldots$ repeating with period 8.

Now let $h(n) = f(n) - g(n)$; $h(n)$ counts permutations of the desired form where n occurs in the middle, sandwiched between $n-1$ and $n-2$. Removing n leaves an acceptable permutation, and any acceptable permutation on $n-1$ symbols can be so produced except those ending in $n-4$, $n-2$, $n-3$, $n-1$. Hence $h(n) = h(n-1) + g(n-1) - g(n-4) = h(n-1) + g(n-2)$; one checks that $h(n)$ modulo 3 repeats with period 24.

Since $1996 \equiv 4 \pmod{24}$, we have $f(1996) \equiv f(4) = 4 \pmod 3$, so $f(1996)$ is not divisible by 3.

Problem 1.5.9. *For natural numbers m, n, show that $2^n - 1$ is divisible by $(2^m - 1)^2$ if and only if n is divisible by $m(2^m - 1)$.*

(1997 Russian Mathematical Olympiad)

Solution. Since

$$2^{kn+d} - 1 \equiv 2^d - 1 \pmod{2^n - 1},$$

we have $2^m - 1$ divides $2^n - 1$ if and only if m divides n. Thus in either case, we must have $n = km$, in which case

$$\frac{2^{km} - 1}{2^m - 1} = 1 + 2^m + \cdots + 2^{m(k-1)} \equiv k \pmod{2^m - 1}.$$

The two conditions are now that k is divisible by $2^m - 1$ and that m is divisible by $m(2^m - 1)$, which are equivalent.

Problem 1.5.10. *Suppose that n is a positive integer and let*

$$d_1 < d_2 < d_3 < d_4$$

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be the four smallest positive integer divisors of n. Find all integers n such that

$$n = d_1^2 + d_2^2 + d_3^2 + d_4^2.$$

(1999 Iranian Mathematical Olympiad)

Solution. The answer is $n = 130$. Note that $x^2 \equiv 0 \pmod 4$ when x is even and that $x^2 \equiv 1 \pmod 4$ when x is odd.

If n is odd, then all the d_i are odd and $n \equiv d_1^2+d_2^2+d_3^2+d_4^2 \equiv 1+1+1+1 \equiv 0 \pmod 4$, a contradiction. Thus, $2|n$.

If $4|n$ then $d_1 = 1$ and $d_2 = 2$, and $n \equiv 1 + 0 + d_3^2 + d_4^2 \not\equiv 0 \pmod 4$, a contradiction. Thus, $4 \nmid n$.

Therefore $\{d_1, d_2, d_3, d_4\} = \{1, 2, p, q\}$ or $\{1, 2, p, 2p\}$ for some odd primes p, q. In the first case, $n \equiv 3 \pmod 4$, a contradiction. Thus $n = 5(1 + p^2)$ and $5|n$, so $p = d_3 = 5$ and $n = 130$.

Problem 1.5.11. *Let p be an odd prime. For each $i = 1, 2, \ldots, p-1$ denote by r_i the remainder when i^p is divided by p^2. Evaluate the sum*

$$r_1 + r_2 + \cdots + r_{p-1}.$$

(Kvant)

Solution. Denote the sum in question by S. Combine the first summand with the last, the second one with the next-to-last, and so on, to get

$$2S = (r_1 + r_{p-1}) + (r_2 + r_{p-2}) + \cdots + (r_{p-1} + r_1). \tag{1}$$

We have $r_i + r_{p-i} \equiv i^p + (p - i)^p \pmod{p^2}$ by the definition of the numbers $r_1, r_2, \ldots, r_{p-1}$. Furthermore, because p is odd,

$$i^p + (p-i)^p = p^p - \binom{p}{1}p^{p-1}i + \binom{p}{2}p^{p-2}i^2 - \cdots + \binom{p}{p-1}pi^{p-1}.$$

Since p is a prime, each binomial coefficient above is divisible by p, which yields the conclusion that $r_i + r_{p-i}$ is divisible by p^2. But $0 < r_i < p^2$, $0 < r_{p-i} < p^2$, because p is a prime (so neither one equals 0), and now we may claim that

$$r_i + r_{p-i} = p^2 \text{ for } i = 1, 2, \ldots, p-1. \tag{2}$$

The equalities (1) and (2) show that

$$S = \frac{p-1}{2}p^2 = \frac{p^3 - p^2}{2}.$$

Problem 1.5.12. *Find the number of integers x with $|x| \leq 1997$ such that 1997 divides $x^2 + (x+1)^2$.*

(1998 Indian Mathematical Olympiad)

Solution. There are 4 such integers. With congruences all taken modulo 1997, we have

$$x^2 + (x+1)^2 \equiv 2x^2 + 2x + 1 \equiv 4x^2 + 4x + 2 \equiv 0,$$

i.e., $(2x+1)^2 \equiv -1$. Since 1997 is a prime of the form $4k+1$, there are exactly two distinct solutions to $u^2 \equiv -1$. Each corresponds to a different solution to $(2x+1)^2 \equiv -1$.

Also, the two solutions to $(2x+1)^2 \equiv -1$ are nonzero since 0 does not satisfy the equation. Therefore, there are exactly two satisfactory integers x from -1997 to -1 and two more from 1 to 1997, for a total of four integer solutions, as claimed.

12.6 Chinese remainder theorem

Problem 1.6.3. *Let $P(x)$ be a polynomial with integer coefficients. Suppose that the integers $a_1, a_2, \ldots, a_n$ have the following property: For any integer x there exists an $i \in \{1, 2, \ldots, n\}$ such that $P(x)$ is divisible by a_i. Prove that there is an $i_0 \in \{1, 2, \ldots, n\}$ such that a_{i_0} divides $P(x)$ for any integer x.*

(St. Petersburg City Mathematical Olympiad)

Solution. Suppose that the claim is false. Then for each $i = 1, 2, \ldots, n$ there exists an integer x_i such that $P(x_i)$ is not divisible by a_i. Hence, there is a prime power $p_i^{k_i}$ that divides a_i and does not divide $P(x_i)$. Some of the powers $p_1^{k_1}, p_2^{k_2}, \ldots, p_n^{k_n}$ may have the same base. If so, ignore all but the one with the least exponent. To simplify notation, assume that the sequence obtained this way is $p_1^{k_1}, p_2^{k_2}, \ldots, p_m^{k_m}$, $m \leq n$ ($p_1, p_2, \ldots, p_m$ are distinct primes). Note that each a_i is divisible by some term of this sequence.

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Since $p_1^{k_1}, p_2^{k_2}, \ldots, p_m^{k_m}$ are pairwise relatively prime, the Chinese Remainder Theorem yields a solution of the simultaneous congruences

$$x \equiv x_1 \pmod{p_1^{k_1}},\ x \equiv x_2 \pmod{p_2^{k_2}}, \ldots, x \equiv x_m \pmod{p_m^{k_m}}.$$

Now, since $P(x)$ is a polynomial with integer coefficients, the congruence $x \equiv x_j \pmod{p_j^{k_j}}$ implies $P(x) \equiv P(x_j) \pmod{p_j^{k_j}}$ for each index $j = 1, 2, \ldots, m$. By the definition of $p_j^{k_j}$, the number $P(x_j)$ is never divisible by $p_j^{k_j}$, $j = 1, 2, \ldots, m$. Thus, for the solution x given by the Chinese Remainder Theorem, $P(x)$ is not divisible by any of the powers $p_j^{k_j}$. And because each a_i is divisible by some $p_j^{k_j}$, $j = 1, 2, \ldots, m$, it follows that no a_i divides $P(x)$ either, a contradiction.

Problem 1.6.4. *For any positive integer set $\{a_1, a_2, \ldots, a_n\}$ there exists a positive integer b such that the set $\{ba_1, ba_2, \ldots, ba_n\}$ consists of perfect powers.*

Solution. There is a finite number of primes $p_1, p_2, \ldots, p_k$ that participate in the prime factorization of $a_1, a_2, \ldots, a_n$. Let

$$a_i = p_1^{\alpha_{i1}} p_2^{\alpha_{i2}} \ldots p_k^{\alpha_{ik}} \text{ for } i = 1, 2, \ldots, n;$$

some of the exponents α_{ij} may be zeros. A positive integer with prime factorization $p_1^{u_1} p_2^{u_2} \ldots p_k^{u_k}$ is a perfect q-th power if and only if all the exponents u_j are divisible by q. Thus it suffices to find positive integers $q_1, q_2, \ldots, q_n$ greater than 1, and nonnegative integers $l_1, l_2, \ldots, l_k$ such that

$$l_1 + \alpha_{11}, l_2 + \alpha_{12}, \ldots, l_k + \alpha_{1k} \text{ are divisible by } q_1,$$

$$l_1 + \alpha_{21}, l_2 + \alpha_{22}, \ldots, l_k + \alpha_{2k} \text{ are divisible by } q_2,$$

$$\ldots$$

$$l_1 + \alpha_{n1}, l_2 + \alpha_{n2}, \ldots, l_k + \alpha_{nk} \text{ are divisible by } q_n.$$

Now it is clear that we have lots of choices; let, for example, q_i be the i-th prime number. As far as l_1 is concerned, the above conditions translate into

$$l_1 \equiv -\alpha_{j1} \pmod{q_j}, \quad j = 1, 2, \ldots, n.$$

This system of simultaneous congruences has a solution by the Chinese Remainder Theorem, because $q_1, q_2, \ldots, q_n$ are pairwise relatively prime. Analogously, each of the systems of congruences

$$l_2 \equiv -\alpha_{j2} \pmod{q_j}, \quad j = 1, 2, \ldots, n$$

$$l_3 \equiv -\alpha_{j3} \pmod{q_j}, \quad j = 1, 2, \ldots, n$$

$$\ldots$$

$$l_k \equiv -\alpha_{jk} \pmod{q_j}, \quad j = 1, 2, \ldots, n$$

is solvable by the same reason. Take $l_1, l_2, \ldots, l_k$ such that all these congruences are satisfied. Multiplying each a_i by $b = p_1^{l_1} p_2^{l_2} \ldots p_k^{l_k}$ yields a set $\{ba_1, ba_2, \ldots, ba_n\}$ consisting of perfect powers (more exactly, ba_i is a perfect q_i-th power).

Remarks. 1) The following problem is a direct consequence of the above result:

Prove that for every positive integer n there exists a set of n positive integers such that the sum of the elements of each of its nonempty subsets is a perfect power.

(Korean proposal for the 33rd IMO)

Indeed, let $\{x_1, x_2, \ldots, x_m\}$ be a finite set of positive integers and $S_1, S_2, \ldots, S_r$ the element sums of its nonempty subsets ($r = 2^m - 1$). Choose a b so that $bS_1, bS_2, \ldots, bS_r$ are all perfect powers. Then the set $\{bx_1, bx_2, \ldots, bx_m\}$ yields the desired example.

2) Another consequence is the following: *There are arithmetic progressions of arbitrary finite length consisting only of powers.* Yet, no such infinite progression exists.

12.7 Numerical systems

Problem 1.7.12. *The natural number A has the following property: the sum of the integers from 1 to A, inclusive, has decimal expansion equal to that of A followed by three digits. Find A.*

(1999 Russian Mathematical Olympiad)

Solution. We know that

$$k = (1 + 2 + \cdots + A) - 1000A$$

$$= \frac{A(A+1)}{2} - 1000A = A\left(\frac{A+1}{2} - 1000\right)$$

is between 0 and 999, inclusive. If $A < 1999$ then k is negative. If $A \geq 2000$ then $\frac{A+1}{2} - 1000 \geq \frac{1}{2}$ and $k \geq 1000$. Therefore $A = 1999$, and indeed $1 + 2 + \cdots + 1999 = 1999000$.

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Problem 1.7.13. *A positive integer is said to be balanced if the number of its decimal digits equals the number of its distinct prime factors. For instance, 15 is balanced, while 49 is not. Prove that there are only finitely many balanced numbers.*

(1999 Italian Mathematical Olympiad)

Solution. Let $p_1 = 2$, $p_2 = 3, \ldots$ be the sequence of primes. If x is balanced and it has n numbers, then

$$10^n \geq p_1 p_2 \ldots p_n \geq 2 \cdot 3 \cdot 5 \ldots (2n-1) > 2 \cdot 2 \cdot 4 \ldots (2n-2) > (n-1)!,$$

which implies that n is bounded and so is x, since $x \leq 10^n$.

Problem 1.7.14. *Let $p \geq 5$ be a prime and choose $k \in \{0, \ldots, p-1\}$. Find the maximum length of an arithmetic progression, none of whose elements contain the digit k when written in base p.*

(1997 Romanian Mathematical Olympiad)

Solution. We show that the maximum length is $p-1$ if $k \neq 0$ and p is $k = 0$. In a p-term arithmetic progression, the lowest nonconstant digit takes all values from 0 to $p-1$. This proves the upper bound for $k \neq 0$, which is also a lower bound because of the sequence $1, \ldots, p-1$. However, for $k = 0$, it is possible that when 0 occurs, it is not actually a digit in the expansion but rather a leading zero. This can only occur for the first term in the progression, so extending the progression to $p+1$ terms would cause an honest zero to appear. Thus the upper bound for $k = 0$ is p, and the sequence $1, p+1, \ldots, (p-1)p+1$ shows that it is also a lower bound.

Problem 1.7.15. *How many 10-digit numbers divisible by 66667 are there whose decimal representation contains only the digits 3, 4, 5, and 6?*

(1999 St. Petersburg City Mathematical Olympiad)

Solution. Suppose that $66667n$ had 10 digits, all of which were 3, 4, 5, and 6. Then

$$3333333333 \leq 66667n \leq 6666666666 \;\Rightarrow\; 50000 \leq n \leq 99999.$$

Now consider the following cases:

(i) $n \equiv 0 \pmod 3$. Then

$$66667n = \frac{2}{3}n \cdot 10^5 + \frac{1}{3}n,$$

the five digits of $3 \cdot \frac{n}{3}$ followed by the five digits of $\frac{n}{3}$. These digits are all 3, 4, 5, or 6 if and only if $\frac{n}{3} = 33333$ and $n = 99999$.

(ii) $n \equiv 1 \pmod 3$. Then

$$66667n = \frac{2}{3}(n-1) \cdot 10^5 + \frac{1}{3}(n+2) + 66666,$$

the five digits of $\frac{2}{3}(n-1)$ followed by the five digits of $\frac{1}{3}(n+2) + 66666$. Because $\frac{1}{3}(n+2) + 66666$ must be between 66667 and 99999, its digits cannot be 3, 4, 5, or 6. Hence there are no satisfactory $n \equiv 1 \pmod 3$.

(iii) $n \equiv 2 \pmod 3$. Let $a = \frac{1}{3}(n-2)$. Then

$$66667n = \left(\frac{2}{3}(n-2) + 1\right) \cdot 10^5 + \frac{1}{3}(n-2) + 33334,$$

the five digits of $x = 2a+1$ followed by the five digits of $y = a+33334$. The units digits in x and y are between 3 and 6 if and only if the units digit in a is 1 or 2. In this case the other digits in x and y are all between 3 and 6 if and only if the other digits in a are 2 or 3. Thus there are thirty-two satisfactory a - we can choose each of its five digits from two options - and each a corresponds to a satisfactory $n = 3a + 2$.

Therefore there is exactly one satisfactory $n \equiv 0 \pmod 3$, and thirty-two satisfactory $n \equiv 2 \pmod 3$ - making a total of thirty-three values of n and thirty-three ten-digit numbers.

Problem 1.7.16. *Call positive integers similar if they are written using the same set of digits. For example, for the set 1, 1, 2, the similar numbers are 112, 121 and 211. Prove that there exist 3 similar 1995-digit numbers containing no zeros, such that the sum of two them equals the third.*

(1995 Russian Mathematical Olympiad)

Solution. Noting that 1995 is a multiple of 3, we might first trying to find 3 similar 3-digit numbers such that the sum of two of them equals the third. There are various digits arrangements to try, one of which is $\overline{abc}+\overline{acb} = \overline{cba}$. The middle column must have a carry or else we would have $c = 0$ and no integer can begin with a 0. If there is a carry, we must have $c = 9$, which implies $a = 4$ by looking at the first column. From the third column, we find $b = 5$ and discover that indeed $459 + 495 = 954$. Now to solve the original problem, simply write $459\ldots459 + 495\ldots495 = 954\ldots954$, where each three-digit number is repeated 1995/3 times.

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Problem 1.7.17. *Let k and n be positive integers such that*

$$(n+2)^{n+2}, (n+4)^{n+4}, (n+6)^{n+6}, \ldots, (n+2k)^{n+2k}$$

end in the same digit in decimal representation. At most how large is k?

(1995 Hungarian Mathematical Olympiad)

Solution. We cannot have $k \geq 5$, since then one of the terms would be divisible by 5 and so would end in a different digit than those not divisible by 5. Hence $k \leq 4$. In fact, we will see that $k = 3$ is best possible.

Since $x^5 \equiv x \pmod{10}$ for all x, $x^x \pmod{10}$ only depends on $x \pmod{20}$. Hence it suffices to tabulate the last digit of x^x for $x = 0, \ldots, 19$ and look for the longest run. For the evens, we get

$$0, 4, 6, 6, 6, 0, 6, 6, 6, 4$$

while for the odds we get

$$1, 7, 5, 3, 9, 1, 3, 5, 7, 9.$$

Clearly a run of 3 is best possible.

Problem 1.7.18. *Let*

$$\prod_{n=1}^{1996} (1 + nx^{3^n}) = 1 + a_1 x^{k_1} + a_2 x^{k_2} + \cdots + a_m x^{k_m},$$

where $a_1, a_2, \ldots, a_m$ are nonzero and $k_1 < k_2 < \cdots < k_m$, Find a_{1996}.

(1996 Turkish Mathematical Olympiad)

Solution. Note that k_i is the number obtained by writing i in base 2 and reading the result as a number in base 3, and a_i is the sum of the exponents of the powers of 3 used. In particular, $1996 = 2^{10}+2^9+2^8+2^7+2^6+2^3+2^2$, so

$$a_{1996} = 10 + 9 + 8 + 7 + 6 + 3 + 2 = 45.$$

Problem 1.7.19. *For any positive integer k, let $f(k)$ be the number of element in the set $\{k+1, k+2, \ldots, 2k\}$ whose base 2 representation has precisely three 1s.*

a) Prove that, for each positive integer m, there exists at least one positive integer k, such that $f(k) = m$.

b) Determine all positive integers m for which there exists exactly one k with $f(k) = m$.

(35th IMO)

Solution. a) Let $g : \mathbb{N} \to \mathbb{N}$ be the function defined as follows: $g(k)$ is the number of elements in the set $\{1, 2, \ldots, k\}$ having three digits 1 in their binary representation. The following equalities are obvious:

$$f(k) = g(2k) - g(k)$$

and

$$f(k+1) - f(k) = g(2k+2) - g(2k) - (g(k+1) - g(k)).$$

The binary representation of $2k+2$ is obtained by adding a final 0 in the binary representation of $k+1$. Thus, we have the following result:

$$f(k+1) - f(k) = \begin{cases} 1 & \text{if binary representation of } 2k+1 \\ & \text{contains three digits 1} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

It proves that the function f increases with at most a unit from k to $k+1$.

Since $g(2^n) = \binom{n}{3}$ and $f(2^n) = \binom{n+1}{3} - \binom{n}{3} = \binom{n}{2}$, it follows that f is unbounded function. If combine with the above property and observe that $f(4) = 1$ one obtains that the range of f is the set of all positive integers.

b) Let suppose that the equation $f(k) = m$ has a unique solution. It follows that

$$f(k+1) - f(k) = f(k) - f(k+1) = 1.$$

By (1), it follows that binary representations of $2k+1$ and $2k-1$ contain three digits 1. Then the binary representation of k contains two digits 1. From $2k - 1 = 2(k-1) + 1$ one obtains that the binary representation of $k-1$ also contains two digits 1. Hence, the last digit of $k-1$ is 1 and the last but one digit is 0. Thus, $k - 1 = 2^n + 1$ and $k = 2^n + 2$, where $n \geq 2$.

For such a number we have:

$$f(2^n + 2) = g(2^{n+1} + 4) - g(2^n + 2) = 1 + g(2^{n+1}) - g(2^n) = 1 + \binom{n}{2}.$$

Thus, we have proved that the equation $f(k) = m$ has unique solution if and only if m is a number of the form $m = 1 + \binom{n}{2}$, $n \geq 2$.

Problem 1.7.20. *For each positive integer n, let $S(n)$ be the sum of digits in the decimal representation of n. Any positive integer obtained by*

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removing several (at least one) digits from the right-hand end of the decimal representation of n is called a stump of n. Let $T(n)$ be the sum of all stumps of n. Prove that $n = S(n) + 9T(n)$.

(2001 Asian Pacific Mathematical Olympiad)

Solution. Let d_i be the digit associated with 10^i in the base 10 representation of n, so that $n = \overline{d_m d_{m-1} \dots d_0}$ for some integer $m \geq 0$ (where $d_m \neq 0$). The stumps of n are $\sum_{j=k}^{m} d_j 10^{j-k}$ for $k = 1, 2, \dots, m$, and their sum is

$$\begin{aligned} T(n) &= \sum_{k=1}^{m} \sum_{j=k}^{m} d_j 10^{j-k} = \sum_{j=1}^{m} d_j \sum_{k=1}^{j} 10^{j-k} \\ &= \sum_{j=1}^{m} d_j \sum_{k=0}^{j-1} 10^k = \sum_{j=1}^{m} d_j \frac{10^j - 1}{10 - 1}. \end{aligned}$$

Hence,

$$\begin{aligned} 9T(n) &= \sum_{j=1}^{m} d_j (10^j - 1) = \sum_{j=1}^{m} 10^j d_j - \sum_{j=1}^{m} d_j \\ &= \sum_{j=0}^{m} 10^j d_j - \sum_{j=0}^{m} d_j = n - S(n), \end{aligned}$$

as desired.

Problem 1.7.21. *Let p be a prime number and m be a positive integer. Show that there exists a positive integer n such that there exist m consecutive zeroes in the decimal representation of p^n.*

(2001 Japanese Mathematical Olympiad)

Solution. It is well-known that if $gcd(s, t) = 1$, then $s^k \equiv 1 \pmod{t}$ for some k: indeed, of all the positive powers of s, some two $s^{k_1} < s^{k_2}$ must be congruent modulo t, and then $s^{k_2 - k_1} \equiv 1 \pmod{t}$.

First suppose that $p \neq 2, 5$. Then $gcd(p, 10^{m+1}) = 1$, so there exists such k that $p^k \equiv 1 \pmod{10^{m+1}}$. Then $p^k = a \cdot 10^{m+1} + 1$, so there are m consecutive zeroes in the decimal representation of p^k.

Now suppose that $p = 2$. We claim that for any a, some power of 2 has the following final a digits: $a - \lceil \log 2^a \rceil$ zeroes, followed by the $\lceil \log 2^a \rceil$ digits of 2^a. Because $gcd(2, 5^a) = 1$, there exists k such that $2^k \equiv 1 \pmod{5^a}$.

Let $b = k + a$. Then $2^b \equiv 2^a \pmod{5^a}$, and $2^b \equiv 0 \equiv 2^a \pmod{2^a}$. Hence, $2^b \equiv 2^a \pmod{10^a}$. Because $2^a < 10^a$, it follows that 2^b has the required property.

Now, simply choose a such that $a - \lceil \log 2^a \rceil \geq m$ (for instance, we could choose $a = \left\lceil \frac{m+1}{1-\log 2} \right\rceil$). Then 2^b contains at least m consecutive zeroes, as desired.

Finally, the case $p = 5$ is done analogously to the case $p = 2$.

Remark. Actually, the property holds for every integer $p \geq 2$. If p is a power of 2, it is trivial. Otherwise, one can prove using Kronecker[1]'s theorem (stating that for $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ the set of $\{n_\alpha\}$ with $n \in \mathbb{N}$ is dense in $[0, 1]$) that the numbers p^n can start with any combination of digits we may need, in particular with $1\underbrace{00\ldots0}_{m\,times}$.

Problem 1.7.22. *Knowing 2^{29} is an 9-digit number whose digits are distinct, without computing the actual number determine which of the ten digits is missing. Justify your answer.*

Solution. It is not difficult to see that, when divided by 9, the remainder is 5. The ten-digit number containing all digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 is a multiple of 9, because the sum of its digits has this property. So, in our nine-digit number, 4 is missing.

Problem 1.7.23. *It is well known that the divisibility tests for division by 3 and 9 do not depend on the order of the decimal digits. Prove that 3 and 9 are the only positive integers with this property. More exactly, if an integer $d > 1$ has the property that $d|n$ implies $d|n_1$, where n_1 is obtained from n through an arbitrary permutation of its digits, then $d = 3$ or $d = 9$.*

Solution. Let d be a k-digit number. Then among the $(k + 2)$-digit numbers starting with 10 there is at least one that is divisible by d. Denote it by $\overline{10a_1a_2\ldots a_k}$. The assumption implies that both numbers $\overline{a_1a_2\ldots a_k10}$ and $\overline{a_1a_2\ldots a_k01}$ are divisible by d, and then so is their difference. This difference equals 9 and the proof is finished, since d may only be some divisor of 9.

Remark. The following problem given in an old Russian Mathematical Olympiad is much more restrictive and difficult:

[1] *Leopold Kronecker* (1823-1891), German mathematician with important contributions in the theory of equations. He made major contributions in elliptic functions and the theory of algebraic numbers.

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Suppose that $d > 1$ *has the property that* $d|n$ *implies* $d|n_1$ *where* n_1 *is obtained from* n *by reversing the order of its digits. Then* $d|99$. Try to solve this problem.

13

Powers of Integers

13.1 Perfect squares

Problem 2.1.14. *Let x, y, z be positive integers such that*

$$\frac{1}{x} - \frac{1}{y} = \frac{1}{z}.$$

Let h be the greatest common divisor of x, y, z. Prove that $hxyz$ and $h(y-x)$ are perfect squares.

(1998 United Kingdom Mathematical Olympiad)

Solution. Let $x = ha$, $y = hb$, $z = hc$. Then a, b, c are positive integers such that $gcd(a, b, c) = 1$. Let $gcd(a, b) = g$. So $a = ga'$, $b = gb'$ and a' and b' are positive integers such that

$$gcd(a', b') = gcd(a' - b', b') = gcd(a', a' - b') = 1.$$

We have

$$\frac{1}{a} - \frac{1}{b} = \frac{1}{c} \Leftrightarrow c(b-a) = ab \Leftrightarrow c(b'-a') = a'b'g.$$

So $g|c$ and $gcd(a, b, c) = g = 1$. Therefore $gcd(a, b) = 1$ and $gcd(b - a, ab) = 1$. Thus $b - a = 1$ and $c = ab$. Now

$$hxyz = h^4abc = (h^2ab)^2 \quad \text{and} \quad h(y-x) = h^2$$

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are both perfect squares, as desired.

Problem 2.1.15. *Let b an integer greater than 5. For each positive integer n, consider the number*

$$x_n = \underbrace{11\ldots1}_{n-1}\underbrace{22\ldots2}_{n}5,$$

written in base b. Prove that the following condition holds if and only if $b = 10$: There exists a positive integer M such that for every integer n greater than M, the number x_n is a perfect square.

(44th IMO Shortlist)

Solution. Assume that $b \geq 6$ has the required property. Consider the sequence $y_n = (b-1)x_n$. From the definition of x_n we easily find that

$$y_n = b^{2n} + b^{n+1} + 3b - 5.$$

Then $y_n y_{n+1} = (b-1)^2 x_n x_{n+1}$ is a perfect square for all $n > M$. Also, straightforward calculation implies

$$\left(b^{2n+1} + \frac{b^{n+2}+b^{n+1}}{2} - b^3\right)^2 < y_n y_{n+1} < \left(b^{2n+1} + \frac{b^{n+2}+b^{n+1}}{2} + b^3\right)^2.$$

Hence for every $n > M$ there is an integer a_n such that $|a_n| < b^3$ and

$$\begin{aligned} y_n y_{n+1} &= (b^{2n} + b^{n+1} + 3b - 5)(b^{2n+2} + b^{n+2} + 3b - 5) \\ &= \left(b^{2n+1} + \frac{b^{n+1}(b+1)}{2} + a_n\right)^2. \end{aligned} \tag{1}$$

Now considering this equation modulo b^n we obtain $(3b-5)^2 \equiv a_n^2$, so that assuming that $n > 3$ we get $a_n = \pm(3b-5)$.

If $a_n = 3b - 5$, then substituting in (1) yields

$$\frac{1}{4}b^{2n}(b^4 - 14b^3 + 45b^2 - 52b + 20) = 0,$$

with the unique positive integer solution $b = 10$. Also, if $a_n = -3b + 5$, we similarly obtain

$$\frac{1}{4}(b^4 - 14b^3 - 3b^2 + 28b + 20) - 2b^{n+1}(3b^2 - 2b - 5) = 0$$

for each n, which is impossible.

For $b = 10$ it is easy to show that $x_n = \left(\dfrac{10^n + 5}{3}\right)^2$ for all n. This proves the statement.

Second solution. In problems of this type, computing $z_n = \sqrt{x_n}$ asymptotically usually works.

From $\lim\limits_{n\to\infty} \dfrac{b^{2n}}{(b-1)x_n} = 1$ we infer that $\lim\limits_{n\to\infty} \dfrac{b^n}{z_n} = \sqrt{b-1}$. Furthermore, from

$$(bz_n + z_{n+1})(bz_n - z_{n+1}) = b^2x_n - x_{n+1} = b^{n+2} + 3b^2 - 2b - 5$$

we obtain

$$\lim_{n\to\infty}(bz_n - z_{n+1}) = \frac{b\sqrt{b-1}}{2}.$$

Since the z_n's are integers for all $n \geq M$, we conclude that

$$bz_n - z_{n+1} = \frac{b\sqrt{b-1}}{2}$$

for all n sufficiently large. Hence $b-1$ is a perfect square, and moreover b divides $2z_{n+1}$ for all large n. It follows that $b|10$; hence the only possibility is $b = 10$.

Problem 2.1.16. *Do there exist three natural numbers greater than 1, such that the square of each, minus one, is divisible by each of the others?*

(1996 Russian Mathematical Olympiad)

Solution. Such integers do not exist. Suppose $a \geq b \geq c$ satisfy the desired condition. Since $a^2 - 1$ is divisible by b, the numbers a and b are relatively prime. Hence the number $c^2 - 1$, which is divisible by a and b, must be a multiple of ab, so in particular $c^2 - 1 \geq ab$. But $a \geq c$ and $b \geq c$, so $ab \geq c^2$, a contradiction.

Problem 2.1.17. *(a) Find the first positive integer whose square ends in three 4's.*

(b) Find all positive integers whose squares end in three 4's.

(c) Show that no perfect square ends with four 4's.

(1995 United Kingdom Mathematical Olympiad)

Solution. It is easy to check that $38^2 = 1444$ is the first positive integer whose square ends in three 4's. Now let n be any such positive integer. Then $n^2 - 38^2 = (n-38)(n+38)$ is divisible by $1000 = 2^3 \cdot 5^3$. Hence at least one of $n - 38$, $n + 38$ is divisible by 4, and thus both are, since their

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difference is $76 = 4 \cdot 19$. Since $5 \nmid 76$, then 5 divides only one of the two factors. Consequently $n - 38$ or $n + 38$ is a multiple of $4 \cdot 5^3 = 500$, so we have $n = 500k \pm 38$. It is easy to check that the square of all numbers of this form (where k is a positive integer) end in three 4's.

Note that c) follows from Problem **??**.

Problem 2.1.18. *Let m, n be a natural numbers and $m + i = a_i b_i^2$ for $i = 1, 2, \ldots, n$, where a_i and b_i are natural numbers and a_i is squarefree. Find all values of n for which there exists m such that $a_1 + a_2 + \cdots + a_n = 12$.*

(1997 Bulgarian Mathematical Olympiad)

Solution. Clearly $n \leq 12$. That means at most three of the $m + i$ are perfect squares, and for the others, $a_i \geq 2$, so actually $n \leq 7$.

We claim $a_i \neq a_j$ for $i = j$. Otherwise, we would have $m + i = ab_i^2$ and $m + j = ab_j^2$, so $6 \geq n - 1 \geq (m + j) - (m + i) = a(b_j^2 - b_i^2)$. This leaves the possibilities $(b_i, b_j, a) = (1, 2, 2)$ or $(2, 3, 1)$, but both of those force $a_1 + \cdots + a_n > 12$.

Thus the a's are a subset of $\{1, 2, 3, 5, 6, 7, 10, 11\}$. Thus $n \leq 4$, with equality only if $\{a_1, a_2, a_3, a_4\} = \{1, 2, 3, 6\}$. But in that case,

$$(6b_1b_2b_3b_4)^2 = (m + 1)(m + 2)(m + 3)(m + 4) = (m^2 + 5m + 5)^2 - 1,$$

which is impossible. Hence $n = 2$ or $n = 3$. One checks that the only solutions are then

$$(m, n) = (98, 2), (3, 3).$$

Problem 2.1.19. *For each positive integer n, denote by $s(n)$ the greatest integer such that for all positive integer $k \leq s(n)$, n^2 can be expressed as a sum of squares of k positive integers.*

(a) Prove that $s(n) \leq n^2 - 14$ for all $n \geq 4$.

(b) Find a number n such that $s(n) = n^2 - 14$.

(c) Prove that there exist infinitely many positive integers n such that

$$s(n) = n^2 - 14.$$

(33rd IMO)

Solution. (a) Representing n^2 as a sum of $n^2 - 13$ squares is equivalent to representing 13 as a sum of numbers of the form $x^2 - 1$, $x \in \mathbb{N}$, such

as $0, 3, 8, 15, \ldots$ But it is easy to check that this is impossible, and hence $s(n) \leq n^2 - 14$.

(b) Let us prove that $s(13) = 13^2 - 14 = 155$. Observe that

$$\begin{aligned} 13^2 &= 8^2 + 8^2 + 4^2 + 4^2 + 3^2 \\ &= 8^2 + 8^2 + 4^2 + 4^2 + 2^2 + 2^2 + 1^2 \\ &= 8^2 + 8^2 + 4^2 + 3^2 + 3^2 + 2^2 + 1^2 + 1^2 + 1^2. \end{aligned}$$

Given any representation of n^2 as a sum of m squares one of which is even, we can construct a representation as a sum of $m + 3$ squares by dividing the odd square into four equal squares. Thus the first equality enables us to construct representations with $5, 8, 11, \ldots, 155$ squares, the second to construct ones with $7, 10, 13, \ldots, 154$ squares, and the third with $9, 12, \ldots, 153$ squares. It remains only to represent 13^2 as a sum of $k = 2, 3, 4, 6$ squares. This can be done as follows:

$$\begin{aligned} 13^2 = 12^2 + 5^2 &= 12^2 + 4^2 + 3^2 \\ &= 11^2 + 4^2 + 4^2 + 4^2 \\ &= 12^2 + 3^2 + 2^2 + 2^2 + 2^2 + 2^2. \end{aligned}$$

(c) We shall prove that whenever $s(n) = n^2 - 14$ for some $n \geq 13$, it also holds that $s(2n) = (2n)^2 - 14$. This will imply that $s(n) = n^2 - 14$ for any $n = 2^t \cdot 13$.

If $n^2 = x_1^2 + \cdots + x_r^2$, then we have $(2n)^2 = (2x_1)^2 + \cdots + (2x_r)^2$. Replacing $(2x_i)^2$ with $x_i^2 + x_i^2 + x_i^2 + x_i^2$ as long as it is possible we can obtain representations of $(2n)^2$ consisting of $r, r + 3, \ldots, 4r$ squares. This gives representations of $(2n)^2$ into k squares for any $k \leq 4n^2 - 62$. Further, we observe that each number $m \geq 14$ can be written as a sum of $k \geq m$ numbers of the form $x^2 - 1$, $x \in \mathbb{N}$, which is easy to verify. Therefore if $k \leq 4n^2 - 14$, it follows that $4n^2 - k$ is a sum of k numbers of the form $x^2 - 1$ (since $k \geq 4n^2 - k \geq 14$), and consequently $4n^2$ is a sum of k squares.

Remark. One can find exactly the value of $s(n)$ for each n:

$$s(n) = \begin{cases} 1, & \text{if } n \text{ has a prime divisor congruent to 3 mod 4,} \\ 2, & \text{if } n \text{ is of the form } 5 \cdot 2^k,\ k \text{ a positive integer,} \\ n^2 - 14, & \text{otherwise.} \end{cases}$$

Problem 2.1.20. *Let A be the set of positive integers representable in the form $a^2 + 2b^2$ for integers a, b with $b \neq 0$. Show that if $p^2 \in A$ for a prime p, then $p \in A$.*

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(1997 Romanian IMO Team Selection Test)

Solution. The case $p = 2$ is easy, so assume $p > 2$. Note that if $p^2 = a^2 + 2b^2$, then $2b^2 = (p-a)(p+a)$, In particular, a is odd, and since a cannot be divisible by p, $gcd(p-a, p+a) = gcd(p-a, 2p) = 2$. By changing the sign of a, we may assume $p - a$ is not divisible by 4, and so

$$|p + a| = m^2, \quad |p - a| = 2n^2$$

Since $|a| < |p|$, both $p + a$ and $p - a$ are actually positive, so we have $2p = m^2 + 2n^2$, so $p = n^2 + 2(m/2)^2$.

Problem 2.1.21. *Is it possible to find 100 positive integers not exceeding 25000 such that all pairwise sums of them are different?*

(42nd IMO Shortlist)

Solution. Yes. The desired result is an immediate consequence of the following fact applied on $p = 101$.

Lemma. *For any odd prime number p, there exist p nonnegative integers less than $2p^2$ with all pairwise sums mutually distinct.*

Proof. We claim that the numbers $a_n = 2np + (n^2)$ have the desired property, where (x) denotes the remainder of x upon division by p.

Suppose that $a_k + a_l = a_m + a_n$. By the construction of a_i, we have

$$2p(k + l) \leq a_k + a_l \leq 2p(k + l + 1).$$

Hence we must have $k + l = m + n$, and therefore also

$$(k^2) + (l^2) = (m^2) + (n^2).$$

Thus

$$k + l \equiv m + n \quad \text{and} \quad k^2 + l^2 \equiv m^2 + n^2 \pmod{p}.$$

But then it holds that

$$(k - l)^2 = 2(k^2 + l^2) - (k + l)^2 \equiv (m - n)^2 \pmod{p},$$

so $k - l \equiv \pm(m - n)$, which leads to $(k, l) = (m, n)$. This proves the lemma.

Problem 2.1.22. *Do there exist 10 distinct integers, the sum of any 9 of which is a perfect square?*

(1999 Russian Mathematical Olympiad)

Solution. Yes, there do exist 10 such integers. Write $S = a_1 + a_2 + \cdots + a_{10}$, and consider the linear system of equations

$$S - a_1 = 9 \cdot 1^2$$

$$S - a_2 = 9 \cdot 2^2$$

$$\cdots$$

$$S - a_{10} = 9 \cdot 10^2.$$

Adding all these gives

$$9S = 9 \cdot (1^2 + 2^2 + \cdots + 10^2)$$

so that

$$a_k = S - 9k^2 = 1^2 + 2^2 + \cdots + 10^2 - 9k^2.$$

Then all the a_k's are distinct integers, and any nine of them add up to a perfect square.

Problem 2.1.23. *Let n be a positive integer such that n is a divisor of the sum*

$$1 + \sum_{i=1}^{n-1} i^{n-1}$$

Prove that n is square-free.

(1995 Indian Mathematical Olympiad)

Solution. If $n = mp^2$ for some prime p, then

$$1 + \sum_{i=1}^{n-1} i^{n-1} = 1 + \sum_{j=0}^{p-1} \sum_{k=0}^{mp-1} (kp + j)^{n-1}$$

$$\equiv 1 + (mp) \left(\sum_{j=0}^{p-1} j^{n-1} \right) \equiv 1 \pmod{p}$$

and the sum is not even a multiple of p. Hence if the sum is a multiple of n, n must have no repeated prime divisors, or equivalently no square divisors greater than 1.

Remark. The famous Giuga's conjecture states that if $n > 1$ verifies $n | 1 + \sum_{i=1}^{n-1} i^{n-1}$, then n is a prime.

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The reader can prove instead that for any such n we have: for any prime divisor p of n, $p-1|\frac{n}{p}-1$ and $p|\frac{n}{p}-1$.

Problem 2.1.24. *Let n, p be integers such that $n>1$ and p is a prime. If $n|(p-1)$ and $p|(n^3-1)$, show that $4p-3$ is a perfect square.*

(2002 Czech-Polish-Slovak Mathematical Competition)

Solution. From $n|p-1$ it follows $p-1 \geq n$ and $p>n$. Because

$$p|n^3-1=(n-1)(n^2+n+1)$$

we get $p|n^2+n+1$, i.e. $pk=n^2+n+1$ for some positive integer k.

On the other hand $n|p-1$ implies $p \equiv 1 \pmod{n}$ and $pk \equiv k \pmod{n}$. We obtain $n^2+n+1 \equiv k \pmod{n}$, hence $k \equiv 1 \pmod{n}$.

It follows that $p=an+1$, $k=bn+1$ for some integers $a>0$, $b \geq 0$. We can write

$$(an+1)(bn+1)=n^2+n+1,$$

so

$$abn^2+(a+b)n+1=n^2+n+1$$

i.e.

$$abn+(a+b)=n+1.$$

If $b \geq 1$, then $abn+(a+b) \geq n+2>n+1$. So $b=0$, $k=1$, $p=n^2+n+1$. Therefore

$$4p-3=4n^2+4n+4-3=4n^2+4n+1=(2n+1)^2.$$

Problem 2.1.25. *Show that for any positive integer $n>10000$, there exists a positive integer m that is a sum of two squares and such that $0<m-n<3\sqrt[4]{n}$.*

(Russian Mathematical Olympiad)

Solution. We have $a^2<n \leq (a+1)^2$ for some integer $a \geq 100$. If we write $n=a^2+k$, this means that $k \leq 2a+1$. We want $m=a^2+b^2$ for some integer b. The condition $0<m-n<3\sqrt[4]{n}$ becomes $k<b^2<k+3\sqrt[4]{a^2+k}$. We will show that

$$b=\begin{cases} \sqrt{k}+1 & \text{if } k \text{ is a perfect square} \\ \lceil \sqrt{k} \rceil & \text{if } k \text{ is a perfect square} \end{cases}$$

will work.

Note that in both cases, $k < b^2 \leq (\sqrt{k}+1)^2$. Thus we want

$$(\sqrt{k}+1)^2 < k + 3\sqrt[4]{a^2+k}$$

$$2\sqrt{k}+1 < 3\sqrt[4]{a^2+k}$$

$$4k + 4\sqrt{k} + 1 < 9\sqrt{a^2+k}.$$

Since $k \leq 2a+1$, it is sufficient to prove

$$4(2a+1) + 4\sqrt{2a+1} + 1 < 9a$$

$$a > 4\sqrt{2a+1} + 5$$

$$a^2 - 10a + 25 > 16(2a+1)$$

$$a^2 - 42a + 9 > 0.$$

Because $a^2 - 42a + 9 = a(a-42) + 9$, this last inequality is clearly true for $a \geq 100$.

Problem 2.1.26. *Show that a positive integer m is a perfect square if and only if for each positive integer n, at least one of the differences*

$$(m+1)^2 - m, (m+2)^2 - m, \ldots, (m+n)^2 - m$$

is divisible by n.

(2002 Czech and Slovak Mathematical Olympiad)

Solution. First, assume that m is a perfect square. If $m = a^2$, then

$$(m+c)^2 - m = (m+c)^2 - a^2 = (m+c+a)(m+c-a).$$

Clearly, there exists some c, with $1 \leq c \leq n$, for which $m+c+a$ is divisible by n. Thus, one of the given differences is divisible by n if m is a perfect square.

Now, we assume that m is not a perfect square and show that there exists n for which none of the given differences is divisible by n. Clearly, there exist a prime p and positive integer k such that p^{2k-1} is the highest power of p which divides m. We may let $m = bp^{2k-1}$, with b and p being relatively prime. Furthermore, pick $n = p^{2k}$. For the sake of contradiction, assume there exists a positive integer c for which $(m+c)^2 - m$ is divisible by n. By expanding $(m+c)^2 - m$, we note that

$$p^{2k} | (2bcp^{2k-1} + c^2 - bp^{2k-1})$$

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If p^{2k} divides the quantity, then so does p^{2k-1}. Thus, $p^{2k-1}|c^2$ and so $p^k|c$. Let $c = rp^k$. Then, we have

$$p^{2k}|(2brp^{3k-1} + r^2p^{2k} - bp^{2k-1})$$

However, this implies that $p|b$, which contradicts the original assumption that b and p are relatively prime. Therefore, if m is not a perfect square, n may be chose so that none of the given differences are divisible by n. This completes the proof.

13.2 Perfect cubes

Problem 2.2.5. *Find all the positive perfect cubes that are not divisible by 10 so that the number obtained by erasing the last three digits is also a perfect cube.*

Solution. We have $(10m + n)^3 = 1000a^3 + b$, where $1 \leq n \leq 9$ and $b < 1000$.

The equality gives

$$(10m + n)^3 - (10a)^3 = b < 1000,$$

so

$$(10m + n - 10a)[(10m + n)^2 + (10m + n) \cdot 10a + 100a^2] < 1000.$$

As $(10m+n)^2+(10m+n)\cdot 10a+100a^2 > 100$, we obtain $10m+n-10a < 10$, hence $m = a$.

If $m \geq 2$, then $n(300m^2 + 30mn + n^2) > 1000$ false.

Then $m = 1$ and $n(300 + 30n + n^2) < 1000$, hence $n \leq 2$. For $n = 2$, we obtain $12^3 = 1728$ and for $n = 1$ we get $11^3 = 1331$.

Problem 2.2.6. *Find all positive integers n less than 1999 such that n^2 is equal to the cube of the sum of n's digits.*

(1999 Iberoamerican Mathematical Olympiad)

Solution. In order for n^2 to be a cube, n must be a cube itself. Because $n < 1000$ we must have $n = 1^3, 2^3, \ldots,$ or 9^3. Quick checks show that $n = 1$ and $n = 27$ work while $n = 8, 64,$ and 125 don't. As for $n \geq 6^3 = 216$, we have $n^2 \geq 216^2 > 27^2$. However, the sum of n's digits is at most $9 + 9 + 9 = 27$, implying that no $n \geq 6^3$ has the desired property. Thus $n = 1, 27$ are the only answers.

Problem 2.2.7. *Prove that for any non-negative integer n the number*

$$A = 2^n + 3^n + 5^n + 6^n$$

is not a perfect cube.

Solution. We will use modular arithmetic. A perfect cube has the form $7k$, $7k+1$, or $7k-1$, since

$$(7x+1)^3 \equiv (7x+2)^3 = (7x+4)^3 \equiv 1 \pmod 7,$$

and

$$(7x+3)^3 \equiv (7x+5)^3 \equiv (7x+6)^3 \equiv -1 \pmod 7.$$

Now observe that

$$2^6 = 4^3 \equiv 1 \pmod 7$$

$$3^6 = 9^3 \equiv 2^3 \equiv 1 \pmod 7$$

$$5^6 = (-2)^6 = 2^6 \equiv 1 \pmod 7$$

$$6^6 \equiv (-1)^6 \equiv 1 \pmod 7.$$

It follows that $2^{6k} \equiv 3^{6k} \equiv 5^{7k} \equiv 6^{6k} \equiv 1 \pmod 7$.

Denote $a_n = 2^n + 3^n + 5^n + 6^n$ for $n \geq 0$. Set $n = 6k + r$, with $r \in \{0, 1, 2, 3, 4, 5, 6\}$. As $2^n \equiv 2^r \pmod 7$, $3^n \equiv 3^r \pmod 7$, $5^n \equiv 5^r \pmod 7$, and $6^n \equiv 6^r \pmod 7$ we have $a_n \equiv a_r \pmod 7$.

It is easy to observe that $a_0 \equiv a_2 \equiv a_6 \equiv 4 \pmod 7$, $a_1 \equiv a_4 \equiv 2 \pmod 7$ and $a_3 \equiv 5 \pmod 7$. Therefore, a_n is not a perfect cube.

The actual representations are given by (1) and

$$6n + 1 = 6n + 1^3$$
$$6n + 2 = 6(n-1) + 2^3$$
$$6n + 3 = 6(n-4) + 3^3$$
$$6n + 4 = 6(n+1) + (-2)^3$$
$$6n + 5 = 6(n+1) + (-1)^3.$$

Problem 2.2.8. *Prove that any integer is a sum of five cubes.*

Solution. For any integer n we have the identity

$$6n = (n+1)^3 + (n-1)^3 + (-n)^3 + (-n)^3. \tag{1}$$

For an arbitrary integer m we choose the integer v such that $v^3 \equiv m \pmod 6$. It follows that $m - v^3 = 6n$ for some integer n and we apply identity (1).

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Problem 2.2.9. *Show that any rational number can be written as a sum of three cubes.*

Solution. Let n be a rational number. We are looking for a relation of the form

$$a^3(x) + b^3(x) = c^3(x) + \alpha x,$$

where a, b, c are rational functions and $\alpha \in \mathbb{Q}$.

Let $\varepsilon = \cos\dfrac{2\pi}{3} + i\sin\dfrac{2\pi}{3}$. We have

$$a^3(x) + b^3(x) = (a(x) + b(x))(a(x) + \varepsilon b(x))(a(x) + \varepsilon^2 b(x))$$

and consider

$$\begin{cases} a(x) + \varepsilon b(x) = (x - \varepsilon)^3 \\ a(x) + \varepsilon^2 b(x) = (x - \varepsilon^2)^3 \end{cases}$$

Solving the above system in terms of $a(x)$ and $b(x)$ we derive the identity

$$\left(\frac{1 + 3x - x^3}{1 + x + x^2}\right)^3 + \left(\frac{3x^2 + 3x}{x^2 + x + 1}\right)^3 + (x - 1)^3 = 9x,$$

and the desired conclusion follows dividing by 9.

Remark. There are rational numbers which are not sum of two cubes. We suggest to the reader to find a such example.

13.3 k^{th} powers of integers, $k \geq 4$

Problem 2.3.6. *Let p be a prime number and a, n positive integers. Prove that if*

$$2^p + 3^p = a^n,$$

then $n = 1$.

(1996 Irish Mathematical Olympiad)

Solution. If $p = 2$, we have $2^2 + 3^2 = 13$ and $n = 1$. If $p > 2$, then p is odd, so 5 divides $2^p + 3^p$ and so 5 divides a. Now if $n > 1$, then 25 divides a^n and 5 divides

$$\frac{2^p + 3^p}{2 + 3} = 2^{p-1} - 2^{p-2} \cdot 3 + \cdots + 3^{p-1} \equiv p2^{p-1} \pmod 5,$$

a contradiction if $p \neq 5$. Finally, if $p = 5$, then $2^5 + 3^5 = 753$ is not a perfect power, so $n = 1$ again.

Problem 2.3.7. *Let x, y, p, n, k be natural numbers such that*

$$x^n + y^n = p^k.$$

Prove that if $n > 1$ is odd, and p is an odd prime, then n is a power of p.

(1996 Russian Mathematical Olympiad)

Solution. Let $m = gcd(x, y)$. Then $x = mx_1$, $y = my_1$ and by virtue of the given equation, $m^n(x_1^n + y_1^n) = p^k$, and so $m = p^\alpha$ for some nonnegative integer α. It follows that

$$x_1^n + y_1^n = p^{k-n\alpha}. \tag{1}$$

Since n is odd,

$$\frac{x_1^n + y_1^n}{x_1 + y_1} = x_1^{n-1} - x_1^{n-2}y_1 + x_1^{n-3}y_1^2 - \cdots - x_1y_1^{n-2} + y_1^{n-1}. \tag{2}$$

Let A denote the right side of the equation (2). By the condition $p > 2$, it follows that at least one of x_1, y_1 is greater than 1, so since $n > 1$, $A > 1$.

From (1) it follows that $A(x_1 + y_1) = p^{k-n\alpha}$, so, since $x_1 + y_1 > 1$ and $A > 1$, both of these numbers are divisible by p; moreover, $x_1 + y_1 = p^\beta$ for some natural number β. Thus

$$A = x_1^{n-1} - x_1^{n-2}(p^\beta - x_1) + \cdots - x_1(p^\beta - x_1)^{n-2} + (p^\beta - x_1)^{n-1}$$

$$= nx_1^{n-1} + Bp.$$

Since A is divisible by p and x_1 is relatively prime to p, it follows that n is divisible by p.

Let $n = pq$. Then $x^{pq} + y^{pq} = p^k$ or $(x^p)^q + (y^p)^q = p^k$. If $q > 1$, then by the same argument, p divides q. If $q = 1$, then $n = p$. Repeating this argument, we deduce that $n = p^l$ for some natural number l.

Problem 2.3.8. *Prove that a product of three consecutive integers cannot be a power of an integer.*

Solution. Let n be an integer and assume by contradiction that

$$n(n + 1)(n + 2) = x^z$$

for some integers x and z, where $z \geq 2$. We note that $n(n+2) = (n+1)^2 - 1$ and $n + 1$ and $(n + 1)^2 - 1$ are relatively prime. It follows

$$\begin{cases} n + 1 = a^z \\ (n + 1)^2 - 1 = b^z \end{cases}$$

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for some integers a and b. It follows $a^{2z} - b^z = 1$, i.e.

$$(a^2 - b)((a^2)^{z-1} + (a^2)^{z-2}b + \cdots + b^{z-1}) = 1.$$

We get $a^2 - b = 1$, hence $a^2 = b + 1$. The equation $(b+1)^z - b^z = 1$ has unique solution $z = 1$, a contradiction.

Remark. A famous theorem of Erdös and Selfridge, answering a conjecture of more than 150 years, states that the product of consecutive integers is never a power.

Problem 2.3.9. *Show that there exists an infinite set A of positive integers such that for any finite nonempty subset $B \subset A$, $\sum_{x \in B} x$ is not a perfect power.*

(Kvant)

Solution. The set

$$A = \{2^n 3^{n+1} : \ n \geq 1\}$$

has the desired property. Indeed, if $B = \{2^{n_1}3^{n_1+1}, \ldots, 2^{n_k}n_k^{+1}\}$ is a finite subset of A, where $n_1 < \cdots < n_k$, then

$$\sum_{x \in B} x = 2^{n_1}3^{n_1+1}(1 + 2^{n_2-n_1}3^{n_2-n_1} + \cdots + 2^{n_k-n_1}3^{n_k-n_1}) = 2^{n_1}3^{n_1+1}N,$$

where $gcd(N, 2) = gcd(N, 3) = 1$. Taking into account that n_1 and $n_1 + 1$ are relatively prime it follows that $\sum_{x \in B} x$ is not a perfect power.

Problem 2.3.10. *Prove that there is no infinite arithmetic progression consisting only of powers ≥ 2.*

Solution. Assume that we have a such arithmetic progression, $an + b$, $n = 1, 2, \ldots$ It is well known that

$$\sum_{n \geq 1} \frac{1}{an + b} = \infty \tag{1}$$

But on the other hand we have

$$\sum_{n \geq 1} \frac{1}{an + b} \leq \sum_{m,s \geq 2} \frac{1}{m^s} < +\infty,$$

relation contradicting (1).

14

Floor Function and Fractional Part

14.1 General problems

Problem 3.1.10. *Let n be a positive integer. Find with proof a closed formula for the sum:*

$$\left\lfloor \frac{n+1}{2} \right\rfloor + \left\lfloor \frac{n+2}{2^2} \right\rfloor + \cdots + \left\lfloor \frac{n+2^k}{2^{k+1}} \right\rfloor + \ldots$$

(10th IMO)

Solution. We rewrite the equality as

$$\left\lfloor \frac{n}{2} + \frac{1}{2} \right\rfloor + \left\lfloor \frac{n}{2^2} + \frac{1}{2} \right\rfloor + \cdots + \left\lfloor \frac{n}{2^{k+1}} + \frac{1}{2} \right\rfloor + \cdots = n,$$

and use a special case of Hermite's identity ($n = 2$):

$$\left\lfloor x + \frac{1}{2} \right\rfloor = \lfloor 2x \rfloor - \lfloor x \rfloor.$$

This allows us to write the equality as

$$\lfloor n \rfloor - \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{2} \right\rfloor - \left\lfloor \frac{n}{2^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{2^k} \right\rfloor - \left\lfloor \frac{n}{2^{k+1}} \right\rfloor + \cdots = n.$$

The sum telescopes and $\lfloor n/2^{k+1} \rfloor = 0$ for large enough k's.

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Problem 3.1.11. *Compute the sum*

$$\sum_{0\le i<j\le n} \left\lfloor \frac{x+i}{j} \right\rfloor,$$

where x is a real number.

Solution. Denote the sum in question by S_n. Then

$$\begin{aligned} S_n - S_{n-1} &= \left\lfloor \frac{x}{n} \right\rfloor + \left\lfloor \frac{x+1}{n} \right\rfloor + \cdots + \left\lfloor \frac{x+n-1}{n} \right\rfloor \\ &= \left\lfloor \frac{x}{n} \right\rfloor + \left\lfloor \frac{x}{n} + \frac{1}{n} \right\rfloor + \cdots + \left\lfloor \frac{x}{n} + \frac{n-1}{n} \right\rfloor, \end{aligned}$$

and, according to Hermite's identity,

$$S_n - S_{n-1} = \left\lfloor n\frac{x}{n} \right\rfloor = \lfloor x \rfloor.$$

Because $S_1 = \lfloor x \rfloor$, it follows that $S_n = n\lfloor x \rfloor$ for all n.

Problem 3.1.12. *Evaluate the difference between the numbers*

$$\sum_{k=0}^{2000} \left\lfloor \frac{3^k + 2000}{3^{k+1}} \right\rfloor \quad \textit{and} \quad \sum_{k=0}^{2000} \left\lfloor \frac{3^k - 2000}{3^{k+1}} \right\rfloor.$$

Solution. We can write each term of the difference in question as

$$\left\lfloor \frac{1}{3} + v_k \right\rfloor - \left\lfloor \frac{1}{3} - v_k \right\rfloor,$$

where $v_k = 2000/3^{k+1}$. Since $-\lfloor u \rfloor = \lfloor -u \rfloor + 1$ for each nonintegral value of u, and since $\dfrac{1}{3} - v_k$ is never an integer, we have to examine the sum

$$\sum_{k=0}^{2000} \left(\left\lfloor v_k + \frac{1}{3} \right\rfloor + \left\lfloor v_k - \frac{1}{3} \right\rfloor + 1 \right).$$

Taking $n = 3$ and $x = v - \dfrac{1}{3}$ in (1) yields

$$\left\lfloor v + \frac{1}{3} \right\rfloor + \left\lfloor v - \frac{1}{3} \right\rfloor + 1 = \lfloor 3v \rfloor - \lfloor v \rfloor.$$

Hence the desired difference becomes

$$\sum_{k=0}^{2000} \left(\left\lfloor \frac{2000}{3^k} \right\rfloor - \left\lfloor \frac{2000}{3^{k+1}} \right\rfloor \right)$$

and telescopes to

$$\lfloor 2000 \rfloor - \left\lfloor \frac{2000}{3} \right\rfloor + \left\lfloor \frac{2000}{3} \right\rfloor - \left\lfloor \frac{2000}{3^2} \right\rfloor + \cdots = 2000.$$

Problem 3.1.13. *a) Prove that there are infinitely many rational positive numbers x such that:*

$$\{x^2\} + \{x\} = 0,99.$$

b) Prove that there are no rational numbers $x > 0$ such that:

$$\{x^2\} + \{x\} = 1.$$

(2004 Romanian Mathematical Olympiad)

Solution. a) Since $0,99 = \frac{99}{100}$, it is natural to look for a rational x of the form $\frac{n}{10}$, for some positive integer n. It is not difficult to see that $x = \frac{13}{10}$ satisfies the given equality and then that $x = 10k + \frac{13}{10}$ also satisfies the equality for any positive integer k.

b) Suppose that $x = \frac{p}{q}$, with p, q positive integers, $gcd(p, q) = 1$, verifies $\{x^2\} + \{x\} = 1$. We can see that $\frac{p^2 + pq - q^2}{q^2} = x^2 + x - 1 \in \mathbb{Z}$, thus $q | p^2$ and since $gcd(p, q) = 1$, one has $q = 1$. Thus $x \in \mathbb{Z}$ and this is obviously impossible.

Problem 3.1.14. *Show that the fractional part of the number $\sqrt{4n^2 + n}$ is not greater than 0.25.*

(2003 Romanian Mathematical Olympiad)

Solution. From inequalities $4n^2 < 4n^2 + n < 4n^2 + n + 1$ one obtains $2n < \sqrt{4n^2 + n} < 2n + 1$. So, $\lfloor \sqrt{4n^2 + n} \rfloor = 2n$. We have to prove that $\sqrt{4n^2 + n} < 2n + 0.25$.

This is obvious, since by squaring the inequality one obtains:

$$4n^2 + n < 4n^2 + n + \frac{1}{16}.$$

Problem 3.1.15. *Prove that for every natural number n,*

$$\sum_{k=1}^{n^2} \{\sqrt{k}\} \leq \frac{n^2 - 1}{2}.$$

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(1999 Russian Mathematical Olympiad)

Solution. We prove the claim by induction on n. For $n = 1$, we have $0 \leq 0$. Now supposing that the claim is true for n, we prove it is true for $n + 1$.

Each of the numbers $\sqrt{n^2+1}, \sqrt{n^2+2}, \ldots, \sqrt{n^2+2n}$ is between n and $n + 1$. Thus

$$\{\sqrt{n^2+i}\} = \sqrt{n^2+i} - n < \sqrt{n^2+i+\frac{i^2}{4n^2}} - n = \frac{i}{2n}, \quad i = 1, 2, \ldots, 2n.$$

Therefore we have

$$\sum_{k=1}^{(n+1)^2} \{\sqrt{k}\} = \sum_{k=1}^{n^2} \{\sqrt{k}\} + \sum_{k=n^2+1}^{(n+1)^2} \{\sqrt{k}\} < \frac{n^2-1}{2} + \frac{1}{2n}\sum_{i=1}^{2n} i + 0$$

$$= \frac{n^2-1}{2} + \frac{2n+1}{2} = \frac{(n+1)^2-1}{2},$$

completing the inductive step and the proof.

Problem 3.1.16. *The rational numbers $\alpha_1, \ldots, \alpha_n$ satisfy*

$$\sum_{i=1}^{n} \{k\alpha_i\} < \frac{n}{2}$$

for any positive integer k.

(a) Prove that at least one of $\alpha_1, \ldots, \alpha_n$ is an integer.

(b) Do there exist $\alpha_1, \ldots, \alpha_n$ that satisfy

$$\sum_{i=1}^{n} \{k\alpha_i\} \leq \frac{n}{2},$$

such that no α_i is an integer?

(2002 Belarus Mathematical Olympiad)

Solution. (a) Assume the contrary. The problem would not change if we replace α_i with $\{\alpha_i\}$. So we may assume $0 < \alpha_i < 1$ for all $1 \leq i \leq n$. Because α_i is rational, let $\alpha_i = \frac{p_i}{q_i}$, and $D = \prod_{i=1}^{n} q_i$. Because $(D-1)\alpha_i + \alpha_i = D\alpha_i$ is an integer, and α_i is not an integer, $\{(D-1)\alpha_i\} + \{\alpha_i\}$. Then

$$1 > \sum_{i=1}^{n} \{(D-1)\alpha_i\} + \sum_{i=1}^{n} \{\alpha_i\} = \sum_{i=1}^{n} 1 = n$$

contradiction. Therefore, one of the α_i has to be an integer.

(b) Yes. Let $\alpha_i = \frac{1}{2}$ for all i. Then $\sum_{i=1}^{n}\{k\alpha_i\} = 0$ when k is even and $\sum_{i=1}^{n}\{k\alpha_i\} = \frac{n}{2}$ when k is odd.

14.2 Floor function and integer points

Problem 3.2.3. *Prove that*

$$\sum_{k=1}^{n}\left\lfloor\frac{n^2}{k^2}\right\rfloor = \sum_{k=1}^{n^2}\left\lfloor\frac{n}{\sqrt{k}}\right\rfloor$$

for all integers $n \geq 1$.

Solution. Consider the function $f : [1, n] \to [1, n^2]$,

$$f(x) = \frac{n^2}{x^2}$$

Note that f is decreasing and bijective and

$$f^{-1}(x) = \frac{n}{\sqrt{x}}.$$

Using formula in Theorem 3.2.3 we obtain

$$\sum_{k=1}^{n}\left\lfloor\frac{n^2}{k^2}\right\rfloor - \sum_{k=1}^{n^2}\left\lfloor\frac{n}{\sqrt{k}}\right\rfloor = n\alpha(1) - n^2\alpha(1) = 0,$$

hence

$$\sum_{k=1}^{n}\left\lfloor\frac{n^2}{k^2}\right\rfloor = \sum_{k=1}^{n^2}\left\lfloor\frac{n}{\sqrt{k}}\right\rfloor, \quad n \geq 1,$$

as desired.

Problem 3.2.4. *Let* θ *be a positive irrational number. Then, for any positive integer* m,

$$\sum_{k=1}^{m}\lfloor k\theta\rfloor + \sum_{k=1}^{\lfloor m\theta\rfloor}\left\lfloor\frac{k}{\theta}\right\rfloor = m\lfloor m\theta\rfloor.$$

Solution. Consider the function $f : [1, m] \to [\theta, m\theta]$, $f(x) = \theta x$. Because θ is irrational, we have $n(G_f) = 0$ and the conclusion follows from Theorem 3.2.5

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Problem 3.2.5. *Let p and q be relatively prime positive integers and let m be a real number such that $1 \leq m < p$.*

1) If $s = \left\lfloor \frac{mq}{p} \right\rfloor$, then

$$\sum_{k=1}^{\lfloor m \rfloor} \left\lfloor \frac{kq}{p} \right\rfloor + \sum_{k=1}^{s} \left\lfloor \frac{kp}{q} \right\rfloor = \lfloor m \rfloor s.$$

2) (Landau) If p and q are odd, then

$$\sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor + \sum_{k=1}^{\frac{q-1}{2}} \left\lfloor \frac{kp}{q} \right\rfloor = \frac{(p-1)(q-1)}{4}.$$

Solution. 1) Let $f : [1, m] \to \left[\frac{q}{p}, \frac{mq}{p}\right]$, $f(x) = \frac{q}{p}x$. Because $\gcd(p, q) = 1$ and $m < p$, we have $n(G_f) = 0$ and the desired equality follows from Theorem 3.2.1.

2) In the previous identity we take $m = \frac{p}{2}$. It follows that $s = \frac{q-1}{2}$ and the conclusion follows.

14.3 An useful result

Problem 3.3.3. *Let p be an odd prime and let q be an integer that is not divisible by p. Shows that*

$$\sum_{k=1}^{p-1} \left\lfloor (-1)^k k^2 \frac{q}{p} \right\rfloor = \frac{(p-1)(q-1)}{2}.$$

Solution. For $f : \mathbb{Z}_+^* \to \mathbb{R}$, $f(s) = (-1)^s s^2$, conditions i) and ii) in Theorem 3.3.1 are both satisfied. We obtain

$$\sum_{k=1}^{p-1} \left\lfloor (-1)^k k^2 \frac{q}{p} \right\rfloor = \frac{q}{p}(-1^2 + 2^2 - \cdots + (p-1)^2) - \frac{p-1}{2}$$

$$= \frac{q}{p} \cdot \frac{p(p-1)}{2} - \frac{p-1}{2},$$

hence

$$\sum_{k=1}^{p-1} \left\lfloor (-1)^k k^2 \frac{q}{p} \right\rfloor = \frac{(p-1)(q-1)}{2}.$$

Remarks. 1) By taking $q = 1$ we get

$$\sum_{k=1}^{p-1} \left\lfloor (-1)^k \frac{k^2}{p} \right\rfloor = 0.$$

Using now the identity $\lfloor -x \rfloor = 1 - \lfloor x \rfloor$, $x \in \mathbb{R}$, the last display takes the form

$$\sum_{k=1}^{p-1} (-1)^k \left\lfloor \frac{k^2}{p} \right\rfloor = \frac{1-p}{2}$$

2) Similarly, applying Theorem 3.3.1 to $f : \mathbb{Z}_+^* \to \mathbb{R}$, $f(s) = (-1)^s s^4$ yields

$$\sum_{k=1}^{p-1} \left\lfloor (-1)^k k^4 \frac{q}{p} \right\rfloor = \frac{q(p-1)(p^2-p-1)}{2} - \frac{p-1}{2}.$$

Taking $q = 1$ gives

$$\sum_{k=1}^{p-1} \left\lfloor (-1)^k \frac{k^4}{p} \right\rfloor = \frac{(p-2)(p-1)(p+1)}{2}.$$

Problem 3.3.4. *Let p be an odd prime. Show that*

$$\sum_{k=1}^{p-1} \frac{k^p - k}{p} \equiv \frac{p+1}{2} \pmod{p}.$$

Solution. For $f(s) = \dfrac{s^p}{p}$, conditions i) and ii) in Theorem 3.3.1 are also satisfied and for $q = 1$ we have

$$\sum_{k=1}^{p-1} \left\lfloor \frac{k^p}{p^2} \right\rfloor = \frac{1}{p} \sum_{k=1}^{p-1} \frac{k^p}{p} - \frac{p-1}{2}$$

$$= \frac{1}{p} \sum_{k=1}^{p-1} \frac{k^p}{p} - \frac{1}{p^2} \sum_{k=1}^{p-1} k + \frac{1}{p^2} \frac{p(p-1)}{2} - \frac{p-1}{2}$$

$$= \frac{1}{p} \sum_{k=1}^{p-1} \frac{k^p - k}{p} - \frac{1}{p} \cdot \frac{(p-1)^2}{2}.$$

It follows

$$\sum_{k=1}^{p-1} \frac{k^p - k}{p} - \frac{(p-1)^2}{2} = p \sum_{k=1}^{p} \left\lfloor \frac{k^p}{p^2} \right\rfloor,$$

i.e.

$$\sum_{k=1}^{p-1} \frac{k^p - k}{p} \equiv \frac{(p-1)^2}{2} \pmod{p}.$$

The conclusion follows since

$$\frac{(p-1)^2}{2} \equiv \frac{p^2+1}{2} \equiv \frac{p+1}{2} \pmod{p}.$$

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Remarks. 1) For each $k = 1, 2, \ldots, p-1$ denote by r_k the remainder when k^p is divided by p^2. We have

$$k^p = \left\lfloor \frac{k^p}{p^2} \right\rfloor p^2 + r_k, \quad k = 1, 2, \ldots, p-1,$$

hence

$$\sum_{k=1}^{p-1} k^p = p^2 \sum_{k=1}^{p-1} \left\lfloor \frac{k^p}{p^2} \right\rfloor + \sum_{k=1}^{p-1} r_k = -\frac{p^2(p-1)}{2} + \sum_{k=1}^{p-1} r_k + \sum_{k=1}^{p-1} k^p.$$

It follows

$$r_1 + r_2 + \cdots + r_{p-1} = \frac{p^2(p-1)}{2}$$

2) The formula in our problem shows that the sum of the quotients obtained when $k^p - k$ is divided by p (Fermat's Little Theorem) is congruent to $\dfrac{p+1}{2}$ modulo p.

15

Digits of Numbers

15.1 The last digits of a number

Problem 4.1.4. *In how may zeroes can the number* $1^n + 2^n + 3^n + 4^n$ *end for* $n \in \mathbb{N}$*?*

(1998 St. Petersburg City Mathematical Olympiad)

Solution. There can be no zeroes (i.e., $n = 4$), one zero ($n = 1$) or two zeroes ($n = 2$). In fact, for $n \geq 3$, 2^n and 4^n are divisible by 8, while $1^n + 3^n$ is congruent to 2 or 4 mod 8. Thus the sum cannot end in 3 or more zeroes.

Problem 4.1.5. *Find the last* 5 *digits of the number* 5^{1981}*.*

Solution. First, we prove that $5^{1981} = 5^5 \pmod{10^5}$. We have

$$\begin{aligned} 5^{1981} - 5^5 &= (5^{1976} - 1)5^5 = 5^5[(5^8)^{247} - 1] \\ &= \mathcal{M}[5^5(5^8 - 1)] = \mathcal{M}[5^5(5^4 - 1)(5^4 + 1)] \\ &= \mathcal{M}[5^5(5 - 1)(5 + 1)(5^2 + 1)(5^4 + 1)] \\ &= \mathcal{M}5^2 2^5 = \mathcal{M}100,000. \end{aligned}$$

Therefore $5^{1981} = \mathcal{M}100,000 + 5^5 = \mathcal{M}100,000 + 3125$, so 03125 are the last 5 digits of the number 5^{1981}.

Problem 4.1.6. *Consider all pairs* (a, b) *of natural numbers such that the product* $a^a b^b$*, written in base 10, ends with exactly 98 zeroes. Find the pair* (a, b) *for which the product* ab *is smallest.*

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(1998 Austrian-Polish Mathematics Competition)

Solution. Let a_2 be the maximum integer such that $2^{a_2}|a$. Define a_5, b_2, and b_5 similarly. Our taks translates into the following: find a, b such that $\min\{a_5a + b_5b, a_2a + b_2b\} = 98$ and ab is minimal. Since $5|a_5a + b_5b$, $a_5a + b_5b > 98$ and $\min\{a_5a + b_5b, a_2a + b_2b\} = a_2a + b_2b = 98$. Note that if $5|gcd(a, b)$, then $a_2a + b_2b \neq 98$, contradiction. Without loss of generality, suppose that $a_5 \geq 1$ and $b_5 = 0$. Let $a = 2^{a_2}5^{a_5}x$ and $2^{b_2}y$. ($gcd(2, x) = gcd(5, x) = gcd(2, y) = 1$.) Then $a_5a = a_5(2^{a_2}5^{a_5}x) > 98$ and $a_2a = a_2(2^{a_2}5^{a_5}x) \leq 98$. So $a_5 > a_2$. We consider the following cases.

(a) $a_2 = 0$. Then $b_2(2^{b_2}y) = 98$. So $b_2 = 1$, $y = 49$, $b = 98$. Since $a_5(5^{a_5}x) \geq 98$ and x is odd $a = 5^{a_5}x \geq 125$ for $a_5 \geq 3$; $x \geq 3$ and $a \geq 75$ for $a_5 = 2$; $x \geq 21$ and $a \geq 105$ for $a_5 = 1$. Hence for $a_2 = 0$, $b = 98$, $a \geq 75$.

(b) $a_2 \geq 1$. Then $a_5 \geq 2$. We have $2^{a_2}5^{a_5}x \leq 98$ and $5^{a_5}x \leq 49$. Thus $a_5 = 2$, $x = 1$, $a_2 = 1$, $a = 50$. Then $b_2b = 48$. Let $b = 2^{b_2}y$. Then $b_2(2^{b_2}y) = 48$, which is impossible.

From the above, we have $(a, b) = (75, 98)$ or $(98, 75)$.

15.2 The sum of the digits of a number

Problem 4.2.7. *Show that there exist infinitely many natural numbers n such that $S(3^n) \geq S(3^{n+1})$.*

(1997 Russian Mathematical Olympiad)

Solution. If $S(3^n) < S(3^{n+1})$ for large n, we have (since powers of 3 are divisible by 9, as are their digit sums) $S(3^n) \leq S(3^{n+1}) - 9$. Thus $S(3^n) \geq 9(n - c)$ for some c, which is eventually a contradiction since for large n, $3^n < 10^{n-c}$.

Problem 4.2.8. *Do there exist three natural numbers a, b, c such that $S(a + b) < 5$, $S(b + c) < 5$, $S(c + a) < 5$, but $S(a + b + c) > 50$?*

(1998 Russian Mathematical Olympiad)

Solution. The answer is yes. It is easier to find $a+b$, $b+c$, $c+a$ instead. Since $a+b+c$ is an integer, their sum $2(a+b+c)$ must be even; since a, b, c are positive, they must satisfy the triangle inequality. Finally, $a+b+c$ must have a digit sum of at least 51.

This leads to the solution

$$a+b=100001110000, \quad b+c=11110000000, \quad c+a=100000001110.$$

These four numbers have digit sum 4, and

$$a+b+c=105555555555$$

has digit sum 51. We get

$$a=105555555555-11110000000=94445555555$$

$$b=105555555555-100000001110=5555554445$$

$$c=105555555555-100001110000=5554445555.$$

Problem 4.2.9. *Prove that there exist distinct positive integers* $\{n_i\}_{1\le i\le 50}$ *such that*

$$n_1+S(n_1)=n_2+S(n_2)=\cdots=n_{50}+S(n_{50}).$$

(1999 Polish Mathematical Olympiad)

Solution. We show by induction on k that there exist positive integers $n_1,\ldots,n_k$ with the desired property. For $k=1$ the statement is obvious. For $k>1$, let $m_1<\cdots<m_{k-1}$ satisfy the induction hypothesis for $k-1$. Note that we can make all the m_i arbitrarily large by adding some large power of 10 to all of them, which preserves the described property. Then, choose m with $1\le m\le 9$ and $m\equiv m_1+1 \pmod 9$. Observing that $S(x)\equiv x \pmod 9$, we have $m_1-m+S(m_1)-S(m)+11=9l$ for some integer l. By choosing the m_i large enough we can ensure $10^l>m_{k-1}$. Now let $n_i=10^{l+1}+m_i$ for $i<k$ and $n_k=m+10^{l+1}-10$. It is obvious that $n_i+S(n_i)=n_j+S(n_j)$ for $i,j<k$, and

$$n_1+S(n_1)=(10^{l+1}+m_1)+(1+S(m+1))=(m_1+S(m_1)+1)+10^{l+1}$$

$$=(9l+S(m)+m-10)+10^{l+1}=(m+10^{l+1}-10)+(9l+S(m))=n_k+S(n_k),$$

as needed.

Problem 4.2.10. *The sum of the decimal digits of the natural number* n *is* 100*, and that of* $44n$ *is* 800*. What is the sum of the digits of* $3n$*?*

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(1999 Russian Mathematical Olympiad)

Solution. The sum of the digits of $3n$ is 300.

Suppose that d is a digit between 0 and 9, inclusive. If $d \leq 2$ then $S(44d) = 8d$, and if $d = 3$ then $S(8d) = 6 < 8d$. If $d \geq 4$, then $44d \leq 44(9)$ has at most 3 digits so that $S(44d) \leq 27 < 8d$.

Now write $n = \sum n_i \cdot 10^i$, so that the n_i are the digits of n in base 10. Then

$$\sum 8n_i = S(44n) \leq \sum S(44n_i \cdot 10^i)$$

$$= \sum S(44n_i) \leq \sum 8n_i,$$

so equality must occur in the second inequality – that is, each of the n_i must equal 0, 1, or 2. Then each digit of $3n$ is simply three times the corresponding digit of n, and $S(3n) = 3S(n) = 300$, as claimed.

Alternative solution. Using properties 3, 5, we have

$$S(3n) \leq 3S(n) = 300$$

and

$$800 = S(11 \cdot 3n + 11n) \leq S(11 \cdot 3n) + S(11n)$$

$$\leq S(11)S(3n) + S(11)S(n) = 2S(3n) + 200,$$

from where $S(3n) \geq 300$. Thus, $S(3n) = 300$.

Problem 4.2.11. *Consider all numbers of the form $3n^2 + n + 1$, where n is a positive integer.*

(a) How small can the sum of the digits (in base 10) of such a number be?

(b) Can such a number have the sum of its digits (in base 10) equal to 1999?

(1999 United Kingdom Mathematical Olympiad)

Solution. (a) Let $f(n) = 3n^2 + n + 1$. When $n = 8$, the sum of the digits of $f(8) = 201$ is 3. Suppose that there was some m such that $f(m)$ had a smaller sum of digits. Then the last digit of $f(m)$ must be either 0, 1, or 2. Because $f(n) \equiv 1 \pmod 2$ for all n, $f(m)$ must have units digit 1.

Because $f(n)$ can never equal 1, this means we must have $3m^2 + m + 1 = 10^k + 1$ for some positive integer k, and $m(3m + 1) = 10^k$. Because m and $3m + 1$ are relatively prime, and $m < 3m + 1$, we must either have

$(m, 3m+1) = (1, 10^k)$ – which is impossible – or $(m, 3m+1) = (2^k, 5^k)$. For $k = 1$, $5^k \neq 3 \cdot 2^k + 1$; for $k > 1$, we have

$$5^k = 5^{k-2} \cdot 25 > 2^{k-2} \cdot (12+1) \geq 3 \cdot 2^k + 1.$$

Therefore, $f(m)$ can't equal $10^k + 1$, and 3 is indeed the minimum value for the sum of digits.

(b) Consider $n = 10^{222} - 1$.

$$f(n) = 3 \cdot 10^{444} - 6 \cdot 10^{222} + 3 + 10^{222}.$$

Thus, its decimal expansion is

$$2\underbrace{9\ldots9}_{221}5\underbrace{0\ldots0}_{221}3,$$

and the sum of digits in $f(10^{222} - 1)$ is 19999.

Problem 4.2.12. *Consider the set A of all positive integers n with the following properties: the decimal expansion contains no 0, and the sum of the (decimal) digits of n divides n.*

(a) Prove that there exist infinitely many elements in A with the following property: the digits that appear in the decimal expansion of A appear the same number of times.

(b) Show that for each positive integer k, there exists an element in A with exactly k digits.

(2001 Austrian-Polish Mathematics Competition)

Solution. (a) We can take $n_k = \underbrace{11\ldots1}_{3^k \text{ times}}$ and prove by induction that $3^{k+2} | 10^{3^k} - 1$. Alternatively, one can observe that

$$10^{3^k} - 1 = (10-1)(10^2+10+1)(10^{2\cdot3}+10^3+1)\ldots(10^{2\cdot3^{k-1}}+10^{3^{k-1}}+1)$$

and that $9 | 10^{-1}$ and $3 | 10^{2\cdot3^i} + 10^{3^i} + 1$ for $0 \leq i \leq k-1$.

(b) We will need the following lemmas.

Lemma 1. *For every $d > 0$ there exists a d-digit number that contains only ones and twos in its decimal expansion and is a multiple of 2^d.*

Proof. Exactly in the same way as in the proof of Theorem 1.7.1 one can prove that any two d-digit numbers which have only ones and twos give different residues mod 2^d. Since there are 2^d such numbers, one of them is a multiple of 2^d. □

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Lemma 2. *For each $k > 2$ there exists $d \leq k$ such that the following inequality holds: $k + d \leq 2^d \leq 9k - 8d$.*

Proof. For $3 \leq k \leq 5$, $d = 3$ satisfies the inequalities. For $5 \leq k \leq 10$, $d = 4$ satisfies the inequalities. We will show that $d = \lfloor \log_2 4k \rfloor$ satisfies for all $k > 10$. If $k > 3$, then $\log_2 4k \leq 2^k$, so $d < k$. Additionally, $k + d \leq 2k \leq 2^d$. If $k > 10$, then $16k^2 \leq 2^k$, so $4k \leq 2^{k/2} \leq 2^{5k/8}$, $d \leq \log_2 4k \leq \frac{5}{8}n$ and $8k - 8d \geq 4k \geq 2^d$. □

Now, return to the original problem. For $k = 1$, $n = 1$ has the desired property. For $k = 2$, $n = 12$ has the desired property. Now, for each $k > 2$ we have some number d satisfying the condition Lemma 2. Consider a k-digit integer n such that the last d digits of n have the property described in the first Lemma. We can choose each of the other digits of n to be any number between zero and nine. We know that the sum of the last d digits of n is between d and $2d$, and we can choose the sum of the other $k-d$ digits to be any number between $k-d$ and $9(k-d)$. Since $k-d+2d \leq 2^d \leq 9(k-d)+d$, we can choose the other digits such that the sum of the digits of n is 2^d. This completes the proof because n is a multiple of 2^d.

Remark. A number divisible by the sum of its digits is called a *Niven*[1] *number*. It has been proved recently that the number of Niven numbers smaller than x is $\left(\frac{14}{27}\log 10 + o(1)\right)\frac{x}{\log x}$. The courageous reader may try to prove that there are arbitrarily long sequences of consecutive numbers which are not Niven numbers (which is easily implied by the above result; yet there is an elementary proof of the last assertion). For more details one can read the article "*Large and small gaps between consecutive Niven numbers*", Journal of Integer Sequences, Vol.6(2003), by J.-M. Koninck and N. Doyon.

15.3 Other problems involving digits

Problem 4.3.3. *A wobbly number is a positive integer whose digits in base 10 are alternately non-zero and zero, the units digit being non-zero. Determine all positive integers which do not divide any wobbly number.*

(35th IMO Shortlist)

[1] *Ivan Niven* (1915-), Canadian mathematician with contributions in the ones of Diophantine approximation, the study of irrationality and transcendence of numbers, and combinatorics.

Solution. If n is a multiple of 10, then the last digit of any multiple of n is 0. Hence it is not wobbly. If n is a multiple of 25, then the last two digits of any multiple of n are 25, 50, 75 or 00. Hence it is not wobbly. We now prove that these are the only numbers not dividing any wobbly number.

We first consider odd numbers m not divisible by 5. Then $gcd(m, 10) = 1$, and we have $gcd((10^k - 1)m, 10) = 1$, for any $k \geq 1$. It follows that there exists a positive integer l such that $10^l \equiv 1 \pmod{(10^k - 1)m}$, and we have $10^{kl} \equiv 1 \pmod{(10^k - 1)m}$. Now

$$10^{kl} - 1 = (10^k - 1)(10^{k(l-1)} + 10^{k(l-2)} + \cdots + 10^k - 1).$$

Hence $x_k = 10^{k(l-1)} + 10^{k(l-2)} + \cdots + 10^k + 1$ is a multiple of m for any $k \geq 1$. In particular, x_2 is a wobbly multiple of m. If m is divisible by 5, then $5x_2$ is a wobbly multiple of m.

Next, we consider powers of 2. We prove by induction on t that 2^{2t+1} has a wobbly multiple w_t with precisely t non-zero digits. For $t = 1$, take $w_1 = 8$. Suppose w_t exists for some $t \geq 1$. Then $w_t = 2^{2t+1}d$ for some d. Let $w_{t+1} = 10^{2t}c + w_t$ where $c \in \{1, 2, 3, \ldots, 9\}$ is to be chosen later. Clearly, w_{t+1} is wobbly, and has precisely $t+1$ non-zero digits. Since $w_{t+1} + 2^{2t}(5^{2t}c + 2d)$, it is divisible by 2^{2t+3} if and only if $5^{2t}c + 2d \equiv 0 \pmod 8$ or $c \equiv 6d \pmod 8$. We can always choose c to be one of 8, 6, 4 and 2 in order to satisfy this congruence. Thus the inductive argument is completed. It now follows that every power of 2 has a wobbly multiple.

Finally, consider numbers of the form $2^t m$, where $t \geq 1$ and $gcd(m, 10) = 1$. Such a number has $w_t x_{2t}$ as a wobbly multiple.

Problem 4.3.4. *A positive integer is called monotonic if its digits in base 10, read from left right, are in nondecreasing order. Prove that for each $n \in \mathbb{N}$, there exists an n-digit monotonic number which is a perfect square.*

(2000 Belarussian Mathematical Olympiad)

Solution. Any 1-digit perfect square (namely, 1, 4, or 9) is monotonic, proving the claim for $n = 1$. We now assume $n > 1$.

If n is odd, write $n = 2k - 1$ for an integer $k \geq 2$, and let

$$x_k = (10^k + 2)/6 = 1\underbrace{66\ldots6}_{k-2}7.$$

Then

$$x_k^2 = \frac{10^{2k} + 4 \cdot 10^k + 4}{36} = \frac{10^{2k}}{36} + \frac{10^k}{9} + \frac{1}{9}. \tag{1}$$

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Observe that

$$\frac{10^{2k}}{36} = 10^{2k-2}\left(\frac{72}{36}+\frac{28}{36}\right)$$

$$= 2\cdot 10^{2k-2} + 10^{2k-2}\cdot\frac{7}{9} = 2\underbrace{77\ldots7}_{2k-2}+\frac{7}{9}.$$

Thus, the right-hand side of (1) equals

$$\left(2\underbrace{77\ldots7}_{2k-2}+\frac{7}{9}\right)+\left(\underbrace{11\ldots1}_{k}+\frac{1}{9}\right)+\frac{1}{9} = 2\underbrace{77\ldots7}_{k-2}\underbrace{88\ldots8}_{k-1}9,$$

an n-digit monotonic perfect square.

If n is even, write $n = 2k$ for an integer $k \geq 1$, and let

$$y_k = \frac{10^k+2}{3} = \underbrace{33\ldots3}_{k-1}4.$$

Then

$$y_k^2 = \frac{1}{9}(10^{2k}+4\cdot 10^k+4) = \frac{10^{2k}}{9}+4\cdot\frac{10^k}{9}+\frac{4}{9}$$

$$= \left(\underbrace{11\ldots1}_{2k}+\frac{1}{9}\right)+\left(\underbrace{44\ldots4}_{k}+\frac{4}{9}\right)+\frac{4}{9} = \underbrace{11\ldots1}_{k}\underbrace{55\ldots5}_{k-1}6,$$

an n-digit monotonic perfect square. This completes the proof.

16

Basic Principles in Number Theory

16.1 Two simple principles

Problem 5.1.7. *Let $n_1 < n_2 < \cdots < n_{2000} < 10^{100}$ be positive integers. Prove that one can find two nonempty disjoint subsets A and B of $\{n_1, n_2, \ldots, n_{2000}\}$ such that*

$$|A| = |B| \quad , \sum_{x \in A} x = \sum_{x \in B} x, \quad \text{and} \quad \sum_{x \in A} x^2 = \sum_{x \in B} x^2.$$

(2001 Polish Mathematical Olympiad)

Solution. Given any subset $S \subseteq \{n_1, n_2, \ldots, n_{2000}\}$ of size 1000, we have

$$0 < \sum_{x \in S} x < 1000 \cdot 10^{100},$$

$$0 < \sum_{x \in S} x^2 < 1000 \cdot 10^{200}.$$

Thus, as S varies, there are fewer than $(1000 \cdot 10^{100})(1000 \cdot 10^{200}) = 10^{306}$ values of $\left(\sum_{x \in S} x, \sum_{x \in S} x^2 \right)$.

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Because $\sum_{k=0}^{2000} \binom{2000}{k} = 2^{2000}$ and $\binom{2000}{1000}$ is the biggest term in the sum, $\binom{2000}{1000} > \frac{2^{2000}}{2001}$. There are

$$\binom{2000}{1000} > \frac{2^{2000}}{2001} > \frac{10^{600}}{2001} > 10^{306}$$

distinct subsets of size 1000. By the Pigeonhole Principle, there exist distinct subsets C and D of size 1000, such that $\sum_{x\in C} x^2 = \sum_{x\in D} x^2$ and $\sum_{x\in C} x = \sum_{x\in D} x$. Removing the common elements from C and D yields sets A and B with the required properties.

Problem 5.1.8. *Find the greatest positive integer n for which there exist n nonnegative integers $x_1, x_2, \ldots, x_n$, not all zero, such that for any sequence $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ of elements $\{-1, 0, 1\}$, not all zero, n^3 does not divide $\varepsilon_1 x_1 + \varepsilon_2 x_2 + \cdots + \varepsilon_n x_n$.*

(1996 Romanian Mathematical Olympiad)

Solution. The statement holds for $n = 9$ by choosing $1, 2, 2^2, \ldots, 2^8$, since in that case

$$|\varepsilon_1 + \cdots + \varepsilon_9 2^8| \leq 1 + 2 + \cdots + 2^8 < 9^3.$$

However, if $n = 10$, then $2^{10} > 10^3$, so by the Pigeonhole Principle, there are two subsets A and B of $\{x_1, \ldots, x_{10}\}$ whose sums are congruent modulo 10^3. Let $\varepsilon_i = 1$ if x_i occurs in A but not in B, -1 if x_i occurs in B but not in A, and 0 otherwise; then $\sum \varepsilon_i x_i$ is divisible by n^3.

Problem 5.1.9. *Given a positive integer n, prove that there exists $\varepsilon > 0$ such that for any n positive real numbers $a_1, a_2, \ldots, a_n$, there exists $t > 0$ such that*

$$\varepsilon < \{ta_1\}, \{ta_2\}, \ldots, \{ta_n\} < \frac{1}{2}.$$

(1998 St. Petersburg City Mathematical Olympiad)

Solution. More generally, we prove by induction on n that for any real number $0 < r < 1$, there exists $0 < \varepsilon < r$ such that for $a_1, \ldots, a_n$ any positive real numbers, there exists $t > 0$ with

$$\{ta_1\}, \ldots, \{ta_n\} \in (\varepsilon, r).$$

The case $n = 1$ needs no further comment.

Assume without loss of generality that a_n is the largest of the a_i. By hypothesis, for any $r' > 0$ (which we will specify later) there exists $\varepsilon' > 0$ such that for any $a_1, \ldots, a_{n-1} > 0$, there exists $t' > 0$ such that

$$\{t'a_1\}, \ldots, \{t'a_{n-1}\} \in (\varepsilon', r').$$

Let N be an integer also to be specified later, A standard argument using the Pigeonhole Principle shows that one of $t'a_n, 2t'a_n, \ldots, Nt'a_n$ has fractional part in $(-1/N, 1/N)$. Let $st'a_n$ be one such term, and take $t = st' + c$ for $c = (r - 1/N)/a_n$. Then

$$ta_n \in (r - 2/N, r).$$

So we choose N such that $0 < r - 2/N$, thus making $\{ta_n\} \in (r - 2/N, r)$. Note that this choice of N makes $c > 0$ and $t > 0$, as well.

As for the other ta_i, for each i we have $k_i + \varepsilon' < t'a_i < k_i + r'$ for some integer k_i, so $sk_i + s\varepsilon' < st'a_i < sk_i + sr'$ and

$$sk_i + \varepsilon' < (st' + c)a_i < sk_i + sr' + \frac{a_i(r - 1/N)}{a_n} \leq sk_i + Nr' + r - 1/N.$$

So we choose r' such that $Nr' - 1/N < 0$, thus making $\{ta_i\} \in (\varepsilon', r)$.

Therefore, letting $\varepsilon = \min\{r - 2/N, \varepsilon'\}$, we have

$$0 < \varepsilon < \{ta_1\}, \{ta_2\}, \ldots, \{ta_n\} < r$$

for any choices of a_i. This completes the inductive step, and the claim is true for all natural numbers n.

Problem 5.1.10. *We have 2^n prime numbers written on the blackboard in a line. We know that there are less than n different prime numbers on the blackboard. Prove that there is a compact subsequence of numbers in that line whose product is a perfect square.*

Solution. Suppose that $p_1, p_2, \ldots, p_m$ $(m < n)$ are primes which we met in the sequence $a_1, a_2, \ldots, a_{2^n}$ written on the blackboard. It is enough to prove that there is a compact subsequence, where each prime occurs even times. Denote c_{ij} the exponent of the prime p_i $(1 \leq i \leq m)$, in the product of the first j numbers $a_1 \ldots a_2 \ldots a_j$ from our sequence. Let d_{ij} be the residue modulo 2 of c_{ij}, then we can write $c_{ij} = 2t_{ij} + d_{ij}$, $d_{ij} \in \{0, 1\}$. Every system $(d_{1j}, d_{2j}, \ldots, d_{mj})$ is formed from m zeros and ones. Number of possible such systems is 2^m which is less than 2^n. Hence by Pigeonhole

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Principle there exist two identical systems.

$$(d_{1k}, d_{2k}, \ldots, d_{mk}) = (d_{1l}, d_{2l}, \ldots, d_{ml}), \quad 1 \leq k < l \leq 2^n$$

We have $d_{ik} = d_{il}$ for $1 \leq i \leq m$ and from here

$$c_{il} - c_{ik} = 2(t_{il} - t_{ik}) + (d_{il} - d_{ik}) = 2(t_{il} - t_{ik})$$

and $c_{il} - c_{ik}$ is divisible by 2 for $1 \leq i \leq m$.

Thus the exponent of the p_i in the product $a_{k+1}a_{k+2}\ldots a_l = \dfrac{a_1a_2\ldots a_l}{a_1a_2\ldots a_k}$ is equal to $c_{il} - c_{ik}$, so every number p_i has an even exponent is the product $a_{k+1}a_{k+2}\ldots a_l$. Hence $a_{k+1}a_{k+2}\ldots a_l$ is the perfect square.

Problem 5.1.11. *Let $x_1 = x_2 = x_3 = 1$ and $x_{n+3} = x_n + x_{n+1}x_{n+2}$ for all positive integers n. Prove that for any positive integer m there is an integer $k > 0$ such that m divides x_k.*

Solution. Observe that setting $x_0 = 0$ the condition is satisfied for $n = 0$.

We prove that there is integer $k \leq m^3$ such that x_k divides m. Let r_t be the remainder of x_t when divided by m for $t = 0, 1, \ldots, m^3 + 2$. Consider the triples (r_0, r_1, r_2), $(r_1, r_2, r_3), \ldots, (r_{m^3}, r_{m^3+1}, r_{m^3+2})$. Since r_t can take m values, it follows by the Pigeonhole Principle that at least two triples are equal. Let p be the smallest number such that triple (r_p, r_{p+1}, r_{p+2}) is equal to another triple (r_q, r_{q+1}, r_{q+2}), $p < q \leq m^3$. We claim that $p = 0$.

Assume by way of contradiction that $p \geq 1$. Using the hypothesis we have

$$r_p \equiv r_{p-1} + r_pr_{p+1} \pmod{m} \text{ and } r_{q+2} \equiv r_{q-1} + r_qr_{q+1} \pmod{m}.$$

Since $r_p = r_q$, $r_{p+1} = r_{q+1}$ and $r_{p+2} = r_{q+2}$, it follows that $r_{p-1} = r_{q-1}$, so $(r_{p-1}, r_p, r_{p+1}) = (r_{q-1}, r_q, r_{q+1})$, which is a contradiction with the minimality of p. Hence $p = 0$, so $r_q = r_0 = 0$, and therefore $x_q \equiv 0 \mod m$.

16.2 Mathematical induction

Problem 5.2.7. *Let p be an odd prime. The sequence $(a_n)_{n\geq 0}$ is defined as follows: $a_0 = 0$, $a_1 = 1, \ldots, a_{p-2} = p-2$ and, for all $n \geq p-1$, a_n is the least positive integer that does not form an arithmetic sequence of length p with any of the preceding terms. Prove that, for all n, a_n is the number obtained by writing n in base $p-1$ and reading the result in base p.*

(1995 USA Mathematical Olympiad)

Solution. Our proof uses the following result.

Lemma. *Let $B = \{b_0, b_1, b_2, \dots\}$, where b_n is the number obtained by writing n in base $p-1$ and reading the result in base p. Then*

(a) for every $a \notin B$, there exists $d > 0$ such that $a - kd \in B$ for $k = 1, 2, \dots, p-1$; and

(b) B contains no p-term arithmetic progression.

Proof. Note that $b \in B$ if and only if the representation of b in base p does not use the digit $p-1$.

(a) Since $a \notin B$, when a is written in base p at least one digit is $p-1$. Let d be the positive integer whose representation in base p is obtained from that of a by replacing each $p-1$ by 1 and each digit other than $p-1$ by 0. Then none of the numbers $a-d$, $a-2d, \dots, a-(p-1)d$ has $p-1$ as a digit when written in base p, and the result follows.

(b) Let $a, a+d, a+2d, \dots, a+(p-1)d$ be an arbitrary p-term arithmetic progression of nonnegative integers. Let δ be the rightmost nonzero digit when d is written in base p, and let α be the corresponding digit in the representation of a. Then $\alpha, \alpha+\delta, \dots, \alpha+(p-1)\delta$ is a complete set of residues modulo p. It follows that at least one of the numbers $a, a+d, \dots, a+(p-1)d$ has $p-1$ as a digit when written in base p. Hence at least one term of the given arithmetic progression does not belong to B. □

Let $(a_n)_{n\geq 0}$ be the sequence defined in the problem. To prove that $a_n = b_n$ for all $n \geq 0$, we use mathematical induction. Clearly $a_0 = b_0 = 0$. Assume that $a_k = b_k$ for $0 \leq k \leq n-1$, where $n \geq 1$. Then a_n is the smallest integer greater than b_{n-1} such that $\{b_0, b_1, \dots, b_{n-1}, a_n\}$ contains no p-term arithmetic progression. By part (i) of the proposition, $a_n \in B$ so $a_n \geq b_n$. By part (ii) of the proposition, the choice of $a_n = b_n$ does not yield a p-term arithmetic progression with any of the preceding terms. It follows by induction that $a_n = b_n$ for all $n \geq 0$.

Problem 5.2.8. *Suppose that x, y and z are natural numbers such that $xy = z^2+1$. Prove that there exist integers a, b, c and d such that $x = a^2+b^2$, $y = c^2+d^2$, and $z = ac+bd$.*

(Euler's problem)

Solution. We prove the claim by strong induction on z. For $z = 1$, we have $(x, y) = (1, 2)$ or (2,1); in the former (resp. latter) case, we can set $(a, b, c, d) = (1, 0, 1, 1)$ (resp. (0,1,1,1)).

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Suppose that the claim is true whenever $z < z_0$, and that we wish to prove it for $(x, y, z) = (x_0, y_0, z_0)$ where $x_0 y_0 = z_0^2 + 1$. Without loss of generality, assume that $x_0 \leq y_0$. Consider the triple $(x_1, y_1, z_1) = (x_0, x_0 + y_0 - 2z_0, z_0 - x_0)$, so that $(x_0, y_0, z_0) = (x_1, x_1 + y_1 + 2z_1, x_1 + z_1)$.

First, using the fact that $x_0 y_0 = z_0^2 + 1$, it is easy to check that $(x, y, z) = (x_1, y_1, z_1)$ satisfies $xy = z^2 + 1$.

Second, we claim that $x_1, y_1, z_1 > 0$. This is obvious for x_1. Next, note that $y_1 = x_0 + y_0 - 2z_0 \geq 2\sqrt{x_0 y_0} - 2z_0 > 2z_0 - 2z_0 = 0$. Finally, because $x_0 \leq y_0$ and $x_0 y_0 = z_0^2 + 1$, we have $x_0 \leq \sqrt{z_0^2 + 1}$, or $x_0 \leq z_0$. However, $x_0 \neq z_0$, because this would imply that $z_0 y_0 = z_0^2 + 1$, but $z_0 \nmid (z_0^2 + 1)$ when $z_0 > 1$. Thus, $z_0 - x_0 > 0$, or $z_1 > 0$.

Therefore, (x_1, y_1, z_1) is a triple of positive integers (x, y, z) satisfying $xy = z^2 + 1$ and with $z < z_0$. By the induction hypothesis, we can write $x_1 = a^2 + b^2$, $y_1 = c^2 + d^2$ and $z_1 = ac + bd$. Then

$$\begin{aligned}(ac + bd)^2 = z_1^2 &= x_1 y_1 - 1 \\ &= (a^2 + b^2)(c^2 + d^2) - 1 \\ &= (a^2 c^2 + b^2 d^2 + 2abcd) + (a^2 d^2 + b^2 c^2 - 2abcd) - 1 \\ &= (ac + bd)^2 (ad - bc)^2 - 1,\end{aligned}$$

so that $|ad - bc| = 1$.

Now, note that $x_0 = x_1 = a^2 + b^2$ and $y_0 = x_1 + y_1 + 2z_1 = a^2 + b^2 + c^2 + d^2 + 2(ac + bd) = (a + c)^2 + (b + d)^2$. In other words, $x_0 = a'^2 + b'^2$ and $y_0 = c'^2 + d'^2$ for $(a', b', c', d') = (a, b, a + c, b + d)$. Then $|a'd' - b'c'| = |ad - bc| = 1$, implying (by logic analogous to the reasoning in the previous paragraph) that $z_0 = a'c' + b'd'$, as desired. This completes the inductive step, and the proof.

Problem 5.2.9. *Find all pairs of sets A, B, which satisfy the conditions:*
(i) $A \cup B = \mathbb{Z}$;
(ii) if $x \in A$, then $x - 1 \in B$;
(iii) if $x \in B$ and $y \in B$, then $x + y \in A$.

(2002 Romanian IMO Team Selection Test)

Solution. We shall prove that either $A = B = \mathbb{Z}$ or A is the set of even numbers and B the set of odd numbers.

First, assume that $0 \in B$. Then we have $x \in B$, $x + 0 \in A$ and so $B \subset A$. Then $\mathbb{Z} = A \cup B \subset A$ and so $A = \mathbb{Z}$. From (ii) we also find that $B = \mathbb{Z}$. Now, suppose that $0 \notin B$, thus $0 \in A$ and $-1 \in B$. Then, using (ii) we find

$-2 \in A$, $-3 \in B$, $-4 \in A$ and by induction $-2n \in A$ and $-2n-1 \in B$, $\forall\ n \in \mathbb{N}$. Of course, $2 \in A$ (otherwise $2 \in B$ and $1 = 2 + (-1) \in A$ and $0 = 1 - 1 \in B$, false) and so $1 = 2 - 1 \in B$. Let $n > 1$ minimal with $2n \in B$. Then $2n-1 \in A$ and $2(n-1) \in B$, contradiction. This shows that $2\mathbb{N} \subset A \setminus B$ and all odd integers are in $B \setminus A$. One can also observe that $-1 \notin A$ (otherwise $-2 \in B$ implies $-1 \in B$ i.e. $-1 \notin A$) and so $A = 2\mathbb{Z}$, $B = 2\mathbb{Z} + 1$.

Problem 5.2.10. *Find all positive integers n such that*

$$n = \prod_{k=0}^{m} (a_k + 1),$$

where $\overline{a_m a_{m-1} \dots a_0}$ is the decimal representation of n.

(2001 Japanese Mathematical Olympiad)

Solution. We claim that the only such n is 18. If $n = \overline{a_m \dots a_1 a_0}$, then let

$$P(n) = \prod_{j=0}^{m} (a_j + 1).$$

Note that if $s \geq 1$ and t is a single-digit number, then $P(10s + t) = (s + 1)P(t)$. Using this we will prove two following statements.

Lemma 1. *If $P(s) \leq s$, then $P(10s + t) \leq 10s + t$.*

Proof. Indeed, if $P(s) \leq s$, then

$$10s + t \geq 10s \geq 10P(s) \geq (t + 1)P(s) = P(10s + t).$$

Equality must fail either in the first inequality (if $t \neq 0$) or in the third inequality (if $t \neq 9$). □

Lemma 2. *$P(n) \leq n + 1$ for all n.*

Proof. We prove this by induction on the number of digits of n. First, we know that for all one-digit n, $P(n) = n+1$. Now suppose that $P(n) \leq n+1$ for all m-digit numbers n. Any $(m+1)$-digit number n is of the form $10s+t$, where s is an m-digit number. Then

$$t(P(s) - 1) \leq 9((s + 1) - 1)$$

$$tP(s) - 10s - t \leq -s$$

$$P(s)(t + 1) - 10s - t \leq P(s) - s$$

$$P(10s + t) - (10s + t) \leq P(s) - s \leq 1,$$

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completing the inductive step. Thus, $P(n) \leq n+1$ for all n. □

If $P(n) = n$, then n has more than one digit and we may write $n = 10s+t$. From the first statement, we have $P(s) \geq s+1$. From the second one, we have $P(s) \leq s+1$. Thus, $P(s) = s+1$. Hence,

$$(t+1)P(s) = P(10s+t) = 10s+t$$

$$(t+1)(s+1) = 10s+t$$

$$1 = (9-t)s.$$

This is possible if $t = 8$ and $s = 1$, so the only possible n such that $P(n) = n$ is 18. Indeed, $P(18) = (1+1)(8+1) = 18$.

Problem 5.2.11. *The sequence $(u_n)_{n\geq 0}$ is defined as follows: $u_0 = 2$, $u_1 = \dfrac{5}{2}$ and*

$$u_{n+1} = u_n(u_{n-1}^2 - 2) - u_1 \text{ for } n = 1, 2, \ldots$$

Prove that $[u_n] = 2^{\frac{2^n-(-1)^n}{3}}$, for all $n > 0$ ($[x]$ denotes the integer part of x).

(18th IMO)

Solution. To start, we compute a few members of the sequence. Write

$$u_1 = \frac{5}{2} = 2 + \frac{1}{2}.$$

Then:

$$u_2 = u_1(u_0^2 - 2) - \left(2 + \frac{1}{2}\right) = \left(2 + \frac{1}{2}\right)(2^2 - 2) - \left(2 + \frac{1}{2}\right) = 2 + \frac{1}{2}$$

$$u_3 = u_2(u_1^2 - 2) - \left(2 + \frac{1}{2}\right) = \left(2 + \frac{1}{2}\right)\left[\left(2 + \frac{1}{2}\right)^2 - 2\right] - \left(2 + \frac{1}{2}\right)$$

$$= \left(2 + \frac{1}{2}\right)\left(2^2 + \frac{1}{2^2}\right) - \left(2 + \frac{1}{2}\right) = \left(2 + \frac{1}{2}\right)\left(2^2 - 1 + \frac{1}{2^2}\right) = 2^3 + \frac{1}{2^3}$$

$$u_4 = \left(2^3 + \frac{1}{2^3}\right)\left[\left(2 + \frac{1}{2}\right)^2 - 2\right] - \left(2 + \frac{1}{2}\right)$$

$$= \left(2^3 + \frac{1}{2^3}\right)\left(2^2 + \frac{1}{2^2}\right) - \left(2 + \frac{1}{2}\right)$$

$$= 2^5 + \frac{1}{2} + 2 + \frac{1}{2^5} - \left(2 + \frac{1}{2}\right) = 2^5 + \frac{1}{2^5}$$

$$u_5 = \left(2^5 + \frac{1}{2^5}\right)\left[\left(2^3 + \frac{1}{2^3}\right)^2 - 2\right] - \left(2 + \frac{1}{2}\right)$$

$$= \left(2^5 + \frac{1}{2^5}\right)\left(2^6 + \frac{1}{2^6}\right) - \left(2 + \frac{1}{2}\right) = 2^{11} + \frac{1}{2^{11}}$$

Taking into account the required result, we claim that $u_n = 2^{a_n} + 2^{-a_n}$, where $a_n = \dfrac{2^n - (-1)^n}{3}$, $\forall\ n \geq 1$. First, we observe that a_n is a positive integer, because $2^n \equiv (-1)^n \pmod 3$.

Then, observe that claimed formula is true for $n = 1, 2, 3, 4, 5$. Using induction and inductive formula which defined u_n we have:

$$u_{n+1} = (2^{a_n} + 2^{-a_n})[(2^{a_{n-1}} + 2^{-a_{n-1}}) - 2] - \left(2 + \frac{1}{2}\right)$$

$$= (2^{a_n} + 2^{-a_n})(2^{2a_{n-1}} + 2^{-2a_{n-1}}) - \left(2 + \frac{1}{2}\right)$$

$$= 2^{a_n + 2a_{n-1}} + 2^{-a_n - 2a_{n-1}} + 2^{2a_{n-1} - a_n} + 2^{a_n - 2a_{n-1}} - 2 - 2^{-1}.$$

We only have to consider the equalities:

$$a_n + 2a_{n-1} = a_{n+1}$$

$$2a_{n-1} - a_n = (-1)^n,$$

which are easy to check. Hence, we obtain the general formula:

$$u_n = 2^{\frac{2^n - (-1)^n}{3}} + \frac{1}{2^{\frac{2^n - (-1)^n}{3}}},\ \forall\ n \geq 1.$$

The required result,

$$[u_n] = 2^{\frac{2^n - (-1)^n}{3}},$$

is now obvious.

Second solution. We have $u_0 \geq 2$, $u_1 \geq \dfrac{5}{2}$. We prove by induction that

$$u_n \geq \frac{5}{2},\ \text{for all } n \geq 1.$$

$$u_{n+1} = u_n(u_{n-1}^2 - 2) - \frac{5}{2} \geq \frac{5}{2}\left(\frac{25}{4} - 2\right) - \frac{5}{2} = \frac{5}{2}\left(\frac{25}{4} - 3\right) > \frac{5}{2}.$$

The equation

$$x + \frac{1}{x} = u_n$$

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has a unique real solution x_n, with $x_n > 1$. Indeed, write the equation under the form

$$x^2 - u_n x + 1 = 0$$

and we observe that $\Delta = u_n^2 - 4 \geq \frac{25}{4} - 4 > 0$. The equation has two positive real solutions, only one being greater than 1.

Therefore, there exists a unique real sequence $(x_n)_{n\geq 1}$ such that $x_n > 1$ and

$$x_n + \frac{1}{x_n} = u_n.$$

Put this formula in the definition for u_{n+1} and obtain

$$x_{n+1} + \frac{1}{x_{n+1}} = x_n x_{n-1}^2 + \frac{1}{x_n x_{n-1}^2} + \left(\frac{x_n}{x_{n-1}^2} + \frac{x_{n-1}^2}{x_n}\right) - \frac{5}{2}.$$

We claim that the sequence $(x_n)_{n\geq 1}$ is uniquely defined by one the conditions:

$$x_{n+1} = x_n x_{n-1}^2 \tag{1}$$

$$\frac{x_{n+1}}{x_{n-1}^2} = 2^{(-1)^{n-1}}. \tag{2}$$

Actually, from condition (1) and $x_1 = 2$, $x_2 = 2$ we deduce

$$x_3 = 2^{1+2} = 2^3, \quad x_4 = 2^{1+2} \cdot 2^{1\cdot 2} = 2^5$$

and generally, $x_n = 2^{\frac{2^n - (-1)^n}{3}}$. After that, the solution follows like in the first part.

16.3 Infinite descent

Problem 5.3.2. *Find all primes p for which there exist positive integers x, y and n such that $p^n = x^3 + y^3$.*

(2000 Hungarian Mathematical Olympiad)

Solution. Observe $2^1 = 1^3 + 1^3$ and $3^2 = 2^3 + 1^3$. We will prove that the only answers are $p = 2$ or $p = 3$. Assume, by contradiction that there exists $p \geq 5$ such that $p^n = x^3 + y^3$ with x, y, n positive integers and n of the smallest possible value. Hence at least one of x and y is greater than 1. We have $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$ with $x + y \geq 3$ and $x^2 - xy + y^2 = (x - y)^2 + xy \geq 2$. It follows that both $x + y$ and $x^2 - xy + y^2$ are divisible by p. Therefore $(x + y)^2 - (x^2 - xy + y^2) = 3xy$ is also divisible by p. However, 3 is not divisible by p, so at least one of x or y must be

divisible by p. As $x+y$ is divisible by p, both x and y are divisible by p. Then $x^3+y^3 \geq 2p^3$ and necessarily $n > 3$. We obtain

$$p^{n-3} = \frac{p^n}{p^3} = \frac{x^3}{p^3} + \frac{y^3}{p^3} = \left(\frac{x}{p}\right)^3 + \left(\frac{y}{p}\right)^3,$$

and this contradicts the minimality of n (see the remark after FMID Variant 1).

16.4 Inclusion-exclusion

Problem 5.4.2. *The numbers from 1 to 1000000 can be colored black or white. A permissible move consists of selecting a number from 1 to 1000000 and changing the color of that number and each number not relatively prime to it. Initially all of the numbers are black. Is it possible to make a sequence of moves after which all of the numbers are colored white?*

(1999 Russian Mathematical Olympiad)

Solution. It is possible. We begin by proving the following lemma:

Lemma. *Given a set S of positive integers, there is a subset $T \subseteq S$ such that every element of S divides an odd number of elements in T.*

Proof. We prove the claim by induction on $|S|$, the number of elements in S. If $|S| = 1$ then let $T = S$.

If $|S| > 1$, then let α be the smallest element of S. Consider the set $S' = S \setminus \{a\}$, the set of the largest $|S|-1$ elements in S. By induction there is a subset $T' \subseteq S'$ such that every element in S' divides an odd number of elements in T'.

If a also divides an odd number of elements in T', then the set $T = T'$ suffices. Otherwise, consider the set $T = T' \cup \{a\}$. a divides an odd number of elements in T. Every other element in T is bigger than a and can't divide it, but divides an odd number of elements in $T' = T \setminus \{a\}$. Hence T suffices, completing the induction and the proof of the lemma. □

Now, write each number $n > 1$ in its prime factorization

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k},$$

where the p_i are distinct primes and the a_i ate positive integers. Notice that the color of n will always be the same as the color of $P(n) = p_1 p_2 \dots p_k$.

Apply the lemma to the set S consisting in all $P(i)$ for $i = 2, 3, \dots, 1000000$ to find a subset $T \subset S$ such that every element of S

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divides an odd number of elements in T. For each $q \in S$, let $t(q)$ equal the number of elements in T that q divides, and let $u(q)$ equal the number of primes dividing q.

Select all the numbers in T, and consider how the color of a number $n > 1$ changes. By the Inclusion-Exclusion Principle, the number of elements in T not relatively prime to n equals

$$\sum_{q|P(n),q>1} (-1)^{u(q)+1} t(q).$$

In particular, if $q|P(n)$ is divisible by exactly $m > 0$ primes, then it is counted $\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots = 1$ time in the sum. (For example, if $n = 6$ then the number of elements in T divisible by 2 or 3 equals $t(2) + t(3) - t(6)$.)

By the definition of T, each of the values $t(q)$ is odd. Because there are $2^k - 1$ divisors $q > 1$ of $P(n)$, the above quantity is the sum of $2^k - 1$ odd numbers and is odd itself. Therefore after selecting T, every number $n > 1$ will switch color an odd number of times and will turn white.

Finally, select 1 to turn 1 white to complete the process.

Note. In fact, a slight modification of the above proof shows that T is unique. With some work, this stronger result implies that there is in essence exactly one way to make all the numbers white up to trivial manipulations.

Second solution. Yes, it is possible. We prove a more general statement, where we replace 1000000 in the problem by some arbitrary positive integer m. We also focus on the numbers divisible by just a few primes instead of all the primes.

Lemma. *For a finite set of distinct primes $S = \{p_1, p_2, \ldots, p_n\}$, let $Q_m(S)$ be the set of numbers between 2 and m divisible only by primes in S. The elements of $Q_m(S)$ can be colored black or white. A permissible move consists of selecting a number in $Q_m(S)$ and changing the color of that number and each number not relatively prime to it. Then it is possible to reverse the coloring of $Q_m(S)$ by selecting several numbers in a subset $R_m(S) \subseteq Q_m(S)$.*

Proof. We prove the lemma by induction on n. If $n = 1$, then selecting p_1 suffices. Now suppose $n > 1$, and assume without loss of generality that the numbers are all black to start with.

Let $T = \{p_1, p_2, \ldots, p_{n-1}\}$, and define t to be the largest integer such that $tp_n \leq m$. We can assume $t \geq 1$ because otherwise we could ignore

p_n and just use the smaller set T, and we'd be done by our induction hypothesis.

Now select the numbers in $R_m(T), R_t(T)$, and $p_n R_t(T) = \{p_n x |\ x \in R_t(T)\}$, and consider the effect of this action on a number y:

- y is not a multiple of p_n. Selecting the numbers in $R_m(T)$ makes y white. If selecting $x \in R_t(T)$ changes y's color, selecting xp_n will change it back so that y will become white.
- y is a power of p_n. Selecting the numbers in $R_m(T)$ and $R_t(T)$ has no effect on y, but each of the $|R_t(T)|$ numbers in $xR_t(T)$ changes y's color.
- $p_n | y$ but y is not a power of p_n. Selecting the numbers in $R_m(T)$ makes y white. Because $y \neq p_n^i$, it is divisible by some prime in T so selecting the numbers in $R_t(T)$ makes y black again. Finally, each of the $|R_t(T)|$ numbers in $xR_t(T)$ changes y's color.

Therefore, all the multiples of p_n are the same color (black if $|R_t(T)|$ is even, white if $|R_t(T)|$ is odd), while all the other numbers in $Q_m(S)$ are white. If the multiples of p_n are still black, we can select p_n to make them white, and we are done. □

We now return to the original problem. Set $m = 1000000$, and let S be the set of all primes under 1000000. From the lemma, we can select numbers between 2 and 1000000 so that all the numbers 2,3,...,1000000 are white. Finally, complete the process by selecting 1.

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Arithmetic Functions

17.1 Multiplicative functions

Problem 6.1.6. *Let f be a function from the positive integers to the integers satisfying $f(m+n) \equiv f(n) \pmod{m}$ for all $m, n \geq 1$ (e.g., a polynomial with integer coefficients). Let $g(n)$ be the number of values (including repetitions) of $f(1), f(2), \ldots, f(n)$ divisible by n, and let $h(n)$ be the number of these values relatively prime to n. Show that g and h are multiplicative functions related by*

$$h(n) = n \sum_{d|n} \mu(d) \frac{g(d)}{d} = n \prod_{j=1}^{k} \left(1 - \frac{g(p_j)}{p_j}\right),$$

where $n = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ is the prime factorization of n.

(American Mathematical Monthly)

Solution. Let m and n be positive integers such that $gcd(m, n) = 1$ and let $1 \leq a \leq m$, $1 \leq b \leq n$. From Chinese Remainder Theorem and the properties of f it follows that $m|f(a)$ and $n|f(b)$ if and only if $mn|f(x)$, where $x = x(a, b)$ is the unique integer such that $x \equiv a \pmod{m}$, $x \equiv b \pmod{n}$, and $1 \leq x \leq \min\{m, n\}$. Thus g is multiplicative. For $d|n$, the number of values of $f(1), \ldots, f(n)$ divisible by d is just $\frac{n}{d} g(d)$. By a

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straightforward inclusion-exclusion count,

$$h(n) = n - \sum_{i=1}^{k} \frac{n}{p_i} g(p_i) + \sum_{1 \le i < j \le k} \frac{n}{p_i p_j} (p_i p_j) - \dots$$

and we get

$$h(n) = n \prod_{j=1}^{k} \left(1 - \frac{g(p_j)}{p_j}\right)$$

Problem 6.1.7. *Define $\lambda(1) = 1$, and if $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, define*

$$\lambda(n) = (-1)^{\alpha_1 + \dots + \alpha_k}$$

1) Show that λ is completely multiplicative.

2) Prove that

$$\sum_{d|n} \lambda(d) = \begin{cases} 1 & \textit{if } n \textit{ is a square} \\ 0 & \textit{otherwise} \end{cases}$$

3) Find the convolutive inverse of λ.

Solution. 1) Assume $m = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ and $n = p_1^{\beta_1} \dots p_k^{\beta_k}$, where $\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k \ge 0$. Then $mn = p_1^{\alpha_1 + \beta_1} \dots p_k^{\alpha_k + \beta_k}$ and

$$\lambda(mn) = (-1)^{\alpha_1 + \beta_1 + \dots + \alpha_k + \beta_k} = (-1)^{\alpha_1 + \dots + \alpha_k} (-1)^{\beta_1 + \dots + \beta_k} = \lambda(m)\lambda(n).$$

2) Because λ is multiplicative, according to Theorem 6.1.2, it follows that its summation function Λ has also this property. Therefore, it is sufficient to calculate Λ on a power of a prime. we have

$$\Lambda(p^\alpha) = \Lambda(1) + \Lambda(p) + \dots + \Lambda(p^\alpha) = \begin{cases} 1 & \text{if} \quad \alpha \quad \text{even} \\ 0 & \text{if} \quad \alpha \quad \text{odd} \end{cases}$$

If $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, then $\Lambda(n) = \Lambda(p_1^{\alpha_1}) \dots \Lambda(p_k^{\alpha_k}) = 1$ if all $\alpha_1, \dots, \alpha_k$ are even and 0 otherwise. Hence

$$\Lambda(n) = \begin{cases} 1 & \text{if } n \text{ is a square} \\ 0 & \text{otherwise} \end{cases}$$

3) Let g be the convolution inverse of λ. From Problem 1.6.4.2) it follows that g is multiplicative, hence it is perfectly by its values on powers of primes. From $g * \lambda = \varepsilon$ we get $(g*\lambda)(p) = g(1)\lambda(p) + g(p)\lambda(1) = -1 + g(p) = 0$, i.e. $g(p) = 1$ for any prime p. Also, $(g*\lambda)(p^2) = 0$ implies $1 - 1 + g(p^2) = 0$,

i.e. $g(p^2) = 0$. A simple inductive argument shows that $g(p^\alpha) = 0$ for any positive integer $\alpha \geq 2$. It follows

$$g(n) = \begin{cases} 1 & \text{if} \quad n = 1 \\ 0 & \text{if} \quad p^2 | n \text{ for some prime } p > 1 \\ 1 & \text{if} \quad n = p_1 \dots p_k, \text{ where } p_1, \dots, p_k \text{ are distinct primes,} \end{cases}$$

i.e. $g = \mu^2$, where μ is the Möbius function.

Problem 6.1.8. *Let an integer $n > 1$ be factored into primes: $n = p_1^{\alpha_1} \dots p_m^{\alpha_m}$ (p_i distinct) and let its own positive integral exponents be factored similarly. The process is to be repeated until it terminates with a unique "constellation" of prime numbers. For example, the constellation for 192 is $192 = 2^{2^2 \cdot 3} \cdot 3$ and for 10000 is $10000 = 2^{2^2} \cdot 5^2$. Call an arithmetic function g generally multiplicative if $g(ab) = g(a)g(b)$ whenever the constellations for a and b have no prime in common.*

1) Prove that every multiplicative function is generally multiplicative. Is the converse true?

2) Let h be an additive function (i.e. $h(ab) = h(a) + h(b)$ whenever $gcd(a, b) = 1$). Call a function k generally additive if $k(ab) = k(a) + k(b)$ whenever the constellations for a and b have no prime in common. Prove that every additive function is generally additive. Is the converse true?

(American Mathematical Monthly)

Solution. 1) Let f be multiplicative. If the constellations for a and b have no prime in common, then the same is true of their factorizations, so $f(ab) = f(a)f(b)$. Hence f is generally multiplicative.

The converse is not true. Indeed, define $g(a)$ to the product of all primes in the constellation of a, taken once only, regardless of how many times they appear in the constellation. Then g is clearly generally multiplicative, but $g(9) = 6$, $g(2) = 2$ and $g(18) = 6$, so $g(9 \cdot 2) \neq g(9)g(2)$.

2) The statement "additive implies generally additive" can be proved in the same way. If $k(a)$ is the sum of all primes in the constellation of a each taken once only, then k is generally additive but $k(9) = 5$, $k(2) = 2$ and $k(18) = 5$.

17.2 Number of divisors

Problem 6.2.5. *Does there exist a positive integer such that the product of its proper divisors ends with exactly 2001 zeroes?*

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(2001 Russian Mathematical Olympiad)

Solution. Yes. Given an integer n with $\tau(n)$ is equal to

$$\sqrt{\left(\prod_{d|n} d\right)\left(\prod_{d|n}(n/d)\right)} = \sqrt{\prod_{d|n} d(n/d)} = \sqrt{n^{\tau(n)}}.$$

Thus, the product of all proper positive divisors of n equals

$$n^{\frac{1}{2}\tau(n)-1}.$$

If $n = \prod_{i=1}^{k} p_i^{q_i}$ with the p_i's distinct primes and the q_i's positive integers, then $\tau(n) = \prod_{i=1}^{k}(q_i+1)$. Hence, if we set $n = 2^1 \cdot 5^1 \cdot 7^6 \cdot 11^{10} \cdot 13^{12}$, then

$$\frac{1}{2}\tau(n) - 1 = \frac{1}{2}(2 \cdot 2 \cdot 7 \cdot 11 \cdot 13) - 1 = 2001.$$

Thus, the product of the proper divisors of n is equal to $2^{2001} \cdot 5^{2001} \cdot 7^{6\cdot 2001} \cdot 11^{10\cdot 2001} \cdot 13^{12\cdot 2001}$, an integer ending in exactly 2001 zeroes.

Problem 6.2.6. *Prove that the number of divisors of the form $4k+1$ of each positive integer is not less than the number of its divisors of the form $4k+3$.*

Solution. To solve the problem, consider the function

$$f(n) = \begin{cases} 0, & \text{if} \quad n \text{ is even} \\ 1, & \text{if} \quad n \equiv 1 \pmod 4 \\ -1, & \text{if} \quad n \equiv 3 \pmod 4. \end{cases}$$

It follows directly from this definition that $f(n)$ is multiplicative. Now we apply (1). The even divisors of n do not influence its left-hand side. Each divisor of the form $4k+1$ contributes a 1, and each divisor of the form $4k+3$ contributes $a-1$. Consequently, it suffices to prove that the summation function of f, $\sum_{d|n} f(d)$ is nonnegative for each positive integer n.

Take any prime divisor p_i of n. If $p_i \equiv 1 \pmod 4$, then the same congruence holds for all powers of p_i, so the ith factor in the right-hand side of (1) is positive. If p_i is congruent to 3 modulo 4, then so are its odd powers while the even powers are congruent to 1 modulo 4. In this case the ith

factor in the right-hand side has the form $1-1+1-1+\ldots$, and it equals 1 or 0 according as α_i is even or odd. Summing up, we conclude that the sum in question is nonnegative.

Problem 6.2.7. *Let $d_1, d_2, \ldots, d_l$ be all positive divisors of a positive integer. For each $i = 1, 2, \ldots, l$ denote by a_i the number of divisors of d_i. Then*

$$a_1^3 + a_2^3 + \cdots + a_l^3 = (a_1 + a_2 + \cdots + a_l)^2.$$

Solution. We have

$$a_1 + a_2 + \cdots + a_l = \sum_{d|n} \tau(d) = \prod_{i=1}^{k} (1 + \tau(p_i) + \cdots + \tau(p_i^{\alpha_i}))$$

$$a_1^3 + a_2^3 + \cdots + a_l^3 = \sum_{d|n} \tau(d)^3 = \prod_{i=1}^{k} (1 + \tau(p_i)^3 + \cdots + \tau(p_i^{\alpha_i})^3)$$

where $n = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ is the prime factorization of n.

Since

$$1 + \tau(p_i) + \cdots + \tau(p_i^{\alpha_i}) = 1 + 2 + \cdots + (\alpha_i + 1)$$

and

$$1+\tau(p_i)^3+\cdots+\tau(p_i^{\alpha_i})^3 = 1^3+2^3+\cdots+(\alpha+i+1)^3 = [1+2+\cdots+(\alpha+i+1)]^2,$$

the conclusion follows.

For example, if $n = 12$ we have $d_1 = 1$, $d_2 = 2$, $d_3 = 3$, $d_4 = 4$, $d_5 = 6$, $d_6 = 12$; $a_1 = 1$, $a_2 = 2$, $a_3 = 2$, $a_4 = 3$, $a_5 = 4$, $a_6 = 6$ and

$$1^3 + 2^3 + 2^3 + 3^3 + 4^3 + 6^3 = 324 = (1 + 2 + 2 + 3 + 4 + 6)^2.$$

Remark. The above identity shows that solving the equation

$$(x_1 + x_2 + \cdots + x_n)^2 = x_1^3 + x_2^3 + \cdots + x_n^3$$

is positive integers is a very difficult job. If we assume that $x_i \neq x_j$ for $i \neq j$, there are only a few solutions. Try to prove this last assertion.

17.3 Sum of divisors

Problem 6.3.5. *For any $n \geq 2$,*

$$\sigma(n) < n\sqrt{2\tau(n)}.$$

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(1999 Belarusian Mathematical Olympiad)

Solution. Let $d_1, d_2, \ldots, d_{\tau(n)}$ be the divisors of n. They can be rewritten in the form

$$\frac{n}{d_1}, \frac{n}{d_2}, \ldots, \frac{n}{d_{\tau(n)}}.$$

By the Power Mean Inequality,

$$\sigma(n) \leq \sqrt{\tau(n) \sum_{i=1}^{\tau(n)} d_i^2}.$$

Now,

$$\frac{1}{n^2}\left(\sum_{i=1}^{\tau(n)} d_i^2\right) = \sum_{i=1}^{\tau(n)} \frac{1}{d_i^2} \leq \sum_{j=1}^{\tau(n)} \frac{1}{j^2} < \sum_{j=1}^{\infty} \frac{1}{j^2} = \frac{\pi^2}{6}.$$

Hence

$$\sigma(n) \leq \sqrt{\tau(n) \sum_{i=1}^{\tau(n)} d_i} < \sqrt{\tau(n) \frac{n^2\pi^2}{6}} < n\sqrt{2\tau(n)}.$$

Problem 6.3.6. *Find all the four-digit numbers so that when decomposed in prime factors have the sum of the prime factors equal to the sum of the exponents.*

Solution. 1) If the number has at least four prime divisors, then $n \geq 2^{14} \cdot 3 \cdot 5 \cdot 7 > 9999$, a contradiction.

2) If n has 3 prime divisors, these must be 2, 3 or 5. The numbers are

$$2^8 \cdot 3 \cdot 5 = 3840,\ 2^7 \cdot 3^2 \cdot 5 = 5760,\ 2^6 \cdot 3^3 \cdot 5 = 8640 \text{ and } 2^7 \cdot 3 \cdot 5^2 = 9600.$$

3) If n has 2 prime divisors, at least one of them must be 2 or 3. The numbers

$$2^4 \cdot 5^3 = 2000,\ 2^3 \cdot 5^4 = 5000,\ 2^8 \cdot 7 = 1792,\ 2^7 \cdot 7^2 = 6272$$

satisfy the solutions.

4) If n has only one prime factor, then $5^5 = 3125$.

Therefore there are 9 solutions.

Problem 6.3.7. *Let m, n, k be positive integers with $n > 1$. Show that $\sigma(n)^k \neq n^m$.*

(2001 St. Petersburg City Mathematical Olympiad)

Solution. Let $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$. Because $\sigma(n) > n$, if $\sigma(n)^k = n^m$, then $\sigma(n) = p_1^{f_1} p_2^{f_2} \dots p_k^{f_k}$ where $f_i > e_i$. This implies $f_i \geq e_i + 1$, for all i and

$$\begin{aligned}\sigma(n) &\geq p_1^{1+e_1} p_2^{1+e_2} \dots p_k^{1+e_k} > \frac{p_1^{1+e_1}-1}{p_1-1}\frac{p_2^{1+e_2}-1}{p_2-1}\dots\frac{p_k^{1+e_k}-1}{p_k-1} \\ &= (1+p_1+\dots+p_1^{e_1})(1+p_2+\dots+p_2^{e_2})\dots(1+p_k+\dots+p_k^{e_k}) \\ &= \sigma(n).\end{aligned}$$

This is a contradiction.

Remark. Actually, we have shown that for $n > 1$, $\sigma(n)$ has a prime factor different from any prime factor of $n!$.

17.4 Euler's totient function

Problem 6.4.5. *For a positive integer n, let $\psi(n)$ be the number of prime factors of n. Show that if $\varphi(n)$ divides $n-1$ and $\psi(n) \leq 3$, then n is prime.*

(1998 Korean Mathematical Olympiad)

Solution. Note that for prime p, if $p^2 | n$ then $p | \varphi(n)$ but $p \nmid n-1$, contradiction. So we need only show that $n \neq pq$, $n \neq pqr$ for primes $p < q < r$.

First assume $n = pq$, so $(p-1)(q-1) | pq - 1$. Note that $q \geq 3$ implies that the left side is even, so the right is too and p, q are odd. But if $p = 3$, $q = 5$ then

$$\frac{pq-1}{(p-1)(q-1)} < 2;$$

the left side is decreasing in each variable and always > 1 so it cannot be an integer, contradiction.

Now let $n = pqr$. As before p, q, r are odd; if $p = 3$, $q = 7$, and $r = 11$ then

$$\frac{pqr-1}{(p-1)(q-1)(r-1)} < 2$$

and again the left side is decreasing and > 1; this eliminates all cases except where $p = 3$, $q = 5$. Then for $r = 7$ we have

$$\frac{pqr-1}{(p-1)(q-1)(r-1)} < 3$$

so the only integer value ever attainable is 2. Note that $(15r-1)/8(r-1) = 2$ gives $r = 15$ which is not a prime and we have eliminated all cases.

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Remarks. 1) The problem is a direct consequence of Problem 1.1.16.

2) A long stonaling conjecture due to Lehmer asserts that if $\varphi(n)|n-1$, then n is a prime. This has been proved so far for $\psi(n) \leq 14$. The proofs are very long and computational and no further progress has been made on this conjecture.

Problem 6.4.6. *Show that the equation $\varphi(n) = \tau(n)$ has only the solutions $n = 1, 3, 8, 10, 18, 24, 30$.*

Solution. We check directly that the listed integers satisfy the equation and there are no others ≤ 30 with this property. We will prove that for $n \geq 31$, $\varphi(n) > \tau(n)$. For this we consider the multiplicative function $f(n) = \dfrac{\varphi(n)}{\tau(n)}$. If n is a prime, we have $f(n) = \dfrac{n-1}{2}$, hence f increases on the set of primes.

For a prime p, define $S_p = \{p^\alpha | \alpha \geq 1\}$. Because

$$f(p^\alpha) = \frac{p^{\alpha-1}(p-1)}{\alpha+1} \text{ and } \frac{p}{\alpha+2} \geq \frac{2}{\alpha+2} > \frac{1}{\alpha+1},$$

we obtain $f(p^{\alpha+1}) > f(p^\alpha)$, that is f increases on S_p. Using the fact that $\min\limits_{p,\alpha} f(p^\alpha) = f(2) = \dfrac{1}{2}$, it follows that in order to solve the given equation we need to consider the integers p^α with $f(p^\alpha) \leq 2$. These are 2, 3, 4, 5, 8, 9, 16, whose prime factors are only 2, 3, or 5 and the conclusion follows.

Problem 6.4.7. *Let $n > 6$ be an integer and $a_1, a_2, \ldots, a_k$ be all positive integers less than n and relatively prime to n. If*

$$a_2 - a_1 = a_3 - a_2 = \cdots = a_k - a_{k-1} > 0,$$

prove that n must be either a prime number or a power of 2.

(32nd IMO)

Solution. It is given that the reduced system of residues mod$n$ chosen from the set $\{1, 2, \ldots, n-1\}$ is an arithmetic progression. We write it as an increasing sequence $1 = a_1 < a_2 < \cdots < a_k = n-1$.

For a prime number n the reduced system of residues is the sequence $1 < 2 < \cdots < n-1$ and it is an arithmetic progression with ratio 1. If $n = 2^l$ the reduced system of residues is $1 < 3 < 5 < \cdots < 2^l - 1$ and it is an arithmetic progression with ratio 2. The problem asks to prove that only these cases can appear.

Let a_2 be the second member of the progression. Because $a_2 > 1$ is the least positive number relatively prime to n, it is a prime number, say p

and $p > 3$. Then, the ratio of the progression is $a_2 - a_1 = p - 1$ and $a_k = n - 1 = 1 + (k-1)(p-1)$. We obtain a "key" formula:

$$n - 2 = (k-1)(p-1).$$

Remembering the choice of p, n is divisible by 3 and then $n - 2 \equiv 1 \pmod 3$. Thus, by the key formula we cannot have $p \equiv 1 \pmod 3$. Since $p > 3$ we have $p \equiv 2 \pmod 3$. Then $a_3 = 1 + 2(p-1) \equiv 0 \pmod 3$ and this contradicts the supposition that a_3 and n are relatively prime numbers.

17.5 Exponent of a prime and Legendre's formula

Problem 6.5.7. *a) If p is a prime, prove that for any positive integer n,*

$$-\left\lfloor \frac{\ln n}{\ln p} \right\rfloor + n \sum_{k=1}^{\left\lfloor \frac{\ln n}{\ln p} \right\rfloor} \frac{1}{p^k} < e_p(n) < \frac{n}{p-1}.$$

b) Prove that

$$\lim_{n\to\infty} \frac{e_p(n)}{n} = \frac{1}{p-1}.$$

Solution. a) From Legendre's formula,

$$e_p(n) = \sum_{k\geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor \leq \sum_{k\geq 1} \frac{n}{p^k} < n \sum_{j=1}^{\infty} \frac{1}{p^j} = \frac{n}{p-1}.$$

For the left bound note that $\left\lfloor \frac{\ln n}{\ln p} \right\rfloor$ is the least nonnegative integer s such that $n < p^{s+1}$. That is $\left\lfloor \frac{n}{p^k} \right\rfloor = 0$ for $k \geq s+1$. It follows that

$$e_p(n) = \sum_{k=1}^{s} \left\lfloor \frac{n}{p^k} \right\rfloor > \sum_{k=1}^{s} \left(\frac{n}{p^k} - 1 \right) = n \sum_{k=1}^{s} \frac{1}{p^k} - s,$$

and we are done.

b) From the inequalities

$$-\frac{1}{n} \left\lfloor \frac{\ln n}{\ln p} \right\rfloor + \sum_{k=1}^{\left\lfloor \frac{\ln n}{\ln p} \right\rfloor} \frac{1}{p^k} < \frac{e_p(n)}{n} < \frac{1}{p-1}$$

and the fact that

$$\lim_{n\to\infty} \frac{1}{n} \left\lfloor \frac{\ln n}{\ln p} \right\rfloor = 0 \quad \text{and} \quad \lim_{n\to\infty} \sum_{k=1}^{\left\lfloor \frac{\ln n}{\ln p} \right\rfloor} \frac{1}{p^k} = \frac{1}{p-1},$$

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the desired formula follows.

Problem 6.5.8. *Show that for all nonnegative integers m, n the number*

$$\frac{(2m)!(2n)!}{m!n!(m+n)!}$$

is also an integer.

(14th IMO)

Solution. It is sufficient to prove that for any prime number p

$$e_p(2m) + e_p(2n) \geq e_p(m) + e_p(n) + e_p(m+n).$$

Again, it is sufficient to prove that for all $i, j \geq 1$, the following inequality holds:

$$\left\lfloor \frac{2m}{p^i} \right\rfloor + \left\lfloor \frac{2n}{p^i} \right\rfloor \geq \left\lfloor \frac{m}{p^i} \right\rfloor + \left\lfloor \frac{n}{p^i} \right\rfloor + \left\lfloor \frac{m+n}{p^i} \right\rfloor.$$

It follows from a more general result.

Lemma. *For any real numbers a, b*

$$\lfloor 2a \rfloor + \lfloor 2b \rfloor \geq \lfloor a \rfloor + \lfloor b \rfloor + \lfloor a+b \rfloor.$$

Proof. Let $a = \lfloor a \rfloor + x$, $b = \lfloor b \rfloor + y$ where $0 \leq x, y < 1$. If $x + y < 1$ we have $\lfloor a+b \rfloor = \lfloor a \rfloor + \lfloor b \rfloor$ and the required inequality becomes:

$$\lfloor 2a \rfloor + \lfloor 2b \rfloor \geq 2(\lfloor a \rfloor + \lfloor b \rfloor).$$

In this form, it is obvious.

Let $1 \leq x + y < 2$. Then $2x \geq 1$ or $2y \geq 1$. Let $2x \geq 1$. Then

$$\lfloor 2a \rfloor = 2\lfloor a \rfloor + 1 \quad \text{and} \quad \lfloor a+b \rfloor = \lfloor a \rfloor + \lfloor b \rfloor + 1.$$

Thus:

$$\lfloor 2a \rfloor + \lfloor 2b \rfloor = 2\lfloor a \rfloor + 1 + \lfloor 2b \rfloor \geq 2\lfloor a \rfloor + 1 + 2\lfloor b \rfloor = \lfloor a \rfloor + \lfloor b \rfloor + \lfloor a+b \rfloor.$$

The other cases follow in a similar way.

Problem 6.5.9. *Prove that $\dfrac{(3a+3b)!(2a)!(3b)!(2b)!}{(2a+3b)!(a+2b)!(a+b)!a!(b!)^2}$ is an integer for any positive integers a, b.*

(American Mathematical Monthly)

Solution. First, let us clearify something. When we write

$$\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \dots,$$

we write in fact $\sum_{k\geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor$ and this sum has clearly a finite number of non-zero terms. Now, let us take a prime p and let us apply Legendre's formula formula as well as the first observations. We find that

$$v_p((3a+3b)!(2a)!(3b)!(2b)!) = \sum_{k\geq 1} \left(\left\lfloor \frac{3a+3b}{p^k} \right\rfloor + \left\lfloor \frac{2a}{p^k} \right\rfloor + \left\lfloor \frac{3b}{p^k} \right\rfloor + \left\lfloor \frac{2b}{p^k} \right\rfloor \right)$$

and also

$$v_p((2a+3b)!(a+2b)!(a+b)!a!(b!)^2)$$
$$= \sum_{k\geq 1} \left(\left\lfloor \frac{2a+3b}{p^k} \right\rfloor + \left\lfloor \frac{a+2b}{p^k} \right\rfloor + \left\lfloor \frac{a+b}{p^k} \right\rfloor + \left\lfloor \frac{a}{p^k} \right\rfloor + 2\left\lfloor \frac{b}{p^k} \right\rfloor \right)$$

Of course, it is enough to prove that for each $k \geq 1$ the term corresponding to k in the first sum is greater than or equal to the term corresponding to k in the second sum. With the substitution $x = \frac{a}{p^k}$, $y = \frac{b}{p^k}$, we have to prove that for any nonnegative real numbers x, y we have

$$\lfloor 3x+3y \rfloor + \lfloor 2x \rfloor + \lfloor 3y \rfloor + \lfloor 2y \rfloor \geq \lfloor 2x+3y \rfloor + \lfloor x+2y \rfloor + \lfloor x+y \rfloor + \lfloor x \rfloor + 2\lfloor y \rfloor.$$

This isn't easy, but with another useful idea the inequality will become easy. The idea is that

$$\lfloor 3x+3y \rfloor = 3\lfloor x \rfloor + 3\lfloor y \rfloor + \lfloor 3\{x\} + 3\{y\} \rfloor$$

and similar relations for the other terms of the inequality. After this operation, we see that it suffices to prove the inequality only for $0 \leq x, y < 1$. Because we can easily compute all terms, after splitting in some cases, so that to see when $\lfloor 2\{x\} \rfloor$, $\lfloor 3\{y\} \rfloor$, $\lfloor 2\{y\} \rfloor$ are 0, 1 or 2.

Problem 6.5.10. *Prove that there exists a constant c such that for any positive integers a, b, n that verify $a! \cdot b! | n!$ we have $a + b < n + c \ln n$.*

(Paul Erdös)

Solution. This time the second formula for $e_p(n)$ is useful. Of course, there is no reasonable estimation of this constant, so we should better see what happens if $a! \cdot b! | n!$. Then $e_2(a) + e_2(b) \leq e_2(n!)$, which can be

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translated as $a-S_2(a)+b-S_2(b) \leq n-S_2(n) < n$. So, we have found almost exactly what we needed: $a+b < n+S_2(a)+S_2(b)$. Now, we need another observation: the sum of digits of a number A when written in binary is at most the number of digits of A in base 2, which is $1+\lfloor \log_2 A \rfloor$ (this follows from the fact that $2^{k-1} \leq A < 2^k$, where k is the number of digits of A in base 2). So, we have the estimations $a+b < n+S_2(a)+S_2(b) \leq n+2+\log_2 ab \leq n+2+2\log_2 n$ (since we have of course $a,b \leq n$). And now the conclusion is immediate.

Problem 6.5.11. *Prove that the equation*

$$\frac{1}{10^n} = \frac{1}{n_1!} + \frac{1}{n_2!} + \cdots + \frac{1}{n_k!}$$

does not have integer solutions such that $1 \leq n_1 < n_2 < \cdots < n_k$.

(Tuymaada Olimpiad)

Solution. Suppose we have found a solution of the equation and let us consider

$$P = n_1! n_2! \ldots n_k!.$$

We have

$$10^n((n_1+1)\ldots(n_k-1)n_k + \cdots + (n_{k-1}+1)\ldots(n_k-1)n_k + 1) = n_k!$$

which shows that n_k divides 10^n. Let us write $n_k = 2^x \cdot 5^y$. First of all, suppose that x, y are positive. Thus,

$$(n_1+1)\ldots(n_k-1)n_k + \cdots + (n_{k-1}+1)\ldots(n_k-1)n_k + 1$$

is relatively prime with 10 and it follows that $e_2(n_k) = e_5(n_k)$. This implies of course that $\left\lfloor \frac{n_k}{2^j} \right\rfloor = \left\lfloor \frac{n_k}{5^j} \right\rfloor$ for all j (because we clearly have $\left\lfloor \frac{n_k}{2^j} \right\rfloor > \left\lfloor \frac{n_k}{5^j} \right\rfloor$) and so $n_k \leq 3$. A verification by hand shows that there is no solution in this case.

Next, suppose that $y = 0$. Then

$$(n_1+1)\ldots(n_k-1)n_k + \cdots + (n_{k-1}+1)\ldots(n_k-1)n_k + 1$$

is odd and thus $e_2(n_k) = n \leq e_5(n_k)$. Again this implies $e_2(n_k) = e_5(n_k)$ and we have seen that this gives no solution. So, actually $x = 0$. A crucial observation is that if $n_k > n_{k-1}+1$, then

$$(n_1+1)\ldots(n_k-1)n_k + \cdots + (n_{k-1}+1)\ldots(n_k-1)n_k + 1$$

is again odd and thus we find again that $e_2(n_k) = n \leq e_5(n_k)$, impossible. So, $n_k = n_{k-1} + 1$. But then, taking into account that n_k is a power of 5, we deduce that

$$(n_1 + 1) \dots (n_k - 1)n_k + \dots + (n_{k-1} + 1) \dots (n_k - 1)n_k + 1$$

is congruent to 2 modulo 4 and thus $e_2(n_k) = n+1 \leq e_5(n_k)+1$. It follows that $\left\lfloor \frac{n_k}{2} \right\rfloor \leq 1 + \left\lfloor \frac{n_k}{5} \right\rfloor$ and thus $n_k \leq 6$. Since n_k is a power of 5, we find that $n_k = 5$, $n_{k-1} =$ and a quick research of all possibilities shows that there are no solutions.

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More on Divisibility

18.1 Fermat's Little Theorem

Problem 7.1.11. *Let $3^n - 2^n$ be a power of a prime for some positive integer n. Prove that n is a prime.*

Solution. Let $3^n - 2^n = p^\alpha$ for some prime p and some $\alpha \geq 1$, and let q be a prime divisor of n. Assume that $q \neq n$; then $n = kq$, where $k > 1$. Since $p^\alpha = 3^{kq} - 2^{kq} = (3^k)^q - (2^k)^q$, we observe that p^α is divisible by $3^k - 2^k$. Hence $3^k - 2^k = p^\beta$ for some $\beta \geq 1$. Now we have

$$p^\alpha = (2^k + p^\beta)^q - 2^{kq}$$

$$= q2^{k(q-1)}p^\beta + \frac{q(q-1)}{2}2^{k(q-2)}p^{2\beta} + \cdots + p^{q\beta}.$$

Since $\alpha > \beta$ (because $p^\beta = 3^k - 2^k$ is less than $p^\alpha = 3^{kq} - 2^{kq}$), it follows that p^α is divisible by a power of p at least as great as $p^{\beta+1}$. Then the above equality implies that p divides $q2^{k(q-1)}$. On the other hand, p is obviously odd and hence it divides q. Being a prime, q must be then equal to p. Therefore $n = kq = kp$ and $p^\alpha = (3^p)^k - (2^p)^k$ is divisible by $3^p - 3^p$, implying $3^p - 2^p = p^\gamma$ for some $\gamma \geq 1$. In particular, we infer that $3^p \equiv 2^p \pmod{p}$. Now, observing that $p \neq 2, 3$, we reach a contradiction with Fermat's Little Theorem, by which

$$3^p \equiv 3 \pmod{p}, \quad 2^p \equiv 2 \pmod{p}.$$

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Problem 7.1.12. *Let $f(x_1, \ldots, x_n)$ be a polynomial with integer coefficients of total degree less than n. Show that the number of ordered n-tuples $(x_1, \ldots, x_n)$ with $0 \leq x_i \leq 12$ such that $f(x_1, \ldots, x_n) \equiv 0 \pmod{13}$ is divisible by 13.*

(1998 Turkish Mathematical Olympiad)

Solution. (All congruences in this problem are modulo 13.) We claim that

$$\sum_{x=0}^{12} x^k \equiv 0 \text{ for } 0 \leq k < 12.$$

The case $k = 0$ is obvious, so suppose $k > 0$. Let g be a primitive root modulo 13; then the numbers $g, 2g, \ldots, 12g$ are $1, 2, \ldots, 12$ in some order, so

$$\sum_{x=0}^{12} x^k \equiv \sum_{x=0}^{12} (gx)^k = g^k \sum_{x=0}^{12} x^k;$$

since $g^k \not\equiv 1$, we must have $\displaystyle\sum_{x=0}^{12} x^k \equiv 0$. This proves our claim.

Now let $S = \{(x_1, \ldots, x_n) | 0 \leq x_i \leq 12\}$. It suffices to show that the number of n-tuples $(x_1, \ldots, x_n) \in S$ with $f(x_1, \ldots, x_n) \not\equiv 0$ is divisible by 13, since $|S| = 13^n$ is divisible by 13. Consider the sum

$$\sum_{(x_1, \ldots, x_n) \in S} (f(x_1, \ldots, x_n))^{12}.$$

This sum counts the number of n-tuples $(x_1, \ldots, x_n) \in S$ such that $f(x_1, \ldots, x_n) \not\equiv 0$, since by Fermat's Little Theorem

$$(f(x_1, \ldots, x_n))^{12} \equiv \begin{cases} 1, & \text{if} \quad f(x_1, \ldots, x_n) \not\equiv 0 \\ 0, & \text{if} \quad f(x_1, \ldots, x_n) \equiv 0. \end{cases}$$

On the other hand, we can expand $(f(x_1, \ldots, x_n))^{12}$ in the form

$$(f(x_1, \ldots, x_n))^{12} = \sum_{j=1}^{N} c_j \prod_{i=1}^{n} x_i^{e_{ji}}$$

for some integers N, c_j, e_{ji}. Since f is a polynomial of total degree less than n, we have $e_{j1} + e_{j2} + \cdots + e_{jn} < 12n$ for every j, so for each j there exists an i such that $e_{ji} < 12$. Thus by our claim

$$\sum_{(x_1, \ldots, x_n) \in S} c_j \prod_{i=1}^{n} x_i^{e_{ji}} = c_j \prod_{i=1}^{n} \sum_{x=0}^{12} x_i^{e_{ji}} \equiv 0$$

since one of the sums in the product is 0. Therefore

$$\sum_{(x_1,\ldots,x_n)\in S} (f(x_1,\ldots,x_n))^{12} = \sum_{(x_1,\ldots,x_n)\in S} \sum_{j=1}^{N} c_j \prod_{i=1}^{n} x_i^{e_{ji}} \equiv 0,$$

so the number of $(x_1,\ldots,x_n)$ such that $f(x_1,\ldots,x_n) \not\equiv 0 \pmod{13}$ divisible by 13 and we are done.

Problem 7.1.13. *Find all pairs (m,n) of positive integers, with $m,n \geq 2$, such that $a^n - 1$ is divisible by m for each $a \in \{1,2,\ldots,n\}$.*

(2001 Romanian IMO Team Selection Test)

Solution. The solution is the set of all $(p, p-1)$, for odd primes p. The fact that all of these pairs are indeed solutions follows immediately from Fermat's Little Theorem. Now we show that no other solutions exist.

Suppose that (m,n) is a solution. Let p be a prime dividing m. We first observe that $p > n$. Otherwise, we could take $a = p$, and then $p^n - 1$ would not be divisible by p, and alone m. Then because $n \geq 2$, we have $p \geq 3$ and hence p is odd.

Now we prove that $p < n+2$. Suppose on the contrary that $p \geq n+2$. If n is odd, then $n+1$ is even and less than p. Otherwise, if n is even, then $n+2$ is even and hence less than p as well, because p is odd. In either case, there exists an even d such that $n < d < p$ with $\frac{d}{2} \leq n$. Setting $a = 2$, $\frac{d}{2}$ in the given condition, we find that

$$d^n \equiv 2^n \left(\frac{d}{2}\right)^n \equiv 1 \cdot 1 \equiv 1 \pmod{m},$$

so that $d^n - 1 \equiv 0 \pmod{m}$ as well. Because $n < d < p < m$, we see that $1, 2, \ldots, n, d$ are $n+1$ distinct roots of the polynomial congruence $x^n - 1 \equiv 0 \pmod{p}$. By Lagrange's Theorem, however, this congruence can have at most n roots, a contradiction.

Thus, we have sandwiched p between n and $n+2$, and the only possibility is that $p = n+1$. Therefore, all solutions are of the form $(p^k, p-1)$ with p an odd prime. It remains to prove that $k = 1$. Using $a = n = p-1$, it suffices to prove that

$$p^k \nmid ((p-1)^{p-1} - 1).$$

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Expanding the term $(p-1)^{p-1}$ modulo p^2, and recalling that p is odd, we have

$$\begin{aligned}(p-1)^{p-1} &= \sum_{i=0}^{p-1} \binom{p-1}{i}(-1)^{p-1-i}p_i \\ &\equiv \binom{p-1}{0}(-1)^{p-1} + \binom{p-1}{1}(-1)^{p-2}p \\ &\equiv 1-p(p-1) \\ &\equiv 2 \not\equiv \pmod{p^2}.\end{aligned}$$

It follows immediately that k cannot be greater than 1, completing the proof.

Problem 7.1.14. *Let p be a prime and b_0 an integer, $0 < b_0 < p$. Prove that there exists a unique sequence of base p digits $b_0, b_1, b_2, \ldots, b_n, \ldots$ with the following property: If the base p representation of a number x ends in the group of digits $b_n b_{n-1} \ldots b_1 b_0$ then so does the representation of x^p.*

Solution. We are looking for a sequence $b_0, b_1, b_2, \ldots, b_n, \ldots$ of base p digits such that the numbers $x_n = b_0 + b_1 p + \cdots + b_n p^n$ and x_n^p are congruent modulo p^{n+1} for each $n = 0, 1, 2, \ldots$ Of course, the choice of the first term b_0 is predetermined, and given in the problem statement; let us note that the numbers $x_0 = b_0$ and x_0^p are congruent modulo p by Fermat's Little Theorem. Suppose that the base p digits $b_1, b_2, \ldots, b_n$ are already chosen in such a way that $x_n^p \equiv x_n \pmod{p^{n+1}}$. We shall prove that there is a unique digit b_{n+1} such that

$$(x_n + b_{n+1}p^{n+1})^p \equiv x_n + b_{n+1}p^{n+1} \pmod{p^{n+2}};$$

this proves the existence and the uniqueness at the same time. Since

$$(x_n + b_{n+1}p^{n+1})^p = x_n^p + \binom{p}{1}x_n^{p-1}b_{n+1}p^{n+1} + Cp^{n+2}$$

for some integer constant C, and since $\binom{p}{1}$ is divisible by p, we get

$$(x_n + b_{n+1}p^{n+1})^p \equiv x_n^p \pmod{p^{n+2}}.$$

Hence b_{n+1} should satisfy the congruence

$$x_n^p - x_n - b_{n+1}p^{n+1} \equiv 0 \pmod{p^{n+2}}. \tag{1}$$

By the induction hypothesis, the number $x_n^p - x_n$ is divisible by p^{n+1}. This implies that its $(n+2)$nd base p digit (from the right to left) is

indeed the only choice for b_{n+1} such that (1) holds. The inductive proof is complete.

Problem 7.1.15. *Determine all integers $n > 1$ such that $\frac{2^n+1}{n^2}$ is an integer.*

(31st IMO)

Solution. We will prove that the problem has only solution $n = 3$. First, observe that n is odd number. Then, we prove that $3|n$.

Let p be the least prime divisor of n. Since $n^2|2^n+1, 2^n+1 \equiv 0 \pmod{p}$ and $2^{2n} \equiv 1 \pmod{p}$. By Fermat's Little Theorem, $2^{p-1} \equiv 1 \pmod{3}$. Then $2^d \equiv 1 \pmod{p}$, where $d = gcd(p-1, 2n)$. By the definition of p, d has no prime divisor greater than 2, which shows that $d = 2$. It follows $p = 3$.

Let $n = 3^k m$, where $k \geq 1$ and $(3, m) = 1$. Using the identity

$$x^{3^k} + 1 = (x+1)(x^2 - x + 1)(x^{2\cdot 3} - x^3 + 1) \dots (x^{2\cdot 3^{k-1}} - x^{3^{k-1}} + 1)$$

we obtain the decomposition:

$$2^{3^k m}+1 = (2^m+1)(2^{2m}-2^m+1)(2^{2\cdot 3m}-2^{3m}+1) \dots (3^{2\cdot 3^{k-1}m}-2^{3^{k-1}m}+1). \tag{1}$$

Since $2^{2s} - 2^s + 1 \equiv 3 \pmod{9}$ for s of the form 3^j we obtain in (1) that

$$3^k|(2^{2m} - 2^m + 1)(2^{2\cdot 3m} - 2^{3m} + 1) \dots (2^{2\cdot 3^{k-1}m} - 2^{3^{k-1}m} + 1)$$

but 3^{k+1} does not divides the product. Therefore, $3^k|2^m + 1$. Since 3 does not divide m and

$$2^m + 1 = 3^m - \binom{m}{1}3^{m-1} + \dots - \binom{m}{m-1}3$$

we obtain $k = 1$.

Now, we have $n = 3m$ and $9m^2|2^{3m} + 1$. We repeat, in some way, the starting argument. Take q the least prime divisor of m, $2^{6m} \equiv 1 \pmod{q}$ and $2^{q-1} \equiv 1 \pmod{q}$, $\delta = gcd(6m, q-1)$. By the definition of q we can have $\delta = 1, 2, 3$ or 6 and we also have $2^\delta \equiv 1 \pmod{q}$. Thus q can be chosen among prime divisors of the numbers 3, 7, 63. Since $q > 3$, we can have only $q = 7$. Returning to $m^2|2^{3m} + 1$, we obtain $49|2^{3m} + 1$. But we have $2^{3m} + 1 \equiv 2 \pmod{7}$ and we get a contradiction.

Thus, $m = 1$ and $n = 3$.

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Problem 7.1.16. *Let p be a prime number. Prove that there exists a prime number q such that for every integer n, the number $n^p - p$ is not divisible by q.*

(44th IMO)

Solution. Suppose that for every prime q, there exists an n for which $n^p \equiv p \pmod{q}$. Assume that $q = kp + 1$. By Fermat's Little Theorem we deduce that $p^k \equiv n^{kp} = n^{q-1} \equiv 1 \pmod{q}$, so $q|p^k - 1$.

It is known that any prime q such that $q|\dfrac{p^p - 1}{p - 1}$ must satisfy $q \equiv 1 \pmod{p}$. Indeed, from $q|p^{q-1}-1$ it follows that $q|p^{gcd(p,q-1)}-1$; but $q \nmid p-1$ because $\dfrac{p^p - 1}{p - 1} \equiv 1 \pmod{p-1}$, so $gcd(p, q-1) \neq 1$. Hence $gcd(p, q-1) = p$. Now suppose q is any prime divisor of $\dfrac{p^p - 1}{p - 1}$. Then $q|gcd(p^k-1, p^p-1) = p^{gcd(p,k)} - 1$, which implies that $gcd(p, k) > 1$, so $p|k$. Consequently $q \equiv 1 \pmod{p^2}$. However, the number $\dfrac{p^p - 1}{p - 1} = p^{p-1} + \cdots + p + 1$ must have at least one prime divisor that is not congruent to 1 modulo p^2. Thus we arrived at a contradiction.

Remark. Taking $q \equiv 1 \pmod{p}$ is natural, because for every other q, n^p takes all possible residues modulo p (including p too). Indeed, if $p \nmid q - 1$, then there is an $r \in \mathbb{N}$ satisfying $pr \equiv 1 \pmod{q-1}$; hence for any a the congruence $n^p \equiv a \pmod{q}$ has the solution $n \equiv a^r \pmod{q}$.

The statement of the problem itself is a special case of the Chebotarev theorem.

Problem 7.1.17. *Prove that for any $n > 1$ we cannot have $n|2^{n-1} + 1$.*

(Sierpinski)

Solution. Although very short, the proof is tricky. Let $n = \prod_{i=1}^{s} p_i^{k_i}$ where $p_1 < \cdots < p_s$ are prime numbers. The idea is to look at $v_2(p_i - 1)$. Choose that p_i which minimizes this quantity and write $p_i = 1 + 2^{r_i} m_i$ with m_i odd. Then of course we have $n \equiv 1 \pmod{2^{m_i}}$. Hence we can write $n - 1 = 2^m t$. We have $2^{2^m t} \equiv -1 \pmod{p_i}$ thus we surely have $-1 \equiv 2^{2^m t m_i} \equiv 2^{(p_i-1)t} \equiv 1 \pmod{p_i}$ (the last congruence being derived from Fermat's theorem). Thus $p_i = 2$, which is clearly impossible.

Problem 7.1.18. *Prove that for any natural number n, $n!$ is a divisor of*

$$\prod_{k=0}^{n-1}(2^n-2^k).$$

Solution. So, let us take a prime number p. Of course, for the argument to be non-trivial, we take $p \leq n$ (otherwise doesn't divide $n!$). First, let us see what happens with $p=2$. We have

$$e_2(n) = n - S_2(n) \leq n-1$$

and also

$$v_2\left(\prod_{k=0}^{n-1}(2^n-2^k)\right) = \sum_{k=0}^{n-1} v_2(2^n-2^k) \geq n-1$$

(since 2^n-2^k is even for $k \geq 1$), so we are done with this case. Now, let us assume that $p>2$. We have $p|2^{p-1}-1$ from Fermat's theorem, so we also have $p|2^{k(p-1)}-1$ for all $k \geq 1$. Now,

$$\prod_{k=0}^{n-1}(2^n-2^k) = 2^{\frac{n(n-1)}{2}}\prod_{k=1}^{n}(2^k-1)$$

and so, from the above remarks we infer that

$$v_2\left(\prod_{k=0}^{n-1}(2^n-2^k)\right) = \sum_{k=1}^{n} v_2(2^k-1)$$

$$\geq \sum_{1\leq k(p-1)\leq n} v_2(2^{k(p-1)}-1) \geq card\{k|1 \leq k(p-1) \leq n\}$$

Since

$$card\{k|1 \leq k(p-1) \leq n\} = \left[\frac{n}{p-1}\right],$$

we have found that

$$v_2\left(\prod_{k=0}^{n-1}(2^n-2^k)\right) \geq \left[\frac{n}{p-1}\right].$$

But we know that

$$e_2(n) = \frac{n-s_p(n)}{p-1} \leq \frac{n-1}{p-1} < \frac{n}{p-1}$$

and since $e_2(n)$ is an integer, we must have

$$e_2(n) \leq \left[\frac{n}{p-1}\right].$$

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From these two inequalities, we conclude that

$$v_2\left(\prod_{k=0}^{n-1}(2^n-2^k)\right) \geq e_2(n)$$

and now the problem is solved.

18.2 Euler's Theorem

Problem 7.2.5. *Prove that, for every positive integer n, there exists a polynomial with integer coefficients whose values at $1, 2, \ldots, n$ are different powers of 2.*

(1999 Hungarian Mathematical Olympiad)

Solution. It suffices to prove the claim when $n \geq 4$, because the same polynomials which works for $n \geq 4$ works for $n \leq 3$. For each $i = 1, 2, \ldots, n$, consider the product $s_i = \prod_{\substack{j=1 \\ j\neq i}}^{n}(i-j)$. Because $n \geq 4$, one of the terms $i - j$ equals 2 and s_i is even. Thus, we can write $s_i = 2^{q_i}m_i$ for positive integers q_i, m_i with m_i odd. Let L be the least common multiple of all the q_i, and let $r_i = L/q_i$. For each i, there are infinitely many powers of 2 which are congruent to 1 modulo $|m_i^{r_i}|$. (Specifically, by Euler's Theorem, $2^{\phi(|m_i^{r_i}|)j} \equiv 1 \pmod{|m_i^{r_i}|}$ for all $j \geq 0$. Thus there are infinitely many integers c_i such that $c_i m_i^{r_i} + 1$ is a power of 2. Choose such c_i, and define

$$P(x) = \sum_{i=1}^{n} c_i \left(\prod_{\substack{j=1 \\ j\neq i}}^{n}(x-j)\right)^{r_i} + 2^L.$$

For each k, $1 \leq k \leq n$, in the sum each term $\left(\prod_{\substack{j=1 \\ j\neq i}}^{n}(x-j)\right)^{r_i}$ vanishes for all $i \neq k$. Then

$$P(k) = c_k\left(\prod_{\substack{j=1 \\ j\neq k}}^{n}(k-j)\right)^{r_i} + 2^L = 2^L(c_k m_k^{r_k} + 1),$$

a power of 2. Moreover, by choosing the c_i appropriately, we can guarantee that these values are all distinct, as needed.

Problem 7.2.6. *Let $a > 1$ be an odd positive integer. Find the least positive integer n such that 2^{2000} is a divisor of $a^n - 1$.*

(2000 Romanian IMO Team Selection Test)

Solution. Since a is odd, $(a, 2^k) = 1$, for any $k \geq 0$. Hence, by Euler's Theorem, $a^{\varphi(2^k)} \equiv 1 \pmod{2^k}$. Since $\varphi(2^k) = 2^{k-1}$ and we are looking for the least exponent n such that $a^n \equiv 1 \pmod{2^{2000}}$ it follows that n is a divisor of $2^{1999} = \varphi(2^{2000})$.

If $a \equiv 1 \pmod{2^{2000}}$ it follows that $n = 1$. We shall omit this case.

Consider the decomposition:

$$a^{2^m} - 1 = (a-1)(a+1)(a^2+1)(a^{2^2}+1)\dots(a^{2^{m-1}}+1).$$

Assume $a \equiv 1 \pmod{2^s}$ and $a \not\equiv 1 \pmod{2^{s+1}}$, where $2 \leq s \leq 1999$. That is, $a = 2^s b + 1$, where b is an odd number. Equivalently, a has the binary representation

$$a = 1\dots1\underbrace{00\dots1}_{s\ digits}.$$

It is easy to show that for any integer x, $x^2 + 1$ is not divisible by 4. Then, by the above decomposition $a^{2^m} - 1$ is divisible by 2^{s+m} and it is not divisible by 2^{s+m+1}. Hence, the required number is 2^{2000-s}.

Assume that $a \equiv -1 \pmod{2^s}$ and $a \not\equiv -1 \pmod{2^{s+1}}$, where $s \geq 2$. Equivalently, a has the binary representation

$$a = 1\dots0\underbrace{11\dots1}_{s\ digits}.$$

Like before, $a - 1$ is divisible by 2 and nondivisible by 2^2 and $a^{2^k} + 1$ is divisible by 2 and nondivisible by 2^2, $\forall\ k \geq 1$. From the above decomposition $a^{2^m} - 1$ is divisible by 2^{s+m} and nondivisible by 2^{s+m+1}. Hence, in this case, the required exponent is $n = 2^{1999-s}$ when $s < 1999$ and $n = 2$, when $s \geq 1999$.

Problem 7.2.7. *Let $n = p_1^{r_1}\dots p_k^{r_k}$ be the prime factorization of the positive integer n and let $r \geq 2$ be an integer. Prove that the following are equivalent:*

(a) The equation $x^r \equiv a \pmod{n}$ has a solution for every a.

(b) $r_1 = r_2 = \dots = r_k = 1$ and $(p_i - 1, r) = 1$ for every $i \in \{1, 2, \dots, k\}$.

(1995 UNESCO Mathematical Contest)

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Solution. If (b) holds, then $\varphi(n) = (p_1 - 1)\dots(p_k - 1)$ is coprime to r, thus there exists s with $rs \equiv 1 \pmod{\phi(n)}$, and the unique solution of $x^r \equiv a \pmod{n}$ is $a = x^s$. Conversely, suppose $x^r \equiv a \pmod{n}$ has a solution for every a; then $x^r \equiv a \pmod{p_i^{r_i}}$ also has a solution for every a. However, if $r_1 > a$ and a is a number divisible by p but not by p^2, then x^r cannot be congruent to a, since it is not divisible by p unless x is divisible by p, in which case it is already divisible by p^2. Hence $r_1 = 1$.

Let $d = (p_i - 1, r)$ and put $m = (p_i - 1)/d$. If $x^r \equiv a \pmod{p_i}$ and $a \not\equiv 0$, then

$$a^m \equiv x^{rm} = x^{p_i - 1} \equiv 1 \pmod{p_i}.$$

However, if a is a primitive root of p_i, then this only occurs for $m \equiv 0 \pmod{p_i - 1}$, which implies $d = 1$. Hence $r_i = 1$ and $(r, p_i - 1) = 1$, as desired.

18.3 The order of an element

Problem 7.3.6. *Find all ordered triples of primes (p, q, r) such that*

$$p|q^r + 1, q|r^p + 1, r|p^q + 1.$$

(2003 USA IMO Team Selection Test)

Solution. It is quite clear that p, q, r are distinct. Indeed, if for example $p = q$, then the relation $p|q^r + 1$ is impossible. We will prove that we cannot have $p, q, r > 2$. Suppose this is the case. The first condition $p|q^r + 1$ implies $p|q^{2r} - 1$ and so $o_p(q)|2r$. If $o_p(q)$ is odd, it follows that $p|q^r - 1$, which combined with $p|q^r + 1$ yields $p = 2$, which is impossible. Thus, $o_p(q)$ is either 2 or $2r$. Could we have $o_p(q) = 2r$? No, since this would imply that $2r|p - 1$ and so $0 \equiv p^q + 1 \pmod{r} \equiv 2 \pmod{r}$, that is $r = 2$, false. Therefore, the only possibility is $o_p(q) = 2$ and so $p|q^2 - 1$. We cannot have $p|q - 1$, because $p|q^r + 1$ and $p \neq 2$. Thus, $p|q + 1$ and in fact $p|\frac{q+1}{2}$. In the same way, we find that $q|\frac{r+1}{2}$ and $r|\frac{p+1}{2}$. This is clearly impossible, just by looking at the largest among p, q, r. So, our assumption was wrong and indeed one of the three primes must equal 2. Suppose without loss of generality that $p = 2$. Then q is odd, $q|r^2 + 1$ and $r|2^q + 1$. Similarly, $o_r(2)|2q$. If $q|o_r(2)$, then $q|r - 1$ and so $q|r^2 + 1 - (r^2 - 1) = 2$, which contradicts the already established result that q is odd. Thus, $o_r(2)|2$ and

$r|3$. As a matter of fact, this implies that $r = 3$ and $q = 5$, yielding the triple (2,5,3). It is immediate to verify that this triple satisfies all conditions of the problem. Moreover, all solutions are given by cyclic permutations of the components of this triple.

Problem 7.3.7. *Find all primes p, q such that $pq|2^p + 2^q$.*

Solution. Note that $(p, q) = (2, 2), (2, 3), (3, 2)$ satisfy this property and let us show that there are no other such pairs. Assume, by contradiction, that $p \neq 2$ and $q \neq 2$. Write $p - 1 = 2^l n$, $q - 1 = 2^k m$, where l, k are odd positive integers. Because $pq|2^p + 2^q$, using Fermat's Little Theorem, we obtain $0 \equiv 2^p + 2^q \equiv 2^p + 2 \pmod{q}$. It follows $2^{p-1} \equiv -1 \pmod{q}$. If we denote $x = 2^n$, then we have $x^{2^l} \equiv -1 \pmod{q}$, hence $o(x) = 2^{l+1}$ (since $x^{2^{l+1}} \equiv 1 \pmod{q}$ and $x^{2^l} \not\equiv 1 \pmod{q}$). It follows $2^{l+1} = o_q(x)|\varphi(q) = q - 1 = 2^k m$, i.e. $l + 1 \leq k$.

In similar way we can prove that $k + 1 \leq l$ and we get $l \leq k - 1 \leq l - 2$, a contradiction. Therefore, it is necessary to have $p = 2$ or $q = 2$. If, for example, $q = 2$, then $p|2^p + 2^q = 2^p + 2^2$, $0 \equiv 2^p + 2^2 \equiv 2 + 2^2 = 6 \pmod{p}$, and we get $p \in \{2, 3\}$.

Problem 7.3.8. *Prove that for any positive integer n, $3^n - 2^n$ is not divisible by n.*

Solution. Assume by contradiction that $n|3^n - 2^n$ for some positive integer n. Let us denote by p the smallest prime divisor of n. Since $n|3^n - 2^n$, it follows that $p \geq 5$. Consider a positive integer a such that $2a \equiv 1 \pmod{p}$. From $3^n \equiv 2^n \pmod{p}$ we obtain $(3a)^n \equiv 1 \pmod{p}$. Let us denote $d = o_p(3a)$. It follows $d|p - 1$ and $d|n$. But $d < p$ and $d|n$ implies $d = 1$, because the minimality of p. We get $3a \equiv 1 \pmod{p}$ and $2a \equiv 1 \pmod{p}$, i.e. $a \equiv 0 \pmod{p}$, contradiction with $2a \equiv 1 \pmod{p}$.

Problem 7.3.9. *Find all positive integers m, n such that $n|1 + m^{3^n} + m^{2 \cdot 3^n}$.*

(Bulgarian IMO Team Selection Test)

Solution. From $n|1+m^{3^n}+m^{2 \cdot 3^n}$ it follows $n|m^{3^{n+1}}-1$, hence $d = o_n(m)$ divides 3^{n+1}, i.e. $d = 3^k$ for some positive integer k. If $k \leq n$, then $d|3^n$ implies $n|m^{3^n} - 1$. Combining with $n|1 + m^{3^n} + m^{2 \cdot 3^n}$ it follows $n = 3$. If $k \geq n + 1$, then $d = 3^{n+1}$ and $d|\varphi(n)$ implies $d < n$, impossible since $3^{n+1} > n$. Therefore $n = 3$ and, consequently $m \equiv 1 \pmod{3}$.

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Problem 7.3.10. *Let $a, n > 2$ be positive integers such that $n|a^{n-1} - 1$ and n does not divide any of the numbers $a^x - 1$, where $x < n - 1$ and $x|n - 1$. Prove that n is a prime number.*

Solution. Denote $d = o_n(a)$. Since $n|a^{n-1} - 1$ it follows $d|n - 1$. If $d < n-1$, then we contradict the hypotheses that n does not divide $a^d - 1$. Hence $d \geq n - 1$ and consequently $d = n - 1$.

On the other hand, we have $d|\varphi(n)$, hence $n-1|\varphi(n)$. Taking into account that $\varphi(n) \leq n - 1$, we find $\varphi(n) = n - 1$ and it follows that n must be a prime number.

Problem 7.3.11. *Find all prime numbers p, q for which the congruence*

$$\alpha^{3pq} \equiv \alpha \pmod{3pq}$$

holds for all integers α.

(1996 Romanian Mathematical Olympiad)

Solution. Without loss of generality assume $p \leq q$; the unique solution will be (11,17), for which one may check the congruence using the Chinese Remainder Theorem. We first have $2^{3pq} \equiv 2 \pmod 3$, which means p and q are odd. In addition, if α is a primitive root mod p, then $\alpha^{3pq-1} \equiv 1 \pmod p$ implies that $p - 1$ divides $3pq - 1$ as well as $3pq - 1 - 3q(p - 1) = 3q - 1$, and conversely that $q - 1$ divides $3p - 1$. If $p = q$, we now deduce $p = q = 3$, but $4^{27} \equiv 1 \pmod{27}$, so this fails. Hence $p < q$.

Since p and q are odd primes, $q \geq p + 2$, so $(3p - 1)/(q - 1) < 3$. Since this quantity is an integer, and it is clearly greater than 1, it must be 2. That is, $2q = 3p + 1$. On the other hand, $p - 1$ divides $3q - 1 = (9p + 1)/2$ as well as $(9p + 1) - (9p - 9) = 10$. Hence $p = 11$, $q = 17$.

Remark. An integer n such that $a^n \equiv a \pmod n$ for all integers a is called a *Carmichael number*. Very recently W.R. Alford, A. Granville, C. Pomerance [Annals Math., 1994, 703-722] proved that there are infinitely many Carmichael numbers. Using the ideas outlined in this solution of the above problem, one can show that n is a Carmichael number if and only if it is of the form $p_1p_2 \dots p_k$, with p_i different prime numbers such that $p_i - 1|n - 1$ for all $i = 1, 2, \dots, k$.

18.4 Wilson's Theorem

Problem 7.4.5. *Let p be an odd prime. Prove that*

$$1^2 \cdot 3^2 \dots (p - 2)^2 \equiv (-1)^{\frac{p+1}{2}} \pmod p$$

and

$$2^2 \cdot 4^2 \ldots (p-1)^2 \equiv (-1)^{\frac{p+1}{2}} \pmod{p}.$$

Solution. Using Wilson's Theorem, we have $(p-1)! \equiv -1 \pmod{p}$, hence

$$(1 \cdot 3 \ldots (p-2))(2 \cdot 4 \ldots (p-1)) \equiv -1 \pmod{p}.$$

On the other hand,

$$1 \equiv -(p-1) \pmod{p},\ 3 \equiv -(p-3) \pmod{p}, \ldots,$$

$$p-2 \equiv -(p-(p-2)) \pmod{p},$$

therefore

$$1 \cdot 3 \ldots (p-2) \equiv (-1)^{\frac{p-1}{2}} (2 \cdot 4 \ldots (p-1)) \pmod{p}$$

and the conclusion follows.

Problem 7.4.6. *Show that there do not exist nonnegative integers k and m such that $k! + 48 = 48(k+1)^m$.*

(1996 Austrian-Polish Mathematics Competition)

Solution. Suppose such k, m exist. We must have $48|k!$, so $k \geq 7 = 6$; one checks that $k = 6$ does not yield a solution, so $k \geq 7$. In that case $k!$ is divisible by 32 and by 9, so that $(k! + 48)/48$ is relatively prime to 6, as then is $k + 1$.

If $k+1$ is not prime, it has a prime divisor greater than 3, but this prime divides $k!$ and not $k! + 48$. Hence $k + 1$ is prime, and by Wilson's Theorem $k! + 1$ is a multiple of $k + 1$. Since $k! + 48$ is as well, we find $k + 1 = 47$, and we need only check that $46!/48 + 1$ is not a power of 47. We check that $46!/48 + 1 \equiv 29 \pmod{53}$ (by cancelling as many terms as possible in 46! before multiplying), but that 47 has order 12 modulo 53 and that none of its powers is congruent to 29 modulo 53.

Problem 7.4.7. *For each positive integer n, find the greatest common divisor of $n! + 1$ and $(n+1)!$.*

(1996 Irish Mathematical Olympiad)

Solution. If $n + 1$ is composite, then each prime divisor of $(n + 1)!$ is a prime less than n, which also divides $n!$ and so does not divide $n! + 1$. Hence $f(n) = 1$. If $n + 1$ is prime, the same argument shows that $f(n)$ is

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a power of $n+1$, and in fact $n+1|n!+1$ by Wilson's Theorem. However, $(n+1)^2$ does not divide $(n+1)!$, and thus $f(n) = n+1$.

Problem 7.4.8. *Let $p \geq 3$ be a prime and let σ be a permutation of $\{1, 2, \ldots, p-1\}$. Prove that there are $i \neq j$ such that $p|i\sigma(i) - j\sigma(j)$.*

(1986 Romanian IMO Team Selection Test)

Solution. Assume by contradiction that p does not divide $i\sigma(i) - j\sigma(j)$ for any $i, j = 1, 2, \ldots, p-1, i \neq j$. Then, the integers $i\sigma(i)$, $i = 1, 2, \ldots, p-1$, are all not divisible by p and give distinct residues modulo p. We have

$$\prod_{i=1}^{p-1}(i\sigma(i)) \equiv \prod_{i=1}^{p-1} i = (p-1)! \equiv -1 \pmod{p}.$$

On the other hand, $\prod_{i=1}^{p-1}(i\sigma(i)) = \prod_{i=1}^{p-1}((p-1)!)^2 \equiv 1 \pmod{p}$, a contradiction.

19

Diophantine Equations

19.1 Linear Diophantine equations

Problem 8.1.4. *Solve in integers the equation*

$$(x^2+1)(y^2+1)+2(x-y)(1-xy)=4(1+xy).$$

Solution. The equation is equivalent to

$$x^2y^2-2xy+1+x^2+y^2-2xy+2(x-y)(1-xy)=4,$$

or

$$(xy-1)^2+(x-y)^2+2(x-y)(1-xy)=4.$$

Hence $(1-xy+x-y)^2=4$ and, consequently, $|(1+x)(1-y)|=2$.

We have two cases:

I. $(1+x)(1-y)=2$. Then

a) $1+x=2$, $1-y=1$, so $x=1,\ y=0$.

b) $1+x=-2$, $1-y=-1$, so $x=-3,\ y=2$.

c) $1+x=1$, $1-y=2$, so $x=0,\ y=-1$.

d) $1+x=-1$, $1-y=-2$, so $x=-2,\ y=3$.

II. $(1+x)(1-y)=-2$. Then

a) $1+x=2$, $1-y=-1$, so $x=1,\ y=2$.

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b) $1+x=-2$, $1-y=1$, so $x=-3,\ y=0$.
c) $1+x=1$, $1-y=-2$, so $x=0,\ y=3$.
d) $1+x=-1$, $1-y=2$, so $x=-2,\ y=-1$.

Problem 8.1.5. *Determine the side lengths of a right triangle if they are integers and the product of the legs' lengths equals three times the perimeter.*

(1999 Romanian Mathematical Olympiad)

Solution. Let a, b, c be the lengths of triangle's sides. We have

$$a^2 = b^2 + c^2,$$

and

$$bc = 3(a+b+c).$$

Let $P = a+b+c$. Then $bc = 3P$ and

$$b^2 + c^2 = (b+c)^2 - 2bc = (P-a)^2 - 6P = P^2 + a^2 - 2aP - 6P.$$

It follows that

$$a^2 = P^2 + a^2 - 2aP - 6P,$$

so

$$P = 2a + 6,$$

that is,

$$a = b + c - 6.$$

We have then:

$$b^2 + c^2 = b^2 + c^2 + 2bc - 12b - 12c + 36$$

if and only if

$$bc - 6b - 6c + 18 = 0,$$

that is

$$(b-6)(c-6) = 18.$$

Analyzing the ways in which 18 can be written as a product of integers, we find the following solutions:

$$(a,b,c) \in \{(25,7,24), (25,24,7), (17,8,15), (17,15,8), (15,9,12), (15,12,9)\}.$$

Problem 8.1.6. *Let a, b and c be positive integers, each two of them being relatively prime. Show that $2abc - ab - bc - ca$ is the largest integer which cannot be expressed in the form $xbc + yca + zab$ where x, y and z are nonnegative integers.*

(24th IMO)

Solution. We will solve the problem in two steps.

First step. The number $2abc - ab - bc - ca$ cannot be expressed in the required form. Assume contrary, that:

$$2abc - ab - bc - ca = xbc + yca + zab,$$

where $x, y, z \geq 0$. Then, one obtains the combination

$$2abc = bc(x+1) + ca(y+1) + ab(z+1)$$

where $x+1 > 0$, $y+1 > 0$, $z+1 > 0$. It leads to the divisibility $a|bc(x+1)$.

Since a is relatively prime to b and c, a divides $x+1$ and then $a \leq x+1$. Using similar arguments, $b \leq y+1$ and $c \leq z+1$. Thus, $2abc = bc(x+1) + ca(y+1) + ab(z+1) \geq 3abc$. This is a contradiction.

Second step. Any number N, $N > 2abc - ab - bc - ca$, can be expressed in the form $N = xbc + yca + zab$.

First, observe that $2abc - ab - bc - ca + 1 > 0$. It follows by the following argument:

$$\frac{1}{abc}(2abc - ab - bc - ca + 1) = 2 - \frac{1}{a} - \frac{1}{b} - \frac{1}{c} + \frac{1}{abc} > 2 - \frac{1}{1} - \frac{1}{2} - \frac{1}{3} + \frac{1}{abc} > 0.$$

Going further, we have two situations. When $N \equiv 0 \pmod{abc}$, since $N = abcq$, we may consider the combination $N = (ab)cq + bc \cdot 0 + ca \cdot 0$. It is of required form, under notations $x = y = 0$ and $z = cq$.

Suppose that $N \not\equiv 0 \pmod{abc}$. Because $gcd(bc, a) = 1$, the congruence

$$xbc \equiv N \pmod{a}$$

has a solution x_0, $0 < x_0 < a$. Similarly, the congruences

$$yca \equiv N \pmod{b}$$

$$zab \equiv N \pmod{c}$$

have solutions y_0, z_0, respectively, $0 < y_0 < b$, $0 < x_0 < c$.

Let $A = x_0bc + y_0ca + z_0ab$. Then we have:

$$A \equiv x_0bc \equiv N \pmod{a}, \quad A \equiv N \pmod{b}, \quad A \equiv N \pmod{c}.$$

Since a, b, c are pairwise respectively prime, we obtain $A \equiv N \pmod{abc}$.

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The number A is a combination of required form. Since $x_0 \leq a-1$, $y_0 \leq b-1$, and $z_0 \leq c-1$ it follows that $A \leq 3abc-bc-ca-ab$. Using $A \equiv N \pmod{abc}$, we may write $N = A + kabc$. In this sum $k \geq 0$, since $N > 2abc-bc-ca-ab$. Therefore we found for N, as it was required, the expression

$$N = (x_0 + ka)bc + y_0ca + z_0ab$$

where $x_0 + ka \geq 0$, $y_0 \geq 0$, $z_0 \geq 0$.

Remark. One can prove that if $a_1, a_2, \ldots, a_k \in \mathbb{Z}$ are positive integers such that $gcd(a_1, \ldots, a_k) = 1$, then any sufficiently large n is a linear combination with nonnegative coefficients of $a_1, \ldots, a_k$. It is not known the smallest such n for $k \geq 4$. This is the famous problem of Frobenius.

19.2 Quadratic Diophantine equations

19.2.1 Pythagorean equations

Problem 8.2.2. *Find all Pythagorean triangles whose areas are numerically equal to their perimeters.*

Solution. From (3), the side lengths of such a triangle are

$$k(m^2-n^2), \quad 2kmn, \quad k(m^2+n^2).$$

The condition in the problem is equivalent to

$$k^2mn(m^2-n^2) = 2km(m+n),$$

which reduces to

$$kn(m-n) = 2.$$

A simple case analysis shows that the only possible triples (k, m, n) are (2,2,1), (1,3,2), (1,3,1), yielding the Pythagorean triangles $6-8-10$ and $5-12-13$.

Problem 8.2.3. *Prove that for every positive integer n there is a positive integer k such that k appears in exactly n nontrivial Pythagorean triples.*

(American Mathematical Monthly)

Solution. We will prove by induction that 2^{n+1} appears in exactly n Pythagorean triples. The base case $n = 1$ holds for $(3, 2^2, 5)$ is the only such triple. Assume that (x_k, y_k, z_k), where $x_k = u_k^2 - v_k^2$, $y_k = 2u_kv_k$, $z_k = u_k^2 + v_k^2$, $k = 1, \ldots, n$ are the n triples containing 2^{n+1}. Then $(2x_k, 2y_k, 2z_k)$,

$k = 1, \ldots, n$ are n non-primitive Pythagorean triples containing 2^{n+2} and $(2^{2n+2} - 1, 2^{n+2}, 2^{2n+2} + 1)$ is the only such primitive triple.

No other triple with this property exists. Indeed, if $(u^2 - v^2, 2uv, u^2 + v^2)$ were a triple containing 2^{n+2}, then we would have the following cases:

i) $u^2 + v^2 = 2^{n+2}$. Simplifying by the greatest possible power of 2 we get $a^2 + b^2 = 2^k$ where a and b ate not both even. Then the left-hand side is congruent to 1 or 2 (mod 4), while the right-hand side is 0 (mod 4), a contradiction.

ii) $2uv = 2^{n+2}$. We simplify again by the greatest power of 2 and obtain $ab = 2^s$, where $a > b$ are not both even and $s \geq 1$. It follows that $a = 2^s$ and $b = 1$, yielding the triple generated by $(2^{2s} - 1, 2^{s+1}, 2^{2s} + 1)$ multiplied by a power of 2, that is clearly among the non-primitive triples $(2x_k, 2y_k, 2z_k)$.

iii) $u^2 - v^2 = 2^{n+2}$. Simplifying again by the greatest power of 2 we arrive at $a^2 - b^2 = 2^t$, where a and b are not both even and $t \geq 3$. If one of a and b are even, then the left-hand side is odd, while the right-hand side is even, a contradiction. If a and b are both odd, then $a - b = 2$ and $a + b = 2^{t-1}$, yielding $a - 2^{t-2}$ and $b = 2^{t-2} - 1$. Again, we get a triple generated by $(2^t, 2(2^{2t-4} - 1), 2(2^{2t-4} + 1))$ multiplied by a power of 2, which is clearly already among a non-primitive triple $(2x_k, 2y_k, 2z_k)$.

19.2.2 Pell's equation

Problem 8.2.6. *Let p be a prime number congruent to 3 modulo 4. Consider the equation*

$$(p + 2)x^2 - (p + 1)y^2 + px + (p + 2)y = 1.$$

Prove that this equation has infinitely many solutions in positive integers, and show that if $(x, y) = (x_0, y_0)$ is a solution of the equation in positive integers, then $p|x_0$.

(2001 Bulgarian Mathematical Olympiad)

Solution. We show first that $p|x$. Substituting $y = z + 1$ and rewriting, we obtain

$$x^2 = (z - x)((p + 1)(z + x) + p).$$

Let $q = gcd(z-x, (p+1)(z+x)+p)$. Then $q|x$, therefore $q|z$, and therefore $q|p$. On the other hand, $q \neq 1$, because otherwise both factors on the right hand side must be perfect squares, yet $(p+1)(z+x)+p \equiv 3 \pmod 4$. Thus $q = p$ and $p|x$ as desired.

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Now, write $x = px_1$ and $z = pz_1$ to obtain

$$x_1^2 = (z_1 - x_1)((p+1)(z_1 + x_1) + 1).$$

By what we showed above, the two terms on the right are coprime and must be perfect squares. Therefore, for some a, b we have

$$z_1 - x_1 = a^2, \quad (p+1)(z_1 + x_1) + 1 = b^2, \quad x_1 = ab.$$

The above equality implies

$$b^2 = (p+1)((a^2 + 2ab) + 1),$$

i.e.

$$(p+2)b^2 - (p+1)(a+b)^2 = 1.$$

Vice versa, given a and b satisfying the last equation, there exists a unique pair (x_1, y_1) satisfying the equation above, and hence a unique pair (x, y) satisfying the original equation.

Thus, we reduced the original equation to a "Pell-type" equation. To get some solutions, look at the odd powers of $\sqrt{p+2}+\sqrt{p+1}$. It follows easily that

$$(\sqrt{p+2} + \sqrt{p+1})^{2k+1} = m_k\sqrt{p+2} + n_k\sqrt{p+1}$$

for some positive integers m_k, n_k. Then

$$(\sqrt{p+2} - \sqrt{p+1})^{2k+1} = m_k\sqrt{p+2} - n_k\sqrt{p+1},$$

and, multiplying the left and right sides gives

$$(p+2)m_k^2 - (p+1)n_k^2 = 1.$$

Clearly, $n_k > m_k$, so setting $b_k = m_k$, $a_k = n_k - m_k$ gives a solution for (a, b). Finally, it is easy to see that the sequences $\{m_k\}$, $\{n_k\}$ are strictly increasing, so we obtain infinitely many solutions this way.

Problem 8.2.7. *Determine all integers a for which the equation*

$$x^2 + axy + y^2 = 1$$

has infinitely many distinct integer solutions (x, y).

(1995 Irish Mathematical Olympiad)

Solution. The equation has infinitely many solutions if and only if $a^2 \geq 4$. Rewrite the given equation in the form

$$(2x + ay)^2 - (a^2 - 4)y^2 = 4.$$

If $a^2 < 4$, the real solutions to this equation form an ellipse and so only finitely integer solutions occur. If $a = \pm 2$, there are infinitely many solutions, since the left side factors as $(x \pm y)^2$. If $a^2 > 4$, then $a^2 - 4$ is not a perfect square and so the Pell's equation $u^2 - (a^2 - 4)v^2 = 1$ has infinitely many solutions. But setting $x = u - av$, $y = 2v$ gives infinitely many solutions of the given equation.

Problem 8.2.8. *Prove that the equation*

$$x^3 + y^3 + z^3 + t^3 = 1999$$

has infinitely many integral solutions.

(1999 Bulgarian Mathematical Olympiad)

Solution. Observe that $(m-n)^3+(m+n)^3 = 2m^3+6mn^2$. Now suppose we want a general solution of the form

$$(x, y, z, t) = \left(a - b, a + b, \frac{c}{2} - \frac{d}{2}, \frac{c}{2} + \frac{d}{2}\right)$$

for integers a, b and odd integers c, d. One simple solution to the given equation is $(x, y, z, t) = (10, 10, -1, 0)$, so we try setting $a = 10$ and $c = -1$. Then

$$(x, y, z, t) = \left(10 - b, 10 + b, -\frac{1}{2} - \frac{d}{2}, -\frac{1}{2} + \frac{d}{2}\right)$$

is a solution exactly when

$$(2000 + 60b^2) - \frac{1 + 3d^2}{4} = 1999, \qquad \text{i.e. } d^2 - 80b^2 = 1.$$

The second equation is a Pell's equation with solution $(d_1, b_1) = (9, 1)$. We can generate infinitely many more solutions by setting

$$(d_{n+1}, b_{n+1}) = (9d_n + 80b_n, 9b_n + d_n) \text{ for } n = 1, 2, 3, \ldots$$

This can be proven by induction, and it follows from a general recursion

$$(p_{n+1}, q_{n+1}) = (p_1p_n + q_1q_nD, p_1q_n + q_1p_n)$$

for generating solutions to $p^2 - Dq^2 = 1$ given a nontrivial solution (p_1, q_1).

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A quick check also shows that each d_n is odd. Thus because there are infinitely many solutions (b_n, d_n) to the Pell's equation (and with each d_n odd), there are infinitely many integral solutions

$$(x_n, y_n, z_n, t_n) = \left(10 - b_n, 10 + b_n, -\frac{1}{2} - \frac{d_n}{2}, -\frac{1}{2} + \frac{d_n}{2}\right)$$

to the original equation.

19.2.3 Other quadratic equations

Problem 8.2.11. *Prove that the equation*

$$x^2 + y^2 + z^2 + 3(x + y + z) + 5 = 0$$

has no solutions in rational numbers.

(1997 Bulgarian Mathematical Olympiad)

Solution. Let $u = 2x + 3$, $v = 2y + 3$, $w = 2z + 3$. Then the given equation is equivalent to

$$u^2 + v^2 + w^2 = 7.$$

It is equivalent to ask that the equation

$$x^2 + y^2 + z^2 = 7w^2$$

has no nonzero solutions in integers; assume on the contrary that (x, y, z, w) is a nonzero solution with $|w| + |x| + |y| + |z|$ minimal. Modulo 4, we have $x^2 + y^2 + z^2 \equiv 7w^2$, but every perfect square is congruent to 0 or 1 modulo 4. Thus we must have x, y, z, w even, and $(x/2, y/2, z/2, w/2)$ is a smaller solution, contradiction.

Remark. Try to prove the following theorem of Davenport and Cassels: *for $n \in \mathbb{Z}$, the equation $x^2 + y^2 + z^2 = n$ has rational solutions if and only if it has integer solutions.* There is a beautiful elementary geometric proof. Try to find it!

Problem 8.2.12. *Find all integers x, y, z such that $5x^2 - 14y^2 = 11z^2$.*

(2001 Hungarian Mathematical Olympiad)

Solution. The only solution is (0,0,0).

Assume, for sake of contradiction, that there is a triple of integers $(x, y, z) \neq (0, 0, 0)$ satisfying the given equation, and let $(x, y, z) = (x_0, y_0, z_0)$ be a nonzero solution that minimizes $|z + y + z| > 0$.

Because $5x_0^2 - 14y_0^2 = 11z_0^2$, we have

$$-2x_0^2 \equiv 4z_0^2 \pmod 7,$$

or $x_0^2 \equiv -2z_0^2 \equiv 5z_0^2 \pmod 7$. Therefore, we have $z_0 \equiv 0 \pmod 7$, because otherwise we have

$$5 \equiv (x_0 z_0^{-1})^2 \pmod 7,$$

which is impossible because 5 is not a quadratic residue modulo 7. (The squares modulo 7 are 0, 1, 2 and 4.)

It follows that x_0 and z_0 are divisible by 7, so that $14y^2 = 5x^2 - 11z^2$ is divisible by 49. Therefore, $7|y_0$. Then $\left(\frac{x_0}{7}, \frac{y_0}{7}, \frac{z_0}{7}\right)$ is also a solution, but $\left|\frac{x_0}{7} + \frac{y_0}{7} + \frac{z_0}{7}\right| < |x_0 + y_0 + z_0|$, contradicting the minimality of (x_0, y_0, z_0).

Therefore, our original assumption was false, and the only integer solution is (0,0,0).

Problem 8.2.13. *Let n be a nonnegative integer. Find the nonnegative integers a, b, c, d such that*

$$a^2 + b^2 + c^2 + d^2 = 7 \cdot 4^n.$$

(2001 Romanian JBMO Team Selection Test)

Solution. For $n = 0$, we have $2^2 + 1^2 + 1^2 + 1^2 = 7$, hence $(a, b, c, d) = (2, 1, 1, 1)$ and all permutations. If $n \geq 1$, then $a^2+b^2+c^2+d^2 \equiv 0 \pmod 4$, hence the numbers have the same parity. We analyze two cases.

a) The numbers a, b, c, d are odd. We write $a = 2a' + 1$, etc. We obtain:

$$4a'(a' + 1) + 4b'(b' + 1) + 4c'(c' + 1) + 4d'(d' + 1) = 4(7 \cdot 4^{n-1} - 1).$$

The left hand side of the equality is divisible by 8, hence $7 \cdot 4^{n-1} - 1$ must be even. This happens only for $n = 1$. We obtain $a^2 + b^2 + c^2 + d^2 = 28$, with the solutions (3,3,3,1) and (1,1,1,5).

b) The number a, b, c, d are even. Write $a = 2a'$, etc. We obtain

$$a'^2 + b'^2 + c'^2 + d'^2 = 7 \cdot 4^{n-1},$$

so we proceed recursively.

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Finally, we obtain the solutions $(2^{n+1}, 2^n, 2^n, 2^n)$, $(3\cdot 2^n, 3\cdot 2^n, 3\cdot 2^n, 2^n)$, $(2^n, 2^n, 2^n, 5\cdot 2^n)$, $n \in \mathbb{N}$, and the respective permutations.

Problem 8.2.14. *Prove that the equation*

$$x^2 + y^2 + z^2 + t^2 = 2^{2004},$$

where $0 \le x \le y \le x \le t$, *has exactly two solutions in the set of integers.*

(2004 Romanian Mathematical Olympiad)

Solution. The solutions are $(0, 0, 0, 2^{1002})$ and $(2^{1001}, 2^{1001}, 2^{1001}, 2^{1001})$.

In order to prove the statement, let (x, y, z, t) be a solution. Observe that for odd a we have $a = 4n \pm 1$, and a^2 gives the remainder 1 when divided by 8. Thus the equation has no solution with an odd component.

We thus must have $x = 2x_1$, $y = 2y_1$, $z = 2z_1$, $t = 2t_1$, where $0 \le x_1 \le y_1 \le z_1 \le t_1$ are integers and $x_1^2+y_1^2+z_1^2+t_1^2 = 2^{2002}$. By the same argument $x_1 = 2x_2$, $y_1 = 2y_2$, $z_1 = 2z_2$, $t_1 = 2t_2$, where $0 \le x_2 \le y_2 \le z_2 \le t_2$ are integers and $x_2^2 + y_2^2 + z_2^2 + t_2^2 = 2^{2000}$.

Recursively, $x = 2^{2001}a$, $y = 2^{2001}b$, $z = 2^{2001}c$, $t = 2^{2002}d$, where $0 \le a \le b \le c \le d$ are integers and $a^2 + b^2 + c^2 + d^2 = 4$. This relation simply implies the conclusion.

Problem 8.2.15. *Let* n *be a positive integer. Prove that the equation*

$$x + y + \frac{1}{x} + \frac{1}{y} = 3n$$

does not have solutions in positive rational numbers.

Solution. Suppose $x = \dfrac{a}{b}$, $y = \dfrac{c}{d}$ satisfies the given equation, where $(a, b) = (c, d) = 1$. Clearing denominators,

$$(a^2 + b^2)cd + (c^2 + d^2)ab = 3nabcd.$$

Thus, $ab|(a^2 + b^2)cd$ and $cd|(c^2 + d^2)ab$. Now $(a, b) = 1$ implies $(a, a^2 + b^2) = (a, b^2) = 1$, so $ab|cd$; likewise, $cd|ab$, and together these give $ab = cd$. Thus,

$$a^2 + b^2 + c^2 + d^2 = 3nab.$$

Now each square on the left is congruent to either 0 or 1 modulo 3. Hence, either all terms are divisible by 3 or exactly one is. The first case is impossible by the assumption $(a, b) = (c, d) = 1$, and the second is impossible because $ab = cd$.

19.3 Nonstandard Diophantine equations

19.3.1 Cubic equations

Problem 8.3.5. *Find all triples (x, y, z) of natural numbers such that y is a prime number, y and 3 do not divide z, and $x^3 - y^3 = z^2$.*

(1999 Bulgarian Mathematical Olympiad)

Solution. We rewrite the equation in the form

$$(x - y)(x^2 + xy + y^2) = z^2.$$

Any common divisor of $x - y$ and $x^2 + xy + y^2$ also divides both z^2 and $(x^2 + xy + y^2) - (x + 2y)(x - y) = 3y^2$. Because z^2 and $3y^2$ are relatively prime by assumption, $x - y$ and $x^2 + xy + y^2$ must be relatively prime as well. Therefore, both $x - y$ and $x^2 + xy + y^2$ are perfect squares.

Writing $a = \sqrt{x - y}$, we have

$$x^2 + xy + y^2 = (a^2 + y)^2 + (a^2 + y)y + y^2 = a^4 + 3a^2y + 3y^2$$

and

$$4(x^2 + xy + y^2) = (2a^2 + 3y)^2 + 3y^2.$$

Writing $m = 2\sqrt{x^2 + xy + y^2}$ and $n = 2a^2 + 3y$, we have

$$m^2 = n^2 + 3y^2$$

or

$$(m - n)(m + n) = 3y^2,$$

so $(m - n, m + n) = (1, 3y^2)$, $(y, 3y)$, or $(3, y^2)$.

In the first case, $2n = 3y^2 - 1$ and $4a^2 = 2n - 6y = 3y^2 - 6y - 1$. Hence, $a^2 \equiv 2 \pmod 3$, which is impossible.

In the second case, $n = y < 2a^2 + 3y = n$, a contradiction.

In the third case, we have $4a^2 = 2n - 6y = y^2 - 6y - 3 < (y - 3)^2$. When $y \geq 10$ we have $y^2 - 6y - 3 > (y - 4)^2$. Hence, we must actually have $y = 2, 3, 5$, or 7. In this case we have $a = \dfrac{\sqrt{y^2 - 6y - 3}}{2}$, which is real only when $y = 7$, $a = 1$, $x = y + a^2 = 8$, and $z = 13$. This yields the unique solution $(x, y, z) = (8, 7, 13)$.

Problem 8.3.6. *Find all the positive integers a, b, c such that*

$$a^3 + b^3 + c^3 = 2001.$$

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(2001 Junior Balkan Mathematical Olympiad)

Solution. Assume without loss of generality that $a \leq b \leq c$.

It is obvious that $1^3 + 10^3 + 10^3 = 2001$. We prove that $(1, 10, 10)$ is the only solution of the equation, except for its permutations.

We start proving a useful

Lemma. *Suppose n is an integer. The remainder of n^3 when divided by 9 is $0, 1$ or -1.*

Indeed, if $n = 3k$, then $9|n^3$ and if $n = 3k \pm 1$, then $n^3 = 27k^3 \pm 27k_9^2k \pm 1 = \mathcal{M}9 \pm 1$.

Since $2001 = 9 \cdot 222 + 3 = \mathcal{M}9 + 3$, then $a^3 + b^3 + c^3 = 2001$ implies $a^3 = \mathcal{M}9 + 1$, $b^3 = \mathcal{M}9 + 1$ and $c^3 = \mathcal{M}9 + 1$, hence a, b, c are numbers of the form $\mathcal{M}3 + 1$. We search for a, b, c in the set $\{1, 4, 7, 10, 13, \dots\}$.

If $c \geq 13$ then $c^3 \geq 2197 > 2001 = a^3 + b^3 + c^3$, which is false. If $c \leq 7$ then $2001 = a^3 + b^3 + c^3 \leq 3 \cdot 343$ and again is false. Hence $c = 10$ and consequently $a^3 + b^3 = 1001$. If $b < c = 10$ then $a \leq b \leq 7$ and $1001 = a^3 + b^3 \leq 2 \cdot 7^3 = 2 \cdot 343$, a contradiction. Thus $b = 10$ and $a = 1$.

Therefore $(a, b) \in \{(1, 10, 10), (10, 1, 10), (10, 10, 1)\}$.

Problem 8.3.7. *Determine all ordered pairs (m, n) of positive integers such that*

$$\frac{n^3 + 1}{mn - 1}$$

is an integer.

(35th IMO)

Solution. Let $\dfrac{n^3 + 1}{mn - 1} = k$, k a positive integer.

From $n^3 + 1 = k(mn - 1)$, one obtains $k + 1 = n(km - n^2)$. Thus, n divides $k + 1$ and by noting $km - n^2 = q$ one has $k = nq - 1$. Using this form of k we have

$$n^3 + 1 = (nq - 1)(mn - 1) \Leftrightarrow n(mq - n) = m + q.$$

Since $m+q > 0$ it follows that $x = mq-n > 0$. Thus we have the system:

$$\begin{cases} xn = m + q \\ x + n = mq \end{cases}$$

By adding these equations we obtain:

$$xn + mq = x + n + m + q \Leftrightarrow xn + mq - x - n - m - q + 2 = 2 \Leftrightarrow$$

$$(x-1)(n-1)+(m-1)(q-1)=2.$$

The equation

$$(x-1)(n-1)+(m-1)(q-1)=2$$

has only finite number of positive integer solutions. These are listed bellow:

1) $x=1,\ m-1=2,\ q-1=1\ \Rightarrow\ x=1,\ m=3,\ q=2\ \Rightarrow\ m=3,\ n=5.$
2) $x=1,\ m-1=1,\ q-1=2\ \Rightarrow\ m=2,\ n=5.$
3) $n=1,\ m-1=2,\ q-1=1\ \Rightarrow\ n=1,\ m=3.$
4) $n=1,\ m-1=1,\ q-1=2\ \Rightarrow\ n=1,\ m=2.$
5) $m=1,\ x-1=2,\ n-1=1\ \Rightarrow\ m=1,\ n=2.$
6) $m=1,\ x-1=1,\ n-1=2\ \Rightarrow\ m=1,\ n=3.$
7) $q=1,\ x-1=1,\ n-1=2\ \Rightarrow\ n=3,\ m=5.$
8) $q=1,\ x-1=2,\ n-1=1\ \Rightarrow\ n=2,\ m=5.$
9) $x-1=n-1=m-1=q-1=1\ \Rightarrow\ m=n=2.$

Thus, we have obtained the following nine pairs (m,n): (5,3), (3,5), (5,2), (2,5), (3,1), (1,3), (2,1), (1,2), (2,2). All pairs are solutions of the problem.

19.3.2 High-order polynomial equations

Problem 8.3.12. *Prove that there are no positive integers x and y such that*

$$x^5+y^5+1=(x+2)^5+(y-3)^5.$$

Solution. Notice that $z^5\equiv z\pmod{10}$, hence $x+y+1\equiv(x+2)+(y-3)\pmod{10}$, impossible.

Problem 8.3.13. *Prove that the equation $y^2=x^5-4$ has no integer solutions.*

(1998 Balkan Mathematical Olympiad)

Solution. We consider the equation mod 11. Since

$$(x^5)^2=x^{10}\equiv 0 \text{ or } 1 \pmod{11}$$

for all x, we have $x^5\equiv -1,\ 0,$ or $1\pmod{11}$, so the right-hand side is either 6, 7, or 8 modulo 11. However, all squares are 0, 1, 3, 4, 5, or 9 modulo 11, so the equation $y^2=x^5-4$ has no integer solutions.

Problem 8.3.14. *Let $m,n>1$ be integer numbers. Solve in positive integers the equation*

$$x^n+y^n=2^m.$$

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(2003 Romanian Mathematical Olympiad)

Solution. Let $d = gcd(x, y)$ and $x = da$, $y = db$, where $(a, b) = 1$. it is easy to see that a and b are both odd numbers and $a^n + b^n = 2^k$, for some integer k.

Suppose that n is even. As $a^2 \equiv b^2 \equiv 1 \pmod 8$, we have also $a^n \equiv b^n \equiv 1 \pmod 8$. As $2^k = a^n + b^n \equiv 2 \pmod 8$, we conclude $t = 1$ and $u = v = 1$, thus $x = y = d$. The equation becomes $x^n = 2^{m-1}$ and it has an integer solution if and only if n is a divisor of $m - 1$ and $x = y = 2^{\frac{m-1}{n}}$.

Consider the case when n is odd. From the decomposition

$$a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \cdots + b^{n-1}),$$

we easily get $a + b = 2^k = a^n + b^n$. In this case $a = b = 1$, and the proof goes on the line of the previous case.

To conclude, the given equations have solutions if and only if $\dfrac{m-1}{n}$ is an integer and in this case $x = y = 2^p$.

Problem 8.3.15. *For a given positive integer m, find all pairs (n, x, y) of positive integers such that m, n are relatively prime and $(x^2+y^2)^m = (xy)^n$, where n, x, y can be represented in terms of m.*

(1995 Korean Mathematical Olympiad)

Solution. If (n, x, y) is a solution, then the AM-GM inequality yields

$$(xy)^n = (x^2 + y^2)^m \geq (2xy)^m > (xy)^m,$$

so $n > m$. Let p be a common prime divisor of x and y and let $p^a \| x$, $p^b \| y$. Then $p^{(a+b)n} \| (xy)^n = (x^2 + y^2)^m$. Suppose $b > a$. Since $p^{2a} \| x^2$, $p^{2b} \| y^2$, we see that $p^{2a} \| x^2 + y^2$ and $p^{2am} \| (x^2 + y^2)^m$. Thus $2am = (a+b)n > 2an$ and $m > n$, a contradiction. Likewise, $a > b$ produces a contradiction, so we must have $a = b$ and $x = y$. This quickly leads to $x = 2^t$ for some integer t and all solutions are of the form

$$(n, x, y) = (2t + 1, 2^t, 2^t)$$

for nonnegative integers t.

19.3.3 Exponential Diophantine equations

Problem 8.3.19. *Determine all triples (x, k, n) of positive integers such that*

$$3^k - 1 = x^n.$$

(1999 Italian Mathematical Olympiad)

Solution. All triples of the form $(3^k - 1, k, 1)$ for positive integers k, and (2,2,3).

The solutions when $n = 1$ are obvious. Now, n cannot be even because then 3 could not divide $3^k = (x^{\frac{n}{2}})^2 + 1$ (because no square is congruent to 2 modulo 3). Also, we must have $x \neq 1$.

Assume that $n > 1$ is odd and $x \geq 2$. Then $3^k = (x+1)\sum_{i=0}^{n-1}(-x)^i$, implying that both $x + 1$ and $\sum_{i=0}^{n-1}(-x)^i$ are powers of 3. Because $x + 1 \leq x^2 - x + 1 \leq \sum_{i=0}^{n-1}(-x)^i$ we must have $0 \equiv \sum_{i=0}^{n-1}(-x)^i \equiv n \pmod{x+1}$, so that $x + 1|n$. Specifically, this means that $3|n$.

Writing $x' = x^{\frac{n}{3}}$, we have $3^k = x'^3 + 1 = (x' + 1)(x'^2 - x' + 1)$. As before $x' + 1$ must equal some power of 3, say 3^t. Then $3^k = (3^t - 1)^3 + 1 = 3^{3t} - 3^{2t+1} + 3^{t+1}$, which is strictly between 3^{3t-1} and 3^{3t} for $t > 1$. Therefore we must have $t = 1$, $x' = 2$, and $k = 2$, giving the solution $(x, k, n) = (2, 2, 3)$.

Problem 8.3.20. *Find all pairs of nonnegative integers x and y which satisfy the equation*

$$p^x - y^p = 1$$

where p is a given odd prime.

(1995 Czech-Slovak Match)

Solution. If (x, y) is a solution, then

$$p^x = y^p + 1 = (y + 1)(y^{p-1} - \cdots + y^2 - y + 1)$$

and so $y + 1 = p^n$ for some n. If $n = 0$, then $x = y = 0$ and p may be arbitrary. Otherwise,

$$p^x = (p^n - 1)^p + 1$$

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$$= p^{np} - p \cdot p^{n(p-1)} + \binom{p}{2} p^{n(p-2)} + \cdots - \binom{p}{p-2} p^{2n} + p \cdot p^n.$$

Since p is a prime, all of the binomial coefficients are divisible by p. Hence all terms are divisible by p^{n+1}, and all but the last by p^{n+2}. Therefore the highest power of p dividing the right side is p^{n+1} and so $x = n+1$. We also have

$$0 = p^{np} - p \cdot p^{n(p-1)} + \binom{p}{2} p^{n(p-2)} + \cdots - \binom{p}{p-2} p^{2n}.$$

For $p = 3$ this reads $0 = 3^{3n} - 3 \cdot 3^{2n}$, which only occurs for $n = 1$, yielding $x = y = 2$. For $p \geq 5$, the coefficient $\binom{p}{p-2}$ is not divisible by p^2, so every term but the last on the right side is divisible by p^{2n+2}, while the last term is not. Since the terms sum to 0, this is impossible.

Hence the only solutions are $x = y = 0$ for all p and $x = y = 2$ for $p = 3$.

Problem 8.3.21. *Let x, y, z be integers with $z > 1$. Show that*

$$(x+1)^2 + (x+2)^2 + \cdots + (x+99)^2 \neq y^z.$$

(1998 Hungarian Mathematical Olympiad)

Solution. We prove the statement by contradiction. Suppose, on the contrary, that there are integers x, y, z such that $z > 1$, and

$$(x+1)^2 + (x+2)^2 + \cdots + (x+99)^2 = y^z.$$

We notice that

$$\begin{aligned} y^z &= (x+1)^2 + (x+2)^2 + \cdots + (x+99)^2 \\ &= 99x^2 + 2(1+2+\cdots+99)x + (1^2 + 2^2 + \cdots + 99^2) \\ &= 99x^2 + \frac{2 \cdot 99 \cdot 100}{2} x + \frac{99 \cdot 100 \cdot 199}{6} \\ &= 33(3x^2 + 300x + 50 \cdot 199), \end{aligned}$$

which implies that $3|y$. Since $z \geq 2$, $3^2 | y^z$, but 3^2 does not divide $33(3x^2 + 300x + 50 \cdot 199)$, contradiction. So our assumption in fact must be false and the original statement in the problem is correct.

Problem 8.3.22. *Determine all solutions (x, y, z) of positive integers such that*

$$(x+1)^{y+1} + 1 = (x+2)^{z+1}.$$

(1999 Taiwanese Mathematical Olympiad)

Solution. Let $a = x + 1$, $b = y + 1$, $c = z + 1$. Then $a, b, c \geq 2$ and

$$a^b + 1 = (a + 1)^c$$

$$((a + 1) - 1)^b + 1 = (a + 1)^c.$$

Taking the equations mod $(a + 1)$ yields $(-1)^b + 1 \equiv 0$, so b is odd.

Taking the second equation mod $(a + 1)^2$ after applying the binomial expansion yields

$$\binom{b}{1}(a + 1)(-1)^{b-1} + (-1)^b + 1 \equiv 0 \pmod{(a + 1)^2}$$

so $(a+1)|b$ and a is even. On the other hand, taking the first equation mod a^2 after applying the binomial expansion yields

$$1 \equiv \binom{c}{1}a + 1 \pmod{a^2},$$

so c is divisible by a and is even as well. Write $a = 2a_1$ and $c = 2c_1$. Then

$$2^b a_1^b = a^b = (a + 1)^c - 1 = ((a + 1)^{c_1} - 1)((a + 1)^{c_1} + 1).$$

It follows that $gcd((a + 1)^{c_1} - 1, (a + 1)^{c_1} + 1) = 2$. Therefore, using the fact that $2a_1$ is a divisor of $(a + 1)^{c_1} - 1$, we may conclude that

$$(a + 1)^{c_1} - 1 = 2a_1^b$$

$$(a + 1)^{c_1} + 1 = 2^{b-1}.$$

We must have $2^{b-1} > 2a_1^b \Rightarrow a_1 = 1$. Then these equations give $c_1 = 1$ and $b = 3$. Therefore the only solution is $(x, y, z) = (1, 2, 1)$.

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20

Some special problems in number theory

20.1 Quadratic residues. Legendre's symbol

Problem 9.1.7. *Let $f, g : Z^+ \to Z^+$ functions with the properties:*
i) g is surjective;
ii) $2f^2(n) = n^2 + g^2(n)$ for all positive integers n.
If, moreover, $|f(n) - n| \leq 2004\sqrt{n}$ for all n, prove that f has infinitely many fixed points.

(2005 Moldavian IMO Team Selection Test)

Solution. Let p_n be the sequence of prime numbers of the form $8k + 3$ (the fact that there are infinitely many such numbers is a trivial consequence of Dirichlet's theorem, but we invite the reader to find an elementary proof). It is obvious that for all n we have

$$\left(\frac{2}{p_n}\right) = (-1)^{\frac{p_n^2-1}{8}} = -1.$$

Using the condition i) we can find x_n such that $g(x_n) = p_n$ for all n. It follows that $2f^2(x_n) = x_n^2 + p_n^2$, which can be rewritten as $2f^2(x_n) \equiv x_n^2 \pmod{p_n}$. Because $\left(\frac{2}{p_n}\right) = -1$, the last congruence shows that $p_n | x_n$ and $p_n | f(x_n)$. Thus there exist sequences of positive integers a_n, b_n such that $x_n = a_n p_n$, $f(x_n) = b_n p_n$ for all n. Clearly, ii) implies the relation

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$2b_n^2 = a_n^2 + 1$. Finally, using the property $|f(n) - n| \leq 2004\sqrt{n}$ we infer that

$$\frac{2004}{\sqrt{x_n}} \geq \left| \frac{f(x_n)}{x_n} - 1 \right| = \left| \frac{b_n}{a_n} - 1 \right|,$$

that is

$$\lim_{n\to\infty} \frac{\sqrt{a_n^2+1}}{a_n} = \sqrt{2}.$$

The last relation immediately implies that $\lim\limits_{n\to\infty} a_n = 1$. Therefore, starting from a certain rank, we have $a_n = 1 = b_n$ that is $f(p_n) = p_n$. The conclusion now follows.

Problem 9.1.8. *Suppose that the positive integer a is not a perfect square. Then $\left(\frac{a}{p}\right) = -1$ for infinitely many primes p.*

Solution. One may assume that a is square-free. Let us write $a = 2^e q_1 q_2 \dots q_n$, where q_i are different odd primes and $e \in \{0, 1\}$. Let us assume first that $n \geq 1$ and consider some odd distinct primes $r_1, \dots, r_k$ each of them different from $q_1, \dots, q_n$. We will show that there exists a prime p, different from $r_1, \dots, r_k$, such that $\left(\frac{a}{p}\right) = -1$. Let s be a non quadratic residue modulo q_n.

Using the Chinese Remainder Theorem, we can find a positive integer b such that

$$\begin{cases} b \equiv 1 \pmod{r_i}, \ 1 \leq i \leq k \\ b \equiv 1 \pmod{8}, \\ b \equiv q_i, \ 1 \leq i \leq n-1 \\ b \equiv s \pmod{q_n} \end{cases}$$

Now, write $b = p_1 \dots p_m$ with p_i odd primes, not necessarily distinct. Using the quadratic reciprocity law, it follows immediately that

$$\prod_{i=1}^{m} \left(\frac{2}{p_i}\right) = \prod_{i=1}^{m} (-1)^{\frac{p_i^2-1}{8}} = (-1)^{\frac{b^2-1}{8}} = 1$$

and

$$\prod_{j=1}^{m} \left(\frac{q_i}{p_j}\right) = \prod_{j=1}^{m} (-1)^{\frac{p_j-1}{2}\cdot\frac{q_i-1}{2}} \left(\frac{p_j}{q_i}\right) = (-1)^{\frac{q_i-1}{2}\cdot\frac{b-1}{2}} \left(\frac{b}{q_i}\right) = \left(\frac{b}{q_i}\right)$$

for all $i \in \{1, 2, \dots, n\}$. Hence

$$\prod_{i=1}^{m} \left(\frac{a}{p_i}\right) = \left[\prod_{j=1}^{m} \left(\frac{2}{p_j}\right)\right]^2 \prod_{i=1}^{n}\prod_{j=1}^{m} \left(\frac{q_i}{p_j}\right)$$

$$= \prod_{i=1}^{n} \left(\frac{b}{q_i}\right) = \left(\frac{b}{q_n}\right) = \left(\frac{s}{q_n}\right) = -1.$$

Thus, there exists $i \in \{1, 2, \ldots, m\}$ such that $\left(\frac{a}{p_i}\right) = -1$. Because $b \equiv 1 \pmod{r_i}$, $1 \leq i \leq k$ we also have $p_i \in \{1, 2, \ldots\} \setminus \{r_1, \ldots, r_k\}$ and the claim is proved.

The only remaining case is $a = 2$. But this one is very simple, since it suffices to use Dirichlet's theorem to find infinitely many primes p such that $\frac{p^2-1}{8}$ is odd.

Problem 9.1.9. *Suppose that $a_1, a_2, \ldots, a_{2004}$ are nonnegative integers such that $a_1^n + a_2^n + \cdots + a_{2004}^n$ is a perfect square for all positive integers n. What is the minimal number of such integers that must equal 0?*

(2004 Mathlinks Contest)

Solution. Suppose that $a_1, a_2, \ldots, a_k$ are positive integers such that $a_1^n + a_2^n + \cdots + a_k^n$ is a perfect square for all n. We will show that k is a perfect square. In order to prove this, we will use the above result and show that $\left(\frac{k}{p}\right) = 1$ for all sufficiently large prime p. This is not a difficult task. Indeed, consider a prime p, greater than any prime divisor of $a_1 a_2 \ldots a_k$. Using Fermat's little theorem, $a_1^{p-1} + a_2^{p-1} + \cdots + a_k^{p-1} \equiv k \pmod{p}$, and since $a_1^{p-1} + a_2^{p-1} + \cdots + a_k^{p-1}$ is a perfect square, it follows that $\left(\frac{k}{p}\right) = 1$. Thus k is a perfect square. And now the problem becomes trivial, since we must find the greatest perfect square smaller than 2004. A quick computation shows that this is $44^2 = 1936$ and so the desired minimal number is 68.

Problem 9.1.10. *Find all positive integers n such that $2^n - 1 | 3^n - 1$.*

(American Mathematical Monthly)

Solution. We will prove that $n = 1$ is the only solution to the problem. Suppose that $n > 1$ is a solution. Then $2^n - 1$ cannot be a multiple of 3, hence n is odd. Therefore, $2^n \equiv 8 \pmod{12}$. Because any odd prime different from 3 is of one of the forms $12k \pm 1$, $12k \pm 5$ and since $2^n - 1 \equiv 7 \pmod{12}$, it follows that $2^n - 1$ has at least a prime divisor of the form $12k \pm 5$, call it p. Obviously, we must have $\left(\frac{3}{p}\right) = 1$ and using the quadratic reciprocity law, we finally obtain $\left(\frac{p}{3}\right) = (-1)^{\frac{p-1}{2}}$. On the other

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hand $\left(\frac{p}{3}\right) = \left(\frac{\pm 2}{3}\right) = -(\pm 1)$. Consequently, $-(\pm 1) = (-1)^{\frac{p-1}{2}} = \pm 1$, which is the desired contradiction. Therefore the only solution is $n = 1$.

Problem 9.1.11. *Find the smallest prime factor of* $12^{2^{15}} + 1$.

Solution. Let p be this prime number. Because $p|12^{2^{16}} - 1$, we find that $o_p(12)|2^{16}$. We find that $o_p(12) = 2^{16}$ and so $2^{16}|p-1$. Therefore $p \geq 1+2^{16}$. But it is well-known that $2^{16}+1$ is a prime (and if you do not believe, you can check; it is not that difficult). So, we might try to see if this number divides $12^{2^{15}} + 1$. Let $q = 2^{16} + 1$. Then

$$12^{2^{15}} + 1 = 2^{q-1} \cdot 3^{\frac{q-1}{2}} + 1 \equiv 3^{\frac{q-1}{2}} + 1 \pmod{q}.$$

It remains to see whether $\left(\frac{3}{q}\right) = -1$. The answer is positive (use the quadratic reciprocity law), so indeed $3^{\frac{q-1}{2}} + 1 \equiv 0 \pmod 2$ and $2^{16} + 1$ is the smallest prime factor of the number $12^{2^{15}} + 1$.

20.2 Special numbers

20.2.1 Fermat's numbers

Problem 9.2.4. *Find all positive integers* n *such that* $2^n - 1$ *is a multiple of* 3 *and* $\dfrac{2^n - 1}{3}$ *is a divisor of* $4m^2 + 1$ *for some integer* m.

(1999 Korean Mathematical Olympiad)

Solution. The answer is all $n = 2^k$ where $k = 1, 2, \ldots$

First observe that $2 \equiv -1 \pmod 3$. Hence $3|2^n - 1$ if and only if n is even.

Suppose, by way of contradiction, that $l \geq 3$ is a positive odd divisor of n. Then $2^l - 1$ is not divisible by 3 but it is a divisor of $2^n - 1$, so it is a divisor of $4m^2 + 1$ as well. On the other hand, $2^l - 1$ has a prime divisor p of the form $4r + 3$. Then $(2m)^2 \equiv -1 \pmod{4r+3}$, but we have that a square cannot be congruent to -1 modulo a prime of the form $4r + 3$ (see also Problem 1 in Section 7.1).

Therefore, n is indeed of the form 2^k for $k \geq 1$. For such n, we have

$$\frac{2^n - 1}{3} = (2^{2^1} + 1)(2^{2^2} + 1)(2^{2^3} + 1) \ldots (2^{2^{k-1}} + 1).$$

The factors on the right side are all relatively prime since they are Fermat numbers. Therefore by the Chinese Remainder Theorem, there is a positive

integer c simultaneously satisfying

$$c \equiv 2^{2^{i-1}} \pmod{2^{2^i}+1} \text{ for all } i = 1, 2, \ldots, k-1$$

and $c \equiv 0 \pmod 2$. Putting $c = 2m$, $4m^2+1$ is a multiple of $\dfrac{2^n-1}{3}$, as desired.

Problem 9.2.5. *Prove that the greatest prime factor of f_n, $n \geq 2$, is greater than $2^{n+2}(n+1)$.*

(2005 Chinese IMO Team Selection Test)

Solution. From Problem 9.2.3 we can write

$$f_n = \prod_{i=1}^{s} (1 + 2^{n+2} r_i)^{k_i}, \tag{1}$$

where $p_i = 1 + 2^{n+2} r_i$ are distinct primes and $k_i \geq 1$. Taking relation (1) modulo 4^{n+2} it follows

$$0 \equiv \sum_{i=1}^{s} k_i r_i \pmod{2^{n+2}},$$

hence

$$\sum_{i=1}^{s} k_i r_i \geq 2^{n+2}.$$

From (1) it is clear that

$$f_n \geq (1 + 2^{n+2})^{k_1 + \cdots + k_s},$$

hence

$$k_1 + \cdots + k_s \leq \frac{\lg(1 + 2^{2^n})}{\lg(1 + 2^{n+2})}.$$

It follows

$$2^{n+2} \leq \left(\max_{1 \leq i \leq s} \right) \sum_{j=1}^{s} k_j \leq \left(\max_{1 \leq i \leq s} r_i \right) \frac{\lg(1 + 2^{2^n})}{\lg(1 + 2^{n+2})}.$$

Assume that $\left(\max_{1 \leq i \leq s} \right) \leq n$. Applying the last inequality we get

$$2^{n+2} \leq n \frac{\lg(1 + 2^{2^n})}{\lg(1 + 2^{n+2})} < n \frac{\lg(1 + 2^{2^n})}{(n+2)\lg 2},$$

i.e.

$$\frac{n+2}{n} \cdot 2^{n+2} < \log_2(1 + 2^{2^n}),$$

hence $2^{2^{n+2}} < 1 + 2^{2^n}$, a contradiction. Therefore $\max\limits_{1 \leq i \leq s} r_i \geq n+1$, and $\max\limits_{1 \leq i \leq s} p_i > 2^{n+2}(n+1)$.

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20.2.2 Mersenne's numbers

Problem 9.2.7. *Let P^* denote all the odd primes less than 10000, and suppose $p \in P^*$. For each subset $S = \{p_1, p_2, \ldots, p_k\}$ of P^*, with $k \geq 2$ and not including p, there exists a $q \in P^* \setminus S$ such that*

$$(q+1)|(p_1+1)(p_2+1)\ldots(p_k+1).$$

Find all such possible values of p.

(1999 Taiwanese Mathematical Olympiad)

Solution. Direct calculation shows that the set T of Mersenne primes less that 10000 is

$$\{M_2, M_3, M_5, M_7, M_{13}\} = \{3, 7, 31, 127, 8191\}.$$

$2^{11} - 1$ is not prime, it equals $23 \cdot 89$. We claim this is the set of all possible values of p.

If some prime p is not in T, then look at the set $S = T$. Then there must be some prime $q \notin S$ less than 10000 such that

$$(q+1)|(M_2+1)(M_3+1)(M_5+1)(M_7+1)(M_{13}+1) = 2^{30}.$$

Thus, $q+1$ is a power of 2 and q is a Mersenne prime less than 10000, and therefore $q \in T = S$, a contradiction.

On the other hand, suppose p is in T. Suppose we have a set $S = \{p_1, p_2, \ldots, p_k\} \subseteq P^*$ not including p, with $k \geq 2$ and $p_1 < p_2 < \cdots < p_k$. Suppose, by way of contradiction, that for all $q \in P^*$ such that $(q+1)|(p_1+1)\ldots(p_k+1)$, we have $q \in S$. Then

$$4|(p_1+1)(p_2+1) \;\Rightarrow\; M_2 \in S$$

$$8|(M_2+1)(p_2+1) \;\Rightarrow\; M_3 \in S$$

$$32|(M_2+1)(M_3+1) \;\Rightarrow\; M_5 \in S$$

$$128|(M_2+1)(M_5+1) \;\Rightarrow\; M_7 \in S$$

$$8192|(M_3+1)(M_5+1)(M_7+1) \;\Rightarrow\; M_{13} \in S.$$

Then p, a Mersenne prime under 10000, must be in S, a contradiction. Therefore there is some prime $q < 10000$ not in S with $q+1|(p_1+1)\ldots(p_k+1)$, as desired. This completes the proof.

20.2.3 Perfect numbers

Problem 9.2.9. *Prove that if n is an even perfect number, then $8n+1$ is a perfect square.*

Solution. From Problem 1, we have $n = \dfrac{m(m+1)}{2}$ for some positive integer m, hence

$$8n+1 = 4m(m+1)+1 = (2m+1)^2.$$

Problem 9.2.10. *Show that if k is an odd positive integer, then $2^{k-1}M_k$ can be written as the sum of the cubes of the first $2^{\frac{k-1}{2}}$ odd positive integers. In particular, any perfect number has this property.*

Solution. Standard summation formulas verify that

$$\sum_{i=1}^{n}(2i-1)^3 = n^2(2n^2-1).$$

With $n = 2^{\frac{k-1}{2}}$, the right-hand side becomes $2^{k-1}(2^k-1)$, that is $2^{k-1}M_k$ and we are done.

20.3 Sequences of integers

20.3.1 Fibonacci and Lucas sequences

Problem 9.3.5. *Determine the maximum value of m^2+n^2, where m and n are integers satisfying $1 \le m, n \le 1981$ and $(n^2-mn-m^2)^2 = 1$.*

(22nd IMO)

Solution. Let S be the set of pairs (n,m) of positive integers satisfying the equation

$$(x^2-xy-y^2)^2 = 1. \tag{1}$$

If $n = m$, then $n = m = 1$. Hence $(1,1) \in S$. It is clear that (1,0) and (0,1) are also solutions to the equation (1).

We will consider solutions (n,m) with distinct components. Using Fermat's method of infinite descent we obtain the following important result on the set S.

Lemma. *If (n,m) is a positive solution to the equation (1) and $n \neq m$, then $n > m > n-m$ and $(m, n-m)$ is also a solution to (1).*

Proof. From $n^2-nm-m^2 = \pm 1$, we obtain

$$n(n-m) = m^2 \pm 1 > 0.$$

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Thus, $n > m$. Also from $n^2 - nm - m^2 = \pm 1$, we obtain

$$m^2 - m(n-m) - (n-m)^2 = m^2 + mn - n^2 = \mp 1.$$

Apply first part to solution $(m, n-m)$ and obtain $m > n - m$.

From the Lemma we deduce that any pair $(n, m) \in S$ gives rise to a pair $(m, n-m) \in M$, which gives rise to a pair $(a+b, a) \in M$. In this way by descending method $(n, m) \to (m, n-m)$ or by ascending method $(a, b) \to (a+b, a)$, we obtain new solutions of the equation. The ascending and descending methods are reverse to one another.

By applying the descending method to a pair $(n, m) \in S$ we can have only finitely many steps, because $n - m < m$. Hence, in a finite number of steps we obtain a pair with $n = m$, that is the pair (1,1). Thus, all solutions $(n, m) \in S$ are obtained from the pair (1,0) by applying the ascending method:

$$(1,0) \to (1,1) \to (2,1) \to (3,2) \to (5,3) \to \ldots$$

The components of all such pairs are Fibonacci numbers F_n. In this way the ascending transformation is exactly the following:

$$(F_n, F_{n-1}) \to (F_{n+1}, F_n).$$

Thus, to obtain the solution (n, m) with maximum value of $n^2 + m^2$ we consider the members of the Fibonacci sequence, not exceeding 1981:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597.$$

So, the required maximum is $987^2 + 1597^2$.

Remark. Fibonacci numbers F_n have the property:

$$F_{n+1}^2 - F_n F_{n+1} - F_n^2 = \pm 1, \text{ for all } n \geq 0.$$

To prove it for $n = 0$ or $n = 1$ is equivalent to see that $(1, 0) \in S$ and that $(1, 1) \in S$. Further, we can use induction. The relation

$$F_{n+1}^2 - F_n F_{n+1} - F_n^2 = \pm 1$$

implies

$$F_{n+2}^2 - F_{n+1}F_{n+2} - F_{n+1}^2 = (F_{n+1} + F_n)^2 - F_{n+1}(F_{n+1} + F_n) - F_{n+1}^2$$

$$= -(F_{n+1}^2 - F_n F_{n+1} - F_n^2) = \mp 1.$$

Problem 9.3.6. *Prove that for any integer $n \geq 4$, F_n+1 is not a prime.*

Solution. We have the identity

$$F_n^4 - 1 = F_{n-2}F_{n-1}F_{n+1}F_{n+2} \tag{1}$$

Assume that $F_n + 1$ is a prime for some positive integer $n \geq 4$. Using (1), it follows that $F_n + 1$ divides at least one of the integers F_{n-2}, F_{n-1}, F_{n+1}, F_{n+2}. Since $F_n + 1$ is greater than F_{n-2} and F_{n-1}, it follows that $F_n + 1$ divides F_{n+1} or F_{n+2}. But $F_{n+1} < 2F_n$ and $F_{n+2} < 4F_n$, hence $F_n + 1$ cannot divides F_{n+1} or F_{n+2}, and the desired conclusion follows.

Problem 9.3.7. *Let k be an integer greater than 1, $a_0 = 4$, $a_1 = a_2 = (k^2 - 2)^2$ and*

$$a_{n+1} = a_n a_{n-1} - 2(a_n + a_{n-1}) - a_{n-2} + 8 \text{ for } n \geq 2.$$

Prove that $2 + \sqrt{a_n}$ is a perfect square for all n.

Solution. The Fibonacci numbers are involved here again but it is much harder to guess how they are related to the solution.

Let λ, μ be the roots of the equation $t^2 - kt + 1 = 0$. Notice that $\lambda + \mu = k$, $\lambda\mu = 1$. Amending the Fibonacci sequence by setting $F_0 = 0$, we claim that

$$a_n = (\lambda^{2F_n} + \mu^{2F_n})^2 \text{ for } n = 0, 1, 2, \ldots$$

This is readily checked for $n = 0, 1, 2$. Assume it holds for all $k \leq n$. Note that the given recurrence can be written as

$$a_{n+1} - 2 = (a_n - 2)(a_{n-1} - 2) - (a_{n-2} - 2),$$

and that $a_k = (\lambda^{2F_k} + \mu^{2F_k})^2$ is equivalent to $a_k - 2 = \lambda^{4F_k} + \mu^{4F_k}$. Using the induction hypothesis for $k = n - 2, n - 1, n$, we obtain

$$\begin{aligned} a_{n+1} - 2 &= (\lambda^{4F_n} + \mu^{4F_n})(\lambda^{4F_{n-1}} + \mu^{4F_{n-1}}) - (\lambda^{4F_{n-2}} + \mu^{4F_{n-2}}) \\ &= \lambda^{4(F_n+F_{n-1})} + \mu^{4(F_n+F_{n-1})} + \lambda^{4(F_{n-1}+F_{n-2})}\mu^{4F_{n-1}} \\ &\quad + \mu^{4(F_{n-1}+F_{n-2})}\lambda^{4F_{n-1}} - (\lambda^{4F_{n-2}} + \mu^{4F_{n-2}}) \\ &= \lambda^{4F_{n+1}} + \mu^{4F_{n+1}} + (\lambda\mu)^{4F_{n-1}}(\lambda^{4F_{n-2}} + \mu^{4F_{n-2}}) - (\lambda^{4F_{n-2}} + \mu^{4F_{n-2}}). \end{aligned}$$

Since $\lambda\mu = 1$, it follows that

$$a_{n+1} = 2 + \lambda^{4F_{n+1}} + \mu^{4F_{n+1}} = (\lambda^{2F_{n+1}} + \mu^{2F_{n+1}})^2$$

and the induction is complete.

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Now

$$2 + \sqrt{a_n} = 2 + \lambda^{2F_n} + \mu^{2F_n} = (\lambda^{F_n} + \mu^{F_n})^2.$$

Since

$$(\lambda^{m-1} + \mu^{m-1})(\lambda + \mu) = (\lambda^m + \mu^m) + \lambda\mu(\lambda^{m-2} + \mu^{m-2}),$$

we have

$$\lambda^m + \mu^m = k(\lambda^{m-1} + \mu^{m-1}) - (\lambda^{m-2} + \mu^{m-2}),$$

leading to an easy proof by induction that $\lambda^m + \mu^m$ is an integer for all nonnegative integers m. The solution is complete.

20.3.2 Problems involving linear recursive relations

Problem 9.3.12. *Let a, b be integers greater than 1. The sequence x_1, $x_2, \ldots$ is defined by the initial conditions $x_0 = 0$, $x_1 = 1$ and the recursion*

$$x_{2n} = ax_{2n-1} - x_{2n-2}, \quad x_{2n+1} = bx_{2n} - x_{2n-1}$$

for $n \geq 1$. Prove that for any natural numbers m and n, the product $x_{n+m}x_{n+m-1}\ldots x_{n+1}$ is divisible by $x_m x_{m-1}$.

(2001 St. Petersburg City Mathematical Olympiad)

Solution. We will show that $x_m | x_{km}$, and then show that $gcd(x_m, x_{m-1}) = 1$.

First, consider our sequence modulo x_m for some m. Each x_{k+1} is uniquely determined by x_k, x_{k-1} and the parity of k. Express each x_i as a function $f_i(a, b)$. We have $x_i \equiv f_i(a, b)x_1 \pmod{x_m}$. Suppose $x_r \equiv 0 \pmod{x_m}$ for some r. Since each term is a linear combination of two preceding ones,

$$x_{i+r} \equiv f_i(a, b)x_{r+1} \pmod{x_m} \text{ if } m \text{ is even}, \tag{1}$$

$$x_{i+r} \equiv f_i(b, a)x_{r+1} \pmod{x_m} \text{ if } m \text{ is odd}. \tag{2}$$

Now we need to prove the following statement.

Lemma. *The function $f_i(a, b)$ is symmetric for any odd i.*

Proof. We will prove also that $f_i(a, b)$ is symmetric function multiplied by a. Now, we are to prove that $f_{2k-1}(a, b)$ is symmetric and $f_{2k-2}(a, b) = ag_{2k-2}(a, b)$, where g_{2k-2} is symmetric too, for any positive integer k. Proceed by induction on k. For $k = 1$ we have $f_1(a, b) = 1$ and $g_0(a, b) = 0$.

Suppose that $f_{2k-1}(a,b)$ is symmetric and $f_{2k-2}(a,b) = ag_{2k-2}(a,b)$ where $g_{2k-2}(a,b)$ is symmetric too. Then we can write

$$\begin{aligned} f_{2k}(a,b) = x_{2k} &= ax_{2k-1} - x_{2k-2} \\ &= a(x_{2k-1} - g(a,b)) \\ &= a(f_{2k-1}(a,b) - g_{2k-2}(a,b)) \end{aligned}$$

and

$$\begin{aligned} f_{2k+1}(a,b) = x_{2k+1} &= abx_{2k-1} - bx_{2k-2} - x_{2k-1} \\ &= abx_{2k-1} - abq - x_{2k-1} \\ &= (ab-1)f_{2k-1}(a,b) - abg_{2k-2}(a,b). \end{aligned}$$

It shows that f_{2k+1} and g_{2k} are symmetric too and completes the step of induction. □

Now we are to prove that $x_m | x_{km}$. Proceed by induction on k. For $k = 1$ this statement is true. Let $x_m | x_{km}$. Then from (1) and (2) putting $r = km$ and $i = m$, we obtain the following. If km is even, then

$$x_{m(k+1)} \equiv f_m(a,b)x_{km+1} \equiv x_m x_{km+1} \equiv 0 \pmod{x_m}.$$

For km odd m is odd too and $f_m(a,b) = f_m(b,a)$. Hence, we have

$$x_{m(k+1)} \equiv f_m(b,a)x_{km+1} \equiv f_m(a,b)x_{km+1} \equiv x_m x_{km+1} \equiv 0 \pmod{x_m}.$$

So, for each nonnegative integers k, m $x_m | x_{km}$.

Since the product $x_{n+1}x_{n+2}\dots x_{n+m}$ has m terms, one of their indices is divisible by m and another's index is divisible by $m-1$. Thus both x_m and x_{m-1} divide the product. If we can show that x_m is relatively prime to x_{m-1}, we would be done. We will prove this by induction. For the base case, x_0 is relatively prime to x_1. Now, $x_{2n} = ax_{2n-1} - x_{2n-2}$. Any prime factor common to x_{2n} and x_{2n-1} must also divide x_{2n-2}, but because x_{2n-2} is relatively prime to x_{2n-1}, there is no such prime factor. A similar argument holds for x_{2n+1} because $x_{2n+1} = bx_{2n} - x_{2n-1}$. Thus $x_m x_{m-1} | (x_{n+1}x_{n+2}\dots x_{n+m})$.

Problem 9.3.13. *Let m be a positive integer. Define the sequence $\{a_n\}_{n\geq 0}$ by $a_0 = 0$, $a_1 = m$ and $a_{n+1} = m^2 a_n - a_{n-1}$ for $n \geq 1$. Prove that an ordered pair (a,b) of nonnegative integers, with $a \leq b$, is a solution of the equation*

$$\frac{a^2+b^2}{ab+1} = m^2$$

if and only if $(a,b) = (a_n, a_{n+1})$ for some $n \geq 0$.

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(1998 Canadian Mathematical Olympiad)

Solution. The "if" direction of the claim is easily proven by induction on n; we prove the "only if" direction by contradiction. Suppose, on the contrary, that there exist pairs satisfying the equation but not of the described form; let (a, b) be such a pair with minimal sum $a + b$. We claim that $(c, a) = (m^2a - b, a)$ is another such a pair but with smaller sum $c + a$, which leads to a contradiction.

(a) $a = 0$. Then $(a, b) = (0, m) = (a_0, a_1)$, a contradiction.

(b) $a = m$. Then $(a, b) = (m, m^3) = (a_1, a_2)$, a contradiction.

(c) $a = 1$. Then $b \geq 1 = 1$ and $(b + 1)|(b^2 + 1)$; but $(b + 1)|(b^2 - 1)$, thus $(b + 1)|[(b^2 + 1) - (b^2 - 1)] = 2$. We have $b = 1$, thus $m = 1$ and $(a, b) = (1, 1) = (a_1, a_2)$, a contradiction.

(d) $2 \leq a < m$. Rewrite $(a^2 + b^2)/(ab + 1) = m^2$ as

$$b^2 - m^2ab + a^2 - m^2 = 0,$$

we know that $t = b$ is a root of the quadratic equation

$$t^2 - m^2at + a^2 - m^2 = 0. \tag{1}$$

Thus $m^4a^2+4m^2-4a^2$ the discriminant of the equation must be a perfect square. But

$$(m^2a + 1)^2 = m^4a^2 + 2m^2a + 1$$

$$> m^4a^2 + 4m^2 - 4a^2 > (m^2a)^2$$

for $2 \leq a < m$. So the discriminant cannot be a perfect square, a contradiction.

(e) $a > m$. Again $t = b$ is a root of (1). It is easy to check that $t = m^2a - b = c$ also satisfies the equation. We have $bc = a^2 - m^2 > 0$; since $b \geq 0$, $c > 0$. Since $a > 0$ and $c > 0$, $ac + 1 > 0$, we have

$$\frac{c^2 + a^2}{ca + 1} = m^2.$$

Since $c > 0$, $b \geq a$ and $bc = a^2 - m^2 < a^2$, we have $c < a$. Thus (c, a) is a valid pair. Also, it cannot be of the form (a_n, a_{n+1}) or else

$$(a, b) = (a_{n+1}, m^2a_{n+1} - a_n) = (a_{n+1}, a_{n+2}).$$

But then, $c + a < a + a \leq b + a$, as desired.

From the above, we see that our assumption is false. Therefore every pair satisfying the original equation must be of the described form.

Problem 9.3.14. *Let b, c be positive integers, and define the sequence $a_1, a_2, \ldots$ by $a_1 = b$, $a_2 = c$, and*

$$a_{n+2} = |3a_{n+1} - 2a_n|$$

for $n \geq 1$. Find all such (b, c) for which the sequence $a_1, a_2, \ldots$ has only a finite number of composite terms.

(2002 Bulgarian Mathematical Olympiad)

Solution. The only solutions are (p, p) for p not composite, $(2p, p)$ for p not composite, and $(7, 4)$.

The sequence $a_1, a_2, \ldots$ cannot be strictly decreasing because each a_n is a positive integer, so there exists a smallest $k \geq 1$ such that $a_{k+1} \geq a_k$. Define a new sequence $b_1, b_2, \ldots$ by $b_n = a_{n+k-1}$, so $b_2 \geq b_1$, $b_{n+2} = |3b_{n+1} - 2b_n|$ for $n \geq 1$, and $b_1, b_2, \ldots$ has only a finite number of composite terms. Now, if $b_{n+1} \geq b_n$,

$$b_{n+2} = |3b_{n+1} - 2b_n| = 3b_{n+1} - 2b_n = b_{n+1} + 2(b_{n+1} - b_n) \geq b_{n+1},$$

so by induction $b_{n+2} = 3b_{n+1} - 2b_n$ for $n \geq 1$.

Using the general theory of linear recurrence relations (a simple induction proof also suffices), we have

$$b_n = A \cdot 2^{n-1} + B$$

for $n \geq 1$, where $A = b_2 - b_1$, $B = 2b_1 - b_2$. Suppose (for contradiction) that $A \neq 0$. Then b_n is an increasing sequence, and, since it contains only finitely many composite terms, $b_n = p$ for some prime $p > 2$ and some $n \geq 1$. However, then $b_{n+l(p-1)}$ is divisible by p and thus composite for $l \geq 1$, because

$$b_{n+l(p-1)} = A \cdot 2^{n-1} \cdot 2^{l(p-1)} + B \equiv A \cdot 2^{n-1} + b \equiv 0 \pmod{p}$$

by Fermat's Little Theorem. This is a contradiction, so $A = 0$ and $b_n = b_1$ for $n \geq 1$. Therefore b_1 is not composite; let $b_1 = p$, where $p = 1$ or p is prime.

We now return to the sequence $a_1, a_2, \ldots$, and consider different possible values of k. If $k = 1$, we have $a_1 = b_1 = b_2 = a_2 = p$, so $b = c = p$ for

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p not composite are the only solutions. If $k > 1$, consider that $a_{k-1} > a_k$ by the choice of k, but $a_{k+1} = |3a_k - 2a_{k-1}|$, and $a_{k+1} = b_2 = b_1 = a_k$, so $a_{k+1} = 2a_{k-1} - 3a_k$, and thus $a_{k-1} = 2p$. For $k = 2$, this means that $b = 2p$, $c = p$ for p not composite are the only solutions. If $k > 2$, the same approach yields

$$a_{k-2} = \frac{3a_{k-1} + a_k}{2} = \frac{7}{2}p,$$

so $p = 2$. For $k = 3$, this gives the solution $b = 7$, $c = 4$, and because $\frac{3 \cdot 7 + 4}{2}$ is not an integer, there are no solutions for $k > 3$.

Remark. The reader may try to prove the following more general statement: *Let $f \in \mathbb{Z}[X_1, \ldots, X_k]$ be a polynomial and $F(n) = f(n, 2^n, 3^n, \ldots, (k-1)^n)$, $n \geq 1$. If $\lim_{n\to\infty} F(n) = \infty$, then the set of primes dividing terms of sequence $(F(n))_{n\geq 1}$ is infinite.*

20.3.3 Nonstandard sequences of integers

Problem 9.3.21. *Let $\{a_n\}$ be a sequence of integers such that for $n \geq 1$*

$$(n-1)a_{n+1} = (n+1)a_n - 2(n-1).$$

If 2000 divides a_{1999}, find the smallest $n \geq 2$ such that 2000 divides a_n.

(1999 Bulgarian Mathematical Olympiad)

Solution. First, we note that the sequence $a_n = 2n - 2$ works. Then writing $b_n = a_n - (2n-2)$ gives the recursion

$$(n-1)b_{n+1} = (n+1)b_n.$$

For $n \geq 2$, observe that

$$b_n = b_2 \prod_{k=2}^{n-1} \frac{k+1}{k-1} = b_2 \frac{\prod_{k=3}^{n} k}{\prod_{k=1}^{n-2} k} = \frac{n(n-1)}{2} b_2.$$

Thus when $n \geq 2$, the solution to the original equation of the form

$$a_n = 2(n-1) + \frac{n(n-1)}{2}c$$

for some constant c. Plugging in $n = 2$ shows that $c = a_2 - 2$ is an integer.

Now, because $2000|a_{1999}$ we have

$$2(1999-1)+\frac{1999\cdot 1998}{2}c\equiv 0$$

implies $-4+1001c\equiv 0$, hence $c\equiv 4 \pmod{2000}$.

Then $2000|a_n$ exactly when

$$2(n-1)+2n(n-1)\equiv 0 \pmod{2000} \Leftrightarrow$$

$$(n-1)(n+1)\equiv 0 \pmod{1000}.$$

$(n-1)(n+1)$ is divisible by 8 exactly when n is odd, and it is divisible by 125 exactly when either $n-1$ or $n+1$ is divisible by 125. The smallest $n\geq 2$ satisfying these requirements is $n=249$.

Problem 9.3.22. *The sequence $(a_n)_{n\geq 0}$ is defined by $a_0=1$, $a_1=3$ and*

$$a_{n+2}=\begin{cases} a_{n+1}+9a_n & \text{if } n \text{ is even,} \\ 9a_{n+1}+5a_n & \text{if } n \text{ is odd.} \end{cases}$$

Prove that

(a) $\sum_{k=1995}^{2000} a_k^2$ is divisible by 20,

(b) a_{2n+1} is not a perfect square for every $n=0,1,2,\ldots$

(1995 Vietnamese Mathematical Olympiad)

Solution. (a) We will first prove the sum is divisible by 4, then by 5. Note that $a_{n+2}\equiv a_{n+1}+a_n \pmod 4$ whether n is odd or even. The sequence modulo 4 thus proceeds 1, 3, 0, 3, 3, 2, 1, 3,... in a cycle of 6, so the sum of squares of any six consecutive terms in congruent to $1^2+3^2+0^2+3^2+3^2+2^2\equiv 0 \pmod 4$.

Now let us work modulo 5, in which case $a_{n+2}\equiv a_{n+1}+4a_n$ if n is even and $a_{n+2}\equiv 4a_{n+1}$ if n is odd. Hence the sequence modulo 5 proceeds 1, 3, 2, 3, 1, 4, 3, 2, 4, 1, 2, 3,... in a cycle of 8 beginning with a_2. This means

$$a_{1995}^2+\cdots+a_{2000}^2\equiv a_3^2+\cdots+a_8^2\equiv 3^2+1^2+4^2+3^2+2^2+4^2\equiv 0 \pmod 5.$$

(b) Notice that $a_{2n+1}\equiv 5a_{2n-1} \pmod 9$. Since $a_1=3$, by induction $a_{2n+1}\equiv 3 \pmod 9$ for all n. However, no perfect square is congruent to 3 modulo 9, since any square divisible by 3 is also divisible by 9. Hence a_{2n+1} is not a square.

Problem 9.3.23. *Prove that for any natural number $a_1>1$, there exists an increasing sequence of natural numbers $a_1,a_2,\ldots$ such that $a_1^2+a_2^2+\cdots+a_k^2$ is divisible by $a_1+a_2+\cdots+a_k$ for all $k\geq 1$.*

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(1995 Russian Mathematical Olympiad)

Solution. We will prove in fact that any finite sequence $a_1, \ldots, a_k$ with the property can be extended by a suitable a_{k+1}. Let $s_k = a_1 + \cdots + a_k$ and $t_k = a_1^2 + \cdots + a_k^2$. Then we are seeking a_{k+1} such that $a_{k+1} + s_k | a_{k+1}^2 + t_k$. This is clearly equivalent to $a_{k+1} + s_k | s_k^2 + t_k$. Why not, then, choose $a_{k+1} = s_k^2 - s_k + t_k$? Certainly this is greater than a_k and ensures that the desired property is satisfied.

Problem 9.3.24. *The sequence $a_0, a_1, a_2, \ldots$ satisfies*

$$a_{m+n} + a_{m-n} = \frac{1}{2}(a_{2m} + a_{2n})$$

for all nonnegative integers m and n with $m \geq n$. If $a_1 = 1$, determine a_n.

(1995 Russian Mathematical Olympiad)

Solution. The relations $a_{2m} + a_{2m} = 2(a_{2m} + a_0) = 4(a_m + a_m)$ imply $a_{2m} = 4a_m$, as well as $a_0 = 0$. Thus we compute $a_2 = 4$, $a_4 = 16$. Also, $a_1 + a_3 = (a_2 + a_4)/2 = 10$ so $a_3 = 9$. At this point we guess that $a_i = i^2$ for all $i \geq 1$.

We prove our guess by induction on i. Suppose that $a_j = j^2$ for $j < i$. Then the given equation with $m = i - 1$, $j = 1$ gives

$$\begin{aligned} a_i &= \frac{1}{2}(a_{2i-2} + a_2) - a_{i-2} \\ &= 2a_{i-1} + 2a_1 - a_{i-2} \\ &= 2(i^2 - 2i + 1) + 2 - (i^2 - 4i + 4) = i^2. \end{aligned}$$

Problem 9.3.25. *The sequence of real numbers $a_1, a_2, a_3, \ldots$ satisfies the initial conditions $a_1 = 2$, $a_2 = 500$, $a_3 = 2000$ as well as the relation*

$$\frac{a_{n+2} + a_{n+1}}{a_{n+1} + a_{n-1}} = \frac{a_{n+1}}{a_{n-1}}$$

for $n = 2, 3, 4, \ldots$ Prove that all the terms of this sequence are positive integers and that 2^{2000} divides the number a_{2000}.

(1999 Slovenian Mathematical Olympiad)

Solution. From the recursive relation it follows that $a_{n+2}a_{n-1} = a_{n+1}^2$ for $n = 2, 3, \ldots$ No term of our sequence can equal 0, and hence it is possible to write

$$\frac{a_{n+2}}{a_{n+1}a_n} = \frac{a_{n+1}}{a_n a_{n-1}}$$

for $n = 2, 3, \ldots$ It follows by induction that the value of the expression

$$\frac{a_{n+1}}{a_n a_{n-1}}$$

is constant, namely equal to $\frac{a_3}{a_2 a_1} = 2$. Thus $a_{n+2} = 2a_n a_{n+1}$ and all terms of the sequence are positive integers.

From this new relation, we also know that $\frac{a_{n+1}}{a_n}$ is an even integer for all positive integers n. Write

$$a_{2000} = \frac{a_{2000}}{a_{1999}} \frac{a_{1999}}{a_{1998}} \cdots \frac{a_2}{a_1} a_1.$$

In this product each of the 1999 fractions is divisible by 2, and $a_1 = 2$ is even as well. Thus a_{2000} is indeed divisible by 2^{2000}.

Problem 9.3.26. *Let k be a fixed positive integer. We define the sequence $a_1, a_2, \ldots$ by $a_1 = k + 1$ and the recursion $a_{n+1} = a_n^2 - ka_n + k$ for $n \geq 1$. Prove that a_m and a_n are relatively prime for distinct positive integers m and n.*

Solution. We claim that

$$a_n = \prod_{i=0}^{n-1} a_i + k, \quad n > 0,$$

assuming that $a_0 = 1$. Since $a_{j+1} - k = a_j(a_j - k)$, we have

$$a_n - k = \prod_{j=1}^{n-1} \frac{a_{j+1} - k}{a_j - k} = \prod_{j=1}^{n-1} a_j,$$

which is what we wanted.

Therefore, we have that $a_n \equiv k \pmod{a_i}$ for $i < n$. Hence, if there exist integers $d > 1$, $x, y \geq 1$ such that $d|a_x$ and $d|a_y$, d divides k. We now show that for $i > 0$, $a_i \equiv 1 \pmod{k}$ by induction on i. For the base case, $a_1 = k + 1 \equiv 1 \pmod{k}$. Now assume that $a_i \equiv 1 \pmod{k}$. Then, $a_{i+1} \equiv a_i^2 - ka_i + k \equiv a_i^2 \equiv 1 \pmod{k}$. Thus, because all common divisors d of a_x and a_y must be divisors of k, we have $a_x \equiv 1 \pmod{d}$ and $a_y \equiv 1 \pmod{d}$. Therefore, no such divisors exist and a_i is relatively prime to a_j for all $i, j > 0$, as desired.

Problem 9.3.27. *Suppose the sequence of nonnegative integers $a_1, a_2, \ldots, a_{1997}$ satisfies*

$$a_i + a_j \leq a_{i+j} \leq a_i + a_j + 1$$

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for all $i, j \geq 1$ with $i + j \leq 1997$. Show that there exists a real number x such that $a_n = \lfloor nx \rfloor$ for all $1 \leq n \leq 1997$.

(1997 USA Mathematical Olympiad)

Solution. Any x that lies in all of the half-open intervals

$$I_n = \left[\frac{a_n}{n}, \frac{a_n + 1}{n} \right), \quad n = 1, 2, \ldots, 1997$$

will have the desired property. Let

$$L = \max_{1 \leq n \leq 1997} \frac{a_n}{n} = \frac{a_p}{p} \quad \text{and} \quad U = \min_{1 \leq n \leq 1997} \frac{a_n + 1}{n} = \frac{a_q + 1}{q}.$$

We shall prove that

$$\frac{a_n}{n} < \frac{a_m + 1}{m},$$

or, equivalently,

$$ma_n < n(a_m + 1) \tag{$*$}$$

for all m, n ranging from 1 to 1997. Then $L < U$, since $L \geq U$ implies that $(*)$ is violated when $n = p$ and $m = q$. Any point x in $[L, U)$ has the desired property.

We prove $(*)$ for all m, n ranging from 1 to 1997 by strong induction. The base case $m = n = 1$ is trivial. The induction step splits into three cases. If $m = n$, then $(*)$ certainly holds. If $m > n$, then the induction hypothesis gives $(m-n)a_n < n(a_{m-n}+1)$, and adding $n(a_{m-n}+a_n) \leq na_m$ yields $(*)$. If $m < n$, then the induction hypothesis yields $ma_{n-m} < (n-m)(a_m+1)$, and adding $ma_n \leq m(a_m + a_{n-m} + 1)$ gives $(*)$.

Problem 9.3.28. *The sequence $\{a_n\}$ is given by the following relation:*

$$a_{n+1} = \begin{cases} \dfrac{a_n - 1}{2}, & \text{if} \quad a_n \geq 1, \\ \dfrac{2a_n}{1 - a_n}, & \text{if} \quad a_n < 1. \end{cases}$$

Given that a_0 is a positive integer, $a_n \neq 2$ for each $n = 1, 2, \ldots, 2001$, and $a_{2002} = 2$. Find a_0.

(2002 St. Petersburg City Mathematical Olympiad)

Solution. Answer: $a_0 = 3 \cdot 2^{2002} - 1$.

We will first show that this value actually satisfies the condition $a_{2002} = 2$ and $a_i \neq 2$ for any $i < 2002$. Applying the first rule, $a_{n+1} = \dfrac{a_n - 1}{2002}$ times

will show that a_{2002} is in fact 2 and a_i is actually greater than 2 for all $i < 2002$.

Lemma. *For $n \leq 2000$, if a_n is not an integer, $a_n = p_n/q_n$, where p_n and q_n are positive odd integers, $(p, q) = 1$ and $q > 1$.*

Proof. We proceed by using induction. Case case: $n = 2000$. a_{2000} is either $5/7$ or $1/5$. Suppose $a_{n+1} = p/q$. Then:

Case 1. $a_n = 2p/q + 1$. Then $a_n = (2p + q)/q$. Suppose $d|(2p + q)$ and $d|q$. Then $d|(2p + q - q)$ or $d|2p$. Since q is odd, d must also be odd. So $(d, 2) = 1$ and therefore $d|p$. Because $d|p$, $d|q$, and $(p, q) = 1$, d must equal 1. Therefore $(2p+q, q) = 1$ and since $2p+q$a and q are both odd as well, a_n satisfies the conditions stated in the lemma, thus completing the inductive step.

Case 2. $a_n = \dfrac{p}{q} \Big/ \left(2 + \dfrac{p}{q}\right) = \dfrac{p}{2q + p}$. Again suppose d divides both the numerator p and denominator $2p+q$. Then $d|(2p+q-p)$ or $d|2q$. But since p is odd, d is odd, and so $d|q$. Because $(p, q) = 1$, d is equal to 1 and so $(p, 2q + p) = 1$. Since p and $2q + p$ are odd and are relatively prime, a_n satisfies the conditions and completes the proof.

We must now only consider the case where $a_{2001} = 1/2$. In this case, a_{2000} is either 2 or $5/7$. If it is 2, the conditions of the problem are violated. The lemma says that from the $5/7$, we will never see an integer value for any previous term.

Problem 9.3.29. *Let $x_1 = x_2 = x_3 = 1$ and $x_{n+3} = x_n + x_{n+1}x_{n+2}$ for all positive integers n. Prove that for any positive integer m there is an integer $k > 0$ such that m divides x_k.*

Solution. Observe that setting $x_0 = 0$ the condition is satisfied for $n = 0$.

We prove that there is integer $k \leq m^3$ such that x_k divides m. Let r_t be the remainder of x_t when divided by m for $t = 0, 1, \ldots, m^3 + 2$. Consider the triples (r_0, r_1, r_2), $(r_1, r_2, r_3), \ldots, (r_{m^3}, r_{m^3+1}, r_{m^3+2})$. Since r_t can take m values, it follows by the Pigeonhole Principle that at least two triples are equal. Let p be the smallest number such that triple (r_p, r_{p+1}, r_{p+2}) is equal to another triple (r_q, r_{q+1}, r_{q+2}), $p < q \leq m^3$. We claim that $p = 0$.

Assume by way of contradiction that $p \geq 1$. Using the hypothesis we have

$$r_{p+2} \equiv r_{p-1} + r_p r_{p+1} \pmod{m}$$

and

$$r_{q+2} \equiv r_{q-1} + r_q r_{q+1} \pmod{m}.$$

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Since $r_p = r_q$, $r_{p+1} = r_{q+1}$ and $r_{p+2} = r_{q+2}$, it follows that $r_{p-1} = r_{q-1}$ so $(r_{p-1}, r_p, r_{p+1}) = (r_{q-1}, r_q, r_{q+1})$, which is a contradiction with the minimality of p. Hence $p = 0$, so $r_q = r_0 = 0$, and therefore $x_q \equiv 0 \pmod{m}$.

Problem 9.3.30. *Find all infinite bounded sequences $a_1, a_2, \ldots$ of positive integers such that for all $n > 2$,*

$$a_n = \frac{a_{n-1} + a_{n-2}}{gcd(a_{n-1}, a_{n-2})}.$$

(1999 Russian Mathematical Olympiad)

Solution. The only such sequence is 2,2,2,... which clearly satisfies the given condition.

Let $g_n = gcd(a_n, a_{n+1})$. Then g_{n+1} divides both a_{n+1} and a_{n+2}, so it divides $g_n a_{n+2} - a_{n-1} = a_n$ as well. Thus g_{n+1} divides both a_n and a_{n+1}, and it divides their greatest common divisor g_n.

Therefore, the g_i form a nonincreasing sequence of positive integers and eventually equal some positive constant g. At this point, the a_i satisfy the recursion

$$ga_n = a_{n-1} + a_{n-2}.$$

If $g = 1$, then $a_n = a_{n-1} + a_{n-2} > a_{n-1}$ so the sequence is increasing and unbounded.

If $g \geq 3$, then

$$a_n = \frac{a_{n-1} + a_{n-2}}{g} < \frac{a_{n-1} + a_{n-2}}{2} \leq \max\{a_{n-1}, a_{n-2}\}.$$

Similarly, $a_{n+1} < \max\{a_{n-1}, a_n\} \leq \max\{a_{n-2}, a_{n-1}\}$, so that

$$\max\{a_n, a_{n+1}\} < \max\{a_{n-2}, a_{n-1}\}.$$

Therefore the maximum values of successive pairs of terms form an infinite decreasing sequence of positive integers, a contradiction.

Thus $g = 2$ and eventually we have $2a_n = a_{n-1} + a_{n-2}$ or $a_n - a_{n-1} = -\frac{1}{2}(a_{n-1} - a_{n-2})$. This implies that $a_i - a_{i-1}$ converges to 0 and that the a_i are eventually constant as well. From $2a_n = \frac{a_{n-1} + a_{n-2}}{gcd(a_{n-1}, a_{n-2})}$ this constant must be 2.

Now if $a_n = a_{n+1} = 2$ for $n > 1$, then $gcd(a_{n-1}, a_n) = gcd(a_{n-1}, 2)$ either equals 1 or 2. Now

$$2 = a_{n+1} = \frac{a_{n-1} + a_n}{gcd(a_{n-1}, 2)},$$

implying either that $a_{n-1} = 0$, which is impossible, or that $a_{n-1} = 2$. Therefore all the a_i equal 2, as claimed.

Problem 9.3.31. *Let $a_1, a_2, \ldots$ be a sequence of positive integers satisfying the condition $0 < a_{n+1} - a_n \leq 2001$ for all integers $n \geq 1$. Prove that there exist an infinite number of ordered pairs (p, q) of distinct positive integers such that a_p is a divisor of a_q.*

(2001 Vietnamese Mathematical Olympiad)

Solution. Obviously, if $(a_n)_n$ is such a sequence, so is $(a_{n+k})_n$ for all k. Thus it suffices to find $p < q$ such that $a_p | a_q$. Observe that from any 2001 consecutive natural numbers, at least one is a term of the sequence. Now, consider the table:

$$\begin{array}{cccc} a_1 + 1 & a_1 + 2 & \ldots & a_1 + 2001 \\ a_1 + 1 + x & a_1 + 2 + x_1 & \ldots & a_1 + 2001 + x_1 \\ a_1 + 1 + x_1 + x_2 & a_1 + 2 + x_1 + x_2 & \ldots & a_1 + 2001 + x_1 + x_2 \\ \vdots & & & \end{array}$$

where

$$x_1 = \prod_{i=1}^{2001} (a_1 + i), \quad x_2 = \prod_{i=1}^{2001} (a_1 + i + x_1), \quad x_3 = \prod_{i=1}^{2001} (a_1 + x_1 + x_2 + i)$$

and so on. Observe then that if x, y are on the same column, then $x|y$ or $y|x$. Now, look at the first 2002 lines. We find in this 2002×2001 matrix at least 2002 terms of the sequence (at least one on each line), thus there are two terms of the sequence on the same column and one will divide the other.

Problem 9.3.32. *Define the sequence $\{x_n\}_{n \geq 0}$ by $x_0 = 0$ and*

$$x_n = \begin{cases} x_{n-1} + \dfrac{3^{r+1} - 1}{2}, & \text{if} \quad n = 3^r(3k+1), \\ \\ x_{n-1} - \dfrac{3^{r+1} + 1}{2}, & \text{if} \quad n = 3^r(3k+2), \end{cases}$$

where k and r are nonnegative integers. Prove that every integer appears exactly once in this sequence.

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(1999 Iranian Mathematical Olympiad)

Solution. We prove by induction on $t \geq 1$ that

(i) $\{x_0, x_1, \ldots, x_{3^t-2}\} = \left\{-\dfrac{3^t-3}{2}, -\dfrac{3^t-5}{2}, \ldots, \dfrac{3^t-1}{2}\right\}$.

(ii) $x_{3^t-1} = -\dfrac{3^t-1}{2}$.

These claims imply the desired result, and they are easily verified for $t = 1$. Now supposing they are true for t, we show they are true for $t+1$.

For any positive integer m, write $m = 3^r(3k+s)$ for nonnegative integers r, k, s, with $s \in \{1, 2\}$, and define $r_m = r$ and $s_m = s$.

Then for $m < 3^t$, observe that

$$r_m = r_{m+3^t} = r_{m+2\cdot 3^t} \quad \text{and} \quad s_m = s_{m+3^t} = r_{m+2\cdot 3^t},$$

so that

$$x_m - x_{m-1} = x_{3^t+m} - x_{3^t+m-1} = x_{2\cdot 3^t+m} - x_{2\cdot 3^t+m-1}.$$

Setting $m = 1, 2, \ldots, k < 3^t$ and adding the resulting equations, we have

$$x_k = x_{3^t+k} - x_{3^t}$$

$$x_k = x_{2\cdot 3^t+k} - x_{2\cdot 3^t}.$$

Now, setting $n = 3^t$ in the recursion and using (ii) from the induction hypothesis, we have $x_{3^t} = 3^t$, and

$$\{x_{3^t}, \ldots, x_{2\cdot 3^t-2}\} = \left\{\frac{3^t+3}{2}, \ldots, \frac{3^{t+1}-1}{2}\right\},$$

$$x_{2\cdot 3^t-1} = \frac{3^t+1}{2}.$$

Then setting $n = 2 \cdot 3^t$ in the recursion we have $x_{2\cdot 3^t} = -3^t$, giving

$$\{x_{2\cdot 3^t}, \ldots, x_{3^{t+1}-2}\} = \left\{-\frac{3^{t+1}-3}{2}, \ldots, \frac{3^t+1}{2}\right\}$$

$$x_{2\cdot 3^{t+1}-1} = -\frac{3^{t+1}-1}{2}.$$

Combining this with (i) and (ii) from the induction hypothesis proves the claims for $t+1$. This completes the proof.

Second solution. For $n_i \in \{-1, 0, 1\}$, let the number

$$[n_m n_{m-1} \ldots n_0]$$

in base $\overline{3}$ equal $\sum_{i=0}^{m} n_i \cdot 3^i$. It is simple to prove by induction on k that the base $\overline{3}$ numbers with at most k digits equal

$$\left\{-\frac{3^k-1}{2}, -\frac{3^k-3}{2}, \ldots, \frac{3^k-1}{2}\right\},$$

which implies that every integer has a unique representation in base $\overline{3}$.

Now we prove by induction on n that if $n = a_m a_{m-1} \ldots a_0$ in base 3, then $x_n = [b_m b_{m-1} \ldots b_0]$ in base $\overline{3}$, where $b_i = -1$ if $a_i = 2$ and $b_i = a_i$ for all other cases.

For the base case, $x_0 = 0 = [0]$. Now assume the claim is true for $n-1$. If $n = a_m a_{m-1} \ldots a_{r+1} 1 \underbrace{00\ldots0}_{r}$, then

$$\begin{aligned} x_n &= x_{n-1} + \frac{3^{r+1}-1}{2} \\ &= [b_m b_{m-1} \ldots b_i 0 \underbrace{-1-1\cdots-1}_{r}] + [\underbrace{11\ldots1}_{r+1}] \\ &= [b_m b_{m-1} \ldots b_i 1 \underbrace{00\ldots0}_{r}]. \end{aligned}$$

If instead $n = a_m a_{m-1} \ldots a_i 2 \underbrace{00\ldots0}_{r}$, then

$$\begin{aligned} x_n &= x_{n-1} + \left(-\frac{3^{r+1}+1}{2}\right) \\ &= [b_m b_{m-1} \ldots b_i 1 \underbrace{-1-1\cdots-1}_{r}] + [-1 \underbrace{11\ldots1}_{r+1}] \\ &= [b_m b_{m-1} \ldots b_i - 1 \underbrace{00\ldots0}_{r}]. \end{aligned}$$

In either case, the claim is true for n, completing the induction.

To finish the proof, note that every integer appears exactly once in base $\overline{3}$. Thus each integer appears exactly once in $\{x_n\}_{n\geq 0}$, as desired.

Problem 9.3.33. *Suppose that $a_1, a_2, \ldots$ is a sequence of natural numbers such that for all natural numbers m and n, $gcd(a_m, a_n) = a_{gcd(m,n)}$. Prove that there exists a sequence $b_1, b_2, \ldots$ of natural numbers such that $a_n = \prod_{d|n} b_d$ for all integers $n \geq 1$.*

(2001 Iranian Mathematical Olympiad)

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Solution. For each n, let $\text{rad}(n)$ denote the largest square-free divisor of n (i.e. the product of all distinct prime factors of n). We let b_n equal to the ratio of the following two numbers:

- E_n, the product of all $a_{n/d}$ such that d is square-free, divides n, and has an even number of prime factors.
- O_n, the product of all $a_{n/d}$ such that d is square-free, divides n, and has an odd number of prime factors.

Lemma 1. $\displaystyle\prod_{d|a_n} b_d = a_n$.

Proof. Fix n, and observe that $\displaystyle\prod_{d|n} b_n$ equals

$$\frac{\displaystyle\prod_{d|n} E_d}{\displaystyle\prod_{d|n} O_d}. \qquad (*)$$

In the numerator of $(*)$, each E_d is the product of a_m such that $m|d$. Also, $d|n$, implying that the numerator is the product of various a_m such that $m|n$. For fixed m that divides n, how many times does a_m appears in the numerator $\displaystyle\prod_{d|n} E_d$ of $(*)$?

If a_m appears in E_d and $d|n$, then let $t = d/m$. By the definition of E_d, we know that (i) t is square-free and (ii) t has an even number of prime factors. Because $d|n$ and $t = d/m$, we further know that (iii) t divides n/m.

Conversely, suppose that t is any positive integer satisfying (i), (ii), and (iii), and write $d = tm$. By (iii), d is a divisor of n. Also, t is square-free by (i), is a divisor of d, and has an even number of prime factors by (ii). Thus, a_m appears in E_d.

Suppose that n/m has l distinct prime factors. Then it has $\binom{l}{0} + \binom{l}{2} + \ldots$ factors t satisfying (i), (ii), and (iii), implying that a_m appears in the numerator of $(*)$ exactly

$$\binom{l}{0} + \binom{l}{2} + \ldots$$

times. Similarly, a_m appears in the denominator of $(*)$ exactly

$$\binom{l}{1} + \binom{l}{3} + \ldots$$

times. If $m < n$, then $l \geq 1$ and these expressions are equal, so that the a_m's in the numerator and denominator of $(*)$ cancel each other out. If $m = n$, then $l = 0$, so that a_n appears in the numerator once and in the denominator zero times. Therefore,

$$\prod_{d|n} b_d = \frac{\prod_{d|n} E_d}{\prod_{d|n} O_d} = a_n,$$

as desired. □

Lemma 2. *For any integer α that divides some term in $a_1, a_2, \ldots$, there exists an integer d such that*

$$\alpha | a_n \;\Leftrightarrow\; d|n.$$

Proof. Of all the integers n such that $\alpha|a_n$, let d be the smallest.

If $\alpha|a_n$, then $\alpha|gcd(a_d, a_n) = a_{gcd(d,n)}$. By the minimal definition of d, $gcd(d, n) \geq d$. But $gcd(d, n)|n$ as well, implying that $gcd(d, n) = d$. Hence, $d|n$.

If $d|n$, then $gcd(a_d, a_n) = a_{gcd(d,n)} = a_d$. Thus, $a_d|a_n$. Because $\alpha|a_d$, it follows that $\alpha|a_n$ as well. □

Lemma 3. *For each positive integer n, $b_n = E_n/O_n$ is an integer.*

Proof. Fix n. Call an integer d a top divisor (resp. a bottom divisor) if $d|n$, n/d is square-free, and n/d has an even (resp. odd) number of prime factors. By definition, E_d is the product of a_d over all top divisors d, and O_d is the product of a_d over all bottom divisors d.

Fix any prime p. We show that p divides E_n at least as many times as it divides O_n. To do this, it suffices to show the following for any positive integer k:

(1) The number of top divisors d with $a_{n/d}$ divisible by p^k is greater than or equal to the number of bottom divisors d with $a_{n/d}$ divisible by p^k.

Let k be any positive integer. If p^k divides none of $a_1, a_2, \ldots$, then (1) holds trivially. Otherwise, by the previous lemma, there exists an integer d_0 such that

$$p^k | a_m \;\Leftrightarrow\; d_0|m.$$

Hence, to show (1) it suffices to show:

(2) The number of top divisors d such that $d_0|(n/d)$, is greater than or equal to the number of bottom divisors d such that $d_0|(n/d)$.

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If $d_0 \nmid n$, then (2) holds because d_0 does not divide n/d for any divisor d of n, including top or bottom divisors.

Otherwise, $d_0|n$. For which top and bottom divisors d does d_0 divide n/d? Precisely those for which d divides n/d_0. If n/d_0 has $l \geq 1$ distinct prime factors, then there are as many top divisors with this property as there are bottom divisors, namely

$$\binom{l}{0} + \binom{l}{2} + \cdots = 2^{l-1} = \binom{l}{1} + \binom{l}{3} + \dots$$

If instead $d_0 = n$ and $l = 0$, then the top divisor 1 is the only value d with $d|(n/d_0)$. In either case, there are at least as many top divisors d with $d|(n/d_0)$ as there are bottom divisors with the same property. Therefore, (2) holds. This completes the proof. □

Therefore, $a_n = \prod_{d|n} b_d$, and $b_n = E_n/O_n$ is an integer for each n.

Alternative solution. (Gabriel Dospinescu) Let us define $b_1 = a_1$ and $b_n = \dfrac{a_n}{\operatorname*{lcm}\limits_{\substack{d|n\\ d\neq n}} a_d}$ for $n > 1$. Of course, if $d|n$, then $a_d|a_n$ and so $\operatorname*{lcm}\limits_{\substack{d|n\\ d\neq n}} a_d|a_n$ and $b_n \in \mathbb{Z}$.

Now comes the hard part, proving that $\prod_{d|n} b_d = a_n$, which is the same as

$$\prod_{\substack{d|n\\ d\neq n}} b_d = \operatorname*{lcm}_{\substack{d|n\\ d\neq n}} a_d. \tag{1}$$

We will prove (1) by strong induction. For $n = 1$ it is clear

Now, for all $d|n$, $d \neq n$, by inductive hypothesis we have

$$a_d = \prod_{d'|d} b_{d'} \Big| \prod_{\substack{d|n\\ d\neq n}} b_d,$$

thus $\prod\limits_{\substack{d|n\\ d\neq n}} b_d$ is a multiple of $\operatorname*{lcm}\limits_{\substack{d|n\\ d\neq n}} a_d$. It remains to prove that $\prod\limits_{\substack{d|n\\ d\neq n}} b_d \Big| \operatorname*{lcm}\limits_{\substack{d|n\\ d\neq n}} a_d$.

The essential observation is:

Lemma. *If $gcd(b_u, b_v) > 1$, then $u|v$ or $v|u$.*

Proof. We may assume that $u < v$. Assume that u does not divide v. Then

$$b_u = \frac{a_u}{\operatorname*{lcm}\limits_{\substack{d|u\\ d\neq u}} a_d} \Big| \frac{a_u}{a_{gcd(u,v)}}$$

(since $gcd(u,v) \neq u$) and also

$$b_v | \frac{a_v}{a_{gcd(u,v)}}.$$

But then

$$gcd(b_u, b_v) | gcd\left(\frac{a_u}{a_{gcd(u,v)}}, \frac{a_v}{a_{gcd(u,v)}}\right) = \frac{gcd(a_u, a_v)}{a_{gcd(u,v)}}.$$

Now, let p be a prime number and let $i_1 < i_2 < \cdots < i_k$ such that $i_q | n$, $i_q \neq n$ for all q and b_{i_q} has the exponent of p equal to $e_q \geq 1$ and any b_d with $d|n$, $d \neq n$, $d \neq i_q$ for all q is not a multiple of p. From the lemma we find $i_1 | i_2 | \ldots | i_k$. But then from the inductive hypothesis we have

$$a_{i_k} = \prod_{d|i_k} b_d$$

is a multiple of $b_{i_1} b_{i_2} \ldots b_{i_k}$, thus $\underset{\substack{d|n \\ d \neq n}}{lcm}\, a_d$ has the exponent of p greater than or equal to the exponent of p in $\prod_{\substack{d|n \\ d \neq n}} b_d$. This ends the solution.

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21

Problems Involving Binomial Coefficients

21.1 Binomial coefficients

Problem 10.1.7. *Show that the sequence*

$$\binom{2002}{2002}, \binom{2003}{2002}, \binom{2004}{2002}, \dots$$

considered modulo 2002*, is periodic.*

(2002 Baltic Mathematical Competition)

Solution. We will show that the sequence, taken modulo 2002, has period $m = 2002 \cdot 2002!$. Indeed,

$$\begin{aligned}\binom{x+m}{2002} &= \frac{(x+m)(x-1+m)\dots(x-2001+m)}{2002!} \\ &= \frac{x(x-1)\dots(x-2001)+km}{2002!} \\ &= \frac{x(x-1)\dots(x-2001)}{2002!} + 2002k \\ &\equiv \binom{x}{2002} \pmod{2002}.\end{aligned}$$

Problem 10.1.8. *Prove that*

$$\binom{2p}{p} \equiv 2 \pmod{p^2}$$

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for any prime number p.

Solution. A short solution uses the popular Vandermonde identity

$$\sum_{i=0}^{k} \binom{m}{i}\binom{n}{k-i} = \binom{m+n}{k}.$$

Set $m = n = k = p$ to get

$$\binom{2p}{p} = \binom{p}{0}\binom{p}{p} + \binom{p}{1}\binom{p}{p-1} + \cdots + \binom{p}{p-1}\binom{p}{1} + \binom{p}{p}\binom{p}{0}.$$

The first and the last term on the right-hand side equal 1. Since p is a prime, it divides each binomial coefficient $\binom{p}{k}$ for $1 \leq k \leq p-1$. So each of the remaining terms is divisible by p^2, and hence $\binom{2p}{p}$ is congruent to 2 modulo p^2, as required.

Problem 10.1.9. *Let k, m, n be positive integers such that $m+k+1$ is a prime number greater than $n+1$. Let us denote $C_s = s(s+1)$. Show that the product*

$$(C_{m+1} - C_k)(C_{m+2} - C_k)\ldots(C_{m+n} - C_k)$$

is divisible by $C_1C_2\ldots C_n$.

(18th IMO)

Solution. We use the identity:

$$C_p - C_q = p(p+1) - q(q+1) = (p-q)(p+q+1),$$

which is valid for all positive integers p and q. Then one has:

$$C_{m+i} - C_k = (m-k+i)(m+k+i+1),\ \forall\ i = 1, 2, \ldots, n.$$

For the given products we obtain respectively, the formulas

$$(C_{m+1} - C_k)\ldots(C_{m+n} - C_k) = \prod_{i=1}^{n}(m-k+i)\prod_{i=1}^{n}(m+k+1+i)$$

$$C_1C_2\ldots C_n = n!(n+1)!$$

Their quotient is the product of two rational fractions:

$$\frac{\prod_{i=1}^{n}(m-k+i)}{n!} \quad \text{and} \quad \frac{\prod_{i=1}^{n}(m+k+1+i)}{(n+1)!}$$

It is known that the product of any consecutive integers is divisible by $n!$ and their quotient is zero or a binomial coefficient, possible multiplied by -1. In our case we have

$$\frac{1}{n!}\prod_{i=1}^{n}(m-k+i)=\binom{m-k+n}{n}.$$

For the second fraction, a factor is missing to the numerator. We support our argument by using that $m+k+1$ is a prime number greater than $n+1$:

$$\frac{1}{(n+1)!}\prod_{i=1}^{n}(m+k+1+i)=\frac{1}{m+k+1}\cdot\frac{1}{(n+1)!}\prod_{i=0}^{n}(m+k+1+i)$$

$$=\frac{1}{m+k+1}\binom{m_k+n+1}{n+1}.$$

The binomial coefficient $\binom{m+k+n+1}{n+1}$ is an integer number which is divisible by $m+k+1$, then our number is integer.

Problem 10.1.10. *Let n,k be arbitrary positive integers. Show that there exists positive integers $a_1>a_2>a_3>a_4>a_5>k$ such that*

$$n=\pm\binom{a_1}{3}\pm\binom{a_2}{3}\pm\binom{a_3}{3}\pm\binom{a_4}{3}\pm\binom{a_5}{3}.$$

(2000 Romanian IMO Team Selection Test)

Solution. For fixed k, choose $m>k$ such that $n+\binom{m}{3}$ is an odd number. This is possible after considering the parity of n. If n is an odd number, take $m\equiv 0 \pmod 4$ and if n is an even number take $m\equiv 3 \pmod 4$.

Since $n+\binom{m}{3}$ is an odd number, we express it under the form

$$m+\binom{m}{3}=2a+1.$$

Then use the identity:

$$2a+1=\binom{a}{3}-\binom{a+1}{3}-\binom{a+2}{3}+\binom{a+3}{3}$$

and obtain:

$$n=\binom{a}{3}-\binom{a+1}{3}-\binom{a+2}{3}+\binom{a+3}{3}-\binom{m}{3}.$$

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Notice that for large m we may insure that

$$a = \frac{n - 1 + \binom{m}{3}}{2} > m$$

yielding the desired representation.

Problem 10.1.11. *Prove that if n and m are integers, and m is odd, then*

$$\frac{1}{3^m n} \sum_{k=0}^{m} \binom{3m}{3k} (3n-1)^k$$

is an integer.

(2004 Romanian IMO Team Selection Test)

Solution. Let $\omega = e^{\frac{2\pi i}{3}}$. Then

$$3 \sum_{k=0}^{m} \binom{3m}{3k} (3n-1)^k$$

$$= (1 + \sqrt[3]{3n-1})^{3m} + (1 + \omega \sqrt[3]{3n-1})^{3m} + (1 + \omega^2 \sqrt[3]{3n-1})^{3m}. \qquad (1)$$

The right side of the above equality is the sum of the $3m$-th power of the roots x_1, x_2, x_3 of the polynomial

$$(X-1)^3 - (3n-1) = X^3 - 3X^2 + 3X - 3n.$$

Let $s_k = x_1^k + x_2^k + x_3^k$. Then $s_0 = s_1 = s_2 = 3$ and

$$s_{k+3} = 3s_{k+2} - 3s_{k+1} + 3ns_k. \qquad (2)$$

It follows by induction that each s_k is an integer divisible by $3^{\left[\frac{k}{3}\right]+1}$. A repeated application of (2) yields

$$s_{k+7} = 63ns_{k+2} - 9(n^2 - 3n - 3)s_{k+1} + 27n(2n+1)s_k.$$

Since $s_3 = 9n$, it follows inductively that s_{6k+3} is divisible by $3^{2k+2}n$ for all nonnegative integers k, and the conclusion follows by (1).

Problem 10.1.12. *Show that for any positive integer n the number*

$$\sum_{k=0}^{n} \binom{2n+1}{2k+1} 2^{3k}$$

is not divisible by 5.

(16th IMO)

Solution. Let us consider the binomial formula:

$$(1+2\sqrt{2})^{2n+1} = (1+2^{\frac{3}{2}})^{2n+1} = \sum_{i=0}^{2n+1} \binom{2n+1}{i} 2^{\frac{3i}{2}}$$

$$= \sum_{i=0}^{n} \binom{2n+1}{2i} 2^{3i} + \sum_{i=0}^{n} \binom{2n+1}{2i+1} 2^{3i} \cdot 2^{\frac{3}{2}} = a_n + b_n\sqrt{8},$$

where

$$a_n = \sum_{i=0}^{n} \binom{2n+1}{2i} 2^{3i} \quad \text{and} \quad b_n = \sum_{i=0}^{n} \binom{2n+1}{2i+1} 2^{3i}.$$

In a similar way,

$$(1-2^{\frac{3}{2}})^{2n+1} = a_n - b_n\sqrt{8}.$$

After multiplying these two equalities we obtain $-7^{2n+1} = a_n^2 - 8b_n^2$. If $b_n \equiv 0 \pmod 5$ the above equality gives $a_n^2 \equiv -2 \pmod 5 \equiv 3 \pmod 5$. Since 3 is not a perfect square modulo 5, we obtain a contradiction.

Problem 10.1.13. *Prove that for a positive integer k there is an integer $n \geq 2$ such that $\binom{n}{1}, \ldots, \binom{n}{n-1}$ are all divisible by k if and only if k is a prime.*

Solution. If k is a prime we take $n = k$ and the property holds (see property 7)). There are k's for which $\binom{n}{1}, \ldots, \binom{n}{n-1}$ are not all divisible by k, for any $n \geq 2$. Indeed, for $k = 4$,

$$\binom{n}{1} + \cdots + \binom{n}{n-1} = 2^n - 2 \equiv -2 \pmod 4 \text{ for all } n \geq 2.$$

We prove that the set of positive integers k for which the claim holds is exactly the set of primes.

Suppose now that k is not a prime. Then consider two cases:

(a) $k = p^r$, where p is a prime and $r > 1$. We find a value of i for which the statement does not hold.

Suppose, on the contrary, that there is a positive integer n such that for all $1 \leq i \leq n-1$, $\binom{n}{i}$ is divisible by p^r. Clearly, n is divisible by p^r, and we write $n = p^\alpha\beta$ for some β with $gcd(\beta, p) = 1$. Take $i = p^{\alpha-1}$. Then

$$\binom{n}{i} = \prod_{j=0}^{p^{\alpha-1}-1} \frac{\beta p^\alpha - j}{p^{\alpha-1} - j}$$

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If $j = 0$, then $\dfrac{\beta p^\alpha - j}{p^{\alpha-1} - j} = \beta p$. If $gcd(j, p) = 1$, then both the above numerator and the denominator are coprime with p. In all other cases, we write $j = \delta p^\gamma$ for some δ coprime with p and $\gamma \leq \alpha - 2$. Thus,

$$\frac{\beta p^\alpha - j}{p^{\alpha-1} - j} = \frac{\beta p^\alpha - \delta p^\gamma}{p^{\alpha-1} - \delta p^\gamma} = \frac{p^\gamma(\beta p^{\alpha-\gamma} - \delta)}{p^\gamma(p^{\alpha-\gamma-1} - \delta)}$$

Now, since $\alpha-\gamma-1 \geq 1$, we have $\beta p^{\alpha-\gamma} - \delta$ and $p^{\alpha-\gamma-1} - \delta$ coprime with p. In this case, the power of p in the above numerator and the denominator is γ, and the power of p in the above product of fractions, which is an integer, is 1. This contradicts the assumption that $p^r | n$.

(b) k is divisible by at least two distinct primes p, q. Assume by contradiction that there is a positive integer n as required. Then n is divisible by pq and we can write $n = p^\alpha \beta$ where $gcd(p, \beta) = 1$ and $\beta > 1$ (since q divides β). Take $i = p^\alpha$. Then

$$\binom{n}{i} = \prod_{j=0}^{p^\alpha - 1} \frac{\beta p^\alpha - j}{p^\alpha - j}.$$

When $j = 0$, $\dfrac{\beta p^\alpha - j}{p^\alpha - j} = \beta$ is coprime with p. In all other cases, both the numerator and the denominator of $\dfrac{\beta p^\alpha - j}{p^\alpha - j}$ are either coprime with p or are divisible by the same power of p, and therefore the product of those fractions is not divisible by p. But p divides k, and hence $\binom{n}{i}$ is not divisible by k, contrary to our assumption.

Therefore the only positive integers k for which the claim holds are the primes.

21.2 Lucas' and Kummer's Theorems

Problem 10.2.4. *Let p be an odd prime. Find all positive integers n such that $\binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n-1}$ are all divisible by p.*

Solution. Express n in base p: $n = n_0 + n_1 p + \cdots + n_m p^m$, where $0 \leq n_0, n_1, \ldots, n_m \leq p-1$ and $n_m \leq 0$. We also write $k = k_0 + k_1 p + \cdots + k_m p^m$, where $0 \leq k_0, k_1, \ldots, k_m \leq p-1$, where k_m can be zero. From Lucas' Theorem we have

$$\binom{n}{k} \equiv \prod_{j=0}^{m} \binom{n_j}{k_j} \pmod{p}.$$

For $n = p^m$, the property clearly holds. Assume by way of contradiction that $n \neq p^m$. If $n_m > 1$, then letting $k = p^m < n$, we have

$$\binom{n}{k} \equiv n_m \cdot \underbrace{1 \ldots 1}_{m-1\,times} \equiv n_m \not\equiv 0 \pmod{p},$$

a contradiction.

Problem 10.2.5. *Let p be a prime. Prove that p does not divide any of $\binom{n}{1}, \ldots, \binom{n}{n-1}$ if and only if $n = sp^k - 1$ for some positive integer k and some integer s with $1 \leq s \leq p-1$.*

Solution. If n is of the form $sp^k - 1$, then its representation in base p is

$$n = \overline{(s-1)\underbrace{(p-1)\ldots(p-1)}_{k\,times}}.$$

For $1 \leq i \leq n-1$, $i = i_0 + i_1 p + \cdots + i_m p^m$, where $0 \leq i_h \leq p-1$, $h = 1, \ldots, m-1$ and $0 \leq i_m \leq s-1$. Because p is a prime, it follows that p does not divide either $\binom{p-1}{i_h}$ or $\binom{s-1}{i_m}$. Applying Lucas' Theorem, we obtain that p does not divide $\binom{n}{i}$, for all $i = 1, \ldots, n-1$.

Conversely, if n cannot be written in the form $sp^k - 1$, then $n_j < p-1$ for some $0 \leq j \leq m-1$, where $\overline{n_0 n_1 \ldots n_m}$ is the representation of n in base p. For

$$i = \overline{(p-1)\ \underbrace{0 \ldots 0}_{j-1\,times}}$$

in base p, applying again Lucas' Theorem, we have

$$\binom{n}{i} \equiv 0 \pmod{p}.$$

Problem 10.2.6. *Prove or disprove the following claim: For any integer $k \geq 2$, there exists an integer $n \geq 2$ such that the binomial coefficient $\binom{n}{i}$ is divisible by k for any $1 \leq i \leq n-1$.*

(1999 Hungarian-Israel Mathematical Competition)

Solution. The statement is false. To prove this, take $k = 4$ and assume by contradiction that there exists a positive integer $n > 1$ for which $\binom{n}{i}$ is divisible by 4 for every $1 \leq i \leq n-1$. Then

$$0 \equiv \sum_{i=1}^{n-1} \binom{n}{i} = 2^n - 2 \equiv -2 \pmod{4},$$

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a contradiction.

Remark. As we have already seen in Problem 10.1.13, the set of all such integers k is precisely the set of primes. Here we give an argument based on Kummer's Theorem.

Now, suppose the claim holds for some $k > 1$ with the number n. If some prime p divides k, the claim must also hold for p with the number n. Thus n must equal a prime power p^m where $m \geq 1$. Then $k = p^r$ for some $r \geq 1$ as well, because if two primes p and q divided k then n would equal a perfect power of both p and q, which is impossible.

Choose $i = p^{m-1}$. Kummer's Theorem states that $p^t | \binom{n}{i}$ if and only if t is less than or equal to the number of carries in the addition $(n-i)+i$ in base p. There is only such carry, between the p^{m-1} and p^m places:

$$\begin{array}{rccccccc} & 1 & & & & & & \\ & & 1 & 0 & 0 & \dots & 0 \\ + & & p-1 & 0 & 0 & \dots & 0 \\ \hline & 1 & 0 & 0 & 0 & \dots & 0 \end{array}$$

Thus, we must have $r \leq 1$ and k must be prime, as claimed.

(Alternatively, for $n = p^m$ and $i = p^{m-1}$ we have

$$\binom{n}{i} = \prod_{j=0}^{p^{m-1}-1} \frac{p^m - j}{p^{m-1} - j}.$$

When $j = 0$ then $\dfrac{p^m - j}{p^{m-1} - j} = p$. Otherwise, $0 < j < p^{m-1}$ so that if $p^t < p^{m-1}$ is the highest power of p dividing j, then it is also the highest power of p dividing both $p^m - j$ and $p^{m-1} - j$. Therefore $\dfrac{p^m - j}{p^{m-1} - j}$ contributes one factor of p to $\binom{n}{i}$ when $j = 0$ and zero factors of p when $j > 0$. Thus p^2 does not divide binomi, and hence again $r \leq 1$.)

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Miscellaneous Problems

Problem 11.6. *Let a, b be positive integers. By integer division of a^2+b^2 to $a+b$ we obtain the quotient q and the remainder r. Find all pairs (a, b) such that $q^2 + r = 1977$.*

(19th IMO)

Solution. There are finitely many possibilities to obtain $1977 = q^2 + r$. Since 1977 is not a perfect square, $0 < r < a + b$. Also, $q \leq [\sqrt{1977}] = 44$. From $a^2 + b^2 = q(a + b) + r$, we obtain:

$$q = \left[\frac{a^2+b^2}{a+b}\right] \geq \frac{a^2+b^2}{a+b} - 1 \geq \frac{1}{2}(a+b) - 1 > \frac{r}{2} - 1.$$

Suppose $q \leq 43$. Then $r = 1977 - q^2 \geq 1977 - 43^2 = 128$ and $43 \geq q > \frac{r}{2} - 1 \geq 63$, contradiction.

We obtained $q = 44$ and $r = 1977 - 44^2 = 41$. To finish, we have to solve in integer numbers the equation

$$a^2 + b^2 = 44(a + b) + 41.$$

Write it under the form:

$$(a - 22)^2 + (b - 22)^2 = 1009.$$

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It is not difficult to have all pairs of perfect squares having their sum 1009. There exists only the representation $1009 = 28^2 + 15^2$. Then the solutions are $a = 50$, $b = 37$ or $a = 37$, $b = 50$.

Problem 11.7. *Let m, n be positive integers. Show that $25^n - 7^m$ is divisible by 3 and find the least positive integer of the form $|25^n - 7^m - 3^m|$, when m, n run over the set of non-negative integers.*

(2004 Romanian Mathematical Regional Contest)

Solution. Because $25 \equiv 1 \pmod 3$ and $7 \equiv 1 \pmod 3$, it follows that $25^n - 7^m \equiv 0 \pmod 3$.

For the second part of the problem, we first remark that if m is odd, then any number $a = 25^n - 7^m - 3^m$ is divisible by 15. This follows from the first part together with

$$7^m + 3^m \equiv 2^m + (-2)^m \equiv 0 \pmod 5.$$

Moreover, for $m = n = 1$ one obtains $25 - 7 - 3 = 15$.

Assume now that m is even, say $m = 2k$. Then

$$7^m + 3^m = 7^{2k} + 3^{2k} \equiv ((-3)^{2k} + 3^{2k}) \pmod{10}$$

$$\equiv 2 \cdot 9^k \pmod{10} \equiv \pm 2 \pmod{10} \equiv 2 \text{ or } 8 \pmod{10}.$$

So, the last digit of the number $25^n - 7^m - 3^m$ is either 3 or 7. Because the number $25^n - 7^m - 3^m$ is divisible by 3, the required number cannot be 7. The situation $|25^n - 7^m - 3^m| = 3$ also cannot occur, because $25^n - 7^m - 3^m \equiv 1 \pmod 8$.

Problem 11.8. *Given an integer d, let*

$$S = \{m^2 + dn^2 | m, n \in \mathbb{Z}\}.$$

Let $p, q \in S$ be such that p is a prime and $r = \dfrac{q}{p}$ is an integer. Prove that $r \in S$.

(1999 Hungary-Israel Mathematical Competition)

Solution. Note that

$$(x^2 + dy^2)(u^2 + dv^2) = (xu \pm dyv)^2 + d(xv \mp yu)^2.$$

Write $q = a^2 + db^2$ and $p = x^2 + dy^2$ for integers a, b, x, y. Reversing the above construction yields the desired result. Indeed, solving for u and

v after setting $a = xu + dyv$, $b = xv - yu$ and $a = xu - dyv$, $b = xv + yu$ gives

$$u_1 = \frac{ax - dby}{p}, \quad v_1 = \frac{ay + bx}{p},$$
$$u_2 = \frac{ax + dby}{p}, \quad v_2 = \frac{ay - bx}{p}.$$

Note that

$$(ay + bx)(ay - bx) = (a^2 + db^2)y^2 - (x^2 + dy^2)b^2 \equiv 0 \pmod{p}.$$

Hence p divides one of $ay + bx$, $ay - bx$ so that one of v_1, v_2 is an integer. Without loss of generality, assume that v_1 is an integer. Because $r = u_1^2 + dv_1^2$ is an integer and u_1 is rational, u_1 is an integer as well and $r \in S$, as desired.

Problem 11.9. *Prove that every positive rational number can be represented in the form*

$$\frac{a^3 + b^3}{c^3 + d^3}$$

where a, b, c, d are positive integers.

(1999 IMO Shortlist)

Solution. We firstly claim if m, n are positive integers such that the rational number $r = \frac{m}{n}$ belongs to the interval (1,2) then r can be represented in the form

$$\frac{a^3 + b^3}{c^3 + d^3}.$$

This can be realized by taking $a^2 - ab + b^2 = a^2 - ad + d^2$, i.e. $b + d = a$ and $a + b = 3m$, $a + d = 2a - b = 3n$, that is $a = m + n$, $b = 2m - n$, $d = 2n - m$.

We will prove now the required conclusion. If $s > 0$ is a rational number take positive integers p, q such that $q < \frac{p^3}{q^3}s < 2$. There exists positive integers a, b, d such that $\frac{p^3}{q^3}s = \frac{a^3 + b^3}{a^3 + d^3}$, whence $s = \frac{(aq)^3 + (bq)^3}{(ap)^3 + (bp)^3}$.

Problem 11.10. *Two positive integers are written on the board. The following operation is repeated: if $a < b$ are the numbers on the board, then a is erased and $ab/(b-a)$ is written in its place. At some point the numbers on the board are equal. Prove that again they are positive integers.*

(1998 Russian Mathematical Olympiad)

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Solution. Call the original numbers x and y and let $L = \text{lcm}(x, y)$. For each number n on the board consider the quotient L/n; during each operation, the quotients L/b and L/a become L/b and $L/a - L/b$. This is the Euclidean algorithm, so the two equal quotients would be $gcd(L/b, L/a)$ and the two equal numbers on the board are $L/gcd(L/x, L/y)$. But $gcd(L/x, L/y) = 1$, because otherwise x and y would both divide $L/gcd(L/x, L/y)$ and L would not be a least common multiple. So, the two equal numbers equal $L = \text{lcm}(x, y)$, an integer.

Second solution. Again, let x and y be the original numbers and suppose both numbers eventually equal N. We prove by induction, on the number of steps k before we obtain (N, N), that all previous numbers divide N. Specifically, $x|N$, so N must be an integer.

The claim is clear for $k = 0$. Now assume that k steps before we obtain (N, N), the numbers on the board are $(c, d) = (N/p, N/q)$ for some integers $p < q$. Then reversing the operation, the number erased in the $(k + 1)$st step must be $cd/(c+d) = N/(p+q)$ or $cd/(c-d) = N/(q-p)$, completing the inductive step.

Problem 11.11. *Let $f(x) + a_0 + a_1x + \cdots + a_mx^m$, with $m \geq 2$ and $a_m \neq 0$, be a polynomial with integer coefficients. Let n be a positive integer, and suppose that:*

i) $a_2, a_3, \ldots, a_m$ are divisible by all the prime factors of n;

ii) a_1 and n are relatively prime.

Prove that for any positive integer k, there exists a positive integer c such that $f(c)$ is divisible by n^k.

(2001 Romanian IMO Team Selection Test)

Solution. Consider any integers c_1, c_2 such that $c_1 \not\equiv c_2 \pmod{n^k}$. Observe that if $n^k|st$ for some integers s, t where t is relatively prime to n, then $n^k|s$. In particular, $n^k \nmid (c_1 - c_2)t$ if t is relatively prime to n.

Note that

$$f(c_1) - f(c_2) = (c_1 - c_2)a_1 + \sum_{i=2}^{m} a_i(c_1^i - c_2^i)$$

$$= (c_1 - c_2) \underbrace{\left(a_1 + \sum_{i=2}^{m} \left(a_i \sum_{j=0}^{i-1} (c_1^j c_2^{i-1-j}) \right) \right)}_{t}.$$

For any prime p dividing n, p divides $a_2, \ldots, a_m$ but not a_1. Hence, p does not divide the second factor t in the expression above. This implies that t is relatively prime to n, so n^k does not divide the product $(c_1 - c_2)t = f(c_1) - f(c_2)$.

Therefore, $f(0), f(1), \ldots, f(n^k - 1)$ are distinct modulo n^k, and one of them, say $f(c)$, must be congruent to 0 modulo n^k. That is, $n^k | f(c)$, as desired.

Problem 11.12. *Let x, a, b be positive integers such that $x^{a+b} = a^b b$. Prove that $a = x$ and $b = x^x$.*

(1998 Iranian Mathematical Olympiad)

Solution. If $x = 1$, then $a = b = 1$ and we are done. So we may assume $x > 1$. Write $x = \prod_{i=1}^{n} p_i^{\gamma_i}$, where the p_i are the distinct prime factors of x. Since a and b divide x^{a+b}, we have $a = \prod p_i^{\alpha_i}$ and $b = \prod p_i^{\beta_i}$ for some nonnegative integers α_i, β_i.

First suppose that some β_i is zero, that is, p_i does not divide b. Then the given equation implies that $\gamma_i(a + b) = \alpha_i b$, so that $(\alpha_i - \gamma_i)b = a\gamma_i$. Now $p_i^{\alpha_i}$ divides a but is coprime to b, so $p_i^{\alpha_i}$ divides $\alpha_i - \gamma_i$ also. But $p_i^{\alpha_i} > \alpha_i$ for $\alpha_i > 0$, contradiction. We conclude that $\beta_i > 0$.

Now from the fact that

$$\gamma_i(a + b) = \beta_i + b\alpha_i$$

and the fact that p^{β_i} does not divide β_i (again for size reasons), we deduce that p^{β_i} also does not divide a, that is, $\alpha_i < \beta_i$ for all i and so a divides b. Moreover, the equation above implies that a divides β_i, so we may write $b = c^a$ with $c \geq 2$ a positive integer.

Write $x/a = p/q$ in lowest terms (so $gcd(p, q) = 1$). Then the original equation becomes $x^a p^b = bq^b$. Now p^b must divide b, which can only occur if $p = 1$. That is, x divides a.

If $x \neq a$, then there exists i with $\alpha_i \geq \gamma_i + 1$, so

$$\gamma_i(a + b) = \beta_i + \alpha_i b \geq (\gamma_i + 1)b$$

and so $\gamma_i a > b$. On the other hand, a is divisible by $p_i^{\gamma_i}$, so in particular $a \geq \gamma_i$. Thus $a^2 > b = c^a$, or $\sqrt{c} < a^{1/a}$; however, $a^{1/a} < \sqrt{2}$ for $a \geq 5$, so this can only hold for $c = 2$ and $a = 3$, in which case $b = 8$ is not divisible by a, contrary to our earlier observation.

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Thus $x = a$, and from the original equation we get $b = x^x$, as desired.

Problem 11.13. *Let m, n be integers with $1 \leq m < n$. In their decimal representations, the last three digits of 1978^m are equal, respectively, to the last three digits of 1978^n. Find m and n such that $m + n$ is minimal.*

(20th IMO)

Solution. Since 1978^n and 1978^m agree in their last three digits, we have

$$1978^n - 1978^m = 1978^m(1978^{n-m} - 1) \equiv 0 \pmod{10^3}.$$

From the decomposition $10^3 = 2^3 \cdot 5^3$ and since $1978^{n-m} - 1$ is odd we obtain $2^3 | 1978^m$. From $1978 = 2 \cdot 989$, it follows $m \geq 3$.

Let us write $m + n = (n - m) + 2m$. Our strategy is to minimize $m + n$ by taking $m = 3$ and seek the smallest value of $n - m$ such that

$$1978^{n-m} \equiv 1 \pmod{5^3}.$$

Since $(1978, 5) = 1$, the problem is to find the order h of the residue-class $1978 \pmod{125}$. It is known that the order h of an inversible residue-class modulo m is a divisor of $\varphi(m)$, where φ is the Euler function. In our case,

$$\varphi(125) = 5^2(5 - 1) = 100.$$

Hence, $h|100$. From $1978^h \equiv 1 \pmod{125}$ we also have $1978^h \equiv 1 \pmod 5$. But $1978^h \equiv 3^h \pmod 5$. Since the order of the residue-class $3 \pmod 5$ is 4, it follows $4|h$. Using the congruence $1978 \equiv -22 \pmod{125}$ we obtain:

$$1978^4 \equiv (-22)^4 \equiv 2^4 \cdot 11^4 \equiv 4^2 \cdot 121^2$$

$$\equiv (4 \cdot (-4))^2 \equiv (-1)^2 \equiv 256 \equiv 6 \not\equiv 1 \pmod{125}.$$

So we rule out the case $h = 4$. Because $h|100$, the next possibilities are $h = 20$ or $h = 100$. By a standard computation we have:

$$1978^{20} \equiv 6^5 \equiv 2^5 \cdot 3^5 \equiv 32 \cdot (-7) \equiv -224 \equiv 26 \pmod{125} \not\equiv 1 \pmod{125}.$$

Hence we necessarily have: $h = m - n = 100$ and $n + m = 106$.

Glossary

Arithmetic function
A function defined on the positive integers and which is complex valued.

Arithmetic-Geometric Means Inequality
If n is a positive integer and $a_1, a_2, \ldots, a_n$ are nonnegative real numbers, then

$$\frac{1}{n}\sum_{i=1}^{n} a_i \geq (a_1 a_2 \cdots a_n)^{1/n},$$

with equality if and only if $a_1 = a_2 = \cdots = a_n$. This inequality is a special case of the **power mean inequality**.

Base b representation
Let b be an integer greater than 1. For any integer $n \geq 1$ there is a unique system $(k, a_0, a_1, \ldots, a_k)$ of integers such that $0 \leq a_i \leq b-1$, $i = 0, 1, \ldots, k$, $a_k \neq 0$ and

$$n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0.$$

Beatty's Theorem
Let α and β be two positive irrational real numbers such that

$$\frac{1}{\alpha} + \frac{1}{\beta} = 1.$$

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The sets $\{\lfloor\alpha\rfloor, \lfloor 2\alpha\rfloor, \lfloor 3\alpha\rfloor, \dots\}$, $\{\lfloor\beta\rfloor, \lfloor 2\beta\rfloor, \lfloor 3\beta\rfloor, \dots\}$ form a partition of the set of positive integers.

Bernoulli's Inequality
For $x > -1$ and $a > 1$,

$$(1+x)^a \geq 1 + ax,$$

with equality when $x = 0$.

Bezout's Identity
For positive integers m and n, there exist integers x and y such that $mx + by = gcd(m, n)$.

Binomial Coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

the coefficient of x^k in the expansion of $(x+1)^n$.

Binomial Theorem
The expansion

$$(x+y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y + \cdots + \binom{n}{n-1}xy^{n-1} + \binom{n}{n}y^n$$

Canonical factorization
Any integer $n > 1$ can be written uniquely in the form

$$n = p_1^{\alpha_1} \dots p_k^{\alpha_k},$$

where $p_1, \dots, p_k$ are distinct primes and $\alpha_1, \dots, \alpha_k$ are positive integers.

Carmichael's integers
The composite integers n satisfying $a^n \equiv a \pmod{n}$ for any integer a.

Complete set of residue classes modulo n
A set S of integers such that for each $0 \leq i \leq n-1$ there is an element $s \in S$ with $i \equiv s \pmod{n}$.

Congruence relation
Let a, b, and m be integers, with $m \neq 0$. We say that a and b are congruent modulo m if $m|a-b$. We denote this by $a \equiv b \pmod{m}$. The relation "$\equiv$" on the set $\mathbb{Z}$ of integers is called the congruence relation.

Convolution product
The arithmetic function defined by

$$(f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right),$$

where f and g are two arithmetic functions.

Division Algorithm
For any positive integers a and b there exists a unique pair (q, r) of non-negative integers such that $b = aq + r$ and $r < a$.

Euclidean Algorithm
Repeated application of the Division Algorithm:

$$\begin{aligned}
&m = nq_1 + r_1,\ 1 \le r_1 < n,\\
&n = r_1q_2 + r_2,\ 1 \le r_2 < r_1,\\
&\dots\\
&r_{k-2} = r_{k-1}q_k + r_k,\ 1 \le r_k < r_{k-1},\\
&r_{k-1} = r_kq_{k+1} + r_{k+1},\ r_{k+1} = 0
\end{aligned}$$

This chain of equalities is finite because $n > r_1 > r_2 > \cdots > r_k$.

Euler's Theorem
Let a and m be relatively prime positive integers. Then

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

Euler's totient function
The function φ defined by $\varphi(m)$ = the number of all positive integers n less than m that are relatively prime to m.

Factorial base expansion
Every positive integer k has a unique expansion

$$k = 1! \cdot f_1 + 2! \cdot f_2 + 3! \cdot f_3 + \cdots + m! \cdot f_m,$$

where each f_i is an integer, $0 \le f_i \le i$ and $f_m > 0$.

Fermat's Little Theorem (F.L.T.)

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Let a be a positive integer and let p be a prime. Then

$$a^p \equiv a \pmod{p}.$$

Fermat's numbers
The integers $f_n = 2^{2^n} + 1$, $n \geq 0$.

Fibonacci sequence
The sequence defined by $F_0 = 1$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for every positive integer n.

Floor function
For a real number x there is a unique integer n such that $n \leq x < n + 1$. We say that n is the greatest integer less than or equal to x or the floor of x and we denote $n = \lfloor x \rfloor$.

Fractional part
The difference $x - \lfloor x \rfloor$ is called the fractional part of x and is denoted by $\{x\}$.

Fundamental Theorem of Arithmetic
Any integer n greater than 1 has a unique representation (up to a permutation) as a product of primes.

Hermite's Identity
For any real number x and for any positive integer n,

$$\lfloor x \rfloor + \left\lfloor + \frac{1}{n} \right\rfloor + \left\lfloor + \frac{2}{n} \right\rfloor + \cdots + \left\lfloor + \frac{n-1}{n} \right\rfloor = \lfloor nx \rfloor.$$

Legendre's formula
For any prime p and any positive integer n,

$$e_p(n) = \sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor.$$

Legendre's function
Let p be a prime. For any positive integer n, let $e_p(n)$ be the exponent of p in the prime factorization of $n!$.

Legedre's symbol
Let p be an odd prime and let a be a positive integer not divisible by p. The Legendre's symbol of a with respect to p is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue mod } p \\ -1 & \text{otherwise} \end{cases}$$

Linear Diophantine equation
An equation of the form

$$a_1x_1 + \cdots + a_nx_n = b,$$

where $a_1, a_2, \ldots, a_n, b$ are fixed integers.

Linear recurrence of order k
A sequence $x_0, x_1, \ldots, x_2, \ldots$ of complex numbers defined by

$$x_n = a_1x_{n-1} + a_2x_{n-2} + \cdots + a_kx_{n-k}, \quad n \geq k$$

where $a_1, a_2, \ldots, a_k$ are given complex numbers and $x_0 = \alpha_0$, $x_1 = \alpha_1, \ldots,$ $x_{k-1} = \alpha_{k-1}$ are also given.

Lucas' sequence
The sequence defined by $L_0 = 2$, $L_1 = 1$, $L_{n+1} = L_n + L_{n-1}$ for every positive integer n.

Mersenne's numbers
The integers $M_n = 2^n - 1$, $n \geq 1$.

Möbius function
The arithmetic function μ defined by

$$\mu(n) = \begin{cases} 1 & \text{if} \quad n = 1, \\ 0 & \text{if} \quad p^2|n \text{ for some prime } p > 1, \\ (-1)^k & \text{if} \quad n = p_1 \ldots p_k, \text{ where } p_1, \ldots, p_k \text{ are distinct primes} \end{cases}$$

Möbius inversion formula
Let f be an arithmetic function and let F be its summation function. Then

$$f(n) = \sum_{d|n} \mu(d) F\left(\frac{n}{d}\right)$$

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Multiplicative function
An arithmetic function $f \neq 0$ with the property that for any relative prime positive integers m and n,

$$f(mn) = f(m)f(n).$$

Number of divisors
For a positive integer n denote by $\tau(n)$ the number of its divisors. It is clear that

$$\tau(n) = \sum_{d|n} 1$$

Order modulo m
We say that a has order d modulo m, denoted by $o_m(a) = d$, if d is the smallest positive integer such that $a^d \equiv 1 \pmod{m}$.

Pell's equation
The quadratic equation $u^2 - Dv^2 = 1$, where D is a positive integer that is not a perfect square.

Perfect number
An integer $n \geq 2$ with the property that the sum of its divisors is equal to $2n$.

Prime Number Theorem
The relation

$$\lim_{n\to\infty} \frac{\pi(n)}{\dfrac{n}{\log n}} = 1,$$

where $\pi(n)$ denotes the number of primes $\leq n$.

Prime Number Theorem for arithmetic progressions
Let $\pi_{r,a}^{(n)}$ be the number of primes in the arithmetic progression $a, a+r, a+2r, a+3r, \ldots$, less than n, where a and r are relatively prime. Then

$$\lim_{n\to\infty} \frac{\pi_{r,a}(n)}{\dfrac{n}{\log n}} = \frac{1}{\varphi(r)}$$

This was conjectured by Legendre and Dirichlet and proved by de la Vallée Poussin.

Pythagorean equation
The Diophantine equation $x^2 + y^2 = z^2$.

Pythagorean triple
A triple of the form $(m^2 - n^2, 2mn, m^2 + n^2)$, where m and n are positive integers.

Quadratic residue mod m
Let a and m be positive integers such that $gcd(a, m) = 1$. We say that a is a quadratic residue mod m if the convergence $x^2 \equiv a \pmod{m}$ has a solution.

Quadratic Reciprocity Law of Gauss
If p and q are distinct odd primes, then

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$$

Sum of divisors
For a positive integer n denote by $\sigma(n)$ the sum of its positive divisors including 1 and n itself. It is clear that

$$\sigma(n) = \sum_{d|n} d$$

Summation function
For an arithmetic function f the function F defined by

$$F(n) = \sum_{d|n} f(d)$$

Wilson's Theorem
For any prime p, $(p-1)! \equiv -1 \pmod{p}$.

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References

[1] Andreescu, T.; Feng, Z., *101 Problems in Algebra from the Training of the USA IMO Team*, Australian Mathematics Trust, 2001.

[2] Andreescu, T.; Feng, Z., *102 Combinatorial Problems from the Training of the USA IMO Team*, Birkhäuser, 2002.

[3] Andreescu, T.; Feng, Z., *103 Trigonometry Problems from the Training of the USA IMO Team*, Birkhäuser, 2004.

[4] Feng, Z.; Rousseau, C.; Wood, M., *USA and International Mathematical Olympiads 2005*, Mathematical Association of America, 2006.

[5] Andreescu, T.; Feng, Z.; Loh, P., *USA and International Mathematical Olympiads 2004*, Mathematical Association of America, 2005.

[6] Andreescu, T.; Feng, Z., *USA and International Mathematical Olympiads 2003*, Mathematical Association of America, 2004.

[7] Andreescu, T.; Feng, Z., *USA and International Mathematical Olympiads 2002*, Mathematical Association of America, 2003.

[8] Andreescu, T.; Feng, Z., *USA and International Mathematical Olympiads 2001*, Mathematical Association of America, 2002.

Troy University Resources

[9] Andreescu, T.; Feng, Z., *USA and International Mathematical Olympiads 2000*, Mathematical Association of America, 2001.

[10] Andreescu, T.; Feng, Z.; Lee, G.; Loh, P., *Mathematical Olympiads: Problems and Solutions from around the World, 2001–2002*, Mathematical Association of America, 2004.

[11] Andreescu, T.; Feng, Z.; Lee, G., *Mathematical Olympiads: Problems and Solutions from around the World, 2000–2001*, Mathematical Association of America, 2003.

[12] Andreescu, T.; Feng, Z., *Mathematical Olympiads: Problems and Solutions from around the World, 1999–2000*, Mathematical Association of America, 2002.

[13] Andreescu, T.; Feng, Z., *Mathematical Olympiads: Problems and Solutions from around the World, 1998–1999*, Mathematical Association of America, 2000.

[14] Andreescu, T.; Kedlaya, K., *Mathematical Contests 1997–1998: Olympiad Problems from around the World, with Solutions*, American Mathematics Competitions, 1999.

[15] Andreescu, T.; Kedlaya, K., *Mathematical Contests 1996–1997: Olympiad Problems from around the World, with Solutions*, American Mathematics Competitions, 1998.

[16] Andreescu, T.; Kedlaya, K.; Zeitz, P., *Mathematical Contests 1995–1996: Olympiad Problems from around the World, with Solutions*, American Mathematics Competitions, 1997.

[17] Andreescu, T.; Enescu, B., *Mathematical Olympiad Treasures*, Birkhäuser, 2003.

[18] Andreescu, T.; Gelca, R., *Mathematical Olympiad Challenges*, Birkhäuser, 2000.

[19] Andreescu, T., Andrica, D., *An Introduction to Diophantine Equations*, GIL Publishing House, 2002.

[20] Andreescu, T.; Andrica, D., *360 Problems for Mathematical Contests*, GIL Publishing House, 2003.

[21] Andreescu, T.; Andrica, D.; Feng, Z., *104 Number Theory Problems From the Training of the USA IMO Team*, Birkhäuser, 2006 (to appear).

[22] Djuikić, D.; Janković, V.; Matić, I.; Petrović, N., *The IMO Compendium*, A Collection of Problems Suggested for the International Mathematical Olympiads: 1959-2004, Springer, 2006.

[23] Doob, M., *The Canadian Mathematical Olympiad 1969–1993*, University of Toronto Press, 1993.

[24] Engel, A., *Problem-Solving Strategies*, Problem Books in Mathematics, Springer, 1998.

[25] Everest, G., Ward, T., *An Introduction to Number Theory*, Springer, 2005.

[26] Fomin, D.; Kirichenko, A., *Leningrad Mathematical Olympiads 1987–1991*, MathPro Press, 1994.

[27] Fomin, D.; Genkin, S.; Itenberg, I., *Mathematical Circles*, American Mathematical Society, 1996.

[28] Graham, R. L.; Knuth, D. E.; Patashnik, O., *Concrete Mathematics*, Addison-Wesley, 1989.

[29] Gillman, R., *A Friendly Mathematics Competition*, The Mathematical Association of America, 2003.

[30] Greitzer, S. L., *International Mathematical Olympiads, 1959–1977*, New Mathematical Library, Vol. 27, Mathematical Association of America, 1978.

[31] Grosswald, E., *Topics from the Theory of Numbers*, Second Edition, Birkhäuser, 1984.

[32] Hardy, G.H.; Wright, E.M., *An Introduction to the Theory of Numbers*, Oxford University Press, 5th Edition, 1980.

[33] Holton, D., *Let's Solve Some Math Problems*, A Canadian Mathematics Competition Publication, 1993.

Troy University Resources

[34] Kedlaya, K; Poonen, B.; Vakil, R., *The William Lowell Putnam Mathematical Competition 1985–2000*, The Mathematical Association of America, 2002.

[35] Klamkin, M., *International Mathematical Olympiads, 1978–1985*, New Mathematical Library, Vol. 31, Mathematical Association of America, 1986.

[36] Klamkin, M., *USA Mathematical Olympiads, 1972–1986*, New Mathematical Library, Vol. 33, Mathematical Association of America, 1988.

[37] Kürschák, J., *Hungarian Problem Book, volumes I & II*, New Mathematical Library, Vols. 11 & 12, Mathematical Association of America, 1967.

[38] Kuczma, M., *144 Problems of the Austrian–Polish Mathematics Competition 1978–1993*, The Academic Distribution Center, 1994.

[39] Kuczma, M., *International Mathematical Olympiads 1986–1999*, Mathematical Association of America, 2003.

[40] Larson, L. C., *Problem-Solving Through Problems*, Springer-Verlag, 1983.

[41] Lausch, H. *The Asian Pacific Mathematics Olympiad 1989–1993*, Australian Mathematics Trust, 1994.

[42] Liu, A., *Chinese Mathematics Competitions and Olympiads 1981–1993*, Australian Mathematics Trust, 1998.

[43] Liu, A., *Hungarian Problem Book III*, New Mathematical Library, Vol. 42, Mathematical Association of America, 2001.

[44] Lozansky, E.; Rousseau, C. *Winning Solutions*, Springer, 1996.

[45] Mordell, L.J., *Diophantine Equations*, Academic Press, London and New York, 1969.

[46] Niven, I., Zuckerman, H.S., Montgomery, H.L., *An Introduction to the Theory of Numbers*, Fifth Edition, John Wiley & Sons, Inc., New York, Chichester, Brisbane, Toronto, Singapore, 1991.

[47] Savchev, S.; Andreescu, T. *Mathematical Miniatures*, Anneli Lax New Mathematical Library, Vol. 43, Mathematical Association of America, 2002.

[48] Shklarsky, D. O; Chentzov, N. N; Yaglom, I. M., *The USSR Olympiad Problem Book*, Freeman, 1962.

[49] Slinko, A., *USSR Mathematical Olympiads 1989–1992*, Australian Mathematics Trust, 1997.

[50] Szekely, G. J., *Contests in Higher Mathematics*, Springer-Verlag, 1996.

[51] Tattersall, J.J., *Elementary Number Theory in Nine Chapters*, Cambridge University Press, 1999.

[52] Taylor, P. J., *Tournament of Towns 1980–1984*, Australian Mathematics Trust, 1993.

[53] Taylor, P. J., *Tournament of Towns 1984–1989*, Australian Mathematics Trust, 1992.

[54] Taylor, P. J., *Tournament of Towns 1989–1993*, Australian Mathematics Trust, 1994.

[55] Taylor, P. J.; Storozhev, A., *Tournament of Towns 1993–1997*, Australian Mathematics Trust, 1998.

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